Affirmed Networks Gateway Administration Guide

Release 9.4.0.0
Updated: October 2019

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Affirmed Networks Gateway Administration Guide, Release 9.4.0.0
Part Number: 550-01087 Revision: A

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Preface

This guide describes Control and User Plane Separation (CUPS), 5G Non-Standalone and (5G-NSA) New Radio (NR) support.

This guide also describes an overview of the Affirmed Networks Serving Gateway (SGW), Packet Data Network Gateway (PGW), Gateway GPRS support node (GGSN), and Serving GPRS Support Node (SGSN) services running on the Mobile Content Cloud.

This guide also describes the gateway features and functions and the CLI objects and commands (in alphabetical order) used to configure gateway services. Some objects or commands are supported on specific gateways as noted.

For information on accessing the Command Line Interface (CLI), see the *Affirmed Networks CLI Reference Guide*. For information on statistics, see *Acuitas Performance Statistics*.

Note: *The Affirmed Networks Mobile Content Cloud™ (MCC) is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation team staff, and other qualified service personnel responsible for provisioning and monitoring the system. The audience should be familiar with Long Term Evolution (LTE), 3G, 4G, and 5G networks.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

This guide contains the following information:

Chapter/Appendix	Describes
Chapter 1, CUPS Overview	The Control and User Plane Separation (CUPS) features and functions.
Chapter 2, CUPS PFCP Overview	An overview of the Packet Forwarding Control Protocol (PFCP) objects and interfaces that support MCC CUPS.
Chapter 3, 5G NSA NR Overview	The Affirmed Networks 5G Non-Standalone (5G-NSA) New Radio (NR) solution.
Chapter 4, TAM Overview	An overview of the TAM solution.
Chapter 5, ISM VM Overview	The ISM MCC VMs and East/West (E/W) traffic reduction in the MCC.
Chapter 6, Gateway Overview	An overview of the Serving Gateway (SGW), Packet Data Network Gateway (PGW), Networks Gateway GPRS support node (GGSN), and Serving GPRS Support Node (SGSN).
Chapter 7, Geographic Redundancy Parameters	The geographic-redundancy solution supported on the MCC and geographic redundancy groups.
Chapter 8, Network Context Parameters	The network-context objects and parameters supported on the MCC such as arp, dns, ip-route, ip-subpools, loopback-ip, and nat-pool.
Chapter 9, Service Construct Parameters	The service-construct objects and parameters supported on the MCC such as advice-of-charge, application-rule-group, content-insertion, data-record-templates, offline-charging, online-charging, and tcp-optimization-profile.
Chapter 10, Services Parameters	The service objects and parameters supported on the MCC such as rating-group, quality-of-service, steering-policy, and wap-gateway.
Chapter 11, Workflow Parameters	The workflow objects and parameters supported on the MCC such as control-profile, data-profile, and subscriber-analyzer.
Chapter 12, Zone Parameters	The zone objects and parameters supported on the MCC such as gateway profiles for apn, ggns, mcc-gigw, pgw, and gateway common.
Chapter 13, CUPS Sx Interface Chapter 16, Gn/Gp Interface Chapter 17, S5/S8 and S4/S11 Interfaces Chapter 14, S2a Interface Chapter 15, S2b Interface Chapter 18, S6b Interface	The message flows, configuring and monitoring, and message definitions used to control and monitor the following interfaces: Sx, Gn/Gp, S5/S8, S2a, S2b, and S6b.

Features in this Release

This release includes the following new features:

Note: *For a complete list of the features supported in this release, see the software release notes. In addition to the information in this guide, see the additional Affirmed Networks documentation listed in the Affirmed Networks Documentation Library. For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, see Affirmed Networks Terms and Acronyms.*

Feature	Description
Enhancements for flow-idle-timeout-filter See service-construct flow-idle-timeout-filter (page 9-28).	The flow-idle-timeout-filter now supports the following configuration options: source IP address range, destination IP address range, and IP type. These new options are in addition to the existing destination port range and IP protocol options. To configure the new options: `an3000-mcm-slot7cpu1(config)#` **`service-construct flow-idle-timeout-filter <`*`name`*`> ip-version [v4-only|v4-or-v6|v6-only] network-ip-address-end <`*`address`*`> network-ip-address-start <`*`address`*`> network-ip-v6-address-end <`*`address`*`> network-ip-v6-address-start <`*`address`*`> ue-ip-address-end <`*`address`*`> ue-ip-address-start <`*`address`*`> ue-ip-v6-address-end <`*`address`*`> ue-ip-v6-address-start <`*`address`*`>`**
Support for PGW selection of serving network MCC/MNC and TMO-Transparent Data See Configuring PGW Selection of MCC/MNC Values (page 12-45) and Configuring PGW Selection of a TMO-Transparent Data Value (page 12-47).	This feature lets the PGW select MCC/MNC (PLMN ID) values in a configured priority order by using the GTP message from the SGSN/SGW, a sig-info-table, or the RADIUS Access Accept Message from the RADIUS Authentication Server. This feature also lets you configure the selection of TMO-Transparent Data values on the PGW when the RADIUS Access-Accept Message does not return a transparent data value. The PGW selects a value from a sig-info-table with PLMN ID mappings or from a sig-info-table with SGSN/SGW subnet mappings. This is enabled at the APN level.
Support for OSPFv2 adjacency establishment between MCC shared-subnet interfaces and a peer router See network-context ip-shared-subnet (page 8-10).	This feature allows MCC interfaces that share an IP subnet to establish OSPFv2 adjacency with one peer router that is in the same subnet and same OSPF area and is not an Affirmed Networks device. The peer router must have only one IP interface in the same subnet as the MCC shared-subnet interfaces (many-to-one connectivity). The priority for all OSPF interfaces should be set to zero to support this feature.

Feature	Description
Support for mapping a virtual APN to multiple real APNs See workflow control-profile (page 11-5).	This feature adds support for mapping a virtual APN to a list of real APNs, and evenly distributing sessions among the real APNs. The virtual APN and the real APNs can be in the same network context or in different nework contexts. The real APNs can all be in the same network context or in different network contexts. The real APNs should have UE IP pools of the same size. Configure the mapping by using the new **mapped-apn-name-list** command: `an3000-mcm-slot7cpu1(config)#` **workflow control-profile <*name*> mapped-apn-name-list <*list*>** where: **<*list*>** is a list of real APNs to map to. You can list up to 128 APNs. To support this feature, you must also have APN mapping control set to explicit: `an3000-mcm-slot7cpu1(config)#` **workflow control-profile <*name*> apn-mapping-control explicit**
Support for reporting performance statistics for the indivdual APNs that are mapped to an APN See workflow control-profile (page 11-5).	This feature adds support for reporting performance statistics for the individual APNs that are mapped to an APN. By default, this feature is not enabled. Prior to this release, APN performance statistics could only be reported for the mapped APN. ***Note:*** *The following configuration maps an APN to an APN:* `an3000-mcm-slot7cpu1(config)#` **workflow control-profile <*name*> apn-mapping-control explicit mapped-apn-name <*name*>** Configure statistics reporting for the individual, original APNs by using the new **apn-statistics-control** command: `an3000-mcm-slot7cpu1(config)#` **workflow control-profile <*name*> apn-statistics-control [original|mapped]** where: **original** — Specifies that statistics are reported for individual APNs. **mapped** — (default) Specifies that statistics are reported for the mapped APN. To view APN performance statistics, use the following command: `an3000-mcm-slot7cpu1(config)#` **show zone default gateway statistics cluster-level call-performance apn <*name*>** If **apn-statistics-control** is set to **original**, detailed APN performance statistics are shown for the individual, original APN. APN statistics for the mapped APN will still show statistics for IP address allocation activities, which are performed by the mapped APN (for example, `remote-ipv4-reserve-successes` and `static-ipv4-reserve-successes`). If **apn-statistics-control** is set to **mapped**, detailed APN performance statistics are shown for the mapped APN. APN statistics for an individual, original APN will still show statistics for UE connection activities, which are performed before APN mapping takes place (for example, `create-session-attempts`, `total-redirected-apn-subscriber-count`, and `create-session-accepts-apn-matching-type-exact-match`).

Feature	Description
Support for load balancing high-throughput sessions across ISMs, HD-Cores, and proxy instances for Workflow VMs. See zone gateway apn (page 12-7)	This feature allows load balancing of high throughput sessions across Workflow VMs. Load balancing is performed across ISMs, HD-Cores, and proxy instances. This feature is supported for CUPS running with TAM enabled on ISMs. Prior to this release, load balancing was based only on the session count. To enable this feature on an APN, use the new **is-high-throughput-apn** command: an3000-mcm-slot7cpu1(config)# **zone default gateway apn <*apn-name*> is-high-throughput-apn true** This specifies that all sessions associated with this APN are considered high throughput and are load balanced across ISMs, HD-Cores, and proxy instances.
Support for fragmentation of tunneled IPv6 packets exceeding the MRU size at the Gn interfaces. See network-context fragment-tunneled-ipv6 (page 8-6).	This feature adds support for fragmenting tunneled IPv6 packets that egress through Gn interfaces and exceed the MRU size. Prior to this release, MCC supported fragmentation of only IPv4 packets egressing the Gn interfaces. To enable or disable fragmentation of IPv6 packets that egress Gn interfaces (disabled is the default), use the new **fragment-tunneled-ipv6** command in the Gn network context: an3000-mcm-slot7cpu1(config)# **network-context <*name*> fragment-tunneled-ipv6 [true|false]**
Support for ALGs without NAT See Application Layer Gateway (page 10-24).	This feature adds support for the following ALGs when NAT is not performed on the source IP address and port for sent IP packets and is not performed on the destination IP address and port of received IP packets. ■ FTP — Supported for IPv4 and IPv6 ■ RTSP — Supported for IPv4 ■ SIP — Supported for IPv4 and IPv6 ■ TFTP — Supported for IPv4 and IPv6 To apply the ALGs to IP packets when NAT is not performed, use the **subscriber-firewall-flow** command: an3000-mcm-slot7cpu1(config)# **services subscriber-firewall subscriber-firewall-flow <*name*> action pass alg [ftp|rtsp|sip|tftp]** where: ■ **action pass** specifies that NAT is not performed on the packets. ■ **alg [ftp|rtsp|sip|tftp]** specifies the ALGs that should be applied to the packets.

Feature	Description

Support for IP address filters
 	
See service-construct ipv4-filter-list (page 9-57).
	This feature adds support for the ipv4-filter-list object, which filters IPv4 packets by matching one or more ranges of IP addresses.
	
	Prior to this release, you had to use the packet-filter object to filter packets by IP address ranges. The ipv4-filter-list is more efficient than the packet-filter when you are filtering only on IP addresses and a service-rule would require hundreds of packet-filters.
	
	You can configure 10 ipv4-filter-lists. An ipv4-filter-list supports IPv4. IPv6 is *not* currently supported.
	
	To configure an ipv4-filter-list:
	
	`an3000-mcm-slot7cpu1(config)# service-construct ipv4-filter-list <name> filter-type [network|ue] <CIDR range>`
	
	where:
	
	■ **`filter-type [network|ue]`** — Specifies whether the filter is for a network initiated connection or for a user equipment initiated connection.
	
	■ **`<CIDR range>`** — Specifies one or more CIDR ranges for the IP addresses, for example, 3.1.1.0/24. You can enter 4000 CIDR ranges for each ipv4-filter-list.
	
	To include the ipv4-filter-list in a service-rule:
	
	`an3000-mcm-slot7cpu1(config)# service-construct service-rule <name> ip-filter-list <name>`

Support for service-rule logical OR for matching conditions See service-construct service-rule (page 9-72).	This feature adds the new **`evaluation-mode`** parameter to specify how many of the configured classifier types must be matched for a service-rule to be considered a match: `an3000-mcm-slot7cpu1(config)# service-construct service-rule <name> evaluation-mode [AND | OR]` where: ■ **`AND`** – Each type of the configured classifiers must be matched. ■ **`OR`** – At least one type of the configured classifiers must be matched. Prior to this release, only the AND behavior existed.
Support for releasing connection after 30 seconds when TCP RST is received	This feature releases a connection after 30 seconds when TCP RST is received, even if the **`nat-flow-binding-time`** option under **`services subscriber-firewall subscriber-firewall-flow`** is configured for this connection.

<table>
<tr><th>Feature</th><th>Description</th></tr>
<tr><td>Added support for throttling control plane GTP messages on the PGW and SGW when the system is in an overload state

See overload-control max-arp-priority-level (page 12-56).</td><td>Added support for throttling control plane GTP messages on the PGW and SGW when the system is in an overload state. Throttling is based on configurable ARP priority levels. Emergency call setups are not affected by this feature.

To configure throttling, use the new overload-control max-arp-priority-level command:

an3000-mcm-slot7cpu1(config)# zone default gateway profile gateway-common overload-control max-arp-priority-level minor <<i>priority</i>> overload-control max-arp-priority-level major <<i>priority</i>> overload-control max-arp-priority-level critical <<i>priority</i>>

where <<i>priority</i>> is a bearer's ARP priority level.

When the system is in an overload state, the thresholds are used as follows:

■ Create session requests are rejected if the default bearer ARP priority level is greater than the max-arp-priority-level.

■ Create bearer requests and bearer resource command messages are rejected if the session does not have any bearers with an ARP priority level that is less than or equal to the max-arp-priority-level.

<i>Note: All new session setups are rejected when the system is in overload state and the MCD table sync to standby process is in progress.</i>

An overload condition can be any of the following. If more than one overload condition exists, the highest overload state is used for throttling.

■ PGW and SGW tasks exceed the CPU overload states:
– minor — 80 percent
– major — 88 percent
– critical — 95 percent

■ The number of PGW and SGW create-session-requests exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command:

an3000-mcm-slot7cpu1(config)# zone default gateway threshold node-level call-performance pdngw create-session-requests-[1|5|15]-minute-average major-enter <<i>int</i>> minor-enter <<i>int</i>> critical-enter <<i>int</i>> admin-state enabled

■ The node-level memory consumption percentage exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command:

an3000-mcm-slot7cpu1(config)# zone default system threshold node-level memory-stats csm-avg-[1|5|15]min-memory-util minor-enter <<i>int</i>> major-enter <<i>int</i>> critical-enter <<i>int</i>> admin-state enabled</td></tr>
</table>

Feature	Description

Added support for throttling control plane GTP messages on the PGW and SGW when the system is in an overload state.
(*continued*)
	■ The node-level CPU consumption percentage exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command:
	`an3000-mcm-slot7cpu1(config)# ` **`zone default system threshold node-level cpu-stats csm-avg-[1|5|15]min-cpu-util minor-enter <int> major-enter <int> critical-enter <int> admin-state enabled`**
	The following command displays the new overload statistics:
	`an3000-mcm-slot7cpu1# ` **`show zone default gateway statistics cluster-level call-performance [pgw|sgw] | include overload`**
	`num-create-session-rejects-overload`
	`num-create-bearer-rejects-overload`
	The following alarms are generated when the system is in an overload state:
	`anBladeOverload`
	`anThresholdCrossingAlarm`

Support for Multi-Access Edge Computing local breakout for CUPS See zone gateway user-plane-profile (page 12-84).	Added support for Multi-Access Edge Computing (MEC) local breakout on a standalone CUPS SGW. Local breakout allows traffic that only needs local applications to be serviced at the operator edge site. This reduces latency for local applications, and reduces back-haul requirements. Configure local breakout on a CUPS standalone SGW CPF: Enable local breakout in a user-plane-profile. `an3000-mcm-slot7cpu1(config)# ` **`zone default gateway user-plane-profile <name> mec-local-breakout enable`** Configure a user-plane-selector type of subscriber-analyzer that specifies the user-plane-profile that enables local breakout. `an3000-mcm-slot7cpu1(config)# ` **`workflow subscriber-analyzer <name> analyzer-type user-plane-selector priority <priority> action user-plane-profile <name> key <list of keys>`** When a key in this subscriber-analyzer is matched, the data-profile from the matching workflow-profile type of subscriber-analyzer is sent to the UPF. Data packets matching the data-profile undergo a local breakout.
SGX Downlink Data Notification message to MME always provides IMSI See Downlink Data Notification (page 17-87).	SGW will always include the IMSI value in the Downlink Data Notification message if the MME provided the IMSI in the Create Session Request.

Feature	Description
Added support for specifying how the IP type is chosen when PGW-originated messages are sent to the ePDG See zone gateway pgw (page 12-39) and Configuring the S2b Interface (page 15-11).	This release adds CLI parameters that let you specify how the IP type (IPv4 or IPv6) is chosen when the PGW originates messages that are sent to the ePDG. In previous releases, the PGW always used IPv6 when it originated messages that it sent to the ePDG. To configure how the IP type is chosen, use the following command: an3000-mcm-slot7cpu1(config)# **zone default gateway pgw <*gateway-name*> gtpc-endpoint-params dual-stack-behavior [initial-ip-type-used-by-peer|ipv4|ipv6]** where: ■ **initial-ip-type-used-by-peer** – The PGW should use the IP type that was used in the Create Session Request from the ePDG. ■ **ipv4** – The PGW should use IPv4. ■ **ipv6** – The PGW should use IPv6.
Enhancements for ip-address-allocation-mode See zone gateway apn (page 12-7).	The ip-address-allocation-mode now supports the new options ipv4v6-prefer-v4 and ipv4v6-prefer-v6. The ip-address-allocation-mode specifies how the type of allocated address (IPv4 or IPv6) is chosen when the UE requests IPV4IPV6. To configure the new options: an3000-mcm-slot7cpu1(config)# **zone default gateway apn <*name*> ip-address-allocation-mode [ipv4|ipv6|ipv4v6|ipv4v6-prefer-v4|ipv4v6-prefer-v6]** New options: ■ **ipv4v6-prefer-v4** – Allocate an IPv4 address if it is available; otherwise allocate an IPv6 address if it is available. If neither type of address is available, the session fails. ■ **ipv4v6-prefer-v6** – Allocate an IPv6 address if it is available; otherwise allocate an IPv4 address if it is available. If neither type of address is available, the session fails. Options that already existed: ■ **ipv4** – Allocate an IPv4 address if one is available. If an IPv4 address is not available, the session fails. ■ **ipv6** – Allocate an IPv6 address if one is available. If an IPv6 address is not available, the session fails. ■ **ipv4v6** – If a DAF is received from the UE, allocate both an IPv4 and an IPv6 address. If a DAF is not received, allocate an IPv4 address if it is available; otherwise allocate an IPv6 address if it is available. If neither type of address is available, the session fails. ***Note:*** *If the UE requests IPv4 or requests IPv6, that type of address is allocated if it is available, regardless of the ip-address-allocation-mode. If the requested type of IP address is not available, the session fails.*
Added support for specifying whether to include the MS IP address in the redirect URL See service-construct advice-of-charge http (page 9-4).	This feature adds a new configuration parameter under the advice-of-charge object to specify whether to include the MS IP address in the redirect URL. an3000-mcm-slot7cpu1(config)# **service-construct advice-of-charge <*name*> http include-msip [true|false]**

Feature	Description
Support for monitoring the different establishment states of TCP and UDP flows accepted by a subscriber firewall flow See Subscriber Firewall Monitoring of Flow Establishment States (page 10-26).	TCP and UDP flows accepted by a subscriber-firewall-flow can be monitored for different establishment states. Separate timers for each establishment state are used to remove flows and to drop unexpected TCP messages. The timers are used as follows: **tcp-open-timeout** — Specifies the timeout period between each message involved in the three-way TCP connection establishment. During this period, if the expected message is not received, the flow is removed. **tcp-established-timeout** — Specifies the timeout period for no flow activity after the three-way handshake is complete. If no flow activity is detected during this period, the flow is removed. **tcp-close-timeout** — Specifies the timeout period between each message involved in at the four-way TCP close operation, starting with receipt of the initial FIN from either direction. During this period, if the expected message is not received, the message is removed. **udp-open-timeout** — Specifies the timeout period between receiving the initial UDP packet and receiving the response UDP packet. If the response UDP packet is not received during the timeout period, the flow is removed. **udp-established-timeout** — Specifies the timeout period for no flow activity after the response UDP is received. If no flow activity is detected during this period, the flow is removed. In addition, a non-configurable internal timer removes a flow 30 seconds after receiving an RST. The timers can be configured in two ways: ■ Global — The same global tracking profile is used to control the timers for any subscriber-firewall-flow that enables global control. ■ Local — A subscriber-firewall-flow specifies a particular tracking profile to control timers. To globally configure monitoring of the flow establishment states: **1** Enable global control for the tracking profile selection. `an3000-mcm-slot7cpu1(config)#` **`services subscriber-firewall subscriber-firewall-flow`** ***`<name>`*** **`flow-state-tracking-profile-selection global`** **2** Configure a flow-timeout-filter. See service-construct flow-idle-timeout-filter (page 9-28). **3** Assign the flow-timeout filter to the flow-idle-timeout-rule that defines the timeout values. `an3000-mcm-slot7cpu1(config)#` **`service-construct flow-idle-timeout-rule`** ***`<name>`*** **`flow-idle-timeout-filter`** ***`<name>`*** **4** Configure the timeout values for the connection states in a flow-idle-timeout-rule. `an3000-mcm-slot7cpu1(config)#` **`service-construct flow-idle-timeout-rule`** ***`<name>`*** **`tcp-open-timeout`** ***`<seconds>`*** **`tcp-established-timeout`** ***`<seconds>`*** **`tcp-close-timeout`** ***`<seconds>`*** **`udp-open-timeout`** ***`<seconds>`*** **`udp-established-timeout`** ***`<seconds>`***

Feature	Description
Support for monitoring the different establishment states of TCP and UDP flows accepted by a subscriber firewall flow (*continued*)	**5** Assign the flow-idle-timeout-rule to a flow-idle-timeout-profile. `an3000-mcm-slot7cpu1(config)#` **`flow-idle-timeout-profile <name> flow-idle-timeout-rule <name>`** **6** Assign the flow-idle-timeout-profile to the data profile. `an3000-mcm-slot7cpu1(config)#` **`workflow data-profile <name> flow-idle-timeout-profile <name>`** All subscriber firewall flows that enable global monitoring use this flow-idle-timeout-profile. To locally configure monitoring of the flow establishment states: **1** Enable local control for the tracking profile selection. `an3000-mcm-slot7cpu1(config)#` **`services subscriber-firewall subscriber-firewall-flow <name> flow-state-tracking-profile-selection local`** **2** Configure the timeout values for the connection states in a flow-state-tracking-profile. `an3000-mcm-slot7cpu1(config)#` **`flow-state-tracking-profile <name> tcp-open-timeout <seconds> tcp-established-timeout <seconds> tcp-close-timeout <seconds> udp-open-timeout <seconds> udp-established-timeout <seconds>`** **3** Assign the flow-state-tracking-profile to the subscriber-firewall-flow. `an3000-mcm-slot7cpu1(config)#` **`services subscriber-firewall subscriber-firewall-flow <name> flow-state-tracking-profile <name>`**
Increased maximum MTU and MRU to 9100 bytes on all IP interfaces See network-context ip-interface (page 8-7) and the *Affirmed Networks System Administration Guide*.	IP interfaces on the MCC can now support an MTU and MRU as large as 9100 bytes. Prior to this release, the maximum MTU/MRU value was 1800. To configure the MTU/MRU size, first set the maximum packet size of the cluster with the new **`max-packet-size`** command: `an3000-mcm-slot7cpu1(config)#` **`cluster <id> max-packet-size <bytes>`** This value has a default of 2048, and a range of 2048 through 9216. This value must be at least as large as the MTU value you want to use for the IP interfaces, and you must reboot the cluster. Next, set the MTU for IP interfaces: `an3000-mcm-slot7cpu1(config)#` **`network-context <name> ip-interface mtu <bytes>`** This value has a default of 1500, and a range of 576 through 9100.
Added support for the Sd interface between the PCRF and the GiGW (acting as TDF) See Sd Interface (page 20-1).	The Sd interface is located between the PCRF and the Traffic Detection Function (TDF) and is used to detect traffic and report information on the detected application traffic. This feature adds support for the Sd interface between the PCRF and the GiGW (acting as a TDF) as defined in 3GPP 29.212, Release 15.3.0. The GiGW has a new Diameter session-learner interface used to establish and terminate TDF sessions to the PCRF.

Feature	Description
Increase the number of supported service-rules See service-construct service-rule (page 9-72)	Increased the number of supported service-rules from 4096 to 16000.

Conventions

This guide uses the following conventions, when applicable:

<table>
<tr><th>Description</th><th>Convention</th><th>Examples</th></tr>
<tr><td>Command names, keywords, dialog boxes, buttons, icons, menus</td><td>**Times New Roman bold font**</td><td>Use **configure_ha** to configure...
Enter **y**.
To continue, press **Enter**.</td></tr>
<tr><td>Selecting a menu item</td><td>**Menu** => **Option**</td><td>Select **Security** => **Group Management**.</td></tr>
<tr><td>Variable parameters for which you supply values</td><td>***<courier bold italic blue font>***</td><td>**configure_dr convert slave --masterhost <host IP>**</td></tr>
<tr><td>User input</td><td>**Courier bold blue font**</td><td>**cd /opt/Affirmed/NMS/bin**</td></tr>
<tr><td>Screen messages, system output, sample source code</td><td>Courier regular blue font</td><td>Checking for License...</td></tr>
<tr><td>Required keywords and arguments in square brackets</td><td>**[ ]**</td><td>**cluster [id] node [id] reboot**</td></tr>
<tr><td>Optional keywords and arguments in curly brackets</td><td>**{ }**</td><td>***{cdr-file-name}***</td></tr>
<tr><td>Keywords separated by the pipe symbol. Requires you to specify only one item from the set.</td><td>|</td><td>Is this correct? [y|n]</td></tr>
<tr><td>Keywords representing a single element in angle brackets</td><td>**< >**</td><td>***<host name>***</td></tr>
<tr><td>Asterisk after a CLI object indicates a required object.</td><td>*****</td><td>**name***</td></tr>
<tr><td>Variables, book and chapter titles, file path names, new terms, and emphasized text</td><td>*Times New Roman italic font*</td><td>See the *Acuitas User's Guide*.</td></tr>
</table>

Tip: Helpful suggestions.

Note: *Additional information that may apply to the subject text.*

CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the Adobe Reader ®. To download the latest version of the Adobe Reader software from the Adobe web site, go to http://get.adobe.com/reader/

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

Technical Assistance Center

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E-mail: prodsupport@affirmednetworks.com

24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838

Chapter 1

CUPS Overview

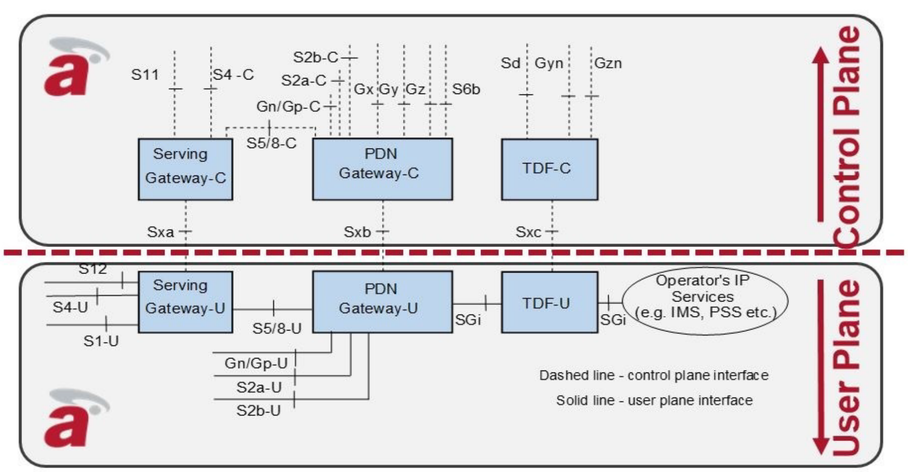
Affirmed Networks supports Control and User Plane Separation (CUPS) in its MCC virtualized Evolved Packet Core (EPC). CUPS is an LTE enhancement that separates the *Control Plane Function (CPF)* and *User Plane Function (UPF)* into separate “nodes.”

With an increasing number of VoLTE-capable terminals, multiple PDN connections are needed to support both Internet connections and external networks such as IP Multimedia Subsystem (IMS). This demand requires more deployment flexibility and optimization for PDN connections.

CUPS extends LTE by providing options for flexible network deployment through distributed or centralized functions, optimized PDN connections, and independent scaling of functions in the CPF and UPF. To support increased data traffic, Operators can add UPF nodes without adding more CPF nodes.

Figure 1-1 shows the CUPS reference model for standalone SGW and PGW in roaming and non-roaming environments.

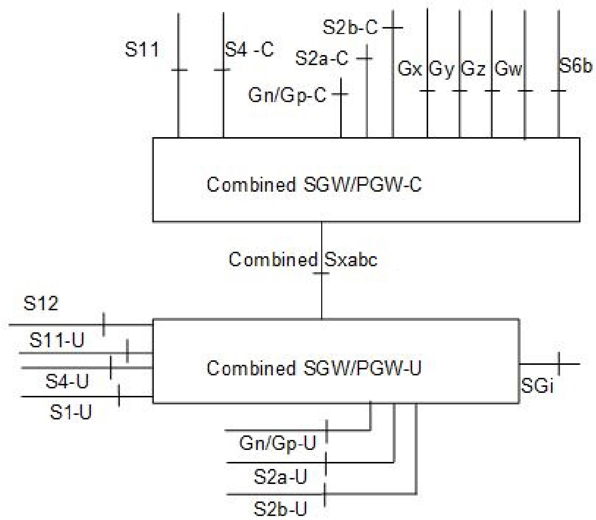
Figure 1-1 CUPS 3GPP - Standalone SGW and PGW



Note: *The Traffic Detection Function (TDF) is embedded in the PGW-C/U node. It is not supported as a standalone node.*

Figure 1-2 shows the CUPS reference model (3GPP Technical Specification 23.214) for the collocated/combined SGW and PGW SAEGW (System Architecture Evolution Gateway).

Figure 1-2 CUPS 3GPP - Collocated SGW/PGW (SAEGW)



CUPS Features

MCC CUPS supports the following features:

- Flexible deployment and independent scaling of CPFs and UPFs. A single CPF can operate with multiple UPFs.

Note: *This release does not support CUPS interoperability with third-party CPFs and UPFs. Session-based call flows use Affirmed proprietary vendor-specific Information Elements (IEs).*

- Efficient delivery of UPF data and reduces latency on application services, for example, by selecting UPF nodes that are closer to the RAN or applicable to a particular UE usage/type
- Inter-working with network entities that are outside the EPC, for example, IMS and in roaming situations

- *Packet Forwarding Control Protocol* (PFCP) signaling on the Sx interfaces between the SGW-C/U, PGW-C/U, and SAEGW-C/U
- Support for default and dedicated bearers for both standalone SGW/PGW and SAEGW
- Support for value added services
- Offline charging support on Ga, Gz and Local interfaces
- Online charging support on Gy interface
- RADIUS Authentication and Accounting
- Dynamic and static Policy Control and Charging (PCC) rules and usage monitoring (key) on the Gx interface with all existing triggers, including Time and Volume, using vendor-specific IEs.
- vProbe iEDR support of session and bearer records on the CPF and flow and HTTP records on the UPF
- Control plane packet mirroring and user plane packet mirroring that are independent
- CALEA Lawful Intercept with the option of sending X3 intercepted packets either from the CPF or UPF
- Traffic Anchoring Mode (TAM) for SSM/ISM VM personality support on the UPF
- UPFs including traffic detection, packet buffering, end-marker, and GTP-U Tunnel Endpoint Identifier (TEID) allocation
- Local High Availability (HA) on CPF and UPF
- Support for all mobility procedures including inter RAT handovers except for 2G
- CUPS and non-CUPS sessions operating concurrently in the same MCC control-plane cluster.

Note: *Running concurrent sessions assumes appropriate configuration for both CUPS and non-CUPS. A non-CUPS session is established when no user-plane selector matches the session setup. An ip-sub-pool must be configured and not associated with any user-plane-profiles.*

Sx Nodes and Interfaces

CUPS separates the Serving Gateway (SGW) and PDN Gateway (PGW) CPF and UPF into the following separate nodes/functions. The C/U nodes are connected through the Sx interfaces.

- Sxa Interface – Standalone SGW-C to SGW-U.
- Sxbc Interface – Standalone PGW-C to PGW-U. The Traffic Detection Function (TDF) is embedded in the PGW-C/U node. It is not supported as a standalone node; therefore, the Sxc interface functions as part of the Sxb interface.
- Sxabc Interface – Collocated SAEGW-C to SAEGW-U.

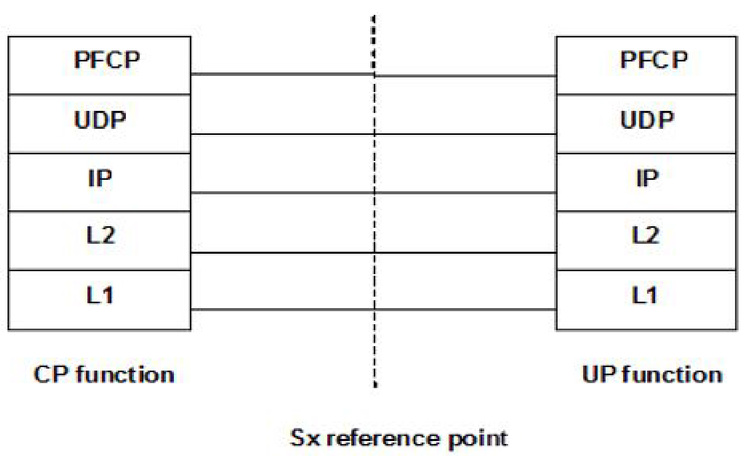
Note: *The SAEGW is a combined PGW/SGW.*

CUPS uses the Packet Forwarding Control Protocol (PFCP) for Sx interface communications between the Sx nodes. CUPS/PFCP provides:

- Reliable delivery/retry of PFCP messages to peers, including PFCP Session Establishment Request/Response and Modification Request/Response, Report from UPF and Report Response messages
- PFCP interface message-level statistics for CPF and UPF
- PFCP Sx protocol logging – PFCP protocol logging captures the node-related messages and session-related messages sent or received by the PGW-C, SGW-C, PGW-U and SGW-U (CPF and UPF)

CPFs and UPFs use the standard port # 8805 which is the destination port for PFCP packets.

Figure 1-3 PFCP Over UDP



Sx Associations and Node Procedures

An Sx association is set up between the control plane function (CPF) and user plane function (UPF) prior to establishing Sx sessions. Only one Sx association is setup between a given pair of CPF and UPF. The Sx association procedures include the following:

- **Sx Association Setup** – Initiated by either the CPF or the UPF.

  The Sx Association Setup Request includes:

  – Node ID (IPv4/IPv6), which identifies the Sx association.

  – List of optional features that the node supports. The UP supports TRST, FTUP, TREU and EMPU.

- **Sx Association Update** – The UPF initiates an Sx Association Update Request to modify an existing Sx association when the UPF is administered down.

  All sessions on the user plane are deleted if the `user-plane-service admin-state` is disabled.

- **Sx Association Release** – The CPF initiates an Sx Association Release Request to release the Sx association if the associated user-plane-profile is administered down.

  The CPF deletes the sessions associated with the user-plane and subsequently sends the Association Release Request after the configured pfcp-guard-time-interval.

  Upon receiving the Sx Association Release Request, the UPF deletes all pending Sx sessions and responds back with an Sx Association Release Response with a successful cause code.

  If the user-plane-service in the UPF is administered down, the UPF sends the CPF an Sx Association Update Request to notify the CPF to send an Sx Association Release Request. The Sx Association Update Request is populated with the UPF's configured pfcp-guard-time-interval. Upon receiving the Sx Association Update Request, the CPF starts deleting the sessions and after the received guard timer expires, sends an Sx Association Release Request to the UPF.

- **Heartbeat Messaging** – The CPF/UPF exchange Heartbeat Request/Response messages to check if a PFCP peer is alive once the Sx association between the CPF and UPF is established. The CPF will periodically send the Heartbeat Request to the UPF according to what is configured for the **`pfcp-heartbeat-periodic-interval`**.

For more information, see Sx Node Related Procedures (page 13-5).

Sx sessions support the following procedures:

- Sx Session Establishment, Modification, and Deletion procedures

- Sx Session Report procedure to report traffic usage or specific events, for example, a DL data packet arrival or start of an application

The CPF node sends the Sx Session Request message to the UPF node, and the UPF node responds with the Sx Session Response message.

For more information, see Chapter 13, CUPS Sx Interface.

Initiating an Association Between PFCP Peers

A PFCP peer association is created for every peer on the PFCP peer list that is associated with a gateway service. The following configuration initiates the association process on the PGW/SGW UPF.

On the CPF:

- The peer is configured using **`zone default gateway pfcp-peer`**.
- An association is made between the *pfcp-peer* in the SGW *sxa-pfcp-peer-list* with the PGW *sxb-pfcp-peer-list*.

  This configuration brings up two associations from the CPF perspective:
  - Sxa peer (used for standalone SGW sessions)
  - Sxb peer (used for standalone PGW sessions and collocated SGW+PGW sessions

On the UPF:

- The peer is configured using **`zone default gateway pfcp-peer`**.
- The *pfcp-peer* is associated with the user-plane-service *pfcp-peer-list*.

  This configuration brings up two associations from the UPF perspective with two peers to the CPF MCC:
  - Sxa peer (used for standalone SGW sessions)
  - Sxb peer (used for standalone PGW sessions and collocated SGW+PGW sessions)

Control Plane Function

The CPF supports the following features and functions:

Note: *For CUPS sessions, all signaling statistics that pertain to GTP-C are available only on the CPF. On the UPF, these statistics are displayed as zero.*

Note: *For CUPS sessions, data-usage statistics, such as call-performance uplink/downlink packet/byte counts, deep-packet-inspection statistics, and any other data profile services statistics (for example, proxy, content-filtering, and other VAS service statistics) are available only on the UPF (see* *page 1-7**). On the CPF, these statistics are displayed as zero.*

- Session/bearer resource management by establishing, modifying, and deactivating bearers (default and dedicated)
- Packet forwarding in the UPF is specified in the PFCP vendor-specific IEs in place of PDRs, FARs, QERs, and URRs.
- UE IP address management:
  - Allocates IP addresses from local IP pool and remote IP pool
- UE mobility:
  - Changes target GTP-U endpoint within 3GPP accesses
- PCC-related functions:
  - Binds bearer (Bearer QoS and TFT)
  - Supports uplink (UL) and downlink (DL) service-level charging (online and offline, per charging key)
  - Monitors usage
  - Reports events and detects applications
  - Redirects
  - Supports predefined PCC rules activation and deactivation
  - Supports PCC
- S1 release, buffering, and downlink data notification (failed paging, abnormal radio link release, number/fraction of packets/bytes dropped at SGW)
- Termination of CPF protocols, including GTP-C and Diameter (Gx, Gy, Gz interfaces) and RADIUS Authentication and Accounting
- Time and Volume accounting interfacing OFCS through interfaces (TS 32.240)
- Lawful Interception support on X1, X2, and X3 interfaces

User Plane Function

The UPF supports the following features and functions:

Note: *For CUPS sessions, all signaling statistics that pertain to GTP-C are available only on the CPF (see page 1-6). On the UPF, these statistics are displayed as zero.*

Note: *For CUPS sessions, data-usage statistics, such as call-performance uplink/downlink packet/byte counts, deep-packet-inspection statistics, and any other data profile services statistics (for example, proxy, content-filtering, and other VAS service statistics) are available only on the UPF. On the CPF, these statistics are displayed as zero.*

- 3GPP standard features, such as Policy Control and Charging (PCC), Lawful Intercept, and so on. See also PFCP 3GPP Compliance (page 1-11).
- PCC functions:
  - Detects service (DPI, IP-5-tuple)
  - Verifies and maps UL bearer binding of DL traffic to bearers
  - Supports UL and DL service-level gating
  - Enforces UL and DL service-level MBR
  - Supports UL and DL service-level charging (online and offline, per charging key)
  - Monitors usage
  - Reports events and detects applications
  - Redirects
  - Activates and deactivates predefined PCC rules
- Bearer/APN (UL/DL APN-AMBR, bearer MBR, non-GBR bearer on Gn/Gp interface)
- Packet forwarding and packet marking at the transport level
- UE mobility by sending the end marker after switching path to target node
- Time and Volume accounting per UE and bearer
- Lawful Interception
- Allocation of GTP-u F-TEID (Fully Qualified Tunnel Endpoint Identifier)

User Plane Selection

The CPF node (SGW-C, PGW-C, SAEGW-C) is primarily responsible for selecting its user-plane counterpart (SGW-U, PGW-U, SAEGW-U). During the initial connection establishment between the CPF and UPF functions, the CPF discovers the UPF's capabilities, for example:

- Dual Connectivity with New Radio (DCNR) type
- UE Location

Operators can configure the criteria for selecting a UE. The following factors also affect UPF selection:

- A UE is served by one SGW-C, however multiple SGW-Us can be selected for different PDN connections. The MME/SGSN provides the location of the UE. The CPF can use these parameters to select the correct UPF which is close to the UE's point of attachment.
- UPF selection uses existing subscriber analyzer keys such as APN (for SAEGW), IMSI, PLMN ID, TAI, and so on) and additional UE capabilities key for support of DCNR (Dual Connectivity with New Radio) type.
- The combined SGW-C/PGW-C nodes select a UPF (SGW-U/PGW-U pair) for the requested APN from all candidate node pairs instead of independently selecting the SGW-U first and then selecting the PGW-U.

CALEA Lawful Intercept Support

CUPS MCC supports CALEA Lawful Intercept through the Split X3 LI Interworking Function (SX3LIF) as defined in 3GPP TS 33.107 Section 12.9 "Functional Requirements for LI in case of Control and User Plane Separation."

CUPS-MCC Lawful Intercept functionality supports:

- Lawful Intercept for a split system where the CPF and UPF are both Affirmed Networks MCC gateways
- Support on UPF for sending X3 intercepted packets directly to a Law Enforcement Monitoring Facility (LEMF)
- GTP-U encapsulated CALEA packets for both CUPS and Non-CUPS sessions

For more information, see the *Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide*.

CUPS Configuration Overview

The Affirmed CLI provides commands to configure the Sx interfaces and PFCP objects for CUPS. **After the initial configuration using the CLI, Affirmed Networks recommends using Acuitas CUPS Manager to replicate the required configuration from the CPF to the UPF**. For example, for CUPS support on Gx and Gy interfaces, the user-plane cluster configuration must match the configuration for the control-plane cluster, including data profiles, charging-instances, billing-plans, rating-groups, service-rules, and packet-filters. For information, see the *Acuitas User's Guide.*

See Chapter 2, CUPS PFCP Overview for CLI command configuration information.

Note: *In cases where the cluster is configured as a CPF as well as a UPF, both the User Plane Profile and the User Plane Service are configured on the same cluster.*

Acuitas EMS CUPS Support

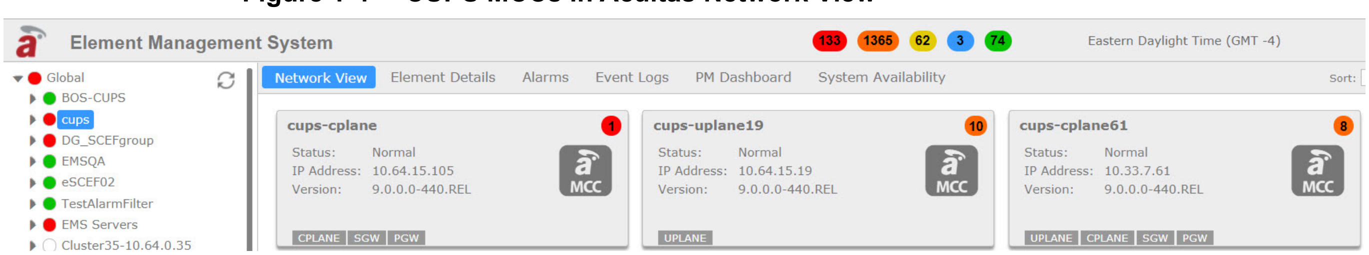
Acuitas includes the CUPS Manager application that manages Control and User Plane Separation (CUPS) MCC devices.

Use the CUPS Manager to:

- Configure the required Sx interfaces and Packet Forwarding Control Protocol (PFCP) objects on the Control Plane and User Plane.
- View the grouping for a Control Plane and associated User Plane(s).
- Perform an audit report of the configuration on a Control Plane/User Plane pair to identify differences.
- Replicate common configuration from the Control Plane to the User Plane.

In Acuitas, users can register CPF and UPF clusters, which appear as standard MCCs in the Network View. The device icon indicates whether the MCC is a CPF (CPLANE) or UPF (UPLANE). For more information, see the *Acuitas User's Guide*.

Figure 1-4 CUPS MCCs in Acuitas Network View



PFCP 3GPP Compliance

Table 1-1 describes the supported 3GPP PFCP compliance.

Table 1-1 Packet Forwarding Control Protocol Compliance

Type	Procedure/Feature	TS29.244 Compliance
PFCP Node-Related Procedures		
	Heartbeat Procedure	Full
	Load Control Procedure	None/Not Supported
	Overload Control Procedure	None/Not Supported
	PFCP Packet Flow Description Management Procedure	None/Not Supported
	PFCP Association Setup Procedure	Full
	PFCP Association Update Procedure	Full
	PFCP Association Release Procedure	Full
	PFCP Node Report Procedure	None/Not Supported
Sx Session-Related Procedures and Functions		
	PFCP Session Establishment Procedure	Full*
	PFCP Session Modification Procedure	Full*
	PFCP Session Deletion Procedure	Full*
	PFCP Session Report Procedure	Full*
	Downlink Data Buffering in C-Plane	None
	Sxa Downlink Data Notification Delay	None
	Sxa Downlink Buffering Duration	None
	Sxb Traffic Steering	Full*
	Sxc Traffic Steering	Full*

Table 1-1 Packet Forwarding Control Protocol Compliance (Continued)

User Plane (UPF) Procedures and Functions		
	Sxa, Sxb F-TEID Allocation	Full
	Sxb, Sxc PFD Management Procedure	None/Not Supported
	Sxb, Sxc Traffic Redirection Enforcement	Full
	Send End Marker Packets	Full
	IP Address Allocation on C-Plane	Full

* Implemented using Affirmed Networks proprietary vendor-specific IEs.

CUPS PFCP Overview

This chapter describes an overview of the procedures used to configure the Sx interfaces and objects that use the Packet Forwarding Control Protocol (PFCP) to provision Control and User Plane Separation (CUPS) in the MCC Evolved Packet Core (EPC).

CPF and UPF Configuration Objects

The following diagrams show the CPF and UPF configuration objects and their associations, respectively.

Figure 2-1 shows the CPF configuration objects.

Figure 2-1 CPF Configuration Objects

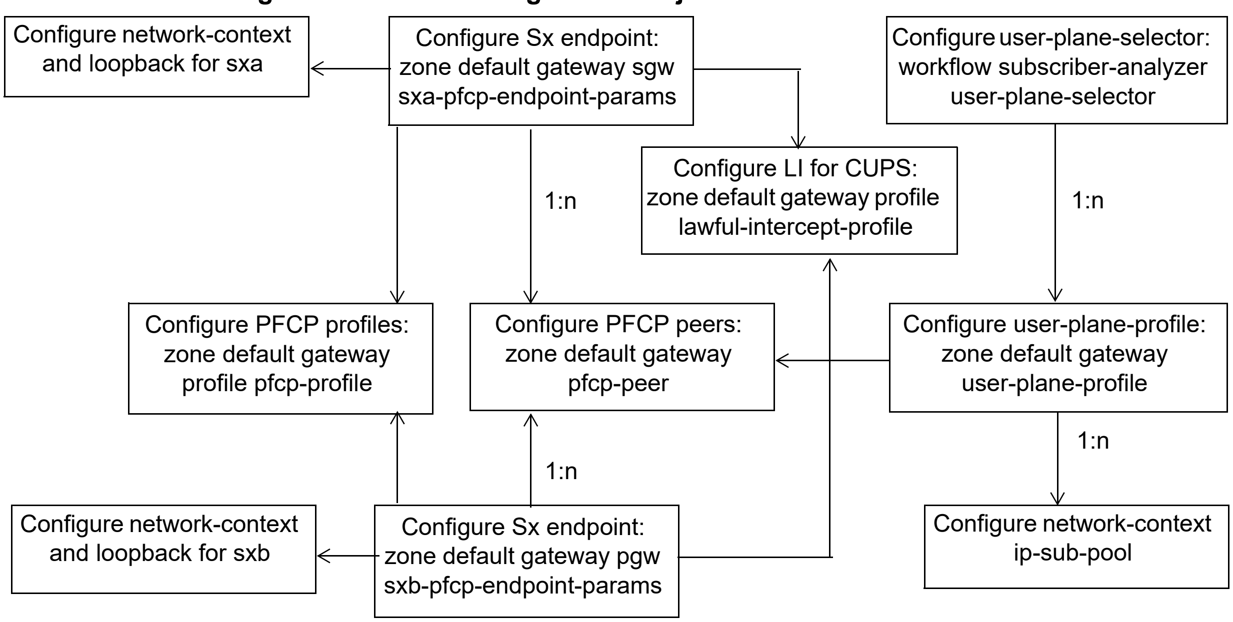
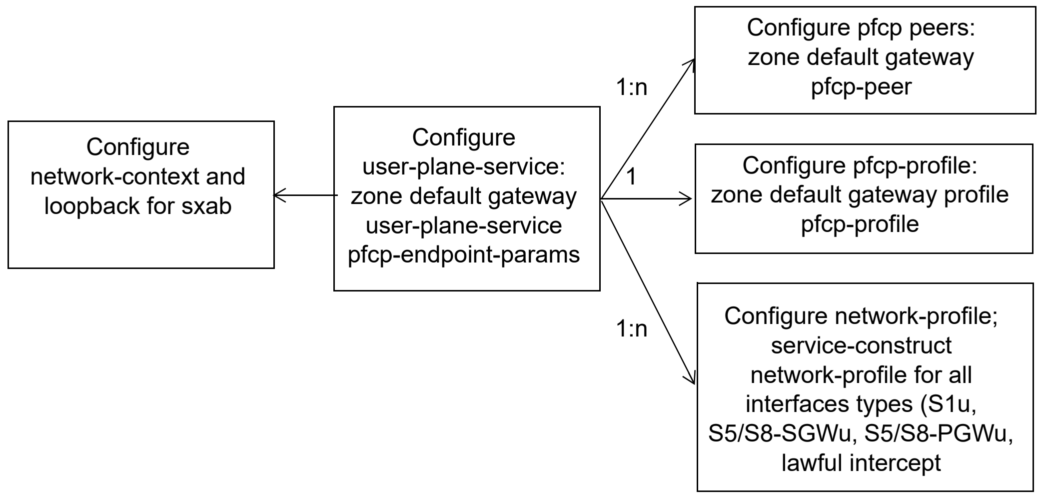


Figure 2-2 shows the UPF configuration objects.

Figure 2-2 UPF Configuration Objects

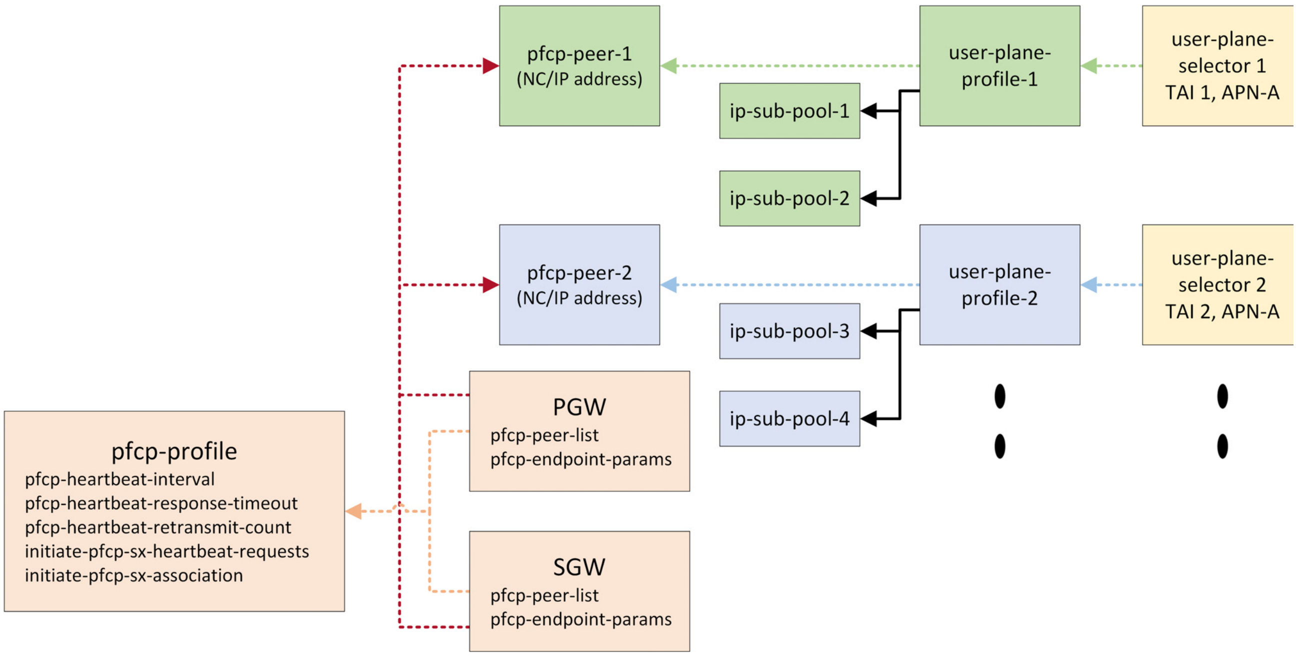


IP-Sub-Pool Allocation

In CUPS, IP sub-pools are configured on the CPF and UPFs. The IP sub-pool in the CPF is divided among the UPFs. The CPF contains a superset of the IP sub-pools from all UPFs. The CPF allows this division by linking the associated IP sub-pools with the user-plane-profile configuration.

UE IP addresses are allocated based on the selected UPF, and the UE IP sub-pool. Figure 2-3 shows an example of how IP sub-pools are allocated and linked to configuration objects.

Figure 2-3 IP Sub-pool Division and Allocation



CUPS Configuration Procedures

This section describes an overview of the procedures used to configure the Sx interfaces and PFCP objects on the CPF and UPF. For CLI object descriptions, see the following:

- Chapter 8, Network Context Parameters
- Chapter 11, Workflow Parameters
- Chapter 12, Zone Parameters

Configuring Sx Interfaces

Configure the Sx interface:

Tip: See Chapter 8, Network Context Parameters for object descriptions.

---

1. Create the network-contexts on the CPF and on the UPF.
2. Configure the network-context and loopback IPs for the Sx interfaces (sxa/sxb/sxab) and specify **yes** to enable GTP-U services for the loopback address(es) defined for the Sx network-context.

```
an3000-mcm-slot8cpu1(config)# network-context <name> loopback-ip
<interface-name> ip-address <ip-address> gtp-u yes
```

Note: Sx interfaces do not support dual-stack IPv4 and IPv6 addresses or secondary loopbacks.

---

3 Configure the Sx ip-interface parameters.

```
an3000-mcm-slot8cpu1(config)# network-context <name> ip-interface
<interface-name> port <port#> mtu <value> ip-address <ip-address>
prefix-length <value> vlan-tag <value> secondary-ip-address
<ip-address> secondary-prefix-length <value>
```

4 Configure the routing protocol (OSPF, BGP, etc.).

Configuring PFCP Peers

Configure PFCP peers:

- On the CPF, a PFCP peer configuration points to the UPF PFCP endpoint, which is associated with SGW and PGW services. See Figure 2-1 on page 2-1.
- On the UPF, the PFCP peer configurations point to the SGW Sxa PFCP endpoint and PGW Sxb PFCP endpoint, which are associated with the user-plane-service. See Figure 2-2 on page 2-2.

Tip: See zone gateway pfcp-peer (page 12-38) for object descriptions.

```
an3000-mcm-slot8cpu1(config)# zone default gateway pfcp-peer <name>
network-context <name> pfcp-version <pfcp-version> peer-ip-address
<ip-address>
```

For example:

```
zone default gateway pfcp-peer SXAB-V4-PEER1 network-context SX
pfcp-version pfcpv1 peer-ip-address 10.33.150.61

zone default gateway pfcp-peer SXAB-V6-PEER1 network-context SX
pfcp-version pfcpv1 peer-ip-address 2001:470:8865:d13d:10:33:150:61
```

Show PFCP Peer Information

To display PFCP peer status/information:

```
an3000-mcm-slot7cpu1# show zone default gateway pfcp-peer-status
<gateway-name>

Possible match completions:

functions-supported        Functions Supported

local-ip-address           Local IP Address, must be a valid IPV4 or
IPV6 address

network-context            Network Context name

peer-association-status    PFCP Peer Association Status
```

```
peer-heartbeat-status     PFCP Peer Heartbeat Status
peer-name                 PFCP Peer Name
peer-recovery-value       Peer Recovery Value
pfcp-version              PFCP version of interface with peer
```

To display PFCP peer information for a gateway instance:

```
an3000-mcm-slot7cpu1# show running-config zone <name> gateway
pfcp-peer-status <gateway-name> <peer-ip-address>
```

To display the PFCP peer with which the CPF or UPF communicates:

```
an3000-mcm-slot7cpu1# show running-config zone <name> gateway
pfcp-peer <name>
```

To display the list of PFCP peer names selected from a peer list:

```
an3000-mcm-slot7cpu1# show running-config zone <name> gateway pgw
<name> [sxb-pfcp-peer-list | sxa-pfcp-peer-list] <name>
```

To display the list of PFCP peer names selected from a peer list:

```
an3000-mcm-slot7cpu1# show running-config zone <name> gateway
user-plane-service <name> pfcp-peer-list <name>
```

Configuring PFCP Profiles

The pfcp-profile configuration specifies the timers, counts, and interval settings for the Sx associations and inter-node messaging and heartbeat monitoring.

Tip: See zone gateway profile pfcp-profile (page 12-63) for object descriptions.

Create and configure the PFCP profile:

```
an3000-mcm-slot7cpu1(config)# zone default gateway profile
pfcp-profile <profile-name>
```

Configuring PGW and SGW

Use the following commands to configure and associate PFCP objects and Sx interfaces and endpoints on the SGW-C and PGW-C.

Note: *MCC CUPS supports Lawful Intercept and the Gateway Lawful-Intercept-Profile. For configuration information, see the Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide.*

Tip: See zone gateway (page 12-3), zone gateway pgw (page 12-39), zone gateway sgw (page 12-69), and zone gateway profile pfcp-profile (page 12-63) for object descriptions.

Configure PFCP on the gateway services CPF:

1 Configure the PFCP (Sx) network-context and GTP-C endpoints on the SGW and PGW.

```
an3000-mcm-slot7cpu1(config)# zone default gateway [sgw|pgw] <name>
```

2 Complete the following values:

- s5-s8-interface-network-context: [GI,GN,GN3,**SX**,...]
- home-plmnid-list: The value must be one of: home_list roaming_list
- gtpc-endpoint-params network-context: [GI,GN,GN3,**SX**,...]
- gtpc-endpoint-params loopback-ip: [sx_v4_lp,sx_v6_lp,sxb_v4_lp,**sxb_v6_lp**]

3 Configure the sxa interface parameters on the SGW and sxb interface parameters on the PGW.

```
an3000-mcm-slot7cpu1(config)# zone default gateway sgw <name>
sxa-interface
```

```
an3000-mcm-slot7cpu1(config)# zone default gateway pgw <name>
sxb-interface
```

4 Specify the Sx interface endpoint parameters.

```
an3000-mcm-slot7cpu1(config)# zone default gateway [sgw|pgw] <name>
an3000-mcm-slot7cpu1(config-sgw/pgw-name)# sxa-pfcp-endpoint-params
```

5 Specify the pfcp-profile to be used by the SGW and PGW service.

```
an3000-mcm-slot7cpu1(config)# zone default gateway [sgw|pgw] <name>
an3000-mcm-slot7cpu1(config-sgw/pgw-name)# pfcp-profile <name>
```

6 Configure the PFCP peer list(s) on the SGW and PGW.

```
an3000-mcm-slot7cpu1(config)# zone default gateway [sgw|pgw] <name>
[sxb-pfcp-peer-list| sxa-pfcp-peer-list] <list-name> user-description
```

This configuration supports two associations from the CPF: Sxa peer (for standalone SGW sessions) and Sxb peer (for standalone PGW sessions and collocated SAEGW sessions).

Configuring User-Plane Profiles

Use the following commands to create a user-plane-profile on the CPF and specify the IP sub-pools and PFCP peer to associate with this profile.

Configure user-plane-profiles:

Tip: See zone gateway user-plane-profile (page 12-84) for object descriptions.

```
an3000-mcm-slot7cpu1(config)# zone default gateway user-plane-profile
<profile-name>
```

Configuring the User-Plane-Selector

The user-plane-selector is configured on the CPF in workflow subscriber-analyzer. It supports existing subscriber analyzer keys and the additional key for UE support for Dual Connectivity with New Radio (DCNR).

The user-plane-selector specifies whether a session is a CPF/CUPS session or a regular integrated non-CUPS session. This configuration applies to both the collocated and standalone PGW service, as well as standalone SGW service.

Tip: See workflow subscriber-analyzer (page 11-19) for object descriptions.

Configure the user-plane-selector analyzer-type:

1. Create the user-plane-selector.

   ```
   an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <name>
   ```

2. Set the objects as follows:
   - action: **user-plane-profile**
   - analyzer-type: **user-plane-selector**
   - key: **ue-capabilities dcnr**
   - priority: set a value from **1-4096** (default is 10).

Configuring User-Plane Services

Use the following commands to configure a user-plane service for each UFP. The user-plane-service is similar to gateway services on the CPF. This configuration also requires a network-profile.

Tip: See Chapter 9, service-construct network-profile and zone gateway user-plane-service (page 12-85) for object descriptions.

Configure a user-plane-service on the UPF:

1. Configure a user-plane-service.

```
an3000-mcm-slot7cpu1(config)# zone default gateway user-plane-service <name>
```

2 Create a minimum of one network-profile for each of the network interface types and specify its network context.

```
an3000-mcm-slot7cpu1(config)# service-construct network-profile <name>
```

```
an3000-mcm-slot7cpu1(config-network-profile-name)# interface-type [lawful-intercept | s1u-gtpu | s5s8-gtpu-pgw | s5s8-gtpu-sgw]
```

```
an3000-mcm-slot7cpu1(config-network-profile-newnetprof)# network-context sxab
```

Configuring a Gateway Agent Task on User Plane

Use the following commands to start the gateway agent tasks on the DCM nodes on a user plane cluster. Execute this command on clusters where the user-plane service is configured.

Enable gateway agent tasks on the user plane:

```
an3000-mcm-slot7cpu# enable-gw-agent
```

Disable (stop) a running gateway agent on the control plane:

Note: *Do not execute this command in clusters where the user-plane-service is configured.*

```
an3000-mcm-slot7cpu# disable-gw-agent
```

CPF and UPF Common Configuration

After the initial CUPS configuration using the CLI, the common configuration must be replicated from the CPF to the UPF. Affirmed Networks recommends using Acuitas CUPS Manager to replicate the required configuration as described in the *Acuitas User's Guide*.

The following tables list the configuration options on the CPF that are replicated on the UPF during the replication process.

Table 2-1 Common Service-Construct Configuration

Command
service-construct advice-of-charge <*name*>
service-construct advice-of-charge <*name*> charging-notification-list <mscc-result-code>
service-construct aoc-hardware-id-format <*name*>
service-construct application-rule-group

Table 2-1 Common Service-Construct Configuration (Continued)

Command
service-construct ca-certificate-list <*name*>
service-construct ca-certificate-list <*name*> ca-certificate <*name*>
service-construct content-insertion content-insertion-profile
service-construct content-insertion insertion-object-list
service-construct data-record-template http-proxy http-transaction
service-construct diffserv-profile dscp-mapping <*code*>
service-construct domain-rule-group <*name*>
service-construct edr-flow-network-stats-profile <*name*>
service-construct edr-interface-profile <*name*>
service-construct edr-interface-profile <*name*> edr-service-profile <*name*>
service-construct edr-server-profile <*name*>
service-construct edr-session-data-stats-profile <*name*>
service-construct host-list
service-construct host-list ip-address
service-construct http-header-enrichment
service-construct http-rule-group <*name*>
service-construct http-url-rewrite <*name*>
service-construct ip-list
service-construct ldap profile
service-construct mms mmsc-group
service-construct packet-filter
service-construct pdngw-list
service-construct protocol-application-rule-group <*name*>
service-construct quality-of-service <*name*>
service-construct service-rule
service-construct session-rule-group
service-construct session-rule-group rule
service-construct service-rule packet-filter

Table 2-1 Common Service-Construct Configuration (Continued)

Command
service-construct sms smsc-group
service-construct static-dns-map
service-construct static-dns-map host
service-construct tcp-optimization-profile
service-construct time-of-day-rule-group rule
service-construct tls-rule-group <*name*>
service-construct tls-rule-group <*name*> rule <*name*>
service-construct ua-profile <*name*>
service-construct url-list

Table 2-2 Common Services Configuration

Command
services affirmed-apps fair-bandwidth-usage-service-profile
services charging billing-plan
services charging charging-instance
services charging rating-group
services content-cache
services content-cache instance <*name*>
services content-filter instance
services crbn-profile <*name*>
services dns-proxy
services dns-proxy dns-ip-flow
services dns-proxy dns-policy
services dns-proxy instance
services dns-proxy static-dns-map
services dns-proxy tunnel-detection-profile
services http-proxy
services http-proxy abr-optimization-profile

Table 2-2 Common Services Configuration (Continued)

Command
services http-proxy encryption-profile
services http-proxy explicit-proxy-profile
services http-proxy http-ip-flow
services http-proxy http-request-policy
services http-proxy http-response-policy
services http-proxy https-profile
services http-proxy http-transaction-record-profile
services http-proxy instance
services http-proxy fraud-detection-profile
services http-proxy sps-attribute
services http-proxy sps-profile
services http-proxy steering-group
services http-proxy tcp-splicing-policy
services http-proxy tcp-splicing-profile
services http-proxy toll-free-profile
services http-proxy web-image-optimization profile
services http-server instance
services metering geo-redundant-group
services metering metering-instance
services quality-of-service qos-flow
services quality-of-service qos-policy
services sip-proxy instance
services steering steering-flow
services steering steering-policy
services subscriber-firewall subscriber-firewall-policy
services video-adaptation instance
services wap-gateway instance

Table 2-3 Common Network-Context Configuration

Command
network-context <*name*>

Table 2-4 Common Workflow Configuration

Command
workflow data-profile <*name*>
workflow data-profile <*name*> geo-redundant-group <*name*>
workflow data-profile <*name*> metering-service <*name*>

Table 2-5 Common Zone Configuration

Command
zone <*name*> gateway apn <*name*>
zone <*name*> gateway apn <*name*> associated-ue-pool <*name*>
zone <*name*> gateway apn <*name*> charging <*name*>
zone <*name*> gateway enhanced-s-list <*name*>
zone <*name*> gateway filter-id-pco-action-association-table <*name*>
zone <*name*> gateway pcscf-table <*name*>
zone <*name*> gateway profile qci-profile <*name*>
zone <*name*> gateway service-rule-pco-action-association-table <*name*>

Table 2-6 Other Configuration

Command
flow-idle-timeout-profile <*name*>
geographic-redundancy group <*name*>
infrastructure file-management profile <*type*> <*prefix*>
networking-profiles neighbor-discovery-profile <*name*>

5G NSA NR Overview

This chapter provides an overview of the Affirmed Networks 5G Non-Standalone (5G-NSA) New Radio (NR) solution.

The Affirmed Networks 5G-NSA NR solution supports 5G networks and devices over an existing 4G infrastructure. The 5G-NSA NR solution uses a dual-connectivity (DC) approach that allows 5G-enabled devices to connect to 5G frequencies over 4G LTE networks to reap improvements in data throughput while still using 4G for non-data duties such as communicating with cell towers and servers. The 5G-NSA DC approach achieves high per-user throughput, mobility robustness, and load balancing.

The 5G-NSA NR solution provides increased data bandwidth and connection reliability using two radio frequency ranges, one of which is referred to as Frequency Range 1 (aka New Radio) (450 MHz - 6,000 MHz). 5G-NSA NR offers simplified and improved efficiency that lowers costs and improves throughput performance to the edge of the network.

The Affirmed Networks 5G-NSA NR solution leverages the Evolved Packet Core (EPC) network to provide a cost-effective path. This solution allows 5G NR devices to connect to an existing LTE network that serves as an anchor for this DC approach. The 4G LTE network provides control functions for the 5G NR. The increased coverage for the NR boosts the user plane capacity during peak/high volume periods.

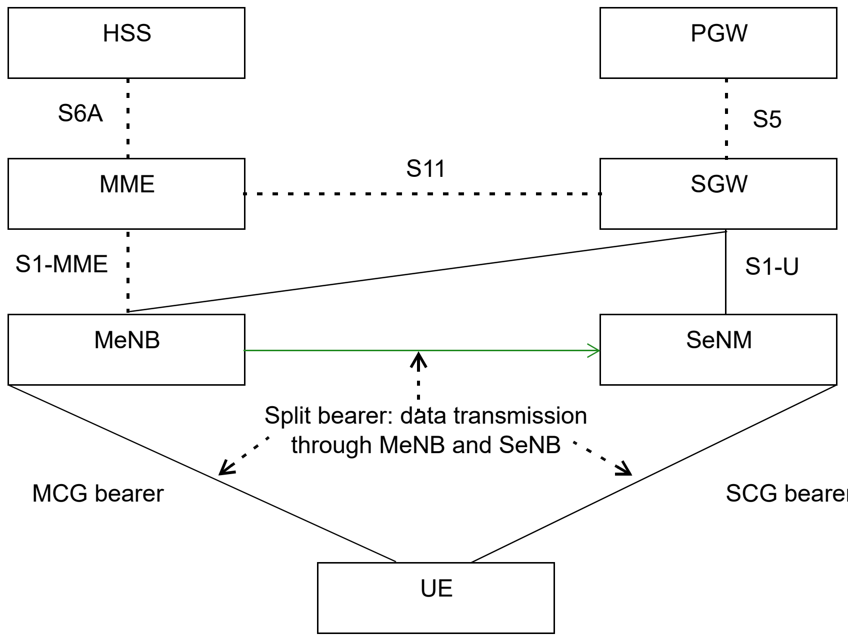
The Affirmed Networks 5G-NSA NR solution is the first of a three-part 3GPP-defined approach to evolve to full 5G implementation. The second phase in this evolutionary approach upgrades existing Evolved Node Bs (eNBs) to support both NextGen Core and EPC interfaces. In the third phase, the solution upgrade path provides for connecting a 5G NR to a Next Generation 5G Core (the *standalone* approach).

Dual Connectivity Overview

The Affirmed Networks 5G-NSA NR dual connectivity approach allows 5G devices to connect to 5G frequencies over 4G LTE networks. This provides improvements in data throughput while using 4G for non-data duties. UEs eligible to connect to the 5G frequencies and 5G-NSA NR capabilities must provide Dual Connectivity with New Radio (DCNR) capability.

DCNR allows a UE to connect to a Master Cell Group (MCG) and a Secondary Cell Group (SCG) via a master eNB (MeNB) and secondary eNB (SeNB) that each operate on different carrier frequencies and are interconnected by traditional backhaul links (known as the X2 interface) ().

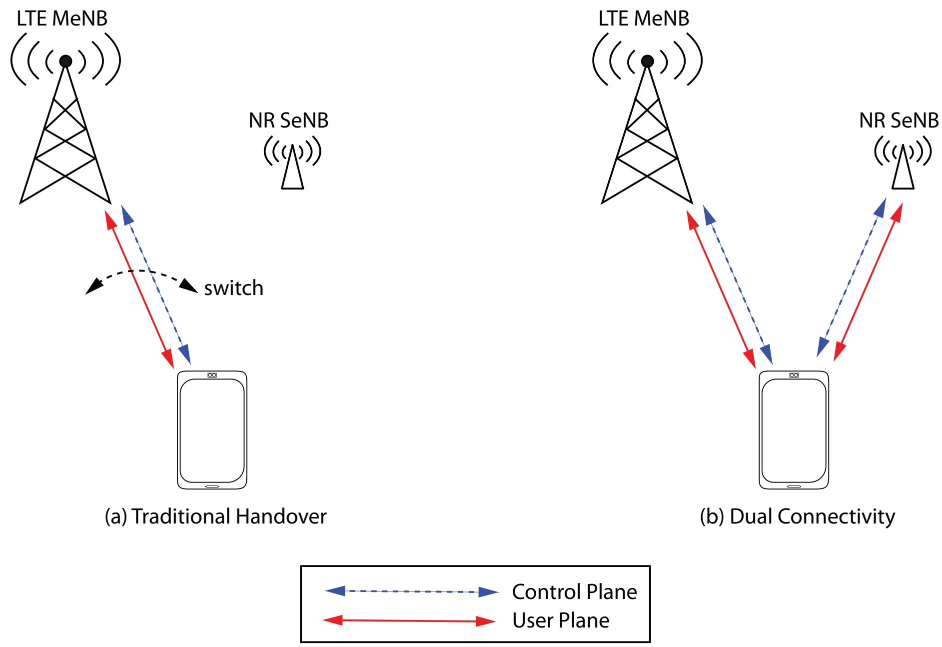
Figure 3-1 5G-NSA NR DCNR



An eNB checks whether the UE is NR-capable. The UE itself broadcasts whether it is capable of DCNR, so if the UE is allowed to use NR, the eNB establishes DCNR as a secondary radio access type (RAT). The Mobility Management Entity (MME) uses support for DCNR for PGW/SGW selection when the UE indicates support for NR and there is no access restriction for NR for the UE.

For a UE to be able to be DCNR, it typically needs separate protocol stacks (RLC and MAC); one for MeNB and one for SeNB, and the UE needs two radios to fully connect to both the MeNB and the SeNB (Figure 3-2). This solution allows user plane aggregation in which a single UE can receive a single flow over multiple RATs or different flows on different RATs in a traditional handover.

Figure 3-2 UE Connectivity Solution Enabled by DCNR



The MCG radio bearer is served only by the MeNB while an SCG radio bearer is served only by the SeNB. The increase in per-user data and significant increase in per-user throughput is accomplished by aggregating radio resources from at least two eNBs. The SGW supports a Modify Bearer Request procedure to switch the user plane between the MCG bearer and the SCG bearer.

A third type of bearer (*split bearer*) can be established in this type of configuration. The MeNB can use the split bearer to split the traffic and forward the packets to the MeNB radio link control (RLC) and/or the SeNB RLC. The split bearer allows network operators to provide transmission diversity to forward the same packets to both MeNB RLC and SeNB RLC to enhance system reliability or forward different packets to MeNB RLC and SeNB RLC to increase per-user throughput.

The MeNB handles the control plane to provide a more robust system by maintaining the radio resource control (RRC) connections. RRC provides such functions as broadcasting of acquisition and reference signal and system information, configuration of lower-layer protocols, mobility management, and measurement and configuration reporting. This means that the MeNB controls the DC configuration, generating and sending all RRC messages to the UE. Thus, the UE receives all messages sent only from the MeNB and the UE only responds back to the MeNB. The transmission of RRC messages is not supported over the SeNB.

QoS Enhancements and Latency

QoS enhancements and latency improvements in 5G-NSA NR include:

- The SGW and PGW support high bit rates for the APN-AMBR (access point name - aggregated maximum bit rate) that are received during session activation and modification procedures.
- The PGW supports QoS bit rates up to 20 Gbps for APN-AMBR, GBR (guaranteed bit rate), and MBR (maximum bit rate) on the Gx interface.
- The maximum bit rates required to be supported are 20 Gbps for downlink and 10 Gbps for uplink per UE.
- The SGW and PGW support new QoS Class Identifier (QCI) values received in GPRS Tunneling Protocol – C-Plane (GTP-C) messages to support low latency.
- 5G NSA QCIs in 3GPP TS 23.203 support ultra low latency: QCI 80, QCI 82 and QCI 83.

Data Volume Reporting

The SGW and PGW support receiving uplink and downlink data volumes reported per bearer for the secondary NR RAT (Radio Access Technology).

The MME sends the received volume reports to the SGW in various GTP-C messages/procedures. The SGW processes and optionally forwards this information to the PGW.

The SGW receives NR data volume information from the MME during X2 handover, S1 handover, S1 release and bearer deactivation procedures. Secondary RAT usage is reported in the SGW and PGW GTTP (ans1) and LOCAL CDRs (asn1, csv).

New fields are introduced in CDRs as well as on the Gz interface to report secondary RAT usage. For details on charging enhancements, refer to 3GPP TS 32.251, 3GPP TS 32.298 and 3GPP TS 32.299.

For more information about 5G-NSA CDRs, see the *Affirmed Networks Charging Data Records Specification*.

User Plane Selection

The SGW receives UE capability information from the MME and forwards the relevant information to the PGW. This includes UE support for Dual Connectivity with NR (DCNR). The SGW is enhanced to support the ability to forward received DCNR-related parameters to PGWs.

The SGW/PGW/SAE-GW select an SGW-U/PGW-U/SAE GW-U (SGW-User plane/PGW-User plane/SAE User plane Gateway) according to the capabilities of the UE. For a UE using 5G NR RAT, a user plane that supports the higher bit rates is selected.

Subscriber analyzer key(s) are available for the user plane. The subscriber analyzer framework includes the ue-capabilities key that can be set to indicate support for DCNR as follows:

```
workflow subscriber-analyzer ANALYZE2 key key1 ue-capabilities dcnr
```

Tip: See workflow subscriber-analyzer (page 11-19).

Sending Volume Reports to the SGW in GTP-C Messages/Procedures

In addition to sending secondary RAT usage data to the SGW/PGW, the SGW is enhanced to support receiving volume reports in GTP-C messages/procedures from the MME.

Chapter 4

TAM Overview

This section provides an overview of TAM (Traffic Anchoring Mode).

TAM is supported on the MCC or CUPS user plane platform for SAEGW traffic that minimizes E/W upstream/downstream traffic that reaches the WSM in a UE session and anchors it directly to a VM without any E/W hops. TAM is supported on SSM/ISM VMs to reduce E/W hops for all user traffic. When TAM is combined with SSM/ISM under normal operating conditions, all user traffic across E/W VAS interfaces is reduced. TAM is supported in both CUPS and non-CUPS deployments.

TAM reduces the hardware footprint, lowers latency and throughput, and increases the efficiency of I/O power to maximize packet core utilization on WSMs. With TAM, no existing MCC functionality is lost and all other services (for example, CALEA, IPSec tunnels, GiGW, etc.) remain fully functional, although they are not optimized for E/W minimization. Single Hop Border Gateway Protocol (SH-BGP) and Multi Hop Border Gateway Protocol (MH-BGP) is supported in TAM mode.

When TAM is disabled, upstream UE packets received on any of the Gn interfaces traverse the E/W fabric to the WSM where the UE session is anchored. If those UE packets need proxy service, the packets take another fabric hop to reach the ASM to undergo VAS processing because the GTP-U loopback route used is for GTP-U traffic that is advertised on all interfaces using routing protocols with equal cost. UE sub-pool routes are also announced on all interfaces in the Gi side with the same cost. Therefore, downstream packets can be received on any of the Gi interfaces and need to traverse the E/W fabric to be processed by the ASM or WSM. Encrypted packets may take an additional hop if the tunnel endpoint is anchored on a different SSM/WSM. Therefore, without TAM, the hardware footprint is larger, latency and throughput are increased, and I/O power is less efficient when trying to maximize packet core utilization on WSMs.

Multi Hop Border Gateway Protocol

Traffic Anchoring Mode (TAM) supports Multi Hop Border Gateway Protocol (MH-BGP). For each session, routes are advertised with the lower cost for IP addresses anchored on the local TAM and with a higher cost for other addresses. Although MH-BGP is designed to distribute traffic over all nodes, when MH-BGP is configured on TAM, only one MH-BGP session per TAM node (ISM/SSM) is supported.

To configure TAM with MH-BGP:

1. Configure one loopback per TAM node.
2. Configure static routes from the TAM node to point to the loopback of the MH-BGP peer.
3. Configure one MH-BGP session per TAM node.
4. Configure the loopback as the update-source interface for the MH-BGP peer.

Note: *When TAM is configured with MH-BGP, it behaves similar to SH-BGP.*

TAM Features

TAM provides the following benefits and supports the following features:

- When all SSMs are fully functional, there is no UE E/W traffic.
- UE traffic continues to work without loss of service in the event of an unlikely SSM/ISM failover, even if there is E/W traffic.
- Anchored to an SSM/ISM VM.
- When TAM is combined with the ISM VM, E/W VAS traffic is reduced.
- TAM is supported with the GGSN, standalone SGW, standalone PGW and SAEGW functions.
- Local and remote UE IP allocation.
- E/W traffic minimization works with CGNAT in guaranteed, deterministic, and statistical modes.
- No loss of existing functionality in TAM mode.
- Works on both CUPS and non-CUPS architectures.
- E/W traffic is expected for IPSec, GiGW, WAG and GRE VPNs.
- SH-BGP and MH-BGP in TAM mode.
- Manual scale-out/scale-in.
- Bidirectional Flow Detection (BFD).
- BGP-MPLS (Border Gateway Protocol-Multiprotocol Label Switching) VPNs.

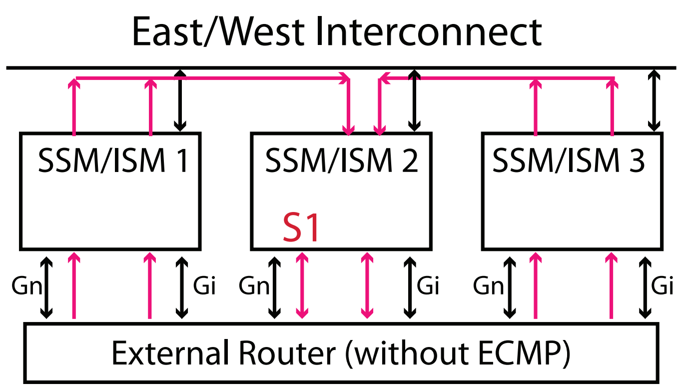
- Existing SSM/ISM redundancy mechanism.
- Remote UE address allocation is supported with restrictions.
- Guaranteed and deterministic NAT in NAT44, NAT64, 1:1 and N:1
- Statistical NAT with TAM. mode.

Packet Flow

The following figures compare the packet flow of traffic when TAM is enabled versus disabled.

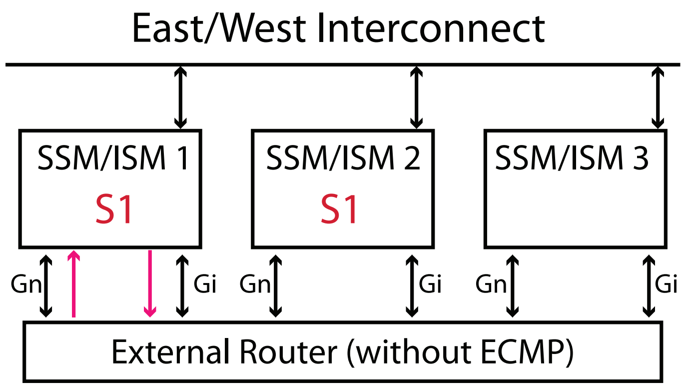
When TAM is disabled, packets from both Gn and Gi interfaces enter through any of the interfaces across the SSMs/ISMs and take an E/W fabric hop to the SSM/ISM where the session is anchored. Any packet being transmitted from the session uses a local interface on that SSM/ISM. Figure 4-1 shows how packet traffic traverses E/W between SSM/ISM pairs.

Figure 4-1 Packet Flow Example - TAM Disabled



However, when TAM is enabled (Figure 4-2), the ingress packets on both Gn and Gi interfaces occur on the interfaces on the SSM/ISM in which the session is anchored. This eliminates the E/W packet flow.

Figure 4-2 Packet Flow Example - TAM Enabled

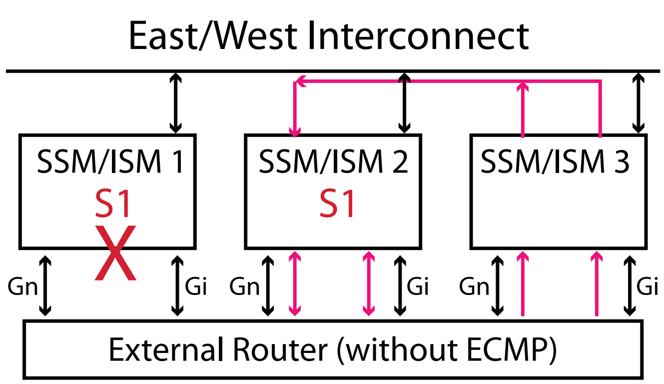


Failure Behavior When TAM is Enabled

This section describes how the MCC reroutes traffic during an ISM or interface failure.

In the event of an SSM/ISM failover, UE traffic continues to work without loss of service, even if there is E/W traffic. If an SSM/ISM or interface fails with TAM enabled, (Figure 4-3), an external router reroutes the traffic through less preferred interfaces on any/all of the SSM/ISMs. This failure results in E/W traffic until the primary SSM/ISM recovers. After the original SSM/ISM recovers, traffic switches to the optimized path without E/W traffic.

Figure 4-3 Packet Flow Example - SSM/ISM Failure with TAM Enabled



Configuring TAM

To configure TAM:

1 Enable TAM:

```
an3000-mcm-slot7cpu1(config)# tam policy mode enabled-for-cluster
```

Note: *Changing the mode triggers an immediate cluster reboot.*

---

2 Configure the TAM profile, which includes a set of loopback-ip addresses and ip-subpools anchored to a cluster/node.

```
an3000-mcm-slot7cpu1(config)# tam profile <clusterId> <nodeSlot>
ip-subpool <network-context> <ip-subpool>
an3000-mcm-slot7cpu1(config)# tam profile <clusterId> <nodeSlot>
loopback-ip <network-context> <loopback-ip>
an3000-mcm-slot7cpu1(config)# commit
The following warnings were generated:
  'tam policy':
Changing TAM Mode to "Enabled" will force a cluster reboot.
```

where:

Object	Description
clusterId	Cluster to which the profile is anchored
nodeSlot	Node to which the profile is anchored.
ip-subpool	List of ip-subpools anchored by this profile. Adding or removing an ip-subpool when TAM is active will immediately result in routing changes.
ip-subpoolName	IP subpool name to which the profile is anchored.
loopback-ip	Loopback IP address to which the profile is anchored.
network-contextName	Context name.
loopback-ipValue	Loopback IP address value.

To view the TAM statistics:

```
an3000-mcm-slot7cpu1# show tam
```

To view a TAM policy configuration:

```
an3000-mcm-slot7cpu1# show running-config tam
```

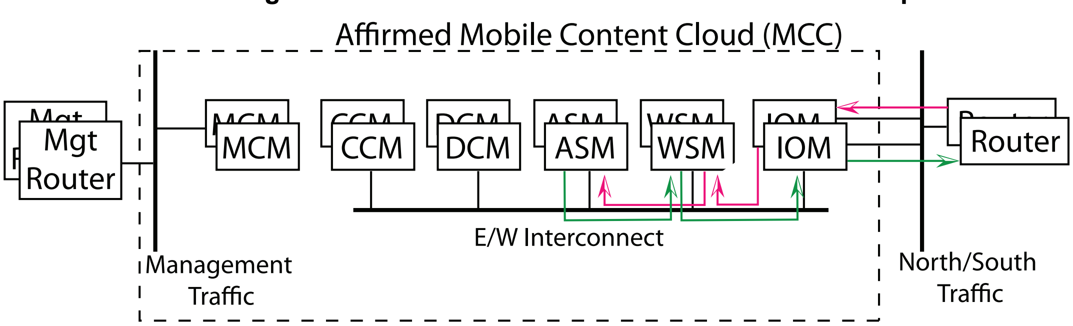
ISM VM Overview

This chapter provides an overview of the Integrated Services Module (ISM) MCC VMs. This chapter also describes the building block strategies and solutions to effectively reduce East/West (E/W) traffic in the MCC.

With the use of more advanced CPUs, the bottleneck in MCC processing is in E/W network traffic using ISM VMs. The ISM VMs minimize inter-VM proxy user plane traffic (also known as East/West (E/W) traffic) during non-failure modes of operation. The ISM VMs minimize multiple hops of upstream and downstream user packets within the MCC platform. Each packet hop decreases data throughput and increases end-to-end delays. ISM VMs significantly speed up traffic throughput by minimizing E/W traffic hops. The solution also reduces the server footprint and results in a lower cost per bit.

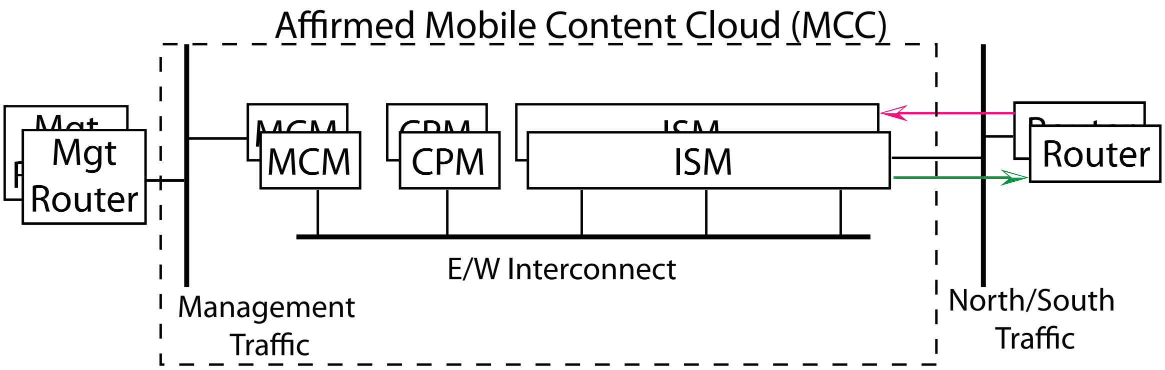
Generally, E/W traffic results from distribution of different functions across the MCC or from packets arriving in a processing card (for example, an I/O Module (IOM) or Workflow Services Module (WSM)) that must navigate to the processing node (for example, the Advanced Services Module (ASM) that can service the packet and forward it to the proper destination).

Figure 5-1 MCC ASM/WSM/IOM Packet Flow Example



The ISM minimizes E/W traffic by co-locating workflow and VAS processing for traffic sessions in the same VM. ISM transforms the SSM and ASM functions into one VM. In an integrated-services mode cluster, the ISM reduces the number of E/W hops by 1 for VAS traffic (Figure 5-2). Also see Building Block Strategies and Node Personalities (page 5-2).

Figure 5-2 MCC IOM/ISM Packet Flow Example



For examples of common traffic pattern problems that can be treated using the ISM, see Examples of Common E/W Traffic Pattern Issues (page 5-3).

Building Block Strategies and Node Personalities

Node personality defines the services that are started on a given node in a cluster. A node can be configured to use a specific personality, each of which runs a predefined collection of functionality sets. This feature allows Operators to run specialized card/nodes with their desired services and increases MCC performance.

To configure personalities, first specify a building block strategy for the cluster. The strategy specifies the supported combinations of node personalities to build the cluster.

The ISM supports the integrated-services building block strategy.

Note: *See the Affirmed Networks System Administration Guide to configure the building block strategy and personality for the ISM.*

ISM Failover and Resilience

If an ISM failure occurs, new workflow HTTP sessions are anchored elsewhere but existing HTTP sessions are lost.

Without TAM, gn or gi packets could arrive in an ISM that does not anchor the workflow session; in this case, packets are transported using E/W data fabric to the ISM anchoring the session. This results in an increase in ASM-bound E/W traffic.

However, TAM can manipulate routing metrics to steer traffic to the correct workflow session anchoring the ISM. TAM minimizes E/W upstream/downstream traffic that reaches the WSM in a UE session and anchors it directly to a VM without any E/W hops. Under normal operating conditions, the combination of TAM with ISM reduces user traffic across all E/W VAS interfaces. For more information, see Example 3: Forwarding Packets from WSM to ASM (page 5-5).

If a proxy failure occurs, the ISM operates in oversubscribed conditions but provides service to the IP addresses and TCP port blocks assigned to the local set of proxies. If all of the proxies fail, service continues and a different ISM processes the load that would otherwise have been locally processed.

Configuring CPU Resource Allocation for ISM

The appropriate assignment of available cores to the functions of I/O, workflow, and proxies allows the MCC to meet traffic patterns in the network.

Higher numbers use more CPU resources to provide increased I/O bandwidth. The configured value for the **vcpu-allocation-profile** (between 1 - 10 with a default value of 6) determines the number of cores allocated to the I/O and workflow side versus the proxy functions in the ISM. This parameter indicates the number of cores that the system allocates to a target machine for background cores, master cores, kernel NIC interface (KNI) processing cores, handling cores, and allocates the remaining system cores to be available for proxy processing.

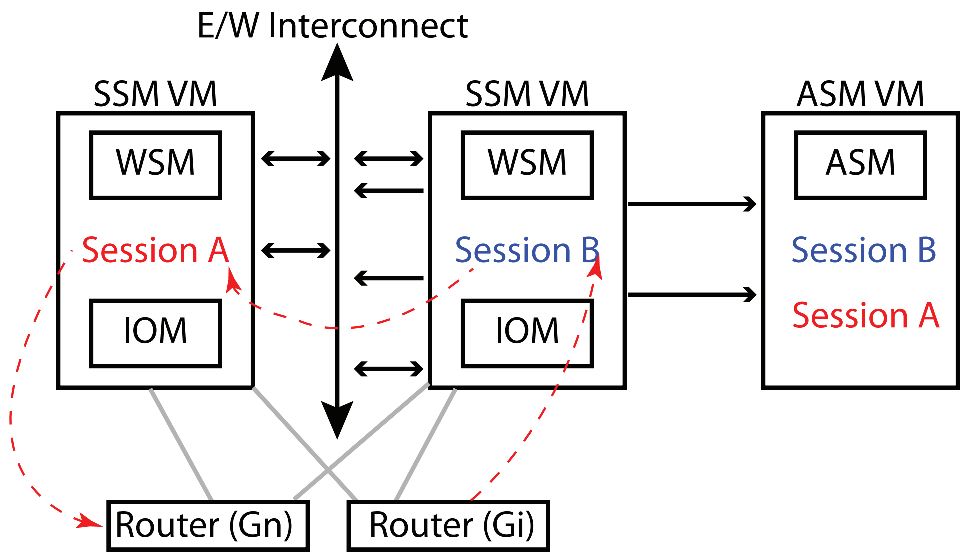
Tip: See the *Affirmed Networks System Administration Guide* to configure the **vcpu-allocation-profile**.

Examples of Common E/W Traffic Pattern Issues

This section describes examples of E/W traffic pattern issues that may arise in the MCC. The ISM is able to avoid these types of traffic issues by reducing the hops, raising throughput, and reducing server requirements.

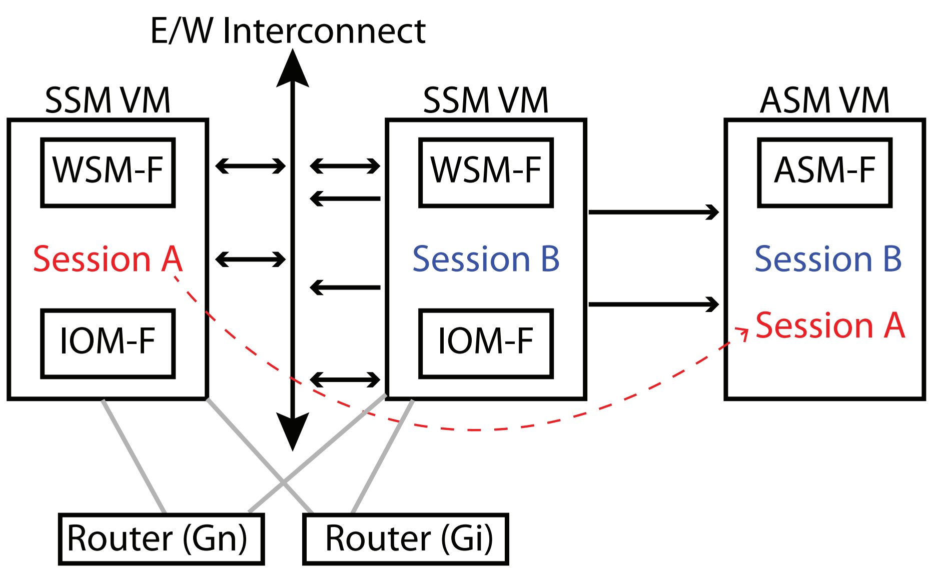
- Forwarding Packets from SSM to SSM — In Figure 5-3, a packet arrives in the SSM where the workflow session for the packet is not local and is hosted in a different Subscriber Services Module (SSM). This necessitates E/W packet forwarding of the packets to the other SSM to be processed.

Figure 5-3 Example 1: Forwarding Packets from SSM to SSM



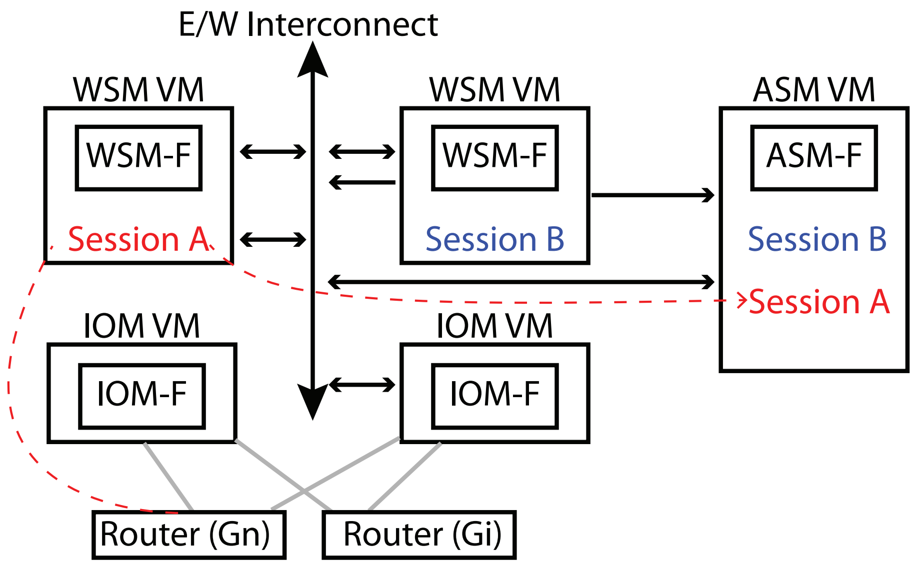
- Forwarding Packets from SSM to ASM — In Figure 5-4, a packet arrives in the SSM where the workflow session is local, but since it is part of the VAS profile it must be forwarded in an E/W direction to the ASM for processing.

Figure 5-4 Example 2: Forwarding Packets from SSM to ASM



- Forwarding Packets from WSM to ASM — In Figure 5-5, a packet arrives in the WSM where the workflow session is local but since it is part of a VAS profile it must be forwarded to the ASM. This necessitates the E/W flow of the packets to the ASM for processing.

Figure 5-5 Example 3: Forwarding Packets from WSM to ASM



Chapter 6

Gateway Overview

This section provides an overview of the following supported Affirmed Networks gateways:

Serving Gateway

This section provides an overview of the Affirmed Networks Serving Gateway (SGW).

The SGW is the point of interconnect between the radio-side and the EPC. This gateway serves the UE by routing incoming and outgoing IP packets. It is logically connected to the PGW.

The SGW acts as the mobility anchor for the user plane during inter-eNodeB handovers and as the anchor for mobility between LTE and other 3GPP technologies (terminating S4 interface and relaying the traffic between 2G/3G systems and PGW). For idle state UEs, the SGW terminates the downlink data path and triggers paging when downlink data arrives for the UE. It manages and stores UE contexts, for example parameters of the IP bearer service, and network internal routing information. The SGW also performs replication of the user traffic in case of lawful interception.

Supported Interfaces

The SGW consists of:

- S11 control plane stack to support S11 interface with MME
- S5/S8 control and data plane stacks to support S5/S8 interface with PGW
- S1 data plane stack to support S1 user plane interface with eNodeB
- S4 data plane stack to support S4 user plane interface between the SGSN and SGW of the eNodeB.The integrated control plane stack for these interfaces consists of IP, UDP, eGTP-C.
- S1-U interface with eNodeB and S5/S8 data plane interface with PGW. The integrated data plane stack for these interfaces consists of IP, UDP, eGTP-U.

The SGW transmits and receives data packets from the *Enhanced Universal Terrestrial Radio Access Network* (EUTRAN) to/from the PDN gateway and acts as a demarcation point between the RAN network and core network. The SGW also acts as termination point from the packet data network to the EUTRAN.

As per the 3GPP standard specifications, the SGW function requires support of a number of interfaces to other entities in the network. These interfaces are 3GPP-defined GTP protocol variants. Table 6-1 lists the supported SGW interfaces.

Table 6-1 SGW Interface Support

Interface	Node-Node	Protocol	Plane
S1-U	SGW- eNB	GTPv1-U	User Plane
S4	SGW - SGSN	GTPv1-U	
		GTPv2-C	Control Plane
S5 S8	SGW - PGW	GTPv1-U	User Plane
		GTPv2-C	Control Plane
S11	SGW - MME	GTPv2-C	Control Plane
S12	UTRAN and SGW for user plane tunneling when direct tunnel is established	GTPv1-U	User Plane
Gz	SGW - OFCS	Diameter	User Plane
Ga	SGW - CFG	GTP’	User Plane
X1_1	X1_1 interface to LI administrator (ADMF)	Proprietary	User Plane

Table 6-1 SGW Interface Support

Interface	Node-Node	Protocol	Plane
X2	X2 interface to LI Mediation/Delivery Function (MF/DF) for Interception Related Information (IRI)	3 GPP 33.108 HI2 format	User Plane
X3	X3 interface to LI Mediation/Delivery Function (MF/DF) for Content of Communication (CC) data	3 GPP 33.108 HI3 format	User Plane

Packet Data Network Gateway

This section provides an overview of the Affirmed Networks Packet Data Network Gateway (PGW).

The PGW is the point of interconnect between the Evolved Packet Core (EPC) and the external IP networks. The PGW routes packets to and from the Packet Data Network (PDN). The PGW also performs various functions such as IP address / IP prefix allocation or policy control and charging. In a mobile network, a major function of the PGW is to allocate IP addresses to the user equipment during default bearer setup.

In the Long Term Evolution (LTE) architecture for the Evolved Packet Core (EPC), the PGW acts as an anchor for user plane mobility. User traffic can be filtered at the PGW for quality-of-service (QoS) differentiation among multiple packet flows. The PGW also collects charging information and forwards Charging Data Records (CDRs) for processing.

The PGW provides connectivity from the UE to the external PDN by acting as the point of exit and entry of traffic for the UE. A UE may have simultaneous connectivity with more than one PGW for accessing multiple PDNs. The PGW performs policy enforcement, packet filtering for each user, charging support, lawful interception, and packet screening. Another key role of the PGW is to act as the anchor for mobility between 3GPP and non-3GPP technologies such as WLAN and 3GPP2 (CDMA 1X and EvDO).

Supported Interfaces

The PGW consists of S5/S8 control and data plane stacks to support the S5/S8 interface with the Serving Gateway (SGW).

- The integrated *control* plane stack for the S5/S8 interfaces consists of IP, UDP, eGTP-C.
- The integrated *data* plane stack for the S5/S8 interface consists of IP, UDP, eGTP-U.

As per the 3GPP standard specifications, the PGW function requires support of a number of interfaces to other entities in the network. Table 6-2 lists the supported PGW interfaces.

Table 6-2 PGW Interface Support

Interface	Node-Node	Protocol
S5 S8	PGW - SGW	GTPv1-U
		GTPv2-C
S2b	ePDG - PGW	GTPv1-U GTPv2-C
S2a	TWAG - PGW	GTPv2-C
S6b	PGW - AAA	Diameter
Ga	PGW - CGF	GTP'
Gi (Sgi)	PGW - external nw	IP
Gn	PGW - SGSN	GTPv1-C GTPv0 GTPv1-U
Gp	PGW - SGSN	GTPv1-C
		GTPv1-U
Bp	Used to transport CDRs from the gateway (PGW/SGW/GGSN) to the Billing Domain using SFTP.	SFTP
Gx	PGW - PCRF	Diameter
Gy and DCCA variants	PGW - OCS	Diameter
Gz Rf	PGW - OFCS	Diameter
X1_1	X1_1 interface to LI administrator (ADMF)	Proprietary
X2	X2 interface to LI Mediation/Delivery Function (MF/DF) for Interception Related Information (IRI)	3 GPP 33.108 HI2 Format
X3	X3 interface to LI Mediation/Delivery Function (MF/DF) for Content of Communication (CC) data	3 GPP 33.108 HI3 Format

GGSN Support

When enabled, a PGW can also function as a GGSN and can be connected to an SGSN and process 2G and 3G session requests. The PGW supports internetworking with pre-release 8 SGSN, which means an SGSN can directly connect to a PGW and establish 3G sessions. The PGW supports session creation and handoff between 3G and 4G networks.

GiLAN-Gateway

This section provides an overview of the GiLAN-Gateway (GiGW).

The GiGW is deployed between the Evolved Packet Core PDN Gateway (PGW/GGSN) and the external IP network. The GiGW communicates with the PDNGW/GGSN over the Gi-access interface and communicates towards external IP networks with the Gi-core interface. The GiGW provides subscriber-aware and subscriber-unaware value added services (VAS) without enabling MCC gateway services. This enables Operators to use their incumbent gateway while taking advantage of VAS. VAS includes enhanced charging, content filtering, content caching, DPI, video optimization, content proxy, etc. The GiGW can be deployed in mobile networks, fixed-line networks, and WiFi networks.

The GiGW also acts as the Traffic Detection Function (TDF), and supports the Sd interface to establish and terminate TDF sessions to the PCRF.

The MCC GiGW supports:

- Redundancy, distributed architecture, fault containment
- Capability to support multiple GiGW services simultaneously
- Capability to support GiGW and GGSN/SGW/PGW service simultaneously
- Hardware agnostic and supports virtualization

The GiGW supports the following gateway session learner interfaces in subscriber-aware mode. See service-construct interface gw-session-learner (page 9-44).

- radius-server – The GiGW works as a RADIUS server and learns the session through received RADIUS messages and creates a GiGW session.
- radius-proxy – The GiGW works as a proxy for RADIUS clients and servers and learns the session by receiving RADIUS request messages from the client and response messages from the server and creates a GiGW session.
- gigw-ldap-client – The GiGW uses the IP packet as the trigger to go to LDAP to learn the subscriber information and set up the GiGW session.

- gigw-ip-local – The GiGW uses the IP packet as the trigger and creates the GiGW session using the local configuration.
- sd-diameter-client – The GiGW uses the Sd interface to establish and terminate TDF sessions to the PCRF.

Supported Interfaces

The GiGW supports the following interfaces:

- GI-Access – Between GiGW and (PGW/GGSN/other GWs)
- GI-Core – Between external Internet network and GiGW
- GiGW-AAA – Between GiGW and AAA server for RADIUS proxy mode
- Gx – Between GiGW and PCRF (partially compliant)
- Gy – Between GiGW and OCS Server (partially compliant)
- Local CDR – GiGW and local CDR server
- Sd – Between GiGW and PCRF (simultaneous support for the Sd and the Gx interface is not supported)

Gateway GPRS Support Node

This section provides an overview of the Affirmed Networks Gateway GPRS support node (GGSN).

The GGSN is a main component of the GPRS network. The GGSN is responsible for the internetworking between the GPRS network and external packet switched networks, such as the Internet and X.25 networks.

The GGSN is the anchor point that enables the mobility of the user terminal in the GPRS/UMTS networks. It maintains the routing necessary to tunnel the protocol data units (PDUs) to the SGSN that services a particular MS (mobile station).

The GGSN converts the GPRS packets coming from the SGSN into the appropriate packet data protocol (PDP) format (for example, IP or X.25) and sends them out on the corresponding PDN. In the other direction, PDP addresses of incoming data packets are converted to the GSM address of the destination user. The readdressed packets are sent to the responsible SGSN. For this purpose, the GGSN stores the current user profile and SGSN address in the location register. The GGSN is responsible for IP address assignment and is the default router for the connected user equipment (UE).

The GGSN also performs authentication and charging functions. Other functions include subscriber screening, IP pool management and address mapping, QoS, and PDP context enforcement. In an LTE scenario, the GGSN functionality moves to SAE gateway (with SGSN functionality working in MME). See Serving GPRS Support Node and MME (page 6-8) for more information.

Supported Interfaces

Table 6-3 lists the supported GGSN interfaces.

Table 6-3 GGSN Interface Support

Interface Name	Node-Node	Protocol
Ga	GGSN - CGF	GTP'
Gi (Sgi)	GGSN - external nw	IP
Gn	GGSN - SGSN	GTPv1-C GTPv0-C
		GTPv1-U GTPv0-U
Gp	GGSN - SGSN	GTPv1-C GTPv0-C
		GTPv1-U GTPv0-U
Gc	GGSN - HLR	MAP
Gx	GGSN - PCRF	Diameter
Gy	GGSN - OCS	Diameter
Gz	GGSN - OFCS	Diameter
		GTP
Accounting Server	GGSN - AcctSvr	RADIUS (proprietary)
Auth Server	GGSN - AuthSvr	RADIUS (proprietary)
X1_1	X1_1 interface to LI administrator (ADMF)	Proprietary
X2	X2 interface to LI Mediation/Delivery Function (MF/DF) for Interception Related Information (IRI)	Proprietary

Table 6-3 GGSN Interface Support

Interface Name	Node-Node	Protocol
X3	X3 interface to LI Mediation/Delivery Function (MF/DF) for Content of Communication (CC) data	Proprietary

The GGSN also supports a direct tunnel where the user bearer path is setup by the SGSN but the bearer traffic is bypassed from the SGSN. A direct tunnel is created from the GGSN to the RAN network. Apart from normal Gn/Gp PDP context establishment (where the bearer path included GERAN/UTRAN-SGSN-GGSN-PDN), the GGSN functionality supports direct tunnel mechanisms where the bearer path includes GERAN/UTRAN- GGSN-PDN (bypassing the SGSN in bearer path) and the SGSN is only involved in control plane PDP context setup/modification and teardown. The GGSN also supports GTPv0 users when a GTPv0-capable SGSN is connected to the GGSN.

Direct Tunneling

The direct tunnel service enables an SGSN to establish a direct user plane tunnel between the Radio Network Controller (RNC) and a GGSN. The SGSN functions as the gateway between the RNC and the core network. It handles both signaling traffic (to keep track of the location of mobile devices) and the actual data packets being exchanged between a mobile device and the Internet.

The functionality is implemented in the existing network through software enhancements and requires no additional hardware. Direct tunneling services make the network more efficient in its handling of data packets. Also see Serving GPRS Support Node and MME (page 6-8) for more information.

Serving GPRS Support Node and MME

This section provides an overview of the Affirmed Networks Serving GPRS Support Node (SGSN).

The SGSN is responsible for the delivery of data packets to and from the mobile stations within its geographical service area. The SGSN is responsible for packet routing and transfer, mobility management (attach/detach and location management), logical link management, and authentication and charging functions. The location register of the SGSN stores location information (for example, current cell, current VLR) and user profiles for example, IMSI, address(es) used in the packet data network) of all registered GPRS users.

The Affirmed Networks Service GPRS Support Node / Mobility Management Entity is a fully Virtualized Network Function (VNF) according the ETSI NFV standards. It is composed of six Virtual Machines (VM) which can be dimensioned independently according to the traffic mix.

The Affirmed Networks SGSN/MME is a 3GPP Release 11 compliant SGSN and MME that provides mobility and session management operations in separated or combined modes for 2G, 3G, 3G+ and 4G mobile data subscribers. The SGSN/MME is fully virtualized and ETSI NFV compliant and can be deployed on dedicated servers or in the Cloud.

C-SGN

The Affirmed Networks Cellular Internet of Things (CIoT) Serving Gateway Node (C-SGN) is a fully virtualized EPC node that is optimized to support the C-IOT network. It is a combined EPC node that collocates EPS entities in the control and user planes (MME, S-GW, P-GW). The external C-SGN interfaces are the respective EPC entities supported by the C-SGN, such as the MME, S-GW, and P-GW.

Operators can deploy the C-SGN in the following configurations:

- Combined MME, S-GW, and P-GW
- Combined MME and S-GW. The C-SGN works with an external PGW for small data communication between the CIoT devices and their corresponding application servers.

On top of the existing EPC capabilities of the C-SGN, it also provides and supports the following optimization for the CIoT devices:

- Narrow band eNodeBs. The C-SGN connects to the NB-eNodeB via the S1-lite interface.
- Control plane CIoT EPS optimization for small data transmission. The C-SGN supports small data transmission using the NAS signaling layer. This is also referred to as DoNAS (Data over Non Access Stratum).
- Security procedures for efficient small data transmission. The small data transmission is integrity protected and encrypted.
- SMS without combined attach for NB-IoT only UEs. The CIoT device can perform SMS without performing combined attach procedure.
- Non-IP data transmission via SGi tunneling. Added support for new PDN type used to deliver non-IP data.
- Attach without PDN connectivity. The UE can attach without activating a PDN session.

- Power savings. The C-SGN supports many features to allow the CIoT devices to conserve battery, for example, Power Saving Mode, Extended DRX, Paging Optimization, etc.

Chapter 7

Geographic Redundancy Parameters

This section describes the geographic redundancy solution supported on the MCC.

Access	config
Guidelines	■ Geographic redundancy is not supported on gtpv0 sessions. ■ A different subscriber analyzer must be used for each geographic redundancy group. ■ Only one geographic redundancy group is allowed per network-context with security enabled.

The geographic redundancy solution is based on a collaborative system of two geographically-separated clusters or cluster functioning as standby partners for each other. A geographic redundancy group has a designated primary node and a designated secondary node (where the node is the cluster). Each group represents a set of services with an active representation on the primary node and a standby representation on the secondary node. If the active node goes out of commission, the services on the standby node can be enabled so these services can resume.

Geographic redundancy paired nodes communicate their specific compatibility version to each other to determine whether to block synchronization to an incompatible version. When the system detects a version mismatch, the geographic redundancy link stays down. The system generates a log message in *var/log/messages* indicating the geographic redundancy version mismatch.

The following alarms support this feature:

- anGeoRedundancyLinkDown
- anGeoRedundancyLinkVersionMismatch

See the *Affirmed Networks SNMP MIB and Alarm Reference Guide* for a description of the alarms.

Supported HA Modes

The geographic redundancy group (*geo-group*) supports the following HA modes:

- *warm-manual* – No session synchronization occurs between nodes. Failovers are Operator controlled.
- *hot-auto* – Synchronization of per-session data occurs between nodes. Failovers are automatically triggered based on network health monitoring results. In addition, failovers can be Operator controlled (using revert). In hot-auto mode, an automatic failback can occur from the secondary active node to the primary active node under similar circumstances (network fault or secondary node failure). The Operator can also trigger a failback to the primary active node (using the revert command).

  This section describes how to perform a failover of a geo-group in hot-auto mode. In hot-auto mode, a geo-group automatically fails over based on network health monitoring results from the configured external interfaces. However, for planned events such as a software upgrade or system maintenance, Operators can optionally perform a manual failover prior to the event to reduce the failover duration.

Note: *You must issue both commands on the primary cluster.*

  Use the following command to change the primary cluster to standby and the secondary cluster to active:

```
an3000-mcm-slot7cpu1# geographic-redundancy group <group-name>
failover
```

  Use the following command to revert the primary cluster to active and the secondary cluster to standby. The optional parameter `force` overrides a validation test of the operational status of the actively monitored geographic redundancy links.

```
an3000-mcm-slot7cpu1# geographic-redundancy group <group-name>
revert {force}
```

- *hot-manual* – Synchronization of per-session data occurs between nodes. Failovers are Operator controlled. In hot-manual mode, the Operator can monitor for geographic redundancy alarms and manually control the failover.

A geo-group configured as either hot-auto or hot-manual allows for synchronization of session states between the NE pair. The active node synchronizes session data associated with the geo-group with that of its peer. The primary node is the Operator's intended active module based on network design. When the primary node is active, it advertises lower-cost routes in comparison with the secondary node, thus causing control and data traffic to flow towards the primary node under normal network conditions. In hot-auto mode, the secondary node independently determines whether the network is directing traffic towards it, and fails over to the active state only when external servers are available. The surrounding NEs select the lower cost routes, thus directing traffic to the primary node under normal conditions. In the event of a network outage, some or all of the surrounding NEs may lose routes to the primary node and thus begin forwarding traffic to the secondary node.

When the secondary node detects network reachability on some or all of its shared IP interfaces, it implies a partial or full outage of either the primary node or a network fault. In this scenario, the active node is not functional. In a hot-auto geo-group, the primary node fails over to standby and the secondary node fails over to active. The primary node withdraws its routes from the network and later re-advertises using a high-cost metric (as it becomes standby). The secondary node (now active) advertises lower-cost routes, achieves connectivity with its neighboring NEs (if the nature of the network outage allows it), and resumes processing using the data synchronized with the former primary node.

Support for Geographic Redundancy on CUPS

Geographic redundancy is supported in the Control and User Plane Separation (CUPS) Control Plane Function (CPF).

The CPFs are configured with the same Sxa/b PFCP endpoint parameters, and their loopback-IPs are the same as the loopback-IPs for the associated geographic redundancy groups (*geo-groups*).

The two CPFs have the same PFCP peer which points to the same User Plane Function (UPF). If the geo-group on CPF1 has an active runtime-state, CPF1 will communicate with the UPF. If the geo-group runtime-state is active on both CPF1 and CPF2, only the CPF on the primary geo-group will communicate with the UPF.

Sessions are preserved across the CPFs. Before and after a switchover, the CPF points to the same session on the UPF. PFCP peer status (associations and heartbeat messaging) is also maintained before and after a switchover.

The CUPS geographic redundant configuration is the same as for non-CUPS, with the geo-group name specified in the workflow subscriber-analyzer object.

For information about configuring CUPS, see CUPS Configuration Procedures (page 2-3).

Using the CLI to Configure Geographic Redundancy Groups

Note: *Affirmed Networks recommends using the* ***Acuitas Geographic Redundancy Manager*** *application to configure and manage geographic redundancy groups. See the Acuitas User's Guide for details. Also see* *Using Acuitas to Configure Geographic Redundancy Groups (page 7-17)*

For geographic redundancy to work correctly, you must replicate a number of configuration objects from the primary geographic redundancy node to the secondary node. These objects include:

- Configuration objects that reference a geo-redundancy-group (*geo-group*) object (see Replicating Config Objects (page 7-5)).
- Configuration objects that *do not* reference a geo-group object (see Config Objects that do not Reference a Geo-Redundancy Group (page 7-8)).

In addition, each node must contain a set of unique configuration objects (not replicated) for geographic redundancy to function properly (see Unique Configuration Objects (not Replicated)). Some of these objects reference the geo-redundancy-group object.

geographic-redundancy-group

Previously, on a standby geographic redundancy cluster (*geo-site cluster*), if a session restoration failed, the session was never restored. Even though the session existed on the active geographic redundancy link, the failed session was lost if a failover occurred.

MCC includes geographic-redundant sync session enhancements to restore sessions on the standby geo-site cluster that failed to restore during session setup. If a standby geo-redundant session fails to restore, the geo-redundant client sends a request to the geo-redundant server to send the full MCD row for the session. When the full row is received, the session is restored on the standby geo-site cluster.

If session restoration fails due to configuration errors or resource failures for a long time (> 24 hours), the session is not restored.

Configure the objects as follows:

Table 7-1 geographic-redundancy-group

Object	Description
admin-state	Set the admin-state to enabled or disabled.
failed-session-resync-max-duration-geo-standby	Set the maximum time duration in which the failed session on the geo-redundant standby should attempt to be restored. Options include: 0 to 86400 (default) seconds (24 hours). If the value is set to 0, no attempt is made to restore/resync the failed session on the geo-redundant standby for that geo-group. Also, any existing failed sessions marked for restoration are cleared.
group-id*	Configure unique group id [1..7], must match with peer.
manual-state*	Set the HA state to active or standby.
mode	Configure the instance HA mode. Options include: warm-manual, hot-auto, hot-manual.

Table 7-1 geographic-redundancy-group

Object	Description
name*	Configure the redundancy name.
role*	Configure the instance HA backup role. Options include: primary or secondary.

Prerequisites

Before setting up a geo-group:

- Verify that the system time between the two nodes is synchronized.
- Disable the Admin State of the Peer objects in the external interface entities you want placed in the geo-group:
  - AAA Interface Group > Diameter > Node Admin > Peer > Admin State > disabled
  - AAA Interface Group > GTPP > Node Admin > Peer > Admin State > disabled
  - AAA Interface Group > Radius Client > Server Group > Peer > Admin State > disabled

Configuring Objects

All of the following configuration objects associated with the primary geographic redundancy node must be replicated on the secondary node. On each node, the entities must have matching values, including entity name.

Replicating Config Objects

All of the following configuration objects associated with the primary geographic redundancy node must be replicated on the secondary node. On each node, the entities must have matching values, including entity name.

ue-pool

All ue-pools used to create sessions to be synchronized must reference the geo-redundancy-group object. The ip-pools under these ue-pools are advertised by BGP with an appropriate metric.

For example:

```
an3000-mcm-slot8cpu1(config)# network-context Customer_VRF_1 ue-pool
uepool_1 geo-redundancy-group GEO-GRP-1
```

loopback-ip (GGSN)

All loopbacks used by GGSNs must be included. By adding it to the geo-redundancy-group object, this loopback IP is advertised by BGP with an appropriate metric.

For example:

```
an3000-mcm-slot8cpu1(config)# network-context GN loopback-ip ggsn_int
geo-redundancy-group GEO-GRP-1
```

ggsn, pgw, sgw

If the primary geo-redundancy-group node has any GGSN/PGW/SGW configured and sessions could be established using them, they must all be replicated on the secondary node.

For example:

```
an3000-mcm-slot8cpu1(config)# zone default gateway pgw NYpgw
geo-redundant-group GEO-GRP-1
```

loopback-ip (AAA/Diameter)

By adding this loopback to the geo-redundancy-group object, the IP address is advertised by BGP with an appropriate metric. This loopback is used for AAA/Diameter interfaces.

For example:

```
an3000-mcm-slot8cpu1(config)#  network-context BOS loopback-ip vip
geo-redundancy-group GEO-GRP-1
```

If *any* of the following interface objects are associated with the primary geographic redundancy node, you must replicate them on the secondary node. On each node, the entities must have matching values, including entity name.

OCS Interface

This command establishes a *shared* OCS interface. Only the active geographic redundancy node is connected to this OCS server.

For example:

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin OCS dwr-retry-count 1
dwr-retry-timer 1 dwr-timer 15 geo-redundant-group GEO-GRP-1
net-health-report-enabled true link-type shared origin-host
ocs-client.aricent.com origin-realm affirmed.ocs.local loopback-ip vip
port 30464 network-context BOS peer an3000p.ocs.affirmed.local realm
affirmed.ocsserver primary-ip-address 179.1.1.2 port 5151 admin-state
enabled role primary
```

PCRF Interface

This command establishes a *shared* PCRF interface. Only the active geographic redundancy node is connected to this PCRF server.

For example:

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin PCRF dwr-retry-count 1
dwr-retry-timer 1 dwr-timer 15 geo-redundant-group GEO-GRP-1
net-health-report-enabled true link-type shared origin-host
pcrf-client.aricent.com origin-realm affirmed.pcrf.local loopback-ip
vip port 35432 network-context BOS peer an3000pcrf.affirmed.local
realm affirmed.pcrfserver primary-ip-address 179.1.1.2 port 4444
admin-state enabled
```

RADIUS Authentication Interface

This command establishes a *shared* RADIUS authentication interface. Only the active geographic redundancy node is connected to this RADIUS server.

For example:

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group RADIUS1
geo-redundant-group GEO-GRP-1 remote-health-check true
server-group-type Authorization net-health-report-enabled true
link-type shared peer RadPeer1 loopback-ip vip local-port 62234
remote-ip-address 179.1.1.2 remote-port 8675 secret-key secret
network-context BOS server-dead-num-attempts 500
server-dead-timeout-value 30 server-resp-timeout-value 1
```

RADIUS Accounting Interface

This command establishes a *shared* RADIUS accounting interface. Only the active geographic redundancy node is connected to this RADIUS server.

For example:

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group RADIUS2
geo-redundant-group GEO-GRP-1 remote-health-check true
server-group-type Accounting net-health-report-enabled true link-type
shared peer RadPeer2 loopback-ip vip local-port 62535
remote-ip-address 179.1.1.2 remote-port 8677 secret-key secret
network-context BOS
```

data-profile

Any data-profile to be used to create a session must reference the geo-group.

For example:

```
an3000-mcm-slot8cpu1(config)# workflow data-profile DataProf1
geo-redundant-group GEO-GRP-1
```

subscriber-analyzer

Any subscriber-analyzer used to create a session must reference the geo-group.

For example:

```
an3000-mcm-slot8cpu1(config)# workflow subscriber-analyzer Analyze1
geo-redundant-group GEO-GRP-1 event bearer-setup action data-profile
DataProf1 control-profile CtrlProf1
```

Config Objects that do not Reference a Geo-Redundancy Group

When a session is created on the active geographic redundancy node, multiple configuration objects are typically used. It is expected the standby geographic redundancy node contains the same configuration objects to activate the synchronized session. The creation of a session may involve the following objects:

- network-context(s) used by VRF, gateway
- APN(s) used by gateway
- packet-filter(s) used by service-rule
- service-rule(s) used by qos-flow and rating-group
- qos-flow(s) used by qos-policy
- qos-policy(s) used by data-profile
- rating-group(s) used by billing-plan
- billing-plan(s) used by charging-instance
- charging-instance(s) used by APN
- QosProfDefault used by control-profile
- control-profile referenced by the subscriber-analyzer
- accounting-service
- authorization-service
- online-charging
- data-record-template used by online-charging, accounting-service, and authorization-service

In addition, if an object listed in Replicating Config Objects (page 7-5) refers to another object, that object must also be replicated. For example, the gateway object refers to a gateway-profile and gtp-peer-list, which must also be replicated.

Unique Configuration Objects (not Replicated)

loopback-ip (geo-redundancy)

This object is the local endpoint used by the local geographic redundancy node. The other peer must have its own configuration with a local loopback.

For example:

```
an3000-mcm-slot8cpu1(config)#  network-context Geo_peer_connect
loopback-ip peer1 ip-address 9.9.9.30
```

ip-interface and service-endpoint

This object is the interface through which the loopback addresses communicate. The peer geographic redundancy node requires a similar configuration.

For example:

```
an3000-mcm-slot8cpu1(config)#  network-context Geo_peer_connect
ip-interface H vlan-tag 92 port vth-4-1 ip-address 120.1.1.3
prefix-length 29
```

```
an3000-mcm-slot8cpu1(config)# network-context Geo_peer_connect
service-endpoint GEO-RED-SVR-1 local-loopback-address peer1
local-port-range-base 4500 local-port-range-numports 128
```

group name and group-id

Primary and secondary geographic redundancy nodes must use the same group name and group-id value. One node is configured as primary/active and the other is configured as secondary/standby. Hot-auto mode is used for automatic failovers in the event the active cluster fails. In contrast, if the active cluster fails with hot-manual, an alarm is generated, but the geographic redundancy nodes do not failover. Acuitas can be used to initiate a manual failover or the CLI can be used to switch the manual-state for each geographic redundancy node.

For example:

```
an3000-mcm-slot8cpu1(config)# geographic-redundancy group GEO-GRP-1
admin-state enabled group-id 3 mode hot-auto role secondary
manual-state standby
```

Peer Node

This object is used to establish the connection to the peer geographic redundancy node. There are three interfaces for connectivity-monitoring-params; both geographic redundancy nodes must use the same settings. A minimum of one interface must be set to enabled. Each node identifies and establishes connectivity with its peer node by configuring the remote-ip-address. The peer node would have a similar configuration pointing to the local loopback-ip.

For example:

```
an3000-mcm-slot8cpu1(config)# service-construct geo-redundancy-server
GEO-GRP-1 network-context Geo-peer-connect remote-ip-address
10.10.10.30 remote-base-port 4500 failover-soak-interval 180
route-metric-offset 10 service-endpoint GEO-RED-SVR-1
connectivity-monitoring-params monitor-OCS enabled ;
connectivity-monitoring-params monitor-PCRF enabled ;
connectivity-monitoring-params monitor-AAA disabled
```

OCS Health Interface

Each geographic redundancy node contains a private OCS health interface. It is expected the loopback addresses used for this interface do not reference the geo-group and do not get modified route metrics.

For example:

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin OCShealth geo-redundant-group
GEO-GRP-1 net-health-report-enabled true link-type private
dwr-retry-count 1 dwr-retry-timer 1 dwr-timer 15 origin-host
ocs-client.aricent.net origin-realm affirmed.ocs.localhealth
loopback-ip health port 33464 network-context BOS peer
an3000p.ocs.affirmed.local realm affirmed.ocsserver primary-ip-address
179.1.1.2 port 5160 admin-state enabled role primary
```

PCRF Health Interface

Each geographic redundancy node contains a private PCRF health interface. It is expected the loopback addresses used for this interface do not reference the geo-group and do not get modified route metrics.

For example:

```
an3000-mcm-slot8cpu1(config)#  service-construct interface
aaa-interface-group diameter node-admin PCRFhealth dwr-retry-count 1
dwr-retry-timer 1 dwr-timer 15 geo-redundant-group GEO-GRP-1
net-health-report-enabled true link-type private origin-host
pcrf-client.aricent.net origin-realm affirmed.pcrf.localhealth
loopback-ip health port 33593 network-context BOS peer
an3000pcrf.affirmed.local realm affirmed.pcrfserver primary-ip-address
179.1.1.2 port 4003 admin-state enabled role primary
```

RADIUS Authorization Health Interface

Each geographic redundancy node contains a private RADIUS authorization health interface. It is expected the loopback addresses used for this interface do not reference the geo-group and do not get modified route metrics.

For example:

```
an3000-mcm-slot8cpu1(config)#  service-construct interface
aaa-interface-group radius-client server-group RADIUS1health
geo-redundant-group GEO-GRP-1 remote-health-check true
server-group-type Authorization net-health-report-enabled true
```

```
link-type private peer RadPeer1 loopback-ip health local-port 61234
remote-ip-address 179.1.1.2 remote-port 6675 secret-key secret
network-context BOS server-dead-num-attempts 500
server-dead-timeout-value 30 server-resp-timeout-value 1
```

RADIUS Accounting Health Interface

Each geographic redundancy node contains a private RADIUS accounting health interface. It is expected the loopback addresses used for this interface do not reference the geo-group and do not get modified route metrics.

For example:

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group radius-client server-group RADIUS2health
geo-redundant-group GEO-GRP-1 remote-health-check true
server-group-type Accounting net-health-report-enabled true link-type
private peer RadPeer2 loopback-ip health local-port 61535
remote-ip-address 179.1.1.2 remote-port 6677 secret-key secret
network-context BOS server-dead-num-attempts 500
server-dead-timeout-value 30 server-resp-timeout-value 1
```

Geographic Redundancy Operation Modes

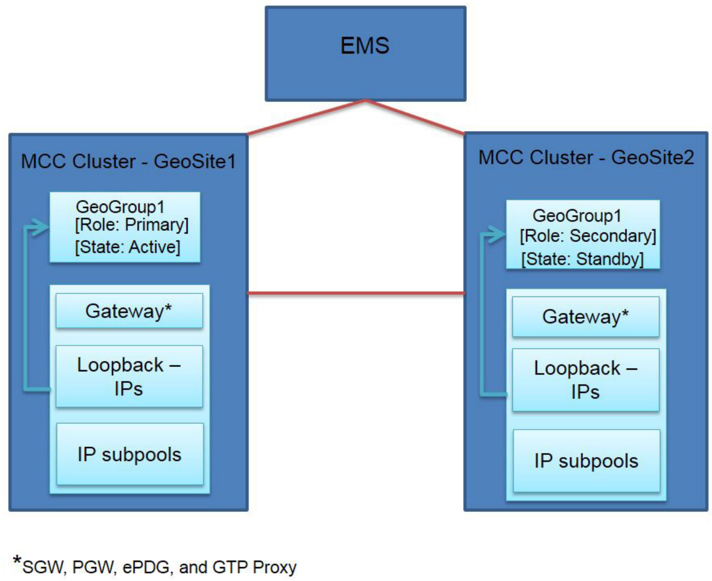
MCC supports geographic redundancy in active/standby and active/active modes.

Active/standby geographic redundancy operates at the geographic-redundancy group (*geo-group*) level in two MCC clusters located at different geographic locations. One cluster contains the primary/active node, and the other cluster contains the secondary/standby node.

A geographical redundancy link (*geo-link*) interface connects the two clusters, which are managed through the management IP.

Figure 7-1 shows an active/standby cluster configuration.

Figure 7-1 Active/Standby Mode

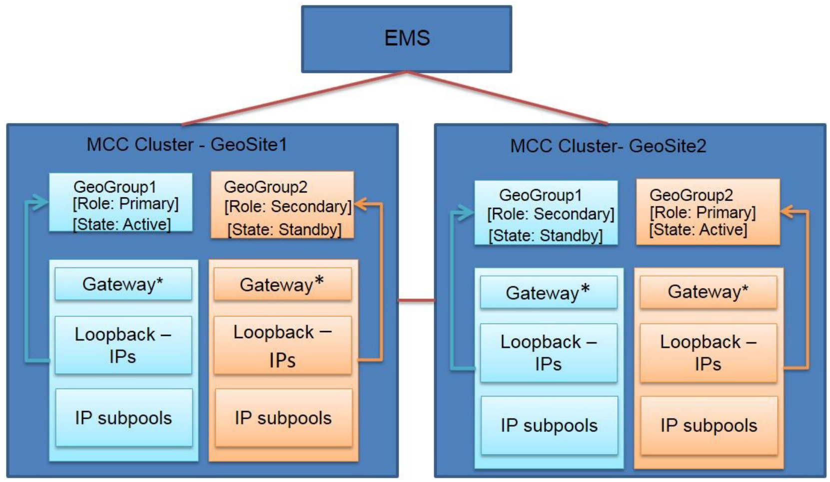


Active/active geographic redundancy operates at the MCC cluster level with the geographic redundancy clusters (*geo-site clusters*) located at different geographic locations. Each geo-site cluster has a primary geo-group and a secondary geo-group; however, the nodes in the same geo-group are located in different geo-site clusters.

In Figure 7-2, for example:

- Geo-site cluster 1 contains GeoGroup1 primary/active and GeoGroup2 secondary/standby.
- Geo-site cluster 2 contains GeoGroup1 secondary/standby and GeoGroup2 primary/active.

Figure 7-2 Active/Active Mode



*SGW, PGW, ePDG, and GTP Proxy

Each geo-group is associated with a different network context.

A geographical redundancy link (*geo-link*) interface connects the two clusters, which are managed through the management IP.

Both active and standby geo-site clusters have Gn/Gi/AAA connectivity in steady-state control, and data traffic flows only through the *active* geo-site cluster.

Geo-clusters support high availability (HA) in hot-auto, hot-manual, and warm manual modes.

- In hot-auto mode, failovers are automatically triggered based on network health monitoring results.
- In hot-manual mode, failovers are Operator controlled upon detection of alarms reported by the clusters for outages.
- In warm manual mode, session-synchronization does not occur between nodes, and failovers are Operator controlled.

For more information, see Supported HA Modes.

Prerequisites

- Geo-site cluster configurations do not need to match. However, you can use the Acuitas Geographic Redundancy Manager application to compare and configure both clusters. See Using Acuitas to Configure Geographic Redundancy Groups.

- For active/active mode, you configure two geographic-redundancy groups per cluster. Objects required: loopback IPs, network contexts, subscriber analyzers, and services. See Using the CLI to Configure Geographic Redundancy Groups.
- Only one geographic-redundancy group is allowed per network context with security enabled.
- Each geographic-redundancy group requires a different subscriber analyzer.
- For ePDG:
  - Use a different subscriber analyzer for ePDG sessions (one per geo-group) for colocated ePDG+PGW deployments
  - Use a separate network context for SWu interfaces (only one geo-group in a network context with security enabled)
- Both clusters should be of similar capacity, and combined capacity should not exceed engineering limits, for example, 70% of CPU.
- Both geo-groups' capacity should not exceed 2M sessions, and transactions should not exceed 80% of CPU.
- Geographic redundancy is not supported for GiGW/WAG sessions and GTP-V0 sessions.

Switchover Process

In active/active geographic redundancy, the following events occur during a switchover:

- When the active geo-site cluster fails, the standby geo-site cluster detects the failure (geo-link goes down) within approximately seven seconds.
- The standby geo-site cluster starts network health-checking for the enabled OCS/AAA/PCRF/SWm interfaces. Once the health checks pass on all enabled interfaces, the standby geo-site cluster changes its runtime state to *active*, less the route convergence time — approximately seven seconds.

  The outage time during a switchover is approximately the same as the route convergence time.
- New route metrics are advertised.
- Sessions are restored completely and data is sent for the restored sessions.

  Session state information is synchronized from the active to standby geo-site cluster for each geo-group. On the standby geo-site cluster, sessions are stored in expanded format to reduce switchover time.
- A geographic redundancy switchover should be transparent to peer nodes.

The following events do *not* trigger a switchover:

- The active geo-site cluster has an outage toward the Gi interface.
- The active geo-site cluster loses connectivity with the EMS.

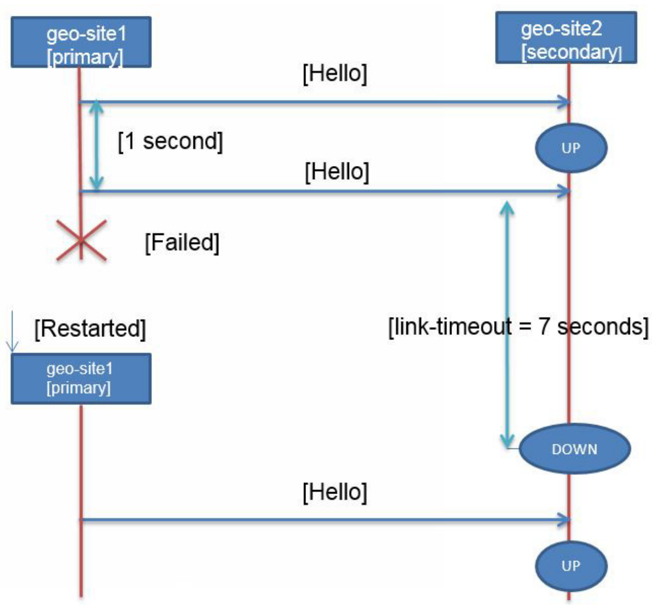
After a switchover occurs, the next switchover will not be initiated until the failure-soak-period time (configurable) expires. This avoids back-to-back switchovers.

Health Checks and Failure Detection

Geo-site clusters support health checks between clusters. Each geo-site cluster performs peer health-checking by sending Hello packets every second. Geo-link failure detection time is seven seconds (default). If a geo-site cluster does not receive the peer's Hello packet within seven seconds, the geo-link is declared *down.*

Shared health-checking on the standby geo-site cluster is activated only after detecting a failure on the active geo-site cluster (through the geo-link or shared health-checking on the active geo-site cluster).

Figure 7-3 Hello Packets Sent between Geo-Site Clusters



Operators must enable at least one interface for geo-redundancy health-checks on AAA (SWm on ePDG), PCRF, or OCS interfaces or RMG, as well. For enabled interfaces, health checks are performed using both private IP addresses and shared IP addresses. Shared IP addresses are advertised by both geo-site clusters using different route metrics. Private IP addresses are advertised by the owned geo-site cluster only.

Within each interface, Operators can select which server-group(s) to include in the geographic redundancy network health-check. Declaring a server-group "unhealthy," means that all the servers in a server-group are also unhealthy.

Network health checks combined with different route-advertised metrics prevent "split brain" scenarios. Split brain refers to a condition where traffic passes through only one geo-site cluster, for example, if a geo-link goes down. A geo-link going down may not trigger a switchover.

Geo-site Cluster Roles and Runtime State Changes

The following use-case examples describe geo-site cluster roles and runtime state changes that occur under certain conditions.

Active geo-site cluster reboot/failure

When the geo-link is *down*, and when all shared network health-checks on the standby geo-site cluster are *up*, the standby geo-site cluster has the active runtime state.

Partial network failures toward OCS/PCRF/AAA/SWm interfaces

When the geo-link is *up*, and all of the following conditions exist, the standby geo-site cluster has the active runtime state:

- One or more health-check-enabled interfaces' shared health-checking is *down* on the active site
- The same interface(s)' shared health-checking is *up* on the standby site
- The local network health-checking is *up* for the remaining health-monitoring-enabled interface(s) on the standby site

Geo-link network failure and partial network failures toward monitored OCS/PCRF/AAA/SWm interfaces

When the geo-link is *down*, and one or more health-check-enabled interfaces' shared network health-checking is *down* on the active geo-site cluster and remains in this state for the failure-soak-period, the active geo-site cluster transitions to the standby runtime state.

Route Metrics

Both geo-site clusters advertise routes for the loopback IPs and UE pools associated with their respective geo-groups but use different metrics depending on each geo-site's role (primary/secondary) and runtime state (active/standby).

Table 7-2 shows the route metrics used for advertising the routes. The lower route metric is the preferred route. Routing metrics support both BGP and OSPF routing protocols.

Table 7-2 Geo-Site Cluster Routing Metrics

Geo-Site Role	Geo-Site Runtime State	Route Metric
Primary	Active	$1
Primary	Standby	$2
Secondary	Active	$3
Secondary	Standby	$4

External nodes do not know which geo-site cluster is active nor the one with which they are communicating. The routing protocols (BGP/OSPF) route/manage this traffic based on the route metric.

Configuration Guidelines

You configure geographic redundancy and the geographic-redundancy group instances as described in Using the CLI to Configure Geographic Redundancy Groups or using the Acuitas Geographic Redundancy Manager application as described in the *Acuitas User's Guide*.

Using Acuitas to Configure Geographic Redundancy Groups

Note: *See the Acuitas User's Guide for details. Set the peer and interface admin status to* ***disabled*** *prior to configuring geographic redundancy attributes on the server-group.*

Use the Acuitas Geographic Redundancy Manager application to manage and monitor Geographic Redundancy Groups as follows:

- Configure geographic redundancy group instances and specify what entities you want placed in the geographic-redundancy group. The user can replicate these entities and their associated configuration data from the active system to the standby system.
- Compare configuration data from the active to the standby node for a selected geographic-redundancy group.
- Replicate configuration data from the active to the standby node for a selected geographic-redundancy group.
- Monitor the health of the geographic-redundancy group.

- Perform a switchover so the standby node takes over service from the previously active node.

Chapter 8

Network Context Parameters

This section describes the network-context objects and parameters supported on the MCC.

Access	config
Guidelines	■ If a gateway is used in conjunction with an http-proxy on the same network-context, a separate loopback(s) must be provisioned for gateway interfaces and http-proxy. ■ Network-context is required to separate AAA traffic on GGSN. Use normal network-context. ■ Operator must configure a separate network-context for an enterprise customer that uses their own DHCP allocation scheme or AAA server (for example, DHCP allocation is performed in an enterprise network). Since DHCP control traffic is required for this VRF, Operators must use the normal network-context and must not use the lightweight network-context. ■ If enterprise customers require separate VRFs for each customer and most customers use local address allocation schemes, Affirmed Networks recommends using a lightweight network-context. ■ Operators cannot modify the vrf-lightweight object or convert a normal network-context to a lightweight network-context or vice versa. This object must be set to either normal or lightweight during the initial creation and cannot be modified. To modify this attribute, you must delete the current configuration for that network-context and add a new network-context. ■ After upgrading, all existing network-contexts are converted to normal network-contexts. If an Operator requires a lightweight network-context, they must remove the current configuration for that network-context and add it as a lightweight network-context.
	The following guidelines apply to mapped UE pools: ■ A mapped-ue-pool-network-context can not have any UE or NAT pools of its own. ■ A network-context that defines a pool of its own can not reference another network-context's pools. ■ A network-context containing any mapped-ue-pool-network-contexts is implicitly a "shared" UE pool network-context. ■ You cannot remove a network-context that is currently sharing its UE pools with one or more mapped-ue-pool-network-contexts.

The following guidelines apply to IP sub pool resizing

Behavior during the modification procedure may differ depending on the type of modification as well as the utilization of the IP sub pools during the modification (IPs assigned to sessions).

Use the following guidelines during the modification procedure:

- The modified IP address range for an ip-sub-pool must overlap with the current range. The system rejects non-overlapping modifications. In that case, create a new ip-sub-pool.
- The new IP address range must not overlap with the IP address range of any existing IP sub pools in that network-context, otherwise the system rejects the modification.
- Increasing the IP address range of an ip-sub-pool takes effect immediately. The system preserves the current sessions and reflects the operating range of the ip-sub-pool modified values immediately.
- The system does not allow any new modification to an IP address range while a modification is in progress.
- During the creation or modification of other IP sub pools, the system checks both the new and the original IP address range of the ip-sub-pool under modification to prevent any overlapping IP address ranges within the same network-context.
- *Shrinking the IP address range of an ip-sub-pool that is free* – results in the modification being completed immediately and the system reflects the new reduced range in the status.
- *Shrinking the IP address range of an ip-sub-pool that is in use* – results in a `modification in progress` state. The operating range becomes the new modified range. That is, new sessions will only be assigned IP addresses from this new modified range. The system preserves the sessions that were in the original range (before the modification) and releases the IP addresses as the sessions are deleted. After all the sessions that were using the IP address range that was shrunk in the last modification are deleted (and the IP addresses are free), the system flags the modification process as `complete`.

network-context

This section describes how to configure network-context parameters.

Configuring Objects

Configure the objects as follows:

Table 8-1 network-context

Object	Description
access-interface	List of access side IP interfaces. See network-context access-interface (page 8-4).
arp	Configure static ARP entries. See network-context arp (page 8-4).

Table 8-1 network-context

Object	Description
block-gtpu-error-pkt	Whether to block sending gtpu error packets when TEID is not found. Options include true or false.
control-dscp-value	Diffserv code point value with which to mark control IP packets on this network context. See Configuring DSCP Values for Control Plane Traffic (page 17-37).
dns	Configure DNS parameters. See network-context dns (page 8-5).
header-sanity-check	Enable or disable validation of TCP and IP header fields. See network-context header-sanity-check (page 8-6).
interested-geo-red-grps	Geored grp list length ranges from 0 to max number of Geo Red Grps. See network-context interested-geo-red-grps (page 8-7).
ip-interface	Configure IP interface parameters. See network-context ip-interface (page 8-7).
ip-route	Configure static IP routes. See network-context ip-route (page 8-10).
ip-shared-subnet	Enable/disable to allow multiple interfaces in the same IP subnet range in a network context.
ip-sub-pool	IP sub pool associated with this network context. See network-context ip-sub-pool (page 8-11).
loopback-ip	List of loopback IP addresses associated with this network context. See network-context loopback-ip (page 8-15).
mapped-ue-pool-network-context	Network Context containing UE Pools to be shared with this network context. See network-context mapped-ue-pool-network-context (page 8-16).
name*	Enter a unique name for this network context.
nat-pool	NAT pool configuration. See network-context nat-pool (page 8-18).
neighbor-discovery-profile	Reference an existing neighbor-discovery-profile. See network-context neighbor-discovery-profile (page 8-25).
number-of-security-services	Enter the number of security tasks servicing this network context.
pass-local-ike-packets	Configure whether packets will be passed with or without the application of any Security Policy Database (SPD) rules. Options include true|false. See network-context pass-local-ike-packets (page 8-26).
security	Enable or disable security firewall operations for this network context. See network-context security (page 8-27).
service-endpoint	Configure a service endpoint on this network context. See network-context service-endpoint (page 8-28).

Table 8-1 network-context

Object	Description
tcp-mss-adjust	TCP MSS adjustment to override the MTU in TCP connections. Overridden by ip-interface tcp-mss-adjust if also set. See network-context tcp-mss-adjust (page 8-29).
tunnel	Configure tunnels. See network-context tunnel (page 8-29).
tunnel-fragmentation-type	Whether to fragment tunneled packets in the inner or outer IP header if necessary. Options include inner|outer.
ue-pool	UE pool configuration. See network-context ue-pool (page 8-32).
vrf-lightweight	Used only for VPN VRF lightweight purposes only. Set to true to use lightweight or false to use normal. See network-context vrf-lightweight (page 8-33). ***Note:*** *VAS Services/BGP/OSPF cannot run in this context*

network-context access-interface

This section describes how to configure the list of access-side IP interfaces.

Configuring Objects

Configure the IP interface that is designated as an access-interface.

network-context arp

This section describes how to configure static Address Resolution Protocol (ARP) entries.

Configuring Objects

Configure the objects as follows:

Table 8-2 network-context arp

Object	Description
ip-addr*	IP address of the target for which this ARP entry is added
ip-interface*	Interface on which this arp entry is configured
mac-address*	MAC address of the target IP endpoint

network-context dns

This section describes how to configure DNS parameters. The MCC HTTP proxy is capable of using either an internal or external DSN-based node type for host name resolution. The DNS query-attempt and query-timeout are configurable. The DNS client uses the configured loopback IP address from which to source internally generated DNS requests. A loopback IP address is **required** for the DNS client inside the network-context.

> ***Note:*** *Modifying any of the network-context DNS attributes results in the MCC reestablishing connections to all servers within the network-context namespace. This causes all pending queries on the connections to be canceled.*

Configuring Objects

Configure the objects as follows:

Table 8-3 network-context dns

Object	Description
dscp-value	DiffServ Code Point value with which to mark DNS IP packets. Options include: dscp_0_be|dscp_01|dscp_02|dscp_03|dscp_04|dscp_05|dscp_06|dscp_07|dscp_08|dscp_09|dscp_10_af11|dscp_11|dscp_12_af12|dscp_13|dscp_14_af13|dscp_15|dscp_16|dscp_17|dscp_18_af21|dscp_19|dscp_20_af22|dscp_21|dscp_22_af23|dscp_23|dscp_24|dscp_25|dscp_26_af31|dscp_27|dscp_28_af32|dscp_29|dscp_30_af33|dscp_31|dscp_32|dscp_33|dscp_34_af41|dscp_35|dscp_36_af42|dscp_37|dscp_38_af43|dscp_39|dscp_40|dscp_41|dscp_42|dscp_43|dscp_44_va|dscp_45|dscp_46_ef|dscp_47|dscp_48|dscp_49|dscp_50|dscp_51|dscp_52|dscp_53|dscp_54|dscp_55|dscp_56|dscp_57|dscp_58|dscp_59|dscp_60|dscp_61|dscp_62|dscp_63
edns0-udp-packet-size	UDP packet size to use for EDNS0. Options include: 512 .. 4096(1280).
loopback-ipv4	Configure IPv4 loopback address.
loopback-ipv6	Configuration of IPv6 loopback address.
server-address	Configure the IP address for the DNS server. ■ ip-address* – IPv4 or IPv6 DNS server IP address ■ tcp-port – Configure DNS server tcp port ■ udp-port – Configure dns server udp port ■ use-extension-dns – When configured, the MCC uses extension mechanisms for DNS and the packet size is increased from 512 (default) to 1280 bytes. If the server receives a response that edns is not supported, a retry is attempted with edns set to off (default).

Table 8-3 network-context dns

Object	Description
transport-protocol	Configure transport protocol used to connect to the DNS server. Affirmed Networks recommends using the default *udp-tcp*. The MCC falls back to *tcp* only if a failed query with a truncated response is received.
query-attempt	The number of times the resolver sends a query to servers before giving up and returning an error. The default is 2 attempts. ■ When only one server is configured within the network-context, the number of query-attempt queries is sent to the server until the total number of attempts or received responses is received. ■ When more than one server is configured within the network-context, retry is not applied on the server that failed, therefor query-attempt is not used.
query-timeout	The amount of time in seconds the resolver waits for a response from a server before retrying the query via a different server. The default is 5 seconds. The value of the query-timeout is the number of seconds each server is given to respond to a query on the first attempt. (After the first attempt, the timeout algorithm becomes more complicated, but scales linearly with the value of timeout.)

network-context fragment-tunneled-ipv6

When fragment-tunneled-ipv6 is enabled, tunneled IPv6 packets that egress through Gn interfaces and that exceed the MRU size are fragmented. Options include:

- true – Fragment IPv6 tunneled packets in the Gn network context that exceed the MRU size.
- false – (default) Discard IPv6 tunneled packets in the Gn network context that exceed the MRU size and set the ICMP error type to “ICMP6_PACKET_TOO_BIG”.

network-context header-sanity-check

When header-sanity-check is enabled, MCC validates TCP and IP header fields (checks for malformed packets) and drops packets that do not concur. This feature provides protection against distributed denial-of-service (DDoS) attacks. Options include:

true – Validate TCP and IP header fields. You can only configure `true` when `network-context <name> security admin-state enabled` is configured.

false – (default) Do not validate TCP and IP header fields.

network-context interested-geo-red-grps

Geored grp list length ranges from 0 to max number of Geo Red Grps. See service-construct geo-redundancy-server (page 9-32). Also see the *Affirmed Networks ePDG Administration Guide*.

Configuring Objects

Configure the objects as follows:

Table 8-4 network-context interested-geo-red-grps

Object	Description
geored-grp-name*	Geo redundancy group that this network-context is part of.

network-context ip-interface

This section describes how to configure the ip-interface parameters. The system supports IPv4 and IPv6 loopback addresses within each network-context. This interface type is not associated with a physical port. I/O redundancy is provided by redundant layer-3 connections and rerouting. This requires that all network services be configured on IPv4 or IPv6 loopback addresses rather than IP interfaces associated with physical ports. The system uses routing protocols to announce reachability of these addresses and the addresses remain reachable to outside systems as long as one physical network interface remains reachable.

Note: *The system does not support configuring multiple overlapping IP interfaces within a single network-context.*

Configuring Objects

Configure the objects as follows:

Table 8-5 network-context ip-interface

Object	Description
admin-state	Configure the admin state of the IP interface.
diffserv-action	DiffServ action for the IP interface. Options include packet-inspection and trust-dscp.

Table 8-5 network-context ip-interface

Object	Description
forwarding-mode	IP interface forwarding mode. ***Note:*** *This feature is supported only on the SA-GiGW and is not supported for Subscriber Unaware GiGW (SU-GiGW). In addition, IP non-transparent mode is not supported.* The SA-GiGW is deployed as a network service header (NSH) unaware service function (SF). The MCC designates an IP interface as a return-to-sender (RTS) interface through the network-context configuration. User traffic received on an RTS IP-interface is returned through the same interface. The forwarding-mode object allows the Operator to tag IP interfaces as an RTS interface. Options include: ip-fib (default) and return-to-sender.
ip-address	Address of the IP interface on which OSPF is enabled. This feature enables support of multiple IP addresses per IP interface.
mtu	IP interface MTU. The default value is 1500, and you can specify a value from 576 through 9100. To specify a value larger than 1800, you must first configure the **`cluster`** ***`<id>`*** **`max-packet-size`** to be at least as large as the **`mtu`** value, and you must reboot the cluster. MCC supports inner fragmentation of IP packets on the S1-U interface. Before adding the GTP-U header and outer IP header, MCC determines if an IP packet fits within the Maximum Transmission Unit (MTU) after those headers are added. If it does not fit, MCC fragments the inner IP packet and adds the GTP-U and outer IP header to each fragment before the SGW/PGW sends them to the eNB. This object ensures the eNB does not have to reassemble fragmented IP packets from the SGW/PGW. Instead, the eNB forwards the IP fragments inside of the tunnel to the UE, which performs the reassembly.
port	Port on which this IP interface is configured.
prefix-length	Prefix length of the IP address.
secondary-ip-address	Secondary address of the IP interface.
secondary-prefix-length	Secondary prefix length of the IP interface.

Table 8-5 network-context ip-interface

Object	Description
startcapture	The capture begins when the Operator executes the startcapture command. This command controls the following parameters: ■ duration – capture duration (in seconds). The default is 1 second. ■ count – packet count (number of packets captured). The default is 1000 packets. The capture continues until the end of the duration or when the max packet count is reached, or when the Operator executes the stopcapture (page 8-9) action command. For more information, see startcapure (page 8-9). ***Note:*** *The system supports one packet capture per IP interface.*
stopcapture	Ends the packet capture session.
tcp-mss-adjust	For more information, see network-context tcp-mss-adjust (page 8-29).
tunnel	Name of the tunnel to which this interface points.
vlan-tag	vlan tag (if any). This feature enables management interfaces to be configured on a VLAN. Use vlan-tag to identify each VLAN with a unique name and numeric tag when more than one VLAN exists. The VLAN tag is 2 bytes in length. The last 12 bits of the tag is reserved for a VLAN identifier (VID), which enables the creation of 4096 VLANs. The default is 0 and 4095 is reserved

startcapure

Note: *The system supports one packet capture per IP interface.*

The packet capture feature is useful in troubleshooting potential problems by capturing IP packets on an IP interface. Operators can start and stop the capture using a capture action command and associated parameters on the IP interface.

MCC stores log messages in var/log/messages to indicate that the first packet is captured. The messages log is not generated without traffic to capture. The log indicates the first packet captured to avoid overwhelming the log file.

MCC captures the packets and stores them as pcapng capture files in *var/log/eventlog*. The file name indicates the network-context, ip-interface, and MAC address of the physical interface where the packets were captured, and the starting timestamp for the capture. For example:

```
<ncID>_<ipID>_<mac>_<starttime>.pcapng
```

network-context ip-route

This section describes how to configure IP static routes. Options include:

Configuring Objects

Configure the objects as follows:

Table 8-6 network-context ip-route

Object	Description
dest*	IP address of the route destination
next-hop*	IP address of the nexthop gateway
preference	Route preference (higher number indicates lower preference of the route)
prefix*	IP prefix length of the route destination
tag	Specify a tag for this route <1-4294967295>

network-context ip-shared-subnet

This section describes how to configure the ip-shared-subnet with this network-context. Previously, MCC required that IP-interfaces be on unique IP subnets within the cluster. This feature supports multiple IP interfaces sharing the same IP subnet within a single network-context.

This feature is supported on both IPv4 and IPv6 IP addresses and helps to achieve IP address conservation and L3 level bundling (similar to LAG at L2 level). This feature is supported on TWAG, ePDG, GiLAN, and SAEGW with and without NAT, and proxy with and without steering and IP transparency.

Configuring Objects

Configure the objects as follows:

Table 8-7 network-context ip-shared-subnet

Object	Description
ip-shared-subnet	Enable/disable to allow multiple interfaces in the same IP subnet range in a network context. This feature supports the following networking/routing functionality with a shared subnet: ■ Interface routes ■ Static routes ■ Single HOP static BFD sessions ■ MH-BGP with / without MPLS ■ Multi active routing without MPLS ■ OSPFv2 with support for establishing OSPFv2 adjacency with one peer router that is in the same subnet and same OSPF area and is not an Affirmed Networks device. The peer router must have only one IP interface in the same subnet as the MCC shared-subnet interfaces (many-to-one connectivity). Set the priority for all OSPF interfaces to zero to support OSPFv2 adjacency. ■ The priority for all OSPF interfaces should be set to zero to support this feature The following limitations apply to this feature: ■ OSPFv3 is not supported with shared subnets for network-contexts. ■ The same family type (IPv4/IPv6) on both primary and secondary IP interfaces is not supported

network-context ip-sub-pool

This section describes how to configure the ip-sub-pool associated with this network-context. If OSPF is enabled, IP subpool addresses are advertised over all IP interfaces in the network-context making the addresses reachable on all of the IP interfaces. Assign the ip-sub-pool to a nat-pool.

Configuring Objects

Configure the objects as follows:

Table 8-8 network-context ip-sub-pool

Object	Description
admin-state	Administrative status: disabled, enabled (default), or maintenance. NAT treats disabled and maintenance states the same. A sub-pool in maintenance or disabled state will not allow any new port chunks to be allocated from any of its addresses. Operators can change the state at any time with the exception of the disabled to maintenance.

Table 8-8 network-context ip-sub-pool

Object	Description
advertise	Indication to dynamic routing protocols whether this pool should be advertised or not: yes or no.
advertise-host-based-routes	Indication to GW whether subscriber assigned from this pool should be advertised. This object controls the advertisement of UE routes during subscriber attachment, instead of advertising routes during configuration. Advertises the UE static IPv4 address as a /32 host route and the IPv6 address as a /64 route on the Gi network after the IP address is allocated to the UE. After the UE session is deleted, the UE IP address is released to the pool and the corresponding host-based route is withdrawn. This feature enables the PGW to handle a maximum of 10,000 UE routes. After the subscriber attaches to the network and MCC assigns the IP address, the PGW installs the host-based route corresponding to the UE IP address (/32 for IPv4 and /64 for IPv6) to the NetAgent for advertisement. ***Note:*** *This feature does not apply to delegated IPv6 prefixes.* ***Note:*** *These attributes are supported during call/session creation only. Any changes to* ***ip-sub-pool advertise*** *or* ***advertise-host-based-routes*** *after call/session creation may result in undefined behavior.* Options include: ■ no (default) – MCC does not advertise the route. ■ yes – MCC advertises the UE IP as a /32 or /64 route received from AAA during session establishment. This parameter can be set to **yes** only when **advertise** is set to **no** and **ip-sub-pool-type** is set to **UE-IP-Pool**.
content-filter-instance	Content filter instance using the IP addresses from the subpool
ip-sub-pool-type	Type of IP sub pool: UE-ip-pool or NAT-ip-pool.
name	Unique name for this IP sub pool. The ip-sub-pool name must be unique across network-contexts.
nat-pool	NAT pool associated with an IP sub-pool of type NAT-IP-Pool.
range-end	Ending value for IP address [v4] or prefix [v4/v6] of this IP sub-pool range. The range can only be modified while the sub-pool is in a disabled state and no connections are using the sub-pool. Operators can't disable the state and change the range in the same commit operation.
range-start	Starting value for IP address [v4] or prefix [v4/v6] of this IP sub-pool range.

Table 8-8 network-context ip-sub-pool

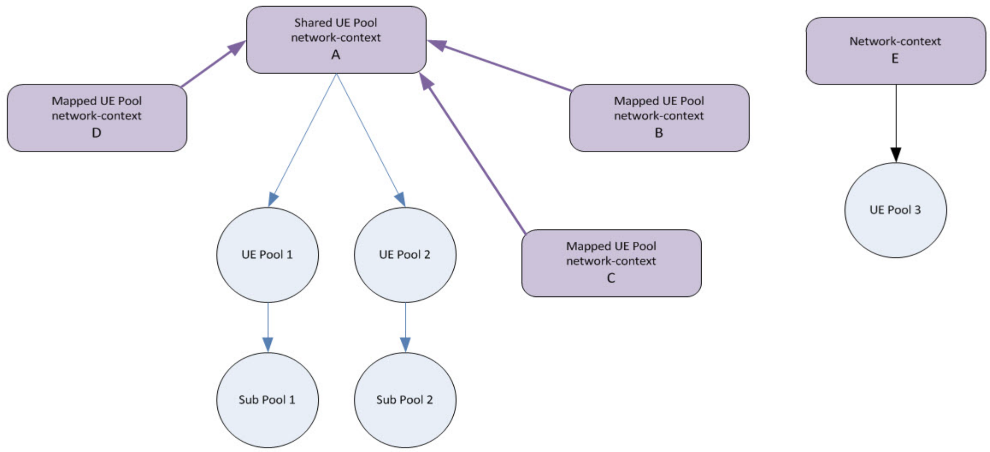
Object	Description
router-ip-address	The router IP address assigned to the DHCP subscriber. If the router-ip-address is in the same range as IP addresses belonging to the ip-sub-pool, then it is automatically reserved. This ensures that the UE IP will not have the same IP as the router IP. MCC also ensures that no IPs in the pool can be a loopback address unless it's a router IP. Affirmed Networks recommends that the ip-sub-pool is set to admin down when the router-ip-address is set, modified, or deleted.
route-summary-prefix-length	Create an aggregate route prefix for peer advertisement, independent of the size of the UE pool or NAT pool. This option is used with the range-start address to advertise just one route for this ip-sub-pool instead of automatically calculating the route or list of routes to exactly match the range specified by range-start and range-end. As long as the prefix masked with the range-start is the same as the range-end, it is used instead of the calculated prefix or set of prefixes that would otherwise be advertised.
route-tag	A unique tag to correlate this sub-pool with route maps.
route-ip	Router IP for DHCP v4 allocation.
suppressicpm	Determines if the forwarding engine will generate icmp responses for packets sent to unallocated addresses: true or false.
ue-pool	UE pool name associated with an IP sub-pool of type UE-IP-Pool.

UE Pools Shared by Multiple Network Context

The shared UE pools feature allows a network-context to reference another network-context resulting in sharing that network context's UE pools in order for the address space to be shared. Sharing or "aliasing" a network-context UE pool among a group of network-contexts results in more efficient MVNO turn-up by eliminating the need to configure new pools for each network-context.

In the example in Figure 8-1,Shared UE Pool network-context A owns UE Pool 1 and UE Pool 2 and their associated Sub Pools. Mapped UE Pool network-context B, C, and D have references to Shared UE Pool network-context A indicating Mapped UE Pool network-context B, C and D can use Shared UE Pool network-context A's UE Pool 1 and 2. Network-context E has its own mutually exclusive pool and therefore can't share the Shared UE Pool network-context A's pools.

Figure 8-1 Shared UE Pools



force-ue-ip-subpool-range-modification

This section describes how to force the completion of the modification process and may impact a large number of subscribers. This command deletes the sessions from the IP address range that is being shrunk, sends a `clear session` request to the peers, and forces the completion of the modification process. For example, if an ip-sub-pool is shrunk and there are sessions using the IP addresses in the shrunk range, Operators can use this command to force those sessions off of the IP address range that is no longer part of that ip-sub-pool.

This feature enables Operators to modify the IP address range associated with an ip-sub-pool that is associated with a UE pool. Both Ipv4 and Ipv6 pools are supported.

Note: *This procedure applies only to IP sub pools associated with UE pools. Although the ip-sub-pool object is used for NAT pools, the modification process for IP sub pools associated with NAT differs from this procedure.*

---

network-context logical-mgmt-ip-address

This section describes how to configure the logical management IP address. MCC supports IPv4/IPv6 addresses. Use the following guidelines:

- MCC supports one management addresses per MCM for IPv6 addresses.
- MCC supports one address type (not a mix of IPv4/IPv6 addresses) for each pair of MCMs or each management interface.
- The management interfaces are not backward compatible. In other words, do not provision an IPv6 management interface and then downgrade to a release that does not support IPv6 management interfaces.

Configuring Objects

Configure the objects as follows:

Table 8-9 network-context logical-mgmt-ip-address

Object	Description
address	Logical management IP address.

network-context loopback-ip

This section describes how to configure a list of loopback IP addresses associated with this network-context. The loopback-ip is used as control and data endpoint addresses. Also see http-request-policy dns-resolution.

Note: *MCC prevents the modification of the loopback ip-address after it is configured. You can create a new loopback with the ip-address and associate it with the application or service.*

Configuring Objects

Configure the objects as follows:

Table 8-10 network-context loopback-ip

Object	Description
allow-network-icmp-to-loopback	Flag to control Network ICMP ECHO is allowed to Loopback Address. Set to false|true. See allow-network-icmp-to-loopback (page 8-16).
allow-subscriber-icmp-to-loopback	Flag to control Subscriber level ICMP ECHO is allowed to Loopback Address. Set to false|true. See allow-subscriber-icmp-to-loopback (page 8-16).
geo-redundancy-group	Geographic Redundancy group,if any, to which this loopback belongs.
gtp-u	Specifies whether to use this loopback address for GTP-U. ■ yes (default) – Use yes only if this loopback address is used as a GTP-U endpoint. ■ no – Use no to disable GTP-U when it is not applicable to a loopback address. This improves security hardening.
ip-address*	IP address of this loopback object
name*	Name of the loopback IP entity

allow-network-icmp-to-loopback

Enables the PGW/GGSN to respond to incoming Internet Control Message Protocol (ICMP) echo requests. If the PGW/GGSN receives an ICMP echo request from a network device, it responds with an ICMP echo reply. Operators can disable this object on a specific IPv4 loopback address and control whether the PGW/GGSN responds to ICMP echo requests from either the subscriber (UE) or the network or both. Note that the PGW/GGSN does not currently initiate ICMP echo requests to the network. The PGW/GGSN only responds to ICMP echo requests to loopback IP addresses associated with the same network-context as network device. The PGW/GGSN does not handle pings originating from the UE to the network or from the network to the UE.

allow-subscriber-icmp-to-loopback

Enables the PGW/GGSN to respond to incoming Internet Control Message Protocol (ICMP) echo requests. If the PGW/GGSN receives an ICMP echo request from a subscriber (UE), it responds with an ICMP echo reply. Operators can enable this object on a specific IPv4 loopback address and control whether the PGW/GGSN responds to ICMP echo requests from either the subscriber (UE) or the network or both. Note that the PGW/GGSN does not currently initiate ICMP echo requests to the subscriber. The PGW/GGSN only responds to ICMP echo requests to loopback IP addresses associated with the same network-context as the subscriber (UE).

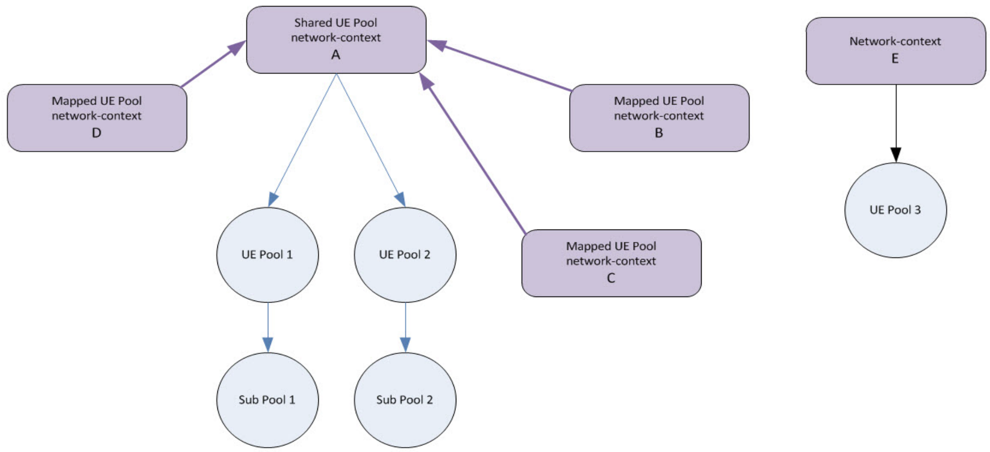
network-context mapped-ue-pool-network-context

This section describes how to configure a network context containing UE pools to be shared with this network context.

The shared UE pools feature allows a network-context to reference another network-context resulting in sharing that network context's UE pools in order for the address space to be shared. Sharing or "aliasing" a network-context UE pool among a group of network-contexts results in more efficient MVNO turn-up by eliminating the need to configure new pools for each network-context.

In the example in Figure 8-2, Shared UE Pool network-context A owns UE Pool 1 and UE Pool 2 and their associated Sub Pools. Mapped UE Pool network-context B, C, and D have references to Shared UE Pool network-context A indicating Mapped UE Pool network-context B, C and D can use Shared UE Pool network-context A's UE Pool 1 and 2. Network-context E has its own mutually exclusive pool and therefore can't share the Shared UE Pool network-context A's pools.

Figure 8-2 Shared UE Pools



network-context mgmt-ip-interface

This section describes how to configure the management IP interface objects. MCC supports IPv4/IPv6 addresses. Use the following guidelines:

- MCC supports one management addresses per MCM for IPv6 addresses.
- MCC supports one address type (not a mix of IPv4/IPv6 addresses) for each pair of MCMs or each management interface.
- The management interfaces are not backward compatible. In other words, do not provision an IPv6 management interface and then downgrade to a release that does not support IPv6 management interfaces.

Configuring Objects

Configure the objects as follows:

Table 8-11 network-context mgmt-ip-interface

Object	Description
chassis*	Chassis number
gateway-ip	IP address of the default gateway
ip-address*	Address of the Mgmt IP interface
port*	Management port id
prefix*	Prefix length of the Mgmt IP address
slot*	MCM slot number

Table 8-11 network-context mgmt-ip-interface

Object	Description
vlan-tag	Mgmt interface VLAN Tag (if any). VLANs enable Ethernet network traffic to be partitioned based on functional requirements while maintaining connectivity across all devices in the network. Use vlan-tag to specify the numeric VLAN tag for this mgmt-ip-interface when it is connected to such an Ethernet network. The tag is a 12-bit identifier (VID), which enables the creation of 4094 VLANs. The default value of 0 disables VLAN tag filtering and the value 4095 is reserved.

network-context nat-pool

NAT pool configuration.

Configuring Objects

Configure the name of the nat-pool associated with the subscriber-firewall-flow.

Configure the objects as follows:

Table 8-12 network-context nat-pool

Object	Description
admin-state	Set the admin status to disabled or enabled. A pool can be disabled at any time. ■ For a **statistical** nat-allocation-policy – Disabling a NAT pool purges all existing flows, thereby releasing all port chunks and prevents any new flows that require NAT. ■ For a **deterministic** or **guaranteed** nat-allocation-policy – Disabling a NAT pool purges all existing sessions and prevents any new session creation that require NAT.
geo-redundancy-group	Geographic Redundancy Group to which this NAT pool belongs. ***Note:*** *If you are modifying this value, you must first disable the NAT pool.*
max-chunks-per-session	Maximum number of port-chunks that can be concurrently assigned to a UE/APN - - 0 for Unlimited 0 .. 512(0) ***Note:*** *The new value applies only to new port chunk allocations. Existing sessions exceeding the new value are not allocated any new port chunks until they have released enough port chunks to drop below the new value.* ***Note:*** *If you are modifying this value, you must first disable the NAT pool.*

Table 8-12 network-context nat-pool

Object	Description
max-users-per-NAT-IP	The maximum number of sessions that can be assigned to a public IP. If not configured, no limit is enforced. A session can use any or all of the available NAT addresses. A NAT pool is considered to be a distributed NAT pool. In this case, port chunk redundancy and flow redundancy are not supported. If configured: ■ NAT pool is considered to be centrally managed. ■ restricts every session to using only one NAT address from the pool and limits the number of sessions that can concurrently share the same NAT address. ***Note:*** *If you are modifying this value, you must first disable the NAT pool.*
min-chunks-per-session	The number of port-chunks that will be retained for this UE/APN once allocated. Not applicable for deterministic nat-allocation-policy. ***Note:*** *If you are modifying this value, you must first disable the NAT pool.*

Table 8-12 network-context nat-pool

Object	Description
nat-allocation-policy See nat-allocation-policy (page 8-22).	Select the allocation policy that controls when port chunks are assigned to UEs when using this NAT pool. Set this option to **statistical** if you intend to configure the Flow Authorization Interface (page 19-1). ■ deterministic – port chunks are assigned in a deterministic way at session establishment. To configure deterministic NAT64, associate an IPv4 nat-ip-sub-pool with an IPv6 ue-ip-sub-pool. ■ guaranteed – port chunks are assigned at session establishment ■ statistical – port chunks are assigned on-demand during connection creation The following options apply to the nat-allocation-policy objects: ■ admin-state – Administrative status ■ geo-redundancy-group – Geographic Redundancy Group this NAT pool belongs to ■ max-chunks-per-session – Maximum number of port-chunks that can be assigned to a UE/APN - 0 for Unlimited ■ max-users-per-NAT-IP – The maximum number of UE/APNs that can be assigned to a public IP - if not configured, there is no limit enforced ■ (statistical) min-chunks-per-session – The number of port-chunks that will be retained for this UE/APN once allocated. ■ nat-port-chunk-size – Number of contiguous ports that will be assigned at a given time from a NAT IP address ■ nat-portchunk-load-balancing-algorithm – Port chunk load balancing algorithm to use for this NAT pool
nat-filter	Select the filtering method for downlink traffic. This determines how strict the MCC is in allowing flows from external networks to an internal endpoint. ■ address-dependent – Downlink traffic from an external IP address is accepted if the internal endpoint has first sent packets to the external endpoint's IP address. ■ address-port-dependent – (Default) Downlink traffic from an external IP address and port is accepted if the internal endpoint has first sent packets to the external endpoint's IP address and port. This is the strictest filtering method. ■ endpoint-independent – Downlink traffic from any external endpoint is accepted if the internal endpoint has first sent packets to any external address. Endpoint-Independent Filtering (EIF) is required for STUN client/server applications. ■ not-set

Table 8-12 network-context nat-pool

Object	Description
nat-hold-timer	Timer value (in seconds) to age out and release inactive NAT address/portchunk allocations from a Statistical NAT pool for a subscriber session0 .. 3600(0)
nat-mode	Setting that controls whether NAT or NAPT is applied for subscribers using this NAT pool. ■ Many-to-One – NAT pool mode where a range of ports in an IP address is assigned to a subscriber for NAPT. Use this options for UEs supporting NAT-T. ■ One-to-One – NAT pool mode where an entire IP is assigned to a subscriber for NAT. Use this options for UEs not supporting NAT-T.
nat-port-chunk-size	Size of contiguous port block that will be assigned from a NAT IP address. Options include: 8|16|32|64|128|256|512 The port block range used by NAPT is 1024-65535. Each workflow module is allocated a subset of this range. In the workflow module, a UE session with active connections is allotted a chunk or chunks of this port block. ***Note:*** *If you are modifying this value, you must first disable the NAT pool.*
nat-portchunk-load-balancing-algorithm not applicable for deterministic nat-allocation-policy	Port chunk load balancing algorithm to use for this NAT pool. Options include: ■ pc-loadbalancing-fill-subpools-sequential – This algorithm tries to pack the port chunk allocations into as few IP addresses as possible, and into as few IP sub-pools as possible, beginning with the first IP sub-pool associated with the nat-pool. ■ pc-loadbalancing-round-robin-subpools – Assigns the port chunks, beginning with the first IP / port chunk and continuing with the next IP sub-pool for the next port chunk allocation request. MCC uses a strict round-robin approach across all IP sub-pools associated with the nat-pool, regardless of the number of port-chunks in use in these IP sub-pools.

Table 8-12 network-context nat-pool

Object	Description
nat64-prefix	IPv6 prefix for this NAT Pool for NAT64 - default is 64::ff9b::/96 defined by RFC 6052. The feature enables the MCC to perform stateful Network Address Translation (NAT) from an IPv6 address to an IPv4 address (NAT64). This allows an IPv6 UE to initiate communications to a IPv4-only server. This feature supports stateful IPv6 to IPv4 translation (RFC 6146) and IPv6 to IPv4 packet translation (RFC 6052). All NAT policies currently supported for NAT44 also apply for NAT64 (statistical, guaranteed, and deterministic). This feature also supports: ■ one-to-one NAT support ■ NAT64 for TCP, UDP, and ICMP (echo-request and echo-reply) ■ Address Dependent Filtering (ADF) ■ Endpoint-Independent Mapping (EIM) ■ Endpoint-Independent Filtering (EIF) ■ Address/Port Dependent Filtering (APDF) ■ Flow Authorization
nat64-prefix-length	Length of the NAT64 IPv6 prefix in use with this NAT Pool. 32|40|48|56|64|96| ***Note:*** *If this object is set to 96, bits 64 to 71 of the nat64-prefix must be 0.*

nat-allocation-policy

MCC supports the statistical, guaranteed, and deterministic modes of NAT operation. In *guaranteed* and *deterministic* mode, port chunks are assigned at session establishment, whereas in *statistical* mode, port chunks are assigned at the time of flow creation. Operators configure which NAT pool(s) to use and the number of port chunks to assign to the session.

Note: *For NAT pools configured as guaranteed or deterministic, set the nat-port-chunk-size to the highest possible value depending on the total ports (and max-chunks-per-session setting) expected to be allocated for every user. For example, if the total ports per user is 640, Affirmed Networks recommends settings the nat-port-chunk-size to 128 and the max-port-chunks-per-user to 5. The optimal setting is to configure the max-port-chunk-size to the highest possible value and as close as possible to the max-ports-per-user.*

Setting the port chunk size to a low value will impact system performance during session setup and modification of NAT pools. If you are modifying a NAT pool configuration with large IP sub pools, a lower port chunk size may cause the configuration modification command to time out.

MCC supports the following modes to configure NAT functionality:

- *Statistical mode* (current implementation) – NAT IP/Port binding is done entirely at the time of NAT flow creation. No port chunks are allocated to the UE during session set-up time. Port bindings occur entirely on an on-demand basis. Fixed mapping of UE IP to NAT IP and port chunk does not occur.

Note: *With Statistical one-to-one NAT pools, each IP sub-pool is allocated based on the max-nat-workflow-modules setting, therefore, each WSM/SSM receives a subset of the IP range from each pool. Depending on the distribution of the sessions, a NAT one-to-one allocation may fail on one WSM/SSM due to lack of resources, even though other WSM/SSMs may have IPs available. The NAT one-to-one pool must be over-provisioned to reduce the possibility of allocation failure.*

- *Guaranteed mode* (new implementation) – For each UE, a fixed number of port chunks (with IP) are allocated during session setup. No dynamic allocation of port chunks occurs during connection set up. The connection fails if no NAT port is available from the initial allocated chunk(s).

  MCC supports intra-cluster NAT port chunk redundancy (between the WSMs/SSMs) for the guaranteed mode of operation. The port chunks assigned to a session are mirrored across the active and standby session. After the WSM failover occurs, the new bindings are allocated from the same port chunk as the original active workflow module. In this case, the actual NAT binding per flow are not preserved.

- *Deterministic* NAT is supported on the MCC for PGW, GGSN, and GiGW (subscriber-aware mode only). Deterministic NAT uses a deterministic algorithm to allocate NAT IP port ranges for an associated ue-ip-sub-pool from a nat-ip-sub-pool.

  NAT IP and port-chunk for a subscriber is allocated during call setup by looking up the UE IP and running a deterministic algorithm with complexity O(1).

  With the implementation of deterministic NAT support, the NAT IP and port chunk always remain the same, provided no changes are made to ue-ip-sub-pool, nat-ip-sub-pool, etc. If a subscriber is disconnected and another subscriber connects with the same UE IP, then NAT IP and port chunk also remain the same for this subscriber.

  This feature enables Operators to:

  – specify a nat-pool to be used for deterministic NAT

  – associate a nat-ip-sub-pool with a ue-ip-sub-pool

  – control NAT subscriber traffic using a NAT IP port-range obtained via deterministic NAT

  – display the allocation for NAT IP port ranges for UE IPs

For example, for a given association of a nat-ip-sub-pool and ue-ip-sub-pool, an Operator can obtain a deterministic map of NAT IPs for a UE-IP anytime without preserving NAT binding records externally, thereby avoiding the need to store NAT binding records.

The following functions are not supported with this feature:

– Deterministic NAT support for GiGW subscriber unaware mode

– Endpoint independence support for deterministic NAT

– ALGs support for deterministic NAT

– NAT IP and UE IP deterministic map is not guaranteed across configuration changes for nat-pool, nat-ip-sub-pool, and ue-ip-sub-pool.

Geo-redundancy Support

MCC supports NAT port chunk redundancy to geo-redundancy groups for the guaranteed mode of operation. The port chunks assigned to a session belonging to a geo-redundant group are mirrored to its standby geo-redundant group in a different cluster. After the active geo-redundant group failover occurs, new flow bindings are allocated from the same port chunks as the original geo-redunant group. In this case, the actual NAT binding per flow are not preserved.

NAT and Endpoint Independent Mapping

The MCC always performs Endpoint Independent Mapping (EIM) when it performs any mode of NAT (guaranteed, statistical, or deterministic). No configuration is needed to enable EIM.

EIM assigns the same external address (NAT IP) and port (NAT Port) for all flows initiated from an internal (UE) IP and port, regardless of the destination IP address and port.

EIM provides an efficient use of NAT ports. EIM and endpoint-independent filtering are required for STUN client/server applications, and allow P2P applications to work seamlessly even when peers are behind the NAT.

Using NAT with Tunneling Protocols

IPsec traffic passes through MCC’s NAT functionality as follows:

- Packet filter is configured for IKE and ESP packets to select the nat-pool for NATíng of IKE and ESP packets
- nat-pool is configured as **One-to-One** in nat-mode
- The ESP traffic is NAT’ed to the same IP address that is used for NATing IKE traffic
- ESP header does not contain port number, therefore only IP address translation is performed

- IKE flow (well-known UDP destination port 500) is NAT'ed, similar to any other UDP packet

MCC applies NAT to IPsec packets without breaking end-to-end IPsec traffic, when the endpoints do not support NAT Traversal (NAT-T). This feature supports NAT-T on IPsec devices that encapsulate ESP packets within a UDP packet (port 4500), therefore no IPsec awareness is required at the NAT device.

The following limitations apply to this feature:

- Supports NAT44 and NAT66; NAT64 is not supported.
- Supports ESP only; AH is not supported
- Supports tunnel mode IPsec; transport mode is not supported
- The endpoints must use port 500 for IKE
- With **One-to-One** nat-mode, the number of UEs that can use IPsec tunnels at the same time is equal to the number of configured NAT IPs

To ensure a successful tunnel setup, an ip-sub-pool with a NAT IPv4 address must have sufficient IPs to support IPsec tunnels from multiple UEs at the same time. Configure the NAT pool as follows:

- Single NAT Pool – A single NAT pool is configured to NAT IPsec traffic from all UEs, regardless of NAT-T support. Operators can also use this configuration if the remote server/gateway does not support NAT-T. The following guidelines apply to single NAT pools:

  To allow for maximum NAT IP reuse, Affirmed Networks recommends setting the **allocation-policy** to **statistical** for NAT44 and **deterministic** for NAT66. Set the **nat-mode** to **One-to-One**. To ensure that NAT bindings remain active during the tunnel lifetime, configure an appropriate **nat-hold-timer** value for statistical NAT. This value must be slightly greater than the period for which the tunnel may remain idle. That is, this value must be greater than the IKE keepalive-timeout etc.

- Multiple NAT Pools – In deployments where the majority of UEs support NAT-T and only a small percentage do not, MCC uses the **Many-to-One** nat-mode to select NAT IPs. Set alg to **ipsecnattassist** to optimize use of public NAT IPv4 addresses. For UEs that do not support NAT-T, MCC uses NAT IPs from the **One-to-One** nat-mode configured as ipsec-natt-assist-nat-pool in the subscriber-firewall-flow.

network-context neighbor-discovery-profile

Reference an existing neighbor-discovery-profile and set parameters such as default-lifetime, reachable-time, MTU, etc in the neighbor discovery router advertisement as well as the number of unsolicited router advertisements to send when a session starts.

Configuring Objects

Configure the objects as follows:

Table 8-13 network-context neighbor-discovery-profile

Object	Description
default-lifetime	The lifetime associated with the default router in units of seconds
name*	Name for this neighbor discovery profile.
number-unsolicited-ra	The number of unsolicited router advertisements the MCC would send when a UE attaches to the network. 0 indicates to send continuously spaced by ra-interval
ra-mtu	MTU for ipv6 Neighbor Discovery Router Advertisements
reachable-time	The time, in milliseconds, a neighbor is considered reachable after receiving a reachability confirmation. 0 indicates unspecified
retransmit-timer	The time in seconds between retransmitted Neighbor Solicitation messages
suppress-ra	Whether to suppress sending icmpv6 Router Advertisements

network-context pass-local-ike-packets

Determine whether packets will be passed with or without the application of any Security Policy Database (SPD) rules. Options include:

true (default) – pass all local packets without applying a security policy. IKE requests are automatically passed.

false – pass only those local packets that conform to a configured security policy. MCC performs a security policy lookup of the IKE requests.

This feature limits the ability of MCC endpoints to accept IKE packets to initiate an IPsec tunnel to the MCC. To initiate the IPsec tunnel, IKE requests must be qualified by originating from a list of configured IP addresses or IP address range(s). Unqualified IKE packets sent by Wi-Fi calls to the MCC are dropped without IKE processing. This feature provides protection against DOS attacks by using local IP addresses.

To allow only IKE packets, configure specific packet filters to allow only ports 500 and 4500. For example:

- Pass packets without applying an SPD rule:

```
an3000-mcm-slot7cpu1(config)# network-context BOSTON
pass-local-ike-packets <true/false>
```

```
Possible completions:
  <true/false>[true]
```

- Pass packets with a rule and configure an SPD rule:

```
an3000-mcm-slot7cpu1(config)# network-context BOSTON
pass-local-ike-packets false
```

```
an3000-mcm-slot7cpu1(config-network-context-GN)# security policy
rule <network-context> <name> priority <value> action <pass | drop
| reject> clone-reverse <empty> source-packet-filter <name>
protocol <value> ip-start <ip address> ip-prefix-length <value>
destination-packet-filter <name> protocol <value>
```

network-context ping

This section describes how to send ICMP echo messages.

Configuring Objects

Configure the objects as follows:

Table 8-14 network-context ping

Object	Description
count	Number of ping packets to send to target
sourceipaddress	Source IP address to use
targetipaddress*	Address to ping

network-context security

Enable or disable security firewall operations for this network context. If you want to enable the header-sanity-check feature (see network-context header-sanity-check (page 8-6), you must first enable security.

Configuring Objects

Configure the objects as follows:

Table 8-15 network-context security

Object	Description
admin-state	Enable or disable security firewall operations for this network context. Options include: ■ disabled ■ enabled

network-context service-endpoint

The name of this service-endpoint on this network-context must be unique. Configure the service-endpoint on this network-context as follows:

The UDP port range is configurable for geo-redundant links. Based on the configuration, the firewall allows those ports (UDP). Each geo-redundant cluster on either side of the geo-redundant link has its own configurable port range. Although the port range is usually the same, it is not mandatory. For example:

```
an3000-mcm-slot8cpu1# show running-config network-context NYC
service-endpoint
network-context NYC
service-endpoint GEO-SVE-1
  local-loopback-address    GEO-LBK-1
  local-port-range-base     5500
  local-port-range-numports 128
!
service-endpoint GEO-SVE-2
  local-loopback-address    GEO-LBK-2
  local-port-range-base     3000
  local-port-range-numports 128
!
!
```

Configuring Objects

Configure the objects as follows:

Table 8-16 network-context service-endpoint

Object	Description
local-loopback-address*	Loopback address on this network context to use for this service endpoint. The local-loopback-address can be shared across multiple services such as RADIUS, DHCP, etc.
local-port-range-base*	Starting port on the loopback address to use for this service endpoint
local-port-range-numports	Number of ports to reserve starting from the base port for this service endpoint
name*	Name of this service endpoint on this network context. Must be unique

network-context tcp-mss-adjust

Maximum Segment Size (MSS) is part of a “handshake” mechanism whereby each side of a TCP link informs the other of its MSS. This feature protects the RAN from the overhead of potential fragmentation issues. In addition, control traffic is less likely to get lost due to fragmentation along the path.

This feature allows the MCC to modify the TCP MSS that is being negotiated by two TCP endpoints. The ability to configure and override TCP MSS enables Operators to control the MSS for TCP sessions, independent from what the TCP endpoints negotiated.

Operators adjust the TCP MSS setting under the interface and network-context object. If both settings are configured, the interface setting overrides the network-context setting.

When a TCP PDU is intercepted with a SYN or SYN-ACK bit set in the FLAG portion of the TCP header, the MCC checks whether a network-context or interface MSS adjustment is necessary. The MCC lowers the MSS in the header to this amount, if the setting is lower than what is in the header. If the header value is already lower, it flows through unmodified. The end hosts use the lower setting of the two hosts.

network-context tunnel

This section describes how to configure the following tunnel types: GRE, L2TP, UDP, and null tunnels. See Tunneling Overview (page 8-30).

Configuring Objects

Configure the objects as follows:

Table 8-17 network-context tunnel

Object	Description
include-checksum	Should optional checksum field be included in the tunnel packets?
local-address	Local IP address of the tunnel, that is source IP address in the outer packet. Must be a loopback address.
l2tp-peer-ip-address	(*static tunnels only*) Configure the remote IP address (destination) of the tunnel and priority assigned to this IP address (1 - highest, 255 - lowest).
mtu	(*L2TP tunnels only*) Maximum sized IP packet that can be sent over the tunnel without fragmentation. See mtu (page 8-32).

Table 8-17 network-context tunnel

Object	Description
remote-ip-address	Remote IP address (destination) of the tunnel. If not specified, a new GRE tunnel is dynamically created whenever a GRE packet with an unrecognized outer packet source address is received at the local-address loopback interface.
tunnel-password	(*static tunnels only*) Password to be used for setting up L2TP tunnel.
type	Select the tunnel type: Null, GRE, L2TP, UDP. See Tunneling Overview (page 8-30).

Tunneling Overview

This section describes an overview of tunneling, L2TP tunnels, GRE tunnels, and policy-based tunneling. The information in this section conforms to *RFC 2784: Generic Routing Encapsulation.*

For information about using NAT with tunneling protocols, see network-context nat-pool (page 8-18).

Tunneling provides a private path for transporting packets through an otherwise public network by encapsulating packets inside a transport protocol known as an IP encapsulation protocol.

Generic Routing Encapsulation

Generic routing encapsulation (GRE) is an IP encapsulation protocol used to transport packets of many different protocols. The protocol that is carried is called the passenger protocol, and the protocol that is used for carrying the passenger protocol is called the transport protocol. GRE encapsulates packets into IP packets and redirects them to an intermediate host where they are de-encapsulated and routed to their final destination.

Information is sent from one network to the other through a GRE tunnel. GRE tunnels are used to transfer traffic that might not otherwise be able to traverse the network. GRE tunnels can also be used to encapsulate traffic so that the traffic inside the tunnel is unaware of the network topology. In addition, regardless of the number of network hops between the source and destination, the traffic passing over the tunnel sees it as a single hop. Because tunnels hide the network topology, the actual traffic path across a network is also unimportant to the traffic inside the tunnel. GRE tunnels behave as virtual point-to-point links that have two endpoints identified by the tunnel source and tunnel destination addresses at each endpoint.

A control-profile may instruct workflow to bypass routing of egress traffic and force upstream packets into a pre-configured GRE tunnel. This is known as policy-based tunneling or compulsory tunneling. When workflow reaches such a conclusion during its processing of upstream packets, it encapsulates UE packets with a GRE tunnel header and performs a Longest Prefix Match (LPM) on the tunnel IP destination address. The result of this lookup determines the egress interface to which the packet is forwarded.

GRE Keepalive Support

This feature adds support for initiating GRE keepalive request packets on the MCC. The keepalive request packets are initiated per-tunnel towards the tunnel endpoint and are based on the timer configuration. The GRE tunnel is marked as down if a GRE keepalive response is not received for the tunnel and the number of such failures reaches the configurable threshold.

This feature is used on heavyweight tunnels that use the GRE tunnel state updated by GRE keepalives. For example:

- OSPF traffic uses the GRE tunnel state updated by GRE keepalives
- V-tap traffic uses the GRE tunnel state updated by GRE keepalives

This feature also supports GRE keepalive requests initiated from external servers and responds to external servers with a GRE keepalive response.

Special considerations

- This feature supports only IPV4 GRE keepalives.
- Lightweight tunnels are not supported.
- BGP traffic uses the GRE tunnel state updated by GRE keepalives. Keepalive functionality is supported for the same number of supported GRE tunnels. Keepalive functionality is supported only for 128 GRE tunnels.
- The tunnel state updated by keepalives also affects the traffic through the GRE heavyweight interface.
- After an SSM recovery, the timer is restarted and the time elapsed for previous SSM per-tunnel timers is lost.

Layer 2 Tunneling Protocol (L2TP)

This release adds support for the Layer 2 Tunneling Protocol (L2TP). The two endpoints of an L2TP tunnel are:

- *LAC (L2TP Access Concentrator)* — The device that physically terminates a call and initiates the tunnel.

- *LNS (L2TP Network Server)* — The device that terminates and possibly authenticates the Point-to-Point Protocol (PPP) stream.

After a tunnel is established, the network traffic between the peers is bidirectional.The LAC supports PPP regeneration with CHAP or PAP authentication. PPP defines a means of encapsulation to transmit multi-protocol packets over L2 point-to-point links.

The PGW/LAC operates in PPP regeneration mode. The L2TP tunnel attributes are either supplied by RADIUS or a static configuration. The RADIUS Access-Accept message carries one or more L2TP tunnel AVPs which instruct the PGW (along with ue-ip-allocation-mode) to use L2TP for the session. If RADIUS returns multiple L2TP tunnel AVPs, the LAC searches for an LNS.

In a static configuration scenario, MCC uses the tunnel configuration to establish an L2TP tunnel + L2TP session. MCC also uses the statically configured tunnel AVPs, for example, tunnel password, tunnel server endpoint (which is the LNS IP address). For more information on AVPs, see the *Affirmed Networks RADIUS Interface Specification*.

mtu

MCC supports inner fragmentation of IP packets on the S1-U interface. Before adding the GTP-U header and outer IP header, MCC determines if an IP packet fits within the MTU after those headers are added. If it does not fit, MCC fragments the inner IP packet and adds the GTP-U and outer IP header to each fragment before the SGW/PGW sends them to the eNB.

This feature ensures the eNB does not have to reassemble fragmented IP packets from the SGW/PGW. Instead, the eNB can forward the IP fragments inside of the tunnel to the UE, which performs the reassembly.

network-context ue-pool

This section describes how to configure a UE pool name associated with an IP sub-pool of type UE-IP-Pool. Specify whether the IP addresses/prefixes in this sub-pool are locally managed or managed remotely (for example, using RADIUS).

Configuring Objects

Configure the objects as follows:

Table 8-18 network-context ue-pool

Object	Description
allocation-mode	Indicates the address allocation-mode for this UE pool. ■ For locally-managed IP sub-pools, configure the UE pool with the allocation-mode set to local-allocation (default). ■ For static UE IP addressing or RADIUS assigned IP pools, configure the UE pool with the allocation-mode set to remote-allocation. Configure one or more IP sub-pools with the IP ranges corresponding to the static UE IP addresses assigned or to the address range managed by RADIUS. Associate the IP sub-pool(s) with the remote UE pool. ***Note:*** *When RADIUS is managing the UE IP, the remote UE pools/sub-pools are required only if it assigns the IP/prefix. If RADIUS specifies only the sub-pool (name) to use, then it is a locally-managed pool.*
geo-redundancy-group	The geographic redundancy group to which this UE pool belongs.
quarantine-timer	Indicates the quarantine-timer for local UE pools before reusing a released IP address.
use-quarantined-ip-if-none-available	This feature enables the system to re-use IP addresses in the quarantine queue if none are available in the IP sub pools associated with the UE pool.

network-context vrf-lightweight

As part of network context scaling, MCC supports five network-contexts types; *lightweight* and *normal*. MCC supports 128 normal network-contexts and 2000 lightweight network-contexts. Refer to the following table to configure a normal or lightweight network-context based on the network-context scenario.

MCC supports 2K VRFs or VPNs (lightweight network-contexts) and association of 2K VRFs under BGP VPNs. The MCC supports 10K subpools with each subpool having two prefixes. The MCC supports a maximum of 25K Forwarding Information Base (FIB) entries or approximately 12K FIB entries per VRF, including pools, loopbacks, and imported routes.

Network Context Scenario	Normal Network Context	Lightweight Network Context
Content services such as Proxy, VAS services	Mandatory	Not supported
Any kind of gateway control traffic (GTP)	Mandatory	Not supported
BGP/OSPF protocol	Mandatory	Not supported

Network Context Scenario	Normal Network Context	Lightweight Network Context
Interfaces need to be configured	Mandatory	Not supported
BGP VPN VRFs can be configured	Optional - use the lightweight setting when scaling is required	Preferred setting
Enterprise VRFs on Gi side	Optional - use the lightweight setting when scaling is required	Preferred setting
IPSec tunnel required	Mandatory	Not supported
RADIUS	Optional	Optional
DHCP client or relay required	Mandatory	Not supported
Diameter traffic	Mandatory	Not supported

Chapter 9

Service Construct Parameters

This section describes the service construct parameters supported on the MCC. Service constructs are the common building blocks to create service instances used in workflow. Workflow defines the treatment of a service flow across various services. Prior to configuring workflow services, you must first create service constructs using the service-construct command.

Access	config
Guidelines	The following guidelines apply when configuring packet-filters for use in an ACL: ■ Set the initiator parameter to both ■ Set the IP loopback address that you want to filter as the UE IP address ■ Set the subnet address that you want to filter as the network IP address The following guidelines apply to the service construct url-list file format: ■ The file can be imported from the server or local file. If you make changes to the file, you must import the file from the CLI using url-list import to rebuild the database for the new file to take effect. ■ The file should have a Fully Qualified Domain Name (FQDN) for each line for HTTP request URL prefix or exact-match. – The protocol schema is ignored and anything before “//” and including “//” is skipped. Only the URL following “//” is used. – If line contains a star “*”, the entry is added in the database using the string before “*” for URL prefix-match. Otherwise, the entry is added in the database for URL exact-match.

service-construct acl-list

Operators can apply access control list (ACL) filtering to incoming control traffic to restrict or permit access to MCC interfaces.

Configuring Objects

Configure the objects as follows:

Table 9-1 service-construct acl-list

Object	Description
action*	Action to take for ACL filter
name*	Name of the ACL filter
packet-filter*	Packet filter to match
priority	Priority

1 Configure the access control list and associated access control filters.

```
an3000-mcm-slot7cpu1(config)# service-construct acl-list <acl-name>
acl-filter <acl-filter-name> packet-filter <packet-filter-name> action
[drop|allow] priority <access-filter-priority>
```

2 Apply the ACL to a network context(s).

```
an3000-mcm-slot7cpu1(config)# services acl acl-networkcontext
<network-context-name> acl-list <acl-name>
```

Access Control List

An ACL contains one or more access control filters. Each access control filter specifies the following:

Note: *Applying ACL filtering rules may cause unintended results if not used properly. Affirmed Networks recommends contacting the TAC prior to implementation.*

- The criteria that a packet must match. You define the criteria in a packet-filter, which can be based on source and destination IP addresses, source port and destination port, and the networking protocol in use.
- The resulting action, either drop or allow, that is taken when a match is found.
- A value for prioritizing the access control filter within the ACL, evaluated in sequential order, first to last. Once a match is found, no more access control filters are evaluated.

ACL filtering uses an implied "`Allow All Traffic`" filtering statement, which means if a received packet does not match any defined criteria in the ACL, the packet is allowed. However, you can configure ACL filtering to explicitly deny certain types of traffic by configuring a lowest priority access control filter with a drop action. In this case, you must configure higher priority filters that explicitly permit the types of traffic allowed.

> ***Note:*** *Affirmed Networks recommends that you configure specific drop/allow filters to ensure accurate traffic filtering.*

ACLs are applied on a network-context basis, where access control filters apply to all interfaces within a specified network-context. The system supports up to 32 access control filters per ACL.

ACL filtering does not apply to the following traffic types:

- Routing Protocols, such as BGP, OSPF, BFD
- DNS, SSH, IPv6 ND, DHCP, IKE
- Control Protocols, such as RADIUS, Diameter, GTP', and CALEA, that match explicitly registered IP address/port combinations
- Upstream GTP-U and GRE tunneled WAG traffic

service-construct advice-of-charge

Defines the action to take after online quota is exhausted, such as block and return a notification message or redirect subsequent HTTP traffic to a configured URL.

Configuring Objects

Configure the objects as follows:

Table 9-2 service-construct advice-of-charge

Object	Description
aoc-hardware-id-format See service-construct aoc-hardware-id-format (page 9-6).	aoc-hardware-id-format selected from aoc-hardware-id-format list
name*	
charging-notification-list	Notification message to be sent to user when reject. ■ mms-result-code ■ mscc-result-code* ■ notification-text* ■ sms-notification-text
http	Action message to be sent to user. See service-construct advice-of-charge http (page 9-4).

service-construct advice-of-charge charging-notification-list

Configure the notification message to be sent to user when reject.

Configuring Objects

Configure the objects as follows:

Table 9-3 service-construct advice-of-charge charging-notification-list

Object	Description
mms-result-code	MMS error response code
mscc-result-code*	MSCC result code
notification-text*	MMS error response text
sms-notification-text	SMS notification text message

service-construct advice-of-charge http

When used for HTTP redirection, the advice-of-charge object defines the parameters to include in the redirect URL sent to the user. Configure the action message to be sent to the user by specifying a parameterized URL (instead of a static URL).

Configuring Objects

Select true/false to include class, mac, imsi, imei, and msisdn in the redirect url. The class information included in the redirect URL is provided by the RADIUS server and used to identify the subscriber in subsequent RADIUS accounting messages. Select true/false to include-original-url in the redirect URL.

Configure the objects as follows:

Table 9-4 service-construct advice-of-charge http

Object	Description
include-class	Whether to include the class attribute from RADIUS in redirect URL (if available).
include-imei	Whether to include IMEI in redirect URL.
include-imsi	Whether to include IMSI in redirect URL.
include-mac	Whether to include MAC address in redirect URL.
include-msip	Whether to include the MS IP address in the redirect URL.
include-msisdn	Whether to include MSISDN in redirect URL.

Table 9-4 service-construct advice-of-charge http

Object	Description
include-original-url	Whether to include the original requested URL in redirect URL.
redirect-code	Enables the MCC to perform HTTP redirection based on the HTTP request method within the MCC, instead of sending the HTTP request to the server and changing the HTTP response from server to an HTTP redirect message to the UE. The three-way handshake for the TCP connection occurs with the Origin Server. After the handshake, any HTTP request is interpreted by the MCC and MCC sends the redirect message to the UE when redirection is required for that connection. HTTP Request Redirect Code. Options include: ■ `302 found` (default) – The requested resource resides temporarily under a different URL ■ `303 See Other` – The requested page can be found under a different URL ■ `307 Temporary Redirect` – The requested page has moved temporarily to a new URL When the MCC determines that the quota is exhausted for the session and advice-of-charge redirection is enabled for the session, then the MCC responds with an HTTP response message to any HTTP request method from the UE to an origin server. MCC redirects the following HTTP request methods from the UE: ■ GET, POST, HEAD, PUT, DELETE, CONNECT, OPTIONS and TRACE MCC constructs the response packet with the appropriate sequence and acknowledge number and HTTP headers. MCC completes the location header field in the response message with the redirected URL. The redirected URL is constructed either from the configured advice-of-charge or the URL received from the Gateway. MCC sends an RST to the origin server and drops any non-HTTP and UDP traffic.

notification-message – Send a notification message to the client - string.

redirect-url – Redirect the client to a different URL - string. The MCC is able to perform advice-of-charge redirects on L7/DPI hits This feature enables the MCC to issue an HTTP redirect-url to an alternate landing page when a user's credit is exhausted. This feature extends the online charging behavior, allowing an operator to create "free of charge" and "not free of charge" rating zones, and where the HTTP redirect behavior is only invoked when the user request is destined to a "not free of charge" zone. The rating zones can be specified based on Layer 7 criteria, including HTTP request header contents, or by DPI hits triggered by Layer 3/4 criteria. Additionally, this feature allows an Operator to specify the HTTP redirect-url by IMSI range, providing the ability to redirect users that share an APN to separate landing pages based on IMSI (for example, MVNO).

service-construct aoc-hardware-id-format

Configure the advice-of-charge hardware ID format.

Configuring Objects

Configure the aoc-hardware-id-format selected from aoc-hardware-id-format list. MCC supports URL redirection through advice-of-charge and predefined rules with AES-encrypted UE hardware information in the redirection URL.

Note: *Redirection is supported only for HTTP. The aoc-hardware-id-format is supported for offline charging only.*

MCC supports URL redirection through advice-of-charge and predefined rules with AES-encrypted UE hardware information in the redirection URL.

This feature enables Operators to apply the redirection URL configured in advice-of-charge which is associated with a rating-group. The PCRF activates a predefined service-rule configured under rating-group.

This feature enables Operators to configure the encryption-algorithm, encryption-key, delimiter, prepend-string, and enclosure-string to be used in formatting the hardware ID information to be appended in the redirection URL.

MCC uses the advice-of-charge configured in the corresponding rule-group to create the encrypted hardware ID details. MCC applies the AES encryption on the hardware ID and appends the encrypted hardware ID to the redirection URL configured in advice-of-charge. MCC sends the redirected URL to the UE for redirection.

MCC supports a maximum of 4096 configured formats.

Configure the objects as follows:

Table 9-5 service-construct aoc-hardware-id-format

Object	Description
delimiter	Delimiter for customized hardware ID [default "&"]
enclosure-string	Enclosure string for encrypted hardware ID [default "#"]
encryption-algorithm	Encryption algorithm to be used [default "aes-cbc-128"]
encryption-key	Encryption key for encrypted algorithm
prepend-string	Prepend string for encrypted hardware ID [default "?encryptedInfo="]

service-construct ca-certificate-list

Configure a list of trusted certificate authority (CA) certificates.

Configuring Objects

The list can contain a bundle of certificates. The trusted CA certificate list is used to authenticate end-user certificates or network entities. Certificates configured in the list must be present in the */public/content/tls/cacerts/* folder on the MCM. Use fault-management clear to install ca certificates into this folder.

The ca-certificate-list is used by Affirmed Network's applications, including HTTPS. See https-profile for information on using a ca-certificate-list for HTTPS.

service-construct code-mapping-list

Maps codes from one interface to another. Configure the name and user-description for the code-mapping-list.

service-construct code-mapping-list incoming-code

Incoming code that needs to be replaced. Configure the incoming-code and outgoing-code.

service-construct content-insertion

Configure the content-insertion-profile and insertion-object-list.

service-construct content-insertion content-insertion-profile

The Affirmed Networks http-proxy operates in content-insertion mode on the configurable URL/domain. The request-policy and server-instance is associated with a configured content-insertion-profile. The content-insertion-profile supports:

- insertion of local or remote hosted images or videos (for example, advertisements)
- insertion of externally hosted content on ad server via URL links
- banners located at the top/bottom of the page or centrally located overlays

Configuring Objects

Configure the objects as follows:

Table 9-6 service-construct content-insertion content-insertion-profile

Object	Description
admin-state	Enable/disable content insertion profile.
insertion-object-list	Configure insertion-object-list. See service-construct content-insertion insertion-object-list (page 9-8).
insertion-style	Select and configure banner or overlay. ■ banner – Container for banner configuration. Display the banner at the top or bottom. ■ overlay – Container for overlay configuration. ■ close-button – Enable or disable the close-button. ■ timeout – Configure the timeout in seconds (amount of time to keep the overlay).

service-construct content-insertion insertion-object-list

Configure a list of insertion objects used in this insertion profile.

Configuring Objects

Configure the objects as follows:

Table 9-7 service-construct content-insertion insertion-object-list

Object	Description
admin-state	Enable/disable insertion-object.
click-through-url	Click through URL.
object-location	Choose between filename or URL. Enter the filename of the local object or the URL of the remote object.
title	Title for the content insertion object
type	Content insertion object type. Options include: image, link, video.

service-construct control-service-profile

This feature enables Operators to create a dhcp-service-profile for the DHCP client or a gateway acting as a DHCP relay agent and bind it to the DHCP. The DHCP service allows you to configure the DHCP mode (either Client or Relay) and allows you to configure up to four external-dhcp-servers. The external-dhcp-server configuration includes the IP of the external DHCP servers, lease timer, and the capability (V4 or V6 addresses). This feature also enables Operators to enable the domain-name-server and netbios-name-server.

service-construct control-service-profile dhcp-local-server-profile

Configure the DHCP Local Server Profile.

Configuring Objects

Configure the objects as follows:

Table 9-8 service-construct control-service-profile dhcp-local-server-profile

Object	Description
broadcast-address	Broadcast IP Address
dhcp-server-name*	Name of the DHCP server profile
domain-name	Domain Name (DHCP Option 15).
interface-mtu	Interface MTU value
lease-timer	Lease timer value
local-server-ip-address	IP Address of the local DHCP Server
primary-dns-ip-address	IP Address of the primary DNS
prior-subscriber-ip-binding-timer	Time (in minutes) to retain the last assigned IP binding even after the session is deleted. If the session is re-established before the timer expires, the same IP Address is assigned. Once the timer expires, the IP Address is released.
router-ip-address	IP Address of router
secondary-dns-ip-address	IP Address of the secondary DNS
subnet-mask	Subnet Mask

service-construct control-service-profile dhcp-service-profile

Configure the DHCP service profile.

Configuring Objects

Configure the objects as follows:

Table 9-9 service-construct control-service-profile dhcp-service-profile

Object	Description
admin-state	The administrative state of this DHCP service profile

Table 9-9 service-construct control-service-profile dhcp-service-profile

Object	Description
dhcp-network-context*	Network Context containing this DHCP service endpoint
dhcp-service-endpoint*	Reference to a service endpoint object on the Network Context for this DHCP service
dhcp-service-mode	The mode of operation of this DHCP service - Client or Relay. See Acting as DHCP Client (page 9-10) and Acting as DHCP Relay Agent (page 9-10) for more information.
dhcp-service-name*	Name of the DHCP service profile
enable-domain-name-server	Use DNS configuration from external DHCP Server when configured as DHCP Client
enable-netbios-name-server	Use NetBIOS Name Server configuration from external DHCP Server when configured as a DHCP Client

Acting as DHCP Client

The DHCP client feature enables the system to act as a client on behalf of the UE in a network. A DHCP client uses the DHCP protocol to acquire configuration information, such as an IP address, a default route, and one or more DNS server addresses from a DHCP server. The DHCP exchange occurs between the system (acting as a DHCP client) and the external DHCP server. The system, acting as a DHCP client, acquires these parameters and assigns them to the UE.

Acting as DHCP Relay Agent

A DHCP relay agent is any host that forwards DHCP packets between DHCP clients and DHCP servers. DHCP relay agents are used to forward requests and replies between clients and servers when they are not on the same physical subnet. Relay agents receive DHCP messages and then generate a new DHCP message to send out on another interface. The relay agent sets the gateway address to the DHCP server. The exchange occurs between the DHCP server and the UE; the system is in the middle and in the process, learns the UE IP address.

This section describes the DHCP relay-based UE IP allocation function on the system. The system supports end-to-end DHCP in the GGSN or PGW by acting as a DHCP Relay Agent (DRA) as follows:

1. The system acts as a DHCP relay agent and routes the DHCP requests/replies between the UEs and the DHCP server(s).
2. At PDP activation, no IP address is assigned to the UE.
3. After the UE IP configuration is completed by the DHCP, the GGSN initiates a PDP context modification to reflect the newly assigned UE IP address.

4 The GGSN (as a DRA) learns the UE IP address by looking at the DHCP messages (specifically the REQUEST) exchanged between the external DHCP server(s) and the UE.

service-construct control-service-profile dhcp-service-profile external-dhcp-server

Configuration of external DHCP Servers for this service endpoint.

Configuring Objects

Configure the objects as follows:

Table 9-10 service-construct control-service-profile dhcp-service-profile external-dhcp-server

Object	Description
admin-state	Enable/disable the administrative state for this external DHCP server.
ip-address	IP address of the external DHCP server.
name*	Name of the external DHCP server. Must be unique.
server-ip-allocation-type	The type of IP address (V4, V6 or V4V6) that can be allocated by this external DHCP server. Options include: ip-allocation-type-v4, ip-allocation-type-v6, ip-allocation-type-v4v6.

service-construct data-record-template

This section describes how to use data-record-templates to customize messages going out on a particular service interface by disabling optional attributes from messages. Operators can configure templates for auth (RADIUS), charging-local, OCS, OFCS, and PCRF for the GGSN, PGW, SGW, and WAG.

The data record template is also used to define the format for the data record. The data record can be a RADIUS message, Diameter message, CDR in a local CDR file, or NBR (NAT binding record) in a local file. The format is determined by the data fields in the record. For the RADIUS and Diameter data record, Operators can choose to include the fields (optional AVPs) in the messages and filter certain hierarchical AVPs (AVPs with sub-attributes). For the CDR record, Operators can configure additional parameters.

MCC reports the MSISDN in RADIUS accounting records, local CDRs and NBRs when provided in the Calling-Station AVP of the RADIUS Change of Authorization-Request (CoA-Request) message.

Configuring Objects

Dynamic PCC

The dynamic PCC feature implements PCRF-installed PCC rules on the PGW. PCRF can initiate a PCC rule install, modification and, removal. The PCRF can install changing rules on the PGW as follows:

- Through the CCA (Credit Control Accept) in response to any CCR (Credit Control Request) from PGW
- Through the RAR (Re-Auth Request) to PGW at any time.

The AVPs contained in either CCR or CCA are Charging-Rule-Install (for installation and modification) and Charging-Rule-Remove. The AVP Charging-Rule-Definition within Charging-Rule-Install specifies the details of each rule as follows:

charging-rule-name – Used for PCRF to activate a specific PCC rule that is predefined at the PCEF. This is not supported.

charging-rule-definition – The PGW creates or modifies dedicated bearer(s) based on QoS, packet-filter, rating-group and so on that are provided in the Charging-Rule-Definition. Note that the Charging-Rule-Name AVP within the Charging-Rule-Definition must be unique for the service-construct service-rule configuration. The Service-Identifier AVP must have a value greater than 4096 and the value of the rating-group AVP must be previously configured on the MCC as quota-id in services charging rating-group configuration.

MCC supports the Flow-Status AVP under the Charging-Rule-Definition AVP within a CCA or RAR message. The Flow-Status AVP defines whether the service data flow is enabled or disabled and overrides the value configured statically on the gateway. For more information, see the *Affirmed Networks Gx Interface Specification*.

HTTP Transaction Data Records (HTDR)

The http-transaction-record-profile provides information about each HTTP transaction that traverses the NE. The data is represented in comma-separated values format (csv) and written as a string of csv values. For a complete list of supported HTDRs, see the *Affirmed Networks Charging Data Records Specification*.

The DataRecordAgentMCM is the central task that obtains the data records from various tos tasks and logs the tasks to disk. Using sendRaw (no protobuf), the MCC compresses a bundle of data-records at the source (tos tasks) prior to sending the records. This results in the distribution of compression functionality to various sources rather than a centralized task. Compressed data is transferred between tasks and written to the disk on the fly, resulting in increased performance.

Affirmed Networks uses zlib to compress the data buffers on the fly. As a result, the data record file is written to a zlib raw compressed file rather than gzip format. Operators must use the Affirmed Networks uncompress utility to uncompress the data records. See the *Affirmed Networks System Administration Guide* for details.

Each field in CSV format contains the following six configurable parameters:

Alignment	The alignment of the data field. If not configured, the alignment defaults to left aligned.
Format	The format for the data field. If not configured, the format type uses the default.
Length	The fixed length of the data field. The length defaults to zero indicating the data field is written out as it is.
Present	Indicates whether to include the data field in the CDR. By default, this value is set to conditional for all data fields, indicating that the data field will be filled in CDRs when available.
Position	Location of the data field in the CDR. A pre-defined order exists for each data field if positionIndex is not configured.
Padding	The character used for padding when the field has a fixed length.

The HTTP transaction data record includes information such as:

compression-information	Compression and bytes saved, if compression is applied
connection-information	Network context, source and destination address, etc.
minification-information	Information about minification and bytes saved, if applied.
pre-empt-dns-information	Information on pre-emptive DSN such as number of URLs rewritten.
request-information	HTTP request information
response-information	HTTP response information
subscriber-information	IMSI, MSISDN, etc.
time-information	Request timestamp and latency information
video-information	Information about any video served and bytes saved, if applied.
web-image-optimization-information	Information about web image optimization and bytes saved, if applied.

Operators can customize the records in certain positions and can include or omit certain records. The Operator can use this information to:

- Calculate details such as bandwidth savings when features such as Web image optimization is enabled.
- Provide the basis for charging
- Measure analytics

NAT Binding Records

For consumer APNs that utilize NAT, binding records are required for Lawful Intercept purposes. The MCC supports v4-v4 NAT. Each time the MCC assigns a new block of ports on a public IP address to a particular subscriber, a port binding record is generated. Similarly, when the MCC unassigns the block, another binding record is generated. The binding records are stored in a local Usage Data Record (UDR).

NAT Binding Records (NBRs) track the association between the source address/port of a connection/flow initiated by a subscriber in one address space, and its translated source address/port in another address space. The translated address/port is called the NAT source address/port. The NAT source address/port makes it possible to trace the connection back from a destination device in one address space, to the subscriber source device in another address space.

The MCC maintains a log of NBRs to support Lawful Intercept. A NBR, generated by the NAT device prior to connection, is used to identify the binding between the NAT source address/port and the subscriber device. Therefore, the NBR log allows the NAT source address/port to be mapped to the subscriber source device. NBRs can be generated for every source address/port to NAT address/port binding. Affirmed Networks refers to these records as *NAT flow binding records* (NFBRs). Logging every flow binding will generate a large number of events, therefore *NAT port binding records* (NPBRs) are used as an alternative approach without sacrificing the traceability of individual connections.

Affirmed Networks supports local NPBR logging as follows:

- A NAT device assigns a NAT source address and a group of NAT source ports, (otherwise known as a port chunk), to a source device when that device establishes its first connection.
- Subsequent connections from the source device are assigned the same NAT source address but a different NAT source port from the group of ports already allocated to that device. Knowing the range of ports in the assigned port chunk allows any connection using those ports to be mapped back to the device. NPBR logging takes advantage of this behavior by recording only the NAT source port chunk allocations and de-allocations instead of recording every connection binding.
- NPBRs are stored locally in a file. They may be periodically transferred to another device for long-term storage and analysis.

service-construct data-record-template port-nbr

Configure the port NAT binding record template. The port-nbr is referenced by a control-profile. The following objects apply to port-nbr: alignment, format type, length, padding, position index, and present flag.

Configuring Objects

Configure the objects as follows:

Table 9-11 service-construct data-record-template port-nbr

Object	Description
charging-id	Charging ID assigned by PGW to the session.
ield-delimiter	fCharacter used as a field delimiter to separate two fields.
gw-name	Name of gateway.
local-tunnel-address	IP address of local/gateway end of access tunnel carrying session (IPv4 or IPv6).
nat-last-activity-time	Local date/time of last NAT operation for any flow using the port chunk reported in this record.
nat-pool-name	Name of the NAT pool supplying the nat-src-address.
nat-seq-num	Sequence number for the subscriber.
nat-src-address	NAT IPv4 address. Typically a public address.
nat-src-port-block-alloc-flag	NAT source port block allocation flag for the subscriber. 1: allocate; 0: deallocate
nat-src-port-block-end	Last port number of port chunk being allocated/deallocated.
nat-src-port-block-event-time	Timestamp when NAT source port block is allocated.
nat-src-port-block-start	First port number of port chunk being allocated/deallocated.
remote-tunnel-address	IP address of remote end of access tunnel carrying session (IPv4 or IPv6).
sub-imsi	IMSI of UE.
sub-ip-address	IP address of UE (IPv4 or IPv6).
sub-msisdn	MSISDN of UE.
time-zone	Local time zone. The time-zone format is described in TS 29.274. The time-zone field is an ASCII (string) representation of those octets. An empty field indicates the time-zone is unavailable.

Vendor-Specific Attributes

MCC supports the vendor-specific attributes (VSAs) that can be sent in access-request messages to the remote RADIUS server. For details, see the *Affirmed Networks RADIUS Interface Specification*.

DNS and NBNS Configuration Coming From RADIUS

This feature enables the GGSN/PGW to extract primary and secondary DNS and NBNS RADIUS attributes from Access-Accept messages (returned from the RADIUS server) and sends the attributes to the MME (via the SGW) in the PCO of the Create Session Response message.

These RADIUS attributes include the following:

- `msft-primary-dns-address`
- `msft-primary-nbns-address`
- `msft-secondary-dns-address`
- `msft-secondary-nbns-address`

service-construct default-policy-cause-code-list

Configure the default-policy-cause-code-list. The codes being configured must not overlap with any previously configured ranges.

Configuring Objects

Configure the objects as follows:

Table 9-12 service-construct default-poliucy-cause-code-list

Object	Description
default-policy-cause-code-list	Specify the name of result code list.
range	Specify the name of the result code range.
cause-code-range	Specify the low and high end of the result code range. Ensure the result code range does not overlap with any previously configured ranges. To configure a range with a single result code, configure the low and high value with the same code.
retry-secondary	Enable/disable to Retry Secondary Server/Peer - Applicable only if secondary peer is configured.

service-construct device-database

Configure the device-database. This feature enables Operators to use device characteristics to configure data services such as data filters used to allow/block traffic for a particular class of devices based on the device-type parameter and also configure differentiated rate-limiting bit rates based on device-type or device screen size.

This feature supports the following device-database characteristics: TAC (Type Allocation Code), device-type, device-make (or brand), device-model, screen-width, and screen-height. The TAC information is extracted from the first 8 digits of the 15/16 digit IMEI. The device-type field indicates the type of device that corresponds to a particular TAC. Options include: handheld, feature-phone, smart-phone, modem, dongle, tablet, vehicle, connected-device, router, module, ebook, and femto-app.

The device-database is in CSV format and must be previously imported into MCC or downloaded from a remote server. See Initiating a Device Database File Download (page 9-18). MCC obtains the database attributes during session establishment. The device-database contains a mapping of the TAC of a device to its specific parameters. The database is either the GSMA device map or a modified version of the GSMA device map and may also contain a subset of GSMA device map fields.

The service-construct session-rule-group rule (page 9-75) objects match the following device-database parameters: TAC, device-type (exact match), make (exact match), model (exact match), screen-width, and screen-height

The content-filter instance accepts a service-rule to which a session-rule-group is associated.

This following guidelines apply to this feature:

- The device-database must be in CSV format.
- The first line in the CSV file must contain the field names since MCC uses this information for parsing. The files are stored in MCC.
- The header information is case sensitive.
- MCC supports a maximum of 300,000 database entries.
- If the database is modified, the existing session’s values are not changed.
- The database supports a single level of lookup.

Configuring Objects

The csv-field-names are case sensitive. The device-database configuration is a singleton object and supports one instance of the device-database at any time. Configure the objects as follows:

Table 9-13 service-construct device-database

Object	Description
name	Enter a name for the device-database.
import	Import the device-database now. Options include: noalarm – Do not raise any alarms when this import fails nolog – Do not issue error logs when this import fails
csv-field-names	Configure strings for csv headings. ■ device-make – Configure a string to indicate the name of the device-make (brand) field ■ device-model – Configure a string to indicate the name of the device-model field ■ device-type – Configure a string to indicate the name of the device-type field ■ screen-height – Configure a string to indicate the name of the screen-height field ■ screen-width – Configure a string to indicate the name of the screen-width field ■ tac* – Configure a string to indicate the name of the TAC field. This field is mandatory as it is the key to the database.
importFrom	■ import-from-local-file – Configure a filename for importing this list from a local file. Configure the particular: *listsfilename,thefileshouldbein/tosdb/content/policylists/<list-type>/'* ■ import-from-server – Configure a rule for importing this list from a file server. – file-download-profile – Configure the file download profile – filename –Configure the server filename, *<string>* *For example, folder/file-$DATE{yyyymmdd}.txt*

Initiating a Device Database File Download

The file download process can occur either locally or from a remote server, but not both simultaneously. Only one device-database can be configured at a time. If the database is modified or removed, all history is cleared. In order to add a different device-database, the first device-database must be deleted.

Remote Server

1. Add a file download profile and specify the retry intervals and the schedule at which the file is downloaded.
2. Specify the full path on the server where the file resides.

3 In the file download profile, specify the server address, user name, and password.

4 Download the file from the remote server using the following command:
`service-construct device-database <database-config-name> import-from-server`

Local Server

1 Copy the file to */public/content/policylists/devicedb*

2 Execute the following import command to trigger the import device-database now:
`service-construct device-database <database-config-name> import`

3 Indicate `noalarm` or `nolog` to prevent an alarm or log from being raised when this import fails.

service-construct dhcp-mapping template attribute

Configure DHCP options.

Configuring Objects

Configure the objects as follows:

Table 9-14 service-construct dhcp-mapping template attribute

Object	Description
from	item to store in outgoing attribute
from-format	formatted string to store in outgoing attribute
name*	DHCP option name
to	item to set from incoming attribute
to-format	regular expression to select strings from incoming attribute
to-list	item(s) to set from regular expression match groups

service-construct diameter-result-code-policy-list policy

List of Result-Code policies. Any two configured policies must not have overlapping result-code values.

Configuring Objects

Configure the objects as follows:

Table 9-15 service-construct diameter-result-code-policy-list policy

Object	Description
rangeName*	Policy Name.
retry-secondary	Enable/disable to Retry Secondary Server/Peer - Applicable for command-level Result-Code lists and only if secondary peer is configured.
subscriber-type	Configure the Subscriber type for which the policy applies - Applicable only for online charging interfaces.
action	Configure the redirect-url to redirect the subscriber.
result-code-range	Diameter Result Code Range - make sure the codes being configured do not overlap with any previously configured ranges. Enter the result code end value and start value.

service-construct diffserv-profile

Configure the diffserv profile.

Configuring Objects

Configure the objects as follows:

Table 9-16 service-construct diffserv-profile

Object	Description
class-mapping	Select the queue priority that cs0 to cs7 should use. Options include: best-effort, bronze, silver, gold.
dscp-mapping	Configure the mapping of DSCP codes to queue priorities. ■ code – dcsp-code ■ queue-priority – Queue priority that a DSCP value should use ■ vlan-priority – 802.1 PCP priority that a DSCP value should use

service-construct domain-rule-group

Domain rule groups are services rules that are used by workflow for classifying certain connections and taking specific actions for certain domains. The domain-rule-group object enables Operators to configure a list of domains for which snooping is enabled. DNS Snooping (DNSS) enables Operators to permit users to access restricted URLs or sites (for example, social media sites such as Facebook or Twitter).

MCC supports the following matching criteria for the domain-rule-group for domain names:

- begins-with (prefix)
- contains (partial)
- ends-with (suffix)
- exact match
- regular expressions

DNS reverse cache is used by the domain rule group. See services dns-proxy in the *Affirmed Networks Value Added Services Administration Guide*.

Configuring Objects

Configure the objects as follows:

Table 9-17 service-construct domain-rule-group

Object	Description
rule	Domain Rule to be configured for the rule group
name	Specify the domain name to match against begins-with contains ends-with exact regex
url-list	Specify a url-list for matching against request URL.

service-construct edr-flow-network-stats-profile

Configure EDR flow profile (maximum 256). This profile carries the configuration required for flow level network-stats-info record. Default values are applied if this profile is not configured.

Configuring Objects

Configure the objects as follows:

Table 9-18 service-construct edr-flow-network-stats-profile

activity-duration-time-interval	Configure time interval to be used for flow activity duration calculation in uplink/downlink direction. 500ms, 1sec, 2sec
include-icmp	Include/exclude (true/false) ICMP traffic for network-stats attributes calculation.
measurement-layer	TCP/IP layer to be used for network-stats attributes calculation. layer3, layer4, layer7
name*	Enter EDR flow profile name.
peak-throughput-time-interval	Configure time interval to be used for flow peak throughput calculation in uplink/downlink direction. 1sec, 2sec, 3sec

service-construct edr-interface-profile

Configure the EDR (Event Data Record) interface profile to generate EDRs.

Configuring Objects

Configure the objects as follows:

Table 9-19 service-construct edr-interface-profile

Object	Description
admin-state	Enable/disable the EDR interface.
flow-min-lifetime	Number of seconds to wait before generating initial Flow event data record following flow creation.
flow-network-stats-profile	EDR flow profile to be applied on attributes reported via flow network statistics.
flow-statistics-interval	Time interval in seconds between Flow Interim event data records.
max-url-length	Length to which the URL field should be truncated (range 10-65535). Truncate the query portion of the URL within the HTTP clientUrl, serverUrl, and/or refererURL parameters to reduce EDR size. You must explicitly configure the max-url-length parameter to truncate the URL. For example: **`service-construct edr-interface-profile EIP max-url-length 255`** The command truncates the record length to the configured value but maintains the last 5 characters of the record (...RPT0=) to retain some file type information visibility.

Table 9-19 service-construct edr-interface-profile

Object	Description
name*	Profile name
session-data-stats-profile	EDR session profile to be applied on attributes reported via session data statistics.
session-statistics-interval	Time interval in seconds between session and bearer interim event data records.

service-construct edr-interface-profile edr-service-profile

Configure the edr-interface-profile to enable/disable the Pilot Packet subtypes. For the primary Pilot Packet server, transport must be set to UDP and encoding must be set to TLV.

***Note:** Because the Pilot Packets are sent by connectionless UDP, the secondary-server is not used, even if configured in the edr-service-profile.*

record-info

Specify the record information to send to the EDR server for bearer, flow, session, and http.

serving-network-info – Enable/Disable pilot serving network information reporting

subscriber-info – Enable/Disable pilot subscriber information reporting

record-type

For session, bearer, and flow EDR records, specify which record types to send to the EDR server. (Skip this step for HTTP transaction EDR records.) Enable/disable the following:

- attempt-records (session and bearer)
- interim-records (session, bearer, flow)
- start-records (session, bearer, flow)
- stop-records (session, bearer, flow)

Configuring Objects

Configure the objects as follows:

Table 9-20 service-construct edr-interface-profile edr-service-profile

Object	Description
admin-state	Enable/Disable the EDR service profile.
bearer-edr-profile	Configure the bearer EDR profile. ■ bearer-info – Enable/disable bearer info. ■ bearer-qos-information – Enable/disable bearer qos information. ■ data-statistics – Enable/disable data statistics. ■ edr-cause-codes – Enable/disable edr cause codes.
flow-edr-profile	Configure the flow EDR profile. ■ data-statistics – Enable/Disable flow data-statistics ■ dpi-info – Enable/Disable flow deep packet inspection reporting ■ nat-binding-info – Flow EDRs include NAT binding information that contains NAT ip-address and NAT port information. Enable this option to report NAT ip-address and NAT port information in flow EDRs. – nat-ip-address – Connection’s nat-ip-address information. – nat-port – Connection’s nat-port information. ■ network-perf-info – Enable/Disable flow network-perf-info ■ subscriber-info – Enable/Disable flow subscriber information reporting ■ tcp-retransmission-info – Enable/Disable flow TCP retransmission statistics reporting ***Note:*** *If an SSM failure occurs, the session is migrated to the backup SSM and new NAT resources are allocated to ongoing connections. Therefore, a single connection may appear with different NAT resources (ip/port) in subsequent flow EDRs.*

Table 9-20 service-construct edr-interface-profile edr-service-profile

Object	Description
http-edr-profile	Configure the http EDR profile. ■ cache-info ■ compression-info ■ connection-info ■ entitlement-info ■ filter-info ■ http-subscriber-info ■ manifest-info ■ minification-information ■ preempt-dns-info ■ record-info ■ request-info ■ response-info ■ tcp-client-connection-info ■ tcp-server-connection-info ■ tcp-splicing-info ■ time-info ■ upd-proxy-info ■ video-info ■ web-image-opt-info
pilot-edr-profile	Configure the pilot EDR profile.
primary-server	Primary EDR server for this EDR service.
secondary-server	Secondary EDR server for this EDR service.
session-edr-profile	Configure the session EDR profile. ■ data-statistics ■ edr-cause-codes ■ gateway-info ■ serving-network-info ■ session-qos-info ■ session-subscriber-info ■ session-user-csg-info

service-construct edr-server-profile

Configure the EDR (Event Data Record) server profiles (maximum 8).

MCC supports sending Pilot Packets to multiple destinations on Session setup/tear-down, RAT-type change, and NAT address/port portion allocation/de-allocation. The Pilot Packet provides key information about a subscriber session, including Subscriber Identity (MSISDN, IMSI, NAI), Subscriber IP address, and Subscriber NAT IP address and port range. This information can be used to enable new services, such as Subscriber Analytics.

A Pilot Packet contains a number of attributes in a string of type-length-values that are sent by UDP transport to multiple destinations.

Configuring Objects

Configure the EDR profile(s) and specify the EDR record types (session, bearer, flow, http transaction) and records (start, stop, interim, attempt) to send the EDR server. Configure the objects as follows:

Table 9-21 service-construct edr-server-profile

Object	Description
admin-state	Enable/disable the EDR service profile.
edr-source-ip*	Source IP address to use to reach the edr server.
encoding	Encoding type. Options include: protobuf, tlv.
ip-address*	IP address of EDR server.
keep-alive	EDR server keepalive interval.
name*	
network-context*	Network Context for this EDR server.
tcp-user-timeout	Number of seconds for tcp user data retransmission timeout.
transport	Transport type.

service-construct edr-session-data-stats-profile

Configure EDR session profile (maximum 256). This profile carries the configuration required for session level data-statistics record. See *Acuitas Performance Statistics*.

service-construct edr-session-data-stats-profile downlink-bucket

Configure downlink bucket (maximum 16). See *Acuitas Performance Statistics*.

service-construct edr-session-data-stats-profile uplink-bucket

Configure uplink bucket (maximum 16). See *Acuitas Performance Statistics*.

service-construct encryption-profile

Configure encryption profile. See the *Affirmed Networks Value Added Services Guide*.

service-construct file-download-profile

This section describes how to define common parameters for downloading files containing policy definitions, such as files defining a list of URLs from file servers. Enable or disable automatic-download and assign a name to this file-download-profile. The profile is referred to by multiple import-from-server (page 9-84) configuration objects.

Configuring Objects

Configure the objects as follows:

Table 9-22 service-construct file-download-profile

Object	Description
automatic-download	Controls the automatic download. ■ When enabled, Operators must configure an automatic-download schedule (days-of-week and time-of-day) or download-interval. In this mode, both manual and automatic download are allowed. The download-interval takes precedence and is therefore used for automatic download. ■ When disabled, Operators must download the policy list manually. This is called “on demand only” mode. In this mode, the days-of-week/time-of-day and download-interval objects are configurable but not mandatory. ■ See service-construct url-list (page 9-83) for information on manually importing the policy list.
days-of-week	Configure file download days of week, Mon|Tue|Wed|Thu|Fri|Sat|Sun
download-interval	This object supports periodical automatic download. When the download-interval is configured, it takes priority over the *time-of-day* and *days-of-week* download schedule. Operators use the download-interval as an alternative way to schedule automatic downloads in minute intervals. A value of 0 indicates no periodical download.
name*	File download profile name.

Table 9-22 service-construct file-download-profile

Object	Description
network-context	Configure file download server network context.
password	Configure file download password.
retain-count	Configure a number of old downloads to retain, <0-30>. The system can retain a maximum of 30 old downloads.
retry-count	Configure file download retry count, <0-10>.
retry-interval	Configure file download retry interval, minutes <1-120>.
server-address	Configure file download server IP address.
time-of-day	Configure file download time, hh:mm '<0-23>:<0-59>'.
username	Configure file download username.

service-construct flow-idle-timeout-filter

The flow-idle-timeout-filter identifies the destination port range, destination IP address range, source IP address range, IP type, and protocol that are used to filter traffic for a flow-idle-timeout-rule. MCC supports a maximum of 1024 entries.

Also see service-construct flow-idle-timeout-rule (page 9-29).

Configuring Objects

Configure the objects as follows:

Table 9-23 service-construct flow-idle-timeout-filter

Object	Description
ip-version	The IP types that are filtered. ■ v4-only ■ v4-or-v6 ■ v6-only
name*	Name for this flow-idle-timeout-filter.
network-ip-address-end	Destination IPv4 address range end value.
network-ip-address-start	Destination IPv4 address range start value.
network-ip-v6-address-end	Destination IPv6 address range end value.
network-ip-v6-address-start	Destination IPv6 address range start value.
network-port-end	Destination port range end value.

Table 9-23 service-construct flow-idle-timeout-filter

Object	Description
network-port-start	Destination port range start value.
protocol	IP protocol values (for example, all: 255, ICMP: 1, TCP: 6, UDP: 17) The default is 255.
ue-ip-address-end	Source IPv4 address range end value.
ue-ip-address-start	Source IPv4 address range start value.
ue-ip-v6-address-end	Source IPv6 address range end value.
ue-ip-v6-address-start	Source IPv6 address range start value.

service-construct flow-idle-timeout-rule

The flow-idle-timeout-rule is used to configure the conn-idle-timeout and priority. Each flow-idle-timeout-rule has a maximum of 1 bound flow-idle-timeout-filter. MCC supports a maximum of 1024 entries.

MCC does not set the conn-idle-timeout dynamically. It takes effect for new connections created after the change is made.

The flow-idle-timeout-rule is also used to configure timeout values for monitoring the different establishment states of TCP and UDP flows that are accepted by a firewall-subscriber-flow. Separate timers for each establishment state are used to remove flows and to drop unexpected TCP messages.

MCC supports the connection idle-timeout based on the destination port range, destination IP address range, source IP address range, IP type, and protocol. This feature enables the Operator to support a different idle timeout for connections related to different applications. For example, this feature provides the ability to configure different connection timeouts for TLS traffic, DNS traffic, etc.

This feature is useful in preventing the exhaustion of resources, such as public IP addresses in NAT deployments when TCP or UDP sessions are unable to close gracefully, resulting in the public IP addresses not being released back to the available addresses pool immediately.

Note: *When a TCP RST is received for a connection, the connection is released after a 30 second timeout, even if the* **`nat-flow-binding-time`** *option under* **`services subscriber-firewall subscriber-firewall-flow`** *is configured for this connection.*

Also see flow-idle-timeout-profile (page 9-31).

Configuring Objects

Configure the objects as follows:

Table 9-24 service-construct flow-idle-timeout-rule

Object	Description
conn-idle-timeout	Connection idle timeout (in seconds), Warning!!! 2160000 is a long timeout of about 25 days, use it judiciously.
flow-idle-timeout-filter	Name of the flow-idle-timeout-filter that specifies the traffic that matches the rule.
name*	Name for this flow-idle-timeout-rule
priority*	Priority of the flow timeout rule, value 1 indicates highest priority.
tcp-close-timeout	This object is used when a subscriber-firewall-flow is monitoring the flow establishment states of accepted traffic. Specifies the timeout period between each message involved in the four-way TCP close operation, starting with receipt of the initial FIN from either direction. During this period, if the expected message is not received, the message is removed. The default is 30 seconds and the range is 10 through 240. See Subscriber Firewall Monitoring of Flow Establishment States (page 10-26).
tcp-established-timeout	This object is used when a subscriber-firewall-flow is monitoring the flow establishment states of accepted traffic. Specifies the timeout period for no flow activity after the three-way handshake is complete. If no flow activity is detected during this period, the flow is removed. The default is 300 seconds and the range is 10 through 2160000. See Subscriber Firewall Monitoring of Flow Establishment States (page 10-26).
tcp-open-timeout	This object is used when a subscriber-firewall-flow is monitoring the flow establishment states of accepted traffic. Specifies the timeout period between each message involved in the three-way TCP connection establishment. During this period, if the expected message is not received, the flow is removed. The default is 30 seconds and the range is 10 through 30. See Subscriber Firewall Monitoring of Flow Establishment States (page 10-26).

Table 9-24 service-construct flow-idle-timeout-rule

Object	Description
udp-established-timeout	This object is used when a subscriber-firewall-flow is monitoring the flow establishment states of accepted traffic. Specifies the timeout period for no flow activity after the response UDP is received. If no flow activity is detected during this period, the flow is removed. The default is 30 seconds and the range is 1 through 2160000. See Subscriber Firewall Monitoring of Flow Establishment States (page 10-26).
udp-open-timeout	This object is used when a subscriber-firewall-flow is monitoring the flow establishment states of accepted traffic. Specifies the timeout period between receiving the initial UDP packet and receiving the response UDP packet. If the response UDP packet is not received during the timeout period, the flow is removed. The default is 30 seconds and the range is 1 through 30. See Subscriber Firewall Monitoring of Flow Establishment States (page 10-26).

flow-idle-timeout-profile

The flow-idle-timeout-profile contains a list of flow-idle-timeout-rules that are listed in order based on priority, with the highest priority node listed first. For example:

```
[ 100, 1, filter(500, 500, 6)]---> [ 50, 2, filter(500, 4500, 6)]---> [
800, 5, filter(100, 25000, 17)]---> NULL
```

MCC creates a new connection for each unique flow and sets a default connection timeout. After the connection is created, workflow manager iterates through the flow-idle-timeout entries and finds the best match for the destination port range, destination IP address range, source IP address range, IP type, and protocol configured in the service-construct flow-idle-timeout-filter (page 9-28). The best match is typically the first match and has the highest priority because of the list ordering. MCC supports a maximum of 1024 entries.

Configure flow-idle-timeout-profile globally under config.

```
an3000-mcm-slot8cpu1(config)# ?

flow-idle-timeout-profile The flow-idle-timeout-profile that is
applied to a certain data profile for traffic matching
```

Attach a flow-idle-timeout-profile to the workflow data-profile.

```
workflow data-profile flow-idle-timeout-profile <name>?

Possible completions:

max-connections            Max connections

multi-protocol-proxy bind http proxy service instance

flow-idle-timeout-policy Flow timeout policy that contains mapping
flow timeout profile
```

service-construct geo-redundancy-server

Configure the geo-redundancy hot-standby server.

Configuring Objects

Configure the objects as follows:

Table 9-25 service-construct geo-redundancy-server

Object	Description
connectivity-monitoring-params	Network connectivity monitoring for automatic failover. Network health checks are supported on AAA, PCRF, OCS, RMG, S6b, STa, and SwM interfaces and use both shared loopback IPs and private loopback IPs. Operators must enable health-check monitoring on a least one interface. This feature provides support of MCC Geographic Redundancy to account for failures of network contexts that do not include an application element that can be monitored for health (for example, Radius/AAA). Examples include SWu and S2b network contexts. Routing health is used as a method for determining whether these network contexts are viable. Border Gateway Protocol (BGP) Peering Interfaces in such network contexts can be monitored for health via the configuration of one or more Routing Monitoring Groups (RMGs). Each of these RMGs can be used as an IP layer network health indicator. This feature is necessary to help monitor BGP peers (as an indicator of network viability) in the absence of a health check of the next-hop for reachability on the SWu and S2b interfaces. Each RMG is configured within a given Geographic Redundancy Group. A given RMG can be associated with one or more network contexts (and each network context can be associated with zero or more RMGs). If any of the sessions in a given RMG is viable, then the RMG is deemed viable; this allows the local cluster to assume the role of the active cluster. Similarly, if none of the sessions in a given RMG are viable, then the RMG is deemed unviable and the local cluster is unfit to be the Active cluster in the Geo Red scheme. Options include: ■ monitor-AAA ■ monitor-OCS ■ monitor-PCRF ■ monitor-RMG ■ monitor-S6B ■ monitor-STA ■ monitor-SWM Also see *Affirmed Networks ePDG Administration Guide*.

Table 9-25 service-construct geo-redundancy-server

Object	Description
dscp-value	DiffServ Code Point value with which to mark all geo packets. This object enables Operators to configure a DSCP value in geo-redundancy packets, allowing the packets to receive a particular forwarding treatment as they are routed. The geo-redundancy admin-state must be disabled prior to changing the dscp-value.
failover-soak-interval	Number of seconds to soak after state change before considering a failover
geo-redundancy-group*	Geographic Redundancy group, this loopback belongs to
link-timeout	Number of seconds of missed keep-alive packets to declare geo link down
network-context*	Network Context containing this Geo Redundancy Server service endpoint
remote-base-port*	Port number for the remote peer in the group
remote-ip-address*	IP Address for the remote peer in the group
service-endpoint*	Reference to a service endpoint object on the Network Context for this Geo Redundancy Server service

service-construct health-check-profile

This feature uses a health-check mechanism to detect if a monitored service is down or overloaded. MCC then uses this information for traffic-handling decisions.

Steering destinations have configurable priorities. When a higher priority destination goes down, MCC switches traffic to a lower-priority destination. MCC generates an event/alarm when it detects that a steering destination goes down and generates another event/alarm when the steering destination comes back online.

The proxy steering group allows peers with the same IP address to be reachable through different network-contexts.

Note: *MCC uses a TCP protocol handshake for health checking; IMCP is not supported.*

Configuring Objects

Configure the health-check-profile. Associate the health-check-profile to a steering-group (see *Affirmed Networks Value Added Services Guide)*. Configure the objects as follows:

Table 9-26 service-construct health-check-profile

Object	Description
admin-state	Enable/disable health check profile.
interval	Time interval between successive health checks in units of ms. 250 .. 300000(5000).
loopback-ip-list	Configure loopback IP list for the health check profile.
max-attempts	Maximum number of attempts for the destination 1.. 5(2)
name*	Health check profile name.
network-context*	Configure network context to use for health check profile.
protocol	Service protocol for the host/IP.
timeout	Timeout for the destination in units of ms. 100 .. 10000(1000)

service-construct host-list

The host-list serves as a whitelist used by the fraud-detection-profile. It contains the IP addresses that represent legitimate proxies. Enter the host list name and host IP address. Add as many IP address as needed.

service-construct http-header-enrichment

Configure HTTP header enrichment. See the *Affirmed Networks Value Added Services Guide*.

service-construct http-rule-group

The http-rule-group object supports protocol decoding. Protocol decoding allows for the extraction of interesting protocol fields (that is, metadata) for use in filtering and/or reporting mechanisms. The system refers to this object by the service-construct service-rule (page 9-72) object to be used as an L7 filter. When the system finds a match, a specific action or service(s) is applied based on where the service-rule is used. For example, the system may terminate, block, redirect, or allow an action for the configured service.

service-construct http-rule-group rule

This object enables Operators to configure a set of rules for matching header attributes in HTTP request and response.

Configuring Objects

Configure the objects as follows:

Table 9-27 service-construct http-rule-group rule

Object	Description
media	Configure the following http-rule-group rule media objects. See media (page 9-36).
accept	Specify a string for matching against HTTP request's accept header.
content-type	Specify a string for matching against HTTP request's content-type header.
host	Specify a string for matching against HTTP request's host header.
host-port-end	Specify a port end value for matching against HTTP request host port.
host-port-start	Specify a port start value for matching against HTTP request host port.
referer	Specify a string for matching against HTTP request's referer field.
request-method	Specify a string for matching against HTTP requests method field. GET|HEAD|POST|PUT|DELETE|TRACE|OPTIONS|CONNECT| For proper operation, service-rules configured with http-rule-groups that filter on the CONNECT request-method must also be configured with at least one packet-filter. IF the service-rule is configured with only an http-rule-group (and no packet-filter), MCC will not detect CONNECT requests.
request-uri	Specify a string for matching against HTTP request's URI field. The URI is an MMSC URL location in the STI HTTP POST message. The system matches the configured URI value with the received value prior to processing the STI packet.
response-header	Configure HTTP response header matching rule.
response-payload	Specify a signature to match against response payload at the specific offset. Options include: offset – Specify an offset in the response payload to look for the signature. signature – Specify a HEX string without whitespace as the signature.

Table 9-27 service-construct http-rule-group rule

Object	Description
response-status-code	Specify a status code for matching against HTTP response's status line. 100|101|200|201|202|203|204|205|206|300|301|302|303|304|305|306|307|400|401|402|403|404|405|406|407|408|409|410|411|412|413|414|415|416|417|500|501|502|503|504|505|
url-list	Specify a url-list for matching against request's URL.
user-agent	Specify a string for matching against HTTP request's user-agent header.

media

Configure the following http-rule-group rule media objects.

- **all-video –** Specify all video type for matching against the response payload.
- **abr-video** – Specify all ABR video types .(HLS, DASH, MSS, HDS) for matching against the response payload.
  - dash – Specify DASH for matching against the response payload.
  - hds – Specify HDS for matching against the response payload.
  - hls – Specify HLS for matching against the response payload.
- **progressive-video** – Specify progressive video types (FLV, MP4, WebM) for matching against the response payload.
  - flv – Specify FLV for matching against the response payload.
  - mp4 – Specify MP4 for matching against the response payload.
  - webm – Specify WebM for matching against the response payload.

service-construct http-url-rewrite

The http-url-rewrite object provides a URL substitution function. The http-url-rewrite profile is referenced by the http-proxy, through the http-request-policy assigned to a proxy instance/ip-flow and applied to the http request if the service-construct service-rule (page 9-72) under the http-request-policy is a match. Rewritten URLs can provide shorter and more relevant-looking links to web pages. Each URL requested in a WSP or HTTP request is checked for a possible substitution against the configured list. The list contains Target URLs mapped, one to one, against substitution URLs. Both the Target URLs and substitution URLs must contain the host name (including a port number) and may contain any number of segments delimited with the "/" character. The substitution is performed before any other URL-dependent action.

service-construct imsi-range

Configure the IMSI range. The start imsi value must be less than the end imsi value.

Configuring Objects

Configure the objects as follows:

Table 9-28 service-construct imsi-range

Object	Description
end*	IMSI range end value.
name*	IMSI range name - <string>.
start*	IMSI range start value.
imsi-range-list*	Contains list of imsi ranges. name* – IMSI range List name - <string>

service-construct interface

The charging, PCRF, accounting, authorization, and NBR interface parameters are tied to a control-profile.

The CDR storage module is responsible for storing CDRs when the remote server is unavailable or congested. When the remote server becomes available, the pending CDRs are sent. This CDR storage feature is enabled by default on Diameter and GTP' offline charging interfaces. The system reserves 20% of the total disk partition size to store CDRs when the remote server is unavailable. Operators must enable this feature for the RADIUS interface under accounting-service. For example:

```
service-construct interface accounting-service ACCT1 is-ofc-enabled
true pdngw radius server-group AAAGROUP2
```

service-construct interface aaa-interface-group

Configure the Accounting Authentication Authorization interface group, including Diameter, GTPP, and RADIUS parameters. Associate this group with a geographic-redundancy group.

The system supports the configuration of multiple server groups, each associated with a distinct interface, online/offline/policy, etc. The server group must have a distinct origin-realm (unique local realm name). While multiple server groups can have the same local host IP address, they must use a unique origin-realm (unique local realm name).

Diameter

The MCC supports interoperability between the PGW and a Diameter Routing Agent (DRA). A DRA provides routing capabilities between a PGW and an OCS when multiple OCS nodes are deployed in a Diameter realm. When the PGW sends the CCR-Initial to a DRA, it includes the DRA host name and realm within the Destination-Host and Destination-Realm AVPs. However, the PGW uses the values from the Origin-Host/Origin-Realm AVPs received in the CCA-Initial response to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-Update, CCR-Terminate, ASA, and RAA messages. This feature ensures the same OCS manages all messages related to a particular session.

The Diameter service interface table is used to configure Diameter services. The system support three Diameter services:

- **Gx for PCRF server communication** – The Gx interface is configured in the pcrf-charging table.
- **Gz for OFCS server communication** – The Gz interface is configured in the offline-charging table.
- **Gy for OCS server communication** – The Gy interface is configured in the online-charging table.

Weighted Round-robin Distribution to Multiple Diameter Peers

Operators can configure the MCC to use a weighted round-robin algorithm to distribute initial session requests among the available Diameter peers within a Diameter server group based on the configured weight assigned to each peer. The Diameter peer with the highest weight (with 10 being the highest and 1 being the lowest) receives the most distribution of new session requests. The MCC only uses the configured weight for the initial peer selection (for example, CCR-Initial).

For example, if Peer A is set to 4 and Peer B is set 2, the MCC will distribute the initial session requests as ABABAA. In this example, the MCC alternates between both peers starting with Peer A because it has the highest weight.

When a Diameter peer is configured with a weight of 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-Update/CCR-Terminate) for existing sessions.

If the Diameter peer with the highest weight is unavailable, the request is sent to the peer with the next highest weight in the same Diameter server group. If multiple Diameter peers are assigned the highest weight (10), the MCC uses a round-robin approach to select from among those peers for new requests.

Operators can configure the MCC to use a weighted round-robin algorithm to select a Diameter peer from among a list of primary peers or from among a list of secondary peers, but not *across* primary and secondary peers.

Use the following command to assign a weight to a Diameter peer:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> weight <value>
```

weight – Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-Update/CCR-Terminate) for existing sessions. *Applies to Gy/Gx/Gz/STa/Swm/S6b interfaces.*

Modifying Default AVP Flag Values for Diameter AVPs

This section describes how to modify the default AVP flag values for Diameter AVPs on a specific Diameter interface (Gx, Gy, Gz, SWm, STa, and S6b).

Diameter AVPs have flags that indicate to the receiver how each AVP must be handled in Diameter messages. An AVP can have one or both of these flags set.

- Mandatory “M” flag – Indicates that support for this AVP is required. If either the AVP or its value is not recognized, then the message is not processed.
- Vendor-specific “V” flag – Indicates that this AVP has the optional Vendor ID field and when set, the AVP belongs to the specific vendor code address space.

For example, if the RAT-Type AVP has the Mandatory flag and Vendor-Specific flag set by default, all Diameter messages must have this AVP, otherwise the message is not processed. Also, when this AVP is sent, the Vendor ID field will be populated with the value associated with the specific Vendor.

With this feature, if the Operator changes the RAT-Type AVP Flag setting so that only the Vendor-Specific flag is set for the Gx interface, then subsequent Gx messages may or may not have this AVP. However, the Vendor ID field will still be populated with the value associated with the specific Vendor when this AVP is sent.

Use the following CLI command to modify the default Diameter AVP flag values.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter avp-flags <entry-name> attribute-id
<attribute-id> vendor-id <vendor-id> [gx|gy|gz|swm|sta|s6b] flag-value
[none|mandatory|vendor-specific|mandatory-and-vendor-
specific]
```

- *avp-flags <entry-name>* – Specifies the configured name for the AVP flag entry. The entry name is a custom name used for the AVP flag entry and is not necessarily the name of the Diameter AVP.
- *attribute-id* – Specifies the Diameter AVP attribute ID.
- *vendor-id* – Specifies the Diameter AVP vendor ID.

> ***Note:*** *The attribute ID and vendor ID are mandatory fields. This feature will only modify the AVP flag value if the attribute ID and vendor ID are correctly provided. If there is no AVP found with the values provided, the configuration will have no effect.*

- *[gx|gy|gz|swm|sta|s6b]* – Specifies the Diameter interface.
- *flag-value* – Specifies the flag value for the Diameter AVP.
  - If an AVP has a non-zero Vendor ID, the possible flag values are vendor-specific and mandatory-and-vendor-specific.
  - If an AVP has a Vendor ID of zero, the possible flag values are none and mandatory.

SCTP Multi-homing Support

This feature adds the ability to configure a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. SCTP multi-homing enables the MCC to switchover to an alternate path in case of a failure on the primary path. The MCC reverts back to the primary path after the primary connection becomes available. The secondary loopback IP address is only configurable for the SCTP transport type. The primary and secondary loopback IP addresses can be IPv4 or IPv6. MCC supports socket creation to be associated to a single network context. SCTP multi-homing will be supported in a single network context.

If using a mix of IPv4/IPv6, ensure the primary loopback IP address is IPv4 and the secondary is IPv6. SCTP Multi-homing will not work if the primary loopback IP address is IPv6 and the secondary is IPv4. For this reason, Affirmed Networks recommends not using a mix of IPv4/IPv6 addresses for SCTP Multi-homing.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
secondary-loopback-ip <loopback-name>
```

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> secondary-loopback-ip <loopback-name>
```

The secondary-loopback-ip parameter is only visible when the local Diameter node transport type is set to SCTP.

The following show commands for the local Diameter node and peer status now display the secondary IP address.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter node-status <diameter-node-name>
```

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-status <diameter-node-name>
<peer-host-name>
```

disconnect-server-admin-state

MCC supports RADIUS Disconnect-Request and Disconnect-Response messages. This feature enables the PGW/GGSN to receive and process unsolicited Disconnect-Request messages from remote RADIUS servers. Disconnect-Request messages are used when the remote RADIUS server wants to terminate a user session after the user session has been authenticated and authorized to access the requested network services. See the *Affirmed Networks RADIUS Interface Guide* for details.

disconnect-local-server-port

Specify the port to which Disconnect-Request messages are sent. The default is 3799.

Ensure that the disconnect local server port does not conflict (same IP and same port) with the local port range of any of the configured RADIUS server peers.

Note: *If multiple RADIUS server peers have the same local IP and the same disconnect local server port, only one RADIUS disconnect server is instantiated.*

See the *Affirmed Networks RADIUS Interface Specification* for details.

Mirroring RADIUS Accounting-Request Messages to GiGW

This feature adds the ability to mirror RADIUS Accounting-Request messages, sent from a RADIUS client, to the Gi gateway (GiGW). The MCC then forwards these Accounting-Request messages to an external RADIUS server. However, RADIUS Accounting-Response messages received from the external RADIUS server are forwarded to the RADIUS client without mirroring the response messages to the GiGW.

Note: *This feature only applies to a GiGW configured with RADIUS Server mode. It does not apply to a GiGW configured with RADIUS Proxy mode.*

Configure a RADIUS server server-group with an external server IP address and port.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group radius-server server-group <sg-name>
ext-server-ip-address <ip-address> server-port <external-port>
```

- **ext-server-ip-address** – Specifies the IP address of an external RADIUS server. Can be IPv4 or IPv6. When this parameter is configured, the system mirrors RADIUS Accounting-Request messages, received from a RADIUS client, to the GiGW and then forwards these Accounting-Request messages to an external RADIUS server.

- **server-port** – When the `ext-server-ip-address` parameter is configured, this parameter specifies the port number of the external RADIUS server to which RADIUS Accounting-Requests are sent when mirrored to the GiGW.

Note: *A RADIUS server server-group can be configured with either a server IP address (*`server-ip-address`*) or an external server IP address (*`ext-server-ip-address`*), but not both.*
Note: *When the* `ext-server-ip-address` *parameter is configured, the* `secret-key` *parameter is not visible.*

Local Node

A Diameter service is defined for each Diameter service interface on the cluster. The Diameter node contains the Diameter identity for the Diameter interface, including local host name, IP address, and port number. The Diameter local node is created and activated when the service interface, offline-charging, online-charging, and pcrf-charging are configured and enabled.

Peers

Each Diameter local node contains a list of peers (Diameter servers) with which the local node communicates. Each peer entry contains a hostname, IP address, and port. Multiple peers can be configured within a Diameter node.

There are two types of peers; *primary peers* are active peers used for Diameter sessions and *secondary peers* or backup peers are used for Diameter sessions only when the active peers are down. When multiple active peers are configured for the Diameter interface, the system uses a round-robin approach to select peers.

GTPP

The system supports the GPRS tunneling protocol prime (GTPP). The system uses the GTPP protocol to transport ASN.1 encoded Charging Data Records (CDRs) such as accounting requests (start/interim/stop) from the gateway agents (SGW, PGW, and GGSN) to the peer Charging Gateway Function (CGF) via the Ga interface. The GTPP protocol is required when the CGF resides outside the nodes that generate CDRs. For more information, see the *Affirmed Networks Charging Data Records Specification*.

The Charging Data Function (CDF) complies with 3GPP 32.295 as follows:

- Uses the UDP as the path protocol
- Maintains the link towards the peer CGF
- Encapsulates the encoded charging records from the gateways into a GTPP Data Transfer Request

radius-client server-group and radius peers

The radius-client server-group is a container for a group or list of RADIUS client server peers. Each radius-client server-group contains a list of peers (RADIUS client servers) with which the system communicates. Each peer entry contains a server name, IP address, port, secret key, and an associated local network-context/IP address/port four-tuple. For details, see the *Affirmed Networks RADIUS Interface Specification*.

Prioritizing RADIUS Server Peer Selection

The *peer-retry-mode* enables Operators to configure the order in which the RADIUS client (running on the MCC GW) selects RADIUS Server peers within a RADIUS server group. Operators assign each RADIUS server peer a priority number (with 1 being the highest priority). If the RADIUS server peer with the highest priority is unavailable, the request is sent to the peer with the next highest priority in the same group. If multiple RADIUS server peers are assigned the highest priority (1), the RADIUS client uses a round-robin approach to select from among those peers for new requests. For details, see the *Affirmed Networks RADIUS Interface Specification*.

session-aware-peer-selection

When this feature is enabled, the system sends all RADIUS Accounting requests (START/INTERIM/STOP) for a given session to the same RADIUS server peer. If the peer becomes unavailable, the system transmits, in a round robin fashion, the subsequent RADIUS Accounting messages to other active peers in the same RADIUS Accounting server group. For details, see the *Affirmed Networks RADIUS Interface Specification*.

RADIUS Health Check and Request Timeout

The system uses the RADIUS Request Timeout feature for failover support and uses the RADIUS Health Check feature for failure detection. For details, see the *Affirmed Networks RADIUS Interface Specification*.

Enhanced RADIUS Health Check

Enhanced RADIUS Health Check (ERHC) verifies that the remote RADIUS server peer is operational prior to declaring it active by periodically sending RADIUS Authorization/Accounting Health Check messages to RADIUS server peers to check whether they are responding.

service-construct interface accounting-service

Configure the RADIUS accounting service which references a RADIUS server group previously configured. See the *Affirmed Networks RADIUS Interface Specification* for details.

service-construct interface authorization-service

Configure the RADIUS authorization service which references a RADIUS server group previously configured. The RADIUS access request record attributes can be controlled by associating the service with an access request template. See the *Affirmed Networks RADIUS Interface Specification* for details.

service-construct interface gw-session-learner

Configure the MCC session learner interface.

MCC supports WAG functionality to handle RADIUS proxy bypass mode. In this mode, DHCP is used to trigger the WAG to fetch subscriber data from an external T-WAP. RADIUS is used to fetch the data, with the WAG acting as the RADIUS client. In an Access Request/Access Accept callflow, both standard and proprietary AVPs transfer data. After the data is fetched and the PDN session is established, the DHCP continues as normal, with WAG acting as the DHCP relay. Use dhcp-release to clear a session.

Configuring Objects

Configure the objects as follows:

Table 9-29 service-construct interface gw-session-learner

Object	Description
gigw-ip-local	Enter a name and configure the admin state of the Session Learner GiGW Interface with IP trigger and local configuration.
gigw-ldap-client	Enter a name and configure the admin state of the Session Learner GiGW Interface with LDAP Client. ldap-client – Configure the ldap-interface-profile for the LDAP Client configuration for IP-LDAP Session Learner Interface.

Table 9-29 service-construct interface gw-session-learner

Object	Description
non-3gpp-access-diameter-client	Configure Session Learner TWAG Interface with Diameter authentication prior to setting up PDN session. ■ access-network-identity – Access Network Identity ■ admin-state – Admin state of Gateway Session Learner Interface ■ clear-session-if-moved – If a session moves from this session learner interface to another, whether the existing call should be cleared and the session re-established ■ name* – Gateway Session Learner - Non 3GPP Access Diameter Client Name diameter-client server-group-mapping - Diameter Client Server Group List - select from configured lists or set default ■ mapped-apn-name ■ priority* – Server group priority. 0 = lowest priority ■ server-group – Diameter Client Server Group - select from configured list radius-server-trigger - RADIUS Server Configuration for Non-3GPP Access Diameter Interface ■ server-group* – RadiusServer Server Group - select from configured list radius-server-trigger radius-mapping <message-type> - RADIUS server attribute mapping template ■ message-type* – RADIUS message type ■ template-name* – RADIUS mapping template name
radius-proxy	MCC RADIUS Proxy Session Learner Interface ■ radius-client server-group-mapping – Radius Client Server Group Mapping - select from configured lists or set default.
radius-server	Gateway RADIUS Server Session Learner Interface
twag-inband-authentication	Configure Session Learner TWAG Interface with no authentication prior to setting up PDN session.
sd-diameter-client	Configure the Sd session learner interface on the GiGW. ■ admin-state – Admin state of Sd session learner interface ■ name* – Name of the Sd diameter client session learner interface ■ sd-interface – Name of the Sd interface

Table 9-29 service-construct interface gw-session-learner

Object	Description
enable-ue-ipv6	MCC supports the IPv4/IPv6 user plane for TWAG and TWAP. The IPv6 user plane is supported only in stateless autoconfig with unsolicited router advertisement support. MCC periodically sends out IPv6 router advertisement messages to advertise the UE IPv6 prefix address of a session. If a session uses an IPv6 address and GRE tunnel, IPv6 router advertisement messages are sent out automatically. IPv4 addresses are always enabled.
dhcp-trigger-tunnel-interface	DHCP trigger interface configuration for PDN session setup. The dhcp-trigger-tunnel-interface points to the tunnel interfaces used by the TWAG (list of tunnels). This is the tunnel interface to the GW (from which the DHCP requests are coming). DHCP requests are associated with the correct TWAG (assuming multiple TWAGs are configured). The new WAG points to the TWAG session learner interface. If no VLANs are provisioned, this represents all untagged traffic. If VLANs are provisioned, then only sessions on the tunnel with the specific VLANs are selected.
vlan	Optional VLAN value.
twag-radius-client	Session Learner TWAG Interface with RADIUS Client

Charging Overview

The system is capable of establishing a subscriber billing profile to track an individual subscriber's usage by obtaining monthly historical usage from the Operator's PCRF system and combining that usage with real-time usage created within the current subscriber session. The system interacts with the PCRF to obtain the subscriber's policy and charging subscription characteristics. Using 4G LTE capabilities, the system manages usage subscription on an individual subscriber or a small group of subscribers using policy and charging functions.

The charging service supports the following features and functions:

- IP Packets by subscriber IP session and IP Packet Type on a real-time basis.
- Traffic/usage based on the type of access network. Mobile access accounting is independently measured and accounted.
- Usage based on the type of access network. Traffic originating on mobile access accounting is independently measured and accounted.
- Usage based on the Mobile Network Code identifier (MNC) to account for both local roaming as well as international roaming.
- Correlates subscriber network usage to subscriber subscription characteristics such as post-paid, pre-paid, hybrid post-paid, and pay-per-use.

- Correlates subscriber network usage on a real-time basis to tiered subscription plans on a per-user basis.
- Provides real time in session subscriber charging notification of tier consumption level when carrier tier level is being approached. This Advice of Charge subscriber notification level is carrier adjustable.
- Captures type of content being consumed on a real time per subscriber/per session/per IP Packet basis.
- Provides the ability to correlate packets and byte counts usage to type of content consumed. The service allows charging for content versus bytes used.
- Identifies carrier content partners and enforces charging rules that either allow or disallow attachment to the carrier's subscribers by traffic type, protocol, or application.
- Mediates charging to both online and offline charging servers.
- Facilitates the creation of real time CDRs and usage logging for customer support and analytics.

service-construct interface ipms-interface server-group-ipms-servers

Configuration of external IPMS Servers for this service endpoint

Configuring Objects

Configure the objects as follows:

Table 9-30 service-construct ipms-interface

Object	Description
name*	Enter a unique name of the external IPMS Server.
remote-ipms-ip-address*	Remote (IPMS Server) IP address.

service-construct interface offline-charging

The offline charging feature enables Operators to configure the offline charging interface, either to a local file or an *Offline Charging System* (OFCS) server. Offline charging enables Operators to charge their customers, in real-time, based on service usage.

An OFCS is used for offline charging and receives charging events from the *Policy and Charging Enforcement Function* (PCEF) and generates CDRs that can be transferred to the billing system. The SGW, PGW, and GGSN log charging events to CDRs when the local CDR is enabled. They can also send the charging events to offline charging servers in real-time using streaming protocols such as Diameter on the Gz interface and GTP' on the Ga interface.

- The Gz interface is the offline (CDR-based) charging interface between the GSN and the OFCS.
- The Ga interface is the offline charging interface and uses GTP` as the transport protocol to upload CDRs from the GSN's (PGW, GGSN, SGW) to the charging gateway function (CGF). In this case, CDRs are encoded in ASN.1 format.

The system is designed to perform offline charging on tiered subscriber billing access plans. The charging service interacts with the Operator's offline charging back-office systems and PCRF systems to recognize and bill for tiered usage on a non-real time basis. For example, the system is able to recognize three tiered billing plans and the associated overage of the plan and use that data to produce call detail records that can be used by the Operator to compile a monthly bill. The system is able to produce periodic CDRs using a 15-minute interval and transmit those CDRs electronically to the Operator's mediation system using the Operator's administrative network. For details, see the *Affirmed Networks Charging Data Records Specification*.

service-construct interface online-charging

An *Online Charging System* (OCS) is a credit management system for pre-paid charging. Typically, the user purchases a fixed amount of usage (called a quota) on their data plan and that value is enforced. The PGW and GGSN interact with the OCS over the Gy interface to check credit and report credit status. The OCS may authorize access to individual services or a group of services by granting credits for authorized IP flows. Usage of resources is granted in different forms (for example, in the form of a certain amount of time, traffic volume, or chargeable events). For more information on online charging, see the *Affirmed Networks Gy Interface Specification*.

service-construct interface pcrf-charging

A *Policy and Charging Rules Function* (PCRF) is a server that deploys a set of Operator-created business rules. Charging control supports both offline (post-paid services) and online charging (pre-paid services and its authorization). The PCRF also controls the applied measurement method (data volume, duration, combined or event based). The 3GPP standards define the Gx reference point as the policy and charging control interface between the GGSN/PGW and the PCRF. The PCRF charging interface configures the Diameter server-group name, data-record-template for AVPs sent in CCR initial, and updates and terminate requests. For more information on the Gx interface, see the *Affirmed Networks Gx Interface Specification*.

pcrf-down-gx-replay

MCC supports diameter credit control failover on the Gx/Gy interfaces based on the result code received in the CCA message. Operators configure a list of result code ranges (3xxx-5xxx) and when a CCA message is received with a non-success result code (other than 2001), the default Gx policy/Gy quota (if enabled on the interface) is used to set up the session if the received result code is present in the result code list. Also, for each result code range in the list, Operators can specify whether the PGW retransmits the request to a secondary server before applying the default Gx policy/Gy quota.

The gx-replay feature is used if all PCRF servers go down after the Gx session is already established and the session using the default policy is terminated before the PCRF comes back up. In this case, if gx-replay is enabled, the GGSN/PGW replays the cached session information via a CCR-Terminate message.

pcrf-override-plmnid

Operators can configure the MCC GGSN/PGW to override the PLMN-ID sent in the GTP-C messages on the S4/S11/S5/S8/Gn/Gp interfaces. If the override PLMN-ID feature is enabled and the PCRF sends the CCA message containing the PLMN-ID within the 3GPP-SGSN-MCC-MNC AVP, the GGSN/PGW replaces the PLMN-ID received in GTP-C messages with the PLMN-ID received from the PCRF.

The GGSN/PGW uses the PLMN-ID received from the PCRF in subsequent messages to the OCS and in local PGW CDRs (CSV and ASN.1 formats) and GTPP CDRs (Ga interface). Furthermore, the PLMN-ID received from the PCRF is also used if no PLMN-ID is received from MME/SGSN.

Operators can enable/disable the override PLMN-ID feature at the PCRF interface as follows:

- enabled – The PLMN-ID received from the PCRF overrides the PLMN-ID received in GTP-C messages and is used in subsequent messages to the OCS and in local PGW CDRs (CSV and ASN.1 formats) and GTPP PGW CDRs (Ga interface).
- disabled – (default) The GGSN/PGW ignores the PLMN-ID received from PCRF and uses the PLMN-ID sent in GTP-C messages.

policy-server-interface

Configure a policy server interface profile.

port-nbr-service

Configure port NAT binding record interface service profile. Select the encode format file name, and file type to be used for encoding the NAT binding records.

For NBR records, configure a port NBR interface service profile of type **`ramdatarecord.`** For example:

```
service-construct interface port-nbr-service <profile name> local
nbr-file-type ramdatarecord nbr-file-name
```

subscriptionid-imsi-format-non-3gpp-access

IMSI format to be sent in Subscription-id AVP for non-3gpp access. Options include imsiformat (default) and imsinaiprefix0format. The IMSI-based NAI format sent in the Subscription-Id AVP is defined in RFC 4282 and formatted in 3GPP TS 23.003.

The Credit-Control-Request (CCR) data record templates (Start/Interim/Event/Stop) contain fields to include/exclude the Subscription-Id AVP based on the Subscription-Id-Type; subscriptionidimsi and subscriptionidmsisdn.

user-name-type

Configure the user-name AVP format used in messages on the Gx, Gy, S6b, and offline charging interfaces (Ga, Gz, and local) if not present in the PCO. This setting overrides the user-name-type present at the APN level. However, if this user-name-type parameter is set to none, MCC uses the APN level user-name type.

Use the following command to configure on the Gx/Gy/S6b interface:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
[pcrf-charging|online-charging|s6b-interface]
<interface-name> user-name-type
[apn|default-username|imsi|imsi-nai-no-prefix|imsi-naiprefix-
0|imsi-plus-apn|imsi-plus-apnnionly|msisdn|msisdnplus-
apn|msisdn-plus-apnnionly|none]
```

Use the following command to configure on offline charging interfaces (applies to Gz, Ga and local CDR interfaces):

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> user-name-type
[apn|default-username|imsi|imsi-nai-no-prefix|imsi-naiprefix-
0|imsi-plus-apn|imsi-plus-apnnionly|msisdn|msisdnplus-
apn|msisdn-plus-apnnionly|none]
```

service-construct interface s6b-interface

Configure the s6b interface profile. See Chapter 18, S6b Interface.

service-construct interface sd-interface

Configure the s6b interface profile. See Sd Interface (page 20-1).

service-construct interface-trigger-template

Configure an interface trigger template and associate it with an accounting, offline, or online charging interface.

service-construct ip-list

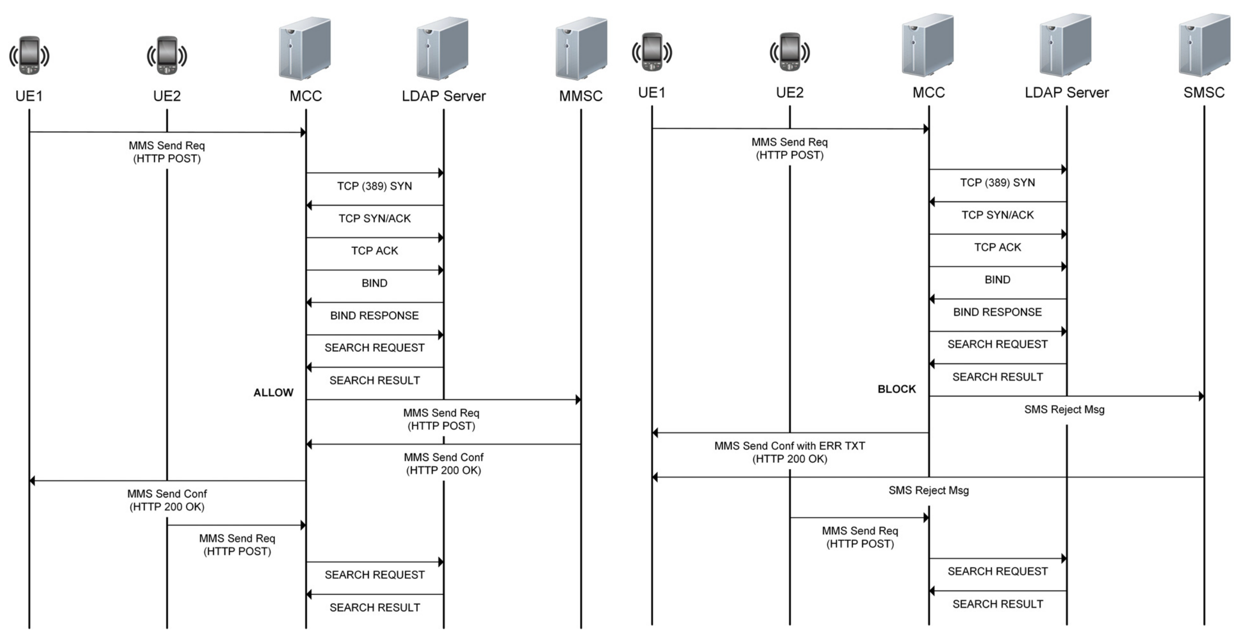
IP address list. Operators can specify a specific IP address range with a start/end value. Operators can use this configured ip-list to configure a list of IP addresses for trusted DNS servers. The dns-proxy does not support this object.

service-construct ldap

The Lightweight Directory Access Protocol (LDAP) is an application protocol for reading and writing directories over an IP network. It uses TCP/IP port 389 and UDP port 389. It is used to query and update an X.500 directory. LDAP is specified in a series of IETFs and RFCs, using the description language ASN.1. Configure the feature-group, profile, and remote-service for this LDAP.

The call flow in Figure 9-1 shows the LDAP interaction with the LDAP server. LDAP is used only if the LDAP profile is included and enabled in the HTTP Proxy service instance.

Figure 9-1 LDAP Allow and Block Call Flow



service-construct ldap feature-group

The Operator creates an LDAP profile containing the list of feature-groups and a list of LDAP servers.

Each LDAP feature-group can have a list of (attribute, value) pairs and associated action. If there is no match in the feature-group, the system uses the action configured in the profile. However, if the LDAP response is no-such-object, the system always uses the block action.

The MM1 Proxy (HTTP Proxy) initiates an LDAP request to the LDAP server with the MSISDN as the key. The result from the LDAP server is then matched against the feature groups configured for that LDAP profile. If a match is found in one of the feature groups, the action configured in that feature-group is taken.

Configuring Objects

Enable and configure LDAP feature-group attributes such as the action to take if a query response matches any attribute configured in the feature-group. Configure the objects as follows:

Table 9-31 service-construct ldap feature-group

Object	Description
admin-state	Enable or disable the admin state for this feature-group.
action	Allow or block service to the user that associated with LDAP search. If the service is set to block, the MCC sends an mms-response message and an sms-notification text message.
attributes	A list of LDAP attributes to be checked upon LDAP response.
enforce	Always check the attributes, even in case of param error.

service-construct ldap profile

Enable and configure LDAP profile attributes such as the default action to take if LDAP failure occurs, feature groups for this profile, network-context, and loopback ip used to connect to the LDAP remote server. In addition, configure the ldap-profile applicable to the instance.

Configuring Objects

Configure the objects as follows:

Table 9-32 service-construct ldap profile

Object	Description
admin-state	Enable or disable this ldap-profile.
default-action	Configure the action to be taken if a permanent LDAP failure occurs. Options include: allow|block.
feature-group	Configure LDAP feature groups for the profile along with a unique priority for each feature-group.
lookup-key-string	The default lookup-key-string is uid. If a lookup-key-string is not configured, the LDAP agent uses the default uid.
lookup-key-type	The LDAP agent applications (that is, GW/http-proxy) set the LDAP query key values based on the configured lookup-key-type in the profile. Key types include ue-msisdn (default) and ue-ip-address. If a lookup-key-type is not configured, the LDAP agent uses the default ue-msisdn.
loopback-ip	Configure the loopback IP address from the configured network-context to be used as the local address, when connecting to the servers. Since a server may be used by different profiles, the same loopback-ip must be used in all profiles that use the same network-context, configured or not.
network-context	Configure network-context used to connect to LDAP remote-server.
remote-server	Configure the LDAP servers for the profile and a priority for each server. ■ priority – Set a priority for the remote-server. If the priority of the servers is the same in a given profile, the LDAP agent performs round-robin load balancing. If the priority of the servers is different, the LDAP agent uses priority-based load balancing. The LDAP agent performs the following actions based on the priority (1..255) of the remote servers in a given profile.
sort-attribute-name	Configure the LDAP attribute name used to sort results. If a sort-attribute-name is not specified, the sort-order must be set to all.

Table 9-32 service-construct ldap profile

Object	Description
sort-order	A specified sorting order is used to sort multiple entries in the response from the server, based on the configured attribute-name. The entries are sorted as follows: ■ all (default) – MCC selects all attributes (up to 100) from matching entries in the order in which they are received. If no sort-attribute-name is configured, all entries are matching entries. ■ ascending-first – MCC selects the first matching entry after all entries are sorted in ascending order based on the sort-attribute-name. ■ descending-first – MCC selects the first matching entry after all entries are sorted in descending order based on the sort-attribute-name. ■ first – MCC selects the first server that has a matching attribute based on the sort-attribute-name.

LDAP Agent Actions

Table 9-33 LDAP Agent Actions Based on Remote Server Priority

LDAP Agent Action	Same Priority	Different Priority	Mix of Priorities
Tries the next active client connection for each LDAP search request in a round-robin fashion. ■ If no active client connections exist, the LDAP agent reverts to the current active client server connection. ■ If all client connections are inactive in a given priority, the LDAP agent tries the next priority servers in the server configuration.	X		X
If all client connections are inactive in entire profile, the LDAP agent retries the servers from the server configuration Entry Index order since priority is same for all servers).	X		
Tries the same active client connection for each LDAP search request as long the server connection is in service and no other server client connections are active in that priority group.		X	X
If the current client connection is inactive, the LDAP agent tries the next priority active client connections.		X	X
If all client connections are inactive, the LDAP agent retries the servers from priority order.		X	X

service-construct ldap remote-server

Enable and configure LDAP remote-server attributes.

Configuring Objects

Configure the objects as follows:

Table 9-34 service-construct ldap remote-server

Object	Description
address	Configure LDAP server IP address
admin-state	Enable/Disable this ldap-server
base-DN	Configure LDAP server base distinguished-name string
bind-DN	Configure LDAP server bind distinguished-name string
bind-retry	Configure LDAP server bind retry
idle-timeout	LDAP server connection idle timeout (in seconds) used to ping the server. If no activity occurs on this server connection for more than this idle timeout value, the system sends an abandon message to the server serving as a ping. The response from server is to reset the timer on monitoring this connection. Otherwise, the connection is closed.
name*	Configure name for this remote-server.
password	Configure LDAP server bind password
port	Configure LDAP server port, default 389
timeout	Configure LDAP server connect timeout value in seconds
version	Configure LDAP server version

service-construct ipv4-filter-list

Filter IPv4 packets by matching on IP addresses only. Using the ipv4-filter-list is more efficient than using a packet-filter when you are filtering only on IP addresses and a service-rule would require hundreds of packet-filters. You can configure 10 ipv4-filter-lists.

An ipv4-filter-list supports IPv4. IPv6 is *not* currently supported.

Configuring Objects

Configure the objects as follows:

Table 9-35 service-construct ipv4-filter-list

Object	Description
<CIDR range>	Enter one or more CIDR ranges for the IP addresses. For example, 3.1.1.0/24. You can enter 4000 CIDR ranges for each ipv4-filter-list.
filter-type	Configure the type of filter: ■ network—Filter a network initiated connection. ■ ue—Filter a user equipment initiated connection.
name*	Name for this ipv4-filter-list.

service-construct loopback-ip-list

Create trusted and untrusted IP lists based on L7 rules. HTTP proxy traffic is sourced from a “trusted” address on the DMZ interface. Non-HTTP proxy traffic is sourced from an “unstrusted” address on the DMZ interface.

The http-proxy instance is typically configured with a list of loopbacks used as source addresses for connections initiated to HTTP origin servers. In this case, MCC uses the instance list loopbacks by default. In some cases, service providers may choose to use a different set of loopbacks (for example, “untrusted” loopbacks) for specific traffic. In this case, an http-request-policy is configured to reference a different list of loopbacks.

If a request matches an http-request-policy with a loopback list, http-proxy initiates the connection from one of the loopbacks from the http-request-policy loopback list. If the loopback-ip-list is not configured or it does not include a specific network-context, http-proxy uses all loopbacks available in the network-context (no loopback constraints). In addition to specifying the loopback-ip-list, the http-request-policy can also specify network-context-override.

service-construct manifest-interface-profile

Configure the MCC gateway to retrieve flow information from a manifest server for ADC reporting over the Gx interface. See the *Affirmed Networks Gx Interface Specification*.

service-construct mms

Configure multimedia messaging service. See the *Affirmed Networks Value Added Services Guide*.

service-construct nbiot-quality-of-service

Configure the NBIoT QoS service construct to configure APN rate control and serving PLMN rate control attributes. Operators can configure APN rate control only, serving PLMN rate control only, or both as part of this configuration.

APN Rate Control

APN rate control enables the enforcement of packet rate control in uplink/downlink data traffic. APN rate control can be used for S11-u or s1-u traffic.

The MCC supports APN rate control for uplink packet control. The PGW applies downlink rate control locally.

If the UE makes a request for APN rate control, then the PGW passes the APN rate parameters configured in the nbiot-quality-of-services profile to the UE in the PCO/ePCO.

Serving PLMN Rate Control

Serving PLMN rate control is similar to APN rate control but applies the control at the multi-PDN connection level.

The MCC supports serving PLMN rate control for uplink/downlink user plane packets. The gateway receives serving PLMN rate control in the session establishment message.

Serving PLMN rate control is sent by the MME to the SGW to enforce rate control for NAS DATA PDUs as part of initial attach, handoff, or UE service request procedures. The serving PLMN rate control is applicable only when S11-U is requested for a PDN connection and control plane optimization is configured.

Note: *Use serving PLMN rate control only if PDN is set with flag* ***`control plane only`****.*

Configuring Objects

Configure the objects as follows:

Table 9-36 service-construct nbiot-quality-of-service

Object	Description
apn-uplink-additional-exception-report	APN Uplink Additional Exception Report
apn-uplink-max-message-size	APN Uplink Maximum Message Size: Bytes
apn-uplink-max-rate	APN Uplink Maximum Rate apn-uplink-rate-measurement-units
apn-uplink-rate-measurement-units	apn-uplink-rate-measurement-units: unrestrict/minute/hour/day/week
serving-plmn-downlink-rate-limit	Serving PLMN Rate Control Downlink Rate Limit: in 6 minute interval
serving-plmn-uplink-rate-limit	Serving PLMN Rate Control Uplink Rate Limit: in 6 minute interval

service-construct network-profile

Configure a network-profile on a User Plane.

Configuring Objects

Configure the objects as follows:

Table 9-37 service-construct network-profile

Object	Description
interface-type*	Interface type. Options include: s1u-gtpu|s5s8-gtpu-sgw|s5s8-gtpu-pgw|lawful-intercept
li-source-ip-address	Loopback IP address name used as source IP for packets sent to SX3LIF IP.
li-source-port	Source UDP port number for packets sent to SX3LIF IP.
name*	Name of this network-profile
network-context*	Name of the network-context associated with this network-profile.

service-construct packet-filter

Define parameters and rules used to filter packets as they travel through the network interface. Filter rules specify the criteria that a packet must match and the resulting action, either block or pass, that is taken when a match is found. Filter rules are evaluated in sequential order, first to last.

Configuring Objects

Configure the objects as follows:

Table 9-38 service-construct packet-filter

Object	Description
initiator	ue initiated or network initiated connection
ip-version	IP version
name*	Name for this packet-filter.
max-uplink-TTL	maximum value of the connection TTL(Time-To-Live) default: 255 See Time to Live (TTL) in HTTP Packet Headers (page 9-62).

Table 9-38 service-construct packet-filter

Object	Description
min-uplink-TTL	minimum value of the connection TTL(Time-To-Live) default: 0 See Time to Live (TTL) in HTTP Packet Headers (page 9-62).
network-ip-address-end	equipment IPv4 address range end (inclusive)
network-ip-address-start	equipment IPv4 address range start (inclusive)
network-ip-v6-address-end	network IPv6 address range end (inclusive)
network-ip-v6-address-start	network IPv6 address range start (inclusive)
network-port-end	network equipment port range end value (valid for protocol = TCP (6), UDP (17), SCTP (132), only). This feature builds the QUIC proxy for splicing the QUIC connection and forces different data rates in a manner similar to HTTP/HTTPS flows. Affirmed Networks recommends configuring the network-port-start and network-port-end as 443. Although 443 is the standard, Affirmed Networks supports other ports and configured ranges.
network-port-start	network equipment port range start value (valid for protocol = TCP (6), UDP (17), SCTP (132), only). This feature builds the QUIC proxy for splicing the QUIC connection and forces different data rates in a manner similar to HTTP/HTTPS flows. Affirmed Networks recommends configuring the network-port-start and network-port-end as 443. Although 443 is the standard, Affirmed Networks supports other ports and configured ranges.
protocol	ip protocol values (for example, all: 255, ICMP: 1, TCP: 6, UDP: 17, SCTP: 132)
ue-ip-address-end	user equipment IPv4 address range end (inclusive)
ue-ip-address-start	user equipment IPv4 address range start (inclusive)
ue-ip-v6-address-end	user equipment IPv6 address range end (inclusive)
ue-ip-v6-address-start	user equipment IPv6 address range start (inclusive)
ue-port-end	user equipment port range end value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)
ue-port-start	user equipment port range start value (valid for protocol = TCP (6), UDP (17), SCTP (132), only)

Time to Live (TTL) in HTTP Packet Headers

The Time to Live (TTL) in HTTP packet headers feature enables Operators to differentiate traffic on combinations of different SPI (packet filter), URL, SNI (as part of the TLS exchange), and Time to Live (TTL).

This feature enables Operators to:

- differentiate charging based on minimum and maximum TTL values by configuring different charging groups with different service rules containing packet filters with TTL values to treat connections with different TTL differently.
- differentiate QoS services based on TTL values by configuring QoS policies with QoS flows having service rules containing packet filters to handle connections with different TTL values differently.

Block UE Spoofing

The MCC supports anti-spoofing of downlink UE packets. This feature enables Operators to block downlink traffic to a UE if the source IP address is from an address within the ranges of the configured UE pools. This feature supports packet-filter parameters used to specify a range of subscribers to be blocked. Operators can specify a UE or network initiated connection (initiator), ip-version, and the protocol (TCP (6), UDP (17), and SCTP (132). Operators can also specify the start and stop range for the following: ue-ip-address, ue-ip-v6-address, network-ip-address, network-ip-v6-address, ue-port, and network-port.

service-construct pdngw-list

Configure the list of PDN gateways. Specify the IP address, description, and weight assigned to the remote PDN gateway.

service-construct plmnid-list

Configure the home list and roaming (visiting) list to add a subscriber to the plmnid-list.

When the PLMN ID (MCC + MNC), received in the subscriber's IMSI, matches a PLMN ID configured in the home-plmnid-list, the configured PLMN ID is sent in the 3GPP-GGSN-MCC-MNC AVP on all external interfaces (RADIUS, Gx, Gy, Gz/Rf). If an entry is not found in the list, the first entry in the home-plmnid-list is used in the 3GPP-GGSN-MCC-MNC AVP.

service-construct protocol-application-rule-group

The protocol-application-rule-group object supports signature analysis. Signature analysis identifies applications and protocols by detecting deterministic and/or heuristic patterns in the bidirectional packet flow. Signature analysis is the key technology that enables workflow to control the application of advanced services on subscriber traffic (for example, charging, per-application filtering, etc.). Using signature analysis, the system identifies applications and protocols by detecting deterministic and/or heuristic patterns in the bidirectional packet flow. Signature analysis provides carriers with the capability to continually recognize "over the top" (OTT) applications wherein identification and control of these applications is key to being able to manage/monetize OTT applications.

The signature analysis feature provides dynamic updates to application profiles to recognize adaptive changes to applications. This feature also integrates the capability into the DPI-workflow subsystem to facilitate carrier billing and content partnerships. Integrated and granular signature analysis enables Operators to control the sessions on their network by providing an enhanced subscriber experience. The system performs this function in-line, coupled with the real-time charging function and enables Operators to identify classes of usage and associate that usage to charging plans. This feature allows Operators to charge special rates on top-up/opt-in services on demand.

Signature analysis provides an integrated model for charging and traffic management for the following application groups:

- business
- conference
- database
- e-commerce
- file-transfer
- gaming
- generic
- industrial
- mail
- messaging
- mobile
- network-management
- peer-to-peer
- remote-control
- sharehosting
- social
- streaming
- tunnel
- voice-over-ip
- web

Configuring Objects

Through the use of the protocol-application-rule-group, filtering can be configured to specify desired behavior (such as QoS, charging, etc.) for:

- All traffic from a subscriber that is currently tethering one or more other devices
- Traffic from only tethered devices
- All traffic from a subscriber device that has a specific Operating System
- Traffic flows from a device (subscriber device or tethered device) that has a specific Operating System

- Various combinations, such as traffic flows that are from a tethered device and have a particular Operating System

service-construct quality-of-service

QoS configures default and network Quality of Service characteristics associated with subscribers. The MCC supports the following QoS functions:

- Allocation and Retention Priority (ARP)
- Uplink and downlink APN Aggregate Maximum Bit Rate (AMBR)
- Uplink and downlink burst size in kilobytes
- Rate measurement units (kbps, mbps, gbps)
- Quality of service Class Identifier (QCI) of the PDP Session
- Dedicated bearer control
- Service Data Flow (SDF) rate enforcement
- Differentiated Services (DiffServ) Code Point (DSCP) marking

In addition, Operators can classify traffic to a service data flow (SDF) based on L3/4 packet filters and L7 application identity and/or extracted HTTP fields.

QoS Overview

This section provides an overview of the 3GPP bearer model used to implement the QoS functions in the Affirmed Networks MCC. The MCC supports “on-the-fly” creation of dedicated bearers.

Bearer Model

A bearer is the basic traffic separation element that enables differential treatment for traffic with differing QoS requirements. Bearers provide a logical, edge-to-edge transmission path with defined QoS between the UE and the PGW. There are two types of bearers: dedicated bearers and default bearers.

- *Default bearer (non-GBR)* – When the UE attaches to the network for the first time, it is assigned the default bearer which remains as long as the UE is attached. The default bearer is assigned best effort service. Each default bearer comes with an IP address. The UE may also have additional default bearers, each with a separate IP address. Default bearers are assigned QCI 5 to 9 (Non- GBR) (see Table 9-39 on page 9-65.

- *Dedicated bearer (GBR or non-GBR)* – Simply stated, a dedicated bearer provides a dedicated tunnel to one or more specific traffic types (for example, VoIP, video, etc). A dedicated bearer acts as an additional bearer on top of the default bearer. A dedicated bearer can be GBR or non-GBR (whereas a default bearer can only be non-GBR). A dedicated bearer uses Traffic Flow Templates (TFTs) to give special treatment to specific services.

  If the PCRF modifies the QCI/ARP values for all charging rules on an existing dedicated bearer, the MCC gateway updates the QCI/ARP value for this dedicated bearer and sends an Update Bearer Request to the MME.

Each bearer is associated with a set of QoS parameters used to describe the properties of the transport channel, including bit rates, packet delay, packet loss, bit error rate, and scheduling policy. A bearer may have two or four QoS parameters, depending on whether it is a real-time or best-effort service:

- QoS Class Indicator (QCI)
- Allocation and Retention Priority (ARP)
- Guaranteed Bit Rate (GBR) – real-time services only
- Maximum Bit Rate (MBR) – real-time services only

QoS Class Indicator (QCI)

The QCI specifies the treatment of IP packets received on a specific bearer. QCI values impact several node-specific parameters, such as link layer configuration, scheduling weights, and queue management. The priority of each QCI is applied when packets are sent across the network. Higher priority packets are transfered before lower priority packets. The packet delay budget associated with each QCI defines an upper boundary for the packet delay between the user equipment and the PCEF. The packet error loss rate defines the percentage of higher layer packets.

Table 9-39 lists the standardized 3GPP QCI types that are supported. You can use any QCI value from 1 through 255 in addition to the standardized values.

Table 9-39 3GPP Standardized QCI Attributes Supported

QCI	Resource Type	Priority	Packet Delay Budget	Packet Error Loss Rate	Example
1	GBR	2	100 ms	10^{-2}	VoIP call
2		4	150 ms	10^{-3}	Video call (live streaming)
3		3	50 ms		Real-time online gaming
4		5	300 ms	10^{-5}	Video streaming (buffered streaming)

Table 9-39 3GPP Standardized QCI Attributes Supported (Continued)

QCI	Resource Type	Priority	Packet Delay Budget	Packet Error Loss Rate	Example
5	Non-GBR	1	100 ms	10^{-3}	IMS signaling
6		6	300 ms	10^{-5}	Video (buffered streaming) TCP-based (for example, www, email, chat, FTP P2P file sharing, progressive video, etc.)
7		7	100 ms		Voice, video (live streaming), interactive gaming
8		8	300 ms	10^{-3}	Video (buffered streaming) TCP-based (for example,www, email, chat, FTP P2P file sharing, progressive video, etc.
9		9		10^{-5}	
65	GBR	0.7	75 ms	10^{-2}	Mission critical user plane push-to-talk voice (for example, MCPTT)
66		2	100 ms		Non-mission-critical user plane push-to-talk voice
69	Non-GBR	0.5	60 ms	10^{-6}	Mission critical delay sensitive signaling (for example, MC-PTT signaling, MC Video signaling)
70		5.5	200 ms		Mission critical data (for example, services are the same as QCI 6/8/9)
80		6.8	10 ms	10^{-6}	Low latency eMBB applications (TCP/UDP-based) (for example, augmented reality)
82	GBR	1.9	10 ms	10^{-4}	Discrete automation, small packets
83		2.2			Discrete automation, big packets

Allocation and Retention Priority

The ARP indicates a priority level for the allocation and retention of bearers. The ARP is used to determine whether new bearer modification or establishment requests should be accepted, considering the current resource situation. The 3GPP standards provide mechanisms to drop or downgrade lower-priority bearers in situations where the network becomes congested. Operators can configure the bearer ARP between 1-15, where 1 corresponds to the highest priority and 15 corresponds to the lowest priority. Bearers with a higher ARP (lower number) are given preference.

Guaranteed Bit Rate and Non-GBR Bearers

There are two major types of guaranteed bit rate bearers:

- *Guaranteed Bit Rate (GBR) bearers* – The GBR denotes the bit rate that can be expected to be provided by a GBR bearer. It is used for real-time services, such as conversational voice and video. A GBR bearer has a minimum amount of bandwidth that is reserved by the network, and always consumes resources in a radio base station regardless of whether it is used or not.

- *Non-GBR bearers* – Non-GBR bearers do not have specific network bandwidth allocation. Non-GBR bearers are mainly used to carry various best-effort data services, such as file downloads, email, and Internet browsing. These bearers experience packet loss when a network is congested. An aggregate maximum bit rate (AMBR) is specified on a per-subscriber basis for all non-GBR bearers.

Service Data Flows

Service Data Flows (SDFs) are an aggregate set of traffic flows that matches a set of filters. SDFs represent the IP packets related to a user service (web browsing, email, etc.). SDFs are bound to specific bearers based on policies defined by the Operator. This binding occurs at the PGW using traffic flow templates (TFT). TFTs contain packet filtering information used to identify and map packets to specific bearers. The filters are configurable by the Operator. The 5-tuple parameters include:

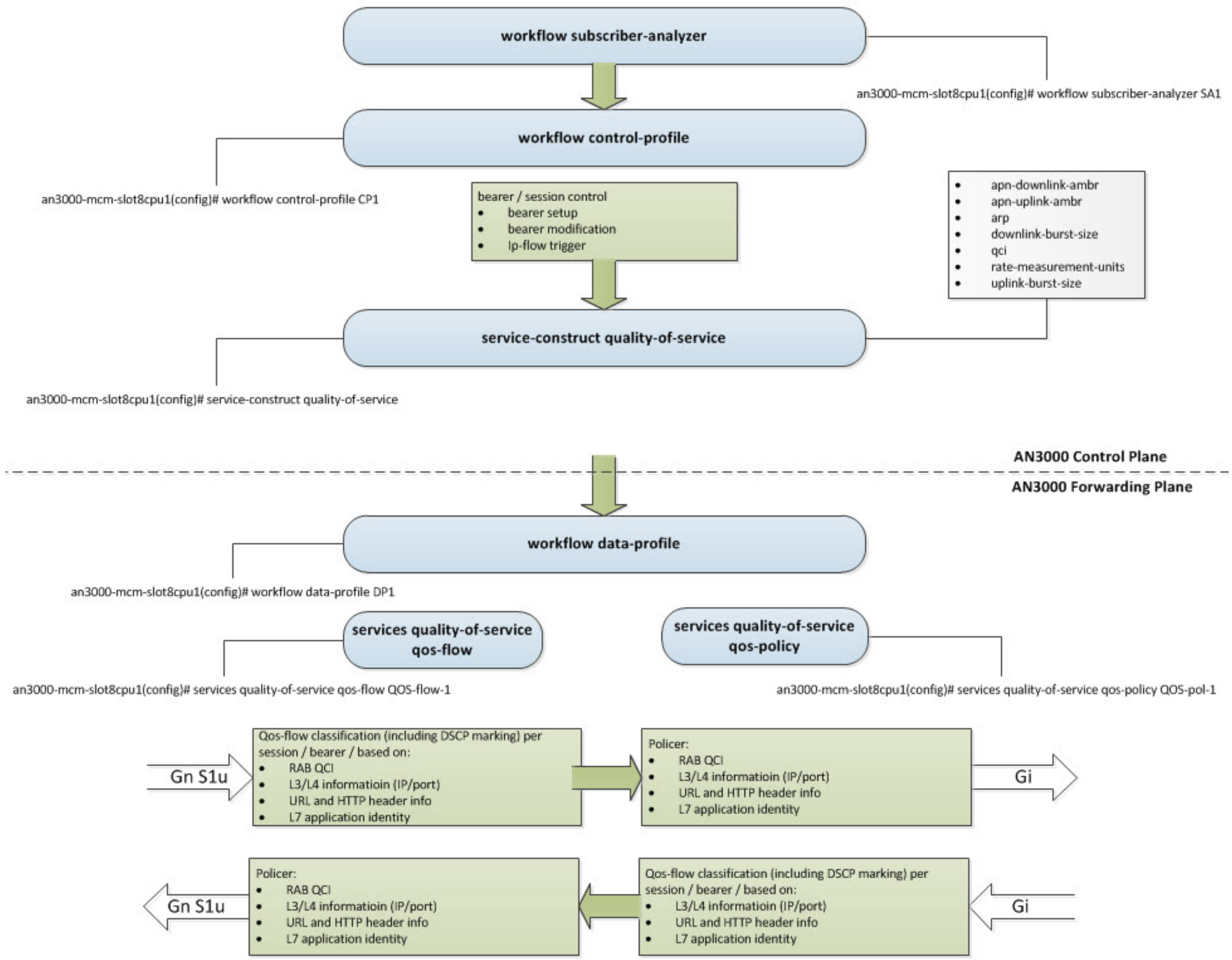
- Source IP address
- Destination IP address
- Source port number
- Destination port number
- Protocol identification (that is, TCP or UDP)

The policy and charging enforcement function (PCEF) in the PGW filters packets coming from external networks using TFTs.

Affirmed Networks QoS Configuration Model

This section describes the Affirmed Networks Quality of Service configuration model. Figure 9-2 shows the QoS workflow.

Figure 9-2 Quality of Service Workflow



Quality of Service

The workflow control-profile references a quality-of-service service-construct that provides the session Allocation and Retention Priority (ARP), uplink and downlink APN Aggregate Maximum Bit Rate (AMBR), uplink and downlink burst size in kilobytes, rate measurement units (kbps, mbps, gbps), and Quality of service Class Identifier (QCI) of the PDP Session. For example:

```
an3000-mcm-slot8cpu1(config)# service-construct quality-of-service QOS-1
Possible completions:
apn-downlink-ambr          Downlink APN Aggregate Maximum Bit Rate
apn-uplink-ambr            Uplink APN Aggregate Maximum Bit Rate
arp                        Allocation & Retention Priority
downlink-burst-size        Downlink burst size; measurement-units: KiloBytes
qci                        Quality-of-service Class Identifier
rate-measurement-units     rate-measurement-units: kbps/mbps/gbps
uplink-burst-size          Uplink burst size; measurement-units: KiloBytes
```

Policy Management

Operators can select the workflow data-profile from a local configuration (based on location, IMSI, and IMEI respectively) from the PCRF.

For QoS functions, the workflow data-profile references a qos-policy service that provides dedicated bearer control, Service Data Flow (SDF) rate enforcement, and Differentiated Services (DiffServ) Code Point (DSCP) marking. For example:

```
an3000-mcm-slot8cpu1(config)# services quality-of-service qos-policy QOS-T
Possible completions:
admin-state               admin state [Enabled/Disabled]
geo-redundant-group       list of Geographic Redundancy Groups that this
qos-policy is associated with
package-id                Identifies policy instance in AAA server
priority-evaluation-mode  when set to flat, qos-flow priority is ignored and
only service-rule priority is used
qci-dscp-map              Quality-of-service Class Identifier Diffserv Code
Point Map
qos-flow                  qos flows - selected from qos-flow list
```

The qos-policy references multiple qos-flow objects (selected from the qos-flow list). The qos-flow, in turn, references service rules that provide the classification filters. Each matching service rule causes the creation of an SDF.

Classification/DSCP Marking

When a packet for a specific subscriber session is received by the MCC, the packet is classified based on the qos-policy from the workflow data-profile. The DSCP value of the packet may be overwritten based on the qos-policy setting.

The DSCP value is obtained by looking up the bearer's QCI in the qos-policy table. Optionally, based on specific flow characteristics, a DSCP override value can be specified to be used instead of the value obtained through bearer QCI mapping. The qos-flow can be classified based on the following information:

- RAB QCI (access interfaces)
- L3/L4 information (IP/Port)
- URL and HTTP header info
- L7 application identity

Bearer Control

The workflow control-profile is responsible for the QoS parameters applied to a bearer/session while the session is created or updated. The workflow control-profile references a quality-of-service service-construct that provides the session AMBR and QCI/ARP of the PDP session.

The setup of dedicated bearers (GBR or non-GBR) is controlled by the QCI and ARP values configured on the qos-flow objects. Each unique pair of QCI/ARP implies a separate bearer. To suppress dedicated bearer creation, the QCI/ARP of the qos-flow must match that of the quality-of-service service-construct, therefore implying that the qos-flows belong to the default bearer.

The qos-flow object contains the enumerated parameter bearer-control with the following values:

default-bearer – Suppresses the QCI and ARP values.

network initiated – If the QCI/ARP is not the same as the default-bearer, the MCC attempts to create a dedicated bearer with the rates specified in the qos-flow.

user-initiated – The qos-flow becomes a filter of allowed dedicated bearers that the user may request. If the user requests a dedicated bearer that does not match the parameters in one of the qos-flows associated with that session, the MCC rejects the user's dedicated bearer request.

Note: *Multiple qos-flows can have the same QCI/ARP. In this case, the rates for the resulting dedicated bearer are the sum of the qos-flows. Rate enforcement is performed at the individual qos-flow level.*

Rate Enforcement

A rate enforcer (policer) categorizes packets into three colors:

Color	Indication
Green	Packet conforms to both the maximum and guaranteed bit rates
Yellow	Packet conforms only to the maximum bit rate
Red	Packet does not conform to the maximum bit rate and packets are dropped

When multiple policers are applied to the same packet, the MCC uses the worst result. For example, if one policer yields green and the other yellow, the packet is marked yellow.

Rate enforcement may be performed on a PDN session (PDP context), bearer (RAB), or Service-Data-Flow (SDF).

When traffic is received on a non-GBR bearer (including the default bearer), it is policed against the session AMBR. Additionally, it is policed against the matching qos-flow, if any.

When traffic is received on a dedicated GBR bearer on a PGW/GGSN, an optimization is made based on the observation that the dedicated GBR bearer rates are the sum of the qos-flows that caused it to be created. In this case, policing is performed at the qos-flow level, only.

service-construct radius-mapping template

Configure the radius-mapping template. Also see the *Affirmed Networks RADIUS Interface Specification*.

Configuring Objects

Configure the objects as follows:

Table 9-40 service-construct radius-mapping template

Object	Description
from	Item to store in outgoing attribute
name*	RADIUS attribute name.
to	Item to set from incoming attribute
to-format	Regular expression to select strings from incoming attribute
to-list	Item(s) to set from regular expression match groups

service-construct realm-list

A pre-configured list of REALM mappings on the WAG is used to steer the EAP request to the correct RADIUS server using the REALM in the NAI of the UE requesting authentication.

service-construct sctp-profile

Configure SCTP profile.

Configuring Objects

Configure the objects as follows:

Table 9-41 service-construct sctp-profile

Object	Description
cookie-life	Maximum lifespan of the Cookie sent in an INIT ACK chunk (unit 100 millisec)

Table 9-41 service-construct sctp-profile

Object	Description
heartbeat-interval	In msecs, this determines the periodic time interval at which the SCTP heart beat messages are sent. Value set to zero would disable SCTP heart beats.
init-retry	Maximum number of attempts to retransmit the init message after which the attempted association is terminated
max-in-streams	Maximum number of incoming SCTP streams.
max-out-streams	Number of outgoing SCTP streams.
name*	SCTP Profile Name - string
path-max-retransmit	Maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive.
path-mtu	Maximum transmission unit (MTU), in bytes, for SCTP streams. Value 0 enables auto-discovery of the path MTU.
rto-initial	In msecs, this defines the initial retransmission time out value. Subsequent SCTP RTO will be recomputed starting from the RTO initial value
rto-max	In msecs, this is the maximum RTO value and any subsequent recomputed RTO may never exceed this maximum value
rto-min	In msecs, this is the minimum RTO value and any subsequent recomputed RTO may never go below this minimum value
sack-period	Selective ACK delay in milliseconds.

service-construct service-rule

The service-rule is associated with a specific service-construct ca-certificate-list (page 9-7), service-construct domain-rule-group (page 9-20), service-construct http-rule-group (page 9-34), and/or service-construct time-of-day-rule-group (page 9-80). Service rules are referenced in other services, such as QoS Flow, Rating Group (charging), Subscriber Firewall Flow, Proxy HTTP IP Flow, and so on.

The identifier does not have to be unique. Operators can configure multiple service-rules with the same service data-flow ID so that multiple PCC rules can map to one service ID.

When generating CDRs, MCC creates a separate service container for each service ID so associating multiple PCC rules (service-rules) with the same service ID reduces the number of service containers created in each CDR.

For a service-rule to match, by default all of the following filter types (if configured) must match. You can specify that only one type of filter type match is needed by using the **evaluation-mode OR** parameter.

- At least one packet-filter or ipv4-filter-list
- At least one time-of-day-rule-group
- At least one http-rule-group
- At least one domain-rule-group
- At least one protocol-application-rule-group or at least one application, as follows:
  - If protocol(s) are specified, at least one must match. However, if the matching protocol has specified subprotocol(s), at least one subprotocol must match.
  - If application(s) are specified, at least one must match.

The service-rule supports the following separate categorization of traffic:

- TCP control traffic – defined as TCP packets with IP total length equal to the IP header plus the TCP header.
- TCP data traffic and non-TCP traffic

This feature requires service-rules to be paired to account for both types of traffic; one is the TCP Control Service Rule and the other is the TCP Data Control Service Rule.

Configuring Objects

Configure the objects as follows:

Table 9-42 service-construct service-rule

Object	Description
admin-state	Enabled/disable service rule option.
correlation-id	Optional non-unique numerical identifier for this service-rule.
domain-rule-group	Optional Domain name rules.
http-rule-group	Optional HTTP protocol rules.
evaluation-mode	Specifies how many of the configured classifier types must be matched for the service-rule to be considered a match: ■ AND – Each type of the configured classifiers must be matched. ■ OR – At least one type of the configured classifiers must be matched.

Table 9-42 service-construct service-rule

Object	Description
ip-filter-list	Name of an ipv4-filter-list, which contains one or more ranges of IP addresses.
include-service-data-flow-id-in-offline-sdc	When disabled, service data containers in offline CDR will be created omitting the service ID field.
install-default-bearer-packet-filters-on-ue	Enable/disable sending the TFT packet filters defined for a service rule to the UE.
name*	Name of this service-rule.
pcc-rule-name	Charging-rule-name from PCRF to support service-rule grouping.
priority	Priority
packet-filter	List of packet-filters selected from packet-filter list.
protocol-application-rule-group	Optional protocol and application rules.
service-activation	Service Activation for this Service Rule. When a billing plan is applied to a session, the MCC: ■ Allocates memory for all rating groups that have at least one service rule configured with the service-activation parameter set to always-on. ■ Does not allocate memory for rating groups that only have service rules configured with the service-activation parameter set to external-activation.
service-data-flow-id*	Mandatory numerical identifier for this service-rule.
session-rule-group	Optional Session name rules.
tcp-control-service-rule	The paired TCP control service-rule.
tcp-filter	Used to split TCP counting into one SDF for control and one for data.
tdf-application-id	Application ID to use in Application-Detection-Information AVP for Application Start/Stop notifications. See the *Affirmed Networks Gx Interface Specification* for details.
time-of-day-rule-group	Optional time-of-day rules.
tls-rule-group	Optional TLS rules.

service-construct session-rule-group rule

The session-rule-group is a list of rule groups that can be configured for RAT type, MBR matching, or both. Video pacing uses a service-rule to match a serving-policy. MCC uses the service-rule session-rule-group to match the session UE rat-type and downlink-mbr.

When HTTP proxy receives an HTTP response from the origin server, MCC adjusts the video-pacing attributes based on the UE rat-type or bandwidth. MCC supports video pacing adjustment according to the rat-type IE in the RADIUS accounting start message and the UE real-time throughput.

The session-rule-group device settings must match the service-construct device-database (page 9-17).

Table 9-43 service-construct session-rule-group rule

Object	Description
congestion-level	Congestion level for applying Optimizations. Options include: mild, moderate, and severe.
device-type	Match on device type from the device DB list. Options include: HANDHELD|FEATURE-PHONE|SMART-PHONE|MODEM|DONGLE|TABLET|VEHICLE|CONNECTED-DEVICE|ROUTER|MODULE|EBOOK|FEMTO-AP|
downlink-mbr	Specify a start and end value in kbps for the downlink maximum bit rate.
max-screen-height	Maximum screen height of the device
max-screen-width	Maximum screen width of the device
min-screen-height	Minimum screen height of the device
min-screen-width	Minimum screen width of the device
name*	Rule name
rat-type	RAT (Radio Technology) used by the mobile device. Options include: E-UTRAN GAN GERAN HSPA-EV NON-3GPP UTRAN VIRTUAL WLAN

service-construct sms

Configure Short Message Service service-construct parameters such as the sms text message, a group of Short Message Service Centers, and a Short Message Service Center profile. In addition, configure the smsc-group applicable to the instance.

A Short Message Service Center (SMSC) is a network element in the mobile telephone network used to deliver SMS messages. Configure the following SMSC parameters:

- SMS notification text message consisting of the cause, brand/market segment, and sms text.
- smsc-group – Configure a group of SMSCs and assign the group to a service rule and http-proxy instance.
- smsc-profile

SMSC Profile

When the LDAP query returns with a negative response or an error and the smsc-group is associated with the HTTP Proxy instance, the MMS message is blocked/not sent to the user. In addition, depending on the configuration, an SMS message can be sent to the user.

The SMS is sent to the user's mobile phone through an SMSC. The SMSCs are configurable through the smsc-profile service-construct. The SMSC configurable parameters are:

- Access username
- Access password
- IP address
- Port

Multiple SMSCs can be configured and grouped together in an SMSC group for redundancy. An SMSC group is then associated with the http-proxy service profile. In addition, SMS text messages can be customized and created in the service-construct. The configurable parameters for an SMS notification text message are:

- *Cause* – Rejection Cause of the MMS message
- *Brand* – Brand number of the UE
- *SMS text message* – Text message to be sent upon rejection of the MMS message. This message differs based on the rejection cause and the brand.

The appropriate SMS message is selected based on the rejection cause returned by the LDAP query and the brand of the UE. The proxy first tries to connect to the SMSC configured and associated with the http-proxy service profile. After a connection failure, other SMSCs (if present) are tried for connectivity. If the connection is successful, they are used for sending the current and subsequent SMS messages. After connecting to the SMSC, a bind transceiver request is sent to the SMSC to initiate a two-way connection with the SMSC. A bind transceiver response is sent from the SMSC corresponding to a success or a failure.

After the bind request succeeds, the SMS message is sent in a submit request PDU to the SMSC that receives it, which delivers it to the appropriate recipient and sends an acknowledgment (not of the delivery of the SMS to the user, but of the success of the submit message). The connection is then closed with an unbind request. For each SMS message sent, a new SMPP connection is established with the SMSC.

service-construct static-dns-map

Configure the static-dns-map. The MCC supports up to 128 instances of static-dns-map, and each static-dns-map supports up to 1024 host names in a fully qualified format. Each host supports up to five IPv4 IP addresses and five IPv6 IP addresses.

Configure a name for this dns-map and associate this static map with a network-context. Specify if the returned host address list should be static or round-robin.

Also see service-construct static-dns-map (page 9-77) to configure a static-dns-map for the http-proxy service.

service-construct subscriber-group

Configure the subscriber-group. This feature enables Operators to rate-limit the bandwidth for a group of subscribers, marked as heavy data users, to a configured aggregate-qos-policy. This ensures that the remaining users have enough bandwidth.

Configuring Objects

Configure the objects as follows:

Table 9-44 service-construct subscriber-group

Object	Description
file-name	Name of file that contains the list of subscriber IDs. The subscriber-group object references a text file containing a list of subscriber IDs (usernames). The subscriber ID text file is stored on the MCM in the */public/gateway/subscriber_group/import* directory. If you modify the Subscriber ID file, issue the following command for the update to take effect. `an3000-mcm-slot7cpu1#` **`service-construct subscriber-group <group-name> update`**
name*	Subscriber group name.
subscriber-id-type	Subscriber ID type (username).

Operators configure an aggregate-qos-policy and associate this qos-policy with the subscriber-group. When the aggregate bandwidth consumption of the subscriber-group exceeds the configured aggregate-qos-policy, MCC rate-limits the bandwidth for the subscriber-group. Also see services quality-of-service qos-policy (page 10-14).

service-construct tcp-optimization-profile

Configure TCP parameters for a profile. Enables Operators to configure the TCP initial receive window and initial congestion window in units of maximum segment size (mss). Operators can improve Web page load times by tuning these TCP parameters.

TCP proxy supports enabling TCP optimization for TCP-based protocols such as SFTP, FTP and HTTPS. The TCP Proxy stitches the client and server TCP connections, enabling Operators to configure the TCP parameters to be used for the client connections separately.

TCP optimization features are applied to data traffic passing through the http-proxy service. These features change the TCP/IP stack to improve the throughput and user experience in the mobile network. This feature enables Operators to configure throughput parameters such as congestion-control algorithm, mms, and send and receive buffer sizes. The changes apply to client TCP connections.

TCP optimization is supported for all TCP connections processed by the http-proxy service. These include HTTP and terminated or originated HTTPS connections. These also include spliced TCP connections for protocols such as SFTP, SCP, FTP and pass-through HTTPS. Different TCP optimization profiles can be applied to client and server TCP connections. Client connections, (both terminated and spliced), are optimized using the configured client's tcp-optimization profile. Similarly, server connections are optimized using the configured server-connections tcp-optimization-profile. The client and server tcp-optimization-profiles are configured on a per-HTTP proxy instance basis. In addition, terminated HTTP/HTTPS connections can be tcp-optimized using configured HTTP request policies.

The system uses the HTTP proxy to optimize mobile content delivery stored at the edge of the network for mobile devices and laptops without using a device-based client. The system provides a significant reduction of radio bandwidth consumption and reduced latency in the core network resulting in an improved user download experience. The system establishes a subscriber policy and billing profile and analyzes the subscriber session at each instance of a new content request. Regardless if Web or video content is cached, the system performs TCP optimization on the link to the mobile device and as a result improves both the user experience and reduces the radio network bandwidth utilization.

Configuring Objects

Configure the objects as follows:

Table 9-45 service-construct tcp-optimization-profile

Object	Description
congestion-control	Configure the TCP congestion control method: cubic|vegas|westwood|reno|bic|highspeed|hybla|htcp|veno|scalable|lp|yeah|illinois|
fast-retransmit-dupacks	The TCP/IP stack enables the Operator to configure the number of duplicate acknowledgement necessary to trigger fast-retransmit algorithm. This feature is tunable on a per-subscriber basis using the subscriber workflow.
initial-congestion-window	The TCP/IP stack allows the Operator to define the initial congestion window for each subscriber. This feature prevents the congestion control algorithm from activating prematurely for wireless links with higher latency.
initial-receive-window	Configure initial TCP receive window in units of mss.
nagle-algorithm	The TCP/IP stack allows the Operator to enable or disable the Nagle algorithm for each subscriber independently. Enabling this feature reduces the overhead in the wireless link and allows combining small packets at the source.
receive-mss	MSS for a connection is negotiated at the time of establishing a TCP session. The TCP/IP stack allows the Operator to set different MSS values for different users based on workflow configuration.
recv-buffer-auto-tuning	Allows the Operator to enable or disable auto-tuning of TCP window. This feature is tunable for each subscriber TCP flow.
recv-buffer-size	The system allows the Operator to define send and receive buffers for each TCP connection for each subscriber.
send-buffer-size	Configure the TCP send buffer size in bytes.
slow-start	The TCP/IP stack for HTTP Proxy provides the capability to enable or disable TCP behavior defined in RFC2861. If this parameter is not configured for a subscriber, the congestion window times out after an idle period, preventing slow start algorithm for persistent connections.
window-size	Configure TCP window size in bytes.

service-construct time-of-day-rule-group

Defines a layer 7 time-of-day-rule-group. This rule group contains time-of-day filters, each keyed by day, start-time, and end-time. If one of the time of day filters matches, then the time of day specification is matched.

service-construct tls-rule-group

Defines a tls-rule-group. The TLS rule group defines a set of rules for matching on SNI (Server Name Indication) and server certificate CN (Common Name) fields of a TLS handshake. When a match is found, specific actions or service(s) are applied, based on where the service-rule is used, such as QoS and charging policies for HTTPS traffic.

Within a rule, you can specify SNI or CN matching or both. If you specify both, each must match in order for the rule to be considered a match.

Both SNI and CN use the following matching criteria:

- begins-with (prefix)
- contains (partial)
- ends-with (suffix)
- exact match
- regular expression

Configuring Objects

Configure the objects as follows:

Table 9-46 service-construct tls-rule-group

Object	Description
server-common-name	Specify a Common Name string for matching against the incoming Common Name.
sni	Specify a Server Name Indication string for matching against the incoming SNI.
sni-url-list	Specify a url-list for matching against the incoming Server Name Indication (SNI). This SNI url-list is supported only for TCP proxy.
alpn	Specify a protocol to match with the negotiated protocol between client and server. Supports the reporting and classification of HTTPS traffic based on ALPN (Application Layer Protocol Negotiation). ALPN allows the application layer to negotiate the protocol used over a secure connection to avoid additional round trips. Incoming connections to tcp-splicing are classified based on the negotiated protocol and are reported through HTDRs and statistics.

service-construct tls-termination-policy

The tls-termination-policy for HTTPS. The tls-termination-policy service-constructs are used to configure policies for terminating Transport Layer Security (TLS) connections on behalf of Operator's or Operator Partners' servers. The policy is applied when the TLS connection's destination IP address matches the network-context and one of the configured server addresses in the tls-termination-policy.

The policy must also include the server's certificate and private key used to authenticate TLS connection on behalf of the server. Affirmed Networks recommends password protecting the private key. In this case, Operators must also configure a key pass phrase.

Server certificates and private keys configured in the tls-termination-policy must exist in the */public/content/tls/servercerts/* folder on the MCM. Use **`filecommand`** to install ca certificates into this folder.

The tls-termination-policy is used by Affirmed Network's applications including HTTPS.

Configuring Objects

Configure the objects as follows:

Table 9-47 service-construct tls-termination-policy

Object	Description
certificate	Configure TLS server certificate file name
default	
ip-address*	Configure IP address of TLS server.
key	Configure TLS server's certificate and private key used to authenticate the TLS connection on behalf of the server.
key-passphrase	Affirmed Networks recommends password protecting the private key. In this case, Operators must also configure a key-passphrase. MCC uses the key-passphrase to decrypt the file.
name*	Configure TLS server virtual name.
network-context	Configure network-context through which TLS server is reachable.

Certificates and Private Keys

Server certificates and private keys configured in the tls-termination-policy must exist in the */public/content/tls/servercerts/* folder on the MCM. Use fault-management clear to install ca certificates into this folder.

Certificates can refer to a file containing a single certificate or certificate chain. The certificate chain must use pem encoding. For example:

```
service-construct tls-termination-policy TLS-TERM-1
 site affirmed.local
   certificate cert_or_cert_chain.pem
```

service-construct ua-profile

Configure a user-agent profile.

Configuring Objects

Configure the objects as follows:

Table 9-48 service-construct ua-profile

Object	Description
name*	Enter the user-agent profile name.
uaProfileDefault	userAgentKey* – Configure the association between User-Agent-Default and UA-Profile file ua-profile-filename – Select a UA Profile that has already been uploaded. The file can be a single file or an archive file (with .tar, .tar.gz,.gz, or .gzip extension).
user-agent-key	Configure the association between User-Agent and UA-Profile file. ua-profile-filename – Select a UA Profile that has already been uploaded. The file can be a single file or an archive file (with .tar, .tar.gz,.gz, or .gzip extension).

service-construct url-list

Configure a list of URLs.

import-as-url-prefixes

Terminator “*” is added at the end of each line in the imported file if it doesn’t already exist. Specify whether imported URLs should be treated as URL prefixes.

import-from-local-file

The url-list database is built after an import action. Operators can upload the local file through Acuitas or using fault-management clear to scp the file to */public/content/policylists/url/userlists/* on the MCC. There is no scheduled automatic import for this mode. When the file on disk changes, you must execute the **`service-construct url-list <name> import`** command to rebuild the database for the new file to take effect.

The file contains a list of URLs in the following format. In the file, each URL is a string in the following format:

http[s]://<domain-name>[:port]/[<url-path>][*]

The port, url-path, and terminating /* are optional. If /* is used, the URL is treated as a URL prefix rather than the exact-match URL.

import-from-server

The url-list database is built after a scheduled or manual download completes.

Configure a rule for importing this list from a file server. The import-from-server object references a file to be imported (see service-construct diffserv-profile on page 9-20) from a file server and configures a rule for importing the file.

The file contains a list of URLs in the following format. In the file, each URL is a string in the following format:

http[s]://<domain-name>[:port]/[<url-path>][*]

The port, url-path, and terminating /* are optional. If /* is used, the URL is treated as a URL prefix rather than the exact-match URL.

This object is referred to by the service-construct http-rule-group (page 9-34) only and used as rules to match HTTP request URL.

Chapter 10

Services Parameters

This section describes the services objects and parameters supported on the MCC. For information on content filtering, http proxy, and http server objects and features, see the *Affirmed Networks Value Added Services Administration Guide*.

Access	config
Guidelines	None

services acl acl-networkcontext

Configure Access Control Policy and Network Context.

Configuring Objects

Configure an ACL filter list to apply and an ACL name.

services affirmed-apps fair-bandwidth-usage-service-profile

Configure Affirmed applications such as fair bandwidth usage parameters.

Configuring Objects

Configure the objects as follows:

Table 10-1 services affirmed-apps fair-bandwidth-usage-service-profile

Object	Description
measurement-units*	measurement units: kb/mb/gb
minor-qos-profile-id	configure minor quality of service profile
minor-volume-threshold	configure minor volume threshold
name*	Fair bandwidth usage service profile name
reset-interval	reset timeout interval, triggers fallback to default parameters

Fair Use Bandwidth Management

Fair Use Bandwidth Management is a category of services aimed at normalizing bandwidth use when monthly subscription levels are breached. Typically, when monthly volumes are breached, the user is rate limited at a throughput defined by the carrier until the end of the subscription period. At the beginning of a new subscription period, the service is restored to the subscribed throughput rate.

The system is designed to handle Fair Use Bandwidth Management principles deployed in European markets. Using the Fair Use Bandwidth Management policy, the system is capable of tracking the bandwidth use of an individual subscriber over a subscription period to determine when the monthly subscription bandwidth management limits are reached.The intent of this policy is to normalize the instantaneous bandwidth available to subscribers that have exceeded their monthly limit by instituting usage controls. Typically, the Operator-based controls are reduced to the subscriber’s real-time use of bandwidth to a parameter not to exceed the instantaneous throughput rate, when the subscriber exceeds monthly usage volumes.

The system is capable of establishing a subscriber billing profile to track the individual subscriber’s usage by obtaining monthly historical usage from the Operator’s PCRF system and combining that usage with the real-time usage created within the current subscriber session. The system interacts with the PCRF to obtain the subscriber’s Policy and Charging subscription characteristics. Using 4G LTE capabilities, the system manages usage subscription on an individual subscriber or a small group of subscribers via Policy and Charging functions.

services charging billing-plan

Defines the charging method to be applied to a list of rating groups. Multiple billing plans can be used to charge differently for home-area, roaming, preferential-roaming, and so on. Each billing-plan specifies interface information (to OCS and OFCS) and reporting parameters.

Select a rating group from the list. Select only service rules used by this billing plan's rating-groups.

Configuring Objects

Configure the objects as follows:

Table 10-2 services charging billing-plan

Object	Description
fraud-charging	Set fraud-charging to true to designate a particular rating-group for charging of fraudulent activity within this billing-plan (see fraud-detection-profile for more information).
priority-override	(optionally) override the service-rule's priority when used with this billing-plan.
weight	Add additional cumulative multiplier to the service-rule when used with this billing-plan.

services charging charging-instance

The charging-instance is made up of multiple billing-plans. Billing plan selection is based on charging characteristics.

Configuring Objects

Configure the objects as follows:

Table 10-3 services charging charging-instance

Object	Description
name*	Name for this charging instance.
priority-evaluation-mode	Set the priority-evaluation-mode. Options include: flat – Service flow priority is ignored and only service-rule priority is used. Any PCC rules take priority. flat-with-pcc-rule-precedence – Any PCC rules in effect have their precedence values compared with the service-rule priority values, and the PCC rule or service-rule with the highest priority (lowest numerical value) is selected. hierarchical – Any PCC rules take priority.
state	Enable/disable the admin state.
geo-redundant-group	List of Geographic Redundancy Groups to which this charging-instance is associated.
billing-plan-map	Name of billing plan and charging choice (charging-characteristic).
charging-choice	Charging choice charging characteristic.
charging-characteristic*	Charging characteristic - wildcard:0 range:1-65535

services charging rating-group

Defines the basis for charging the service data flows. It defines the mapping between usage units (octets, time, or event) and cost. Quota management can occur at the rating-group level or at a higher shared credit-pool level.

The number of rating-groups that can be configured for a single billing plan is based on the following settings:

- Up to 1024 rating-groups – If all the service-rules configured under those rating-groups have the service-activation field set to external-activation.
- Up to 64 rating-groups – If any service-rule configured under those rating-groups has the service-activation field set to always-on.

Note: *Each additional rating-group under a billing plan increases the memory requirement for a session by approximately 2K bytes. This feature was qualified with 256 rating-groups under a billing plan, on a CSM with 120G memory and 100K sessions, each with 30 active rating-groups.*

The number of rating-groups that can be associated with a metering service instance is 256 rating-groups.

Note: *This feature was qualified with 256 rating-groups configured under a metering instance with a maximum of 30 active usage-monitoring groups for a subscriber.*

Configuring Objects

Configure the objects as follows:

charging-method

The rating-group supports the following charging-method options:

- online (page 10-6)
- offline
- local
- nocharge (page 10-6)
- metering (page 10-6)
  - one-time-redirection (page 10-6)

- nometering
- afind

online

When the charging-method is configured for online, the following additional attributes can be configured:

- credit-authorization-event
- quota-black-list-timer
- credit-pool-id

nocharge

The rating-group can be configured with nocharge for prepaid or postpaid users. Any subscriber traffic that uses this rating group is either not charged or CDRs are not generated for this rating group. This feature is useful when an Operator wants to exclude certain flows such as DNS requests or portal requests used to charge subscribers.

metering

The rating-group can also be configured for metering services that operate in parallel with the charging service. The metering service instance contains a set of metering rating groups to convey the usage counts to the Gateway. Each metering rating group has a configurable threshold and an optional AOC landing page URL.

- *service-activation* – Specifies when to use or skip the service-flow for packet classification. This value is set to external-activation for metering rating-groups. Options include:
  - *always-on* – Always use this service flow for classification.
  - *external-activation* – Use the activation state of the service flow to determine whether to use or skip the service flow for packet classification. This setting allows the Gateway to dynamically activate/deactivate rating groups.
- *monitor-key* – Configure with the monitoring key sent by the PCRF.
- *monitor-key-string* – Monitor key for this metering group, in string format.

one-time-redirection

This feature adds the ability to redirect the first HTTP request of a session based on information received from the PCRF as part of the Charging-Rule-Name (CRN) AVP via the CCA-I, CCA-U (including rejoin), or RAR message. Receiving the CRN as part of the Charging-Rule-Definition AVP is also supported.

When the PGW receives the CRN AVP within the CCA or RAR message, the PGW performs a one-time HTTP redirection to the URL configured in the MCC as part of the rating group configuration. The CRN AVP references a predefined service rule configured under the metering rating group and any traffic matching this rule must be redirected to the URL configured in the advice-of-charge (AoC) which is associated with the metering rating group. The redirection is only valid for the first HTTP request of the session and is not valid for subsequent HTTP transactions. Non-HTTP traffic should pass as usual.

This feature adds a new CLI parameter used to enable one-time-redirection on a metering rating group. The default is **disabled**.

```
an3000-mcm-slot7cpu1(config)# services charging rating-group
<rating-group-name> one-time-redirection [enabled|disabled]
```

Note: *Note the following when configuring a metering rating group for one-time HTTP redirection:*

- The `one-time-redirection` parameter is only visible when the `charging-method` parameter for the rating group is set to `metering` and the `advice-of-charge` is configured for the rating group.
- The `service-activation` parameter for the rating group must be set to `external-activation`.
- The service rules associated with the metering rating group used for one-time HTTP redirection must be configured with a high priority to redirect the HTTP traffic.
- A one-time HTTP redirection service rule cannot be associated with multiple rating groups. However, the PGW can support multiple one-time-redirection metering rating groups each with its own set of service rules and AoC.
- The PGW will not remove the service rule that triggered the one-time HTTP redirection or install the same service rule again, if received from the PCRF, until it receives the Charging-Rule-Remove –> Charging-Rule-Name AVP from the PCRF.
- For example, the metering rating group configuration would be similar to the following:

```
an3000-mcm-slot7cpu1(config)# services charging rating-group
metering_rg charging-method metering one-time-redirection enabled
advice-of-charge one_time_aoc service-activation
external-activation service-rule One_time_http_rule
```

HTTPS Redirection for WAG Sessions

HTTPS redirection is supported for WAG sessions. Use the nometering group as an exception list for session-level metering. This feature enables Operators to configure rating group(s) with the charging-method set to **nometering** and add them to the metering instance in the workflow data-profile. Metering semantics are not applied to traffic matching these **nometering** rating groups. Operators can configure a service-rule/packet-filter to match the traffic destined to their Web portal and associate these service rule(s) with the **nometering** rating groups.

Operators can configure multiple service-rules to add a list of IP addresses to the rating group.

Reducing the Load on the Gy Interface

The MCC supports the following methods to help reduce the load on the Gy Interface when a subscriber runs out of credits.

Note: *Operators can use one or both of these methods to help reduce the load on the Gy Interface. If using both methods, Affirmed Networks recommends setting the quota-blacklist-timer to a value of at least 24 hours.*

- *quota-black-list-timer* – The configured quota-black-list-timer parameter specifies how frequently to send grant requests after termination or suspension of a rating-group.

  If a subscriber reaches the credit limit during a session, the MCC device activates the quota blacklist timer, configurable at the rating-group level. This timer prevents the MCC from sending additional quota requests to the OCS for a given rating group until the timer expires. When the timer expires, the MCC device again sends a grant request to the OCS. This timer ensures the MCC device only sends periodic quota requests to the OCS on behalf of this subscriber.

- *Smart recharge* – The MCC uses the re-authorization procedures defined in 3GPP TS 32.299 to support smart recharge.

  If a subscriber reaches the credit limit during a session, the MCC does not send periodic grant requests to the OCS. Instead, after the subscriber's credit limit is increased, the OCS initiates an unsolicited credit re-authorization (for the specific subscriber) by sending a Re-Auth-Request (RAR). Upon receiving this message, the MCC device responds with a Re-Auth-Answer (RAA) and activates the rating groups requested as part of the RAR and starts sending quota requests.

Note: *This method requires that both the OCS and PGW/GGSN support the re-authorization functionality.*

---

services crbn-profile

Configure the charging rule base name profile.

Configuring Objects

Table 10-4 services crbn-profile

Object	Description
charging-instance	Charging instance name.
content-filter	Bind content filter service instance

Table 10-4 services crbn-profile

Object	Description
metering	Metering instance name.
multi-protocol-proxy	HTTP Proxy name.
name*	Name for this crbn profile.
online-interface	Name of the online-charging interface. This overrides the online-charging interface configured for the workflow control profile.
qos-policy	QOS policy name.

services health-check

This feature uses a health-check mechanism to detect if a monitored service is down or overloaded. MCC then uses this information for traffic-handling decisions. Enable the admin-status to perform health checks of external systems. See service-construct health-check-profile (page 9-33).

services quality-of-service

Configure a quality of service policy used to define the QoS to be given to a list of QoS flows, each of which can be prioritized.

The MCC supports the following QoS functions:

- Allocation and Retention Priority (ARP)
- Uplink and downlink APN Aggregate Maximum Bit Rate (AMBR)
- Uplink and downlink burst size in kilobytes
- Rate measurement units (kbps, mbps, gbps)
- Quality of service Class Identifier (QCI) of the PDP Session
- Dedicated bearer control
- Service Data Flow (SDF) rate enforcement
- Differentiated Services (DiffServ) Code Point (DSCP) marking

The quality-of service object supports services quality-of-service qos-flow (page 10-10) and services quality-of-service qos-policy (page 10-14).

services quality-of-service qci-cos-map

This feature maps a QoS Class Identifier (QCI) to an IEEE 802.1Q VLAN Class of Service (CoS) priority for Layer 2 QoS. This configuration applies to all sessions on the cluster.

Configuring Objects

Configure the objects as follows:

Table 10-5 services quality-of-service qci-cos-map

Object	Description
qci-cos-map	QCI value that is mapped. The range is 1 through 255.
cos	802.1Q VLAN CoS value to which the QCI value is mapped. The range is 0 through 7.

services quality-of-service qos-flow

The qos-flow service allows for service data flow-level rate enforcement. Operators can link service rule(s) to the qos-flow to specify the actions to take on traffic that matches at least one of its service rules. For example, you can limit the amount of ICMP traffic to specified uplink and/or downlink rates. QCI/ARP values are compared to the default bearer to determine if a dedicated bearer is required.

Configuring Objects

Configure the objects as follows:

Table 10-6 services quality-of-service qos-flow

Object	Description
admin-state	Set the Admin state to enabled or disabled.

Table 10-6 services quality-of-service qos-flow

Object	Description
aggregate-qos-flow	Set to true or false (default). If you are associating this qos-flow with an aggregate-qos-policy (see Table 10-7 on page 10-15), set this parameter to **true**. The following guidelines apply to this feature: ■ A qos-flow with the aggregate-qos-flow set to **true** can have zero or more service-rules. ■ A data-profile can not be configured with a qos-service with the aggregate-qos-policy set to **true**. ■ Aggregate bandwidth is equally (static) distributed among all enabled SSMs. There is no central entity tracking the overall aggregate bandwidth across all SSMs. ■ A data-profile can be configured with a qos-service and the qos-service can have a mix of aggregate qos-flow(s) and/or non-aggregate qos-flow(s). ■ If the majority of the user application traffic is on fewer SSMs, then the overall per-MCC limit of aggregate bandwidth can not be achieved. ■ The same qos-flow can be configured in multiple qos-policies. The aggregate bandwidth limit is applied per qos-policy. See Bandwidth Sharing on Aggregate Levels (page 10-12).
arp	Allocation and retention priority: higher values are preferred.
bearer-control	This qos-flow is associated with network-initiated, user-initiated, or default bearer.
downlink-gbr	The downlink guaranteed bit rate.
downlink-guaranteed-burst	The downlink guaranteed burst measured in Kbytes.
downlink-max-burst	The downlink maximum burst measured in Kbytes.
downlink-mbr	The downlink maximum bit rate.
dscp	The optional flow-level override to bearer QCI/DSCP mapping.
gate	Set to **block** to block all traffic that does not match the packet filter. Options include: block, pass, pass-downstream, pass-upstream, and terminate-session. MCC supports a **gate** terminate session QoS action when a packet matches a static QoS rule or dynamic QoS rule. When this feature is set, the PCRF removes all the CRBN packages when the user is out of credit. When the MCC detects chargeable traffic matching the QoS flow, it terminates the session with cause code **Reactivation Requested** to the UE. ***Note:*** *This feature is supported only on the SAE-GW, standalone PGW, GGSN, GiGW, and WAG. This feature is not supported on the standalone SGW or ePDG.*
priority	Configure a priority.

Table 10-6 services quality-of-service qos-flow

Object	Description
rate-measurement-units	Configure the measurement units in kbps, mbps, or gbps.
service-activation	Service activation for this rating-group.
service-rule	The service rule selected from the service-rule list.
state	Enable or disable the admin state.
uplink-gbr	The uplink guaranteed bit rate.
uplink-guaranteed-burst	The uplink guaranteed burst measured in kbytes.
uplink-max-burst	The uplink maximum burst measured in kbytes.
uplink-mbr	The uplink maximum bit rate.

Bandwidth Sharing on Aggregate Levels

MCC supports bandwidth limiting per-application or a group of applications to a configured aggregate bandwidth across all subscribers associated with a data-profile. This feature provides the ability to shape traffic at the application or group of applications level among all subscribers on the node.

The configured aggregate bandwidth is equally distributed across all active SSMs. When new SSMs are added or existing SSMs are removed or set to admin disabled, the configured bandwidth is redistributed. This distribution is somewhat static and the fair distribution of bandwidth across all users and SSMs is not guaranteed.

- Scale-in impact: When new VMs of type SSM/WSMs are removed or admin-disabled, the aggregate bandwidth per qos-flow is recomputed based on the currently enabled SSM/WSMs. As a result, the per-SSM/WSM aggregate bandwidth is increased.
- Scale-out impact: When new VMs of type SSM/WSMs are added to the system (cluster), the aggregate bandwidth per qos-flow is recomputed based on the currently enabled SSM/WSMs. As a result, the per-SSM/WSM aggregate bandwidth is reduced.

When two qos policies have the same qos-flow in two different data-profiles, bandwidth is applied per data-profile and per qos-policy. When the same qos-policy is configured in two different data-profiles, subscribers of each data-profile receive the same aggregate bandwidth. Configured aggregate bandwidth is divided equally among all active SSM/WSMs.

Note: *The qos-policy configured in* *services rate-limiting-subscriber-group (page 10-16)* *must have the aggregate-qos-policy set to* ***true*** *and must have only one aggregate qos-flow with no service-rules.*

Deployment Use cases

The following use cases apply to this feature:

- A subscriber wants to restrict P2P application use between 1500 and 0000 hours. During these hours, P2P use is restricted to 32 Kbps per subscriber. In this scenario, a time-of-day rule-group, protocol-application-rule-group, and qos-policy (aggregate) is configured.
- An Operator applies bandwidth throttling for a group of subscribers for tethering service. Subscriber characteristics are sent from the PGW in a RADIUS message during bearer establishment. This triggers the MCC to send a Diameter message to the PCRF so that rules and the subscriber profile can be learned. A policy is created and used when the first uplink packet arrives from the customer belonging to this specific group. A qos-policy (aggregate) and protocol-application-rule-group is used.

Configuration Example

Use the following example to apply the rate-limit feature on all traffic for all subscribers matching the qos-policy configured in the data-profile and matching the aggregate qos-flows within that qos-policy.

```
workflow data-profile dp1
max-connections                 1024
force-packet-drop               false
qos-service                     QosPolicy
priority-evaluation-mode        hierarchical
dpi-pre-detection-max-pkts      0
dpi-pre-detection-max-bytes     0
first-packet-over-volume-quota allow
!

services quality-of-service qos-policy QosPolicy
admin-state              enabled
priority-evaluation-mode hierarchical
dscp-replacement-mode    both
aggregate-qos-policy     false
qos-flow AQF2
  fraud-qos false
!
!
services quality-of-service qos-flow AQF2
admin-state               enabled
rate-measurement-units    kbps
uplink-mbr                2000
downlink-mbr              100000
```

```
uplink-gbr                 0
downlink-gbr               100
uplink-max-burst           128
uplink-guaranteed-burst    128
downlink-max-burst         128
downlink-guaranteed-burst  128
gate                       pass
priority                   99
bearer-control             default-bearer
service-activation         always-on
aggregate-qos-flow         true
service-rule ASR1
!
!

service-construct service-rule ASR1
admin-state                                  enabled
priority                                     100
service-data-flow-id                         1
http-rule-group                              AHRG1
tcp-filter                                   all
service-activation                           always-on
install-default-bearer-packet-filters-on-ue disabled
!

service-construct service-rule ASR1
http-rule-group AHRG1
!

an3000-mcm-slot7cpu1# show running-config service-construct
http-rule-group
service-construct http-rule-group AHRG1
rule HttpRule
  host contains 10.53.1.120
!
```

services quality-of-service qos-policy

The qos-policy defines the QoS to be given to a list of QoS flows, each of which can be prioritized.

Configuring Objects

Configure the objects as follows:

Table 10-7 services quality-of-service qos-policy

Object	Description
admin-state	Enable or disable the admin state.
aggregate-qos-policy	Set the aggregate-qos-policy to true or false (default). Operators configure an aggregate-qos-policy and associate this qos-policy with the subscriber-group. When the aggregate bandwidth consumption of the subscriber-group exceeds the configured aggregate-qos-policy, MCC rate-limits the bandwidth for the subscriber-group. See service-construct subscriber-group (page 9-78). Also see Bandwidth Sharing on Aggregate Levels (page 10-12) and services rate-limiting-subscriber-group (page 10-16).
geo-redundant-group	A list of geographic redundancy groups to which this qos-policy is associated. Associate this qos-policy with a geographic-redundancy group.
package-id	Identifies the policy instance in the AAA server.
priority-evaluation-mode	Set the priority-evaluation-mode. Options include: flat – Service flow priority is ignored and only service-rule priority is used. Any PCC rules take priority. flat-with-pcc-rule-precedence – Any PCC rules in effect have their precedence values compared with the service-rule priority values, and the PCC rule or service-rule with the highest priority (lowest numerical value) is selected. hierarchical – Any PCC rules take priority.
qci-dscp-map	Configure a mapping of a QoS class identifier to a diffserv code point. qci-dscp-map – Enter the QCI value. dscp – Enter the DSCP value: af11|af12|af13|af21|af22|af23|af31|af32|af33|af41|af42|af43|be |cs6|cs7|dscp_01|dscp_02|dscp_03|dscp_04|dscp_05|dscp_06 |dscp_07|dscp_08|dscp_09|dscp_11|dscp_13|dscp_15|dscp_16 |dscp_17|dscp_19|dscp_21|dscp_23|dscp_24|dscp_25|dscp_27 |dscp_29|dscp_31|dscp_32|dscp_33|dscp_35|dscp_37|dscp_39 |dscp_40|dscp_41|dscp_42|dscp_43|dscp_44_va|dscp_45 |dscp_47|dscp_49|dscp_50|dscp_51|dscp_52|dscp_53|dscp_54 |dscp_55|dscp_57|dscp_58|dscp_59|dscp_60|dscp_61|dscp_62 |dscp_63|ef
qos-flow	The qos-flow selected from the qos flow list.
fraud-qos	Set to true to designate a particular qos-flow for handling of fraudulent activity within this qos-policy. See fraud-detection-profile.

services rate-limiting-subscriber-group

Associate the aggregate qos-policy (in Table 10-7 on page 10-15) with the rate-limiting-subscriber-group. Configure the subscriber-group name.

The qos-policy associated with the rate-limiting-subscriber-group must have the aggregate-qos-policy parameter set to **true**.

services smtp-proxy instance

Configure the SMTP proxy for handling SMTP traffic. This feature enables Operators to enable the smtp-proxy and create an instance name for each SMTP proxy.

Configuring Objects

Configure a network-context and loopback address to use for the SMTP proxy instance. Configure the objects as follows:

Table 10-8 services smtp-proxy instance

Object	Description
admin-state	Enable/Disable SMTP Proxy Instance
mms-mm3	MMS MM3 Instance associated with the SMTP Proxy Instance
name*	Content SMTP Proxy Instance Name - string
loopback-ip	Configure Loopback IP address
network-context	Configure network-context to be used for the server instance.
port	Configure Loopback IP port

services steering

Steering allows the normal routing of a packet to be overridden for upstream subscriber traffic based on the configured services steering steering-policy (page 10-19).

Steering allows upstream subscriber traffic to be force-routed to a specific next hop or into a specific tunnel. Steering is compatible with NAT, as an upstream packet can be source NAT'd and steered. Steering is subservient to the HTTP proxy and its dependent services, in other words, if the packet is to be proxied, steering is ignored.

Steering can be based on any of the inputs present on the service-rule:

- L3/4 packet-filter
- time-of-day-rule-group

- protocol-application-rule-group
- L7 http-rule-group

Note: *The steering service does not support steering into IPSEC tunnels.*

VRF Remapping to Support Bi-direction Properties for Steering

This feature allows a ue-pool to be anchored in a base network-context and shared among auxiliary network-contexts. A classification mechanism is also provided to direct appropriate traffic into an auxiliary network-context. To support this feature, the shared-network-contexts is added to the ue-pool. This object is a list of network-contexts that will share this ue-pool. For example:

```
network-context <name> ue-pool shared-network-contexts
```

A new action, nc-steering, is added to steering-flow. For example:

```
services steering steering-flow <name> action nc-steering
```

Use the following example to configure this service:

```
network-context Gi-Enterprise-D
security                    disabled
number-of-security-services 1
vrf-lightweight             false
tunnel-fragmentation-type   inner
block-gtpu-error-pkt        false
ue-pool poolOne
  allocation-mode                      local-allocation
  quarantine-timer                     0
  use-quarantined-ip-if-none-available false
  ue-pool-utilization-threshold        95
  shared-network-contexts Gi-Enterprise-M
ip-sub-pool ueSubPool1
  admin-state      enabled
  ip-sub-pool-type UE-IP-Pool
  ue-pool          poolOne
  range-start      171.16.200.1
  range-end        171.16.200.254
  route-summary-prefix-length 24
  advertise        yes
  route-tag        1
  suppressicmp     false
```

services steering steering-flow

The Layer 3 steering-flow allows for upstream subscriber traffic steering. You can link a service-rule(s) to the steering-flow to specify the actions to take on traffic that matches at least one of its service-rules.

Configuring Objects

Configure the objects as follows:

Table 10-9 services steering steering-flow

Object	Description
action	Action to take for this steering flow. no-steering – No steering for this flow. steering – Operators can specify the network-context and the name of the tunnel or next-hop on which upstream traffic will be sent. nc-steering – For network-context steering. See VRF Remapping to Support Bi-direction Properties for Steering (page 10-17).
admin-state	Configuration that enables or disables this steering flow
name*	Name for this steering flow.
network-context	Network context of the tunnel or next-hop on which upstream traffic will be sent
priority	Priority of this steering flow. The priority is used to resolve conflicts in the case of multiple steering-flows. The priority is hierarchical. In other words, the priority of the higher-level object takes precedence over the lower-level object.
tunnel	Name of the tunnel or next-hop on which upstream traffic will be sent

services steering steering-policy

The Layer 3 steering-policy. Configure the steering-policy name and enable/disable the steering-policy.

Configuring Objects

Enable or disable the steering-policy and enter a name for the steering-policy. Select the name of the steering-flows associated with the steering-policy. The steering-flows must already exist.

services subscriber-firewall

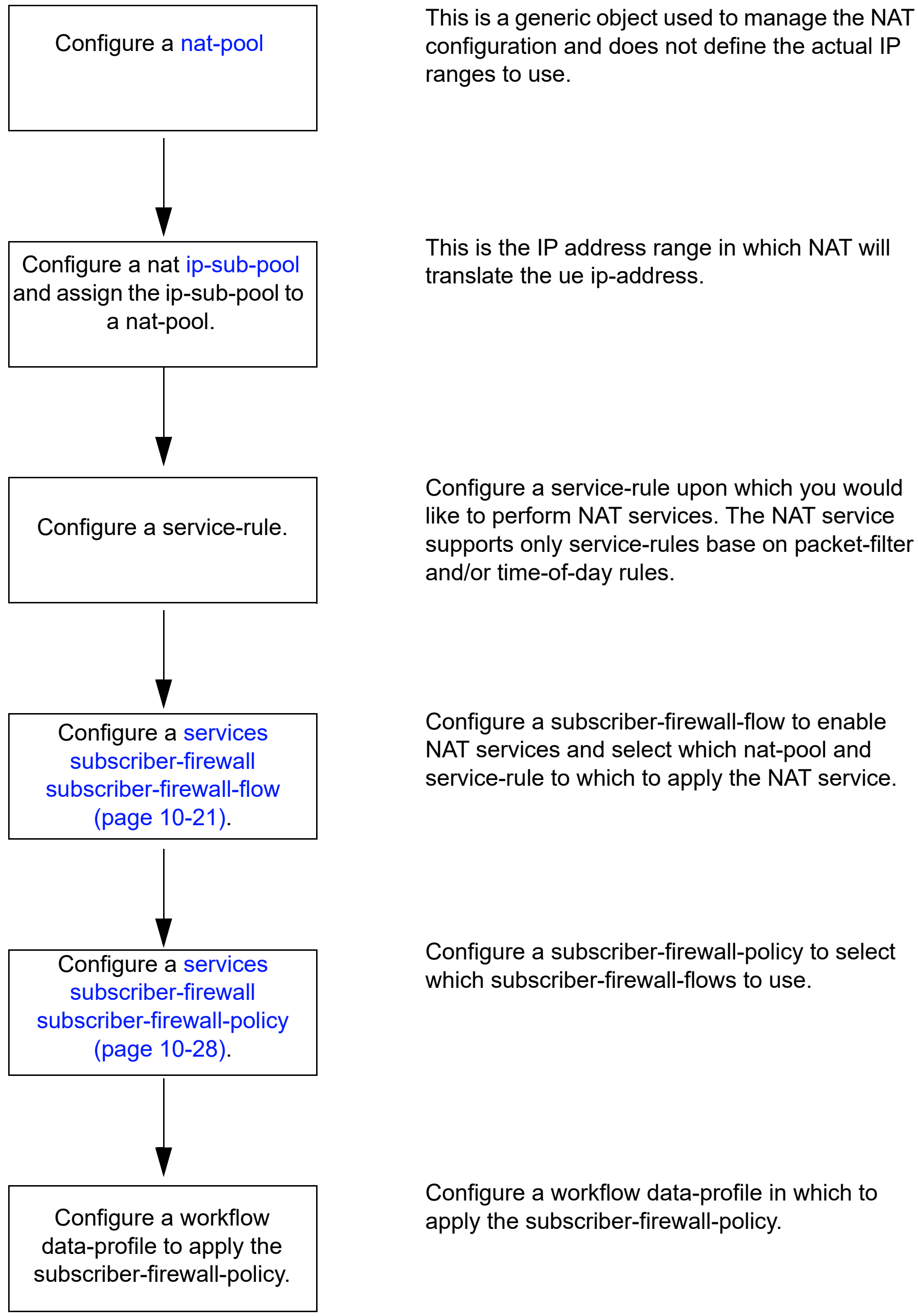
The subscriber-firewall is only applied to subscriber traffic and is configured to identify traffic that should either be dropped or allowed to pass.

Configure subscriber-firewall services. Affirmed Networks subscriber-firewall-flow services support NAT, PASS, and DROP.

In computer networking, *Network Address Translation* (NAT) is the process of modifying IP address information in IP packet headers while in transit across a traffic routing device. NAT is used to change a private UE address to an addresses that can be routed in the network. In addition, with IPv4 addresses at an exhaustion point, NAT allows for reuse of addresses in different realms. Most systems using NAT do so in order to enable multiple hosts on a private network to access the Internet using a single public IP address.

Figure 10-1 show an overview of the steps to enable subscriber firewall services using the CLI.

Figure 10-1 Enabling Subscriber Firewall Services



services subscriber-firewall subscriber-firewall-flow

The subscriber-firewall-flow binds one or more (prioritized) subscriber-firewall-flows and defines the NAT treatment to be applied when associated with a workflow data-profile (or service flow). The subscriber-firewall-flow allows differentiated Network Address and Port Translation treatment of multiple classified traffic flows. Network Address and Port Translation is used to map private UE IP addresses to a limited pool of public IP addresses. For example, certain traffic can be mapped to public addresses from one particular NAT pool, other traffic may be mapped to another NAT pool, and some traffic may not be mapped.

Configure a name for the service-rule associated with this subscriber-firewall flow. Configure a subscriber-firewall policy. Configure a name of a subscriber-firewall flow.

Configuring Objects

Configure the objects as follows:

Table 10-10 services subscriber-firewall subscriber-firewall-flow

Object	Description
action	Action to take for this subscriber-firewall-flow. Options include drop, nat, and pass. ***Note:*** *You must select* ***nat*** *if you intend to configure the Flow Authorization Interface (page 19-1).*
admin-state	Enable or disable this subscriber-firewall-flow.

Table 10-10 services subscriber-firewall subscriber-firewall-flow

Object	Description
alg	Configure firewall-flows with or without alg. See Application Layer Gateway (page 10-24) for a description of each option. Similar to NAT, the firewall has Application Layer Gateways (ALGs) to allow the firewall to work with the given protocol. For example, some protocols open dynamic flows after some negotiation. If a protocol firewall is configured, the ALG for that protocol opens the appropriate port mappings or "pin holes" in the firewall based on that protocol negotiation. ***Note:*** *The ICMP ALG is always enabled.* Options include: ■ ftp – File Transfer Protocol. The alg type of the corresponding subscriber-firewall flow must be configured as ftp. ■ ipsecnattassist – IPsec NAT assist. ■ None – No firewall flow or ALG. ■ pptp – Point-to-Point Tunneling Protocol. ■ rtsp – Real Time Streaming Protocol ■ sip – Session Initiated Protocol. *Note: Not supported for SIP with TLS encryption. This feature is not supported on a UE acting as a SIP server.* ■ tftp – Trivial File Transfer Protocol
flow-state-tracking-profile	Name of the monitoring profile that is used in the monitoring of different establishment states for flows accepted by the subscriber-firewall-flow. Use this object when flow-state-tracking-profile-selection is set to local. See Subscriber Firewall Monitoring of Flow Establishment States (page 10-26).
flow-state-tracking-profile-selection	Method of configuring monitoring of different establishment states for flows accepted by the subscriber-firewall-flow. See Subscriber Firewall Monitoring of Flow Establishment States (page 10-26). ■ global – Use a global monitoring profile. ■ local – Specify a particular monitoring profile. ■ none – (Default) Do not enable monitoring of flow states.
ipsec-natt-assist-nat-pool	Name of the NAT Pool associated with this subscriber-firewall-flow used to NAT IPsec traffic for UEs that do not support NAT-T.
name*	Name of this subscriber-firewall-flow.

Table 10-10 services subscriber-firewall subscriber-firewall-flow

Object	Description
nat-flow-binding-time	The amount of time, in seconds, that the NAT binding is persistent even when the connection for this NAT binding is triggered for closure due to inactivity (no data traffic in either direction). When the configured nat binding timeout is reached, the binding is removed from MCC and the translated IP address becomes available in the free pool. ***Note:*** *When a TCP RST is received, the NAT resource is released after a 30 second timeout.* This feature enables Operators to configure the filter modes at both the network-context nat-pool level and subscriber-firewall-flow level. The filter modes in subscriber-firewall-flow are given priority. If these modes are not configured then the MCC uses the filter modes configured in nat-pool. MCC keeps the address mapping binding (NATed IP) persistent for the nat-binding-timer period by keeping the connection object alive even after the data connection is deleted. This helps in reusing the same NAT'ed IP for the connections, if the connection returns within the nat-binding-time period. ***Note:*** *Setting this timer puts strain on resources.*
nat-flow-filter-mode	NAT filter type for this NAT flow. By default, the nat-filter configured on the nat-pool (see network-context nat-pool on page 8-18) is applied to this flow. If the nat-flow-filter-mode is configured for this flow, it overrides the nat-filter setting configured for the nat-pool. Therefore, set this nat-flow-filter-mode only if a different filtering mode is desired for this flow. ■ *address-dependent* – Filtering mode wherein downlink traffic from [A:*] on nat [ip:port] is accepted if UE has active uplink connection to [A:*] via [ip:port] ■ *address-port-dependent* – Filtering mode wherein downlink traffic from [A:a] on nat [ip:port] is accepted if UE has active uplink connection to [A:a] via [ip:port] ■ *endpoint-independent* – Filtering mode wherein downlink traffic from [*:*] on nat [ip:port] is accepted if UE has active uplink connection to any [*:*] via [ip:port]not-set – default. ■ *not-set (default)*
nat-pool	The name of a nat-pool associated with this subscriber-firewall-flow.
priority	The priority of this subscriber-firewall-flow.
service-rule	The name of a service-rule associated with this subscriber-firewall-flow.

Application Layer Gateway

This section describes the ftp, icmp, ipsecnattassist, pptp, rtsp, sip, and tftp application layer gateway options.

Table 10-11 on page 10-24 summarizes the support for NAT-ALG.

Table 10-11 NAT-ALG Support

NAT/ALG	RTSP[a]	PPTP	TFTP	ICMP	SIP[b]	FTP
44 1:1	Yes	Yes	Yes	Yes	Yes	Yes
44 N:1	Yes	Yes	Yes	Yes	Yes	Yes
66 1:1	No	No	Yes	Yes	No	No
64 1:1	No	No	No	No	Yes	Yes
64 N:1	No	No	No	No	Yes	Yes

[a] Control is supported for TCP and data is supported for UDP.

[b] Control and data are supported for UDP.

Table 10-12 on page 10-24 summarizes the ALG support for non-NATed traffic.

Table 10-12 ALG Support for Non-NATed Traffic

IP version	RTSP	TFTP	SIP	FTP
IPv4	Yes	Yes	Yes	Yes
IPv6	No	Yes	Yes	Yes

ftp

MCC supports the File Transfer Protocol (FTP) for NAT44 and NAT64 as follows:

- NAT44 – supported in many-to-one and one-to-one mode.
- NAT64 – supported in many-to-one and one-to-one mode.

MCC also supports the FTP ALG for non-NATed IPv4 and IPv6 traffic.

MCC monitors FTP commands and responses on the control connection for correct syntaxes. MCC translates the FTP commands and responses and the associated parameters if the FTP client and server operate with different IP versions.

When necessary, MCC rewrites the control packets with the appropriate NAT address and port information and opens corresponding pinholes to allow the establishment of data channel connections. FTP data connections are not proxied; instead, there is a direct connection between the FTP clients (UE) and servers.

If the FTP control session payload is modified, MCC updates the TCP sequence number, acknowledgment number, and checksum.

Note: *The alg type of the corresponding subscriber-firewall flow must be configured as ftp. This feature is not supported for UEs acting as an FTP server. MCC does not support the ALG command that allows FTP clients to query and set the ALG's status.*

---

icmp

The Internet Control Message Protocol (ICMP) is an error-reporting protocol that devices such as routers use to generate error messages to the source IP address when network issues prevent delivery of IP packets. MCC translates ICMP error packets with the appropriate NAT address and port information and updates ICMP error counters. The ICMP application layer gateway is always enabled for subscriber-firewall-flows.

ipsecnattassist

IPSECNATTASSIST is used to select a NAT resource based on whether the UE supports NAT traversal (NAT-T) when NAT is enabled for an end-to-end IPsec tunnel between the UE and VPN gateway/enterprise servers. The ipsecnattassist ALG options is only supported for NAT44.

- UEs supporting NAT-T – A NAT resource is allocated from the Many-to-One nat-pool.
- UEs not supporting NAT-T – A NAT resource is allocated from the one-to-one nat-pool. Use ipsecnattassist ALG when all VPN Gateways/Enterprise servers support NAT-T and a large percentage of UEs support NAT-T (with only a small percentage not supporting NAT-T).

pptp

The Point-to-Point Tunneling Protocol (PPTP) is an obsolete method for implementing virtual private networks. PPTP has many well-known security issues. PPTP is supported on NAT 44 addresses.

rtsp

Real Time Streaming Protocol (RTSP) is used to establish and control media sessions between endpoints. Use RTSP ALG when NAT is enabled for subscribers that use RTSP. The RTSP ALG is also supported for non-NATed IPv4 traffic. RTSP does not stream the media; it communicates with the server that streams the media. Most RTSP servers use RTP with RTCP to deliver streaming data. RTSP protocol messages describe how media streams are transported. When NAT is enabled, RTSP control messages contain private addresses. RTSP ALG understands RTSP protocol messages and allocates NAT bindings and opens pinholes to allow RTP/RTCP streams to connect through the firewall.

sip

Session Initiated Protocol (SIP) works with the TCP and UDP transport types and is supported on NAT44 and NAT64. The SIP ALG is also supported for non-NATed IPv4 and IPv6 traffic. SIP ALG maintains the state of SIP control (based on the transactions) flows based on the call IDs used as part of the SIP control session initiation and deletes pinholes and other allocated NAT resources if transaction failures occur on SIP control messages.

Note: *SIP ALG is not supported for SIP with TLS encryption. This feature is not supported on a UE acting as a SIP server.*

---

tftp

Trivial File Transfer Protocol (TFTP) is a simple protocol for transferring files and is implemented on top of the UDP/IP protocols using well-known port number 69. Use TFTP ALG when NAT is enabled for subscribers that use TFTP. The TFTP ALG is also supported for non-NATed IPv4 and IPv6 traffic. TFTP ALG processes TFTP packets that initiate requests and creates pinholes to allow return packets.

Subscriber Firewall Monitoring of Flow Establishment States

Overview

The MCC can monitor the various establishment states of TCP and UDP flows that are accepted by a subscriber firewall flow. Monitoring is controlled by separate timers for each establishment state.

The timers are used as follows:

- `tcp-open-timeout` — Specifies the timeout period between each message involved in the three-way TCP connection establishment. During this period, if the expected message is not received, the flow is removed. The `tcp-open-timeout` has a default of 30 seconds and a range of 10 through 30.

- **`tcp-established-timeout`** — Specifies the timeout period for no flow activity after the three-way handshake is complete. If no flow activity is detected during this period, the flow is removed. The **`tcp-established-timeout`** has a default of 300 seconds and a range of 10 through 2160000.

- **`tcp-close-timeout`** — Specifies the timeout period between each message involved in at the four-way TCP close operation, starting with receipt of the initial FIN from either direction. During this period, if the expected message is not received, the message is removed. The **`tcp-close-timeout`** has a default of 30 seconds and a range of 10 through 240.

- **`udp-open-timeout`** — Specifies the timeout period between receiving the initial UDP packet and receiving the response UDP packet. If the response UDP packet is not received during the timeout period, the flow is removed. The **`udp-open-timeout`** has a default of 30 seconds and a range of 1 through 30.

- **`udp-established-timeout`** — Specifies the timeout period for no flow activity after the response UDP is received. If no flow activity is detected during this period, the flow is removed. The **`udp-established-timeout`** has a default of 30 seconds and a range of 1 through 2160000.

- RST timeout — Non-configurable internal timer that removes a flow 30 seconds after receiving an RST.

Flow monitoring timers can be controlled by tracking profiles in two ways:

- Local — A subscriber-firewall-flow specifies a particular tracking profile.

- Global —A global tracking profile is used for any subscriber-firewall-flow that enables global control.

Configuring Subscriber Firewall Monitoring of Flow Establishment States Based on Local Tracking Profile

Follow these steps to configure monitoring of the flow establishment states in a subscriber-firewall-flow by specifying a particular tracking profile.

1. Enable local control for the tracking profile selection.

```
an3000-mcm-slot7cpu1(config)# services subscriber-firewall
subscriber-firewall-flow <name> flow-state-tracking-profile-selection
local
```

2. Configure the timeout values for the connection states in a flow-state-tracking-profile.

```
an3000-mcm-slot7cpu1(config)# flow-state-tracking-profile <name>
tcp-open-timeout <seconds> tcp-established-timeout <seconds>
tcp-close-timeout <seconds> udp-open-timeout <seconds>
udp-established-timeout <seconds>
```

3. Assign the flow-state-tracking-profile to the subscriber-firewall-flow.

```
an3000-mcm-slot7cpu1(config)# services subscriber-firewall
subscriber-firewall-flow <name> flow-state-tracking-profile <name>
```

Configuring Subscriber Firewall Monitoring of Flow Establishment States Based on Global Tracking Profile

Follow these steps to configure monitoring of the flow establishment states in a subscriber-firewall-flow by using a global tracking profile.

1 Enable global control for the tracking profile selection.

```
an3000-mcm-slot7cpu1(config)# services subscriber-firewall
subscriber-firewall-flow <name> flow-state-tracking-profile-selection
global
```

2 Configure a flow-timeout-filter. See service-construct flow-idle-timeout-filter (page 9-28).

3 Assign the flow-timeout filter to the flow-idle-timeout-rule that defines the timeout values.

```
an3000-mcm-slot7cpu1(config)# service-construct flow-idle-timeout-rule
<name> flow-idle-timeout-filter <name>
```

4 Configure the timeout values for the connection states in the flow-idle-timeout-rule.

```
an3000-mcm-slot7cpu1(config)# service-construct flow-idle-timeout-rule
<name> tcp-open-timeout <seconds> tcp-established-timeout <seconds>
tcp-close-timeout <seconds> udp-open-timeout <seconds>
udp-established-timeout <seconds>
```

5 Assign the flow-idle-timeout-rule to a flow-idle-timeout-profile.

```
an3000-mcm-slot7cpu1(config)# flow-idle-timeout-profile <name>
flow-idle-timeout-rule <name>
```

6 Assign the flow-idle-timeout-profile to the data profile.

```
an3000-mcm-slot7cpu1(config)# workflow data-profile <name>
flow-idle-timeout-profile <name>
```

All subscriber firewall flows that enable global monitoring use this flow-idle-timeout-profile.

services subscriber-firewall subscriber-firewall-policy

Binds one or more (prioritized) subscriber-firewall-flows and defines the NAT treatment to be applied when associated with a workflow data-profile (or service flow).

Configuring Objects

Configure the objects as follows:

Table 10-13 services subscriber-firewall subscriber-firewall-policy

Object	Description
admin-state	Enables or disables this subscriber-firewall-policy.
subscriber-firewall-flow	subscriber-firewall-policy.

Subscriber Firewall Rules

The rules defined in the subscriber-firewall-policy enable Operators to manage PDN traffic (by assigning traffic to specific NAT IP pools) based on various subscriber-level and flow characteristics, as follows:

- Force all subscriber/flows from a specific Operator to a range of NAT IP addresses.
- Use a specific NAT IP pool for traffic originating from an Operator or based on the destination.

services wap-gateway instance

Configure the WAP gateway service instance.

Configuring Objects

Configure the objects as follows:

Table 10-14 services wap-gateway instance

Object	Description
admin-state	Enable/Disable WTLS
gateway-certificate	Configure the gateway certificate file name
gateway-key	Configure the gateway private key file name

services wap-gateway wap-ip-flow

Configure the wap-ip-flows applicable for the instance.

Configuring Objects

Configure the objects as follows:

Table 10-15 services wap-gateway wap-ip-flow

Object	Description
wtls	Configure Wireless Transport Layer Security (WTLS) parameters such as the gateway certificate file name and the gateway private key file name.
wap-ip-flow	Configure the WAP ip-flow name and priority for the ip-flow.

Wireless Application Protocol

Wireless Application Protocol (WAP) is a technical standard that enables mobile device users to access information over a wireless network. A WAP browser is a web browser for mobile devices such as mobile phones that use WAP.

WAP is designed to work with most wireless networks such as CDPD, CDMA, GSM, PDC, PHS, TDMA, FLEX, ReFLEX, iDEN, TETRA, DECT, DataTAC, Mobitex and GRPS.

WAP is composed of the following layers:

- Wireless Application Environment (WAE)
- Wireless Session Protocol (WSP)
- Wireless Transaction Protocol (WTP)
- Wireless Transport Layer Security (WTLS)
- Wireless Datagram Protocol (WDP)

A WAP gateway (WAPGW) is similar to a proxy and acts like a bridge between mobile devices using the WAP protocol and the Internet. The WAP gateway translates pages into a form suitable for mobile devices.

The system contains a WAP gateway designed to provide WAP clients with access and protocol translation between WAP 1.x and HTTP/1.1. The WAP gateway provides support for packet network access for older WAP 1.x devices until these devices are upgraded or phased out of the network. The WAP gateway implements the WAP 1.x stack toward the client and HTTP stack toward the content server and provides the translation layer between the two interfaces. It supports WAP 1.0 and WAP 1.2 clients with and without WTLS support. The system supports the protocols defined in the WAP standards (OMA).

The WAP gateway passes client traffic to the HTTP proxy service. The system is then able to process content using services such as MMS adaptation, content filtering, and content caching.

The WAP gateway processes WAP requests and translates them to HTTP requests. HTTP responses are translated into WAP messages. Both WAP requests and responses are optimized for mobile networks by:

- binary encoding of headers
- caching of repeating headers and cookies
- using transport layer adaptive to high packet loss and congestion in access radio

WAP1.x Gateway

WAP Gateway is a service implemented in the Affirmed MCC. The service provides data access over WAP1.x protocols to mobile wireless clients.

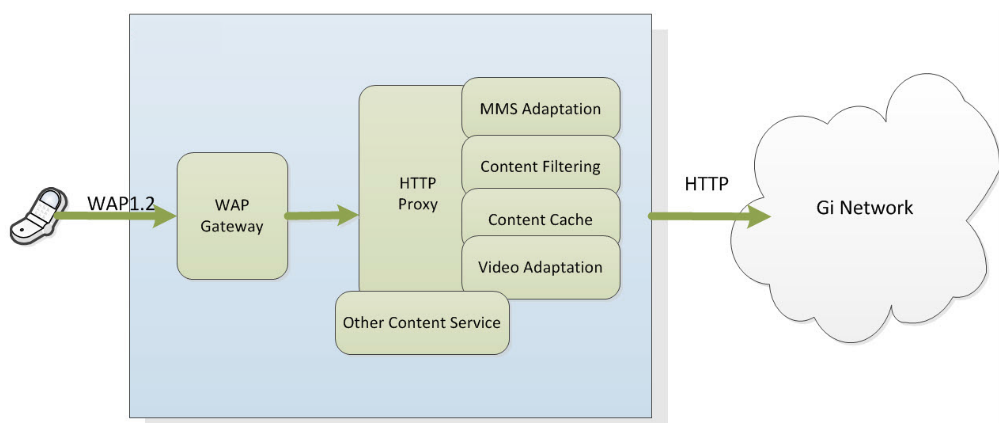
WAP-Gateway provides the following features:

- Efficient (saving wireless network bandwidth) encoding and decoding of messages sent over wireless network
- Protocol translation between WSP and HTTP/1.1
- Protocol translation between HTML content and WML
- Reliable message delivery
- Message segmentation and reassembly
- Connectionless (UDP port 9000) and connection-oriented (UDP port 9001) communication
- Secure communication

WAP Gateway works with the collocated HTTP Proxy to provide value-added services, such as MMS adaptation, content filtering, and content caching. When using the WAP Gateway, Operators configure the HTTP Proxy and optionally any other content services.

Figure 10-2 shows the WAP Gateway interfacing with HTTP (WAP2.0) Proxy.

Figure 10-2 WAP Gateway Interfacing with HTTP (WAP2.0) Proxy



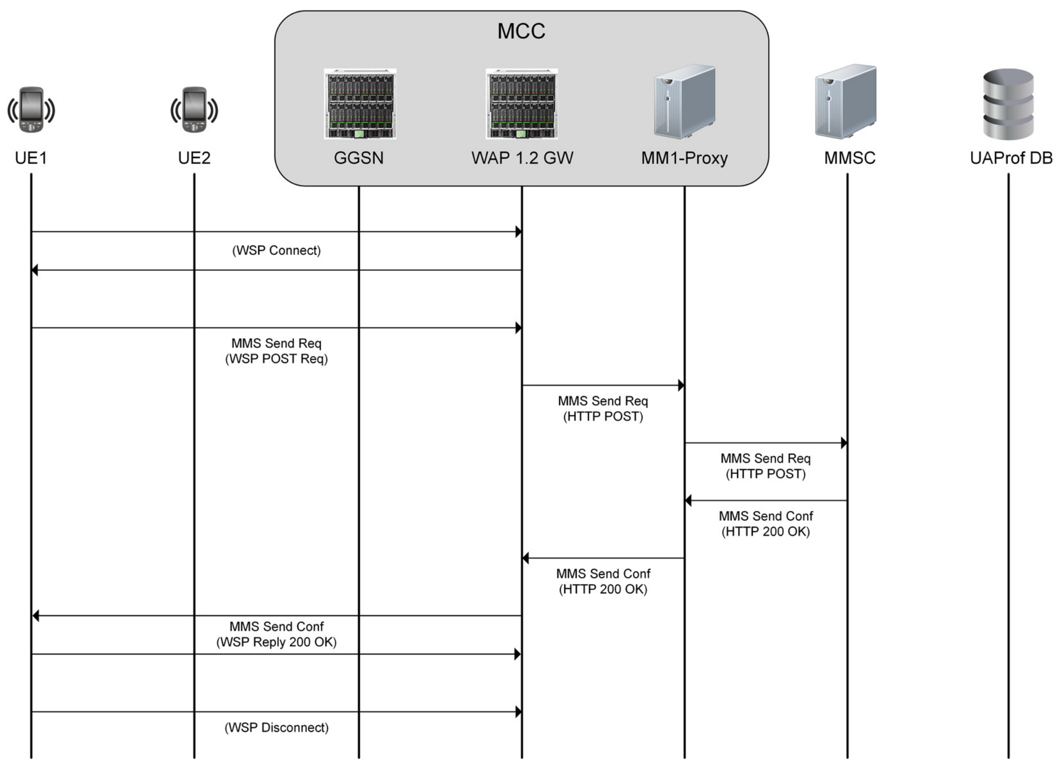
MMS MO Over WAP1.x

Figure 10-3 shows a typical flow for posting an MMS message by a mobile WAP1.x client. The message is posted via WAP1.x Gateway and MM1 proxy to the MMSC.

The sequence of events is as follows:

- The client first connects to WAP-Gateway using WSP Connect and WSP Connect Reply messages. When the connection is created, the connection capabilities (such as maximum message size) are negotiated. The HTTP headers and cookies common to all transactions made by the client are also passed on to the WAP-Gateway.
- The client sends an MMS m-send-req message encapsulated in WSP Post (and WTP Method Invoke) to WAP-Gateway. The WAP-Gateway, after successfully processing WSP Post, translates the message to the HTTP Post and forwards the message via MM1-Proxy to MMSC.
- The MMS m-send-conf message sent by MMSC via MM1-Proxy is received by WAP-Gateway. The HTTP headers of the message are converted to WSP equivalent and the MMSC response is sent to the client (delivered as WTP Result).
- If the WAP1.x client does not issue more transactions, it issues WSP Disconnect to terminate WAP1.x connection with WAP-Gateway.

Figure 10-3 MMS-MO Transaction Over WAP1.2



MMS MT Over WAP1.x

Figure 10-4 shows the typical flow of retrieval of the MMS message by WAP1.x client. The message is retrieved via WAP1.x Gateway and MM1 proxy from the MMSC.

The sequence of events is as follows:

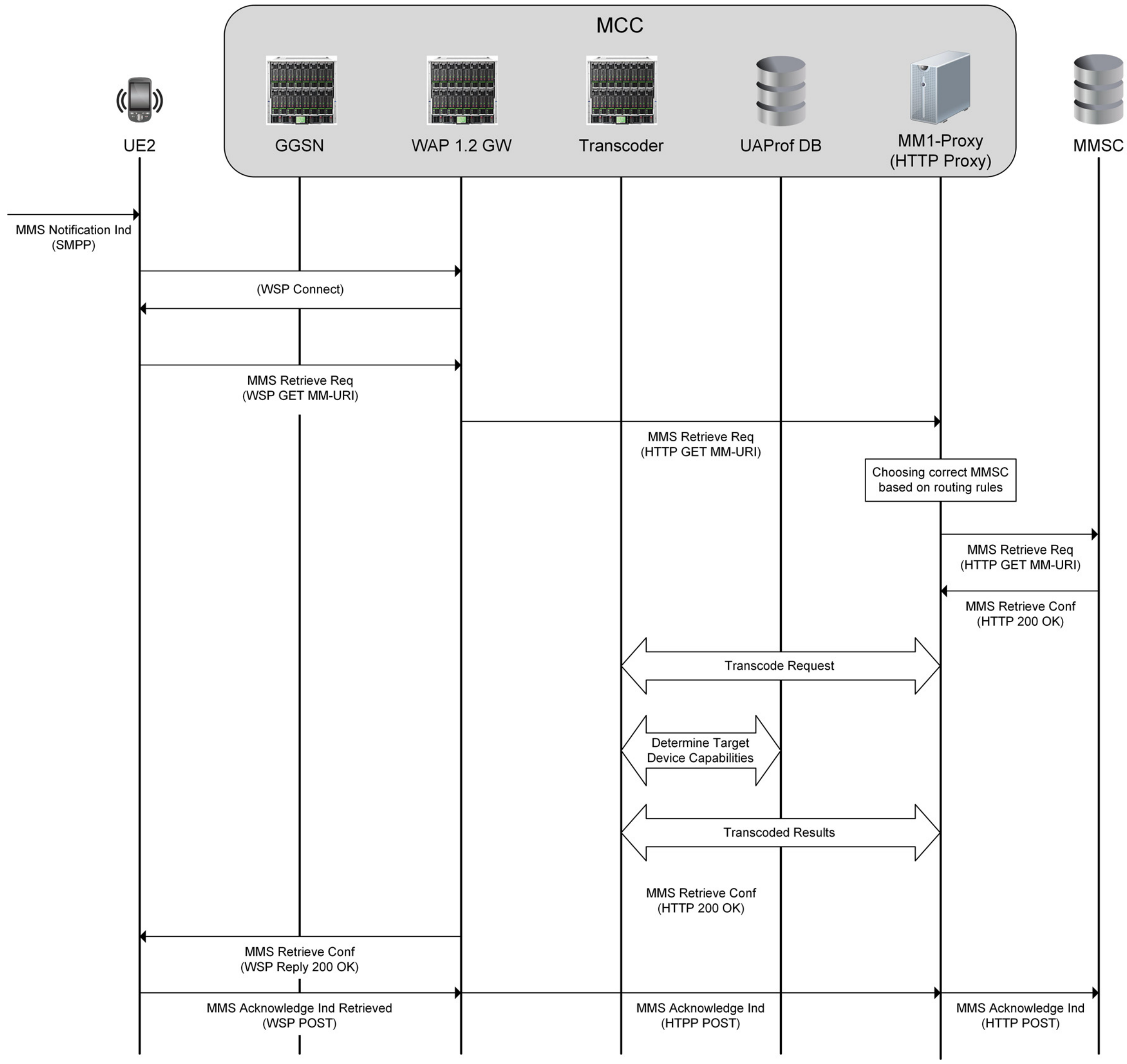
- The client is notified about a pending MMS message with an MMS Notification Indication message sent over SMPP protocol.
- If the client does not have an existing connection to the WAP-Gateway, it creates a new connection using a WSP Connect and WSP Connect Reply message exchange. When the connection is created, the connection capabilities such as maximum message size are negotiated. The HTTP headers and cookies common to all transactions made by the client are also passed on to the WAP-Gateway.

Note: *Due to WTP protocol limitations with standard segmentation and reassembly functions, the maximum message size supported by WAP-Gateway is 300 KB.*

- The client sends a WSP Get (over WTP Method Invoke) to the WAP-Gateway. The WAP-Gateway, after successfully processing the WSP Get, translates the message to an HTTP Get and forwards the message via the MM1-Proxy to the MMSC.
- The MMS m-retrieve-conf message sent by MMSC via the MM1-Proxy is received by the WAP-Gateway. The HTTP headers of the message are converted to a WSP equivalent and the MMSC response is sent to the client (delivered as WTP Result).
- The client posts an MMS m-acknowledge-ind (with WSP Post via WAP-Gateway, MM1-Proxy) to the MMSC to acknowledge successful retrieval of the MMS message.

If the WAP1.x client does not issue more transactions, it issues a WSP Disconnect to terminate the WAP1.x connection with the WAP-Gateway.

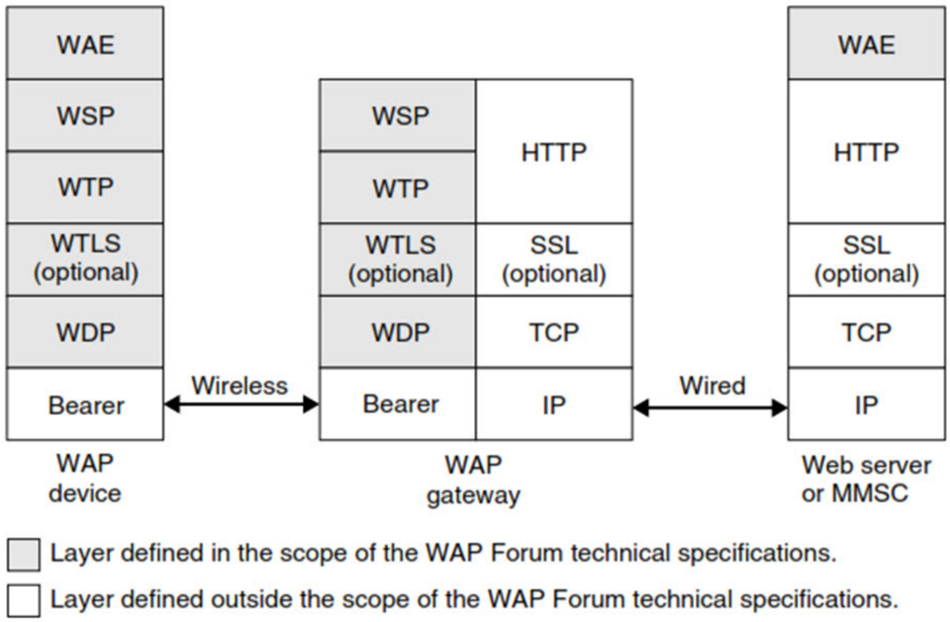
Figure 10-4 MMS-MT Call Flow Over WAP1.x



WAP 1.x Protocol Stack

Figure 10-5 shows the protocol stack of the configuration defined in the WAP specification suite 1.x. This configuration is also supported by the WAP specification suite 2.0 in addition to other configurations. In this configuration, the WAP device communicates with a remote server via an intermediary WAP gateway. The primary function of the WAP gateway is to optimize the transport of content between the remote server and the WAP device. For this purpose, the content delivered by the remote server is converted into a compact binary form by the WAP gateway prior to being transferred over the wireless link. The WAP gateway converts commands conveyed between datagram-based protocols (WSP, WTP, WTLS, and WDP) and protocols commonly used on the Internet (HTTP, SSL, and TCP).

Figure 10-5 WAP 1.x Protocol Stack



The *Wireless Application Environment (WAE)* is a general-purpose application environment in which Operators and service providers can build applications (for example, MMS clients) for a wide variety of wireless platforms.

The *Wireless Session Protocol (WSP)* provides features also available in HTTP (requests and corresponding responses). Additionally, WSP supports long-lived sessions and the possibility to suspend and resume previously established sessions. The WSP requests and corresponding responses are encoded in a binary form for transport efficiency.

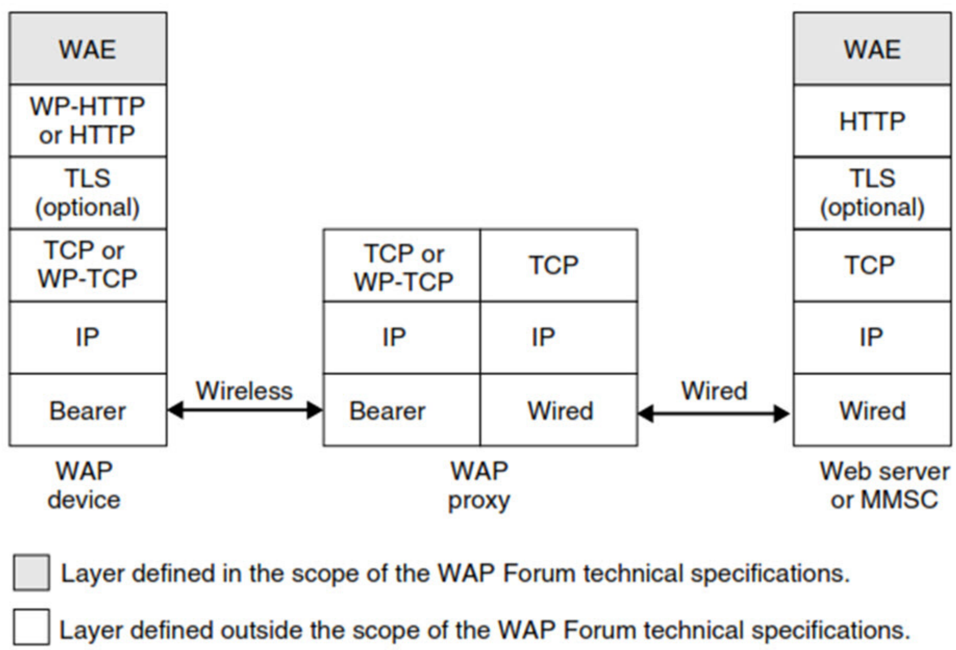
The *Wireless Transaction Protocol (WTP)* is a lightweight transaction-oriented protocol. The WTP improves the reliability over underlying datagram services by ensuring the acknowledgment and re-transmission of data grams. The WTP has no explicit connection set-up or connection release. WTP is a message-oriented protocol and therefore is appropriate in the implementation of mobile services such as browsing. Optionally, Segmentation And Reassembly (SAR) of packets composing a WTP protocol unit is also supported.

The optional *Wireless Transport Layer Security (WTLS)* provides privacy, data integrity, and authentication between applications communicating with the WAP technology. This includes the support of a secure transport service. The WTLS provides operations for the establishment and the release of secure connections. The *Wireless Data Protocol (WDP)* is a general datagram service based on underlying low-level bearers. The WDP offers a level of service equivalent to those offered by the Internet's *User Datagram Protocol (UDP)*. At the bearer level, the connection may be a circuit-switched connection (as found in GSM networks) or a packet-switched connection (as found in GPRS and UMTS networks). Alternatively, the transport of data at the bearer level may be performed over the SMS or over the Cell Broadcast Service.

WAP HTTP Proxy with Wireless Profiled TCP and HTTP

Figure 10-6 shows a configuration in which the WAP device communicates with web servers via an intermediary WAP proxy. The primary role of the proxy is to optimize the transport of content between the fixed Internet and the mobile network. The proxy also acts as a *Domain Name System (DNS)* for mobile devices. In this configuration, Internet protocols are preferred versus legacy WAP protocols to support IP-based protocols in an end-to-end configuration, from the Web server back to the WAP device. Figure 10-6 shows the protocol stack of this configuration as defined in the WAP specification suite 2.0.

Figure 10-6 WAP 2.0 Protocol Stack



The *Wireless Profiled HTTP (WP-HTTP)* is an HTTP profile specifically designed to cope with the limitations of wireless environments. The WP-HTTP is fully inter-operable with HTTP/1.1 and supports message compression.

The optional *Transport Layer Security (TLS)* ensures the secure transfer of content for WAP devices involved in the exchange of confidential information. The *Wireless Profiled TCP (WP-TCP)* offers a connection-oriented service. It is adapted to the limitations of wireless environments but remains inter-operable with existing *Transmission Control Protocol (TCP)* implementations.

WTLS Key Exchange

The Affirmed WAP-Gateway supports RSA anonymous (RSA_anon) mode for a key exchange suite. It is an RSA key exchange without authentication. The gateway sends its RSA public key. The client generates a secret value, encrypts it with the gateway's public key, and sends it to the gateway. The pre-master secret is the secret value appended with the server's public key.

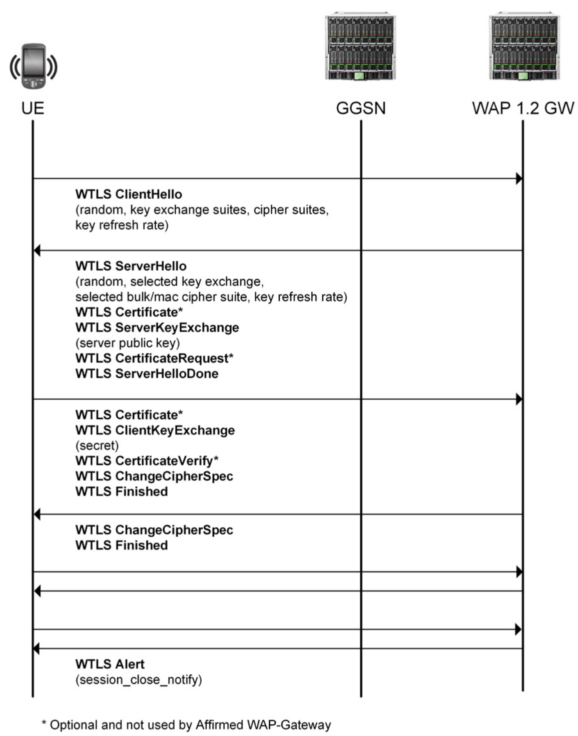
When WTLS is used, the Affirmed WAP-Gateway is configured with X.509 certificate and private key.

The Affirmed WAP-gateway supports a maximum key size of 2048 bits.

Other key exchange algorithms listed by the WTLS specification *are not* supported by the Affirmed WAP-gateway: NULL, shared-secret, Diffie-Hellman: DH_anon (with key sizes 512, 768, 2048), RSA_anon (with key sizes 512, 768), RSA (with key sizes 512, 768, 2048), Elliptic Curve DH: ECDH_anon (with key sizes 113, 131, 2048), ECDH-ECDSA (with key size 2048).

To ensure that keys are not compromised, the keys are periodically refreshed. The key refresh rate is negotiated between client and server when establishing a secure session. Figure 10-7 shows the WTLS message flows.

Figure 10-7 WTLS Message Flows



WTLS Authentication

The WAP1.x WTLS client can optionally authenticate WAP-Gateway using X.509 certificate. This procedure is used for non-anonymous secure session establishment.

The WAP-Gateway returns the WTLS server certificate in the initial handshake response and the certificate is validated by client.

If a WAP-Gateway needs to authenticate WAP1.x client, it sends CertificateRequest in the initial handshake response. The client responds with its Certificate and the CertificateVerify message. The gateway completes authentication by validating client certificate and returned signature in the client's CertificateVerify message. The Affirmed WAP-Gateway does not currently use this procedure, but instead uses the anonymous key exchange suite.

Confidentiality

Confidentiality is supported by encrypting transmitted user data. Received encrypted data from client is decrypted by WTLS layer before passing it up the stack to WTP, WSP layers.

The Affirmed WAP-Gateway supports the following encryption algorithms:

- DES-CBC (56)
- RC5-CBC (40, 56, 128)

The following algorithms listed by WTLS specification *are not* supported by the Affirmed WAP-Gateway:

- NULL
- DES-CBC (40)
- 3DES
- IDEA
- RC5-CBC (64)
- IDEA (not supported due to patent issues)

WTLS Integrity

WTLS supports message integrity through the means of stamping each transmitted packet with a Message Authentication Code (MAC) - hash.

MAC on received message is validated and discarded in the event that MAC is incorrect.

The Affirmed WAP-Gateway supports the following hashing algorithms:

- SHA1 (40, 80, 160)
- MD5 (40, 80, 128)

The SHA1-XOR-40 algorithm listed by the WTLS specification *is not* supported by the Affirmed WAP-Gateway.

Wireless Transport Layer Security

The Affirmed WAP-Gateway supports the Wireless Transport Layer Security (WTLS) protocol. The WTLS protocol is derived from a popular and commonly deployed Transport Layer Security (TLS) protocol. Unlike TLS, which runs over connection-oriented transport, WTLS is adapted to datagram oriented transport (UDP). WTLS also provides for more efficient message encoding.

Similar to non-secure WAP1.x modes, WTLS operates in the following modes:

- Secure connectionless mode (UDP port 9202)
- Secure connection-oriented mode (UDP port 9203).

WTLS supports the following main features:

- Cryptographic algorithm negotiation
- Key exchange
- WAP-gateway, and optionally, client authentication
- Confidentiality (packet encryption)
- Integrity (packet signatures – hashes)
- Compression (not supported by Affirmed WAP-Gateway).

Chapter 11

Workflow Parameters

This section describes the workflow objects and parameters supported on the MCC. Workflow is an application running on the system that defines the treatment of a service flow across various services residing on the system. Workflow allows the Operator to customize a subscriber's packet treatment across various services. Workflow binds a single or set of subscribers to a workflow data profile and/or a workflow control profile. The workflow specifies what event triggers this binding (bearer setup). All CDRs, external communication, and statistics collection occur as if it's on the mapped-APN if configured.

> ***Note:*** *See Acuitas Performance Statistics for details on statistics.*

Access	config
Guidelines	None

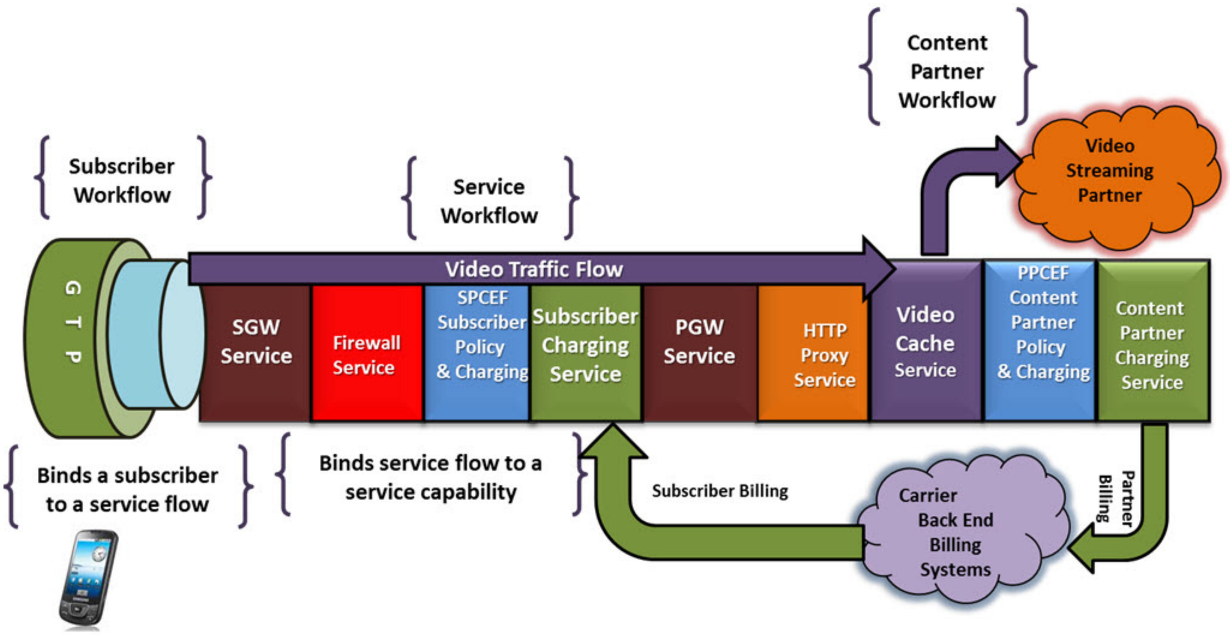
Workflow enables Operators to customize the use of current services (Subscriber Management Gateways – SGW, PGW, GGSN, online and offline charging, accounting, policy, QoS, NAT, HTTP proxy, content filtering, TCP optimization, and caching) and add new services as they become available. Using workflow, Operators can customize each subscriber's packet treatment across various services on a per-flow basis.

As shown in Figure 11-1, workflow simplifies the creation of services in the Operator's network and provides the following carrier benefits:

- Reduces time to market for new services from service conception to service delivery, leading to rapid revenue generation for new services.
- Reduces capital expenditures by reducing network integration expenses.
- Reduces operating expenses by minimizing the need for professional services.
- Allows flexibility in service creation and development at Layers 4 through 7.

- Improves service-level monitoring by creating new service-monitoring processes.
- Simplifies device parameter and programming.
- Supports documented service flows and flow interactions.
- Adapts to existing services using workflow.

Figure 11-1 Workflow



Operators can use workflow to define combinations of treatment for user flows for the following functions, that together form the basis of a rich and dynamic, Operator-defined service:

Note: *Affirmed currently supports the following functions. New functions are added to the same framework as they are supported. For the latest available features and functions, see the Release Notes.*

- MME
- PGW
- SGW
- GGSN
- QoS

- Charging (online, offline or hybrid online / offline)
- Content filtering
- HTTP proxy
- TCP / media delivery optimization
- Traffic steering

Workflow Framework

Workflow uses the subscriber analyzer to define the set of rules to select the subscriber sessions to apply a specific services policy. The subscriber analyzer uses the set of rules that can match signaling attributes for a subscriber session, such as IMSI, IMEI, UE device type, time of day, APN, SGSN, or MME IP address.

Each subscriber analyzer is tied to workflow control profile that defines which external policy databases to consult (including internal or external policy manager) to control the subscriber policy and interpret results. The workflow control profile provides Operators with a powerful mechanism to manage subscriber policy for each individual subscriber using a combination of RADIUS or DIAMETER Authentication Servers, PCRF Servers, LDAP Servers (SPR), or Internal Policy Manager.

This approach provides Operators with the flexibility to control the exact mechanism used to manage the subscriber policy. For example, one workflow control profile may use a RADIUS authentication server to retrieve the name of the subscriber workflow policy, while another workflow control profile may use an external PCRF server, and yet another workflow control profile may look up the subscriber workflow policy from a locally defined database.

The Operator controls the subscriber analyzer filtering rules, such as IMSI and APN, and associated workflow control profiles through workflow configuration objects. In one scenario, the Operator may configure the subscriber policy to be driven entirely by an external PCRF selected using the workflow control profile. In the other scenario, the Operator may provision the system with the internal Policy Manager, or static workflow policies in the absence of a PCRF. The Operator can also choose a hybrid of the above two approaches, thus providing the Operator complete flexibility in defining and using the subscriber policies for managing different service behaviors.

The combination of subscriber analyzer and workflow control profile defines the service function rules for a particular subscriber in the workflow data-profile. The workflow data-profile defines a collection of rules for each service function to be examined and executed on a per-flow basis. Each rule defined in the workflow data-profile can either be pushed to the system by the policy server (internal or external) or predefined. Each rule can be defined based on per-flow filters such as L3/4 Packet Filter – 5 Tuples, L7 filters – HTTP URL / URIs, application being used, time of day, geographic location, etc. In addition, different rules may be effective for different functions for the same subscriber. For example, a charging function may be based on time of day while QoS may be based on a combination of HTTP URL and geographic location.

Workflow has been designed to define services that are both subscriber-aware and content-aware. In addition to providing services that are based on the subscriber parameters described above, services can also be based on content provider policies. For example, a service may not be charged (zero-rated) to a subscriber, but rather charged to the content provider for delivery of the service / content. For example, the charging service may be configured to charge for all HTTP traffic, while the traffic steering service may be configured to steer all uplink and downlink video traffic for a particular content site to either an internal or external video optimization or video caching service.

Workflow Application

Workflow binds a single or set of subscribers to a workflow data profile and/or a workflow control profile. Workflow specifies what event triggers this binding (bearer setup). All CDRs, external communication, and statistics collection occur as if it's on the mapped-APN if configured.

Workflow supports an apn mapping list that allows the Operator to specify criteria to which apn values can be mapped. For example, criteria such as PAP username can be used to select a specific APN to use for the specific subscriber.

workflow apn-mapping-list

The apn-mapping-list allows the user to specify criteria to which the apn values can be mapped. For example, criteria such as PAP username can be used to select a specific APN to use for the specific subscriber.

Tip: Affirmed Networks recommends not mapping an APN to another APN with the same name. Doing so may result in incorrect APN names in CDRs and AAA accounting records. Also see apn-reporting in Table 11-1 on page 11-5.

Note: *APN mapping is not supported on emergency APNs.*

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workflow classify

Initiate workflow classification on described flow.

workflow control-profile

Specifies the back-end interfaces to use for PCRF, authentication, accounting, tunneling, and charging systems. As an alternative, the profile allows Operators to configure subscriber profiles statically rather than using these interfaces. The profile also specifies the mechanism to dynamically obtain the name of a workflow data-profile that has been programmed on the system. Configure default interfaces for online-charging, offline-charging, pcrf-charging, authentication, and accounting. The object supports a maximum of two offline-charging interfaces, one for local CDR and one for streaming to the OFCS server. The control-profile also supports APN mapping parameters for reporting, controlling dedicated bearers, selecting mapping lists, mapping control, and mapping behavior. The MCC supports the creation of "on-the-fly" dedicated bearers.

Configuring Objects

Configure the objects as follows:

Table 11-1 workflow control-profile

Object	Description
allocate-delegated-prefix-from-local-pool	Set to enabled allocate the delegated-IPv6-prefix from local ip-sub-pools. The default is disabled. The MCC allocates a Prefix Delegation (PD) from the dynamic/local pools if RADIUS does not provide PD for the UE in Access-Accept. This feature applies to both PGW and GGSN. The MCC supports only one PD for each PDN session. The PDs are reported in RADIUS Accounting Requests. MCC supports the PDs with the following prefix lengths: /48, /52, /56, /60, /62 and /64. ■ /64 PDs share the pool with UE IPv6-prefixes because the UE IPv6-prefix-length is also 64. ■ /48, /52, /56, /60, and /62 PDs are allocated from the matching prefix length IPv6 pools. That is, a /48 PD is allocated from a /48 pool, a /56 PD is allocated from a a /56 pool, etc. Also see delegated-prefix-length.

Table 11-1 workflow control-profile

Object	Description
allow-dedicated-bearers	Set to enable or disable to indicate whether dedicated bearers/secondary PDP contexts are allowed. Enables you to block a home Public Land Mobile Network (HPLMN) or a UE from initiating dedicated bearers on a per-subscriber basis. Also see the *Affirmed Networks GTP Proxy Administration Guide*.
allow-mapped-apn-based-sub-analyzer	Allows mapped APN based Subscriber Analyzer to be applied to the subscriber. Set this parameter to true to apply the Subscriber Analyzer rules associated with the APN received in the Called-Station-Id attribute to the subscriber.
apn-mapping-control	Control for mapping APN. To enable this feature, set this parameter to external. ■ explicit – Map APN to the mapped-apn-name value or to the mapped-apn-list. ■ external – Map APN using value received from external source. The apn-mapping is done based on received apn-name from RADIUS or in mvno ID in case of Gx. ■ none – No APN Mapping. No apn mapping occurs and the incoming apn is used. ■ pap-username-exclusive – Map APN using PAP user name and apn-mapping-list. The apn-mapping occurs based on the pap-user name received in PCO and the configured workflow apn-mapping-list. If it does not match, MCC rejects the call. ■ pap-username-or-explicit – Map APN using PAP user name and apn-mapping-list if possible, otherwise default to mapped-apn-name. The apn-mapping occurs based on the pap-user name received in PCO and the configured workflow apn-mapping-list. If it does not match, MCC uses the mapped-apn-name configured under control-profile. ***Note:*** *If this parameter is not set to* `external`*, the PGW ignores the new Called-Station-Id AVP and continues the session with the control profile selected initially.*

Table 11-1 workflow control-profile

Object	Description
continued ...	In addition, set the `allow-mapped-apn-based-sub-analyzer` parameter to `true` to apply the Subscriber Analyzer rules associated with the APN received in the Called-Station-Id attribute to the subscriber. ***Note:*** *Only one APN redirection is allowed per subscriber.* The MCC supports the ability to map to an APN using information received from the RADIUS server in the Called-Station-Id attribute and to use the mapped APN information to redirect the subscriber session to a new control-profile. When the PGW/GGSN receives an Access-Accept message containing a Called-Station-Id that is different from the one sent in the Access-Request, the PGW/GGSN maps the subscriber session to the APN received in the Called-Station-Id attribute. In doing so, the subscriber session uses the Subscriber Analyzer and Workflow Control Profile parameters defined for the APN received in the Called-Station-Id attribute. To use RADIUS-based APN mapping/redirection, the APN received in the Called-Station-Id AVP must match an APN configured locally. If the APN received in the Called-Station-Id AVP, within the Access Accept message, is not configured locally on the PGW, the subscriber session is rejected. If the Access-Accept message does not contain a new Called-Station-Id AVP, the subscriber session continues with the control profile selected initially.
apn-mapping-failure-behavior	Behavior of GW if APN remapping fails.
apn-mapping-list	List of mappings to APNs.
apn-reporting	The Access Point Name (APN) to use when reporting APN to external devices. Options include original, mapped, prior-mapped, custom. prior-mapped – If set to prior-mapped, and no prior-mapped APN exists, the system uses the original APN to report CDRs or accounting records. custom – Enables Operators to customize the APN reported on external interfaces. The customized APN is used in the Called-Station-ID AVP and sent in messages on the following interfaces: ■ Access-Request on RADIUS ■ Accounting-Request on Gz, Ga (GTPP), Local, RADIUS ■ Credit-Control-Request (CCR) on Gx and Gy interfaces

Table 11-1 workflow control-profile

Object	Description
apn-statistics-control	Configure how performance statistics are reported for the individual APNs that are mapped to an APN. ■ original — Specifies that statistics are reported for individual APNs. APN statistics for the mapped APN will still show statistics for IP address allocation activities, which are performed by the mapped APN (for example, `remote-ipv4-reserve-successes` and `static-ipv4-reserve-successes`). ■ mapped — (default) Specifies that statistics are reported for the mapped APN. APN statistics for an individual, original APN will still show statistics for UE connection activities, which are performed before APN mapping takes place (for example, `create-session-attempts`, `total-redirected-apn-subscriber-count`, and `create-session-accepts-apn-matching-type-exact-match`). View the APN performance statistics with the command **`show zone default gateway statistics cluster-level call-performance apn`** ***<name>***.
charging-characteristic	Charging characteristic per subscriber.
custom-reporting-apn-name	The value set in the custom-reporting-apn-name parameter is used to fill the Called-Station-Id value in messages to external interfaces. The custom-reporting-apn-name parameter is visible only if apn-reporting is set to custom.
data-profile-reselection-template	Data profile reselection template.
default-accounting-interface	Configure default accounting interface: radius/diameter.
default-advice-of-charge	Advice-of-charge selected from advice-of-charge list.
default-auth-interface	Configure default authorization Interface.
default-home-post-paid-charging-instance	Default postpaid charging service home PLMN - selected from charging services list.
default-home-pre-paid-charging-instance	Default prepaid charging service home PLMN - selected from charging services list.

Table 11-1 workflow control-profile

Object	Description
default-offline-interface	Configure default offline Interface: local/diameter. If the Control Profile with which an offline rating group is associated does not have a `default-offline-interface` configured, the MCC still allows the PDN session to be established and allows offline charging traffic. If the Control Profile has a local CDR interface configured, the MCC will generate CDRs and store them in a local file.
default-online-interface	Configure default online Interface: local/diameter. If the Control Profile with which an online rating group is associated does not have a `default-online-interface` configured, the MCC still allows the PDN session to be established if there is no active online service rule during session setup. In this case, traffic that does not match the online rating group's service rules, such as offline rating group traffic, will be allowed and CDRs will be generated. However, if a rating group has at least one static online service rule (`always-on` rule) or if a predefined/dynamic online service rule is installed by the PCRF during/after PDN session setup, the MCC terminates the PDN session if there is no `default-online-interface` configured.
default-pcrf-interface	Configure default pcrf Interface.
default-port-nbr-interface	Configure default port NAT Binding Record Interface: local.
default-qos-profile	Default quality of service profile supplies UE signaled AMBR if none is requested.
default-roaming-post-paid-charging-instance	Default postpaid charging service visited PLMN - selected from charging services list.
default-roaming-pre-paid-charging-instance	default prepaid charging service visited PLMN - selected from charging services list.
default-rx-interface	Configure default Rx Interface.
default-s6b-interface	Configure the default s6b interface.
default-tunnel-configuration	Default tunnel configuration - selected from a list of tunnel objects.

Table 11-1 workflow control-profile

Object	Description
delegated-prefix-length	Prefix length of the delegated-IPv6-prefix to be assigned from local ip-sub-pools. This object applies to only dynamic/local PD allocation. ■ prefixlength*- 48|52|56|60|62|64| ■ priority* – priority to choose the delegated-IPv6-prefix to be assigned from local ip-sub-pools. range 1 .. 10
dhcp-local-server-admin-mode	Flag to enable/disable dhcp-local-server-mode.
dhcp-local-server-profile	Configure the DHCP server profile if the dhcp-local-server-admin-mode is enabled. These options are only included if the UE requests them in the parameter request list. See dhcp-local-server-profile (page 11-12).
dhcp-service-profile	The DHCP service to use if the UE IP allocation is managed using DHCP.
duplicate-ip-reserve-request-handling	This object is enabled only if the ue-ip-allocation-mode is not set to local-ue-ip-allocation. See duplicate-ip-reserve-request-handling (page 11-12).
edr-interface	Configure the edr interface.
initial-user-unknown-error-handling	Initial diameter user unknown error result code handling.
is-ocs-enabled	OCS enabled per subscriber - true/false.
ldap-interface-profile	Configure LDAP Interface .
manifest-interface-profile	Name of the manifest-interface-profile to use for retrieving flow information from a manifest server for ADC reporting over the Gx interface. See the *Affirmed Networks Gx Interface Specification*.
mapped-apn-name-list	Configure a list of real APNs that a virtual APN is mapped to. You can list up to 128 APNs. Sessions are evenly distributed among these APNs. You must set apn-mapping-control to explicit. The virtual APN and the real APNs can be in the same network context or in different nework contexts. The real APNs can all be in the same network context or in different network contexts. The real APNs should have UE IP pools of the same size.
mapped-apn-name	Mapped APN name to use. You must set apn-mapping-control to explicit.

Table 11-1 workflow control-profile

Object	Description
mvno-profile	Profile for sessions associated with an MVNO and using this control profile.
nbiot-qos-profile	Configure NBIoT profile.
profile-type	Select either default-qos-profile or network-qos-profile.
redirect-on-creation	If redirect URL is provisioned, session will be redirected to the URL immediately.
remote-ip-alloc-fallback-option	Fallback option to use when remote (RADIUS) fails to allocate UE IP. See v4v6-hybrid-allocation-state (page 11-13).
ue-ip-allocation-mode	The UE IP allocation mode enables Operators to specify the IP allocation mode for the sessions associated with the control profile. See ue-ip-allocation-mode (page 11-14).
update-prior-mapped-apn	Specify whether to update the prior mapped APN when APN mapping is performed. By default, this parameter is set to **`true`**. true – (default) If APN mapping is configured on this control profile, the APN mapped prior to the current APN is populated in the prior mapped APN. false – If APN mapping is configured on this control profile, the APN mapped prior to the current APN is not populated in the prior mapped APN.
user-subscription-charging-type	User Subscription Charging Type. This object is used to advertise the charging methods (online/offline) supported on the MCC. The charging capabilities are sent at the command level in CCR-I for each session using online and offline AVPs. By default, MCC supports both online and offline charging (no-subscription). Options include: prepaid-subscription – Prepaid user type. MCC sends online AVP as enabled and offline AVP as disabled postpaid-subscription – Postpaid user type. MCC sends online AVP as disabled and offline AVP as enabled no-subscription – no subscription type. MCC sends online and offline AVP as enabled
v4v6-hybrid-allocation-state	The policy to use for hybrid V4V6 UE IP address allocation - where one address is assigned by an external entity (ex. Radius) and the other is assigned locally. See v4v6-hybrid-allocation-state (page 11-13).

dhcp-local-server-profile

- *broadcast-address* – Broadcast IP Address
- *dhcp-server-name* – Name of the DHCP server profile. Setting or clearing this field causes MCC to add or omit the corresponding DHCP option.
- *interface-mtu* – Interface MTU value. Setting or clearing this field causes MCC to add or omit the corresponding DHCP option.
- *lease-timer* – lease timer value
- *local-server-ip-address* – IP Address of the local DHCP Server
- *primary-dns-ip-address* – IP Address of the primary DNS. Setting or clearing this field causes MCC to add or omit the corresponding DHCP option.
- *prior-subscriber-ip-binding-timer* – Time (minutes) to retain the last assigned IP binding even after the session is deleted. When the ue-ip-allocation-mode is set to local-ue-ip-allocation, this parameter is used to retaining subscriber binding awareness for a configured time even after the session is deleted. If the session is re-established before the timer expires, the same IP address is assigned. After the timer expires, the IP address is released. The UE is identified using its MAC address. This ensures that the same IP is reassigned to the UE even if the session is deleted and re-established within that binding history window.

MCC assigns a specific UE IP in the DHCP request message as long as that UE IP is not assigned to any other UE IP or has a prior subscriber binding associated with another UE.

- *router-ip-address* – IP Address of router. Setting or clearing this field causes MCC to add or omit the corresponding DHCP option.
- *secondary-dns-ip-address* – IP Address of the secondary DNS. Setting or clearing this field causes MCC to add or omit the corresponding DHCP option.
- *subnet-mask* – Subnet Mask. Setting or clearing this field causes MCC to add or omit the corresponding DHCP option.

duplicate-ip-reserve-request-handling

This object is enabled only if the ue-ip-allocation-mode is not set to local-ue-ip-allocation. This object allows the Operator to control the subscriber session setup during a duplicate IP assignment (for example, when an external server (such as RADIUS) assigns a UE IP to a subscriber that is already assigned to a different subscriber).

- *reject-new-session (default)* – Reject the new subscriber session setup and keep the old session.

- *delete-existing-session* – Close the existing session and accept the new session creation request with the duplicate IP assignment.

v4v6-hybrid-allocation-state

The policy to use for hybrid v4v6 UE IP allocation, where one IP address is managed externally (RADIUS) and the second IP address is managed internally by the MCC, when the ue-ip-allocation-mode is set to radius-ue-ip-allocation.

- v*4v6-hybrid-allocation-disabled* – Hybrid UE IP allocation is disabled. In the event that RADIUS returns only one of the IP addresses, the session setup request is rejected.

- *v4v6-hybrid-allocation-local-pool* – If RADIUS allocates one of the IP addresses, MCC attempts to allocate the other IP address from one of the local IP sub pools.

- *v4v6-hybrid-allocation-remote-pool* – If RADIUS allocates one of the IP addresses, MCC attempts to allocate the other IP address from one of the remote IP sub pools.

  This configuration applies only in the case where the UE is requesting a v4v6 IP address and RADIUS allocates only a v4 or a v6 IP address.

  If the RADIUS does not assign any UE IP address, hybrid allocation mode does not apply. Actions are dependent on the remote-ip-alloc-fallback-option.

- *IP Allocation Type* – Defines the type of IP addresses that can be assigned by an external DHCP Server. Options include V4 (default), V6 and both V4 and V6.

- *List of External DHCP Servers* – A repeated field of leaf references to DHCP server objects indicating the DHCP servers to use when the IP Allocation mode is configured to use *dhcp-relay-ue-ip-allocation* or *dhcp-client-ue-ip-allocation*. The DHCP server object is the new object configured in a network-context.

- *remote-ip-alloc-fallback-option* – If the RADIUS server does not allocate an IP address to a user requesting access, this option allows Operators to configure the GGSN/PGW to allocate a UE IP address locally from either the remote UE pool or local UE pool. The system rejects the request if the GGSN/PGW allocates an IP address from the remote UE Pool and later the RADIUS server requests to reserve the same IP address. A quarantine timer is not applied to UE IP addresses allocated from the remote UE Pool.
  - *alloc-remote-pool* – If the RADIUS server fails to send the Framed-Ip-Address attribute or IPv4/IPv6 pool names in the Access-Accept message, the GGSN/PGW allocates the UE IP address from the remote pool.
  - *alloc-local-pool* – If the RADIUS server fails to send the Framed-Ip-Address attribute or IPv4/IPv6 pool names in the Access-Accept message, the GGSN/PGW allocates the UE IP address from the local pool.

   - *alloc-none* – If the RADIUS server fails to send the Framed-Ip-Address attribute or IPv4/IPv6 pool names in the Access-Accept message, the GGSN/PGW rejects the request.

mirrored-accounting-interface-list

A list of mirrored accounting interface names. See the *Affirmed Networks RADIUS Interface Specification* for more information on mirroring.

VoLTE (Proxy-Call Session Control Function)

Operators can configure the GGSN/PGW with the IP address (IPv4 and/or IPv6) of the Proxy-Call Session Control Function (P-CSCF) server used for IP Multimedia Subsystem (IMS) calls. If requested during session creation, the GGSN/PGW sends the P-CSCF address as part of the Protocol Configuration Options (PCO) Information Element (IE) to user equipment (UE) that supports VoLTE. If Operators configure two P-CSCF addresses (primary and secondary P-CSCF server), the GGSN/PGW sends both addresses in the PCO IE. Even if the UE does not request the P-CSCF address, Operators can configure the GGSN/PGW to always send the configured P-CSCF address in the PCO IE as part of the initial PDP context/session response.

Handing of Duplicate IP Requests

The duplicate-ip-reserve-request-handing object is enabled only if the ue-ip-allocation-mode is not set to local-ue-ip-allocation. This object allows the Operator to control the subscriber session setup during a duplicate IP assignment (for example, when an external server (such as RADIUS) assigns a UE IP to a subscriber that is already assigned to a different subscriber).

- *reject-new-session (default)* – Reject the new subscriber session setup and keep the old session.
- *delete-existing-session* – Close the existing session and accept the new session creation request with the duplicate IP assignment.

ue-ip-allocation-mode

The UE IP allocation mode enables Operators to specify the IP allocation mode for the sessions associated with the control profile. Options include:

- *local-ue-ip-allocation (default)* – The UE IP assignment is managed locally using local pools.
- *radius-ue-ip-allocation* – The UE IP allocation is managed from an external server (for both Framed IP behavior as well as for pool name-based assignment).

When UE IP allocation is managed remotely from a RADIUS Server, set the **remote-ip-alloc-fallback-option** to **alloc-local-pool** to apply quarantine to UE IP addresses from the local pool. In this case, if the RADIUS server does not allocate a UE IP address, the GGSN/PGW allocates an IP address locally from the local UE pool and if configured, quarantines these locally-assigned IP addresses. Note that a quarantine timer is not applied to UE IP addresses allocated from the remote UE Pool.

- *dhcp-relay-ue-ip-allocation* – The UE IP is assigned/managed using a DHCP relay, with the UE acting as the DHCP client and the system learning the IP address selected by the UE. The system manages a *remote* UE pool to track the DHCP assigned UE addresses.
- *dhcp-client-ue-ip-allocation* – The UE IP is assigned by the external DHCP server, however the system acts as a DHCP client and performs the DHCP negotiation on behalf of the UE.
- *IP Allocation Type* – Defines the type of IP addresses that can be assigned by an external DHCP Server. Options include V4 (default), V6 and both V4 and V6.
- *List of External DHCP Servers* – A repeated field of leaf references to DHCP server objects indicating the DHCP servers to use when the IP Allocation mode is configured to use *dhcp-relay-ue-ip-allocation* or *dhcp-client-ue-ip-allocation*. The DHCP server object is the new object configured in a network context.
- *remote-ip-alloc-fallback-option* – If the RADIUS server does not allocate an IP address to a user requesting access, this option allows Operators to configure the GGSN/PGW to allocate a UE IP address locally from either the remote UE pool or local UE pool. The system rejects the request if the GGSN/PGW allocates an IP address from the remote UE Pool and later the RADIUS server requests to reserve the same IP address. A quarantine timer is not applied to UE IP addresses allocated from the remote UE Pool.
  - *alloc-remote-pool* – If the RADIUS server fails to send the Framed-Ip-Address attribute or IPv4/IPv6 pool names in the Access-Accept message, the GGSN/PGW allocates the UE IP address from the remote pool.
  - *alloc-local-pool* – If the RADIUS server fails to send the Framed-Ip-Address attribute or IPv4/IPv6 pool names in the Access-Accept message, the GGSN/PGW allocates the UE IP address from the local pool.
  - *alloc-none* – If the RADIUS server fails to send the Framed-Ip-Address attribute or IPv4/IPv6 pool names in the Access-Accept message, the GGSN/PGW rejects the call.

workflow data-profile

Contains a collection of service instances. Configure the services to provide for the PDN session and control resource utilization by the session. When the workflow is established, each associated subscriber's traffic flow is steered toward the collection of associated services. Different subscriber flows can be given different treatment to provide end-to-end service(s).

When dpi-pre-detection is configured, MCC caches the previous matched service flows and rules for a connection for each connection reclassify and directly applies the cached service flows and rules during the connection deletion process to avoid the expensive connection reclassify process that causes the congestion under high traffic load.

Configuring Objects

Configure the objects as follows:

Table 11-2 workflow data-profile

Object	Description
charging	charging service selected from charging services list.
content-cache	bind content cache service instance.
content-filter	bind content filter service instance.
default-crbn-profile	Default crbn-profile to be used when PCRF is unreachable.
dns-proxy	bind dns proxy service instance.
dpi-pre-detection-max-bytes	Allow max number of bytes for DPI pre-protocol and application detection. The default is 0 (disabled).
dpi-pre-detection-max-pkts	Allow max number of packets for DPI pre-protocol and application detection. The default is 0 (disabled).
fair-bandwidth-usage	Bind fair bandwidth service profile.
first-packet-over-volume-quota	Behavior for first packet to exceed volume quota [allow/disallow].
flow-idle-timeout-profile	Flow idle timeout profile which is bound to data profile.
force-packet-drop	Set to true to force all the packets to drop both on the access and core side after workflow processing. Set to true when the system is acting as a probe, and you want it to analyze the data and drop the packets.
geo-redundant-group	List of Geographic Redundancy Groups to which this data-profile is associated. For CUPS, this is the same geo-redundant-group referenced by the ue-pools in the ip-subpools associated with the specified user-plane-profile.

Table 11-2 workflow data-profile

Object	Description
multi-protocol-proxy	Bind http proxy service instance.
http-server	Bind http server instance max-connections.
max-connections	Enter the maximum number of allowed connections.
metering-service	Metering service selected from metering service list.
network-qos-profile	Network quality of service profile overrides UE signaled AMBR.
pcc-rule-bearer-check	Control which bearers are processed by a PCC rule for a service data flow (SDF). Options include: ■ true – The PCC rule processes traffic only for the dedicated bearer identified in the rule. Any traffic arriving on another bearer is processed by an alternate SDF as if the PCC rule is not present. ■ false – (Default) The PCC rule processes traffic for all bearers.
priority-evaluation-mode	Set the priority-evaluation-mode. Options include: flat – Service flow priority is ignored and only service-rule priority is used. Any PCC rules take priority. flat-with-pcc-rule-precedence – Any PCC rules in effect have their precedence values compared with the service-rule priority values, and the PCC rule or service-rule with the highest priority (lowest numerical value) is selected. hierarchical – Any PCC rules take priority.
qos-service	Bind qos service policy.
steering-policy	Layer 3 route steering policy.
subscriber-firewall	Bind subscriber-firewall policy.
wap-gateway	Bind wap gateway service instance.

DPI Pre-Detection

DPI detection for the protocol-application-rule-group requires more than one packet passing through the DPI engine for detection and correlation. This feature enables DPI to identify the application and protocol when quota is exhausted. This feature is enabled by default. As an option, Operators can customize the maximum number of allowable bytes and packets passing through the DPI engine. Similarly, the http-rule-group and TLS-rule-group also use DPI when the quota is exhausted

Switching Data Profiles

The MCC supports switching workflow data-profiles, whereby any services can be switched to a new service.

For example, in the following scenarios, MCC switches the workflow data-profile to match the corresponding new plan dynamically without terminating the subscriber:

- A subscriber's plan quota runs out
- A subscriber purchases optional plan packages
- A subscriber upgrades/downgrades a package
- A policy changes in the network

When workflow data profile switching occurs, the system reclassifies traffic and applies all referenced services supported by the new workflow data profile.

Some services (such as charging instance and QoS Policy) immediately take effect for the next packet. Existing connections are mapped to different QoS flows or rating-groups, based on the new workflow data profile.

Other services (including HTTP proxy and HTTP proxy-dependent services) take effect only for new transactions. Current transactions continue to use the previous data profile information.

When switching to a new charging instance, all existing referenced rating groups in both the previous and new data profile are reauthorized for grants from the OCS server. Rating groups present in the previous data profile but not in the new data profile are closed by sending the FINAL reporting reason to the OCS server with current usage information. These rating groups are deleted from the subscriber and the subscriber data traffic is reclassified and assigned to the new rating groups.

Restrictions to the qos-policy

The following restrictions apply to the qos-policy:

- Data profile switching does not operate properly in the case of network initiated/user initiated dedicated bearers (that is, configured locally to create dedicated bearers from the MCC). Data profile switching operates properly in the case of Dynamic bearers created by PCRF.
- Data profile switching cannot convert a prepaid user to postpaid user. The system is unable to initiate a new diameter session due to switching workflows.

Other services (including HTTP proxy and HTTP proxy-dependent services) take effect only for new transactions. Current transactions continue to use the previous data-profile information.

When switching to a new charging instance, all existing referenced rating groups in both the previous and new data-profile are reauthorized for grants from the OCS server. Rating groups present in the previous data-profile but not in the new data-profile are closed by sending the FINAL reporting reason to the OCS server with current usage information. These rating groups are deleted from the subscriber and the subscriber data traffic is reclassified and assigned to the new rating groups.

workflow data-profile-reselection-trigger-template

This feature provides a template to trigger a data-profile switch and associate the template from the control-profile. MCC switches the data-profile by applying the subscriber-analyzer selection based on RAT type. This feature supports switching of data-profile with and without PCRF on rat-change: This feature also supports 3GPP to non-3GPP handover and vice versa. Enable or disable the rat-change.

workflow reclassify

Initiate workflow reclassification on described flow.

workflow subscriber-analyzer

Specify the data-profile or user-plane-profile to statically bind a service flow to the workflow. Omit the data-profile or user-plane-profile to dynamically obtain the services programmed on the system.

The user-plane-selector is configured on the CPF. It supports existing subscriber analyzer keys and the additional key for UE support for Dual Connectivity with New Radio (DCNR).

The user-plane-selector specifies whether a session is a CPF/CUPS session or a regular integrated non-CUPS session. This configuration applies to both the collocated and standalone PGW service, as well as standalone SGW service.

Configuring Objects

Configure the objects as follows:

Table 11-3 workflow subscriber-analyzer

Object	Description
analyzer-type	Workflow analyzer type. Options include: workflow-profile, peer-selector, and user-plane-selector.
event	Select bearer-setup for this subscriber-analyzer.
geo-redundant-group	Geographic Redundancy Group Name
name*	Enter a name for the subscriber analyzer.
peer-list	Configure GTP peer list.
priority*	priority [1..4096]

Table 11-3 workflow subscriber-analyzer

Object	Description
action	Select workflow data-profile if statically configured on the cluster, otherwise select control-profile. If the data-profile does not exist, the profile is retrieved from a remote server. (CUPS) Select user-plane-profile to associate with this analyzer type. If the specified user-plane-profile has enabled **`mec-local-breakout`** (see zone gateway user-plane-profile (page 12-84)), the data-profile from the matching workflow-profile type of subscriber-analyzer is sent to the UPF. Data packets matching the data-profile undergo a local breakout. Local breakout allows traffic that only needs local applications to be serviced at the operator edge site. This reduces latency for local applications, and reduces back-haul requirements.
key	List of analyzed keys. A key can have multiple attributes. You can configure multiple keys, each identified by name. If any key matches, the subscriber analyzer is selected. A key can specify multiple triggers, all of which must match for a key to match. ***Note**: If a subscriber analyzer is configured with no keys or with keys that have no attributes, the analyzer is ignored and is not part of the selection process during session setup.* See key (page 11-20).

key

You can configure multiple keys, each identified by name. If any key matches, the subscriber analyzer is selected. A key can specify multiple triggers, all of which must match for a key to match. Configure the parameters as follows:

```
Name: key name
<your list of keys>
```

The subscriber-analyzer supports the following analyzed key parameters:

- apn-name — APN name selected from configured APNs
- apn-name-list — Multiple APN names to match
- cell-global-id — Cell global identifier
- charging-characteristics — Charging characteristics information
- credit-control-group — Credit control group value
- epdg-name — ePDG name selected from the configured list
- eutran-cell-global-id — E-Utran cell global identifier

■ ggsn-name	GGSN name -selected from configured list
■ gigw-unknown-subscriber-profile	Subscriber aware GiGW unknown subscriber profile Also see Configurable Treatment for Unknown GiGW Subscribers (page 11-22).
■ gtp-proxy-name	GTP proxy service name
■ home-plmnid	Subscriber's home PLMNID, 5 or 6 digits
■ imei-range	User equipment identifier range
■ imsi-range	International mobile subscriber identity range
■ mcc-name	MCC name selected from configured list
■ msisdn-range	MSISDN range
■ mvno-identifier	MVNO Name or Identifier
■ mvno-sub-identifier	MVNO Sub Name or Sub Identifier
■ name*	Analyzed key set name.
■ pdn-type	Using this key, subscribers with IP and non-IP PDN types can use the same APN but different services based on the workflow control profile. Options include: IPv4, IPv6, IPv4v6, and non-IP.
■ pgw-name	PGW name selected from configured list
■ rat-type	Radio Access Type. Options include: eutran gan geran hspaEvolution utran wlan nbiot-eutran
■ realm-name	Realm name - Domain name portion of the user name.
■ roaming-user	Specify whether user is a roaming user
■ routing-area-id	Routing Area Identifier
■ service-area-id	Service Area Identifier
■ serving-nw-mcc	Serving Network's Mobile Country Code The GTP Proxy service can select any PGW based on a UE's location or classification type. This facilitates topology hiding, simplifies deploying or adding more PGWs, and streamlines the billing-data information required for supporting multiple countries with multiple operators. See the *Affirmed Networks GTP Proxy Administration Guide*.

■ serving-nw-plmnid	Serving network's PLMN ID, 5 or 6 digits. The RADIUS AVP 3GPP-SGSN-Address is received in the Accounting Request and retrieves the serving-nw-plmnid (plmn-id-mcc and plmn-id-mnc) from the SigInfo table. If this serving-nw-plmnid is configured as key for any of the Subscriber Analyzers, then that Subscriber Analyzer is selected for the session. In GiGW, this key is matched against the 3GPP-SGSN-MCC-MNC AVP.
■ sgw-name	SGW name selected from configured list
■ signalling-priority-lapi	Signalling priority indication LAPI. This key is used to define criteria for the subscriber-analyzer selection and configure rules to differentiate the treatment for LAPI PDNs. See Signaling Rejection for UE Groups Based on IMSI or LAPI (page 11-24).
■ tracking-area-id	Tracking area identifier
■ ue-capabilities	UE capabilities
■ ue-ip-address	User equipment IP address
■ user-name-prefix	User name prefix of the user name to match
■ wag-name	WAG name selected from the configured list

Configurable Treatment for Unknown GiGW Subscribers

This feature is supported on the Subscriber Aware GI Gateway (SA-GiGW) configured with RADIUS Server/Proxy as session learner. This feature serves a configurable static Web page to subscribers for which the SA-GiGW does not have any information.

The Boolean type key attribute **`gigw-unknown-subscriber-profile`** allows the Operator to configure a subscriber analyzer intended for SA-GiGW unknown subscribers.

The SA-GiGW may fail to obtain subscriber information in a scenario where the RADIUS Server/Proxy momentarily goes down. Instead of dropping the packets of unknown subscribers, the SA-GiGW serves a static Web page to unknown subscribers, describing the configured reason for access denied. This enables the Operator to communicate the configured reason for access denied to the unknown subscriber. This feature also enables the Operator to perform a certain action on the traffic from the unknown subscriber. The appropriate error message is served by HTTP/WAP services running on the CSM.

Note: *This feature is supported only for SA-GiGW configured with session learner mechanism RADIUS Server/Proxy. If multiple SA-GiGW instances are configured with the same APN, MCC uses the first SA-GiGW instance for unknown subscribers.*

The Boolean type key attribute **`gigw-unknown-subscriber-profile`** allows the Operator to configure a subscriber analyzer intended for SA-GiGW unknown subscribers.

Operators configure a special subscriber analyzer with data profile and key with tuple. The data profile configured under this subscriber analyzer is for *unknown* subscribers.

Operators can also configure a subscriber analyzer for legitimate subscribers. The data profile configured under this subscriber analyzer is used for *known* subscribers.

Mapping to a Virtual APN

Operators can map subscribers from a single common Access Point Name (APN) to individual specific virtual APNs using information received from the PCRF. Virtual APNs enable Operators to apply different service treatments to subscribers that share a common APN.

When the PGW/GGSN receives a Credit-Control-Answer or Re-Auth-Request message (from the PCRF) containing the Mobile Virtual Network Operator (MVNO) Reseller ID and/or the MVNO Sub Class ID vendor-specific attributes, the PGW/GGSN uses these attributes (as subscriber analyzer keys) to map subscriber sessions to specified mapped virtual APNs. In doing so, these subscriber sessions use the workflow control profile parameters defined for the mapped virtual APNs.

Configurable subscriber analyzer keys are used to match the vendor-specific attributes received from the PCRF to a subscriber analyzer:

Identifying Excessive Users

Operators can configure IMSIs identified as excessive users (based on the configured imsi-range) as keys for the subscriber analyzer that point to differentiated QoS policies. During session setup, subscribers identified as excessive may be denied access to certain high bandwidth applications and subjected to different QoS policies used to throttle the subscriber's bit rates.

The system uses workflow workflow data-profile (page 11-16) to point to the qos-flow and qos-policy containing the desired set of rules and bit rates. The qos-policy and qos-flow is associated with a data-profile and subscriber-analyzer.

Signaling Rejection for UE Groups Based on IMSI or LAPI

This feature supports a configurable option to reject signaling from a certain group on UEs based on IMSI. Operators configure a parameter to reject signaling from a set of UEs/USIM based on the IMSI range for a congested APN, when the load reaches beyond a certain configured threshold.

In addition, this feature provides an option to configure a parameter to reject signaling from LAPI (Low Access Priority Indication) devices. Operators configure this feature using the `signalling-prioriry-lapi` key in `subscriber-analyzer`.

This feature is not limited to IMSI (or) LAPI-based differentiation; it can also be used with other subscriber-analyzer rules as long as the desired workflow control-profile is selected. For example, imei-range, rat-type, and pdn-type etc.

The configured IMSI range or LAPI is associated with a critical, major, or minor threshold using `workflow control-profile congestion-reject-level`.

This feature also provides congestion control action enforcement for remapped APNs. The system applies the congestion control action on the final remapped APN, after choosing the subscriber-analyzer and workflow control-profile.

This feature supports:

- A subscriber-analyzer signalling-priority-lapi key
- Configurable APN threshold congestion rejection level in workflow control-profile
- Decoding and encoding signaling priority indication
- Decoding of LAPI received in signaling priority indication IE in create-session-request message
- PGW and APN-level counters for create-session-request attempt, accept, and reject with LAPI bit set in signaling priority indication IE
- Reject create-session-request (or new sessions) at PGW, if selected final APN is in congestion state
- Cause code **APN Congestion (113)** and **PGW Back-off Time** in create-session-response if the new session is rejected due to APN congestion

Note: *The following limitations apply to this feature:*

- The system enforces the apn-congestion-reject action on the final APN after running subscriber-analyzer. The final APN can be original or remapped. In the remapped APN case, the original APN is passed even if it is congested, if the final APN is not in a congested state. Sessions are counted as part of the final APN.
- The system does not decode LAPI in bearer resource command, therefore, for the existing session, after the APN enters into congestion, if a new dedicated bearer activation arrives, the CSGSN does not enforce the apn-reject-congestion-level action.
- Low priority access inclusion in CDRs is not supported.
- Gateway performance statistics for a number of current LAPI PDN sessions are not supported.
- Alarms relating to APN congestion are not supported.

Chapter 12

Zone Parameters

This section describes the zone objects and parameters supported on the MCC. This section describes how to partition resources for gateway and resource-control.

Access	config
Guidelines	MCC supports a single zone for the cluster, called the default zone. Do not configure zones other than the default.

zone content

Use the content object to configure the congestion control and threshold parameters.

Configuring Objects

Configure the admin state and the major/minor/critical parameters for the following thresholds:

- total number of client connections currently in use
- total number of server connections currently in use

Configure the admin state and the major/minor/critical parameters for the number of active tcp connections. Operators can also configure an action (in one, five or fifteen minute increments) to take when a congestion control threshold is exceeded.

About Overload Control and Threshold Monitoring

System overload occurs when a system's resources are unable to handle the offered load. This may result in diminished capacity and poor performance. The system is designed to avoid overload and manage resources using configurable thresholds.

The threshold-monitoring feature allows you to set policies and thresholds and specify how the system reacts when faced with a heavy load condition. Threshold monitoring monitors the system for conditions that could potentially degrade performance when the system is under heavy load. Typically, these conditions are temporary (for example, high CPU or memory utilization) and are quickly resolved. However, continuous or large numbers of these conditions within a specific time interval may have an impact the system's ability to service subscriber sessions. Threshold monitoring helps identify such conditions and invokes policies for addressing the situation.

The system generates alarms and events when configured thresholds are exceeded. Operators can configure thresholds to generate minor, major, or critical alarms. These alarms may be escalated as the measured values increase through the configured thresholds. For example, a value may cross the minor threshold resulting in a minor alarm being generated. The value may continue increasing through the configured thresholds through the major and critical levels. At each escalation point, the severity of the alarm is raised and reported in Acuitas and the CLI interface.

For example, the entry points for each threshold level are defined in the following table. If the CPU occupancy reaches 70% for 30 seconds, the system generates a minor alarm. If the occupancy reaches 80%, the system escalates the alarm to major. If the occupancy reaches 90%, the system escalates the alarm to critical. In this example, the system clears the alarm severity when the occupancy falls below 60% for 30 seconds. The sampling interval defaults to 30 seconds.

Table 12-1 Threshold Monitoring Example

Alarm Severity	Value to Enter Level
Critical	90
Major	80
Minor	70

zone gateway

Configure a specific GW service to be run.

Configuring Objects

Configure the objects as follows:

Table 12-2 zone gateway

Object	Description
admin state	Enable/disable the admin state
apn	Configure access point name
congestion-control	Gateway service congestion control configuration.
connectivity-check	Gateway connectivity check.
debug-stats	Gateway debug statistics.
enhanced-features-list	Enable/disable list of enhanced features. See zone gateway enhanced-features-list (page 12-25).
epdg	Evolved Packet Data Gateway (ePDG) Services Configuration. See the *Affirmed Networks ePDG Administration Guide*.
filter-id-pco-action-association-table	Configure a list of filter-id to pco-action associations.
ggsn	GGSN Service Configuration
gtp-peer	GTP peer that SGW, PGW, or GGSN communicates with.
geo-redundant-group	Geographic Redundancy Group Name
indirect-forwarding	Enable/disable indirect forwarding tunnel
indirect-forwarding-tunnel-max-clearing-state	Enable/disable PDN Session Clearing for indirect forwarding tunnels present longer than the maximum time allowed
indirect-forwarding-tunnel-max-timer-duration	Maximum Timer duration for indirect forwarding tunnels, in seconds. If a PDN session has been present longer than the session maximum timer duration, the PDN session is cleared

Table 12-2 zone gateway

Object	Description
mcc-gigw	Mobile Content Cloud (MCC) GiLAN-Gateway (GiGW). configuration.
name*	SGW name - string
pcrf-table	Configure a list of IPV4 and IPV6 P-CSCF addresses.
pgw	PGW service configuration
profile	Name of Gateway Profile used by this service.
recovery	
roaming-plmnid-list	List of PLMNIDs considered as roaming
service-rule-pco-action-association-table	Configure a list of service-rule/PCO-action associations.
session-blockid-management	teid-quarantine-timer-duration – The duration in seconds for which a session's TEID will not be re-used after the session is deactivated. The current range is 10 to 432000 seconds (5 days) and 4294967295 seconds. The default is 1800 seconds. ■ A high timer duration – Increases the number of quarantined tokens, which increases the probability of exhausting the number of block IDs. ■ 4294967295 – The token released as part of the session is added to the quarantine list and is never released back to the free token list in the session block. See teid-quarantine-timer-duration (page 12-5) for more information.
session-load-balancer	Configure session load balancer configuration parameters. See zone gateway profile tethering-profile (page 12-66).
sgw	SGW service configuration
s1u-interface-network-context*	Context containing the S1-u Interface
sig-info-table	Configure information elements for GTP signaling messages. You can also use sig-info-tables to map a PLMN ID or SGSN/SGW subnet to a transparent data value when the RADIUS Access-Accept Message does not return a transparent data value to the PGW. See Configuring PGW Selection of a TMO-Transparent Data Value (page 12-47) for details.
subscriber	
subscriber-trace	Trace collection entity configuration.
s5-s8-interface-network-context*	Context containing the S5/S8 Interface
state	State of this gateway.

Table 12-2 zone gateway

Object	Description
timestamp	Timestamp
threshold	Overload thresholds for the number of create-session-requests for 1, 5, and 15-minute periods. For details, see threshold (page 12-6).
user-description	User description for this gateway.
wag	

teid-quarantine-timer-duration

When the `teid-quarantine-timer-duration` is configured, MCC adds a token in a block that is released as part of a session release to a quarantine list, instead of freeing the token. When the token appears in the quarantine list, it is not used for new sessions.

When the quarantine duration of a token (*current time – token quarantine time*) exceeds the configured `teid-quarantine-timer-duration`, MCC releases the token from the quarantine list and adds it back to the available tokens in the session block.

When establishing a session, if all free tokens in all blocks of the selected GW and FSAPI for the session are unavailable and no additional blocks exist in the SBM for that georedundant group, MCC removes the oldest token from quarantine list and uses that token for the session TEID. This allows the TEID to be reused only after all other TEIDs are used.

Note: *Note: Since session blocks (from which tokens are allocated for session TEIDs) are local to a given GW task and FSAPI, when all blocks in MCC are allocated (across all GWs and FSAPIs), a token in a session block of a GW task and FSAPI might be reused from the quarantine list even though available tokens exist in other FSAPIs of the same GW task or in other GW tasks.*
Note: *Note: When the GW task fails over in a local cluster, the information for quarantined tokens for all FSAPIs of that GW task is lost, therefor, the TEIDs are reused earlier than expected with actual TEID quarantine functionality.*

If the `teid-quarantine-timer-duration` value is changed back to a definite value, all tokens for which the quarantine duration exceeded the new configured value are released to the free list of corresponding blocks and are available for TEID allocation for new sessions.

For GTP V0 sessions, the tokens are always quarantined for an indefinite time. When all V0 blocks are allocated and all tokens are used, MCC uses the oldest token from the V0 quarantine list for new GTP v0 sessions.

The number of available block IDs is 16320 (*total number of block IDs is 16384 less 64 that are reserved for GTP-V0*).

For a non-default georedundant group, the number of available block IDs is 16384.

Use the following command to monitor block ID availability:

```
an3000-mcm-slot7cpu1# show zone default gateway
block-management-statistics
gateway block-management-statistics
NUM NUM NUM NUM NUM NUM NUM
NUM NUM NUM NUM NUM
GEO GROUP ALLOC ALLOC ALLOC RESERVE RESERVE RESERVE
RELEASE RELEASE RELEASE UPDATE UPDATE UPDATE
NAME REQUEST SUCCESS FAILURE REQUEST SUCCESS FAILURE
REQUEST SUCCESS FAILURE REQUEST SUCCESS FAILURE
----------------------------------------------------------------------
--------------------------------------------------
default 64 64 0 0 0 0 0
0 0 0 0 0
default-v0 0 0 0 0
```

threshold

The threshold specifies a limit on the number of PGW and SGW create-session-requests for 1, 5, or 15 minutes. When this number is exceeded, the system is in overload. These thresholds are configured with the following commands:

```
an3000-mcm-slot7cpu1(config)# zone default gateway threshold
node-level call-performance pdngw
create-session-requests-[1|5|15]-minute-average major-enter <int>
minor-enter <int> critical-enter <int> admin-state enabled
```

See overload-control max-arp-priority-level (page 12-56) for details on throttling create-session-request messages in overload conditions.

zone gateway apn

Configure the Access Point Names that are accessible via local gateway services. Associate the APN with a PGW.

Configuring Objects

Configure the objects as follows:

Table 12-3 zone gateway apn

Object	Description
3gpp-to-non3gpp-handover-delay	Handover to non-3gpp delay timer duration for PDN sessions, in milliseconds. A value of 0 indicates handover is completed immediately without any delay. ***Note:*** *A value of 0 indicates that existing behavior remains active (that is, sending a DBR on an LTE bearer immediately after sending Create Session Response to ePDG or TWAG).* ***Note:*** *Changes to the 3gpp-to-non3gpp-handover-delay value apply to new handovers only.* When a handover is initiated from 3GPP to non-3GPP, packets are occasionally dropped due to a delay of IPSEC establishment between the ePDG and UE. This may result in a few seconds of service interruption. This feature enables Operators to control the handoff completion (that is, when to send a Delete Bearer Request (DBR) to the 3GPP tunnel). In other words, this feature enables Operators to delay the DBR to the 3GPP tunnel until after the MCC receives the first uplink packet on the non-3GPP tunnel, or until a configured timer expires (whichever occurs first). During a handover delay, MCC continues to send downlink packets on the 3GPP tunnel and receive and process uplink packets from both 3GPP and non-3GPP tunnels. ***Note:*** *This configuration is supported for non-CUPS and CUPS non-standard interface deployments, however, this timer is not supported for a CUPS standard PFCP interface.*
admin-state	The APN admin state. Options include disabled, enabled, and maintenance.

Table 12-3 zone gateway apn (Continued)

Object	Description
allow-accounting-on-off-msg	If enabled, the gateway sends Accounting On/Off messages to the Remote RADIUS Server. MCC supports RADIUS Accounting-On and Accounting-Off messages to ensure session information between the GGSN/PGW and the RADIUS server is synchronized. The GGSN/PGW sends Accounting-On messages to the RADIUS server to indicate that the cluster is up after a scheduled reboot. The GGSN/PGW sends Accounting-Off messages to the RADIUS server to indicate that the cluster is going down due to a scheduled reboot. The Accounting-On and Accounting-Off messages are sent on all the configured RADIUS peer interfaces to that remote RADIUS server and for every GGSN/PGW configured on the system.
allow-dedicated-bearers	Set to enable or disable to indicate whether dedicated bearers/secondary PDP contexts are allowed. Enables you to block a home Public Land Mobile Network (HPLMN) or a UE from initiating dedicated bearers on a per-subscriber basis. Also see the *Affirmed Networks GTP Proxy Administration Guide*.
allow-ue-hairpin-traffic	If enabled, the GW allows traffic to hairpin locally from UE-to-UE without going out the Gi interface. In this case, the system receives traffic from a UE for a particular session and makes a forwarding decision based on the destination IP address in the received packet. If the destination IP address is another UE within the UE pools configured on the system, the system forwards that packet back out the Gn interface to the destination UE. If allow-ue-hairpin-traffic is disabled (default), the system drops the packets.
allowed-apn-selection-mode	A policy to authorize apn selection mode during session setup.

Table 12-3 zone gateway apn (Continued)

Object	Description
apn-congestion-cause-code	Configure an apn congestion cause code. When an APN exceeds a configured threshold, new session setup requests for the APN are rejected by the gtp-v2-cause-code – Gtp V2 cause code to be sent in session setup reject message gtp-v1-cause-code – Gtp V0/V1 cause code to be sent in session setup reject message For gtp-v2-cause-code, valid values are 64 - 127 with default: APN Congestion – 113 For gtp-v1-cause-code, valid values are 192 - 228 with default: No Resource Available – 199
apn-congestion-handling	Enable/disable APN congestion handling.
apn-congestion-timer	Enter the timer-unit and timer-value for the critical, major, and minor crossing threshold. This configuration causes the gateway to reject the call with a cause code of 113 EPC timer with configured value if APN is congested. The system detects congestion if the number of calls is greater than or equal to the configured critical threshold. ■ timer-unit – EPC timer-unit: 2-sec/1-min/10-min/1-hrs/10-hrs/infinite ■ timer-value – EPC timer value: 1-31 To specify the APN congestion threshold with a timer-value of 0: ■ critical-enter timer-value <0-31> ■ major-enter timer-value <0-31> ■ minor-enter timer-value <0-31> When the timer-value is 0, the timer value is not encoded in the response.
apn-congestion-threshold	Configure the apn-congestion-threshold. Options include critical, major, and minor. Also see apn-congestion-threshold (page 12-22). Select the threshold type. Options include oc, cl, op, and ap. To enable this feature, enable the apn-congestion-threshold admin-state.
apn-restriction	An apn restriction value that determines whether the subscriber can establish bearers with other APNs.

Table 12-3 zone gateway apn (Continued)

Object	Description
associated-ue-pool	List of UE Pools (and associated IP Sub Pools) to restrict the UE IP allocation for this APN. When local UE pools are associated with the APN, IP allocation is constrained to the IP sub pools associated with those local UE pools. When remote UE pools are associated with the APN, reservation of remotely assigned UE IP (RADIUS, Static, DHCP client, and so on) is constrained to the IP sub pools associated with those remote UE pools. An APN can be configured with either local UE pools or remote UE pools, or both.
block-ue-spoofing	If enabled, all traffic is blocked if the IP address used is not the IP address assigned to the UE. In computer networking, the term IP address spoofing or IP spoofing refers to the creation of IP packets with a forged source IP address with the purpose of concealing the identity of the sender or impersonating another computing system. The system supports an anti-spoofing feature to prevent IP spoofing. If this features is enabled, all traffic is blocked if the IP address used is not the IP address assigned to the UE. The GGSN/PGW verifies the source IP address of the IP packets issued by the UE and compares it against the IPv4 or IPv6 IP address assigned for the session. If packet verification fails, the GGSN/PGW discards the packets and logs the event in the security log against the subscriber information (IMSI/MSISDN).
called-station-id-type	Choose the called-station-id (format) to be used. Options include: apn – The APN must contain both NI and OI . apn-ni-only – The APN must contain the NI portion and may contain the OI portion. If a UE sends the full APN when this option is selected, then gateway selects only NI portion and removes the OI portion. For all subsequent outgoing messages to Gx, Gy, OFCS etc, only the NI portion of the APN is used. apn-fully-formed – Reserved for future use. Currently functions the same as apn option.
calling-station-id-type	Specify whether the PGW reports the IMSI or MSISDN in the Calling Number AVP in incoming call request (ICRQ) messages. Options include imsi and msisdn (default).
charging	The charging instance selected from charging instances list.

Table 12-3 zone gateway apn (Continued)

Object	Description
default-password	The default password at APN level if password is not present in PCO.
default-user-name	Default User Name at APN Level if user name is not present in PCO
delay-tolerant-connection-indication-support	Enable/disable DTCI for the APN. The MME rejects control messages (CBReq/UBReq) with cause code 123 (UE is temporarily not reachable due to power saving) if the UE is in power saving mode. It is used by the MME/SGSN in the Create/Update Bearer Response message to reject the corresponding network-initiated procedures for a Delay Tolerant PDN connection and also request the PGW to hold the network-initiated procedure until it receives the subsequent Modify Bearer Request message with the UASI flag, indicating that the UE is available for end-to-end signaling. The PGW reattempts the messages when the MME informs the UE of its availability. This feature implementation is compliant with Release 13 GTPV2 specification, September, 2016.
dns-resolution-for-pgw-selection	Enable APN DNS resolution for PGW selection
dns-server-configuration	Set the DNS server configuration per APN. ■ primary-ipv4-address – Configure the primary IPv4 server addresses that can be returned to the UE using the Protocol Configuration Options IE in GTP-C signaling messages. ■ secondary-ipv4-address – Configure the secondary IPv4 server addresses that can be returned to the UE using the Protocol Configuration Options IE in GTP-C signaling messages. ■ primary-ipv6-address – Configure the primary IPv6 server addresses that can be returned to the UE using the Protocol Configuration Options IE in GTP-C signaling messages. ■ secondary-ipv6-address – Configure the secondary IPv6 server addresses that can be returned to the UE using the Protocol Configuration Options IE in GTP-C signaling messages.
emergency-apn	Specifies that this APN is to be used exclusively for emergency services.
emergency-inactivity-timer-duration	Duration of the inactivity timer for emergency calls which specifies the time in seconds that the PGW waits to tear down an emergency session after the emergency dedicated bearer is released.

Table 12-3 zone gateway apn (Continued)

Object	Description
emergency-inactivity-timer-state	Enable or disable emergency session inactivity timer.
enhanced-features	Map the configured enhanced-features-list to an APN. Also see zone gateway enhanced-features-list (page 12-25).
epco-nonip-link-mtu-size-non-roaming	Non-IP link MTU size for a non-roaming PDN connection to be sent in the extended PCO IE.
epco-nonip-link-mtu-size-roaming	Non-IP link MTU size for a roaming PDN connection to be sent in the extended PCO IE.
filter-id-pco-action-association-table	Filter id/rating group/pco-action association table. Also see zone gateway filter-id-pco-action-association-table (page 12-27).
gi-access-mode	Mode by which access to the SGi interface is obtained. Options include: epc, non-seamless-wifi-offload.
guard-timeout	Amount of time in milliseconds to wait before retrying a network initiated request after receiving a response indicating temporary reject.
gtpv0-sessions	This options enables the you to block Gtpv0 packets on the PGW. The gtpv0-sessions (enable/disable) configuration option is used to disable Gtpv0 traffic (if required), since Gtpv0 does not support geographic redundancy. The default is enable. The GTP Proxy service enables you to block an SGSN from initiating a GTP-v0 session for an APN either through the initial request or through a fallback from GTP-v1/GTPv2. If a GTP-v0 session setup request is received when support is disabled, the PGW/GTP Proxy sends a `Create PDP Context Reject` response message with the Cause code set to `resource unavailabl`e.
idle-session-clearing-state	Enable or disable the idle-session-clearing-state. The idle-session-clearing-state enables Operators to clear subscribers who are idle for longer than the configured idle-timer-duration. If *idle-session-clearing-state* is enabled and the subscriber remains idle for the duration of the configured *idle-timer-duration*, the subscriber is cleared. For SGWs, the values are defined in gateway-common. For PGWs and GGSNs, the values are defined in the gateway apn object if specified, otherwise the system uses the gateway-common settings. Operators can specify the *idle-timer-duration*.

Table 12-3 zone gateway apn (Continued)

Object	Description
idle-timer-duration	Idle Timer duration for PDN sessions, in seconds. If a PDN session has been IDLE for longer than the idle timer duration, the PDN session is cleared
inactivity-timer-duration	Enables Operators to detect and potentially take action on subscribers that are inactive. Inactive is defined as no user data in the uplink or downlink direction for all PDN sessions and all bearers. For SGWs – The inactivity-timer-duration is defined in the gateway-common profile. For PGWs and GGSNs – The inactivity-timer-duration is defined in the gateway apn object if the value is non-zero, otherwise the value is defined in the gateway-common profile. A value of 0 indicates not specified. The subscriber/PDN session is marked as Idle or Active as follows: ■ Idle – If no activity occurs within the configured inactivity period (in seconds), the subscriber/PDN session is declared Idle. ■ Active – When a subscriber first connects to a GW, the timer is set and the subscriber/PDN session state is set to Active. In addition, if activity is detected on a subscriber/PDN session that had previously been declared Idle, the session is declared Active. ***Note:*** *The value defined in the gateway apn object takes precedence if the value is non-zero. Otherwise, MCC uses the value defined in the gateway-common profile.*

Table 12-3 zone gateway apn (Continued)

Object	Description
ip-address-allocation-mode	Specifies how the type of allocated address (IPv4 or IPv6) is chosen when the UE requests IPV4IPV6. ■ ipv4 – Allocate an IPv4 address if one is available. If an IPv4 address is not available, the session fails. ■ ipv6 – Allocate an IPv6 address if one is available. If an IPv6 address is not available, the session fails. ■ ipv4v6 – If a dual address bearer flag (DAF) is received from the UE, allocate both an IPv4 and an IPv6 address. If a DAF is not received, allocate an IPv4 address if it is available; otherwise allocate an IPv6 address if it is available. If neither type of address is available, the session fails. ■ ipv4v6-prefer-v4 – Allocate an IPv4 address if it is available; otherwise allocate an IPv6 address if it is available. If neither type of address is available, the session fails. ■ ipv4v6-prefer-v6 – Allocate an IPv6 address if it is available; otherwise allocate an IPv4 address if it is available. If neither type of address is available, the session fails. ***Note:*** *If the UE requests IPv4 or requests IPv6, that type of address is allocated if it is available, regardless of the ip-address-allocation-mode. If the requested type of IP address is not available, the session fails.*
is-high-throughput-apn	Enable or disable load balancing of high throughput sessions across Workflow VMs . Load balancing is performed across ISMs, HD-Cores, and proxy instances. This feature is supported for CUPS running with TAM enabled on ISMs. ■ true – All sessions associated with this APN are considered high throughput and are load balanced across ISMs, HD-Cores, and proxy instances. ■ false – (default) Load balancing is based only on the session count.

Table 12-3 zone gateway apn (Continued)

Object	Description
max-pcscf-ipv4-addresses max-pcscf-ipv6-addresses	Number of P-CSCF IP V4 Address to be included in the PCO IE. Number of P-CSCF IP V6 Address to be included in the PCO IE. Enables Operators to allocate a maximum number of P-CSCF IPv4/IPv6 addresses at the APN level to be included in the PCO IE by the PGW. The default is 2. MCC enables Operators to configure a table of P-CSCF IP V4/V6 addresses on an individual basis and associate the addressed to an APN. If the number of P-CSCF IPv4/IPv6 addresses is greater than the IP address configured in the pcscf-table, then MCC uses only the IP addresses in the pcscf-table. This feature is used only if the address-allocation-mode in pcscf-table is set to **round-robin** instead of **weighted-round-robin**.
max-subnets-behind-ue	Maximum number of subnets permitted behind a UE. Setting this parameter to a value of 0 disables the support for routing behind a UE for the APN. This feature supports the enabling/disabling of routing behind a UE on a per-APN basis. The feature enables the MCC to support traffic originating from or destined to IP subnets behind a UE router. The MCC learns about the list of IP subnets behind a UE router via RADIUS messaging at the time of PDN session creation. A maximum of 16 IP subnets is supported behind each UE. When this parameter is set to 0, the MCC ignores any of the framed routes specified in either framed-route or framed-ipv6-route AVPs. Similarly, when the total number of framed routes specified in RADIUS AVPs exceeds the configured value for max-subnets-behind-ue, additional routes are ignored. From a routing perspective, the downstream traffic that is destined to a host residing behind a UE is delivered to the UE, and this UE acts as a gateway and delivers the packets received to the appropriate host. The workflow manager and other workflow service modules treat all traffic originating from a host located behind a UE router similar to upstream traffic received from a UE. Similarly, the MCC treats all downstream traffic destined to a host that resides behind a UE router similar to downstream traffic destined to a UE.

Table 12-3 zone gateway apn (Continued)

Object	Description
nas-identifier	NAS identifier to be populated in RADIUS accounting messages when this apn is being used. When the GGSN/PGW receives a Create PDP Context/Create Session request message for a given APN and if the APN is configured with a NAS-Identifier, the GGSN/PGW uses the configured NAS-Identifier in RADIUS Accounting messages. If the session uses APN mapping, the mapped APN will be used to select the NAS-Identifier. ***Note:*** *When using the* `nas-identifier` *parameter under the APN object, do not configure the* `nas-identifier` *parameter under the RADIUS server group.* ***Note:*** *If you do not configure the NAS-Identifier under the APN object or the RADIUS server group, the MCC gateway sends a string form of the RADIUS server peer's local loopback address in the NAS-Identifier attribute.* You can configure whether to include or exclude the NAS-Identifier attribute in the Accounting-Request (Start/Interim/Stop) data record templates associated with the RADIUS Primary interface and RADIUS Mirrored interface. `an3000-mcm-slot7cpu1(config)#` **`service-construct data-record-template pdngw accounting radius [start|interim|stop] <template-name> <attribute-name> presentFlag [conditional|no|yes]`**
name*	Name of this zone gateway APN.
network-context *	The network-context for this gateway apn.
network-initiated-deactivation-cause	GTP cause code used with network initiated session deactivation. Options include: none and reactivation-requested. When MCC operates as a stand-alone PGW or combined S/PGW, if the default bearer of the IMS PDN is deleted, the PGW sends a delete bearer request with cause reactivation Requested(8). The PGW deletes the bearer regardless of the delete bearer response or delete bearer response cause. Based on the configuration, MCC sends a delete bearer request with a reactivation request. As the default, no cause code is sent. The default bearer (session deactivation) is removed when the Operator initiates **subscriber clear**.

Table 12-3 zone gateway apn (Continued)

Object	Description
num-pgws-to-try-for-session-setup	Number of PGWs to try for reselection. 0 disables PGW reselection.
override-pco-username	Override the PCO CHAP or PAP username and allow user-name-type setting to be used instead. The default value is false. Setting this value to true overrides the PCO fields.
pco-ipv4-link-mtu-size-non-roaming	Send a configurable MTU size for the IPv4 link in the PCO IE of the Create Session Response message to the MME, based on a local PDN connection. MTU size for a non-roaming PDN connection to be sent in the PCO IE.
pco-ipv4-link-mtu-size-roaming	Send a configurable MTU size for the IPv4 link in the PCO IE of the Create Session Response message to the MME, based on a roaming PDN connection. MTU size for a roaming PDN connection to be sent in the PCO IE.
pco-pcscf-fully-qualified-domain-name	Fully qualified domain name of the default P-CSCF server.
pco-nonip-link-mtu-size-non-roaming	Send a configurable MTU size for the non-IP link in the PCO IE of the Create Session Response message to the MME, based on a roaming PDN connection.
pco-username-type	Choose CHAP or PAP username format to be sent to remote peer

Table 12-3 zone gateway apn (Continued)

Object	Description
pcscf-table	Configure list of P-CSCF addresses. The P-CSCF allows Operators to configure a P-CSCF table containing a list of P-CSCF IP address pairs (IPv4 and/or IPv6) on the MCC GGSN/PGW. MCC supports a maximum of 128 IPv4 and 128 IPv6 address pairs in each P-CSCF table. Note: If both an IPv4 and IPv6 address are configured as the primary and secondary P-CSCF addresses and the UE requests only one type of IP address (IPv4 or IPv6), the system returns only the requested IP address type in the PCO IE. ■ address-allocation-mode – Algorithm to use when allocating a pair of P-CSCF addresses from the P-CSCF table. – round-robin – By default, the MCC GGSN/PGW selects the P-CSCF IP address pair in a round-robin fashion – weighted-round-robin – Operators can configure the MCC GGSN/PGW to use a weighted-round-robin algorithm to select the P-CSCF IP address pair. Operators assign a weight to each P-CSCF IP address pair to distribute the IMS calls based on the P-CSCF capacity. ■ ip-v4-list – List of IPv4 address pairs. – primary-ipv4-address – Primary P-CSCF server IPv4 address. – secondary-ipv4-address – Secondary P-CSCF server IPv4 address. – weight – weight allocated for this pair. ■ ip-v6-list – List of IPv6 address pairs. – primary-ipv4-address – Primary P-CSCF server IPv4 address. – secondary-ipv4-address – Secondary P-CSCF server IPv4 address. – weight – weight allocated for this pair. ■ user-description – Description of P-CSCF table.
pdngw-list	List of PDNGWs that support this APN.
pgw-barred-time	Duration for which PGW is barred for selection (minutes). Zero disables barring
pgw-selection-mode	Priority is based on sorted NAPTR hostnames, Weight = 65535-NAPTR preference
qci-arp-stats-config-state	Enable/Disable APN QCI ARP Stats.
qci-profile	QCI Profile

Table 12-3 zone gateway apn (Continued)

Object	Description
select-barred-pgw-when-all-barred	Enable selection of barred PGW when all PGWs are barred,
send-pcscf-address-in-pco-always	Sends configured pair of IPv4 and IPv6 addresses in PCO IE even if not requested by the UE. Each P-CSCF address table is associated with an APN. When the MCC GGSN/PGW receives a Create PDP Context/Session Request for a P-CSCF address, the configured algorithm (round robin or weighted round robin) is used to select a pair of IP addresses from the P-CSCF table with which the APN is associated. Even if the UE does not request the P-CSCF address, Operators can configure the MCC GGSN/PGW to always send the configured P-CSCF IP address pair in the PCO IE as part of the initial Create PDP Context/Session Response message. Select true or false to configure whether the MCC GGSN/PGW always sends the P-CSCF IP address pair in the PCO IE. ■ false – (default) The GGSN/PGW only includes the P-CSCF server IP address(es) in the PCO IE of the Create PDP Context/Session Response message if the UE requests it. ■ true – The GGSN/PGW always includes the P-CSCF server IP address(es) in the PCO IE of the Create PDP Context/Session Response message, even if the UE does not request it.
service-rule-pco-action-association-table	The service-rule/pco-action association.
serving-nw-plmnid	Enable or disable PGW local-mapping control (sig-info-table and default PLMN ID) for the selection of the PLMN ID for an APN. You can enable or disable selection at both the start-session and mid-session. If local mapping is not enabled, then the sgsn-sgw-ip-table option in the prioritized list is skipped and the default PLMN ID value is ignored. ■ local-mapping-start-session [enabled|disabled]– Enable or disable PGW selection of the PLMN ID at session start. ■ local-mapping-mid-session [enabled|disabled]– Enable or disable PGW selection of the PLMN ID for a mid-session update/modify. For details about configuring the selection of MCC/MNC (PLMN ID) values on the PGW, see Configuring PGW Selection of MCC/MNC Values (page 12-45).

Table 12-3 zone gateway apn (Continued)

Object	Description
session-max-clearing-state	Enable or disable PDN session clearing for sessions present longer than the maximum time allowed. This object is used to clear subscribers that have been connected to a gateway for longer than the configured session-max-timer-duration, regardless of the idle/active state. If enabled, Operators can configure the session-max-timer-duration when a subscriber connects to a gateway.
session-max-timer-duration	Maximum timer duration for PDN sessions, in minutes. The range is 2 to 40,320 minutes. If a PDN session exists for longer than the session-max-timer-duration, the PDN session is cleared. This object is used to tear down a subscriber's session that has been connected to a gateway service for longer than the configured duration, regardless of whether the session is inactive. When the timer expires, the subscriber is cleared. This object enables Operators to control the session-max-timer-duration separately for each MVNO and/or configure a separate timer value for each MVNO ID and subclass ID on the PGW. MCC determines the session-max-timer-duration value as follows: ■ PGW/GGSN – MCC uses the timer in the MVNO profile if defined, otherwise MCC uses the timer defined in RADIUS, APN, or the gateway-common profile in that order. ■ SGW – MCC uses the settings defined in the gateway-common profile only if it is a stand-alone gateway. ■ SAEGW (SGW co-located PGW) – The SGW does not run the max-timer-duration for that session. The PGW runs the max-timer-duration for that session.
strip-trailing-null-pco-username-password	This feature is used to suppress trailing NULL characters in Protocol Configuration Options (PCO) PAP username/password fields before sending to the AAA server in the RADIUS Access-Request message. ■ disabled – (default) The GGSN/PGW does not suppress trailing NULL characters in PCO PAP username/password fields before sending to the AAA server in the RADIUS Access-Request message. ■ enabled – The GGSN/PGW suppresses trailing NULL characters in PCO PAP username/password fields before sending to the AAA server in the RADIUS Access-Request message.

Table 12-3 zone gateway apn (Continued)

Object	Description
tethering-control-params	Controls tethering for subscribers on the APN (also see Tethering Control for APNs (page 12-22)). ■ pcrf-down-tethering-entitlement – Enable | Disable tethering if the PCRF is down during session setup. ■ tethering-profile – Name of the tethering profile that specifies the strings that must exist in the acwentitlement VSA received from the PCRF. For information about configuring the tethering profile, see zone gateway profile tethering-profile (page 12-66).
transparent-data	Enable or disable the ability of the PGW to assign a transparent data value for sessions belonging to the APN. local-mapping [enable|disable] – Enable or disable PGW assignment of a transparent data value when the RADIUS Access-Accept Message does not return a transparent data value. The PGW uses sig-info-tables to map a PLMN ID or SGSN/SGW subnet to a transparent data value. See Configuring PGW Selection of a TMO-Transparent Data Value (page 12-47) for details.
uci-change-reporting-action	Configure Closed Subscriber Group (CSG) change reporting: ■ start-reporting-uci-csg – The MME starts reporting user CSG information when the UE accesses, enters and leaves a CSG cell. ■ start-reporting-uci-shc – The MME starts reporting user CSG information when the UE accesses, enters and leaves a subscribed hybrid cell (SHC). ■ start-reporting-uci-uhc – The MME starts reporting user CSG information when the UE accesses, enters and leaves an unsubscribed hybrid cell (UHC). See the *Affirmed Networks Charging Data Records Specification* for details. ***Note:*** *To configure more than one option for this parameter, separate each option by a comma (,).*
uli-change-reporting-action	The uli change reporting action (enum list).
use-aaa-provided-pgw-address-for-non-handover-session	Use AAA provided PGW Address (FQDN or Replacement OI) for DNS for non-handover session.

Table 12-3 zone gateway apn (Continued)

Object	Description
user-name-type	Choose the user-name-type (format) to be used if not present in PCO. Options include: apn|default-username|imsi|imsi-plus-apn|imsi-plus-apnnionly|msisdn|msisdn-plus-apn|msisdn-plus-apnnionly|imsi-nai-prefix-0|imsi-nai-no-prefix| ■ msisdn-plus-apn – The format of username sent is msisdn@apn. ■ default-user-name – The default user name at APN level if user name is not present in PCO. The MCC displays this information if the enum entered for user-name-type is set to default-username. ■ msi-nai-no-prefix – Remove the leading digit in front of the IMSI-NAI in the Username AVP.
user-password-type	Choose user password type to be sent to remote peer

apn-congestion-threshold

The PGW rejects the call if the APN is congested. The PGW sends the EPC (back-off timer) timer IE to the MME in the Create Session Response, so that the MME does not send calls during congestion. After receiving the Create Session Response, the MME may decide to redirect the call to a different PGW. The rejection cause code is set to APN Congested (113).

To enable this feature, enable the apn-congestion-threshold admin-state. By default, the APN state is **APN Not Congested.** The EPC timer value is fixed at 10 seconds. Set the critical, major, and minor crossing threshold.

Threshold detection occurs periodically, every 30 seconds. Therefore, there is a chance that a few calls are accepted, even if the threshold crosses the configured value.

Note: *This release supports critical-enter (crossing critical threshold) only. The APN load is calculated based on number of sessions only. Only the APN received in the session request is used to reject calls. The load on the mapped APNs is not considered. If the Operator configures both the SGW and PGW on the same cluster, then only calls at the PGW are counted against the APN.*

Tethering Control for APNs

Configuring `tethering-control-params` lets you control tethering for subscribers on the APN. The `tethering-control-params` object points to a tethering profile, which in turn identifies the specific strings that need to be in the acwentitlement VSA that is received from the PCRF in order to enable tethering.

You can also enable tethering for an APN regardless of which strings are in the acwentitlement VSA by pointing to a tethering profile that does not specify any strings. For details, see Configuring a Tethering Profile That Always Enables Tethering (page 12-67).

If you do not configure `tethering-control-params`, then tethering is enabled based on the presence of a GTETH, MHS1, or TETH string in the acwentitlement VSA, regardless of the APN for the subscriber, and tethering when the PCRF is down is controlled by `service-construct interface pcrf-charging test pcrf-down-tethering-entitlement [enable|disable]`, as described in the *Affirmed Networks Gx Interface Specification*.

zone gateway congestion-control slot-level

This section describes the congestion control configuration at the slot level. Operators can configure congestion-control parameters such as the admin state and the minor, major, and critical action to take to reduce threshold crossing congestion.

Note: *See Acuitas Performance Statistics for details on statistics.*

Configuring Objects

Configure the objects as follows:

Table 12-4 zone gateway congestion-control slot-level

Object	Description
admin-state	Enable/Disable the administrative state of this congestion configuration.
congestion-control-type	oc|cl|op|
service*	Infrastructure specific service. Options include: memory-management|
table*	Which stats table. Options include: system-memory-table|
action-critical action-major action-minor	Operators can display the cpu-stats and memory-stats. Configure an action (in 1, 5, or 15-minute increments) to take to reduce threshold crossing congestion. Action to take to reduce critical threshold crossing congestion – none|redirect-15-percent|redirect-30-percent|redirect-45-percent|redirect-60-percent|redirect-75-percent|redirect-100-percent|memory-profile|oom-killer|

zone gateway connectivity check

This feature enables Operators to verify that the gateway and external peer configuration is functioning properly without the use of external test equipment. Specifically, this feature enables Operators to generate calls to the gateway Device Under Test (DUT) and verify that all internal and external configurations and interfaces are working properly. This includes verifying the function of the internal configuration for gateway, gateway profile, PLMNID list, workflow, etc., as well as external interfaces to AAA, PCRF, OCS/OFCS, and other devices. The system generates a report indicating the status (success/failure) of each gateway and if necessary, lists the cause of the failure.

To configure the UE information and the DUT:

```
an3000-mcm-slot8cpu1# zone default gateway connectivity-check
subscriber-information imsi 17567547897899
an3000-mcm-slot8cpu1# zone default gateway connectivity-check
subscriber-information imei 917567665558999
an3000-mcm-slot8cpu1# zone default gateway connectivity-check
subscriber-information apn-name anp.name
an3000-mcm-slot8cpu1# zone default gateway connectivity-check
network-information device-under-test pgw
an3000-mcm-slot8cpu1# zone default gateway connectivity-check
network-information test-type setup-teardown
an3000-mcm-slot8cpu1# zone default gateway connectivity-check
network-information pgw-service NYpgw
commit
```

To run the test:

```
an3000-mcm-slot8cpu1# zone default gateway connectivity-check run
```

To check the results:

```
an3000-mcm-slot8cpu1# zone default gateway connectivity-check results
```

Configuring Objects

Configure the objects as follows:

Table 12-5 zone gateway connectivity-check

Object	Description
connectivity-check	The gateway connectivity check feature is supported on the following configurations: ■ SGW ■ PGW ■ GGSN ■ SGW-plus-PGW ■ PGW-3g (PGW services acting as GGSN) ■ unspecified Configure the network-information and subscriber-information for the gateway-connectivity check feature. After configuring this information, you can run the test and check the results. ***Note:*** *The gateway connectivity check feature is supported on IPv4 gateway addresses only. Statistics, logs, and CDRs continue to generate during the gateway connectivity check test. See Acuitas Performance Statistics for details on statistics.*

zone gateway enhanced-features-list

Enable or disable (default) NetLoc and RAN-NAS reporting. This feature supports reporting of the following information in CCR-U/CCR-T messages to the PCRF:

- Access Network Information from the access network in the following AVPs:
  - 3GPP-User-Location-Info AVP
  - 3GPP-SGSN-MCC-MNC AVP
  - 3GPP-MS-TimeZone AVP
  - User-Location-Info-Time AVP
- RAN-NAS-Cause indication – Reports detailed release or failure code information in the RAN-NAS-Release-Cause AVP.

NetLoc and RAN-NAS Release Cause are asserted in the Supported-Features AVP of the CCR-I message to specify that reporting of Access Network Information and RAN/NAS Cause are supported.

This feature allows RAN/NAS cause codes to be sent to the PCRF based on RAN-NAS cause information received in Update Bearer Response messages as well.

With this feature, when the PGW receives an Update Bearer Response with a RAN-NAS Cause indication, the PGW reports detailed release or failure code information in the Charging-Rule-Report –> RAN-NAS-Release-Cause AVP in a CCR-U to the PCRF.

As part of this feature, the RAN/NAS Cause information element (IE) is added to the Bearer Context IE within the Update Bearer Response message on the S5/S8 interface.

To configure this feature, you must use the following existing commands to enable RAN-NAS reporting in the enhanced-feature-list associated with the gateway APN:

Enable RAN-NAS reporting. By default, the parameters in this list are disabled:

```
an3000-mcm-slot7cpu1(config)# zone default gateway
enhanced-features-list <list-name> ran-nas enabled
```

Map the configured enhanced-features-list to an APN:

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name>
enhanced-features <list-name>
```

Configuring Objects

Configure the objects as follows:

Table 12-6 zone gateway filter-id-pco-action-association-table

Object	Description
filter-id	List of filter-id to pco-action associations
user-description	Description of filterId to PcoAction Association table

zone gateway enhanced-features-list

Enable or disable list of enhanced features.

This object enables the “Extended-BW-For-NR” flag in the Supported-Features AVP on the MCC. The “Extended-BW-NR” feature is sent in the Features-List-ID (2), within the Supported-Features AVP, to indicate support for the extended QoS-Information AVP values in Kbps.

Configuring Objects

Configure the objects as follows:

Table 12-7 zone gateway enhanced-features-list

Object	Description
extended-bw-nr	Enable/disable extended BW NR features.
name*	Features list name.
net-loc	Enable/disable to NetLoc feature.
ran-nas	Enable/disable to RanNas feature.

zone gateway epdg

Evolved Packet Data Gateway (ePDG) Service Configuration. See the *Affirmed Networks ePDG Administration Guide*.

zone gateway filter-id-pco-action-association-table

The Operator-specific Protocol Configuration Options (PCO) (UC time-of-day, redirection, SA) mechanism improves user experience and reduces unnecessary network traffic. The UE indicates support for the PCO during the initial LTE/eHRPD attach request. Upon PDN connection, the network responds with a PCO IE to the UE. The UE retrieves the PCO action value and displays a pop-up message to the user if necessary. The pop-up message may include instructions on modifying price plans or services. After parsing the PCO action value and displaying the pop-up message, the UE stops pinging the network until the next user-initiate request.

Operators use this PCO IE feature when the PGW sends a downlink GTP message. The following events may trigger an Operator to set this variable: re-direction rules, UC/hard stop occurs, or if the MCC receives an SA flag from the S6b server or if MCC receives a session termination RAR message from the PCRF server. Configure this feature as follows:

Configure the filter-id to the pco-action table:

```
an3000-mcm-slot7cpu1(config)# zone default gateway
filter-id-pco-action-association-table <table-name> filter-id
<filter-id-name> pco-action <pco-action-value>
```

Configure the service-rule to the pco-action table:

```
an3000-mcm-slot7cpu1(config)# zone default gateway
service-rule-pco-action-association-table <table-name> service-rule
<service-rule-name> pco-action <pco-action-value>
```

Configure the associated filter-id to the pco-action table to APN:

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <consumer>
filter-id-pco-action-association-table <table_name>
```

Configure the associated service-rule to the pco-action table to APN:

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <consumer>
service-rule-pco-action-association-table <table_name>
```

zone gateway ggsn

Configure a specific GGSN service to be run.

Configuring Objects

Configure the objects as follows:

Table 12-8 zone gateway ggsn

Object	Description
apn-list	Select the APN from the list for this GGSN.
distinguished-name	Distinguished name of the gateway. ■ domain-component ■ domain-prefix ■ subnetwork
gn-interface gp-interface	Configure the gn/gp interface parameters. The GTP T3 timeout value and N3 retry value are configurable on a per interface basis (S11, S5/S8, Gn/Gp, S4). ■ t3-timeout — controls the length of time that the gateway waits for an acknowledgment for a message (ms). ■ n3-count — specifies the number of times the gateway will retransmit a message if no acknowledgment is received.
gtpc-endpoint-params	The MCC supports the capability for the GTP-C protocol to transport over both IPv4 and IPv6, including the simultaneous assignment of IPv4 and IPv6 addresses for the peer to use. The following interfaces are supported: Gn/Gp, S1U, S5, S11. Complete the GTPc endpoint parameters for this gateway: ■ loopback-ip – Name of the primary loopback IP address. ■ network-context – Name of the network-context on which this GTPc endpoint is anchored. ■ port – The port to be used with the primary loopback IP address. ■ secondary-loopback-ip– Name of the secondary loopback IP address. ■ secondary-port – The port to be used with the secondary loopback IP Address.

Table 12-8 zone gateway ggsn

Object	Description
gtpu-loopback-list	List of GTP-U loopback addresses. This feature enables Operators to associate a local loopback list to a GGSN or SGW/PGW. See Dual Stack GTP-C (page 12-31) and Dual Stack GTP-U (page 12-31). Define one of the following loopback addresses: ■ S5/S8 or Gn network-context for the GTP-U endpoint on the PGW/GGSN ■ S5/S8 network-context for the GTP-U endpoint on the SGW ■ S1U network-context for the GTP-U endpoint on the SGW ***Note:*** *If the list of specific GTP-U loopback addresses has only IPv4 or IPv6 addresses, only that type is used for the sessions on the GW.*
gtp-peer	Configure the GTP peer with which the PGW, SGW, and GGSN communicate. A peer is identified by its IP address, Network Context ID, and the version of the GTP-C protocol it supports. A peer can be associated only with a gateway instance whose GTP-C endpoint address is in the same network context as the peer. Operators can configure the number of times the GW transmits a message if no acknowledgment is received. Use the following command to display status information for a GTP peer: `an3000-mcm-slot7cpu1#` **`show zone`** ***`<name>`*** **`gateway gtp-peer-status`** To filter the results with a network-context parameter, you must also include the gateway name and gateway peer in the CLI command. For example: `san3000-mcm-slot7cpu1# show` **`zone`** ***`<name>`*** **`gateway gtp-peer-status`** ***`<gateway-name>`*** ***`<peer-ip>`*** ***`<network-context>`***

Table 12-8 zone gateway ggsn

Object	Description
indirect-forwarding	Indirect forwarding can be enabled or disabled (default) for each SGW object. Indirect forwarding is a mechanism used to minimize data disruption during handovers where eNodeB X2 communication does not exist (for example, irat or S1 handovers). For example, during S1-based handovers, indirect forwarding allows the source and destination eNodeBs to transfer user data between each other via the SGWs using indirect forwarding GTP-u tunnels. If SGW relocation occurs, indirect tunnels may be established between the source and destination SGWs in order to pass the user data between the source and destination eNodeBs via the source and destination SGWs. ***Note:*** *The following restrictions apply to indirect forwarding:* ■ *Indirect forwarding between two SGWs hosted on the same MCC cluster is not supported.* ■ *Due to internal restrictions on the number of GTP-U TEIDs Affirmed Networks supports per subscriber, the number of GTP-U tunnels cannot exceed 14. All requests requiring more than 14 locally allocated TEIDs are rejected.* ■ *One user TEID is required for each traffic flow redirected. For example, 11 DL only bearers require 11 TEIDs; 7 UL+DL bearers require 14 TEIDs.* ■ *Bearers created and removed for purposes of indirect forwarding tunnels are not subjected to lawful interception. In other words, no X2 or X3 events are generated for bearers used for indirect forwarding tunnels.* See Figure 12-1 on page 12-32, Figure 12-2 on page 12-32, and Figure 12-3 on page 12-33.
max-clearing-state	By default, the max-clearing-state is enabled with a duration of 30 seconds. The MCC displays this object only if indirect-forwarding-tunnel is enabled. This object overrides the configured values in the GW common profile.
max-timer-duration	By default, the max-timer-duration is 30 seconds. The MCC displays this object only if indirect-forwarding-tunnel is enabled. This object overrides the configured values in the GW common profile.

Table 12-8 zone gateway ggsn

Object	Description
subscriber-bulk-deactivation-rate-control	The subscriber-bulk-deactivation-rate-control parameter provides a mechanism to control the rate at which Gateway services (SGW/PGW/GGSN) deactivate sessions. This feature enables Operators to control the number of subscriber deactivation requests as to not overwhelm the peer NEs with multiple requests. When Operators issue a subscriber-clear command, each Gateway service initiates deactivations for all subscribers belonging to that service, based on the **`subscriber-bulk-deactivation-rate-control`** parameter configured for each GW service. ■ is-static – Specify true or false to indicates the type of deactivation rate control. Specify if the deactivation rate is a constant value per second or is dynamically learned based on the peer response. – true – The GW service deactivates subscribers by sending the configured number of deactivation requests every second. – false – The GW service sends the configured number of initial requests and then dynamically learns the rate by deactivating one subscriber for every subscriber that completes deactivation. ■ rate – Specify the maximum number of subscriber deactivation requests to be sent per second when sending a burst of requests. For dynamic rate control, this is the initial burst rate after which the rate is dynamically learned. Operators can also use the **`subscriber clear-local`** command to clear subscribers locally without informing the peer NEs. This command supports the use of filter parameters. For example, the **`subscriber clear-local gateway-type-sgw`** command clears all subscribers belonging to all SGWs locally.

Dual Stack GTP-C

The MCC supports the capability for the GTP-C protocol to transport over both IPv4 and IPv6, including the simultaneous assignment of IPv4 and IPv6 addresses for the peer to use. The following interfaces are supported: Gn/Gp, S1U, S5, S11.

Dual Stack GTP-U

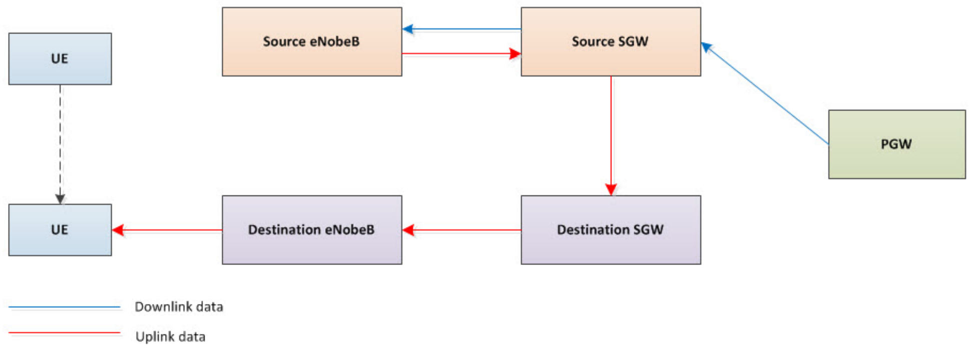
Operators can configure a GW with both IPv4 and IPv6 GTP-U endpoints, with support for sessions that can have an outer encapsulation of either IPv4 or IPv6, depending on the capability of the external network elements. If only an IPv4 or IPv6 address is specified, the remote peer must use that address to send the GTP-U packets.

If both an IPv4 and IPv6 GTP-U address are specified, the remote peer may use either address to send the GTP-U packets. If a GGSN or SGW/PGW does not associate to a list of local loopback endpoints, MCC falls back to the existing algorithm and uses all local loopback endpoint with GTP-U for load balancing.

Indirect Forwarding of Uplink and Downlink Data

Figure 12-1 shows the data path for downlink data using indirect forwarding.

Figure 12-1 Indirect Forwarding of Downlink Data



Note: *If SGW relocation does not occur, then only a single indirect tunnel exists between the source and destination eNodeBs using a single SGW. The SGW(s) chosen for indirect tunneling may not be the same SGW(s) that host the UE itself.*

Figure 12-2 shows the data path for uplink data using indirect forwarding. Indirect data forwarding for uplink data is optional.

Figure 12-2 Indirect Forwarding of Uplink Data

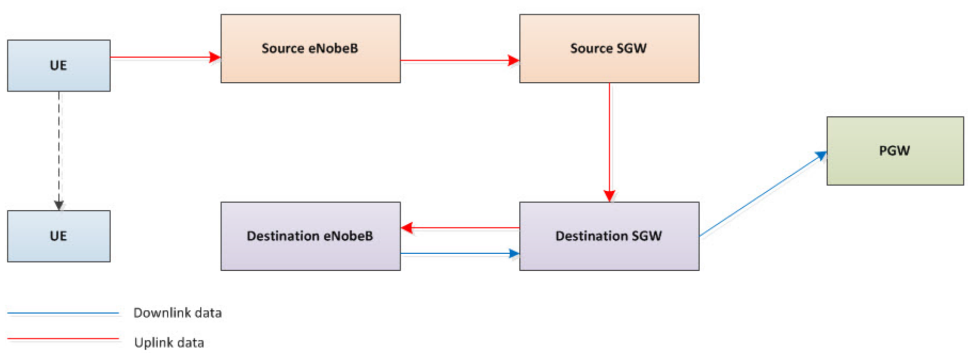
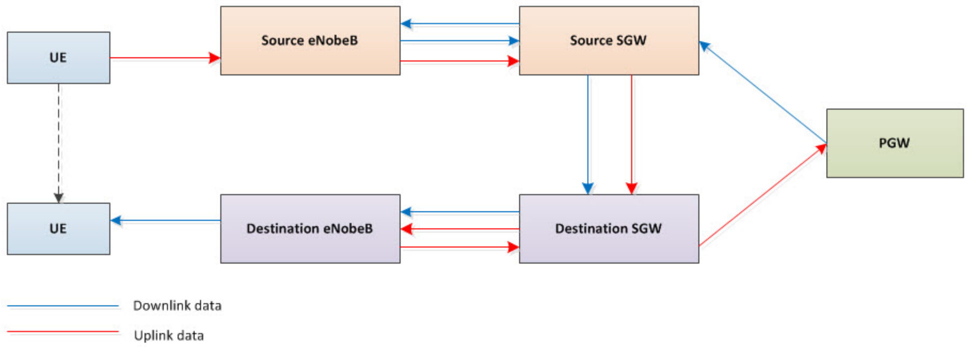


Figure 12-3 shows the data path of indirect tunnels for both uplink and downlink traffic.

Figure 12-3 Indirect Forwarding of Uplink and Downlink Data



In the example in Figure 12-3, each arrow represents 1 – 11 different tunnels; one for each bearer in the original call. Therefor, each indirect tunnel can have up to 22 individual tunnels; 11 for the uplink traffic and 11 for the downlink traffic.

If SGW relocation does not occur, then only one downlink and optionally one uplink indirect tunnel per bearer is created between the source and destination eNodeBs using a single SGW. It is not necessary for the SGW(s) chosen for indirect tunneling to be the same SGW(s) used to host the UE.

Since indirect forwarding tunnel subscribers are expected to be short-lived, MCC supports an override for the max-clearing-state (enabled/disabled) and max-timer-duration.

zone gateway gtp-peer

Configure the GTP with which the SGW, PGW, or GGSN communicates.

Configuring Objects

Configure the objects as follows:

Table 12-9 zone gateway gtp-peer

Object	Description
gtp-version	GTP version of interface with peer is-present-in-local-PLMN specify if peer is present in the same PLMN as the gateway.
network-context	Peer's network context
peer-ip-address	Peer's IP address.

zone gateway mcc-gigw

Mobile Content Cloud (MCC) GiLAN-Gateway (GiGW) configuration. See GiLAN-Gateway (page 6-5) for overview information.

Configuring Objects

Configure the objects as follows:

Table 12-10 zone gateway mcc-gigw

Object	Description
apn-list	List of APN names selected from list.
admin-state	Admin state of this Mobile Content Cloud (MCC) service.
default-apn-name	Subscriber Aware GiGW support for fixed line subscribers. In Subscriber Aware Mode, subscriber session information is learned from RADIUS Accounting messages via the GiGW session learner (configured with either RADIUS Server or RADIUS Proxy mode). Operators configure a default APN for the GiGW session learner in either RADIUS Proxy or RADIUS server mode. The GiGW session learner uses the default APN to set up the GiGW session when the RADIUS Accounting messages do not contain an APN. The PGW/GGSN uses the Realm and Username (NAI) as subscriber analyzer keys to provide differentiated services to fixed line subscribers. For example: `an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <subscriber-analyzer-name> key user-name-prefix <username-prefix> realm-name <realm>`
gateway-profile	Gateway profile used by this service.
home-plmnid-list	List of PLMNIDs considered as home.

Table 12-10 zone gateway mcc-gigw

Object	Description
mode	This feature supports the following modes of operation for the Gi Gateway: ■ Subscriber unaware mode (default) – In this mode, subscriber session information is not learned and all subscribers receive all offered Gi Gateway services. In other words, there are no differentiated services. Gi Gateway services such as DPI, analytics, and video optimization can be enabled in this mode. EDRs are not supported in this mode. See Subscriber Aware Mode (page 12-37). ■ Subscriber aware mode – In this mode, subscriber session information is learned from the RADIUS accounting/authorization interface. Then, using session information, subscriber user profile information is retrieved (from PCRF/AAA/LDAP/OCS/Static Workflow Configuration) and then the system provides all user subscribed value added services. In this mode, all supported Gi Gateway services can be enabled and can provide differentiated services determined by the subscriber user profile. See Subscriber Unaware Mode (page 12-37). ***Note:*** *The MCC "dummy" RADIUS Server does not support AAA functionality, therefore the PGW/GGSN must communicate with the real AAA server for AAA services.*
name*	Name of this mcc-gigw.
network-context	Network Context - selected from list of configured network contexts.
sd-diameter-client	Name of the Sd diameter client session learner interface on the GiGW.
session-idle-timeout	This parameter applies only to subscriber unaware sessions and indicates the session idle timeout in seconds. If the session is idle for this timeout value after deleting the last service-data-flow of the session, then the session is deleted.
sig-info-table	List of sig info tables. Operators perform optimizations (gzip compression in this case) on the GiGW for certain subscribers based on the SGSN MCC-MNC value. However, the SGSN MCC-MNC value is not mandatory and its value is not always sent. In this case, optimization is performed based on the incoming SGSN IP address (or range of SGSN IP addresses). This feature provides a mechanism to auto-fill the SGSN MCC MNC with a pre-configured value based on the SGSN IP address. This feature enables Operators to attach a data-profile to a subscriber based on the SGSN MCC-MNC value.
user-description	Enter a description for this GiGW.

Mirroring RADIUS Accounting-Request Messages

This feature adds the ability to mirror RADIUS Accounting-Request messages, sent from a RADIUS client, to the Gi gateway (GiGW). The MCC then forwards these Accounting-Request messages to an external RADIUS server. However, RADIUS Accounting-Response messages received from the external RADIUS server are forwarded to the RADIUS client without mirroring the response messages to the GiGW. For configuration information, see Mirroring RADIUS Accounting-Request Messages to GiGW (page 9-41).

Content-Filtering Policy

This feature enables the GiGW to apply a content-filtering policy based on the crbn-profiles selected by the PCRF during session setup /update. The GiGW is configured with the PCRF interface with the crbn-mode set to crbn-profile. The Charging-Rule-Base-Name AVP (CRBN) is mapped to the crbn-profile and a service-rule or group of service-rules. The PCRF installs the CRBNs/CRNs in CCA-I messages.

This feature allows a configuration to bind content-filter instances in the crbn-profiles and assign priorities to content-filter instances.

Added support to change the content-filter configuration by installing CRBNs as part of the RAR/CCA-U.

1. Enter a unique priority field in the content-filter instance (*optional*). The default is 1. Affirmed Networks recommends configuring a unique priority to avoid ambiguity in selection.

   ```
   services content-filter instance <cf-instance-name>
   priority            unique priority number
   ```

2. Bind the content-filter in the crbn-profile. In the data-profile, specify the http-proxy if selecting a content-filter instance ID. A content-filter can be configured in the crbn-profile without adding an http-proxy.

   Bind the content-filter instance ID to the crbn-profile (*optiona*l). See services crbn-profile (page 10-8).

   ```
   services crbn-profile <crbn-profile-name>
   content-filter                  bind content filter service instance
   ```

3. Add the content-filter instance currently applied on the session.

   The show subscriber pdn-session command displays only the content-filter selected through the active CRBN profiles and not the content-filter selected/applied through the data-profile.

   ```
   show subscriber pdn-session 131073
   content-filter-instance “”
   ```

Subscriber Aware Mode

In subscriber aware mode, the MCC-GiGW reuses the PDNGW (GGSN, PGW) subscriber architecture for MCC subscribers. The PDN Gateway subscriber state-machine is reused for the MCC subscriber session. This feature introduces a "session learning" phase where session profile information is learned. After the complete session profile is learned, the system triggers the MCC-GiGW session setup and the subscriber state-machine retrieves the user profile from the configured interface (LDAP/PCRF/OCS/RADIUS) and accordingly establishes a workflow session on the fastpath.

RADIUS Proxy Mode

In RADIUS Proxy mode, MCC acts as a proxy between GGSN/PGW and AAA. In this mode, MCC acts as a RADIUS Server for the GGSN/PGW and a RADIUS Client for AAA and is configured accordingly.

Affirmed Networks recommends passing RADIUS accounting messages through the MCC RADIUS Proxy and sending RADIUS authentication messages directly between the GGSN/PGW and AAA server. Affirmed also recommends that the PGW/GGSN wait for an Accounting Start Response message prior to forwarding the user data packets to the MCC.

RADIUS Server Mode

In RADIUS Server mode, MCC acts as a "Dummy" RADIUS Server and receives GGSN/PGW/AAA/DRA RADIUS feed(s) and sends dummy responses (with OK result code) back to the source. If required, dummy responses can be suppressed from the MCC.

Note: *The MCC "dummy" RADIUS Server does not support AAA functionality, therefore the PGW/GGSN must communicate with the real AAA server for AAA services.*

Affirmed Networks recommends providing a RADIUS Accounting feed (no need for RADIUS Auth messages) to the MCC RADIUS Server. Affirmed also recommends that the PGW/GGSN/AAA wait for an Accounting Start Response message prior to forwarding the user data packets to the MCC.

Subscriber Unaware Mode

The subscriber unaware MCC-GiGW is used to enable subscriber unaware VASs (video optimizations, DPI, etc). In this mode, the Operator is not required to supply a RADIUS feed. The workflow sessions are created on the fly upon receiving the first data packet from a subscriber and the same workflow services are applied to all subscribers. EDRs are not supported in this mode.

zone gateway pcscf-table

Configure a list of IPv4 and IPv6 P-CSCF addresses.

Configuring Objects

Configure the objects as follows:

Table 12-11 zone gateway pcscf-table

Object	Description
address-allocation-mode	Algorithm to use when allocating a pair of P-CSCF Addresses from the P-CSCF table. Options include: round-robin, weighted-round-robin
name*	P-CSCF Table name
user-description	Description of P-CSCF Table

zone gateway peer-selection flushBarredPeers

Flush Barred Peers

Configuring Objects

Configure the objects as follows:

Table 12-12 zone gateway peer-selection flushBarredPeers

Object	Description
ipAddress	IP address for this peer.
serviceName	Service name for this peer.

zone gateway pfcp-peer

PFCP Peer with which SGW, PGW, or user plane communicates.

- On the CPF, a PFCP peer configuration points to the UPF PFCP endpoint, which is associated with SGW and PGW services. See Figure 2-1 on page 2-1.
- On the UPF, the PFCP peer configurations point to the SGW Sxa PFCP endpoint and PGW Sxb PFCP endpoint, which are associated with the user-plane-service. See Figure 2-2 on page 2-2.

Configuring Objects

Configure the objects as follows:

Table 12-13 zone gateway pfcp-peer

Object	Description
name*	Name of the PFCP peer. On the CPF, the gateway services point to the PFCP Peer(s) on the UPF. On the UPF, the user plane services point to the PFCP Peer on the CPF.
network-context*	Network context to associate with this peer. The CPF and UPF have different network-contexts.
peer-ip-address*	IP address of this peer.
pfcp-version*	PFCP version on the interface shared with this peer. The default is V1.

zone gateway pgw

Configure a specific PGW service to be run.

Configuring Objects

Configure the objects as follows:

Table 12-14 zone gateway pgw

Object	Description
admin-state	
evolved-arp-support	Enable/disable Evolved Arp Support for 3G Subscribers.
gateway-profile	Name of gateway profile used by this service.
geo-redundant-group	Geographic redundancy group name.
ggsn-support	When enabled, a PGW can function as a GGSN and can be connected to an SGSN and process 3G session requests. The PGW supports internetworking with pre-release 8 SGSN, which means an SGSN can directly connect to a PGW and establish 3G sessions. The PGW supports session creation and hand-off between 3G and 4G networks. This parameter is disabled by default.

Table 12-14 zone gateway pgw

Object	Description
gn-interface gp-interface	■ t3-timeout — controls the length of time that the gateway waits for an acknowledgment for a message (ms). ■ n3-count — specifies the number of times the gateway will retransmit a message if no acknowledgment is received. ■ dscp-value – DiffServ Code Point value with which to mark all IP packets on this interface. – dscp_none (default) – dscp_0_be, dscp_01, dscp_02, dscp_03,...
guard-retry-count	When the guard timer expires, the PGW retransmits the GTP-C request message up to the maximum number of retransmission attempts configured.
guard-timeout	The MCC supports the GTP-C cause code **110** `Temporarily rejected due to handover procedure in progress` on the PGW and SGW as defined in 3GPP TS 29.274 and 3GPP TS 23.401. When an MCC PGW receives a GTP-C Create/Update/Delete Bearer Response message indicating a rejected request due to cause code **110** from an MME/S4-SGSN during PGW-initiated procedures, the PGW starts a locally configured guard timer. Set the guard-timeout at a higher interval than the T3 timeout value.
gtpc-endpoint-params	The GTPc endpoint parameters for this gateway. ■ dual-stack-behavior – When both the PGW and the ePDG are dual stack, specify how the IP type (IPv4 or IPv6) is chosen when the PGW originates messages that it sends to the ePDG. Options are: – initial-ip-type-used-by-peer – The PGW should use the IP type that was used in the Create Session Request from the ePDG. – ipv4 – The PGW should use IPv4. – ipv6 – The PGW should use IPv6. ■ loopback-ip – Name of the primary loopback IP address. ■ network-context – Name of the network-context on which this GTPc endpoint is anchored. ■ port – The port to be used with the primary loopback IP address. ■ secondary-loopback-ip– Name of the secondary loopback IP address. ■ secondary-port – The port to be used with the secondary loopback IP Address.

Table 12-14 zone gateway pgw

Object	Description
gtpu-loopback-list	List of GTP-U loopback addresses. This feature enables Operators to associate a local loopback list to a GGSN or SGW/PGW. Operators can configure a GW with both IPv4 and IPv6 GTP-U endpoints, with support for sessions that can have an outer encapsulation of either IPv4 or IPv6, depending on the capability of the external network elements. If only an IPv4 or IPv6 address is specified, the remote peer must use that address to send the GTP-U packets. If both an IPv4 and IPv6 GTP-U address are specified, the remote peer may use either address to send the GTP-U packets. If a GGSN or SGW/PGW does not associate to a list of local loopback endpoints, MCC falls back to the existing algorithm and uses all local loopback endpoint with GTP-U for load balancing. Define one of the following loopback addresses: ■ S5/S8 or Gn network-context for the GTP-U endpoint on the PGW/GGSN ■ S5/S8 network-context for the GTP-U endpoint on the SGW, ■ S1U network-context for the GTP-U endpoint on the SGW ***Note:*** *If the list of specific GTP-U loopback addresses has only IPv4 or IPv6 addresses, only that type is used for the sessions on the GW.*
home-plmnid-list*	List of PLMNIDs considered as home. MCC checks if the home-plmnid-list entry matches with the IMSI-MCC-MNC. If there is a match, then IMSI-MCC-MNC is sent in the 3GPP-GGSN-MCC-MNC AVP on all the interfaces (RADIUS, Gx, Gy, Gz etc..). If there is no match, MCC reverts to the existing design of sending the first entry in the home-plmnid-list.
lawful-intercept-profile	Name of Lawful Intercept Profile used by this service. See the *Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide*.
name*	Name for this PGW.
pfcp-profile	Name of PFCP profile used with this service.
push-to-db-threshold	Push To Database or Compression threshold 0.. 500000(0).

Table 12-14 zone gateway pgw

Object	Description
qos-update-wait-timer	Timer for configuring a wait time before sending the PGW Initiated QoS Update 5.. 120(60). Configuring the qos-update-wait-timer causes the PGW to wait for a small duration of time in case the MME sends a Modify Bearer Message. This allows Operators to configure AMBR and timer values for QoS upgrades from the PGW. This feature ensures notification to the PCRF if it is subscribed to QoS changes. This feature is useful in the following instances: ■ Sending UB message – All conditions are met for the UB to be sent and a timer is set waiting for a **`Modify Bearer Command`**. When the timer expires without receiving a `Modify Bearer Command`, MCC sends out the UB. ■ Comparison prior to sending UB message – All conditions are met for the UB to be sent and a timer is configured and waiting for a `Modify Bearer Command`. Before the timer expires, MCC receives a `Modify Bearer Command` from the MME. MCC compares the AMBR value in this message with the value configured on the PGW. If the configured value on the PGW is greater, MCC sends a UB with the configured APN AMBR value. ■ PLMN/APN is blacklisted – While checking for conditions during a hand-off, the PLMN ID is found in the blacklisted values in the subscriber-analyzer. This feature has the following limitations: ■ Although the QoS is configured in the control-profile, it must also be explicitly configured in the data-profile. ■ Only APN AMBR values are supplemented by this feature and sent in the Upgrade Bearer Request to the MME.
reject-inter-access-handover-request-no-session	Select true or false to reject Create Session Request with Handover Indication during 3gpp to non-3gpp access and vice-versa.
roaming-plmnid-list	List of PLMNIDs considered as roaming.
s2a-interface-gtpu-network-context	Network context containing the s2a GTP-U Interface.
s2b-interface-gtpu-network-context	Network context containing the s2b GTP-U Interface.
s5-interface s8-interface sxb-interface	Define the s5/s8 interface parameters. The GTP T3 timeout value and N3 retry value are configurable on a per interface basis (S11, S5/S8, Gn/Gp, S4). ■ dscp-value – DiffServ Code Point value with which to mark all IP packets on this Interface ■ t3-timeout — controls the length of time that the gateway waits for an acknowledgment for a message (ms). ■ n3-count — specifies the number of times the gateway will retransmit a message if no acknowledgment is received.

Table 12-14 zone gateway pgw

<table>
<tr><th>Object</th><th>Description</th></tr>
<tr><td>serving-nw-plmnid-mapping</td><td>You can configure the PGW to select the MCC/MNC (PLMN ID). You configure a selection mode and priority order. See Configuring PGW Selection of MCC/MNC Values (page 12-45) for details.

selection-mode – Specify the method for selecting the PLMN ID:

custom – Chooses the PLMN ID from a configured prioritized list of methods that includes a sig-info-table and the GTP message elements <code>serving-nw</code>, <code>rai</code>, and <code>uli</code> that come from the SGSN/SGW at the start of a session or at a mid-session update/modify.
radius-auth – Chooses the PLMN ID from the RADIUS Access Accept message. If the Access Accept message does not include the MCC/MNC values, then the configured or prioritized list of methods is used.

priority-order [sgn-sgw-ip-table | serving-nw | rai | uli] – Specify an ordered list of methods, listing the highest priority first. You must configure a priority-order if you set the selection-mode to custom.

sgn-sgw-ip-table – Uses the <code>plmn-id-mcc</code> and <code>plmn-id-mnc</code> SGSN/SGW subnet mappings in the sig-info-table that is assigned to the PGW.
serving-nw – Uses the serving network value in the GTP message from the SGSN/SGW.
rai – Uses the Routing Area Identity value in the GTP message from the SGSN.
uli – Uses the User Location Information value in the GTP message from the SGSN/SGW.

If you chose the radius-auth selection mode and do not configure a prioritized list, and the RADIUS Access Accept message does not provide the MCC/MNC values, the following default priority order is used: serving-nw, rai, sgsn-sgw-ip-table, uli.
default-plmnid-sgsn – Specify the PLMN ID to use when the methods in the prioritized list do not provide a value.from the SGSN.
default-plmnid-sgw – Specify the PLMN ID to use when the methods in the prioritized list do not provide a value from the SGW.
</td></tr>
<tr><td>sig-info-table</td><td>Configure information elements for GTP signaling messages.</td></tr>
</table>

Table 12-14 zone gateway pgw

Object	Description
subscriber-bulk-deactivation-rate-control	The subscriber-bulk-deactivation-rate-control parameter provides a mechanism to control the rate at which Gateway services (SGW/PGW/GGSN) deactivate sessions. This feature enables Operators to control the number of subscriber deactivation requests as to not overwhelm the peer NEs with multiple requests. When Operators issue a subscriber-clear command, each Gateway service initiates deactivations for all subscribers belonging to that service, based on the **`subscriber-bulk-deactivation-rate-control`** parameter configured for each GW service. ■ is-static – Specify true or false to indicates the type of deactivation rate control. Specify if the deactivation rate is a constant value per second or is dynamically learned based on the peer response. – true – The GW service deactivates subscribers by sending the configured number of deactivation requests every second. – false – The GW service sends the configured number of initial requests and then dynamically learns the rate by deactivating one subscriber for every subscriber that completes deactivation. ■ rate – Specify the maximum number of subscriber deactivation requests to be sent per second when sending a burst of requests. For dynamic rate control, this is the initial burst rate after which the rate is dynamically learned. Operators can also use the **`subscriber clear-local`** command to clear subscribers locally without informing the peer NEs. This command supports the use of filter parameters. For example, the **`subscriber clear-local gateway-type-sgw`** command clears all subscribers belonging to all SGWs locally.
sxb-pfcp-endpoint-params	Specify the sxb PFCP endpoint parameters for this Gateway ■ loopback-ip – Name of the primary loopback IP address. ■ network-context – Name of the network-context on which this endpoint is anchored. ■ secondary-loopback-ip – Name of the secondary loopback IP address.
sxb-pfcp-peer-list	For standalone PGW sessions and collocated SAEGW sessions. List of PFCP peer names - selected from list. ■ name* – Name of peer list. ■ user-description – Description of PLMNID.

Table 12-14 zone gateway pgw

Object	Description
transparent-data	When the RADIUS Access-Accept Message does not return a transparent data value, specify the sig-info-tables that the PGW uses to map a PLMN ID or SGSN/SGW subnet to a transparent data value. See Configuring PGW Selection of a TMO-Transparent Data Value (page 12-47) for details. ■ plmnid-mapping-table – Name of the sig-info-table that assigns a transparent data value based on a PLMN ID. ■ sig-info-table – Name of the sig-info-table that assigns a transparent data value based on the SGSN/SGW subnet. ■ default-value – Default value for the transparent data. This value is used when the RADIUS Access-Accept Message does not return the value and the sig-info-table mappings do not produce a value.
user-description	Description for this PGW.

Configuring PGW Selection of MCC/MNC Values

You can configure the PGW to select the MCC/MNC (PLMN ID) values on the PGW. You configure a selection mode and priority order.

To configure PGW selection of the MCC/MNC:

1 Choose a selection mode:

```
an3000-mcm-slot7cpu1(config)# zone default gateway pgw <name>
serving-nw-plmnid-mapping selection-mode [custom|radius-auth]
```

where:

Parameter	Description
custom	Chooses the PLMN ID from a configured prioritized list of methods that includes a sig-info-table and the GTP message elements `serving-nw`, `rai`, and `uli` that come from the SGSN/SGW at the start of a session or at a mid-session update/modify.
radius-auth	Chooses the PLMN ID from the RADIUS Access Accept message. If the Access Accept message does not include the MCC/MNC values, then the prioritized list of methods is used.

2 Configure the prioritized list of methods for the PLMN ID selection process, listing the highest priority first. You must perform this step if you chose the custom selection-mode.

```
an3000-mcm-slot7cpu1(config)# zone default gateway pgw <name>
serving-nw-plmnid-mapping priority-order
[sgn-sgw-ip-table|serving-nw|rai|uli]
```

where:

Parameter	Description
sgn-sgw-ip-table	Uses the `plmn-id-mcc` and `plmn-id-mnc` SGSN/SGW subnet mappings in the sig-info-table that is assigned to the PGW.
serving-nw	Uses the serving network value in the GTP message from the SGSN/SGW.
rai	Uses the Routing Area Identity value in the GTP message from the SGSN/SGW.
uli	Uses the User Location Information value in the GTP message from the SGSN/SGW.

If you chose the radius-auth selection mode and do not configure a prioritized list, and the RADIUS Access Accept message does not provide the MCC/MNC values, the following default priority order is used: serving-nw, rai, sgsn-sgw-ip-table, uli.

3 Assign the sig-info-table that has the `plmn-id-mcc` and `plmn-id-mnc` SGSN/SGW subnet mappings to the PGW.

```
an3000-mcm-slot7cpu1(config)# zone default gateway pgw <name>
sig-info-table <table-name>
```

4 Configure default PLMN ID values for the SGSN and SGW, which are used when the methods in the prioritized list do not provide a PLMN ID value:

```
an3000-mcm-slot7cpu1(config)# zone default gateway pgw <name>
serving-nw-plmnid-mapping default-plmnid-sgsn <plmnid>
default-plmnid-sgw <plmnid>
```

5 Enable/disable the local-mapping control (sig-info-table and default PLMN ID) for an APN. If local-mapping control is disabled for an APN, then the sgsn-sgw-ip-table option in the prioritized list is skipped and the default PLMN ID value is ignored.

You can enable local-mapping control for a session start and for a mid-session update/modify:

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <name>
serving-nw-plmnid local-mapping start-session [enabled|disabled]
local-mapping mid-session [enabled|disabled]
```

6 To modify the priority order:

a Remove the priority order, then the selection mode, and then commit:

```
an3000-mcm-slot7cpu1(config-pgw-<name>)# no
serving-nw-plmnid-mapping priority-order
```

```
an3000-mcm-slot7cpu1(config-pgw-<name>)# no
serving-nw-plmnid-mapping selection-mode
```

```
an3000-mcm-slot7cpu1(config-pgw-<name>)# commit
```

b Configure the selection-mode and then the prority order as shown in step 1 and step 2.

Configuring PGW Selection of a TMO-Transparent Data Value

You can configure the PGW to assign a transparent data value when the RADIUS Access-Accept Message does not return a transparent data value. The PGW selects a value from a sig-info-table with PLMN ID mappings or from a sig-info-table with SGSN/SGW subnet mappings. This is enabled at the APN level.

The PGW selects a transparent data value in the following order:

- RADIUS Access-Accept Message
- PLMN ID mappings
- SGSN/SGW subnet mappings
- Default value

To configure PGW assignment of the transparent data value:

1. Configure the sig-info-table that assigns a transparent data value based on the PLMN ID:

```
an3000-mcm-slot7cpu1(config)# zone default gateway sig-info-table
<table-name> key-type plmnid sig-info-entry <entry-name> plmnid-key
<plmnid> transparent-data <value>
```

2. Assign the PLMN ID – transparent data mappings sig-info-table to the PGW.

```
an3000-mcm-slot7cpu1(config)# zone default gateway pgw <name>
transparent-data plmnid-mapping-table <table-name>
```

3. Configure the sig-info-table that assigns a transparent data value based on the SGSN/SGW subnet:

```
an3000-mcm-slot7cpu1(config)# zone default gateway sig-info-table
<table-name> key-type ipsubnetmask sig-info-entry <entry-name>
source-ip-subnet <ipaddress> source-ip-mask <netmask> transparent-data
<value>
```

4. Assign the SGSN/SGW subnet – transparent data mappings sig-info-table to the PGW:

```
an3000-mcm-slot7cpu1(config)# zone default gateway pgw <name>
transparent-data sig-info-table <table-name>
```

5. Configure a default value for the transparent data value. This value is used when the RADIUS Access-Accept Message does not return the value and the sig-info-table mappings do not produce a value.

```
an3000-mcm-slot7cpu1(config)# zone default gateway pgw <name>
transparent-data default-value <value>
```

6. Enable or disable PGW assignment of the transparent data value for an APN when the RADIUS Access Accept Message does not return a value:

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <name>
transparent-data local-mapping [enable|disable]
```

zone gateway profile gateway-common

Configure common profiles used by local gateway services. The system uses PLMN ID lists to group PLMN IDs together for common treatment, for example, all expected home PLMN IDs.

The gateway common profile allows the user to specify gateway common attributes such as timer values, retransmission counts, gtp-peer parameters, inactivity and idle timers, and identify home and roaming subscribers.

Configuring Objects

Configure the objects as follows:

Table 12-15 zone gateway profile gateway-common

Object	Description
allow-interrat-handover-with-same-remote-fqteid	Specify whether the PGW should allow an Inter Radio Access Technologies (IRAT) Handover (SGW–>SGSN or SGSN–>SGW) when the PGW receives a Modify Bearer Request from the SGSN with the same TEID and IP Address as the SGW and there is a GTP version change (GTPv1 to GTPv2). enabled – When the PGW receives a Modify Bearer Request from the SGSN with the same TEID and IP Address as the SGW and there is a GTP version change (GTPv1 to GTPv2), the PGW allows the Inter Radio Access Technologies (IRAT) Handover (SGW–>SGSN or SGSN–>SGW). disabled – (default) When the PGW receives a Modify Bearer Request from the SGSN with the same TEID and IP Address as the SGW, the PGW considers the peer as unknown and discards the handover message. ***Note:*** *The allow-interrat-handover-with-same-remote-fqteid parameter does not apply to SGW->SGW handovers with the same TEID and IP address.*
allowed-plmnid	A policy to authorize the subscriber's PLMNID during session setup.
apn-matching-type	Simplifies APN configuration on the MCC by requiring only a single APN to be defined. That single APN is used to authorize session setup requests received with APN names having the same NI but different OI suffixes or no OIs. SGW, PDNGW, WAG, and ePDG services handle all APN names with the same NI but different OI strings, including malformed OI strings using a single APN configuration. The APN name used to report to external entities (for interfaces such as RADIUS, Gx, etc) is the same as the configured APN name. Options include: exact-match (default), ni-only, or ni-and-malformed-oi

Table 12-15 zone gateway profile gateway-common

Object	Description
await-coa-timeout	Timeout value (ms) used by the GW for closed sessions to wait for a CoA. The default is 0 (disabled). 0 .. 3600000(0).
change-notification	GTPv2-C Change Notification (CN) Request and Response messages are supported over the following interfaces as defined in 3GPP TS 29.274 V12.4.0: ■ S4 interface between the SGSN and SGW ■ S11 interface between the MME and SGW ■ S5/S8 interface between the SGW and PGW ***Note:*** *The Change Notification Request/Response message feature is enabled by default. If disabled, the system discards Change Notification messages.*
clear-subscribers-on-path-failure	When the system detects a path failure, it implements call clearing based on the clear-subscribers-on-path-failure setting as follows: ■ true – After the system detects the path failure, all calls that go to that remote peer are actively closed. ■ false – All calls are retained until they are manually cleared, cleared by another remote device, or the path recovers and restart processing clears the calls.
default-gtp-peer-classification	Used to indicate the default peer classification when not enough information is present to classify a remote GTP peer as local (same administrative domain) or remote (different administrative domain).
defer-delete-session-response	Defer sending delete session response until after delete processing is complete.
dhcp-relay-response-timeout	Timeout value used by the gateway (acting as a DHCP Relay) waiting for the user equipment (UE) to complete its DHCP transaction and obtain an IP (ms).
dhcp-server-response-timeout	Timeout value used by the GW (acting as a DHCP server) for DHCP responses from UE (ms).
eps-r99-priority-mapping-high	The ARP priority level range is 1 through 15 for 4G, and is 1 through 3 for 3G. Specify the 4G ARP priority level to which to map the 3G priority level 1, the highest priority level for 3G. By default, the 3G priority level 1 is mapped to the 4G priority level 5. This feature complies with TS 23.401 Annex E. ***Note:*** *The 3G priority value 3, the lowest priority, always maps to the 4G priority level 15.*
eps-r99-priority-mapping-medium	The ARP priority level range is 1 through 15 for 4G, and is 1 through 3 for 3G. Specify the 4G ARP priority level to which to map the 3G priority level 2. By default, the 3G priority level 2 is mapped to the 4G priority level 5. This feature complies with TS 23.401 Annex E. ***Note:*** *The 3G priority value 3, the lowest priority, always maps to the 4G priority level 15.*

Table 12-15 zone gateway profile gateway-common

Object	Description
gtp-echo-interval	The gtp-echo-interval indicates the amount of time (ms) to wait before initiating an Echo request.When Path Management procedures are initiated, Echo requests are transmitted periodically according to the configured value of gtp-echo-interval. GW instances include a Restart Counter in the Recovery IE in GTP-C messages when contacting the peer for the first time or upon a cluster restart.
gtp-echo-response-timeout and gtp-echo-transmit-count	The gtp-echo-response-timeout value indicates the amount of time (ms) to wait before transmitting an Echo request. Path failures are detected if there is no Echo Response received after exhausting *gtp-echo-retransmit-count* retries. Path failures are also detected if the peer changes its Restart Counter value. Upon detection of a path failure, all sessions associated with the peer's IP Address are deleted if *clear-subscribers-on-path-failure* in the gateway common profile is set to *true*.
gtpc-peer-aging-timer	The gtp-c peer aging timer value in seconds. When this timer is configured, MCC deletes GTP-C peer objects when they are not in use. A value of 0 (default) seconds indicates that the feature is disabled and therefore the aging timer does not start for unused GTP-C peers. As a result, peers are never deleted. ***Note:*** *This parameter does not apply to GTP-U peers. GTP-U peers use a hard-coded aging timer of 15 minutes.* SGW and PDNGW tasks create GTP peer objects for every GTP peer with which they communicate for GTP-C and GTP-U traffic. Unlike GTP-U peer objects, which are deleted periodically when they are not in use, GTP-C peer objects do not age and continue to reserve allocated memory indefinitely. GTP-C peer objects contain sequence number token managers that pre-allocate a large number of tokens, contributing to the memory consumption. This feature processes sessions counts and aging peers as follows: ■ Session counts per-peer – Gateway tasks maintain and increment a count of sessions and other internal procedures associated with the GTP-C peer contexts. ■ Aging peers – MCC provides the gtpc-peer-aging-timer parameter in the gateway common-profile to set the periodic timer value to be used to age out the GTP-C peers. When the count of sessions and the other internal procedure references associated with the GTP-C peer reaches 0 (zero), the MCC starts an aging timer for that GTP-C peer with the configured aging time. MCC deletes the GTP-C peer after the timer expires. All statistics associated with a GTP peer are lost when the peer object is deleted. If a new session or a new procedure is associated to a GTP-C peer for which the aging timer is running, then the MCC stops the timer.

Table 12-15 zone gateway profile gateway-common

Object	Description
idle-session-clearing-state	Enable or disable the idle-session-clearing-state. The idle-session-clearing-state enables Operators to clear subscribers who are idle for longer than the configured idle-timer-duration. If *idle-session-clearing-state* is enabled and the subscriber remains idle for the duration of the configured *idle-timer-duration*, the subscriber is cleared. For SGWs, the values are defined in gateway-common. For PGWs and GGSNs, the values are defined in the gateway apn object if specified, otherwise the system uses the gateway-common settings. Operators can specify the *idle-timer-duration*.
inactivity-timer-duration	Enables Operators to detect and potentially take action on subscribers that are inactive. Inactive is defined as no user data in the uplink or downlink direction for all PDN sessions and all bearers. For SGWs – The inactivity-timer-duration is defined in the gateway-common profile. For PGWs and GGSNs – The inactivity-timer-duration is defined in the gateway apn object if the value is non-zero, otherwise the value is defined in the gateway-common profile.The subscriber/PDN session is marked as Idle or Active as follows: A value of 0 indicates not specified. The subscriber/PDN session is marked as Idle or Active as follows: ■ Idle – If no activity occurs within the configured inactivity period (in seconds), the subscriber/PDN session is declared Idle. ■ Active – When a subscriber first connects to a GW, the timer is set and the subscriber/PDN session state is set to Active. In addition, if activity is detected on a subscriber/PDN session that had previously been declared Idle, the session is declared Active. ***Note:*** *The value defined in the gateway apn object takes precedence if the value is non-zero. Otherwise, MCC uses the value defined in the gateway-common profile.*

Table 12-15 zone gateway profile gateway-common

Object	Description
initiate-echo-requests	When GTP peers are configured and associated with a GW instance, the GW instance (by default) begins path management procedures for each peer. The *initiate-echo-requests* field in the gateway-common profile controls this behavior. If this value is set to false, the GW does not send any Echo Request messages to the peer. If this value is set to true for the GTP-C Gateway Profile object, the SGW/PGW/GGSN and GTP Proxy now only initiate GTP-C Echo Requests towards: ■ Configured peers (Static GTP peers) ■ GTP peers added based on the IP address encoded in the Sender F-TEID for Control Plane IE of the Create Session Request/Create PDP Context messages. ***Note:*** *For an SGW, the PGWs do not need to be configured. Since the PGW information is received from the MME, the SGW always allows session setups with PGWs. Path management procedures are initiated only with PGWs that are not collocated with the SGW.*
install-default-bearer-packet-filters-on-ue	Enable/disable sending Traffic Flow Template (TFT) packet filters installed for the default bearer to the UE.
non-matching-plmnid	Subscribers are classified as home or visiting based on the subscriber's PLMNID and the GW PLMNID configuration. The GW compares the PLMNID of the subscriber (show subscriber) to the PLMNIDs in the home and roaming lists and declares the subscriber as follows: ■ home – The subscriber's PLMNID matches the PLMNID in the home list. ■ visiting – The subscriber's PLMNID matches the PLMNID in the roaming list. A default classification in the gateway-common-profile is used to classify subscribers that do not match configured PLMNIDs.
overload-control max-arp-priority-level	Configure the ARP priority levels that are used to throttle control plane GTP messages on the PGW and SGW when the system is under congestion. You define a max-arp-priority-level for a specific type of overload state: ■ minor <*priority*> ■ major <*priority*> ■ critical <*priority*> The default for each state is 15, which causes all messages to be accepted even when there is an overload condition. For details, see overload-control max-arp-priority-level (page 12-56)

Table 12-15 zone gateway profile gateway-common

Object	Description
parallel-pdn-connections	Defines criteria under which parallel PDN connections (same IMSI-APN combination) need to be allowed. This feature adds support for multiple PDN connections/default bearers for the same IMSI/APN combination on the SGW/PGW/SAEGW. ■ allow-different-ebi – A Create Session Request with the same IMSI/APN combination of an existing session will be allowed if the default bearer ID is different from the existing session's default bearer ID. ■ allow-different-ebi-pdntype – A Create Session Request with the same IMSI/APN combination of an existing session will be allowed if the default bearer ID and PDN type are different from the existing session's default bearer ID and PDN Type. ■ disallow – (default) Multiple PDN connections/default bearers for the same IMSI/APN combination are not allowed.
reject-gtpc-messages-from-mismatched-src-and-fteid-ip	Policy to reject GTP-C messages received from source IP address that does not match sender's F-TEID IP address; this is applicable for non-handover messages that modify or delete a session.
reject-gtpc-messages-from-unconfigured-peers	Session requests received from peers that are not configured in the cluster are rejected if this parameter is set to *true*. If set to *false*, sessions are accepted.
session-max-clearing-state	Used to clear subscribers that have been connected (regardless of Idle or Active state) to a GW for longer than the configured session-max-timer-duration. If the session-max-clearing-state is enabled, a timer is set for the configured duration when a subscriber connects to a gateway. When the timer expires, the subscriber is cleared.
session-max-timer-duration	Maximum timer duration for PDN sessions, in minutes. The range is 2 to 40,320 minutes. If a PDN session exists for longer than the session-max-timer-duration, the PDN session is cleared. This object is used to tear down a subscriber's session that has been connected to a gateway service for longer than the configured duration, regardless of whether the session is inactive. When the timer expires, the subscriber is cleared. This object enables Operators to control the session-max-timer-duration separately for each MVNO and/or configure a separate timer value for each MVNO ID and subclass ID on the PGW. MCC determines the session-max-timer-duration value as follows: ■ PGW/GGSN – MCC uses the timer in the MVNO profile if defined, otherwise MCC uses the timer defined in RADIUS, APN, or the gateway-common profile in that order. ■ SGW – MCC uses the settings defined in the gateway-common profile only if it is a stand-alone gateway. ■ SAEGW (SGW co-located PGW) – The SGW does not run the max-timer-duration for that session. The PGW runs the max-timer-duration for that session.

Table 12-15 zone gateway profile gateway-common

Object	Description
stats-sync-interval	A configurable micro-checkpoint sync timer used to sync geo-redundant active and backup clusters so that interim session statistics are synced every stats-sync-interval (at the minimum). The default value of zero disables this feature and causes sessions to sync when any major event occurs on the session (for example, RAN events such as handoff, or offline charging events such as cutting of a CDR).
supported-features-list	Enable/disable various features for the service associated with this profile.
user-description	The user description for this gateway-common profile.

Identifying Roaming Subscribers

Based on the peer's status and the subscriber's local home/visiting status, a subscriber is classified as *Home, Roam-in,* or *Roam-out*. Roaming subscribers are defined as subscribers that traverse a non-home device and are identified by the SGW, PGW, and GGSN as follows:

- SGW determines if the PGW is in the same PLMN (connected via an S5 interface) or a different PLMN (connected via an S8 interface).
- PGW determines if the SGW is in the same PLMN (connected via an S5 interface) or a different PLMN (connected via an S8 interface).
- GGSN determines if the SGSN is in the same PLMN (connected via a Gn interface) or a different PLMN (connected via a Gp interface).

A peer is classified as *local* (same administrative domain) or *remote* (different administrative domain) based on the following criteria:

Classification Based on GTP Configuration

If the GTP peer is configured, the GTP peer is classified as local or remote based on the configuration. The interface to the peer is classified as S5/S8 or Gn/Gp as follows:

GW Type	GTP Peer	Interface Classification
SGW	Local PGW	S5
SGW	Remote PGW	S8
PGW	Local SGW	S5
PGW	Remote SGW	S8
GGSN	Local SGSN	Gn

GGSN	Remote SGSN	Gp

Classification Based on Serving Network in GTP Messages or Default Classification

If the GTP is dynamic, the GTP is classified based on the serving network in the GTP message or default classification as follows:

GW Type	Condition	Interface Classification
SGW	Default Classification	S5/S8 based on configuration
PGW	Serving network in Home PLMNID list	S5
PGW	Serving network in Roaming PLMNID list	S8
PGW	Serving network unknown	S5/S8 based on configuration
GGSN	Serving network in Home PLMNID list	Gn
GGSN	Serving network in Roaming PLMNID list	Gp
GGSN	Serving network unknown	Gn/Gp based on configuration

Classification Based on Interface Classification and the UE Classification

The UE roaming state is based on the interface classification and UE classification as follows:

GW Type	Interface Classification	UE Classification	Roaming State
SGW	S5	Home	N/A (Home)
SGW	S5	Visiting	Roam-in
SGW	S8	Home	Roam-out
SGW	S8	Visiting	Roam-in
PGW	S5	Home	N/A (Home)
PGW	S5	Visiting	Roam-in
PGW	S8	Home	Roam-out
PGW	S8	Visiting	Roam-in
GGSN	Gn	Home	N/A (Home)
GGSN	Gn	Visiting	Roam-in
GGSN	Gp	Home	Roam-out

GGSN	Gp	Visiting	Roam-in

Operators can configure GTP peers on the system and associate that peer with any of the gateway objects. A peer is identified by its IP address, Network Context ID and the version of the GTP-C protocol it supports. A peer can be associated only with a gateway instance whose GTP-C endpoint address is in the same network-context as the peer.

overload-control max-arp-priority-level

The overload-control max-arp-priority-level specifies the ARP priority levels that are used to throttle control plane GTP messages on the PGW and SGW when the system is under congestion.

Note: *Emergency call setups are not affected by this throttling feature.*

You define a max-arp-priority-level for a specific overload state:

- minor <priority>
- major <priority>
- critical <priority>

The default for each overload state is 15, which causes all messages to be accepted even when there is an overload condition.

When the system is overloaded, the thresholds are used as follows:

- Create session requests are rejected if the default bearer ARP priority level is greater than the max-arp-priority-level.
- Create bearer requests and bearer resource command messages are rejected if the session does not have any bearers with an ARP priority level that is less than or equal to the max-arp-priority-level.

Note: *All new session setups are rejected when the system is in overload state and the MCD table sync to standby process is in progress.*

An overload condition can be any of the following. If more than one overload condition exists, the highest overload state is used for throttling.

- PGW and SGW tasks exceed the CPU overload states:
  - minor — 80 percent

- major — 88 percent
- critical — 95 percent

- The number of PGW and SGW create-session-requests exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command:

```
an3000-mcm-slot7cpu1(config)# zone default gateway threshold
node-level call-performance pdngw
create-session-requests-[1|5|15]-minute-average major-enter <int>
minor-enter <int> critical-enter <int> admin-state enabled
```

- The node-level memory consumption percentage exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command:

```
an3000-mcm-slot7cpu1(config)# zone default system threshold
node-level memory-stats csm-avg-[1|5|15]min-memory-util minor-enter
<int> major-enter <int> critical-enter <int> admin-state enabled
```

- The node-level CPU consumption percentage exceeds the configured 1, 5, or 15 minute thresholds. These thresholds are configured with the following command:

```
an3000-mcm-slot7cpu1(config)# zone default system threshold
node-level cpu-stats csm-avg-[1|5|15]min-cpu-util minor-enter <int>
major-enter <int> critical-enter <int> admin-state enabled
```

zone gateway profile gateway-gtpc-peer-load-control-overload-control-profile

The gateway load control and overload control profile.

Configuring Objects

Configure the objects as follows:

Table 12-16 zone gateway profile gateway-gtpc-peer-load-control-overload-control-profile

Object	Description
name*	Profile name
receive-load-control-info-from-local-plmn-gtpc-peer	Allows Gateway to handle received Load Control Information from the GTPC Peer within the Local PLMN.
receive-load-control-info-from-roaming-plmn-gtpc-peer	Allows Gateway to handle received Load Control Information from the GTPC Peer outside of the Local PLMN
receive-overload-control-info-from-local-plmn-gtpc-peer	Allows Gateway to handle received Overload Control Information from the GTPC Peer within the Local PLMN
receive-overload-control-info-from-roaming-plmn-gtpc-peer	Allows Gateway to handle received Overload Control Information from the GTPC Peer outside of the Local PLMN
user-description	User description for this profile.

zone gateway profile gtp-u-path-management-service-profile

The gtp-u-path-management-service-profile is used to maintain data path status and handle path failures. This feature provides visibility into the health of the data path and enables the Operator to clear a PDN connection if a path failure occurs.

The MCC obtains peer connectivity status by sending an Echo Request (gtp-u-echo-request) and processing the Echo Response (gtp-u-echo-response-timeout) or response timeout. If the Echo Response times out, MCC declares a path failure and issues the anGtppPeerDown event. If the Operator sets clear-bearers-on-gtp-u-path-failure to true, the PDN connections on that peer are cleared.

Configuring Objects

See the *Affirmed Networks Gn/Gp Interface Specification* for details.

Configure the objects as follows:

Table 12-17 zone gateway profile gtp-u-path-management-service-profile

Object	Description
clear-bearers-on-gtp-u-path-failure	Policy to clear bearers when GTP-U path failure occurs.
gtp-u-echo-interval	GTP-U Echo Interval - amount of time to wait before initiating an echo request over GTP-U path(ms).
gtp-u-echo-response-timeout	GTP-U Echo Response Timeout Value - amount of time to wait before retransmitting an echo request over GTP-U (ms)
gtp-u-echo-retransmit-count	GTP-U Echo Retransmit Count - number of echo retransmissions to make before declaring a GTP-U path down event request.
initiate-gtp-u-echo-requests	Policy to send periodic echo requests to peers upon learning peer information.

zone gateway profile lawful-intercept-profile

The lawful intercept profile contains LI configuration applied at the gateway. See the *Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide* for details.

Configuring Objects

Configure the objects as follows:

Table 12-18 zone gateway profile lawful-intercept-profile

Object	Description
li-indicator-gx	
name*	Profile name.

zone gateway profile mvno-profile

Configure parameters for sessions associated with an MVNO.

Configuring Objects

Configure the objects as follows:

Table 12-19 zone gateway profile mvno-profile

Object	Description
session-max-clearing-state	Enable/disable PDN session clearing for sessions present longer than the maximum time allowed. This object is used to clear subscribers that have been connected to a gateway for longer than the configured session-max-timer-duration, regardless of the idle/active state. If enabled, Operators can configure the session-max-timer-duration when a subscriber connects to a gateway.
session-max-timer-duration	Maximum timer duration for PDN sessions, in minutes. The range is 2 to 36,000 minutes. If a PDN session exists for longer than the session-max-timer-duration, the PDN session is cleared. This object is used to tear down a subscriber's session that has been connected to a gateway service for longer than the configured duration, regardless of whether the session is inactive. When the timer expires, the subscriber is cleared. This object enables Operators to control the session-max-timer-duration separately for each MVNO and/or configure a separate timer value for each MVNO ID and subclass ID on the PGW. MCC determines the *session-max-timer-duration* value as follows: ■ PGW/GGSN – MCC uses the timer in the MVNO profile if defined, otherwise MCC uses the timer defined in RADIUS, APN, or the gateway-common profile in that order. ■ SGW – MCC uses the settings defined in the gateway-common profile only if it is a stand-alone gateway. ■ SAEGW (SGW co-located PGW) – The SGW does not run the max-timer-duration for that session. The PGW runs the max-timer-duration for that session.

zone gateway profile peer-overload-control-default-profile

Configure the default peer overload control profiles to use for gateway interfaces.

Configuring Objects

Configure the objects as follows:

Table 12-20 zone gateway profile peer-overload-control-default-profile

Object	Description
default-gtp-peer-overload-control-profile	Default peer overload control profile to use for the GTP interfaces
default-gtpp-peer-overload-control-profile	Default peer overload control profile to use for the GTPP interfaces
default-ocs-peer-overload-control-profile	Default peer overload control profile to use for the OCS (Gy) interfaces
default-ofcs-peer-overload-control-profile	Default peer overload control profile to use for the OFCS (Gz) interfaces
default-pcrf-peer-overload-control-profile	Default peer overload control profile to use for the PCRF (Gx) interfaces
default-radius-acct-peer-overload-control-profile	Default peer overload control profile to use for the RADIUS Accounting interfaces
default-radius-auth-peer-overload-control-profile	Default peer overload control profile to use for the RADIUS Authentication interfaces
default-swm-peer-overload-control-profile	Default peer overload control profile to use for the SWm interfaces. See the *Affirmed Networks ePDG Administration Guide*.
default-sta-peer-overload-control-profile	Default peer overload control profile to use for the STa interfaces. See the *Affirmed Networks ePDG Administration Guide*.

Peer Overload Control

The peer overload control function detects overload on peer servers and any potential network issues on the access network to peer servers. Specifically, this feature detects any potential transient failure condition by reducing the message load on the peer interface. After the peer link recovers, the system increases the messaging rate to its original rate. This feature also detects overload on peer servers by enforcing a maximum messaging rate.

The system monitors the health status (up or down) and processing capacity of peer nodes using different signaling interfaces. Operators can configure policing rates for each interface peer. Interfaces may be associated with a peer overload control profile that encapsulates the configuration and threshold for each congestion state.

Each peer node is categorized into different states of congestion based on the size of the outstanding transaction queue and/or failure rates. The system issues an alarm when an interface becomes congested. An interface is declared congested based a failure response with the following status: `System Failure, System Busy, or Remote Peer not Responding`.

During congestion, the messaging rate is reduced to avoid overloading the peer. Signaling messages (for real-time interfaces such as S5/S8 or RADIUS Authentication) are prioritized to maintain system consistency. For example, the SGW gives higher priority to GTP Delete Session request messages towards the PGW to avoid a stale session on the PGW.

For offline interfaces (for example, Gz and Ga), the system buffers CDR messages (both in memory and on disk) to be delivered in order to avoid discards. These buffered messages are retransmitted and the signaling rate is restored to the sustained rate after the congestion state clears. The messaging rate is increased gradually to ensure a sustainable rate. Operators can configure the maximum messaging limit based on the capability of the peer server.

Overload Control Example for Offline Interfaces

In this example, the Gz interface to the OFCS server is heavily congested.

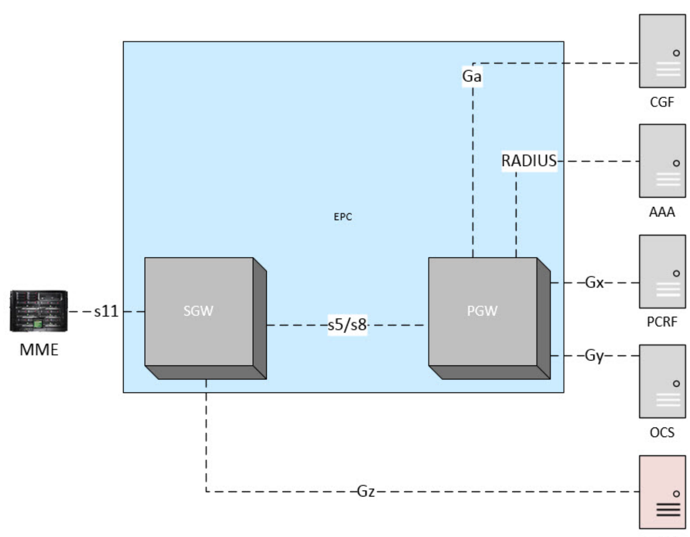
Figure 12-4 Overload Control Example for Offline Interfaces

1 The PGW (and its associated Gz Agent) reduces and stops any messaging to the external OFCS server during congestion.

2 CDRs are queued in memory after the PGW detects that the external OFCS server is heavily congested. This queuing has no impact on session processing.

3 After the number of records queued in memory exceeds a certain threshold, subsequent CDRs are written to disk.

4 After the peer state recovers, CDRs are retransmitted in the order in which they were queued (from both memory and from disk).

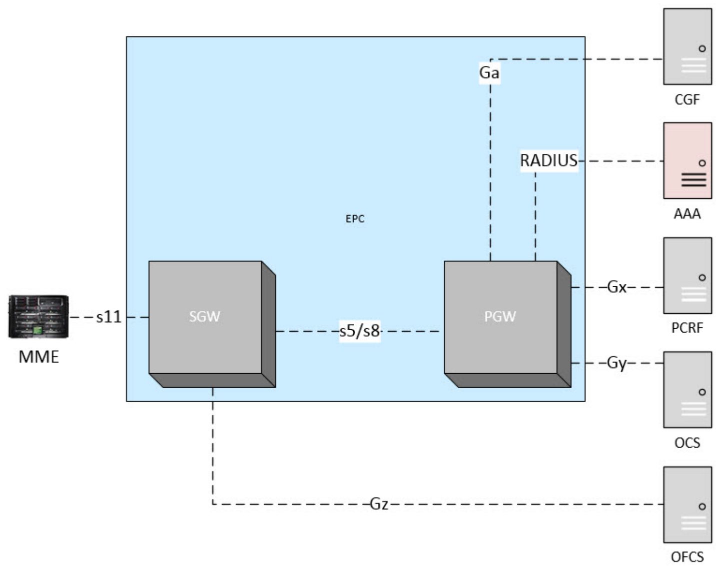


Overload Control Example for Real-Time Interfaces

In this example, the AAA peer for authentication is mildly congested.

Figure 12-5 Overload Control Example for Real-Time Interfaces

1. The PGW reduces the messaging rate (for example to 10%).
2. Some of the session setup requests on the PGW requiring AAA Authentication fail. The PGW sends the `Remote Peer not Responding` failure reason to the SGW. After the failure reaches the MME, it results in reduced signaling to the SGW.
3. The corrective actions are tiered based on the Peer's congestion state.
   - *Mild congestion* – Request for new sessions are rejected to meet the reduced rate.
   - *Heavy congestion* – All Authentication message are rejected.



zone gateway profile pfcp-profile

The pfcp-profile configuration specifies the timers, counts, and interval settings for the Sx associations (Sxa/Sxb/Sxc) and inter-node messaging and heartbeat monitoring.

Configuring Objects

Configure the objects as follows:

Table 12-21 zone gateway profile pfcp-profile

Object	Description
name*	Name of the PFCP profile.
pfcp-association-retry-interval	Amount of time in milliseconds the PGW/SGW UPF waits before re-initiating an Sx association if the association is down.
pfcp-clear-on-session-monitor-timeout	Clear a session when a session monitoring message times out. Session monitoring messages are used when **`pfcp-session-monitor-interval`** is configured. ■ Enabled (default) ■ Disabled

Table 12-21 zone gateway profile pfcp-profile

Object	Description
pfcp-guard-time-interval	Amount of time in milliseconds to wait before tearing down the Sx association on the C-Plane. On the user plane, this value is sent in the Graceful Release Period IE in increments of 2 seconds
pfcp-heartbeat-periodic-interval	Amount of time in milliseconds to wait before retransmitting a heartbeat request.
pfcp-node-response-timeout	Amount of time in milliseconds the PGW/SGW UPF waits before retransmitting a node level message. If it does not receive a response, the message is retransmitted based on the configured pfcp-node-retransmit-count.
pfcp-node-retransmit-count	Number of node-level message retransmissions to send before declaring a path down event request.

Table 12-21 zone gateway profile pfcp-profile

<table>
<tr><th>Object</th><th>Description</th></tr>
<tr><td>pfcp-session-monitor-interval</td><td>
For a given session on the Control Plane Function (CPF) or the User Plane Function (UPF), the amount of time in minutes to wait before checking to see if there is was any activity from its peer PFCP session (on UPF or CPF). The timer begins when a session is first successfully set up.
At the end of the interval, the CPF or UPF on which you configured <code>pfcp-session-monitor-interval</code> checks to see if there were any session-level messages between the CPF and UPF during the interval. If there were messages, that indicates that the session on the peer still exists, and the interval timer restarts.
If there were no session-level messages between the CPF and UPF during the interval, a monitoring message is sent to the peer:

If the interval timer is configured on the CPF, the monitoring message sent from the CPF is a dummy Sx Session Modification Request. If the UPF identifies a matching session, it sends a successful Session Modification Response to the CPF, and the timer restarts. If the UPF does not identify a matching session, it sends a Session Modification Response with a rejection cause code “Session Context Not Found”, and the CPF locally closes the session.
If the interval timer is configured on the UPF, the monitoring message sent from the UPF is a dummy Sx Session Report Request to the CPF. If the CPF identifies a matching session, it sends a successful Session Report Response to the UPF, and the timer restarts. If the CPF does not identify a matching session, it sends a Session Report Response with a rejection cause code “Session Context Not Found”, and the UPF locally closes the session.

For example, if the <code>pfcp-session-monitor-interval</code> is set to 60 on the CPF:

After the session has been set up for 60 minutes, the CPF checks to see if there were any session-level messages between the CPF and UPF during those 60 minutes, and finds no messages.
The CPF sends a dummy Sx Session Modification Request to the UPF.
The UPF looks for a matching session, but does not find one, so it sends a Session Modification Response with a rejection cause code “Session Context Not Found” to the CPF.
The CPF locally closes the session.

Note: If the interval time is 0 (zero), session monitoring is disabled even when pfcp-clear-on-session-monitor-timeout is set to enabled..
</td></tr>
</table>

Table 12-21 zone gateway profile pfcp-profile

Object	Description
pfcp-session-monitor-interval *(continued)*	***Note:*** *When pfcp-clear-on-session-monitor-timeout is disabled, sessions are not closed when there is no response from the peer or the request times out, but monitoring is enabled if the configured interval time is between 60 and 1440 minutes.* ***Note:*** *If the pfcp-session-monitor-interval is set to 0 (zero), session monitoring is disabled for sessions configured after this change. Subsequent changes to the pfcp-session-monitor-interval apply to new sessions. If the pfcp-session-monitor-interval was never disabled, changing the value applies only to existing sessions on the next session monitor timeout for that session.*

zone gateway profile qci-profile

Configure parameters for different qci values. The maximum number of supported qci profiles is 255.

Configuring Objects

Configure the objects as follows:

Table 12-22 zone gateway profile qci-profile

Object	Description
name*	QCI profile name.
user-description	Description of QCI Profile.
qci-config	Configure the following qci configuration parameters: idle-timer-duration – Idle timer duration for a dedicated bearer with this qci, in seconds (value must be multiple of 10); If no activity is detected on the bearer with this qci value within the idle timer duration, the bearer is deactivated. Options include: 30 .. 2160000(300). idle-timer-state – Enable/Disable idle timer. qci* – QCI value to be configured/modified [1 to 255].

zone gateway profile tethering-profile

The tethering-profile is used to enable tethering by specifying the strings that must be present in the acwentitlement VSA sent by the PCRF. A tethering profile is then applied to an APN by specifying the tethering-profile name in **`zone default gateway apn <name>`** (see zone gateway apn (page 12-7)).

Configuring Objects

Configure the objects as follows:

Table 12-23 zone gateway profile tethering-profile

Object	Description
name*	Tethering profile name.
mandatory-tethering-param	Identifies a string or multiple strings that must all be present in the acwentitlement VSA to enable tethering. To specify mutltiple strings, include multiple mandatory-tethering-param entries.
conditional-tethering-param	Identifies a string or multiple strings that enable tethering if at least one of the strings is present in the acwentitlement VSA. To specify mutltiple strings, include multiple conditional-tethering-param entries.

Configuring a Tethering Profile That Always Enables Tethering

If you want an APN to always be enabled for tethering, configure a tethering-profile that does not have any **`mandatory-tethering-param`** or **`conditional-tethering-param`** objects, or that only identifies empty strings (“”).

For example, either of the following tethering profiles results in tethering being enabled, regardless of the strings that are in the acwentitlement VSA:

```
an3000-mcm-slot7cpu1(config)# zone default gateway profile
tethering-profile profile1
```

or

```
an3000-mcm-slot7cpu1(config)# zone default gateway profile
tethering-profile profile2 mandatory-tethering-param “”
conditional-tethering-param “”
```

zone gateway session-load-balancer

Fine tune the distribution of new sessions across multiple (SGW/PDNGW) gateway processes.

Configuring Objects

Configure the objects as follows:

Table 12-24 zone gateway session-load-balancer

Object	Description
admission-control-send-reject	Specify if a create session/pdp context response should be sent with cause code RESOURCE-UNAVAILABLE when admission control rate is hit. ■ disabled – no response is sent. ■ enabled
admission-control-session-setup-rate	Number of create session/pdp context requests allowed per second.
admission-control-state	Enable or disable admission control.
apn-level-admission-control-state	Enable or disable apn level rate control.
max-absolute-load-val-per-gw-process	Maximum absolute load value per gateway process. Overall load is balanced across all gateway tasks when overload thresholds vary. The recommended value for this parameter depends on the allocated memory and vCPUs for CPM/CSM/DCM VM. Contact Affirmed Networks Technical Assistance Center (TAC) for the recommended value.
critical-high-level major-high-level minor-high-level	download-threshold – Critical, major, or minor high load state threshold for switching to lower (down) load state. up-threshold – Critical, major, or minor high load state threshold for switching to higher (up) load state.
critical-low-level major-low-level minor-low-level	download-threshold – Critical, major, or minor low load state threshold for switching to lower (down) load state. up-threshold – Critical, major, or minor low load state threshold for switching to higher (up) load state.

zone gateway sgw

Define the sgw parameters. The GTP t3-timeout and n3-count are configurable on a per-interface basis (S11, S5/S8, Gn/Gp, S4) as described in the following sections.

Configuring Objects

Configure the objects as follows:

Table 12-25 zone gateway sgw

Object	Description
admin-state	Enable or disable the admin state.
apn-list	List apn names - selected from list.
blocked-apn-list	Session setup for an APN with Network Identifier matching any Network Identifier present in this list will be rejected by the SGW. See zone gateway sgw blocked-apn-list (page 12-73)
default-offline-interface	Default offline interface configuration.
distinguished-name	Distinguished Name of the Gateway.
dscp-value	DiffServ Code Point value with which to mark all IP packets on this interface. ■ dscp_none (default) ■ dscp_0_be, dscp_01, dscp_02, dscp_03,...
edr-interface	Configure the edr Interface diameter|gtpp/local.
gateway-profile	Name of Gateway Profile used by this service.
geo-redundant-group	Geographic Redundancy Group Name.
gtp-peer-list	List gtp peer names - selected from list.
gtpc-endpoint-params	GTPc endpoint parameters for this Gateway. See zone gateway sgw gtpc-endpoint-params (page 12-73).
home-plmnid-list	List of PLMNIDs considered as home.
indirect-forwarding	Indirect Forwarding Tunnel Enable/Disable.
indirect-forwarding-tunnel-max-clearing-state	Enable/disable PDN Session Clearing for indirect forwarding tunnels present longer than the maximum time allowed

Table 12-25 zone gateway sgw

Object	Description
indirect-forwarding-tunnel-max-timer-duration	Maximum Timer duration for indirect forwarding tunnels, in seconds. If a PDN session has been present longer than the session maximum timer duration, the PDN session is cleared
override-ecgi-on-s8	Enable the capability to override the ECI.
paging-profile	Configure parameters for paging. See zone gateway sgw paging-profile (page 12-74).
procedure-duration	Procedure duration parameters. See zone gateway sgw procedure-duration (page 12-75).
qos-policy-for-qci-dscp-mapping	qos policy for qci to dscp mapping.
recovery	
roaming-plmnid-list	List of PLMNIDs considered as roaming.
port	The port to be used with the primary loopback IP address.
secondary-loopback-ip	Name of the secondary loopback IP address.
s1u-gtpu-loopback-list	List of GTP-U loopback addresses for s1 interfaces.
s11-interface s8-interface s4-interface s5-interface sxa-interface	■ t3-timeout – Controls the length of time that the gateway waits for an acknowledgment for a message (ms). ■ n3-count – Specifies the number of times the gateway will retransmit a message if no acknowledgment is received. ■ DiffServ Code Point value with which to mark all IP packets on this interface. – dscp_none (default) – dscp_0_be, dscp_01, dscp_02, dscp_03,...

Table 12-25 zone gateway sgw

Object	Description
override-ecgi-on-s8 enabled	Operators can configure the SGW to modify the E-UTRAN cell global identifier (ECGI) sent in the User Location Information (ULI) IE on the S8 interface. If this feature is enabled on the SGW and the SGW determines that the subscriber's PDN connection is remote, the SGW replaces the ECGI information received from the MME with random values and forwards these random ECGI values in the ULI IE to the PGW. The SGW uses these random ECGI values in all subsequent messages with the PGW over the S8 interface. However, the SGW uses the real ECGI values in CDRs and log messages. To determine whether a PDN connection is local or remote, the SGW compares the PGW IP address received in the Create Session Request message with a list of local PGW IP addresses configured on the SGW. If there is not a match and the PGW IP address is not in the list of GTP peers belonging to the local PLMN, the subscriber is considered remote. Verify that the default-gtp-peer-classification is set to remote. See Table 12-15 on page 12-48.
s5-s8-gtpc-endpoint-params	Configure an S5/S8 GTP-C endpoint on the SGW to separate the S1/S11 GTP-C network context from the S5/S8 GTP-C network context. ■ loopback-ip – Name of the primary loopback IP address. ■ network-context – Name of the network-context on which this GTPc endpoint is anchored. ■ port – The port to be used with the primary loopback IP address. ■ secondary-loopback-ip– Name of the secondary loopback IP address. ■ secondary-port – The port to be used with the secondary loopback IP Address.
s5-s8-gtpu-loopback-list	List of GTP-U loopback addresses for S5/S8 interfaces.
state	State of the SGW.

Table 12-25 zone gateway sgw

Object	Description
subscriber-bulk-deactivation-rate-control	Provides a mechanism to control the rate at which Gateway services (SGW/PGW/GGSN) deactivate sessions. This feature enables Operators to control the number of subscriber deactivation requests as to not overwhelm the peer NEs with multiple requests. When Operators issue a subscriber-clear command, each Gateway service initiates deactivations for all subscribers belonging to that service, based on the **subscriber-bulk-deactivation-rate-control** parameter configured for each GW service. ■ is-static – Specify true or false to indicates the type of deactivation rate control. Specify if the deactivation rate is a constant value per second or is dynamically learned based on the peer response. ■ true – The GW service deactivates subscribers by sending the configured number of deactivation requests every second. ■ false – The GW service sends the configured number of initial requests and then dynamically learns the rate by deactivating one subscriber for every subscriber that completes deactivation. ■ rate – Specify the maximum number of subscriber deactivation requests to be sent per second when sending a burst of requests. For dynamic rate control, this is the initial burst rate after which the rate is dynamically learned. Operators can also use the **subscriber clear-local** command to clear subscribers locally without informing the peer NEs. This command supports the use of filter parameters. For example, the **subscriber clear-local gateway-type-sgw** command clears all subscribers belonging to all SGWs locally.
sxa-pfcp-endpoint-params	Specify the sxa PFCP endpoint parameters for this Gateway ■ loopback-ip – Name of the primary loopback IP address. ■ network-context – Name of the network-context on which this endpoint is anchored. ■ secondary-loopback-ip – Name of the secondary loopback IP address.

Table 12-25 zone gateway sgw

Object	Description
sxa-pfcp-peer-list	For standalone SGW sessions. Listof pfcp peer names - selected from list. ■ name* – Name of peer list. ■ user-description – Description of PLMNID.
timestamp	Timestamp
user-description	User description for this gateway.

zone gateway sgw blocked-apn-list

Network Identifier part of APN name. Session setup for an APN with Network Identifier matching any Network Identifier present in this list is replaced by the SGW; affirmed:subpoint zoneGatewaySgwBlockedApnListSP.

zone gateway sgw gtpc-endpoint-params

This sections describes the GTPc endpoint parameters for this gateway.

Configuring Objects

Configure the objects as follows:

Table 12-26 zone gateway sgw gtcp-endpoint-params

Object	Description
loopback-ip	Name of the primary loopback IP address.
network-context	Name of the network-context on which this GTPc endpoint is anchored.
port	The port to be used with the primary loopback IP address.
secondary-loopback-ip	Name of the secondary loopback IP address.
secondary-port	The port to be used with the secondary loopback IP Address.

zone gateway sgw paging-profile

The SGW paging procedure is compliant with 3GPP TS 29.274 V11.0.0 and 3GPP TS 23.401 V11.0.0, including support for the EPS Bearer ID and Allocation/Retention Priority information elements (IEs) in the Downlink Data Notification message.

When the SGW receives a downlink data packet for a bearer associated with a UE in idle mode, the SGW triggers paging with the MME/SGSN by sending a Downlink Data Notification request. The SGW only triggers the paging procedure if the downlink data packet received is the first packet queued for that bearer. To accommodate subscribers who remain in an Idle state for a longer duration in extended paging mode, the SGW supports an increased maximum extended paging downlink buffer size of 100 packets.

Configuring Objects

Configure the objects as follows:

Table 12-27 zone gateway sgw paging-profile

Object	Description
arp-priority-level-threshold	Specifies the Allocation and Retention Policy (ARP) Priority Level value. Bearers with ARP Priority Level values equal to or less than this threshold value are marked as high priority bearers and are not subject to throttling. The SGW only applies throttling parameters received from the MME/SGSN to bearers with ARP Priority Levels greater than this threshold value.
gtp-page-timeout	Specifies the length of time (in milliseconds) to wait for a UE to be paged before discarding buffered data on the SGW. ***Note:*** *If the SGW receives a Downlink Data Notification Acknowledge message from the MME/SGSN indicating that the request has been temporarily rejected, the SGW buffers the downlink data packets for that bearer and waits the configured time limit (gtp-page-timeout) for the MME/SGSN to send the Modify Bearer Request message to continue with the procedure. Otherwise, the SGW releases the buffered downlink data packets after receiving the Delete Session Request message from the MME/SGSN.*

Table 12-27 zone gateway sgw paging-profile

Object	Description
guard-timeout	The MCC supports the GTP-C cause code **110** `Temporarily rejected due to handover procedure in progress` on the PGW and SGW as defined in 3GPP TS 29.274 and 3GPP TS 23.401. When an MCC SGW receives a Downlink Data Notification Acknowledge message indicating a rejected request due to cause code **110** from an MME/S4-SGSN during paging procedures, the SGW starts a locally configured guard timer, buffers all downlink user packets for the given UE, and waits to receive a Modify Bearer Request message. If the SGW does not receive the Modify Bearer Request message when the guard timer expires, the SGW releases the buffered downlink user packets. Set the guard-timeout at a higher interval than the GTP page timeout value.
max-downlink-pkts-to-buffer	Maximum number of downlink data packets to buffer per bearer when UE is being paged. This object is used *only* when the MME does not send the DLBufferPacketCount attribute to the SGW as part of the Downlink Notification Ack message in case of paging procedure.
maxcap-downlink-pkts-to-buffer	Maximum number of downlink data packets to cap when number of page packets are received from MME.
page-default-bearer-only	Specifies whether the SGW only triggers paging when downlink packets arrive on the default bearer. ■ true – The SGW only triggers paging with the MME/SGSN when downlink data packets arrive on the default bearer. ■ false – (default) The SGW triggers paging with the MME/SGSN when downlink data packets arrive on the default bearer and dedicated bearer.

zone gateway sgw procedure-duration

Configure the procedure-duration parameters for bin-count and bin-size.

This features applies to SAEGW procedure-level time histograms and is used to calculate how long a specific SAEGW procedure takes (on average) to complete. This object provides the Operator with the flexibility to configure the bin-size (size of histogram bins in milliseconds) and bin-count (number of bins to display).

For example, if bin-size is 50 ms and bin-count is 100, the histogram bins are 0-49 ms, 50-99 ms ….4950-4999 ms. For each SGW or PGW procedure (initial-attach etc), MCC displays the number of procedures that took time to finish in each bin.

For example:

```
an3000-mcm-slot7cpu1# show zone default gateway procedure-duration-stats

gateway procedure-duration-stats system pgw
```

```
PROCEDURE   DURATION IN
IDENTIFIER  MS           PROCEDURE NAME                          COUNT
------------------------------------------------------------------------
15 3050-3099ms network-initiated-deactivate-subsciber  1
25 0-49ms ocs-update 1834 (number ocs-update procedures that took time in between 0-49ms)
29 0-49ms duration-threshold-timeout 3767
40 0-49ms pcrf-rejoin-timer 1838

gateway procedure-duration-stats service pgw
       PROCEDURE   DURATION IN
NAME   IDENTIFIER  MS           PROCEDURE NAME                          COUNT
-------------------------------------------------------------------------------
MYPGW  15          3050-3099ms  network-initiated-deactivate-subsciber  1
MYPGW  25          0-49ms       ocs-update                              1834
MYPGW  29          0-49ms       duration-threshold-timeout              3767
MYPGW  40          0-49ms       pcrf-rejoin-timer                       1838

gateway procedure-duration-stats instance pgw
                           PROCEDURE   DURATION IN
NAME   CHASSIS  SLOT  INSTANCE  IDENTIFIER  MS          PROCEDURE NAME
COUNT
------------------------------------------------------------------------------------------
--------------
MYPGW  29 1 1 15          3050-3099ms  network-initiated-deactivate-subsciber  1
MYPGW  29 1 1 25          0-49ms       ocs-update                              1834
MYPGW  29 1 1 29          0-49ms       duration-threshold-timeout              3767
MYPGW  29 1 1 40          0-49ms       pcrf-rejoin-timer                       1838
```

Configuring Objects

Configure the objects as follows:

Table 12-28 zone gateway sgw procedure-duration

Object	Description
bin-count	Number of procedure duration histogram bins (between 1-100).
bin-size	Size of each individual procedure duration histogram bin in milliseconds (between 5-100).

zone gateway subscriber-trace

The subscriber trace function supports the ability to record data from signaling messages of an active call on different interfaces dedicated to the events of the traced subscriber. Trace is typically activated for a limited time period and used for analysis purposes.

Trace provides and supports the following troubleshooting activities:

- Determining the root cause of a malfunctioning mobile device
- Advanced troubleshooting activities
- Dropped call analysis
- EPC procedure validation

The subscriber trace function provides the following functions:

- Collects subscriber trace data on all supported interfaces for all GW network elements, SGW, PGW, and GGSN. The trace records contain information elements or signaling messages from control signaling and/or the characteristics of the user data.
- Collects different levels of details of trace data.
- Provides signaling as well as management based activation and deactivation of subscriber trace.
- Formats trace records in XML-based format as specified in TS 32.423 and stores that information in files.
- Transfers subscriber trace data files periodically to an external Trace Collection Entity (TCE) via SFTP.

Configuring Objects

Configure the objects as follows:

Table 12-29 zone gateway subscriber-trace

Object	Description
trace-collection-entity	Trace collection entity to use for this trace rule
trace-record-depth*	Trace depth measurement. Options include: minimum|medium|maximum|minimumWithoutExt|mediumWithoutExt|maximumWithoutExt| See Trace Depth Guidelines (page 12-78).

Trace data is characterized as follows:

Trace scope – Defines the Network Element types and signaling interfaces to trace. Affirmed supports the following NE types and interfaces:

- GGSN: Gn, Gi
- PGW: S5, Gx, S8, SGi
- SGW: S4, S5, S8, S11

Note: *For the Gi and SGi interfaces, the system traces diameter, RADIUS, and DHCP messages; however; user data packets are not traced.*

---

trace-record-depth – Defines the level of detail of trace information which to retrieve. Affirmed supports the following levels:

Note: *The maximum, minimum, and medium also allow for vendor-specific information to be traced and sent to the trace report file.*

Trace Depth Guidelines

Trace Depth Level	Description
Maximum or Maximum without Vendor-Specific Extension	Allows for the retrieval of all signaling interface messages within the trace scope in encoded format.
Minimum or Minimum without Vendor-Specific Extension	Allows for the retrieval of a decoded subset of the IEs contained in the signaling interface messages.
Medium or Medium without Vendor-Specific Extension	Allows for the retrieval of the decoded subset of the IEs contained in the signaling interface messages in the minimum level and a selected set of decoded radio measurement IEs.

Activating a Trace Session

A trace session is activated in an NE for a specific user (for example, IMSI) or mobile type (for example, IMEI) either by management activation or signaling activation. A globally unique ID is configured during the activation for each trace session for identification. A trace session is deactivated by management deactivation or signaling deactivation and a trace reference is used to identify the session to be deactivated. The trace reference is globally unique.

When a trace session is activated, the system creates one or more trace recording sessions within the trace session. Triggering events are configured as part of the activation procedure that determine when to start/stop a trace recording session.

Operators can also activate/deactivate a trace session through signaling messages. The parameters may arrive piggy-backed on control messages or through dedicated signaling messages. The SGW, PGW, and GGSN tasks support receipt of trace parameters in the following GTP-C messages:

- Create Session Request
- Trace Session Activation
- Trace Session Deactivation
- Create PDP Context Request
- Update PDP Context Request

The SGW and PDNGW tasks transfer the trace parameters to other NEs to which they interface according to the trace scope received. If the trace scope received does not include the NE that received the signaling message, the parameters are passed through and the session is not marked as an active trace session on that NE.

Trace Record File format

Affirmed uses the following conventions when the trace event agent task creates a trace data file:

```
<Type><Startdate>.<Starttime>
<SenderType>.<SenderName>.[<TraceReference>].[<TraceRecordingSessionRe
f>]
```

The following table defines the trace data file conventions.

Convention	Description
<Type>	Indicates if the file contains trace data for a single trace recording session or multiple sessions, where: ■ A indicates single trace recording session, single sender NE. ■ B indicates multiple trace recording sessions, single sender NE. ***Note:*** *Affirmed supports Type A files only.*
<Startdate>	Indicates the date of the first record in the trace file. The Startdate field uses the YYMMDD format, where: ■ YYYY indicates the year in four-digit notation. ■ MM indicates the month in two digit notation (01 - 12). ■ DD indicates the day in two digit notation (01 - 31).

Convention	Description
`<Starttime>`	Indicates the time of the first record in the trace file. The Starttime field uses the HHMMshhmm format, where: ■ HH indicates the two digit hour of the day (local time), based on 24 hour clock (00 - 23). ■ MM indicates the two digit minute of the hour (local time), is the sign of the local time differential from UTC (+ or -), in case the time differential to UTC is 0 then the sign may be arbitrarily set to "+" or "-". ■ hh indicates the two digit number of hours of the local time differential from UTC (00-23). ■ mm indicates the two digit number of minutes of the local time differential from UTC (00-59).
`<SenderType>`	The type of NE that recorded and sent the trace file. For example: `SenderType: SGW, SenderName:SGW1`
`<SenderName>`	The identifier of the NE that recorded and sent the trace file.
`[<TraceReference>]`	TraceReference is set if the type field is A. For type B, the trace reference is optional and is used when one trace file is created per trace session with multiple trace recording session. TraceReference is represented in hexadecimal format and is composed the of PLMN ID (MCC, MNC) and Trace ID. The PLMN identity consists of 3 digits for MCC followed by either a filler digit plus 2 digits from MNC (in case of 2 digit MNC) or 3 digits from MNC (in case of a 3 digit MNC). MCC and MNC are in BCD format. For example: If MCC: 405, MNC: 139 octet 1: 0x04 (MCC digit 2, MCC digit 1) octet 2: 0x15 (MNC digit 1, MCC digit 3) octet 3: 0x93 (MNC digit 3, MNC digit 2) For example: If the MNC is two digits (MCC: 405 and MNC 39) octet 1: 0x04 (MCC digit 2, MCC digit 1) octet 2: 0xF5 (MNC digit 1, MCC digit 3) octet 3: 0x93 (MNC digit 3, MNC digit 2) Example of filename: A20090928.2315+0200-SGW.SGW5. 13F23200056.125 This example shows a file produced by SGW <SGW5> on September 28, 2009, first trace record at 23:15 local with a time differential of +2 hours against UTC. The file contains trace data for the trace session with the trace reference 13F232000056 (where MCC is 312, MNC is 23, and Trace ID is 000056, all in hexadecimal format) and for the trace recording session with the reference 125.

For more information on the XML element structure of a trace XML file, see *33GPP TS 32.423– Trace Data Definition and Management*.

Traced Subscriber's Status

The status of individual subscribers is queried to determine the existence and state of the traced subscribers as well as statistics. The status may also be queried per NE. The system displays the following information:

- IMSI or IMEI
- Trace session configuration parameters
- Method of trace session activation
- Trace session state
- TCE interface status
- Statistics related to the interfaces with various configured TCEs configured may be queried to determine the state of the connection/file transfers. Some of the related information includes:
  - IP address of TCE
  - Statistics such as the number of files or trace records transferred, errors encountered during the transfer, etc

---

Note: *See the Acuitas Performance Statistics for details on statistics.*

The Affirmed subscriber trace implementation adheres to the following specifications:

- *3GPP TS 32.421– Trace Concepts and Requirements*
- *32.422– Trace Control and Configuration Management*
- *3GPP TS 32.423– Trace Data Definition and Management*

zone gateway system threshold

Generate a call performance table showing the average number of session setup requests over one, five, or fifteen minutes. Generate CPU threshold statistics or system memory threshold statistics. The proxy optimization overload control features add logic to a proxy task to bypass some images or videos to prevent the system from being overloaded. If the **minor-enter** value is not configured or is disabled, http-proxy uses the default value of 80.

This feature allows Operators to monitor utilization over all DPDK interfaces (both North/South and East/West) at the system level and generate an alarm when interface utilization crosses the configured thresholds.

The anThresholdCrossingAlarm alarm is generated as a minor, major, or critical alarm when utilization crosses thresholds for 1, 5, and 15 minute durations. When enabled, this configuration is applied to all DPDK-enabled interfaces in the cluster.

Note: *CPU usage for http-proxy services will not take effect based on the major-enter or critical-enter values. For example, if major-enter = 60 and minor-enter is not configured, http-proxy uses the default value of 80.*

Enable the optimization CPU overload control feature as follows:

```
(config)# services multi-protocol-proxy optimization-overload-control
admin-state administrative-state [enabled|disabled]
```

For more information on http-proxy, see the *Affirmed Networks Value Added Services Administration Guide*.

The following example applies to node-level thresholds:

1 Use the following command to enable monitoring for each 1, 5 and 15 minute average interface utilization duration:

```
(config)# zone default system threshold node-level intf-stats
[avg-1min-intf-util|avg-5min-intf-util|
avg-15min-intf-util] admin-state [enabled|disabled]
```

2 Use the following command to configure the critical, major, and minor crossing threshold for each duration configured in step 1. You can optionally enter a percentage threshold value to override the default value.

```
(config)# zone default system threshold node-level intf-stats
[avg-1min-intf-util|avg-5min-intf-util|
avg-15min-intf-util] [critical-enter|major-enter|
minor-enter] {percentage}
```

The following example applies to csm slot-level thresholds:

1 Use the following command to enable monitoring for each 1, 5 and 15 minute average interface utilization duration:

```
(config)# zone default system threshold slot-level cpu-stats
[csm-avg-1min-intf-util|csm-avg-5min-intf-util|
csm-avg-15min-intf-util] admin-state [enabled|disabled]
```

2 Use the following command to configure the critical, major, and minor crossing threshold for each duration configured in step 1. You can optionally enter a percentage threshold value to override the default value.

```
(config)# zone default system threshold slot-level cpu-stats
[csm-avg-1min-intf-util|csm-avg-5min-intf-util|
csm-avg-15min-intf-util] [critical-enter|major-enter|
minor-enter] {percentage}
```

Configuring Objects

Configure the objects as follows:

Table 12-30 zone gateway system threshold

Object	Description
admin-state	Enable or disable the administrative state of this threshold
critical-enter	Crossing critical threshold. If the critical-enter value is not configured, http-proxy uses the default value of 97.
major-enter	Crossing major threshold. If the major-enter value is not configured, http-proxy uses the default value of 87.
minor-enter	Crossing minor threshold. If the minor-enter value is not configured, http-proxy uses the default value of 80.
threshold-type	Options include: oc, cl, op, ap.

zone policyServer

This section describes how to configure the roaming profile and PCRF service.

Configuring Objects

Configure the pcrf name and gx interface name. Configure pcrf roaming-list name, plmnid, and type. Configure the IP address range associated with this roaming profile.

zone resource-control

This section describes the hardware resource control configuration.

Configuring Objects

Configure the objects as follows:

Table 12-31 zone resource-control

Object	Description
card-type	Card type to configure hardware resources
cpu-share	Share of the CPU for this service
service-type	Service type

zone gateway user-plane-profile

Configure the user-plane-profile. For overview information, see Chapter 2, CUPS PFCP Overview.

Configuring Objects

Configure the objects as follows:

Table 12-32 zone gateway user-plane-profile

Object	Description
name*	Name of this profile. Each UPF requires a unique user-plane-profile.
admin-state	Admin state of this profile. ■ Enabled (default) ■ Disabled
associated-ue-ip-sub-pool	Name of the UE IP sub-pool and its network-context to associate with this profile. On the CPF, the IP sub-pools are not advertised. On the UPF, the IP sub-pools are advertised and are local to a specific UPF. ■ ipsubpool* – IP Sub Pool for this user plane profile. ■ networkcontext* – Network Context. ■ tam-node-id – Traffic Anchoring Mode (TAM) node ID associated with this IP-sub-pool in the UPF.
mec-local-break-out	Enable or disable Multi-Access Edge Computing (MEC) local breakout for this profile. ■ Enabled ■ Disabled (default) MEC local breakout can be configured on a standalone CUPS SGW CPF. When a user-plane-selector type of workflow subscriber-analyzer is matched and it specifies a user-plane-profile that enables local breakout, the data-profile from the matching workflow-profile type of subscriber-analyzer is sent to the UPF. Data packets matching the data-profile undergo a local breakout. Local breakout allows traffic that only needs local applications to be serviced at the operator edge site. This reduces latency for local applications, and reduces back-haul requirements.

Table 12-32 zone gateway user-plane-profile

Object	Description
pfcp peer	PFCP peer to assign to this profile. ■ mode – PFCP Mode. TAM mode is used to anchor the loopbacks and IP sub-pools on a specific node for Sx sessions. Options include: – pfcp-prime (default) - This option supports the Affirmed Networks proprietary Vendor-specific IEs in the session-related procedures. – pfcp-std (not supported) ■ peer – Name of the peer.

zone gateway user-plane-service

User plane service configuration for each UPF. The user-plane-service is similar to gateway services on the CPF. This configuration also requires a network-profile.

Configuring Objects

Configure the objects as follows:

Table 12-33 zone gateway user-plane-service

Object	Description
admin-state	Administrative state of this service. ■ Enabled (default) ■ Disabled – MCC deletes all sessions on the user plane.
name*	Name of this user-plane-service.
network-profile	Name of the network profile to associate with this user-plane-service. This object is configured using service-construct network-profile. The UPF requires at least one network-profile configuration for each of the following interface types. This enables Operators to choose the network contexts for the various UPF network interfaces. ■ S1-U - GTP-U ■ S5-S8 - GTP-U SGW ■ S5-S8 - GTP-U PGW ■ S5-S8 - Lawful Intercept (LI)

Table 12-33   zone gateway user-plane-service

Object	Description
pfcp-endpoint-params	PFCP endpoint parameters for this service: ■ loopback-ip - Name of the primary loopback IP address ■ network-context - Network context (SX) on which this endpoint is anchored ■ secondary-loopback-ip - Name of the secondary loopback IP address
pfcp-link-down-queue-messages	Enable/disable message buffering. When enabled, MCC buffers data usage messages on the UPF when the PFCP link is down between the UPF and CPF. The UPF transmits the buffered data usage messages (as Sx Session Report messages) when the PFCP link is restored. This allows the CPF to more accurately report data usage for CDRs and offline interfaces even with intermittent PFCP link failures. ■ Enabled ■ Disabled (default)
pfcp-peer-list	PFCP peer for this service. Name* – Name of this pfcp-peer-list user-description – PLMNID description.
pfcp-profile	Name of the PFCP profile used by this service.

zone system threshold node-level

The threshold node-level lets you configure thresholds for the average memory and CPU utilization over one, five, and fifteen minutes.

When these thresholds are exceeded, the PGW/SGW is considered to be in an overload condition. See overload-control max-arp-priority-level (page 12-56) for details on throttling control plane GTP messages on the PGW and SGW when the system is in an overload condition.

Configuring Objects

Configure the objects as follows:

Table 12-34 zone system threshold node-level

Object	Description
memory-stats	Set thresholds (percentages) for the average memory utilization. csm-avg-[1|5|15]min-memory-util – Threshold for average memory utilization over one, five, or fifteen minutes: ■ minor-enter <int> – Utilization percentage that indicates minor overload. ■ major-enter <int> – Utilization percentage that indicates major overload. ■ critical-enter <int>– Utilization percentage that indicates critical overload. ■ admin-state [enabled|disabled]
cpu-stats	Set thresholds (percentages) for the average CPU utilization. csm-avg-[1|5|15]min-memory-cpu – Threshold for average CPU utilization over one, five, or fifteen minutes: ■ minor-enter <int> – Utilization percentage that indicates minor overload. ■ major-enter <int> – Utilization percentage that indicates major overload. ■ critical-enter <int>– Utilization percentage that indicates critical overload. ■ admin-state [enabled|disabled]

zone wag

Wifi Access Gateway (WAG) service configuration.

Configuring Objects

Configure the objects as follows:

Table 12-35 zone wag

Object	Description
admin-state	Admin state of this WLAN Access Gateway (WAG) service
gateway-profile	Name of Gateway Profile used by this service
home-plmnid-list	List of PLMNIDs considered as home
name*	name - string

Table 12-35 zone wag

Object	Description
s2a-interface-gtpu-network-context	Network context containing the s2a GTP-U Interface
user-description	

CUPS Sx Interface

This section contains the following information used to control and monitor the Sx interfaces used when the MCC is deployed as a CUPS system:

Interface Overview

This section describes the Packet Forwarding Control Protocol (PFCP) used between the following interfaces in the Evolved Packet Core (EPC) Control and User Plane Separation (CUPS) architecture:

- Sxa Interface – Standalone SGW-C to SGW-U.
- Sxbc Interface – Standalone PGW-C to PGW-U. The Traffic Detection Function (TDF) is embedded in the PGW-C/U node. It is not supported as a standalone node; therefore, the Sxc interface functions as part of the Sxb interface.
- Sxabc Interface – Collocated SAEGW-C to SAEGW-U.

Note: *The SAEGW is a collocated/combined PGW/SGW.*

PFCP is the signaling protocol used on these reference points. Sx Node Related Procedures support the use of standard PFCP information elements (IEs), as defined in 3GPP TS 29.244 V14.4.0, and Sx Session Related Procedures support the use of PFCP with proprietary vendor-specific IEs for session-related messages.

Figure 13-1 illustrates the PFCP protocol stack for the Sx interface. The UDP Destination Port number for a PFCP Request message is 8805, which is the registered port number for PFCP.

Figure 13-1 PFCP Protocol Stack for the Sx Interface

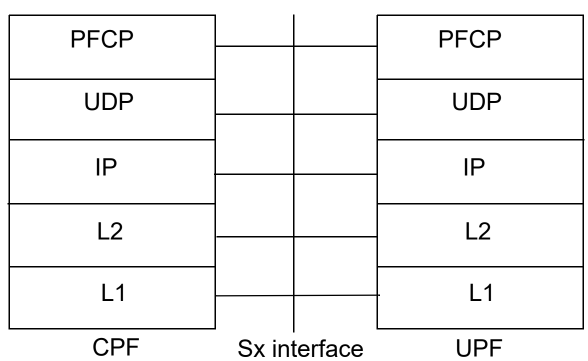
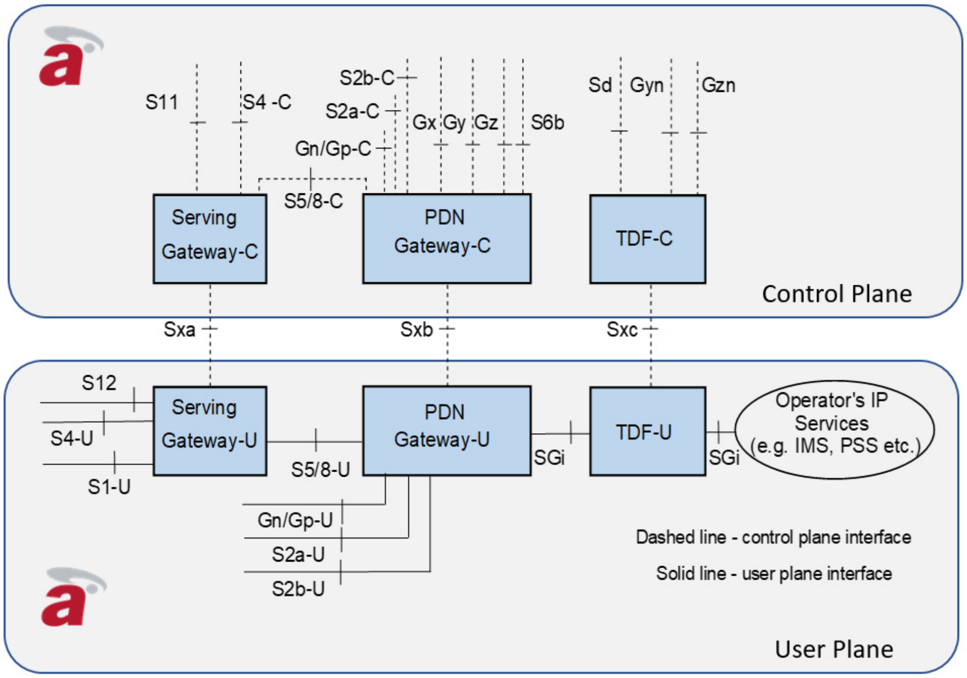


Figure 13-2 illustrates the Sx interfaces used in a CUPS architecture.

Figure 13-2 CUPS Sx Interfaces - Standalone SGW and PGW



References and Specifications

The system supports the following references and specifications for the Sx interface:

- 3GPP TS 23.214 V15.2.0 (2018-03): *Architecture enhancements for control and user plane separation of EPC nodes*
- 3GPP TS 29.244 V14.4.0 (2018-06): *Interface between the Control Plane and the User Plane of EPC Nodes*

Sx Message Definitions

This section lists the supported PFCP message definitions used on the Sx interfaces and lists the IEs contained in each message type. These message definitions are in association with Sx Node Related Procedures and Sx Session Related Procedures, as defined in 3GPP TS 29.244 V14.4.0. Sx Session Related Procedures use PFCP with proprietary vendor-specific IEs.

Table 13-1 lists the supported PFCP messages used on the Sx interfaces.

Table 13-1 Supported PFCP Sx Interface Message Definitions

Message Type	Decimal Code
Sx Node Related Procedures	
Sx Association Setup Request	5
Sx Association Setup Response	6
Sx Association Update Request	7
Sx Association Update Response	8
Sx Association Release Request	9
Sx Association Release Response	10
Sx Heartbeat Request	1
Sx Heartbeat Response	2
Sx Session Related Procedures	
Sx Session Establishment Request	50
Sx Session Establishment Response	51
Sx Session Modification Request	52
Sx Session Modification Response	53
Sx Session Deletion Request	54
Sx Session Deletion Response	55
Sx Session Report Request	56
Sx Session Report Response	57

Sx Node Related Procedures

Sx Node Related Procedures use standard PFCP IEs, as defined in 3GPP TS 29.244 V14.4.0.

Sx Association Setup Request/Response

An Sx association is set up between the Control Plane Function (CPF) and User Plane Function (UPF) prior to establishing Sx sessions. Only one Sx association is setup between a given pair of CPF and UPFs. The Sx Association Setup Request is initiated by either the CPF or the UPF. The Sx Association Setup Response is sent as an answer to an Sx Association Request message. The Sx Association Setup Request/Response includes:

- Node ID (IPv4/IPv6), which identifies the Sx association.
- List of optional features that the node supports, such as TRST, FTUP, TREU and EMPU.

If the peer rejects the request, it sends an Sx Association Setup Response with an appropriate error cause.

Table 13-2 lists the supported Sx Association Setup Request/Response information elements.

Table 13-2 Sx Association Setup Request/Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)
Node ID	Mandatory	This IE contains the unique identifier of the sending node.	Node ID	60
Recovery Time Stamp	Mandatory	This IE contains the time stamp when the node was started. See clause 19A of 3GPP TS 23.007 [24].	Recovery Time Stamp	96
UP Function Features	Conditional	This IE is present if the UPF sends this message and the UPF supports at least one UP feature defined in this IE. When present, this IE indicates the features the UPF supports.	UP Function Features	43
CP Function Features	Conditional	This IE is present if the CPF sends this message and the CPF supports at least one CP feature defined in this IE. When present, this IE indicates the features the CPF supports.	CP Function Features	89

Sx Association Update Request/Response

The UPF initiates an Sx Association Update Request to modify an existing Sx association when the UPF is administered down. Use the following command to change the UPF admin-state:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway user-plane-service <uplane-service-name> admin-state [enabled|disabled]
```

Note: *All sessions on the user plane will be deleted if the* `user-plane-service admin-state` *is* `disabled`*.*

Table 13-3 lists the supported Sx Association Update Request information elements.

Table 13-3 Sx Association Update Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)
Node ID	Mandatory	This IE contains the unique identifier of the sending node.	Node ID	60
UP Function Features	Optional	If present, this IE indicates the supported features when the sending node is the UPF.	UP Function Features	43
Sx Association Release Request	Conditional	This IE is present if the UPF requests the CPF to release the Sx association.	Sx Association Release Request	111
Graceful Release Period	Conditional	This IE is present if the UPF requests a graceful release of the Sx association.	Graceful Release Period	112

Table 13-4 lists the supported Sx Association Update Response information elements.

Table 13-4 Sx Association Update Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)
Node ID	Mandatory	This IE contains the unique identifier of the sending node.	Node ID	60
Cause	Mandatory	This IE indicates the acceptance or rejection of the corresponding request message.	Cause	19
CP Function Features	Optional	If present, this IE indicates the supported features when the sending node is the CPF.	CP Function Features	89

Sx Association Release Request/Response

The CPF initiates an Sx Association Release Request to release the Sx association if the user-plane-profile on the CPF is administered down. Upon receiving the Sx Association Release Request, the UPF deletes all Sx sessions, Sx associations and any related information locally before sending an Sx Association Release Response with a successful cause code.

If the user-plane-service is administered down, the UPF sends the CPF an Sx Association Update Request to notify the CPF to send an Sx Association Release Request to release the Sx association. Upon receiving the Sx Association Update Request, the CPF starts a configurable guard timer, which is included in the Sx Association Update Request from the UPF. When the guard timer expires, the CPF sends an Sx Association Release Request to the UPF and deletes all Sx sessions to the UPF.

Use the following command to configure the guard timer:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile pfcp-profile
<profile-name> pfcp-guard-time-interval <value>
```

Table 13-5 lists the supported Sx Association Release Request information elements.

Table 13-5 Sx Association Release Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)
Node ID	Mandatory	This IE contains the unique identifier of the sending node.	Node ID	60

Table 13-6 lists the supported Sx Association Release Response information elements.

Table 13-6 Sx Association Release Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)
Node ID	Mandatory	This IE contains the unique identifier of the sending node.	Node ID	60
Cause	Mandatory	This IE indicates the acceptance or rejection of the corresponding request message.	Cause	19

Sx Heartbeat Request/Response

The CPF/UPF exchange Heartbeat Request/Response messages to check if the PFCP peer is alive once the Sx association between the CPF and UPF is established. The CPF will periodically send the Heartbeat Request to the UPF according to what is configured for the following command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile pfcp-profile
<profile-name> pfcp-heartbeat-periodic-interval <value>
```

Note: *The* `pfcp-profile` *must be associated with the gateway service (SGW/PGW) on the CPF and the* `user-plane-service` *on the UPF.*

Table 13-7 lists the supported Heartbeat Request/Response information elements.

Table 13-7 Sx Heartbeat Request/Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)
Recovery Time Stamp	Mandatory	This IE contains the time stamp when the node was started. See clause 19A of 3GPP TS 23.007 [24].	Recovery Time Stamp	96

Sx Session Related Procedures

The following Sx Session Related procedures are supported when the MCC is deployed as a CUPS system. Sx Session Related Procedures use PFCP with proprietary vendor-specific IEs. The proprietary vendor specific IEs have been divided into three categories as listed in Table 13-8.

Table 13-8 PFCP Proprietary Vendor-Specific IE Categories

Category	Description
Standard	These IEs are defined in accordance with 3GPP TS 29.244 V14.4.0.
Vendor-Specific (VS)	These IEs correspond to PFCP constructs in some cases but differ in the actual definition.
Vendor Specific - Not Applicable (VS-NA)	These IEs are not applicable for CUPS and are used primarily for Non-CUPS MCC messaging.

Sx Session Establishment Procedure

Used to setup an Sx session between the CPF and UPF and to configure rules in the UPF so that the UPF can provide appropriate service for UE traffic.

- CPF: The CPF initiates an Sx Session Establishment Request to setup an Sx Session for a PDN Connection/IP-CAN Session.
- UPF: When the UPF receives an Sx Session Establishment Request message, it establishes the UPF session based on proprietary vendor-specific IEs.

Table 13-9 lists the supported Sx Session Establishment Request information elements.

Table 13-9 Sx Session Establishment Request Information Elements

Field	Name	Category	Comment
Node ID	Node ID	Standard	This IE contains the unique identifier of the sending node. This IE is equivalent to the IP address of the CP F-SEID IE in 3GPP TS 29.244.
F-SEID	F-SEID	Standard	This IE contains the unique identifier allocated by the CPF identifying the session. This IE is equivalent to the SEID of the CP F-SEID IE in 3GPP TS 29.244.
Enterprise Specific IE	Enterprise Specific IE	VS	The Affirmed Networks Enterprise ID is 39321. See Table 13-11 on page 13-10 for a list of IEs within the Enterprise Specific IE.

Table 13-10 lists the supported Sx Session Establishment Response information elements.

Table 13-10 Sx Session Establishment Response Information Elements

Field	Name	Category	Comment
Node ID	Node ID	Standard	This IE contains the unique identifier of the sending node. This IE is equivalent to the IP address of the CP F-SEID IE in 3GPP TS 29.244.
Cause	Cause	Standard	Indicates the acceptance or the rejection of the corresponding request message.
F-SEID	F-SEID	Standard	This IE contains the unique identifier allocated by the CPF identifying the session. This IE is equivalent to the SEID of the CP F-SEID IE in 3GPP TS 29.244.
Enterprise Specific IE	Enterprise Specific IE	VS	The Affirmed Networks Enterprise ID is 39321.

Enterprise Specific IE Within Sx Session Establishment Request

Within the Sx Session Establishment Request message, the Enterprise Specific IE contains other IEs and is considered a Grouped IE.

Table 13-11 lists the information elements contained in the Enterprise Specific IE.

Table 13-11 Enterprise Specific IEs (within Sx Session Establishment Request)

Field	Name	Category	Comment
WfSessionMcdTableFullSchema	Workflow Session MCD Table Full Schema		
pdn_sess_id	PDN Session ID	VS	Indicates the PGW session ID.
sgw_sess_id	SGW Session ID	VS	Indicates the SGW Session ID.
tid	Transaction ID	VS-NA	
wf_fsapid	Workflow FSAPID	VS-NA	
sender_svc_id	Sender Service ID	VS-NA	
sender_svc_type	Sender Service Type	VS-NA	
changemask	Change Mask	VS-NA	
row_change_id	Row Change ID	VS-NA	
sgw_svc_id	SGW Service ID	VS-NA	
pgw_svc_id	PGW Service ID	VS-NA	
session_flag	Session Flag	VS-NA	
pgw_role	PGW Role	VS	Indicates the role of the PGW.
wf_data_prof_id	Workflow Data Profile ID	VS	Indicates the Data Profile associated with the UPF.
bill_plan_prof_id	Billing Plan Profile ID	VS	Indicates the Billing Plan configuration.
uplink_ambr	Uplink AMBR	VS	Indicates the uplink aggregated maximum bit rate (AMBR). This IE is equivalent to the Create QER –> Maximum Bitrate IE in 3GPP TS 29.244.
dnlink_ambr	Downlink AMBR	VS	Indicates the downlink aggregated maximum bit rate (AMBR). This IE is equivalent to the Create QER –> Maximum Bitrate IE in 3GPP TS 29.244.
uplink_burstsize	Uplink Burst Size	VS-NA	
dnlink_burstsize	Downlink Burst Size	VS-NA	
access_side_brr_type	Access Side Bearer Type	VS	This IE is equivalent to the PDI –> Source Interface IE in 3GPP TS 29.244.
core_side_brr_type	Core Side Bearer Type	VS	This IE is equivalent to the PDI –> Source Interface IE in 3GPP TS 29.244.
imsi	IMSI	VS	Indicates the International Mobile Subscriber Identity (IMSI).

Table 13-11 Enterprise Specific IEs (within Sx Session Establishment Request)

Field	Name	Category	Comment
imei	IMEI	VS	Indicates the International Mobile Equipment Identity (IMEI).
msisdn	MSISDN	VS	Indicates the Mobile Subscriber Integrated Services Digital Network Number (MSISDN).
subscr_ip_allocated	Subscriber IP Allocated	VS	Indicates whether the subscriber was allocated an IP address.
subscr_ncid	Subscriber Network Context ID	VS	
subscr_ip	Subscriber IP	VS	Indicates the subscriber IP value.
type	Type	VS	Indicates the subscriber IP type.
value	Value	VS	Indicates the subscriber IPv4 value.
apn_id	APN ID	VS	Indicates the access point name (APN) ID. This IE is equivalent to the Network Instance IE within the PDI and Create Traffic Endpoint IEs in 3GPP TS 29.244.
home_plmnid_len	Home PLMN ID Length	VS-NA	Indicates the Home Public Land Mobile Network (PLMN) ID length.
sess_creation_time_stamp	Session Creation Timestamp	VS-NA	
pdngw_idle_timer	PGW Idle Timer	VS	
sgw_idle_timer	SGW Idle Timer	VS	
allocation_mode	Allocation Mode	VS	Indicates the local IP allocation mode.
serving_gw_ip	Serving Gateway IP	VS	Indicates the SGW IP value.
type	Type	VS	Indicates the SGW IP type.
v4_value	IPv4 Value	VS	Indicates the SGW IPv4 value.
v6_value	IPv6 Value	VS	Indicates the SGW IPv6 value.
sgw_auto_id	SGW Auto ID	VS-NA	
pgw_auto_id	PGW Auto ID	VS-NA	
is_suspended	Is Suspended	VS-NA	
non_3gpp_sess_type	Non 3GPP Session Type	VS-NA	
pdngw_session_inactivity_timer	PGW Session Inactivity Timer	VS	

Table 13-11 Enterprise Specific IEs (within Sx Session Establishment Request)

Field	Name	Category	Comment
sgw_session_inactivity_timer	SGW Session Inactivity Timer	VS	
geo_group_id	Geo Group ID	VS	Indicates the Geographic Redundancy Group ID.
pdn_type	PDN Type	VS	Indicates the packet data network (PDN) type.
gw_service_params	Gateway Service Parameters	VS	
location_info	Location Information	VS-NA	
bearer_list	Bearer List	VS	Indicates the Bearer List information. See Table 13-12 on page 13-14.
serviceflowlist	Service Flow List	VS	This IE is equivalent to the Create PDR –> Activate Predefined Rule IE in 3GPP TS 29.244.
service_type	Service Type	VS	
service_instance_id	Service Instance ID	VS-NA	
service_flow_id	Service Flow ID	VS	
action	Action	VS	
service_rule_id	Service Rule ID	VS	
service_rule_name	Service Rule Name	VS	
vsa_list	VSA List	VS	Indicates the Value-Added Service (VAS) attribute list.
mapped_attribute_id	Mapped Attribute ID	VS	VAS attribute
vsa_value_list	VSA Value List	VS	VAS attribute
stringval	String Value	VS	VAS attribute
vas_service_rule_list	VAS Service Rule List	VS	VAS attribute
tethering_entitlement	Tethering Entitlement	VS	VAS attribute
rating_group	Rating Group	VS	This IE is equivalent to the Create URR IE in 3GPP TS 29.244.
rating_group_id	Rating Group ID	VS	
charging_type	Charging Type	VS	This IE is equivalent to the Create URR –> Measurement Method IE in 3GPP TS 29.244.

Table 13-11 Enterprise Specific IEs (within Sx Session Establishment Request)

Field	Name	Category	Comment
charging_level	Charging Level	VS	
final_unit_indication	Final Unit Indication	VS	
manage_time_envelopes	Manage Time Envelopes	VS-NA	
charging_method	Charging Method	VS	
charg_service_inst_id	Charging Service Instance ID	VS	
stats_reporting_interval	Statistics Reporting Interval	VS	This IE is equivalent to the Create URR –> Measurement Method IE in 3GPP TS 29.244.
sgw_stats_reporting_interval	SGW Statistics Reporting Interval	VS	This IE is equivalent to the Create URR –> Measurement Method IE in 3GPP TS 29.244.
rg_initial_quota	Rating Group Initial Quota	VS	
charge_rg_initial_quota	Charge Rating Group Initial Quota	VS	Indicates the initial quota for the rating group.
event_logging	Event Logging	VS-NA	
log_policy_mapping	Log Policy Mapping	VS-NA	
time_of_gw_sent	Time of Gateway Sent	VS-NA	
is_pd_capable	Is Prefix Delegation Capable	VS-NA	Indicates if the session is capable of Prefix Delegation for DHCPv6.
active_crbn_profile	Active CRBN Profile	VS	Indicates the active Charging-Rule-Base-Name (CRBN) profile.
crbn_name	CRBN Name	VS	Indicates the name of the active CRBN profile.
bundle_size	Bundle Size	VS-NA	
PgwIpMcdTableFullSchema	PGW IP MCD Table Full Schema	VS	Indicates IP schema information. See Table 13-13 on page 13-16.
PfcpCplaneConfigSchema	PFCP C-Plane Configuration Schema	VS	
default_gy_enabled	Default Gy Enabled	VS	
default_gy_quota	Default Gy Quota	VS	
max_num_default_gy_quota	Maximum Number Default Gy Quota	VS	

Bearer List IE Within Sx Session Establishment Request

The Bearer List IE is contained within the Enterprise Specific IE in the Sx Session Establishment Request message. The Bearer List IE contains other IEs and is considered a Grouped IE.

Table 13-12 lists the information elements contained in the Bearer List IE.

Table 13-12 Bearer List IEs (within Sx Session Establishment Request)

Field	Name	Category	Comment
ebi	EBI	VS	Indicates the logical Evolved Packet System (EPS) Bearer ID (EBI).
changemask	Change Mask	VS-NA	
realebi	Real EBI	VS	
qci	QCI	VS	Indicates the QoS Class Identifier (QCI). This IE is equivalent to the Create QER IE in 3GPP TS 29.244.
arp	ARP	VS	Indicates the Allocation and Retention Priority (ARP). This IE is equivalent to the Create QER IE in 3GPP TS 29.244.
is_default	Is Default	VS	Indicates whether it is a default bearer.
access_ncid	Access Network Context ID	VS	This IE is equivalent to the Create Traffic Endpoint –> Traffic Endpoint ID IE in 3GPP TS 29.244.
access_local_teid	Access Local TEID	VS	Indicates the access local Tunnel End Identifier (TEID). This IE is equivalent to the Create Traffic Endpoint –> Local F-TEID IE in 3GPP TS 29.244.
access_local_ipaddr	Access Local IP Address	VS	This IE is equivalent to the Create Traffic Endpoint –> Local F-TEID IE in 3GPP TS 29.244.
type	Type	VS	Indicates the access local IP type.
v4_value	IPv4 Value	VS	Indicates the access local IPv4 value.
access_local_udpport	Access Local UDP Port	VS	Indicates the access local User Datagram Protocol (UDP) port.
access_remote_teid	Access Remote TEID	VS	This IE is equivalent to the Create FAR –> Forwarding Parameters –> Outer Header Creation IE in 3GPP TS 29.244.
access_remote_ipaddr	Access Remote IP Address	VS	This IE is equivalent to the Create FAR –> Forwarding Parameters –> Outer Header Creation IE in 3GPP TS 29.244.
type	Type	VS	
v4_value	IPv4 Value	VS	

Table 13-12 Bearer List IEs (within Sx Session Establishment Request)

Field	Name	Category	Comment
access_remote_udpport	Access Remote UDP Port	VS	
paging_queue_size	Paging Queue Size	VS	
uplink_gbr	Uplink GBR	VS	Indicates the uplink Guaranteed Bit Rate (GBR). This IE is equivalent to the Create QER –> Guaranteed Bitrate IE in 3GPP TS 29.244.
uplink_gbs	Uplink GBS	VS-NA	Indicates the uplink Guaranteed Bucket Size (GBS).
uplink_mbr	Uplink MBR	VS	Indicates the uplink Maximum Bit Rate (MBR). This IE is equivalent to the Create QER –> Maximum Bitrate IE in 3GPP TS 29.244.
uplink_mbs	Uplink MBS	VS-NA	Indicates the uplink Maximum Bucket Size (MBS).
downlink_gbr	Downlink GBR	VS	Indicates the downlink Guaranteed Bit Rate (GBR). This IE is equivalent to the Create QER –> Guaranteed Bitrate IE in 3GPP TS 29.244.
downlink_gbs	Downlink GBS	VS-NA	Indicates the downlink Guaranteed Bucket Size (GBS).
downlink_mbr	Downlink MBR	VS	Indicates the downlink Maximum Bit Rate (MBR). This IE is equivalent to the Create QER –> Maximum Bitrate IE in 3GPP TS 29.244.
downlink_mbs	Downlink MBS	VS-NA	Indicates the downlink Maximum Bucket Size (MBS).
bearer_flags	Bearer Flags	VS-NA	
data_gtp_version	Data GTP Version	VS	Indicates the GPRS tunneling protocol (GTP) version.
threshold	Threshold	VS	Indicates the interface threshold.
intfType	Interface Type	VS	
volumeThreshold	Volume Threshold	VS	This IE is equivalent to the Create URR –> Volume Threshold IE in 3GPP TS 29.244.
charging_id	Charging ID	VS	

PGW IP MCD Table Full Schema IE Within Sx Session Establishment Request

The PGW IP MCD Table Full Schema IE is contained within the Enterprise Specific IE in the Sx Session Establishment Request message. The PGW IP MCD Table Full Schema IE contains other IEs and is considered a Grouped IE.

Table 13-13 lists the information elements contained in the PGW IP MCD Table Full Schema IE.

Table 13-13 PGW IP MCD Table Full Schema IEs (within Sx Session Establishment Request)

Field	Name	Category	Comment
pdn_sess_id	PDN Session ID	VS	Indicates PGW session ID.
sender_svc_id	Sender Service ID	VS-NA	
imsi	IMSI	VS	Indicates the International Mobile Subscriber Identity (IMSI).
ncid	Network Context ID	VS	Indicates the network context of the IP address subpool.
subscr_ip	Subscriber IP	VS	Indicates the subscriber IP address.
type	Type	VS	
v4_value	IPv4 Value	VS	This IE is equivalent to the UP IP Address IE within the PDI and Create Traffic Endpoint IEs in 3GPP TS 29.244.
apn_id	APN ID	VS	Indicates the access point name (APN) ID. This IE is equivalent to the Network Instance IE within the PDI and Create Traffic Endpoint IEs in 3GPP TS 29.244.
v4_pool_info	IPv4 Pool Information	VS	Indicates the IPv4 sub pool ID.
wf_fsapid	Workflow FASPID	VS-NA	
ipallocated_ncid	IP Allocated Network Context ID	VS	
v4_pool_name	IPv4 Pool Name	VS	Indicates the IPv4 sub pool name.
geored_group_id	Geographic Redundancy Group ID	VS	
access_teid	Access TEID	VS-NA	Indicates the access Tunnel End Identifier (TEID).
cups_session	CUPS Session	VS	Indicates the Control and User Plane Separation (CUPS) session.

Sx Session Modification Procedure

Used to modify an existing Sx session. For example, to configure a new rule, to modify an existing rule, or to delete an existing rule.

- CPF: The CPF initiates an Sx Session Modification Request to modify a session. For example, mobility/GTP procedures or an event from the PCRF/OCS could trigger an Sx Session Modification procedure.

- UPF: When the UPF receives an Sx Session Modification Request, it processes the modification request as directed if the session is present. If the UPF cannot find the session, it sends an Sx Session Modification Response with rejection cause code “Session Context Not Found”.

The Sx Session Modification Request message supports the same IEs as the Sx Session Establishment Request, except the Sx Session Modification Request will only contain those IEs that have changed.

Sx Session Deletion Procedure

Used to delete an existing Sx session between the CPF and the UPF.

- CPF: The CPF initiates an Sx Session Deletion Request to delete an existing session.
- UPF: When the UPF receives an Sx Session Deletion Request, it processes the session deletion request if the session is present, otherwise it sends an Sx Session Deletion Response with rejection cause code “Session Context Not Found”.

Table 13-14 lists the supported Sx Session Deletion Request information elements.

Table 13-14 Sx Session Deletion Request Information Elements

Field	Name	Category	Comment
Enterprise Specific IE	Enterprise Specific IE	VS	The Affirmed Networks Enterprise ID is 39321.
WfSessionMcdTableFull Schema	Workflow Session MCD Table Full Schema		
pdn_sess_id	PDN Session ID	VS	Indicates the PGW session ID.
sgw_sess_id	SGW Session ID	VS	Indicates the SGW session ID.
tid	Transaction ID	VS-NA	
wf_fsapid	Workflow FSAPID	VS-NA	
sender_svc_id	Sender Service ID	VS-NA	
sender_svc_type	Sender Service Type	VS-NA	
changemask	Change Mask	VS-NA	
row_change_id	Row Change ID	VS-NA	
is_suspended	Is Suspended	VS-NA	
time_of_gw_sent	Time of Gateway Sent	VS-NA	

Table 13-14 Sx Session Deletion Request Information Elements

Field	Name	Category	Comment
PfcpCplaneConfig Schema	PFCP C-Plane Configuration Schema	VS	
default_gy_enabled	Default Gy Enabled	VS	
default_gy_quota	Default Gy Quota	VS	
max_num_default_gy_quota	Maximum Number Default Gy Quota	VS	

Table 13-15 lists the supported Sx Session Deletion Response information elements.

Table 13-15 Sx Session Deletion Response Information Elements

Field	Name	Category	Comment
Cause	Cause	Standard	Indicates the acceptance or the rejection of the corresponding request message.
Enterprise Specific IE	Enterprise Specific IE	VS	The Affirmed Networks Enterprise ID is 39321.

Sx Session Report Procedure

Used by the UPF to report information related to the Sx session to the CPF.

- UPF: The UPF initiates an Sx Session Report Request to report information related to an Sx session. An Sx report can be triggered for many scenarios including when the UPF needs to notify the CPF on online/offline data usage.
- CPF: When the CPF receives an Sx Session Report Request, it processes the information being reported as appropriate, otherwise it sends an Sx Session Report Response with rejection cause code “Session context not found”.

Table 13-16 lists the supported Sx Session Report Request information elements.

Table 13-16 Sx Session Report Request Information Elements

Field	Name	Category	Comment
Enterprise Specific IE	Enterprise Specific IE	VS	The Affirmed Networks Enterprise ID is 39321.
BrrStatsRsp	Bearer Statistics Response	VS	Indicates the Bearer Statistics Response from the UPF. This IE is equivalent to the Usage Report IE in 3GPP TS 29.244. See Table 13-18 on page 13-19.

Table 13-17 lists the supported Sx Session Report Response information elements.

Table 13-17 Sx Session Report Response Information Elements

Field	Name	Category	Comment
Enterprise Specific IE	Enterprise Specific IE	VS	The Affirmed Networks Enterprise ID is 39321.

Bearer Statistics Response IE Within Sx Session Report Request

The Bearer Statistics Response IE is contained within the Enterprise Specific IE in the Sx Session Report Request message. The Bearer Statistics Response IE contains other IEs and is considered a Grouped IE.

Table 13-18 lists the information elements contained in the Bearer Statistics Response IE.

Table 13-18 Bearer Statistics Response IEs (within Sx Session Report Request)

Field	Name	Category	Comment
tid	Transaction ID	VS-NA	
sender_id	Sender ID	VS-NA	
pdn_list	PDN List	VS	
pdn_sess_id	PDN Session ID	VS	Indicates PGW session ID.
gw_sess_id	Gateway Session ID	VS-NA	
is_dirty	Is Dirty	VS-NA	
stats	Statistics	VS	This IE is equivalent to the Usage Reporting Rule (URR) statistics in 3GPP TS 29.244.
ul_pkts_out	Uplink Packets Out	VS	
ul_bytes_out	Uplink Bytes Out	VS	
ul_pkts_drop	Uplink Packets Drop	VS	
ul_bytes_drop	Uplink Bytes Drop	VS	
dl_pkts_out	Downlink Packets Out	VS	
dl_bytes_out	Downlink Bytes Out	VS	
dl_pkts_drop	Downlink Packets Drop	VS	
dl_bytes_drop	Downlink Bytes Drop	VS	
brr_list	Bearer List	VS	
brr_id	Bearer ID	VS	

Table 13-18 Bearer Statistics Response IEs (within Sx Session Report Request)

Field	Name	Category	Comment
ebi	EBI	VS	Indicates the logical Evolved Packet System (EPS) Bearer ID (EBI).
is_dirty	Is Dirty	VS-NA	
stats	Statistics	VS	This IE is equivalent to the Usage Reporting Rule (URR) statistics in 3GPP TS 29.244.
ul_pkts_out	Uplink Packets Out	VS	
ul_bytes_out	Uplink Bytes Out	VS	
ul_pkts_drop	Uplink Packets Drop	VS	
ul_bytes_drop	Uplink Bytes Drop	VS	
dl_pkts_out	Downlink Packets Out	VS	
dl_bytes_out	Downlink Bytes Out	VS	
dl_pkts_drop	Downlink Packets Drop	VS	
dl_bytes_drop	Downlink Bytes Drop	VS	
threshold_stats	Threshold Statistics	VS	This IE is equivalent to the Usage Reporting Rule (URR) statistics in 3GPP TS 29.244.
intf_type	Interface Type	VS	
total_bytes	Total Bytes	VS	
rating_group	Rating Group	VS	
rating_group_id	Rating Group ID	VS	
servicerule_list	Service Rule List	VS	
service_type	Service Type	VS	
service_flow_id	Service Flow ID	VS	
service_rule_id	Service Rule ID	VS	
is_dirty	Is Dirty	VS-NA	
ul_pkts_out	Uplink Packets Out	VS	
ul_bytes_out	Uplink Bytes Out	VS	
ul_pkts_drop	Uplink Packets Drop	VS	
ul_bytes_drop	Uplink Bytes Drop	VS	
dl_pkts_out	Downlink Packets Out	VS	
dl_bytes_out	Downlink Bytes Out	VS	

Table 13-18 Bearer Statistics Response IEs (within Sx Session Report Request)

Field	Name	Category	Comment
dl_pkts_drop	Downlink Packets Drop	VS	
dl_bytes_drop	Downlink Bytes Drop	VS	
time_of_first_pkt	Time of First Packet	VS	
time_of_last_pkt	Time of Last Packet	VS	
service_id	Service ID	VS-NA	
wf_session_id	Workflow Session ID	VS-NA	
trigger_info	Trigger Information	VS	
trigger_type	Trigger Type	VS	This IE is equivalent to the Usage Report Trigger IE in 3GPP TS 29.244.
intf_type	Interface Type	VS	
sender_svc_type	Sender SVC Type	VS-NA	
gw_service_id	Gateway Service ID	VS-NA	
fp_gw_service_id	Fastpath Gateway Service ID	VS-NA	

Sx Session Message Flows

This section shows the Sx Session message flows for the Create, Modify, and Delete Session Request/Response messages.

Figure 13-3 Create Session Request/Response Message Flow

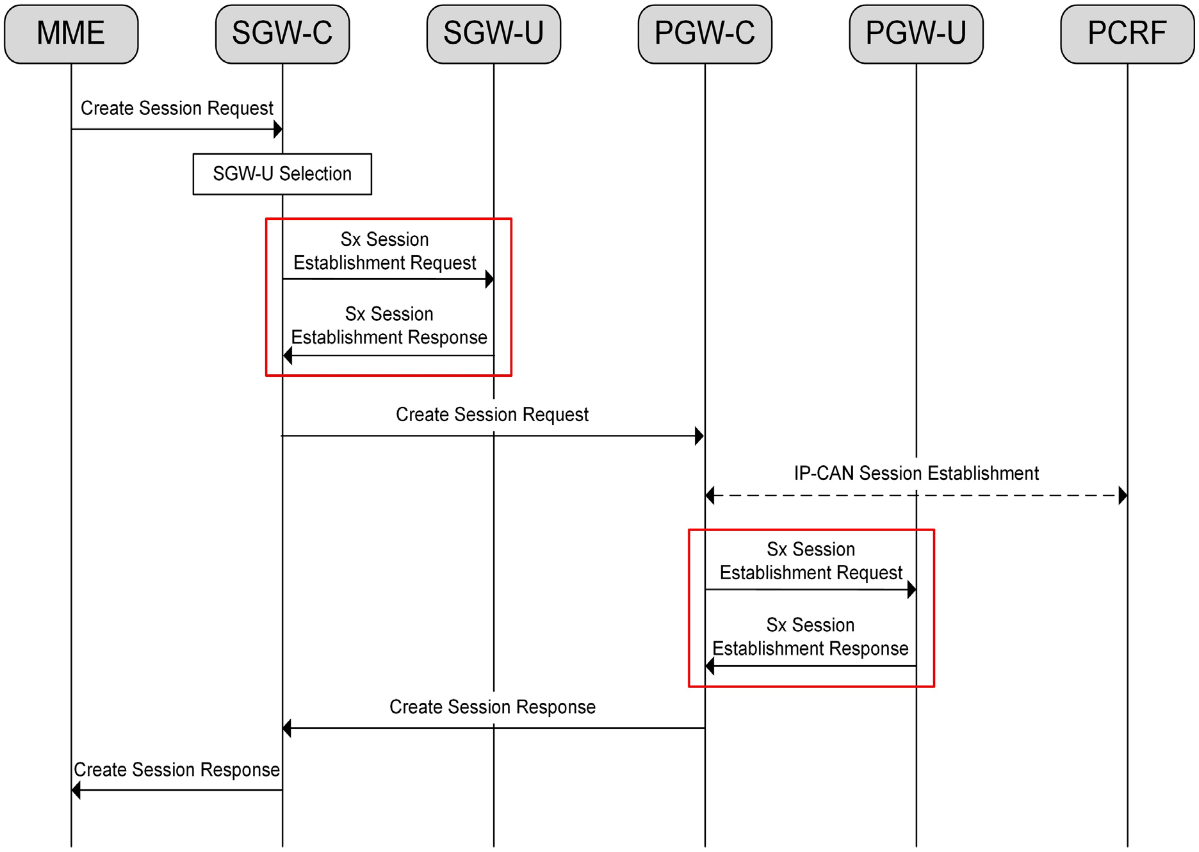


Figure 13-4 Modify Session Request/Response Message Flow

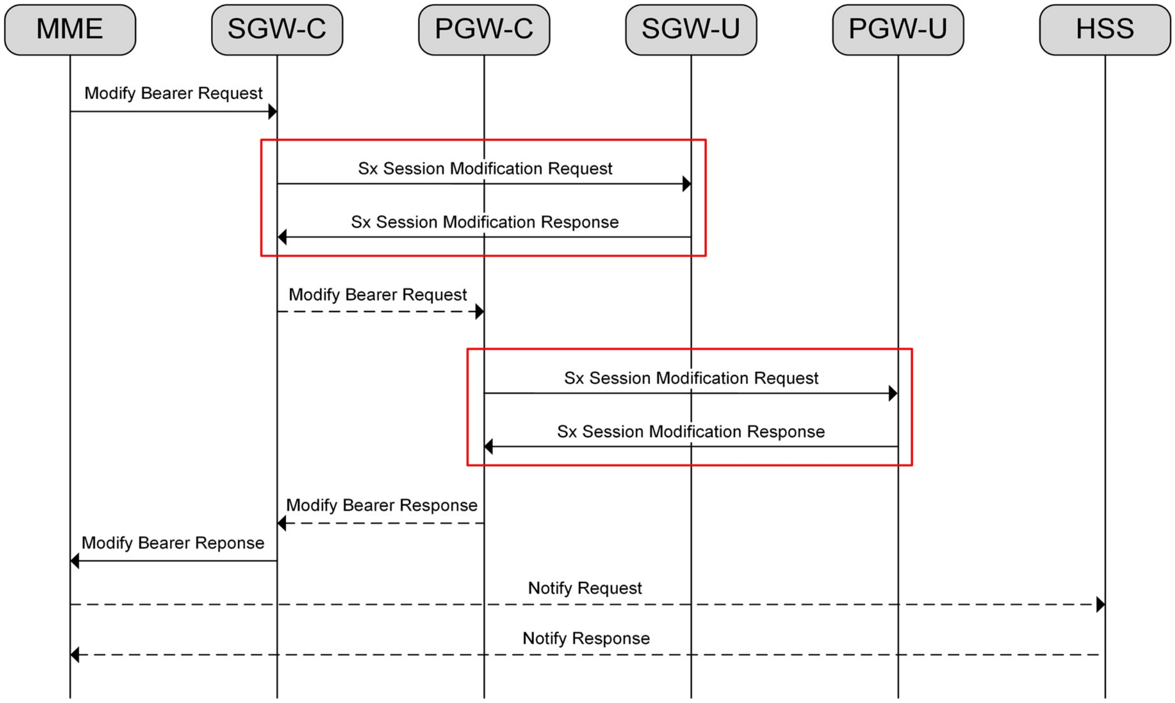
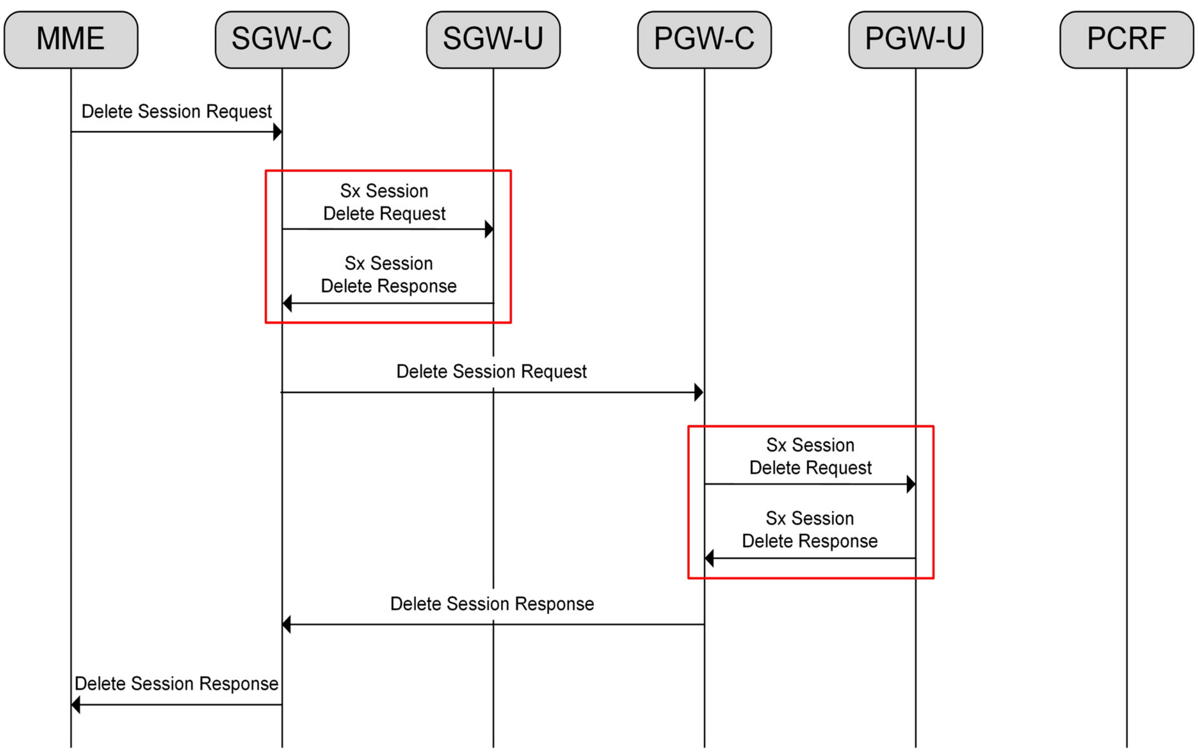


Figure 13-5 Delete Session Request/Response Message Flow



Monitoring the Sx Interface

This section describes how to monitor the operational status of the PFCP peer via the Sx interface and how to display statistical information for Sx communication messages. This section also describes the alarm notification that occurs when the remote PFCP peer goes down. For more information on statistics, see *Acuitas Performance Statistics.*

PFCP Peer Status

Use the following command to display the status information of a specific PFCP peer, such as association and heartbeat status, peer recovery value, and IP addresses.

```
an3000-mcm-slot7cpu1# show zone <name> gateway pfcp-peer-status
<gateway-name> <peer-ip-address>
```

The system displays output similar to the following:

```
gateway pfcp-peer-status PGW 10.33.153.61
peer-name                UPS2-V4-PEER
local-ip-address         10.64.151.105
network-context          SX
pfcp-version             v1
peer-association-status  up
peer-heartbeat-status    up
peer-recovery-value      2018-06-07T10:48:15-0400
functions-supported      408
```

Sx Interface Statistics

You can display statistical information for Sx interface communication messages for a specific PFCP peer as shown in the following example.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
cluster-level pfcp peer <name>
num-pfcp-session-establishment-request-sent
num-pfcp-session-establishment-request-resent
num-pfcp-session-establishment-response-received
num-pfcp-session-establishment-request-timeout
num-pfcp-session-modification-request-sent
num-pfcp-session-modification-request-resent
num-pfcp-session-modification-response-received
num-pfcp-session-modification-request-timeout
num-pfcp-session-deletion-request-sent
num-pfcp-session-deletion-request-resent
num-pfcp-session-deletion-response-received
num-pfcp-session-deletion-request-timeout
```

```
num-pfcp-session-report-request-received
num-pfcp-session-report-response-sent
num-pfcp-session-establishment-request-received
num-pfcp-session-establishment-response-sent
num-pfcp-session-modification-request-received
num-pfcp-session-modification-response-sent
num-pfcp-session-deletion-request-received
num-pfcp-session-deletion-response-sent
num-pfcp-session-report-request-sent
num-pfcp-session-report-request-resent
num-pfcp-session-report-response-received
num-pfcp-session-report-request-timeout
num-heartbeat-request-sent
num-heartbeat-response-received
num-heartbeat-request-timeout
num-heartbeat-request-received
num-heartbeat-response-sent
num-pfcp-association-setup-request-sent
num-pfcp-association-setup-response-received
num-pfcp-association-setup-request-timeout
num-pfcp-association-setup-request-received
num-pfcp-association-setup-response-sent
num-pfcp-version-not-supported-response-sent
num-pfcp-version-not-supported-response-received
num-pfcp-association-update-request-sent
num-pfcp-association-update-response-received
num-pfcp-association-update-request-timeout
num-pfcp-association-update-request-received
num-pfcp-association-update-response-sent
num-pfcp-association-release-request-sent
num-pfcp-association-release-response-received
num-pfcp-association-release-request-timeout
num-pfcp-association-release-request-received
num-pfcp-association-release-response-sent
num-pfcp-session-establish-request-decode-failure
num-pfcp-session-establish-request-encode-failure
num-pfcp-session-establish-response-decode-failure
num-pfcp-session-establish-response-encode-failure
num-pfcp-session-modify-request-decode-failure
num-pfcp-sesion-modify-request-encode-failure
num-pfcp-session-modify-response-decode-failure
num-pfcp-session-modify-response-encode-failure
num-pfcp-session-delete-request-decode-failure
```

```
num-pfcp-session-delete-request-encode-failure
num-pfcp-session-delete-response-decode-failure
num-pfcp-session-delete-response-encode-failure
num-pfcp-session-report-request-decode-failure
num-pfcp-session-report-request-encode-failure
num-pfcp-session-report-response-decode-failure
num-pfcp-session-report-response-encode-failure
num-pfcp-node-level-message-decode-failure
num-pfcp-node-level-message-encode-failure
num-pfcp-session-context-not-found-received
```

PFCP Protocol Logging

PFCP protocol logging captures Sx Node Related messages and Sx Session Related messages sent/received by the PGW-C, SGW-C, PGW-U and SGW-U in the EPC CUPS architecture. The MCC captures the packets and stores them as pcapng capture files in *var/log/eventlog* on both the CPF and UPF. These files can be viewed in Wireshark.

Set parameters for this feature using the following commands:

1. Create a local file in which to store packets.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
eventlog <filename>
```

2. Create a log policy.

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<policy-name> admin-state enabled log-type protocol
protocol-logging-filter-field protocol filter-value pfcp
```

To disable the log policy, change the `admin-state` to `disabled`.

3. Create a sink under the log policy.

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<policy-name> sinks <sinks-name> type file file-name <file-name>
file-format pcapng
```

Note: *Secondary level filters, such as IMSI, IMEI and MSISDN, can be configured on both the CPF and UPF. The Policies/Filters on the CPF and UPF work independently of each other and do not have to match.*

Alarms

A major-level trap similar to the following occurs when a PFCP peer goes down. The alarm clears after connectivity is restored. For more information on alarms, see the *Affirmed Networks SNMP MIB and Alarm Reference Guide*.

`anPfcpPeerDown`

Description:	PFCP Peer down indicates that the remote PFCP Peer is not responding to any of the PFCP messages sent to it.
Trap ID:	321
Status:	Current
Severity:	Major
Auto Clearing:	No
Potential Impact:	Sessions that go to the remote pfcp peer may be cleared, resulting in loss of data connectivity for the affected UEs.
Corrective Action:	Verify that the remote peer is up, running and reachable. Make sure that the local PFCP Endpoint is provisioned correctly and running normally.

Chapter 14

S2a Interface

This section contains the following information used to control and monitor the S2a Interface:

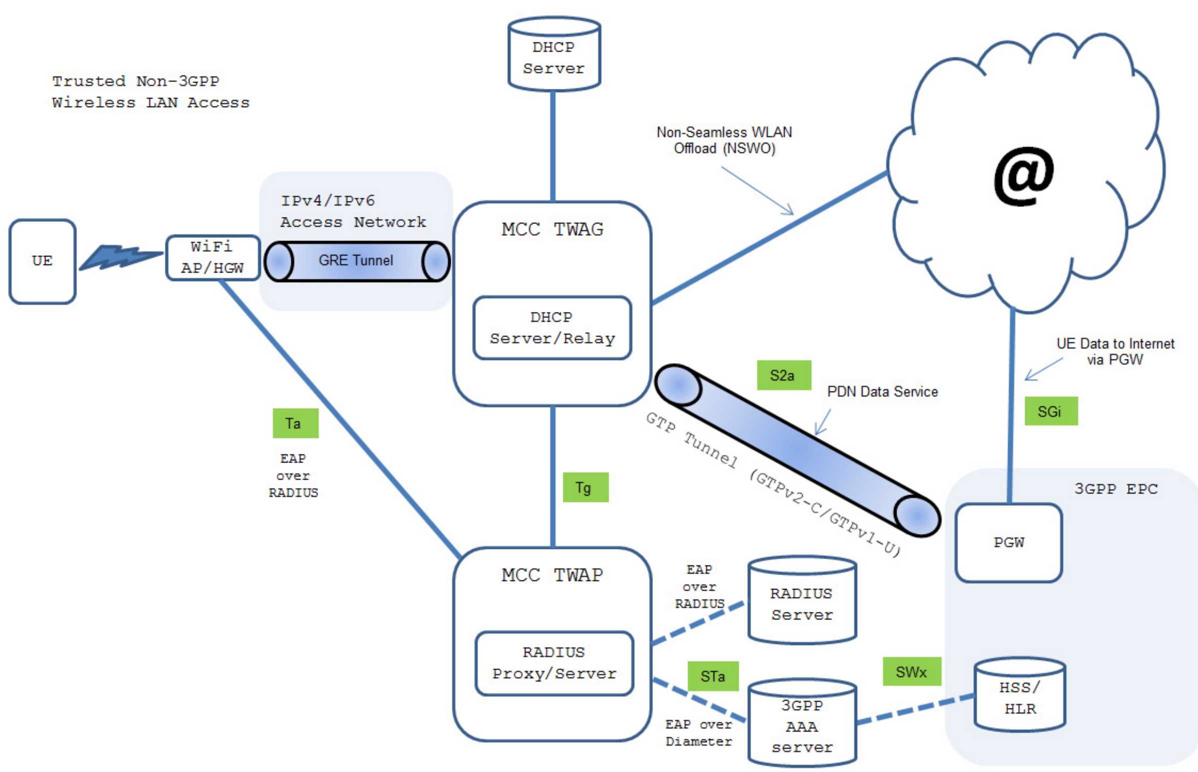
Interface Overview

The S2a interface provides trusted Wireless Local Area Network (WLAN) access to the Evolved Packet Core (EPC) using 3GPP S2a Mobility based on GTP services as defined in 3GPP TR 23.852. The S2a interface is between the MCC Trusted WLAN Access Gateway/Proxy (TWAG/TWAP) and the Packet Data Network Gateway (PGW) and is used for Public Data Network (PDN) session creation and maintenance and to exchange user equipment (UE) data between the WLAN access and the EPC.

This document describes the S2a interface implementation in the *Affirmed Networks Mobile Content Cloud* (MCC) in accordance with 3GPP TS 23.402 V12.0.0. The S2a interface uses the General Packet Radio Service (GPRS) tunneling protocol (GTPv2-C and GTPv1-U) to transfer signaling messages and user data between the TWAG/TWAP and the EPC network.

Figure 14-1 illustrates the position and role of the network components and interfaces for trusted Wi-Fi access to the EPC. Note that while Figure 14-1 shows the TWAG and TWAP as separate entities, they can also be co-located in the same physical location for maximum deployment flexibility.

Figure 14-1 MCC TWAG/TWAP Network Deployment



References and Specifications

The system supports the following references and specifications for the S2a interfaces:

- 3GPP TS 29.274 V11.13.0 (2014-12): *3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3*
- 3GPP TS 29.274 V13.6.0 (2016-06): *3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3*
- 3GPP TS 23.401 V9.3.0 (2009-12): *General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access*
- 3GPP TS 29.281 V9.3.0 (2010-06): *General Packet Radio System (GPRS) Tunnelling Protocol User Plane (GTPv1-U) (Release 9)*
- 3GPP TR 23.852 V12.0.0 (2013-09): *Study on S2a Mobility based on GPRS Tunnelling Protocol (GTP) and Wireless Local Area Network (WLAN) access to the Enhanced Packet Core (EPC) network (SaMOG); Stage 2*
- 3GPP TS 23.402 V12.0.0 (2013-03): *Technical Specification Group Services and System Aspects; Architecture Enhancements for Non-3GPP Accesses*

S2a Interface Message Flows

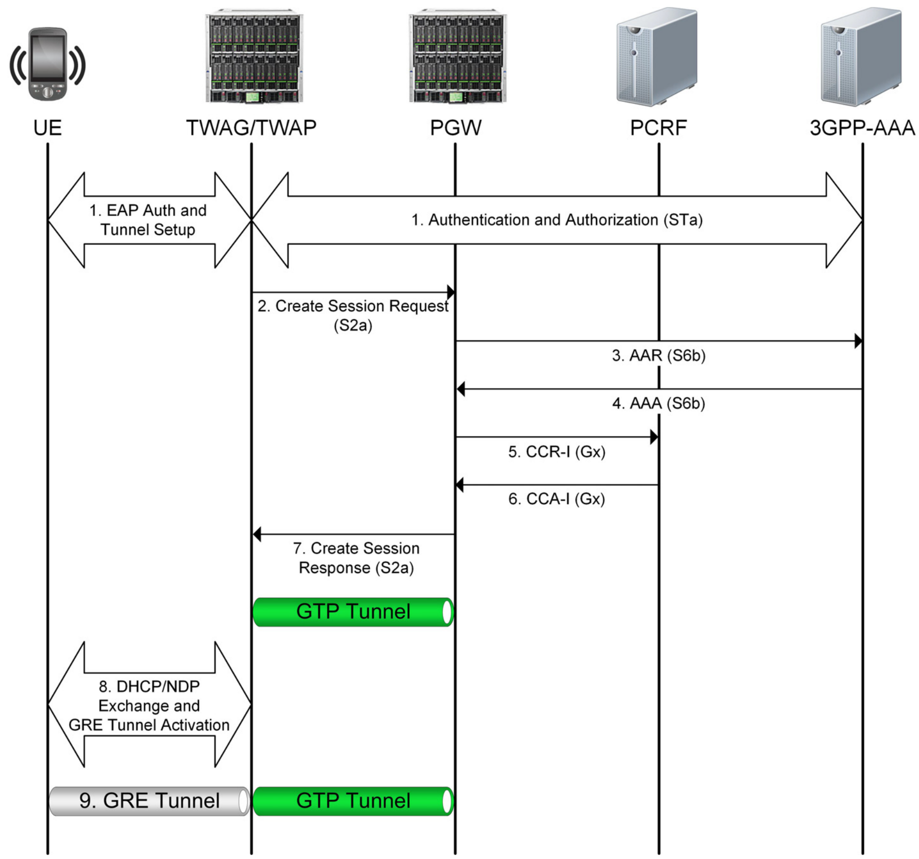
This section provides examples of message flows between the TWAG and PGW over the S2a interface. For more information on S2a message flows, see 3GPP TS 23.402 V12.0.0.

- Establishing the Default Bearer (Initial Attach) (page 14-3)
- Establishing a Dedicated Bearer (page 14-5)
- Wi-Fi to LTE / LTE to Wi-Fi Handovers (page 14-6)

Establishing the Default Bearer (Initial Attach)

Figure 14-2 illustrates an example of the message flow between the TWAG and PGW over the S2a interface for establishing the default bearer (initial attach). The default bearer connects the UE to the PDN and provides the UE with an "always-on" IP connectivity to the PDN.

Figure 14-2 Establishing the Default Bearer (Initial Attach)



1 The UE sends an attachment request to the TWAG/TWAP. The UE is authenticated through Extensible Authentication Protocol (EAP) between the 3GPP AAA Server and TWAG/TWAP. Upon a successful authorization, the 3GPP AAA server returns the APN-AMBR, static QoS profile, and trace information.

2 The TWAG/TWAP sends a Create Session Request to the PGW and sets up a GTP-C tunnel with the PGW for signaling messages. The message includes control and user plane information, such as the UE IMSI, APN, TWAG TEID, PDN address, EPS bearer identity, and default EPS bearer QoS. Additional parameters include the authentication credentials for secondary authentication if the UE provides this information (for example, PAP credentials for L2TP access).

3 The PGW sends an AA-Request to the 3GPP AAA Server with the session information for subscriber authorization. When the UE attaches to the EPC via the S2a interface, the S6b interface is used to update the 3GPP AAA Server with the PGW address and GTPv2 protocol information.

4 The 3GPP AAA Server returns an AA-Answer with the subscriber authorization information.

5 The PGW sends a Credit-Control-Request (CCR) to the PCRF Server.

6 The PCRF Server returns a Credit-Control-Answer (CCA) to the PGW. The PGW creates a new entry in its bearer context table and generates a Charging ID.

7 The PGW returns a Create Session Response with the PGW address and TEIDs (user and control plane), PDN type and address, EPS bearer identity and QoS, APN-AMBR, Charging ID, Cause, and IP address(es) allocated for the UE.

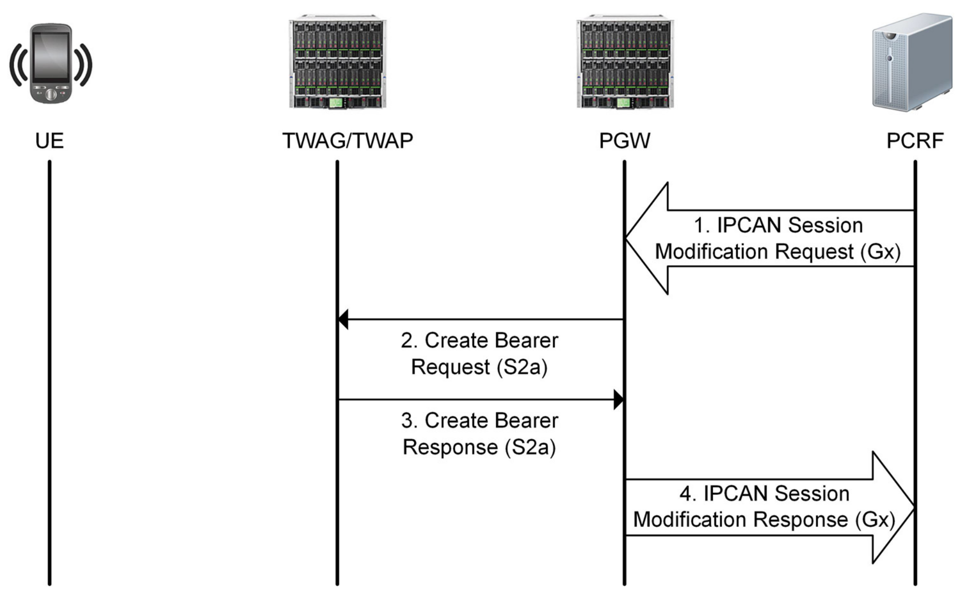
8 Upon a successful Dynamic Host Configuration Protocol/Network Discovery Protocol (DHCP/NDP) exchange, the TWAG/TWAP forwards the allocated IP address to the UE and activates the Generic Routing Encapsulation (GRE) tunnel.

9 The TWAG/TWAP maps the GRE tunnel to the GTP tunnel established with the PGW.

Establishing a Dedicated Bearer

Figure 14-3 illustrates the message flow between the TWAG/TWAP and PGW over the S2a interface for establishing a dedicated bearer. Dedicated bearers must be established when the UE requires a specific QoS for the requested service.

Figure 14-3 Message Flow for Establishing a Dedicated Bearer



1 If dynamic PCC is deployed, the PCRF sends a QoS policy message to the PGW (similar to the PCRF response in PCEF initiated IP-CAN Session Modification up to where the PGW requests IP-CAN bearer signaling). If dynamic PCC is not deployed, the PGW may apply the local QoS policy.

2 The PGW sends a Create Bearer Request to the TWAG/TWAP. The Linked EPS Bearer Identity (LBI) is the EPS Bearer Identity of the default bearer.

3 The TWAG/TWAP selects an EPS Bearer Identity, which has not yet been assigned to the UE, and uses the uplink packet filter to determine the mapping of uplink traffic flows to the S2a bearer. The TWAG/TWAP acknowledges the S2a bearer activation to the PGW by sending a Create Bearer Response.

4 If the dedicated bearer activation is the result of a QoS policy message from the PCRF, the PGW indicates to the PCRF whether the requested QoS policy can be enforced or not, which completes the IP-CAN Session Modification.

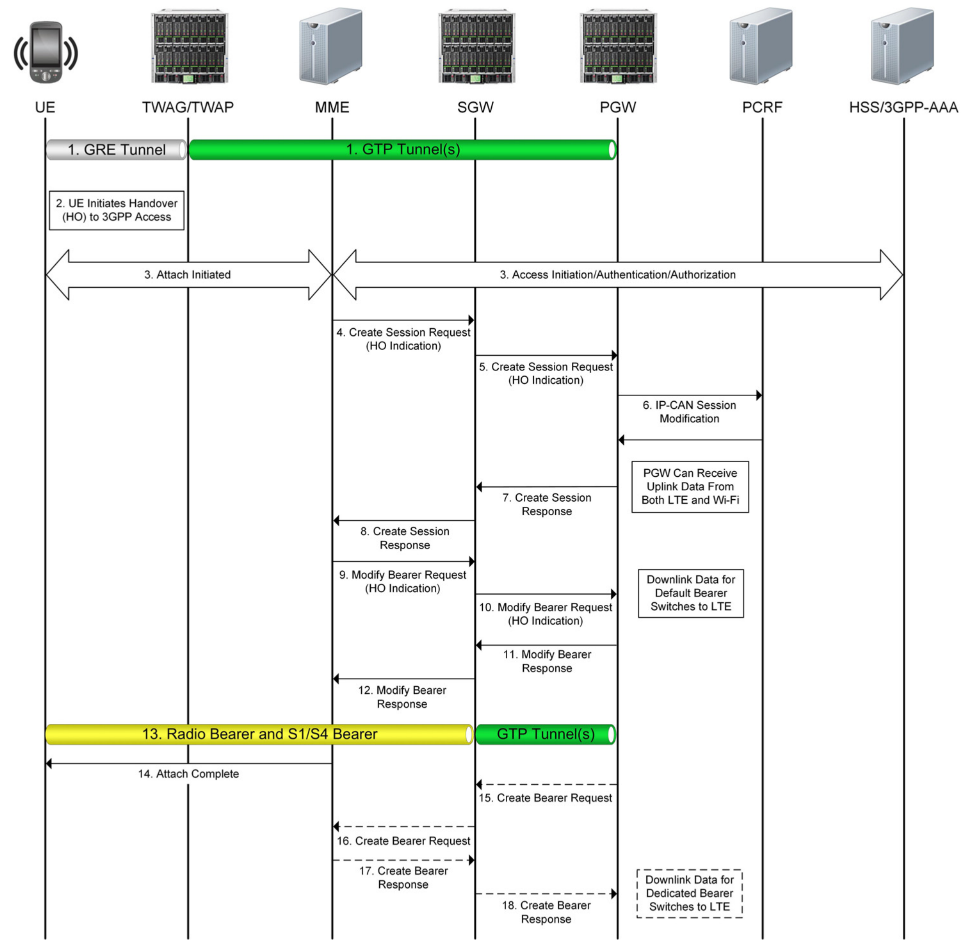
Wi-Fi to LTE / LTE to Wi-Fi Handovers

Over the S2a interface, the TWAG/TWAP and PGW support handovers from the non-3GPP trusted IP access network to the 3GPP Long Term Evolution (LTE) network and vice versa. This section describes the handover message flows.

Note: *If the MCC detects a network-initiated CBR/UBR with CSR collision, the MCC will retry the CBR/UBR bearer operation with the new target node after a successful inter LTE/Wi-Fi handover. In the case of a DBR collision with CSR, the MCC will not establish that bearer on the new target node after a successful inter LTE/Wi-Fi handover.*

Figure 14-4 shows the message flow in a handover from trusted Wi-Fi to LTE.

Figure 14-4 Wi-Fi-to-LTE Handover Message Flow



1. The UE and TWAG/TWAP have an established GRE tunnel, and the TWAG/TWAP and PGW have an established GTP tunnel over the S2a interface.
2. The UE discovers the 3GPP network and sends an Attach-Request to the Mobility Management Entity (MME).
3. The MME may contact the Home Subscriber Server (HSS)/3GPP AAA Server and authenticate the UE. The MME receives information about the PDNs connected to the UE over non-3GPP access in the subscriber data obtained from the HSS.
4. The MME sends a Create Session Request with Handover Indication to the selected Service Gateway (SGW).
5. The SGW sends a Create Session Request with Handover Indication to the PGW. The PGW does not yet change the tunnel from non-3GPP IP access to 3GPP access.
6. The PGW executes a PCEF-Initiated IP CAN Session Modification with the PCRF to obtain the information needed to function as the PCEF for all the active sessions the UE has with the new IP-CAN type as a result of the handover.

7 The PGW responds with a Create Session Response to the SGW, which contains the IP address/prefix that was assigned to the UE while it was connected to the non-3GPP IP access. It also contains the Charging ID previously assigned to the PDN connection in the non-3GPP access, which still applies to the non-3GPP access. At this point, the PGW can receive uplink data from both Wi-Fi and LTE.

8 The SGW returns a Create Session Response to the MME, which includes the IP address of the UE.

9 The MME sends a Modify Bearer Request to the SGW.

10 The SGW sends a Modify Bearer Request to the PGW to instruct the PGW to tunnel packets from non-3GPP IP access to 3GPP access and immediately start routing packets to the SGW for the established default bearer and any dedicated EPS bearers.

Note: *The PGW removes the old PCC rules for the untrusted non-3GPP IP access and applies the new rules for E-UTRAN access for charging. The Charging ID previously used for the PDN connection in the non-3GPP access now only applies to the default bearer in use for E-UTRAN access. If dedicated bearers are created, the PGW assigns a new Charging ID for each bearer*

---

11 The PGW acknowledges by sending the Modify Bearer Response to the SGW.

12 The SGW acknowledges by sending the Modify Bearer Response (EPS Bearer Identity) message to the MME.

13 The S1/S4 bearer setup is complete, radio and access bearers are established in the 3GPP access network, and GTP tunnel(s) are established over the S5 interface. The UE sends and receives data through the E-UTRAN system.

14 The UE attach is complete.

Note: *Steps 15-18 apply only if a dedicated bearer was established before the handover occurred.*

---

15 The PGW sends the Create Bearer Request to the SGW.

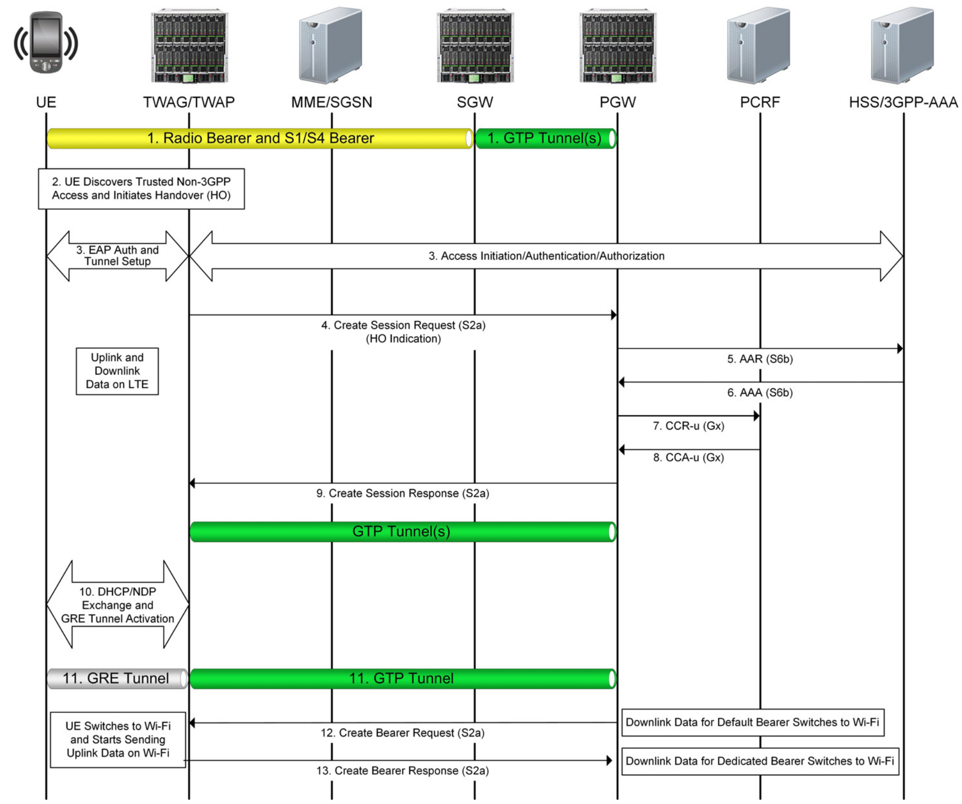
16 The SGW forwards the Create Bearer Request to the MME.

17 The MME returns the Create Bearer Response to the SGW.

18 The SGW forwards the Create Bearer Response to the PGW. The downlink data for the dedicated bearer switches to LTE.

Figure 14-5 shows the message flow in a handover from LTE to trusted Wi-Fi.

Figure 14-5 LTE-to-Wi-Fi Handover Message Flow



1. The UE is initially attached to the 3GPP access network.
2. The UE moves and attaches to a trusted non-3GPP IP access network.
3. The UE sends an attachment request to the TWAG/TWAP. The UE is authenticated through EAP between the 3GPP AAA Server and TWAG/TWAP. After the UE is authenticated, the UE is also authorized for access to the APN.
4. The TWAG/TWAP sends the Create Session Request to the PGW. If the UE supports IP address preservation and includes the address in its information, the TWAG/TWAP sets the Handover Indication in the Create Session Request to allow the PGW to reallocate the same IP address or prefix that was assigned the UE while it was connected to the 3GPP IP access network, and to initiate a PCEF-Initiated IP CAN Session Modification with the PCRF.

Note: *Even though the PGW receives a Create Session Request, the PGW does not re-run the workflow selection, that is, the subscriber-analyzer and the control-profile do not change, because this is the same session for the PGW.*

---

5. The PGW sends an AA-Request to the 3GPP-AAA Server with the session information for subscriber authorization.

6 The 3GPP-AAA Server returns an AA-Answer with the subscriber authorization information.

7 The PGW sends a Credit-Control-Request to the PCRF Server.

8 The PCRF Server returns a Credit-Control-Answer to the PGW. The PGW creates a new entry in its bearer context table and generates a Charging ID.

9 The PGW returns a Create Session Response to the TWAG/TWAP, which contains the IP address and/or the prefix that was assigned to the UE while it was connected to the 3GPP IP access network. The Charging ID provided by the PGW is the Charging ID previously assigned to the default bearer of the PDN connection in the 3GPP access.

10 Upon a successful DHCP/NDP exchange, the TWAG/TWAP forwards the allocated IP address to the UE.

11 The GRE tunnel is established between the UE and TWAG/TWAP, and a GTP tunnel is established between the TWAG/TWAP and PGW.

12 The PGW sends a Create Bearer Request to the TWAG/TWAP. The UE switches to non-3GPP access and starts sending uplink data via Wi-Fi access.

13 The TWAG/TWAP returns the Create Bearer Response to the PGW.

Configuring the S2a Interface

The S2a interface connects the PGW in the EPC with the TWAG in the WLAN. The S2a interface on the PGW uses the GPRS tunneling protocol (GTPv2-C and GTPv1-U) to communicate with a remote TWAG.

This section describes the steps for configuring the S2a interface on the MCC PGW. For information on configuring the S2a interface on the MCC TWAG/TWAP, see the *Affirmed Networks TWAG/TWAP Administration Guide*.

To configure the S2a interface on the MCC PGW, perform the following tasks:

1 Create a network context for the S2a interface and configure the S2a GTP-U and GTP-C interface loopback IP addresses.

```
an3000-mcm-slot7cpu1(config)# network-context <name> loopback-ip
<loopback-name> ip-address <ip-address> gtp-u [yes|no]
```

where

Parameter	Description
network-context	Specifies the name of the S2a network context.
loopback-ip	Specifies the name of the loopback address associated with this network context.
ip-address	Specifies a loopback IP address to use for the service endpoint (S2a interface) on the network context. Can be IPv4/IPv6.
gtp-u	Specifies whether to use this loopback address for GTP-U. ■ **yes** – (Default) Use yes only when configuring the S2a GTP-U interface loopback IP address. ■ **no** – Use no when configuring the S2a GTP-C interface. This disables GTP-U, which improves security hardening.

Note: *For dual-stack GTP-U and GTP-C support, configure the S2a network context associated with the GW object with one or more IPv4 and IPv6 loopback addresses.*

2 Associate the S2a network context containing the GTP-U endpoint with the PGW service.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
s2a-interface-gtpu-network-context <name>
```

3 Configure the Access Point Name(s) (APN) accessible via the PGW.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
network-context <name> admin-state enabled
```

where

Parameter	Description
apn	Specifies the name of the APN.
network-context	Specifies the name of the network context for this gateway APN.
admin-state	Specifies whether to enable or disable the administrative state for this APN. ■ **enabled** ■ **disabled**

4 Configure the service profile for GTP-U data path management used to maintain data path status and handle data path failures.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gtp-u-path-management-service-profile
```

The following table describes the configurable parameters.

Parameter	Description
clear-bearers-on-gtp-u-path-failure	Specifies whether to clear bearers when GTP-U path failure occurs. If set to true, the gateway clears the bearers on the peer when a GTP-U path failure occurs. ■ **false** ■ **true**
gtp-u-echo-interval	Specifies the amount of time to wait before initiating an Echo Request over the GTP-U path (ms).
gtp-u-echo-response-timeout	Specifies the amount of time to wait before retransmitting an Echo Request over the GTP-U path (ms).
gtp-u-echo-retransmit-count	Specifies the maximum number of Echo Request retransmission attempts (not including the initial Echo Request) to make before declaring a GTP-U path down.
initiate-gtp-u-echo-requests	Specifies whether to send periodic GTP-U Echo Requests to peers upon learning peer information. ■ **true** – (default) The gateway initiates GTP-U path management procedures for each peer. ■ **false** – The gateway does not send any GTP-U Echo Request messages over the GTP-U path.

5 Configure the GTP parameters for the gateway using the Gateway Profile object.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gateway-common <gateway-profile-name>
```

The following table describes the configurable parameters.

Parameter	**Description**
`allowed-plmnid`	Specifies the policy to authorize the subscriber's PLMN ID during session setup. ■ `any` ■ `homelist` ■ `home-or-roaming-list`
`await-coa-timeout`	Specifies the timeout value (ms) used by the gateway for closed sessions to wait for a Change of Authorization (CoA).
`change-notification`	Specifies whether to enable or disable GTPv2-C Change Notification Request/Response messages. ■ `enabled` (default) ■ `disabled` ***Note:*** *If disabled, the system discards Change Notification messages.*
`clear-subscribers-on-path-failure`	Specifies the policy to clear the subscriber when a path failure occurs. ■ `true` – After the system detects the path failure, all calls that go to that remote peer are actively closed. ■ `false` – All calls are retained until they are manually cleared, cleared by another remote device, or the path recovers and restart processing clears the calls.
`default-gtp-peer-classification`	Specifies the default peer classification when not enough information is present to classify a remote GTP peer as local (same administrative domain) or remote (different administrative domain). ■ `local` ■ `remote`
`defer-delete-session-response`	Specifies whether to defer sending a delete session response until after the delete processing is complete. ■ `true` ■ `false`
`dhcp-relay-response-timeout`	Specifies the timeout value used by the gateway (acting as a DHCP Relay) waiting for the UE to complete its DHCP transaction and obtain an IP (ms).
`gtp-echo-interval`	Specifies the amount of time to wait before initiating an Echo Request over the GTP-C path (ms).
`gtp-echo-response-timeout`	Specifies the amount of time to wait before retransmitting an Echo Request over the GTP-C path (ms).

Parameter	Description
`gtp-echo-retransmit-count`	Specifies the maximum number of Echo Request retransmission attempts (not including the initial Echo Request) to make before declaring a path down.
`idle-session-clearing-state`	Specifies whether to enable or disable Idle Session Clearing. Idle Session Clearing enables Operators to clear subscribers who are idle for longer than the configured `idle-timer-duration`. ■ **`enabled`** ■ **`disabled`** ***Note:*** *This value is defined in the gateway apn object if specified, otherwise the system uses the* `gateway-common profile` *settings.*
`idle-timer-duration`	Specifies the idle timer duration for sessions, in seconds. If a session has been idle for longer than the idle timer duration, the session is cleared. ***Note:*** *The* `idle-timer-duration` *can also be configured in the gateway apn object. The value defined in the gateway apn object takes precedence if the value is non-zero. Otherwise, the value defined in the* `gateway-common profile` *is used.*
`inactivity-timer-duration`	Specifies the inactivity timer duration for sessions, in seconds. If no activity is detected on the session within the inactivity timer duration, the session is declared idle. ***Note:*** *The* `inactivity-timer-duration` *can also be configured in the gateway apn object. The value defined in the gateway apn object takes precedence if the value is non-zero. Otherwise, the value defined in the* `gateway-common profile` *is used.*
`initiate-echo-requests`	Specifies whether to send periodic GTP-C Echo Requests to peers upon learning peer information. ■ **`true`** – (default) When GTP peers are configured and associated with a gateway instance, the gateway begins path management procedures for each peer. ■ **`false`** – The gateway does not send any Echo Request messages to the peer.
`non-matching-plmnid`	Specifies the policy for how to treat subscribers with a home PLMN ID that is neither in the home nor the roaming PLMN ID lists. ■ **`home`** ■ **`visiting`**
`reject-gtpc-messages-from-unconfigured-peers`	Specifies whether to reject GTP-C messages received from non-configured peers. ■ **`true`** ■ **`false`**

Parameter	Description
session-max-clearing-state	Specifies whether to enable or disable PDN Session Clearing for sessions present longer than the maximum time allowed. ■ **enabled** ■ **disabled** ***Note:*** *For PGWs and GGSNs, this value is defined in the gateway apn object if specified, otherwise the system uses the* `gateway-common` *settings.*
session-max-timer-duration	Specifies the maximum timer duration for PDN sessions, in minutes. If a PDN session has been present longer than the session maximum timer duration, the PDN session is cleared.

6 Configure the S2a GTP-C endpoint used for the control plane handling of signaling messages on the PGW.

Note: *The S2a interface uses the same network context and GTP-C endpoint as the S5/S8 interface on the PGW.*

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
s5-s8-interface-network-context <name> home-plmnid-list <list>
gateway-profile <name> admin-state enabled
reject-inter-access-handover-request-no-session [true|false] apn-list
<apn-name>; gtpc-endpoint-params network-context <name> loopback-ip
<primary-loopback-name> secondary-loopback-ip
<secondary-loopback-name>
```

The following table describes the configurable parameters.

Parameter	Description
s5-s8-interface-network-context	Specify the network context containing the S2a interface. ***Note:*** *The S2a interface uses the same network context as the S5/S8 interface.*
home-plmnid-list	Specify the list of PLMN IDs considered as home.
gateway-profile	Specify the name of gateway profile used by this service.
admin-state	Enable the administrative state for this service.

Parameter	Description
reject-inter-access-handover-request-no-session	Specifies whether the PGW accepts or rejects session requests for Wi-Fi to LTE or LTE to Wi-Fi inter-access handovers. ■ **true** – (default) If no prior subscriber PDN connection exists and the PGW receives a Create Session Request with the handover indication set to true, the PGW rejects the request by responding with a Create Session Response with the Cause code as "Context not found." ■ **false** – The PGW handles the Create Session Request as a normal Create Session Request without the handover request.
apn-list	Specify the list of APN names.
gtpc-endpoint-params Configure the following S2a GTP-C endpoint parameters on the PGW:	
network-context	Specify the name of the S2a network context on which this GTP-C endpoint is anchored.
loopback-ip	Specify the name of the primary loopback IP address (associated with this network context) used for the S2a GTP-C endpoint. Can be IPv4 or IPv6.
secondary-loopback-ip	(Optional) Specify the name of the secondary S2a GTP-C loopback IP address associated with this network context. Can be IPv4 or IPv6. ***Note:*** *For dual-stack GTP-C support, the primary and secondary loopback IP addresses should be different IP address types (IPv4 or IPv6).*

7 Configure the S2a interface parameters.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
s2a-interface t3-timeout <value> n3-count <value> dscp-value <value>
```

where

Parameter	Description
s2a-interface	Specifies the GTP T3 timeout value and N3 retry value on a per interface basis and specifies the Differentiated Services Code Point (DSCP) value. ■ **t3-timeout** – Specifies the length of time to wait for an acknowledgment for a message (ms). ■ **n3-count** – Specifies the number of times the gateway retransmits a message if no acknowledgment is received. ■ **dscp-value** – Specifies the DSCP value with which to mark all IP packets on this interface. For more information, see Configuring DSCP Values for Control Plane Traffic (page 14-18).

8 To define a list of specific loopback addresses (within the S2a network context) for the GTP-U endpoint on the PGW, use the following CLI command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
gtpu-loopback-list <loopback-interface-name>
```

Note: *If the list of specific GTP-U loopback addresses has only IPv4 or IPv6 addresses, only that type is used for the sessions on the gateway.*

---

9 Configure the GTP peers for each gateway.

Operators can configure GTP peers on the system and associate that peer with any of the gateway objects. A peer is identified by its IP address, network context ID and the version of the GTP-C protocol it supports. A peer can be associated only with a gateway instance whose GTP-C endpoint address is in the same network context as the peer.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-peer <peer-name>
gtpversion gtpv2 is-present-in-local-plmn [false|true] network-context
<name> peer-ip-address <ip-address>
```

10 Associate the GTP peer with the gateway instance by adding the peer to the list of GTP peers with which the gateway communicates.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
gtp-peer-list <peer-name>
```

Optional S2a Interface Configuration Tasks

This section describes optional S2a interface configuration tasks.

Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on GTP interfaces using Differentiated Services Code Point (DSCP).

GTP interfaces (Gn/Gp, S4, S11, S5/S8, S2a, S2b) – DSCP value configured at the gateway object level per interface (SGW, PGW, GGSN).

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway [sgw|pgw|ggsn]
<name> [gn-interface|gp-interface|s4-interface|s11-interface|
s5-interface|s8-interface|S2a-interface|s2b-interface]
dscp-value <value>
```

Note: *A DSCP value can only be configured for the S2a and the S2b interface on the PGW. You cannot configure a DSCP value for the S2a interface on the MCC TWAG/TWAP or the S2b interface on the MCC ePDG.*

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>
control-dscp-value <value>
```

Note the following guidelines when using DSCP for control plane traffic:

- The DSCP value configured at the gateway object level overrides the network-context DSCP value.
- If a DSCP value is not configured at the gateway object level for GTP-C interfaces (`dscp-value` = `dscp_none`), the network-context DSCP value is used.

Monitoring the S2a Interface

This section describes how to monitor the operational status of the GTP peer via the S2a interface and how to display statistical information for S2a communication messages. This section also describes the alarm notification that occurs when the remote GTP peer goes down. For more information on statistics, see *Acuitas Performance Statistics.*

GTP Peer Status

Use the following CLI command to display the status information of a specific GTP peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway gtp-peer-status
<gateway-name> <peer-ip-address>
```

Interface Statistics

You can display statistical information for S2a interface communication messages for the grid, cluster, and slot level for a specific GTP peer as shown in the following examples.

Interface Statistics at the Grid Level

Use the following CLI command to display the S2a interface GTP grid-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics grid-level
gtp peer <name>
```

Interface Statistics at the Cluster Level

Use the following CLI command to display the S2a interface GTP cluster-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
cluster-level gtp peer <name> <cluster-id>
```

Interface Statistics at the Node Level

Use the following CLI command to display the S2a interface GTP node-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics node-level
gtp peer <name> <cluster-id> <slot>
```

Alarms

An major-level trap similar to the following occurs when a GTP peer goes down. The alarm clears after connectivity is restored.

```
anGtpPeerDown
```

```
Description: GTP peer down. This indicates that the remote GTP peer is
                    not responding to any GTP messages sent to it. If
                    the peer is statically provisioned, the severity
                    is Major. If the peer is dynamic, the severity is
                    Info.
Trap ID: 60
Status:Current
Severity: Info, Major
Auto Clearing:No
Potential Impact: Sessions that go to the remote peer may be cleared,
                    resulting in loss of data connectivity for the
                    affect UEs
Corrective Action: Verify that the remote peer is up and running.
                    Verify that the remote peer is reachable. Verify
                    that the local GTP endpoint is correctly
                    provisioned and running normally.
```

Note: *When the interface connection to a dynamic GTP peer goes down, the system generates an Info alarm.*

Message Definitions

This section lists the supported GTPv2-C message definitions used on the S2a interface and lists the information elements (IEs) contained in each message type.

GTPv2-C Message Definitions

Table 14-1 describes the supported GTPv2-C message definitions for the S2a interface. For more information on GTPv2-C messages, see 3GPP TS 29.274 V11.13.0.

Table 14-1 Supported GTPv2-C S2a Interface Message Definitions

Message Type	Decimal Code	Reference
Create Session Request (page 14-21)	32	3GPP 29.274 V11.13.0
Create Session Response (page 14-24)	33	3GPP 29.274 V11.13.0
Delete Session Request (page 14-29)	36	3GPP 29.274 V11.13.0
Delete Session Response (page 14-30)	37	3GPP 29.274 V11.13.0
Create Bearer Request (page 14-26)	95	3GPP 29.274 V11.13.0
Create Bearer Response (page 14-28)	96	3GPP 29.274 V11.13.0
Update Bearer Request (page 14-32)	97	3GPP 29.274 V11.13.0
Update Bearer Response (page 14-33)	98	3GPP 29.274 V11.13.0
Delete Bearer Request (page 14-29)	99	3GPP 29.274 V11.13.0
Delete Bearer Response (page 14-31)	100	3GPP 29.274 V11.13.0

Create Session Request

The TWAG/TWAP sends a Create Session Request message to the PGW as part of the Initial Attach in WLAN on GTP S2a procedure.

Table 14-2 lists the supported Create Session Request information elements.

Table 14-2 Create Session Request Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
IMSI	Conditional	The International Mobile Subscriber Identity (IMSI) is included in messages on the S2a interface.	IMSI	0

Table 14-2 Create Session Request Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
MSISDN	Conditional	The TWAN includes the Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) IE on the S2a interface during an Initial Attach in WLAN on GTP S2a if provided by the HSS/AAA.	MSISDN	0
Serving Network	Conditional	This IE indicates the PLMN identity of the selected PLMN used for 3GPP-based access authentication. In roaming scenarios, the selected PLMN is the PLMN of the 3GPP AAA Proxy and in non-roaming scenarios, the selected PLMN is the PLMN of the 3GPP AAA Server.	Serving Network	0
RAT Type	Mandatory	The TWAN sets the RAT Type value to “WLAN” on the S2a interface.	RAT Type	0
Indication Flags	Conditional	Dual Address Bearer Flag: The TWAN sets this flag to 1 on the S2a interface if it supports IPv4 and IPv6 and the PDN Type determined from the user subscription data is set to IPv4v6.	Indication	0
Sender F-TEID for Control Plane	Mandatory	Specifies the Fully Qualified Tunnel Endpoint Identifier (F-TEID) for the GTP-C tunnel.	F-TEID	0
Access Point Name (APN)	Mandatory	None	APN	0
Selection Mode	Conditional	Included on the S2a interface for an Initial Attach in WLAN on GTP S2a. The value is set to “MS or network provided APN, subscription verified”.	Selection Mode	0
PDN Address Allocation (PAA)	Conditional	Included on the S2a interface for an Initial Attach in WLAN on GTP S2a.	PAA	0
Aggregate Maximum Bit Rate (APN-AMBR)	Conditional	Included on the S2a interface for an Initial Attach in WLAN on GTP S2a.	AMBR	0
Bearer Contexts to be created	Mandatory	One bearer is included on the S2a interface for an Initial Attach in WLAN on GTP S2a.	Bearer Context	0
Trace Information	Conditional	Included on the S2a interface if a PGW trace is activated.	Trace Information	0
Recovery	Conditional	Included on the S2a interface if contacting the peer node for the first time.	Recovery	0
TWAN-FQ-CSID	Conditional	The TWAN includes this IE on the S2a interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID	3

Table 14-2 Create Session Request Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
TWAN LDN	Conditional	Sent by the TWAN to the PGW on the S2a interface (see 3GPP TS 32.423 [44]), when contacting the peer node for the first time. Operators can assign a local distinguished name using the following CLI command: an3000-mcm-slot7cpu1(config)# **zone <*name*> gateway wag <*gateway-name*> local-distinguished-name <*name*>**	Local Distinguished Name	3
3GPP AAA Server Identifier	Optional	The TWAN may include this IE on the S2a interface to provide the selected 3GPP AAA Server Identifier to the PGW, as defined in 3GPP TS 29.274 V13.6.0.0. ***Note:*** *If supported, the PGW contacts the 3GPP AAA server (identified by this IE which carries the Origin-Host and Origin-Realm included in the DEA message received by the TWAN over the STa interface) for establishing the S6b session.* If received, the MCC PGW uses the received server identifier (ID) and realm identified in this IE to select the S6b Diameter peer for authentication. If the received server ID and realm do not match a configured S6b peer, the MCC PGW uses the existing peer selection algorithm to select an S6b peer, but encodes the received server ID and realm in the Destination-Host and Destination-Realm AVPs sent in the AA-Request (AAR) message to the peer. If this IE contains either the server ID (and not the realm) or the realm (and not the server ID), the MCC PGW will use only the server ID or only the realm to select an S6b peer. The MCC PGW ignores this IE in the following cases: ■ If the default S6b interface is not configured in the Workflow control profile. ■ If the MCC PGW receives this IE from the TWAG during a 3GPP to non-3GPP handover and if there is already an existing S6b Diameter connection for that session. Ideally, this IE will have the same server ID and realm as the existing S6b connection’s server, but if they are different, the old AAA server should send a redirect indication in the AA-Answer message to the PGW. ***Note:*** *If you do not want the PGW to send the Destination-Host AVP in the initial AAR, suppress the Destination-Host AVP in the S6b AAR data record template.*	Node Identifier	0

Table 14-2 Create Session Request Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Protocol Configuration Options (PCO)	Conditional	Included on the S2a interface if it is available.	PCO	78
Additional Protocol Configuration Options (APCO)	Optional	The TWAN may include this IE on the S2a interface to retrieve additional IP configuration parameters from the PGW (for example, DNS server).	Additional Protocol Configuration Options (APCO)	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Bearer Contexts to be Created Within Create Session Request

Within the Create Session Request message, the Bearer Contexts to be Created IE contains other IEs and is considered a Grouped IE.

Table 14-3 lists the information elements contained in the Bearer Contexts to be Created IE.

Table 14-3 Bearer Contexts IEs (within Create Session Request)

Octet 1	Bearer Context IE Type = 93 (decimal)			
Octets 2 and 3	Length = n			
Octet 4	Spare and Instance Fields			
IE Name	**Requirement**	**Condition**	**IE Type**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	0
S2a-U TWAN F-TEID	Conditional	Included on the S2a interface for an Initial Attach in WLAN on GTP S2a.	F-TEID	6

Create Session Response

The PGW sends a Create Session Response message to the TWAG/TWAP as an answer to a Create Session Request message. The Create Session Response message contains information about the default bearer, such as the UE’s allocated IP address and the tunnel endpoint IP addresses (TEID).

Table 14-4 lists the supported Create Session Response information elements.

Table 14-4 Create Session Response Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	0
Sender F-TEID for Control Plane	Conditional	This IE is sent on the S11/S4 interfaces. For the S2a interface it is not needed because its content would be identical to the IE PGW S2a F-TEID for PMIP based interface or for GTP based Control Plane interface.	F-TEID	
PGW S5/S8/ S2a/S2b F-TEID for PMIP based interface or for GTP based Control Plane interface	Conditional	The PGW includes this IE on the S2a interface during the Initial Attach in WLAN on GTP S2a procedure.	F-TEID	1
PDN Address Allocation (PAA)	Conditional	Included on the S2a interfaces for the Initial Attach in WLAN on GTP S2a procedures.	PAA	0
Aggregate Maximum Bit Rate (APN-AMBR)	Conditional	Included on the S2a interface if the Policy and Charging Rules Function (PCRF) has modified the APN-AMBR.	AMBR	0
Bearer Contexts created	Mandatory	One bearer is included on the S2a interface for the Initial Attach in WLAN on GTP S2a.	Bearer Context	0
PGW-FQ-CSID	Conditional	Included by the PGW on the S2a interface when forwarded by the SGW on the S11 interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID	0
Protocol Configuration Options (PCO)	Conditional	Included on the S2a interface if it is available.	PCO	78
Additional Protocol Configuration Options (APCO)	Optional	The PGW may include this IE on the S2a interface to provide the TWAN with additional IP configuration parameters (for example, DNS server), if a corresponding request was received in the Create Session Request message.	Additional Protocol Configuration Options (APCO)	0

Table 14-4 Create Session Response Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Trusted WLAN IPv4 Parameters	Conditional	The PGW includes this IE on the S2a interface if the PDN Type in the PAA is set to IPv4 or IPv4v6 and includes: ■ The Subnet Prefix Length of the subnet from which the PGW allocates the UE's IPv4 address. ■ The IPv4 Default Router Address which belongs to the same subnet as the IPv4 address allocated to the UE. When the TWAG/TWAP receives a Create Session Response message on the S2a interface containing the Trusted WLAN IPv4 Parameters IE, the TWAG/TWAP extracts the Subnet Prefix Length and IPv4 Default Router Address, contained within the Trusted WLAN IPv4 Parameters IE, and sends this information as part of the Subnet Mask Option (1) and Router Option (3) in the DHCP Offer message.	IPv4 Configuration Parameters (IP4CP)	0

Bearer Contexts Created Within Create Session Response

Within the Create Session Response message, the Bearer Contexts Created IE contains other IEs and is considered a Grouped IE.

Table 14-5 lists the information elements contained in the Bearer Contexts Created IE.

Table 14-5 Bearer Contexts Created IEs (within Create Session Response)

Octet 1	Bearer Context IE Type = 93 (decimal)			
Octets 2 and 3	Length = n			
Octet 4	Spare and Instance Fields			
IE Name	**Requirement**	**Condition**	**IE Type**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	0
Cause	Mandatory	Indicates whether the bearer handling was successful. If not, this IE provides information on the reason.	Cause	0
S2a-U PGW F-TEID	Conditional	Included on the S2a interface during the Initial Attach in WLAN on GTP S2a.	F-TEID	5

Create Bearer Request

The PGW sends a Create Bearer Request message to the TWAG/TWAP to create a new dedicated bearer.

Table 14-6 lists the Create Bearer Request information elements.

Table 14-6 Create Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Linked Bearer Identity (LBI)	Mandatory	Indicates the default bearer associated with the PDN connection.	EBI (EPS Bearer ID)	0
Bearer Contexts	Mandatory	Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	0
PGW-FQ-CSID	Conditional	Included by the PGW on the S2a interface when received by the SGW on the S11 interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Bearer Contexts Within Create Bearer Request

Within the Create Bearer Request message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 14-7 lists the information elements contained in the Bearer Contexts IE.

Table 14-7 Bearer Contexts IEs (within Create Bearer Request)

Octet 1		Bearer Context IE Type = 93 (decimal)		
Octets 2 and 3		Length = n		
Octet 4		Spare and Instance Fields		
IE Name	**Requirement**	**Condition**	**IE Type**	**Instance Value**
EPS Bearer ID	Mandatory	This IE is set to 0.	EBI	0
TFT	Mandatory	This IE can contain both uplink and downlink packet filters to be sent to the UE or the TWAN/ePDG. ***Note:*** *If the size of Traffic Flow Template (TFT) packet filters exceeds the 255 byte limit or if the number of packet filters exceeds 15 in the original Create Bearer/PDP Context Request or Update Bearer/PDP Context Request message, the gateway sends the remaining TFT packet filters in subsequent Update Bearer/PDP Context Request messages.This capability is supported on the PGW for Create Bearer Request messages sent in handover scenarios, such as 3GPP to non-3GPP access and vice versa.*	Bearer TFT	0
S2a-U PGW F-TEID	Conditional	Included on the S2a interface for the user plane.	F-TEID	5

Table 14-7 Bearer Contexts IEs (within Create Bearer Request)

Octet 1	Bearer Context IE Type = 93 (decimal)			
Octets 2 and 3	Length = n			
Octet 4	Spare and Instance Fields			
IE Name	**Requirement**	**Condition**	**IE Type**	**Instance Value**
Bearer Level QoS	Mandatory	None	Bearer QoS	0
Charging Id	Conditional	Included on the S2a interface if available.	Charging Id	0

Create Bearer Response

The TWAG/TWAP sends a Create Bearer Response message to the PGW as an answer to a Create Bearer Request message.

Table 14-8 lists the Create Bearer Response information elements.

Table 14-8 Create Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	0
Bearer Contexts	Mandatory	Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	0
Recovery	Conditional	Included on the S2a interface if the SGW is contacting the PGW for the first time.	Recovery	0
TWAN-FQ-CSID	Conditional	Included by the TWAN on the S2a interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID	3
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Bearer Contexts Within Create Bearer Response

Within the Create Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 14-9 lists the information elements contained in the Bearer Contexts IE.

Table 14-9 Bearer Contexts IEs (within Create Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)			
Octets 2 and 3	Length = n			
Octet 4	Spare and Instance Fields			
IE Name	**Requirement**	**Condition**	**IE Type**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	0
S2a-U TWAN F-TEID	Conditional	Included on the S2a interface.	F-TEID	10
S2a-U PGW F-TEID	Conditional	Used on the S2a interface to correlate the bearers with those in the Create Bearer Request.	F-TEID	11

Delete Session Request

The TWAG/TWAP sends a Delete Session Request message to the PGW as part of the following procedures:

- UE/TWAN Initiated Detach and UE/TWAN Requested PDN Disconnection in WLAN on GTP S2a
- HSS/AAA Initiated Detach in WLAN on GTP S2a

Table 14-10 lists the Delete Session Request information elements.

Table 14-10 Delete Session Request Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Linked EPS Bearer ID	Conditional	Included on the S2a interface to indicate the default bearer associated with the PDN being disconnected except during the TAU/RAU/Handover procedure with an SGW relocation.	EBI	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Delete Bearer Request

The PGW sends a Delete Bearer Request message to the TWAG/TWAP TWAN as part of the PGW Initiated Bearer Resource Allocation Deactivation in WLAN on GTP on S2a procedure.

Table 14-11 lists the Delete Bearer Request information elements.

Table 14-11 Delete Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Linked EPS Bearer ID (LBI)	Conditional	Included on the S2a interface to indicate the default bearer associated with the PDN being disconnected. This IE is present only when the EPS Bearer is not present.	EBI	0
EPS Bearer ID	Conditional	Included on the S2a interface for bearers different from the default one. (for example, for dedicated bearers). In this case, at least one dedicated bearer is included. Several IEs with this type and instance values are included, if necessary, to represent a list of bearers.	EBI	1
PGW-FQ-CSID	Conditional	Included by the PGW on the S2a interface when forwarded by the SGW on the S11 interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID	0
Cause	Conditional	Included on the S2a interface if the message is caused by a handover from non-3GPP to 3GPP. In this case, the Cause value is set to “Access changed from Non-3GPP to 3GPP”.	Cause	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Delete Session Response

The PGW sends a Delete Session Response message to the TWAG/TWAP as an answer to a Delete Session Request message.

Table 14-12 lists the Delete Session Response information elements.

Table 14-12 Delete Session Response Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Cause	Mandatory	None	Cause	0
Recovery	Conditional	Included on S2a interface if contacting the peer for the first time.	Recovery	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Delete Bearer Response

The TWAG/TWAP sends a Delete Bearer Response message to the PGW as an answer to a Delete Bearer Request message.

Table 14-13 lists the Delete Bearer Response information elements.

Table 14-13 Delete Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Linked EPS Bearer ID (LBI)	Conditional	Included on the S2a interface to indicate the default bearer associated with the PDN being disconnected.	EBI	0
Bearer Contexts	Conditional	Used for dedicated bearers. At least one dedicated bearer is present. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	0
Recovery	Conditional	Included on S2a interface if contacting the peer for the first time.	Recovery	0
TWAN-FQ-CSID	Conditional	Included by the TWAN on the S2a interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID	3
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Bearer Contexts Within Delete Bearer Response

Within the Delete Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 14-14 lists the information elements contained in the Bearer Contexts IE.

Table 14-14 Bearer Contexts IEs (within Delete Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)			
Octets 2 and 3	Length = n			
Octet 4	Spare and Instance Fields			
IE Name	**Requirement**	**Condition**	**IE Type**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	0

Update Bearer Request

The PGW sends an Update Bearer Request message to the TWAG/TWAP as part of the following procedures:

- PGW Initiated Bearer Modification
- HSS Initiated Subscribed QoS Modification

Table 14-15 lists the Update Bearer Request information elements.

Table 14-15 Update Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Bearer Contexts	Mandatory	Used for bearers that require QoS/TFT modification. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers. If there is no QoS/TFT modification, only one IE with this type and instance value is included.	Bearer Context	0
Aggregate Maximum Bit Rate (APN-AMBR)	Mandatory	None	AMBR	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Bearer Contexts Within Update Bearer Request

Within the Update Bearer Request message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 14-16 lists the information elements contained in the Bearer Contexts IE.

Table 14-16 Bearer Contexts IEs (within Update Bearer Request)

Octet 1	Bearer Context IE Type = 93 (decimal)			
Octets 2 and 3	Length = n			
Octet 4	Spare and Instance Fields			
IE Name	**Requirement**	**Condition**	**IE Type**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	0
TFT	Conditional	Included on the S2a interface if the message relates to a bearer modification or TFT change.	Bearer TFT	0
Bearer Level QoS	Conditional	Included on the S2a interface if a QoS modification is requested.	Bearer QoS	0

Update Bearer Response

The TWAG/TWAP sends an Update Bearer Response message to the PGW as an answer to an Update Bearer Request message.

Table 14-17 lists the Update Bearer Response information elements.

Table 14-17 Update Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	Instance Value
Cause	Mandatory	None	Cause	0
Bearer Contexts	Mandatory	Used for bearers for which QoS and TFT modification are requested. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	0
Recovery	Conditional	Included on S2a interface if contacting the peer for the first time.	Recovery	0
TWAN-FQ-CSID	Conditional	Included by the TWAN on the S2a interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID	3
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	Vendor Specific

Bearer Contexts Within Update Bearer Response

Within the Update Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 14-18 lists the information elements contained in the Bearer Contexts IE.

Table 14-18 Bearer Contexts IEs (within Update Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)			
Octets 2 and 3	Length = n			
Octet 4	Spare and Instance Fields			
IE Name	**Requirement**	**Condition**	**IE Type**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	0

S2b Interface

This section describes the S2b interface, which is used for transmitting user data traffic and signaling messages between the evolved Packet Data Gateway (ePDG) and the Packet Data Network Gateway (PGW). For detailed information about the ePDG, refer to the *Affirmed Networks ePDG Administration Guide*.

Overview

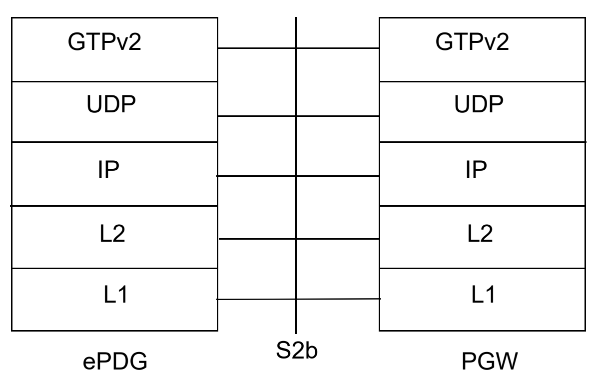
The 3GPP S2b interface provides user plane tunneling and tunnel management with related control and mobility support between the evolved Packet Data Gateway (ePDG) and the Packet Data Network Gateway (PGW). The S2b interface supports the following PGW and ePDG services:

- Provides access to the Evolved Packet Core (EPC) through the ePDG to non-3GPP untrusted access networks
- Supports non-roaming scenarios where the ePDG and PGW are located within the same Public Land Mobile Network (PLMN).
- Supports roaming between an ePDG in a visited PLMN and a PGW in a home PLMN
- Supports handovers between Wi-Fi and LTE networks

This document describes the S2b interface implementation in the *Affirmed Networks Mobile Content Cloud* (MCC) in accordance with 3GPP TS 23.402. The S2b interface uses the GPRS tunneling protocol (GTPv2-C and GTPv1-U) as defined in 3GPP TS 29.274 V12.8.0 and 3GPP TS 29.281 V9.3.0 to transfer signaling messages and user data over the EPC.

Figure 15-1 illustrates the GTPv2 protocol stack for the S2b interface.

Figure 15-1 GTPv2 Protocol Stack for the S2b Interface



S2b Message Flows

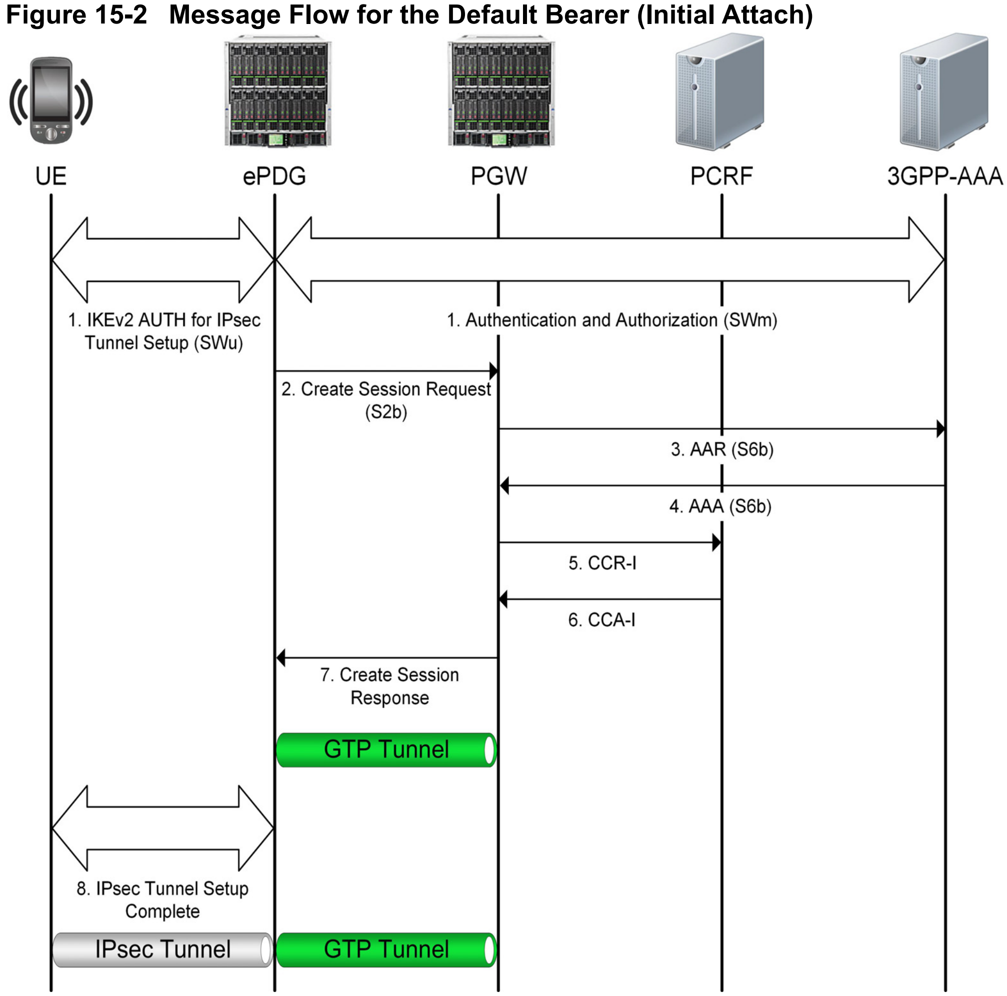
This section shows the basic message flows between the ePDG and the PGW over the S2b interface as defined in 3GPP TS 23.402, Section 7.2.4.

- Establishing the Default Bearer (Initial Attach) (page 15-3)
- Establishing a Dedicated Bearer (page 15-5)

Establishing the Default Bearer (Initial Attach)

Figure 15-2 illustrates an example of the message flow from the UE in the non-3GPP untrusted access network through the ePDG to the Evolved Packet Core (EPC).

Figure 15-2 Message Flow for the Default Bearer (Initial Attach)



1 The user equipment (UE) sends an Attach-Request message to the ePDG. The UE is authenticated through EAP-AKA and IKE signaling between the 3GPP AAA Server and ePDG. Upon a successful authorization, the 3GPP AAA server returns the APN-AMBR, static QoS profile, and trace information.

2 The ePDG sends a Create Session Request message to the PGW and sets up a GTP-C tunnel with the PGW for signaling messages. The message includes UE IMSI, APN, ePDG TEID, PDN address, EPS bearer identity, default EPS bearer QoS, and other control and user plane information. Additional parameters include the authentication credentials for secondary authentication if the UE provides this information (for example, PAP credentials for L2TP access).

3 The PGW sends an AA-Request message to the Diameter 3GPP-AAA Server with the session information for subscriber authorization. When the UE attaches to the EPC via S2b and GTPv2, the S6b interface is used to update the 3GPP AAA Server with the PGW address and GTPv2 protocol information.

4 The Diameter 3GPP-AAA Server returns an AA-Answer with the subscriber authorization information.

5 The PGW sends a Credit Control Request to the PCRF Server.

6 The PCRF Server returns a Credit Control Answer to the PGW. The PGW creates a new entry in its bearer context table and generates a Charging ID.

7 The PGW returns a Create Session Response with the PGW address and TEIDs (user and control plane), PDN type and address, EPS bearer identity and QoS, APN-AMBR, Charging ID, Cause, and IP address(es) allocated for the UE.

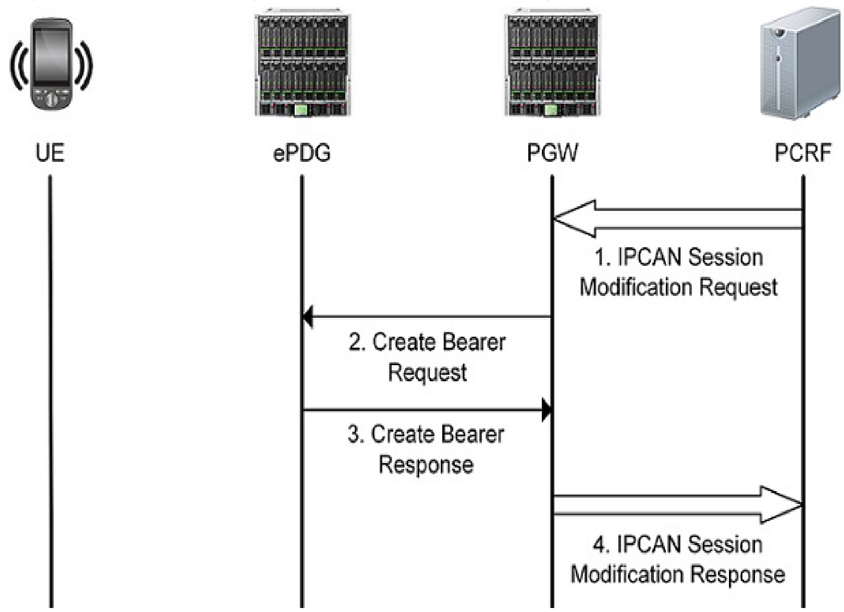
8 The ePDG forwards the allocated IP address to the UE and completes the IPsec tunnel setup.

9 The ePDG maps the UE IPsec tunnel to the GTP tunnel established with the PGW.

Establishing a Dedicated Bearer

Figure 15-3 illustrates the message flow between the ePDG and PGW over the S2b interface for establishing a dedicated bearer. Dedicated bearers must be established when the UE requires a specific QoS for the requested service.

Figure 15-3 Message Flow for Establishing a Dedicated Bearer



1 If dynamic PCC is deployed, the PCRF sends a QoS policy message to the PGW (similar to the PCRF response in PCEF initiated IP-CAN Session Modification up to where the PGW requests IP-CAN bearer signaling). If dynamic PCC is not deployed, the PGW may apply local QoS policy.

2 The PGW sends a Create Bearer Request to the ePDG. The Linked EPS Bearer Identity (LBI) is the EPS Bearer Identity of the default bearer.

3 The ePDG selects an EPS Bearer Identity, which has not yet been assigned to the UE and uses the uplink packet filter to determine the mapping of uplink traffic flows to the S2b bearer. The ePDG acknowledges the S2b bearer activation to the PGW by sending a Create Bearer Response.

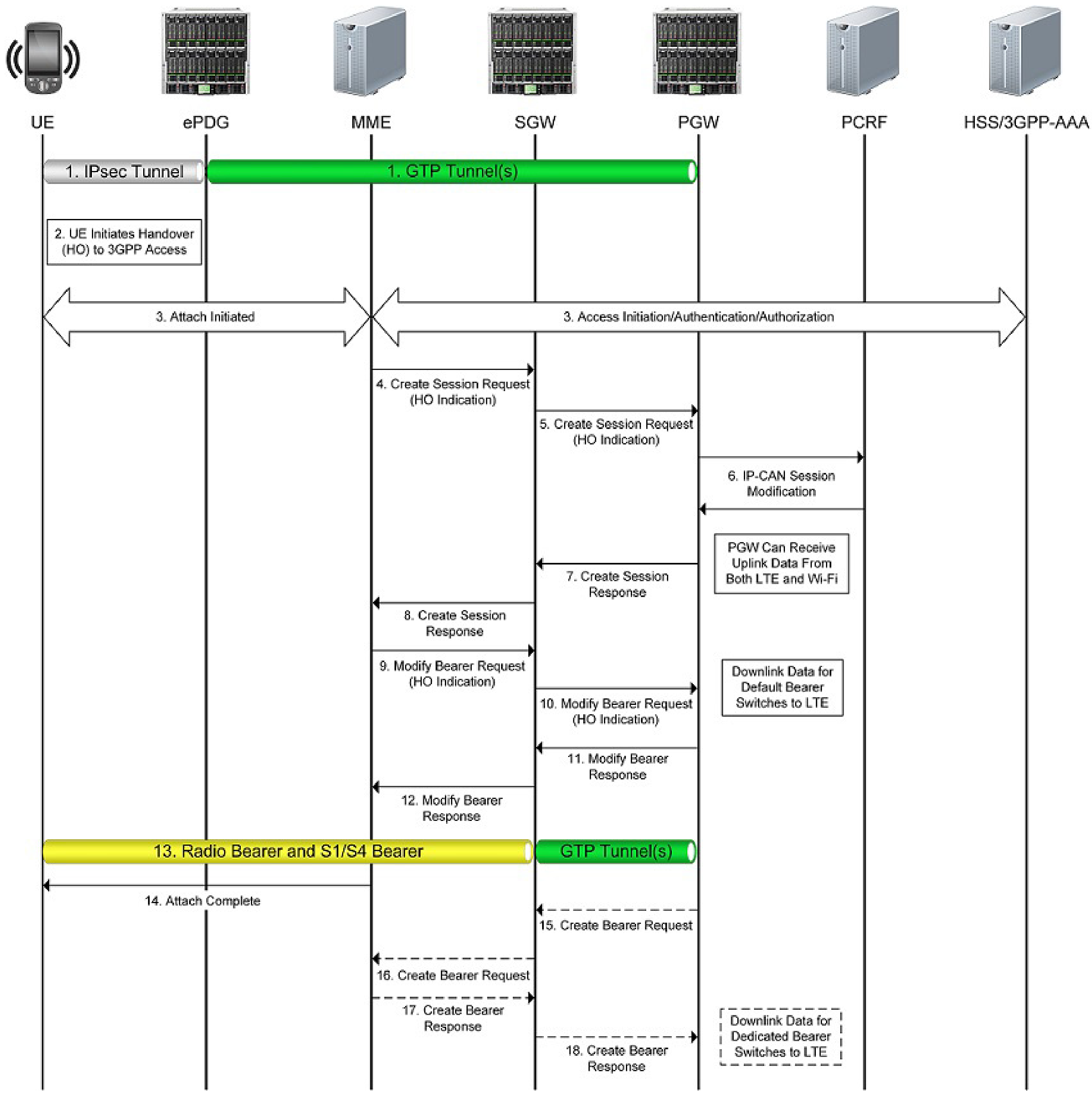
4 If the dedicated bearer activation is the result of a QoS policy message from the PCRF, the PGW indicates to the PCRF whether the requested QoS policy can be enforced or not, which completes the IP-CAN Session Modification.

Wi-Fi to LTE / LTE to Wi-Fi Handovers

Over the S2b interface, the ePDG and PGW support handovers from the non-3GPP untrusted IP access network to the 3GPP LTE macrocellular network and vice versa. This section describes the handover message flows.

Figure 15-4 shows the message flow in a handover from untrusted Wi-Fi to LTE.

Figure 15-4 Wi-Fi-to-LTE Handover Message Flow



1 The UE and ePDG have an established IPsec tunnel, and the ePDG and PGW have an established GTP tunnel over S2b.

2 The UE discovers the 3GPP network and sends the Attach Request to the MME.

3 The MME may contact the HSS/3GPP AAA Server and authenticate the UE. The MME receives information about the PDNs connected to the UE over non-3GPP access in the subscriber data obtained from the HSS.

4 The MME sends a Create Session Request with Handover Indication to the selected SGW.

5 The SGW sends a Create Session Request with Handover Indication to the PGW. The PGW does not yet change the tunnel from non-3GPP IP access to 3GPP access.

6 The PGW executes a PCEF-Initiated IP CAN Session Modification with the PCRF to obtain the information needed to function as the PCEF for all the active sessions the UE has with the new IP-CAN type as a result of the handover.

7 The PGW responds with a Create Session Response to the SGW, which contains the IP address/prefix that was assigned to the UE while it was connected to the non-3GPP IP access. It also contains the Charging ID previously assigned to the PDN connection in the non-3GPP access, which still applies to the non-3GPP access. At this point, the PGW can receive uplink data from both Wi-Fi and LTE.

8 The SGW returns a Create Session Response to the MME, which includes the IP address of the UE.

9 The MME sends a Modify Bearer Request to the SGW.

10 The SGW sends a Modify Bearer Request to the PGW to instruct the PGW to tunnel packets from non-3GPP IP access to 3GPP access and immediately start routing packets to the SGW for the established default bearer and any dedicated EPS bearers.

Note: *The PGW removes the old PCC rules for the untrusted non-3GPP IP access and applies the new rules for E-UTRAN access for charging. The Charging ID previously used for the PDN connection in the non-3GPP access now only applies to the default bearer in use for E-UTRAN access. If dedicated bearers are created, the PGW assigns a new Charging ID for each bearer.*

---

11 The PGW acknowledges by sending the Modify Bearer Response to the SGW.

12 The SGW acknowledges by sending the Modify Bearer Response (EPS Bearer Identity) message to the MME.

13 The S1/S4 bearer setup is complete, radio and access bearers are established in the 3GPP access network, and GTP tunnel(s) are established over S5. The UE sends and receives data through the E-UTRAN system.

14 The UE attach is complete.

15 The PGW sends the Create Bearer Request to the SGW.

Note: *Steps 15-18 apply only if a dedicated bearer was established before the handover occurred.*

---

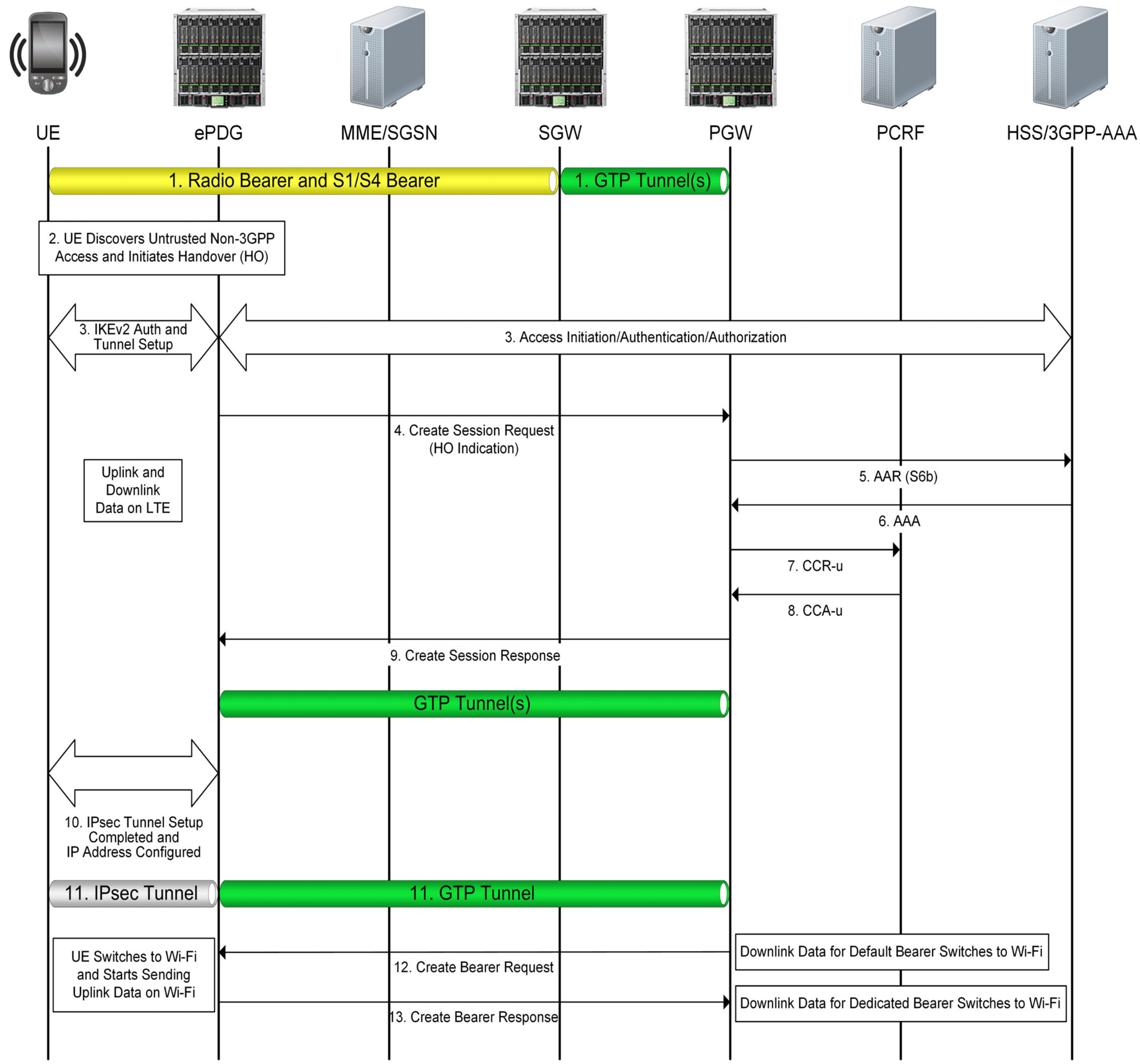
16 The SGW forwards the Create Bearer Request to the MME.

17 The MME returns the Create Bearer Response to the SGW.

18 The SGW forwards the Create Bearer Response to the PGW. The downlink data for the dedicated bearer switches to LTE.

Figure 15-5 shows the message flow in a handover from LTE to untrusted Wi-Fi.

Figure 15-5 LTE-to-Wi-Fi Handover Message Flow



1 The UE is initially attached to the 3GPP access network.

2 The UE moves and attaches to an untrusted non-3GPP IP access network.

3 The UE initiates the IKEv2 tunnel establishment with the ePDG. After the UE is authenticated, the UE is also authorized for access to the APN.

4 The ePDG sends the Create Session Request to the PGW. If the UE supports IP address preservation and includes the address in its information, the ePDG sets the Handover Indication in the Create Session Request to allow the PGW to reallocate the same IP address or prefix that was assigned the UE while it was connected to the 3GPP IP access network, and to initiate a PCEF-Initiated IP CAN Session Modification with the PCRF.

Note: *Even though the PGW receives a Create Session Request, the PGW does not re-run the workflow selection, that is, the subscriber-analyzer and the control-profile do not change, because this is the same session for the PGW.*

---

5 The PGW sends an AA-Request message to the Diameter 3GPP-AAA Server with the session information for subscriber authorization.

6 The Diameter 3GPP-AAA Server returns an AA-Answer with the subscriber authorization information.

7 The PGW sends a Credit Control Request to the PCRF Server.

8 The PCRF Server returns a Credit Control Answer to the PGW. The PGW creates a new entry in its bearer context table and generates a Charging ID.

9 The PGW returns a Create Session Response message to the ePDG, which contains the IP address and/or the prefix that was assigned to the UE while it was connected to the 3GPP IP access network. The Charging ID provided by the PGW is the Charging ID previously assigned to the default bearer of the PDN connection in the 3GPP access.

10 The ePDG and UE continue the IKEv2 exchange and IP address configuration. IPsec tunnel setup is complete.

11 The IPsec tunnel is established between the UE and ePDG, and a GTP tunnel is established between the ePDG and PGW.

12 The PGW sends a Create Bearer Request to the ePDG. The UE switches to non-3GPP access and starts sending uplink data via Wi-Fi access.

13 The ePDG returns the Create Bearer Response to the PGW.

Configuring the S2b Interface

This section describes the steps for configuring the S2b interface on the MCC PGW.

Note: *For information about configuring the S2b interface on ePDG, see the Affirmed Networks Evolved Packet Data Gateway Administration Guide.*

To configure the S2b interface on the MCC PGW, perform the following tasks:

1 Create a network-context for the S2b interface. For information on configuring network-contexts, see the *Network Context Parameters* chapter in this guide.

Note: *For dual-stack GTP-U and GTP-C support, configure the S2b network-context associated with the gateway object with one or more IPv4 and IPv6 loopback addresses.*

---

2 Enable GTP-U services for the loopback address(es) defined for the S2b network-context.

```
an3000-mcm-slot7cpu1(config)# network-context <name> loopback-ip
<loopback-name> ip-address <ip-address> gtp-u yes
```

where

Parameter	Description
network-context	Specifies the name of the network context created for the S2b interface.
loopback-ip	Specifies the name of the loopback address associated with this network context.
ip-address	Specifies the IP address of this loopback object. Can be IPv4/IPv6.
gtp-u yes	Specifies that this loopback address is used for GTP-U.

3 Configure the Access Point Name(s) (APN) accessible via the PGW.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
network-context <name> admin-state enabled
```

4 Configure the service profile for GTP-U data path management used to maintain data path status and handle data path failures.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gtp-u-path-management-service-profile
```

The following table describes the configurable parameters.

Parameter	Description
clear-bearers-on-gtp-u-path-failure	Specifies whether to clear bearers when GTP-U path failure occurs. If set to true, the gateway clears the bearers on the peer when a GTP-U path failure occurs. ■ **false** ■ **true**
gtp-u-echo-interval	Specifies the amount of time to wait before initiating an Echo Request over the GTP-U path (ms).
gtp-u-echo-response-timeout	Specifies the amount of time to wait before retransmitting an Echo Request over the GTP-U path (ms).
gtp-u-echo-retransmit-count	Specifies the number of echo retransmissions to make before declaring a GTP-U path down event request.
initiate-gtp-u-echo-requests	Specifies whether to send periodic GTP-U Echo Requests to peers upon learning peer information. ■ **true** – (default) The gateway initiates GTP-U path management procedures for each peer. ■ **false** – The gateway does not send any GTP-U Echo Request messages over the GTP-U path.

5 Configure the GTP parameters for the gateway using the gateway profile object.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gateway-common <gateway-profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
allowed-plmnid	Specifies the policy to authorize the subscriber's PLMN ID during session setup. ■ **any** ■ **homelist** ■ **home-or-roaming-list**
await-coa-timeout	Specifies the timeout value (ms) used by the gateway for closed sessions to wait for a Change of Authorization (CoA).
change-notification	Specifies whether to enable or disable GTPv2-C Change Notification Request/Response messages. ■ **enabled** (default) ■ **disabled** ***Note:*** *If disabled, the system discards Change Notification messages.*

Parameter	Description
`clear-subscribers-on-path-failure`	Specifies the policy to clear the subscriber when a path failure occurs. ■ `true` – After the system detects the path failure, all calls that go to that remote peer are actively closed. ■ `false` – All calls are retained until they are manually cleared, cleared by another remote device, or the path recovers and restart processing clears the calls.
`default-gtp-peer-classification`	Specifies the default peer classification when not enough information is present to classify a remote GTP peer as local (same administrative domain) or remote (different administrative domain). ■ `local` ■ `remote`
`defer-delete-session-response`	Specifies whether to defer sending a delete session response until after the delete processing is complete. ■ `true` ■ `false`
`dhcp-relay-response-timeout`	Specifies the timeout value used by the gateway (acting as a DHCP Relay) waiting for the UE to complete its DHCP transaction and obtain an IP (ms).
`gtp-echo-interval`	Specifies the amount of time to wait before initiating an Echo Request over the GTP-C path (ms).
`gtp-echo-response-timeout`	Specifies the amount of time to wait before retransmitting an Echo Request over the GTP-C path (ms).
`gtp-echo-retransmit-count`	Specifies the number of echo retransmissions to make before declaring a GTP-C path down event request.
`idle-session-clearing-state`	Specifies whether to enable or disable Idle Session Clearing. Idle Session Clearing enables Operators to clear subscribers who are idle for longer than the configured **idle-timer-duration**. ■ `enabled` ■ `disabled` ***Note:*** *For PGWs and GGSNs, this value is defined in the gateway apn object if specified, otherwise the system uses the gateway-common settings.*
`idle-timer-duration`	Specifies the idle timer duration for sessions, in minutes. If a session has been idle for longer than the idle timer duration, the session is cleared

Parameter	Description
inactivity-timer-duration	Specifies the inactivity timer duration for sessions, in minutes. If no activity is detected on the session within the inactivity timer duration, the session is declared idle. ***Note:*** *For PGWs and GGSNs, this value is defined in the gateway apn object if the value is non-zero, and the gateway-common profile if the value is 0 in the gateway apn object.*
initiate-echo-requests	Specifies whether to send periodic GTP-C Echo Requests to peers upon learning peer information. ■ **true** – (default) When GTP peers are configured and associated with a gateway instance, the gateway begins path management procedures for each peer. ■ **false** – The gateway does not send any Echo Request messages to the peer.
non-matching-plmnid	Specifies the policy for how to treat subscribers with a home PLMN ID that is neither in the home nor the roaming PLMNID lists. ■ **home** ■ **visiting**
reject-gtpc-messages-from-unconfigured-peers	Specifies whether to reject GTP-C messages received from non-configured peers. ■ **true** ■ **false**
session-max-clearing-state	Specifies whether to enable or disable PDN Session Clearing for sessions present longer than the maximum time allowed. ■ **enabled** ■ **disabled** ***Note:*** *For PGWs and GGSNs, this value is defined in the gateway apn object if specified, otherwise the system uses the gateway-common settings.*
session-max-timer-duration	Specifies the maximum timer duration for PDN sessions, in minutes. If a PDN session has been present longer than the session maximum timer duration, the PDN session is cleared.

6 Configure the list of PLMN IDs for the home list:

```
an3000-mcm-slot7cpu1(config)# service-construct plmnid-list
home-list-plmnid <plmnid-list>
```

7 Configure the S2b GTP-C endpoint used for the control plane handling of signaling messages on the PGW.

Note: *The S2b interface uses the same GTP-C endpoint and network context as the S5/S8 interface.*

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
s5-s8-interface-network-context <name>
home-plmnid-list <list-name>
gtpc-endpoint-params network-context <name> loopback-ip
<primary-loopback-name> secondary-loopback-ip
<secondary-loopback-name> dual-stack-behavior
[initial-ip-type-used-by-peer|ipv4|ipv6]
```

```
an3000-mcm-slot7cpu1(config-pgw-pgw1)#
roaming-plmnid-list <list-name>
reject-inter-access-handover-request-no-session [true|false]
gateway-profile <name>
admin-state enabled
apn-list <apn-name>
s2b-interface t3-timeout <value> n3-count <value>
```

where

Parameter	Description
s5-s8-interface-network-context	Specifies the network context containing the S5/S8 interface.
home-plmnid-list	Specifies the list of PLMN IDs considered as home.
gtpc-endpoint-params: Configure the following S2b GTP-C endpoint parameters on the PGW: ***Note:*** *You must use this parameter to configure the S2b GTP-C endpoint on the PGW.*	
dual-stack-behavior	When both the PGW and the ePDG are dual stack, specify how the IP type (IPv4 or IPv6) is chosen when the PGW originates messages that it sends to the ePDG. Options are: ■ initial-ip-type-used-by-peer – The PGW should use the IP type that was used in the Create Session Request from the ePDG. ■ ipv4 – The PGW should use IPv4. ■ ipv6 – The PGW should use IPv6.
network-context	Specifies the name of the S2b network context on which this GTP-C endpoint is anchored.

Parameter	Description
loopback-ip	Specifies the name of the primary loopback IP address (associated with this network context) used for the S2b GTP-C endpoint. Can be IPv4 or IPv6. ***Note:*** *To use this loopback address for GTP-U, set the gtp-u flag to **yes**.*
secondary-loopback-ip	(Optional) Specifies the name of the secondary S2b GTP-C loopback IP address associated with this network context. Can be IPv4 or IPv6. ***Note:*** *For dual-stack GTP-C support, the primary and secondary loopback IP addresses should be different IP address types (IPv4 or IPv6).*
roaming-plmnid-list	Specifies the list of PLMN IDs considered as roaming.
reject-inter-access-handover-request-no-session	Specifies to reject (true) or accept (false) inter-access handovers between Wi-Fi and LTE if the session does not exist (applies to the PGW only). ■ **true** (default) - If set to **true**, no prior subscriber PDN connection exists, and the PGW receives a Create Session Request with the handover indication set to true, the PGW will reject the request by responding with a Create Session Response with the Cause code as "Context not found." ■ **false** - If set to **false**, the Create Session Request will be handled as a normal Create Session Request without the handover request.
gateway-profile	Specifies the name of gateway profile used by the gateway.
admin-state	Specifies to enable or disable the admin-state for this service. ■ **enabled** ■ **disabled**
apn-list	Specifies the list of APN names.
s2b-interface	Specifies the GTP T3 timeout value and N3 retry value on a per interface basis: ■ **t3-timeout** – Specifies the length of time to wait for an acknowledgment for a message (ms). ■ **n3-count** – Specifies the number of times the gateway retransmits a message if no acknowledgment is received.

1 (Optional) To define a list of specific loopback addresses (within the S2b network-context) for the GTP-U endpoint on the PGW, use the following CLI command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
gtpu-loopback-list <loopback-interface-name>
```

***Note:** If the list of specific GTP-U loopback addresses has only IPv4 or IPv6 addresses, only that type is used for the sessions on the gateway.*

---

2 Configure the GTP peers for each gateway.

Operators can configure GTP peers on the system and associate that peer with any of the gateway objects. A peer is identified by its IP address, network context ID and the version of the GTP-C protocol it supports. A peer can be associated only with a gateway instance whose GTP-C endpoint address is in the same network context as the peer.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-peer <peer-name>
gtpversion gtpv2 is-present-in-local-plmn [false|true] network-context
<name> peer-ip-address <ip-address>
```

3 Associate the GTP peer with the gateway instance by adding the peer to the list of GTP peers with which the gateway communicates.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
gtp-peer-list <peer-name>
```

Monitoring the S2b Interface

This section describes how to monitor the operational status of GTP peers via the S2b interface and how to display statistical information for S2b communication messages. For more information on statistics, see the *Acuitas Performance Statistics* guide.

GTP Peer Status

Use the following CLI commands to display the status information of GTP peers.

```
an3000-mcm-slot7cpu1# show zone default gateway gtp-peer-status
```

```
an3000-mcm-slot7cpu1(config)# show zone default gateway
gtp-peer-status
```

```
an3000-mcm-slot7cpu1(config)# show zone default gateway
show-pgw-dns-list <name>
```

The system displays output similar to the following:

```
gateway gtp-peer-status
                                            PEER
GATEWAY   PEER IP      NETWORK  GTP      RESTART  PEER
NAME      ADDRESS      CONTEXT  VERSION  COUNTER  STATE
-----------------------------------------------------------
EPDG      0.0.0.1               gtpv2    0        unknown
EPDG      10.32.36.66  ACCESS   gtpv2    17       up
```

S2b Interface Statistics

You can use the following commands to display statistical information for S2b interface communication messages for all GTP peers or a specified GTP peer.

S2b Interface Statistics (Grid-Level)

The following CLI command displays the S2b interface GTP grid-level statistics.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics grid-level
gtp peer <peer-name>
```

S2b Interface Statistics (Cluster-Level)

The following CLI command displays the S2b interface GTP cluster-level statistics.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level gtp peer <peer-name> <cluster-id>
```

S2b Interface Statistics (Node-Level)

The following CLI command displays the S2b interface GTP node-level statistics.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics node-level
gtp peer <peer-name> <cluster-id> <node>
```

Message Definitions

This section lists the supported GTPv1-U and GTPv2-C message definitions used on the S2b interface and lists the information elements (IEs) contained in each message type.

- GTPv1-U Message Definitions (page 15-18)
- GTPv2-C Message Definitions (page 15-20)

GTPv1-U Message Definitions

Table 15-1 lists the supported GTPv1-U messages for the S2b interface. For more information on GTPv1-U messages, see 3GPP TS 29.281 V9.3.0.

Table 15-1 Supported GTPv1-U S2b Interface Message Definitions

Message Type	Decimal Code	Reference
Echo Request	1	3GPP 29.281 V9.3.0
Echo Response	2	3GPP 29.281 V9.3.0
End Marker	254	3GPP 29.281 V9.3.0
Encapsulated T-PDU	255	3GPP 29.281 V9.3.0

Echo Request

Table 15-2 lists the supported Echo Request information element for GTPv1-U tunnels.

Table 15-2 Echo Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Echo Response

The Echo Response message is sent in response to an Echo Request message. Table 15-3 lists the supported Echo Response information elements for GTPv1-U tunnels.

Table 15-3 Echo Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Recovery	Mandatory	Included on the S2b interface for backward compatibility.	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

End Marker

The End Marker message is sent for each GTP-U tunnel (multiple messages). This message indicates the end of a payload stream on a given tunnel.

Table 15-4 lists the supported End Marker information elements.

Table 15-4 End Marker Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	254	Vendor Specific

Encapsulated T-PDU

The Encapsulated T-PDU message is sent between GTP-U tunnels to identify the source and destination tunnel endpoints.

Table 15-5 lists the supported T-PDU information elements.

Table 15-5 Encapsulated T-PDU Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Source TEID	Mandatory	Contains the Tunnel Endpoint Identifier for the tunnel to which this T-PDU belongs. The TEID is used by the receiving entity to find the PDP context except for: ■ the Echo Request/Response and Supported Extension Header notification messages, where the Tunnel Endpoint Identifier is set to all zeros. ■ the Error Indication message where the Tunnel Endpoint Identifier is set to all zeros.	TEID	255	Vendor Specific

GTPv2-C Message Definitions

Table 15-6 describes the supported GTPv2-C message definitions for the S2b interface. For more information on GTPv2-C messages, see 3GPP TS 29.274 V9.3.0 and 3GPP TS 29.274 V12.4.0.

Table 15-6 Supported GTPv2-C S2b Interface Message Definitions

Message Type	Decimal Code	Reference
Create Session Request (page 15-21)	32	3GPP 29.274 V9.3.0
Create Session Response (page 15-25)	33	3GPP 29.274 V9.3.0
Create Bearer Request (page 15-28)	95	3GPP 29.274 V9.3.0
Create Bearer Response (page 15-29)	96	3GPP 29.274 V9.3.0
Delete Session Request (page 15-33)	36	3GPP 29.274 V9.3.0
Delete Session Response (page 15-34)	37	3GPP 29.274 V9.3.0
Delete Bearer Request (page 15-34)	99	3GPP 29.274 V9.3.0
Delete Bearer Response (page 15-36)	100	3GPP 29.274 V9.3.0
Modify Bearer Response (page 15-37)	35	3GPP 29.274 V9.3.0
Update Bearer Request (page 15-38)	97	3GPP 29.274 V9.3.0
Update Bearer Response (page 15-40)	98	3GPP 29.274 V9.3.0

Create Session Request

The ePDG sends a Create Session Request message to the PGW as part of the following procedures:

- UE-requested Public Data Network (PDN) Connectivity
- UE Initial Attach

Table 15-7 lists the supported Create Session Request information elements.

Table 15-7 Create Session Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
3GPP AAA Server Identifier	Optional	The ePDG/TWAN may include this IE on the S2a/S2b interface to provide the selected 3GPP AAA Server Identifier to the PGW. Note: If supported, the PGW shall contact the 3GPP AAA server (identified by this IE which carries the Origin-Host and Origin-Realm included in the DEA message received by the ePDG/TWAN over SWm or STa interface) for establishing the S6b session.	Node Identifier		0
IMSI	Conditional	Includes the UE's International Mobile Subscriber Identity (IMSI) in the S2b interface message.	IMSI	1	0
MSISDN	Conditional	For the S2b interface, includes the Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) during an attach with GTP and UE initiated connectivity to additional PDN with GTP, if provided by the HSS/AAA.	MSISDN	76	0
Serving Network	Conditional	Included on the Sb2 interface for an attach with GTP, a UE initiated connectivity to additional PDN with GTP, and a handover to untrusted non-3GPP IP access with GTP.	Serving Network	83	0

Table 15-7 Create Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
RAT Type	Mandatory	This IE is set to the 3GPP access type or to the value matching the characteristics of the non-3GPP access the UE uses to attach to the EPC. The ePDG may use the access technology type of the untrusted non-3GPP access network if it is able to acquire it; otherwise it will indicate Virtual as the RAT Type.	RAT Type	82	0
Indication Flags	Conditional	Included on the S2b interface if either of the applicable flags is set to 1: ■ Dual Address Bearer Flag ■ Handover Indication	Indication	77	0
Sender F-TEID for Control Plane	Mandatory	This IE specifies the Fully Qualified Tunnel Endpoint Identifier (F-TEID) for the GTP-C tunnel.	F-TEID	87	0
Access Point Name (APN)	Mandatory	None	APN	71	0
Selection Mode	Conditional	Included on the S2b interface for an initial attach with GTP and a UE initiated connectivity to additional PDN with GTP, and indicates whether a subscribed APN or a non-subscribed APN (chosen by the ePDG) was selected.	Selection Mode	128	0
PDN Address Allocation (PAA)	Conditional	Included on the S2b for an attach with GTP, a UE initiated connectivity to additional PDN with GTP, and a handover to untrusted non-3GPP IP access with GTP. For a handover to untrusted non-3GPP IP access with GTP, the ePDG sets the IPv4 address and/or IPv6 prefix length and IPv6 prefix and Interface Identifier based on the IP address(es) received from the UE.	PAA	79	0
Aggregate Maximum Bit Rate (APN-AMBR)	Conditional	Included on S2b interface for an attach with GTP and UE initiated connectivity to additional PDN with GTP.	AMBR	72	0

Table 15-7 Create Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Bearer Contexts to be created	Mandatory	One bearer is included on the S2b interface during attach with GTP, UE initiated connectivity to additional PDN with GTP, and handover to untrusted non-3GPP IP access with GTP.	Bearer Context	93	0
Trace Information	Conditional	Included on the S2b interface if a PGW trace is activated.	Trace Information	96	0
Recovery	Conditional	Included on the S2b interface if the ePDG is contacting the PGW for the first time.	Recovery	3	0
ePDG-FQ-CSID	Conditional	Included by the ePDG on the S2b interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID	132	2
UE Local IP Address	Conditional/ Optional	Included by the ePDG on the S2b interface during an Initial Attach for emergency session (GTP on S2b). Otherwise included based on local policy. When the PGW receives the IKEv2 Tunnel IP Address IE (UE local IP address) from the ePDG via the S2b interface, the PGW includes this information within the Huawei ePDG UE Local IP vendor-specific attribute (VSA) and sends this VSA in all subsequent RADIUS Accounting-Request messages (Start/Interim/Stop) for all bearers belonging to that PDN session. For details, see the *Affirmed Networks RADIUS Interface Specification*.	IP Address	74	0

Table 15-7 Create Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Additional Protocol Configuration Options (APCO)	Conditional/ Optional	Included on the S2b interface for multiple authentications and performs the corresponding procedures as specified for PAP authentication of the UE with external networks as defined in 3GPP TS 33.402 [50]. If the UE requests the DNS IPv4/IPv6 address in the Configuration Payload (CFG_REQ) during the IPsec tunnel establishment procedure (as specified in 3GPP TS 33.402 [50]), and if the ePDG supports the Additional Protocol Configuration Options IE, the ePDG may include this IE and correspondingly set the "DNS Server IPv4/v6 Address Request" parameter as defined in 3GPP TS 24.008 [5].	Additional Protocol Configuration Options (APCO)		
UE UDP Port	Conditional/ Optional	Included by the ePDG on the S2b interface if NAT is detected, UDP encapsulation is used, and the UE Local IP Address is present.	Port Number	126	0
Private Extension	Optional	Includes vendor or operator specific information. ■ AP_MAC_Address	Private Extension	255	Vendor Specific

Bearer Contexts to be Created Within Create Session Request

Within the Create Session Request message, the Bearer Contexts to be Created IE contains other IEs and is considered a Grouped IE.

Table 15-8 lists the information elements contained in the Bearer Contexts to be Created IE.

Table 15-8 Bearer Contexts IEs (within Create Session Request)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0

Table 15-8 Bearer Contexts IEs (within Create Session Request)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
S2b-U ePDG F-TEID	Conditional	Included on the S2b interface for an attach with GTP, a UE initiated connectivity to additional PDN with GTP, and a handover to untrusted non-3GPP IP access with GTP.	F-TEID	87	5

Create Session Response

The PGW sends a Create Session Response message to the ePDG as an answer to a Create Session Request message. The Create Session Response message contains information about the default bearer, such as the UE's allocated IP address and the tunnel endpoint IP addresses (TEID).

Table 15-9 lists the supported Create Session Response information elements.

Table 15-9 Create Session Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure. 3GPP TS 23.401 [3] and 3GPP TS 23.060 [35] specify the handling of cases when the UE has requested IPv4v6 PDN Type, but PGW restricts the usage of IPv4v6 PDN Type.	Cause	2	0
Sender F-TEID for Control Plane	-	Identical to the PGW S5/S8/ S2a/S2b F-TEID for PMIP based interface or for GTP based Control Plane interface IE.	-	-	-
PGW S5/S8/ S2a/S2b F-TEID for PMIP based interface or for GTP based Control Plane interface	Conditional	Included on the S2b interface for an attach with GTP, UE initiated connectivity to additional PDN with GTP, and handover to untrusted non-3GPP IP access with GTP procedures.	F-TEID	87	1

Table 15-9 Create Session Response Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
PDN Address Allocation (PAA)	Conditional	Included on the S2b interface for an attach with GTP, UE initiated connectivity to additional PDN with GTP, and handover to untrusted non-3GPP IP access with GTP procedures.	PAA	79	0
Aggregate Maximum Bit Rate (APN-AMBR)	Conditional	Included on the S2b interface if the Policy and Charging Rules Function (PCRF) has modified the APN-AMBR.	AMBR	72	0
Bearer Contexts created	Mandatory	One bearer is included on the S2b interface during attach with GTP, UE initiated connectivity to additional PDN with GTP, and handover to untrusted non-3GPP IP access with GTP.	Bearer Context	93	0
PGW-FQ-CSID	Conditional	Included by the PGW on the S2b interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID		0
Additional Protocol Configuration Options (APCO)	Conditional/ Optional	If the PGW supports the APCO IE, and the PGW has received the APCO IE with the "DNS IPv4/IPv6 Server Address Request" in the Create Session Request over S2b, the PGW may include this IE for the S2b interface with the "DNS IPv4/IPv6 Server Address" parameter as specified in 3GPP TS 24.008 [5].	Additional Protocol Configuration Options (APCO)		0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
PGW's Node Level Load Control Information[a]	Optional	The ePDG supports the reception and handling of load control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting load control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Load Control Information	

Table 15-9 Create Session Response Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
PGW's Node Level Overload Control Information[b]	Optional	The ePDG supports the reception and handling of overload control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting overload control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Overload Control Information	0

[a] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of load control.

[b] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of overload control.

Bearer Contexts Created Within Create Session Response

Within the Create Session Response message, the Bearer Contexts Created IE contains other IEs and is considered a Grouped IE.

Table 15-10 lists the information elements contained in the Bearer Contexts Created IE.

Table 15-10 Bearer Contexts Created IEs (within Create Session Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling was successful. If not, this IE provides information on the reason.	Cause	2	0
S2b-U PGW F-TEID	Conditional	Included on the S2b interface for an attach with GTP, UE initiated connectivity to additional PDN with GTP, and a handover to untrusted non-3GPP IP access with GTP.	F-TEID	87	4

Create Bearer Request

The PGW sends a Create Bearer Request message to the ePDG to create a new dedicated bearer.

Table 15-11 lists the Create Bearer Request information elements.

Table 15-11 Create Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Linked Bearer Identity (LBI)	Mandatory	Indicates the default bearer associated with the PDN connection.	EBI (EPS Bearer ID)	73	0
Bearer Contexts	Mandatory	Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0
PGW-FQ-CSID	Conditional	Included by the PGW on the S2b interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID		0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
PGW's Node Level Load Control Information[a]	O	The ePDG supports the reception and handling of load control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting load control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Load Control Information	
PGW's Node Level Overload Control Information[b]	O	The ePDG supports the reception and handling of overload control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting overload control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Overload Control Information	0

[a] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of load control.

[b] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of overload control.

Bearer Contexts Within Create Bearer Request

Within the Create Bearer Request message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 15-12 lists the information elements contained in the Bearer Contexts IE.

Table 15-12 Bearer Contexts IEs (within Create Bearer Request)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
EPS Bearer ID	Mandatory	This IE is set to 0.	EBI	73	0
TFT	Mandatory	This IE can contain both uplink and downlink packet filters to be sent to the UE or the ePDG.	Bearer TFT	84	0
S2b-U PGW F-TEID	Conditional	This IE is included on the S2b interface as a user plane IE.	F-TEID		4
Bearer Level QoS	Mandatory	None	Bearer QoS	80	0
Charging Id	Conditional	This IE is included on the S2b interface.	Charging Id	94	0

Create Bearer Response

The ePDG sends a Create Bearer Response message to the PGW as an answer to a Create Bearer Request message.

Table 15-13 lists the Create Bearer Response information elements.

Table 15-13 Create Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0
Bearer Contexts	Mandatory	Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0

Table 15-13 Create Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Recovery	Conditional	Included on the S2b interface if the ePDG is contacting the PGW for the first time.	Recovery	3	0
ePDG-FQ-CSID	Conditional	The ePDG includes this IE on the S2b interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID		2
UE Local IP Address	Conditional/ Optional	Included by the ePDG on the S2b interface. When the PGW receives the IKEv2 Tunnel IP Address IE (UE local IP address) from the ePDG via the S2b interface, the PGW includes this information within the Huawei ePDG UE Local IP vendor-specific attribute (VSA) and sends this VSA in all subsequent RADIUS Accounting-Request messages (Start/Interim/Stop) for all bearers belonging to that PDN session. For details, see the *Affirmed Networks RADIUS Interface Specification*.	IP Address	74	0
UE UDP Port	Conditional/ Optional	Included by the ePDG on the S2b interface if NAT is detected and UDP encapsulation is used.	Port Number	126	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Create Bearer Response

Within the Create Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 15-14 lists the information elements contained in the Bearer Contexts IE.

Table 15-14 Bearer Contexts Information Elements within Create Bearer Response

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0
NAS Cause	Conditional	If the bearer creation failed, the ePDG shall include this IE on the S2b interface to indicate the cause of the bearer creation failure, if available and if this information is permitted to be sent to the PGW operator according to the ePDG operator's policy. When present, the IE shall be encoded as a Diameter or an IKEv2 cause. ***Note:*** *The ePDG does not exchange signalling with the 3GPP AAA Server in this procedure. The ePDG may include an internal failure cause for the bearer creation failure. The protocol type used to encode the internal failure cause is implementation specific.*	RAN/NAS Cause		0

Table 15-14 Bearer Contexts Information Elements within Create Bearer Response (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
RAN/NAS Cause	Conditional	If the bearer creation failed, the MME shall include this IE on the S11 interface to indicate the RAN cause and/or the NAS cause of the bearer creation failure, if available and if this information is permitted to be sent to the PGW operator according to MME operator's policy. If both a RAN cause and a NAS cause are generated, then several IEs with the same type and instance value shall be included to represent a list of causes. The SGW shall include this IE on the S5/S8 interface if it receives it from the MME.	RAN/NAS Cause		0
	Conditional	If the bearer creation failed, the TWAN shall include this IE on the S2a interface to indicate the cause of the bearer creation failure, if available and if this information is permitted to be sent to the PGW operator according to the TWAN operator's policy. When present, the IE shall be encoded as a Diameter or an ESM cause. See NOTE 2.			
	Conditional	If the bearer creation failed, the ePDG shall include this IE on the S2b interface to indicate the cause of the bearer creation failure, if available and if this information is permitted to be sent to the PGW operator according to the ePDG operator's policy. When present, the IE shall be encoded as a Diameter or an IKEv2 cause. See NOTE 3.			
S2b-U PGW F-TEID	Conditional	This IE is included on the S2b interface. It is used to correlate the bearers with those in the Create Bearer Request.	F-TEID		9

Table 15-14 Bearer Contexts Information Elements within Create Bearer Response (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
S2b-U ePDG F-TEID	Conditional	This IE is included on the S2b interface.	F-TEID		8

Delete Session Request

The ePDG sends a Delete Session Request message to the PGW requesting session termination.

Table 15-15 lists the Delete Session Request information elements.

Table 15-15 Delete Session Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Linked EPS Bearer ID	Conditional	Included on the S2b interface to indicate the default bearer associated with the PDN being disconnected.	EBI	73	0
UE Local IP Address	Conditional/ Optional	Included by the ePDG on the S2b interface. When the PGW receives the IKEv2 Tunnel IP Address IE (UE local IP address) from the ePDG via the S2b interface, the PGW includes this information within the Huawei ePDG UE Local IP vendor-specific attribute (VSA) and sends this VSA in all subsequent RADIUS Accounting-Request messages (Start/Interim/Stop) for all bearers belonging to that PDN session. For details, see the *Affirmed Networks RADIUS Interface Specification*.	IP Address	74	0
UE UDP Port	Conditional/ Optional	Included by the ePDG on the S2b interface if NAT is detected and UDP encapsulation is used.	Port Number	126	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Delete Session Response

The PGW sends a Delete Session Response message to the ePDG as an answer to a Delete Session Request message.

Table 15-16 lists the Delete Session Response information elements.

Table 15-16 Delete Session Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
Recovery	Conditional	Included on S2b interface if the ePDG is contacting the PGW for the first time.	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
PGW's Node Level Load Control Information[a]	O	The ePDG supports the reception and handling of load control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting load control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Load Control Information	
PGW's Node Level Overload Control Information[b]	O	The ePDG supports the reception and handling of overload control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting overload control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Overload Control Information	0

[a] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of load control.

[b] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of overload control.

Delete Bearer Request

The PGW sends a Delete Bearer Request message to the ePDG.

Table 15-17 lists the Delete Bearer Request information elements.

Table 15-17 Delete Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Linked EPS Bearer ID (LBI)	Conditional	Included on the S2b interface to indicate the default bearer associated with the PDN being disconnected. This IE is present only when the EPS Bearer is not present.	EBI	73	0
EPS Bearer ID	Conditional	Included on the S2b interface to indicate the dedicated bearer to be deleted. Several IEs with this type and instance values are included, if necessary, to represent a list of bearers.	EBI	73	1
PGW-FQ-CSID	Conditional	This IE is included on the S2b interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID		0
Cause	Conditional	This IE is included on the S2b interface if the message is caused by handover from non-3GPP to 3GPP. In this case the Cause value is set to “Access changed from Non-3GPP to 3GPP.”	Cause	2	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
PGW’s Node Level Load Control Information[a]	O	The ePDG supports the reception and handling of load control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting load control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Load Control Information	
PGW’s Node Level Overload Control Information[b]	O	The ePDG supports the reception and handling of overload control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting overload control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Overload Control Information	0

[a] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of load control.

[b] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of overload control.

Delete Bearer Response

The ePDG sends a Delete Bearer Response message to the PGW as an answer to a Delete Bearer Request message.

Table 15-18 lists the Delete Bearer Response information elements.

Table 15-18 Delete Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
Linked EPS Bearer ID (LBI)	Conditional	Included on the S2b interface to indicate the default bearer associated with the PDN being disconnected.	EBI	73	0
Bearer Contexts	Conditional	Used for dedicated bearers. At least one dedicated bearer is present. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0
Recovery	Conditional	Included on S2b interface if the ePDG is contacting the PGW for the first time.	Recovery	3	0
ePDG-FQ-CSID	Conditional	The ePDG includes this IE on the S2b interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID		2
UE Local IP Address	Conditional/ Optional	Included by the ePDG on the S2b interface. When the PGW receives the IKEv2 Tunnel IP Address IE (UE local IP address) from the ePDG via the S2b interface, the PGW includes this information within the Huawei ePDG UE Local IP vendor-specific attribute (VSA) and sends this VSA in all subsequent RADIUS Accounting-Request messages (Start/Interim/Stop) for all bearers belonging to that PDN session. For details, see the *Affirmed Networks RADIUS Interface Specification*.	IP Address	74	0

Table 15-18 Delete Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
UE UDP Port	Conditional/ Optional	Included by the ePDG on the S2b interface if NAT is detected and UDP encapsulation is used.	Port Number	126	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Delete Bearer Response

Within the Delete Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 15-19 lists the information elements contained in the Bearer Contexts IE.

Table 15-19 Bearer Contexts IEs (within Delete Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0

Modify Bearer Response

The PGW sends a Modify Bearer Response message to the ePDG on the S2b interface as part of the procedures listed for the Modify Bearer Request.

Table 15-20 lists the Modify Bearer Response information elements.

Table 15-20 Modify Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
PGW's Node Level Load Control Information[a]	O	The ePDG supports the reception and handling of load control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting load control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Load Control Information	
PGW's Node Level Overload Control Information[b]	O	The ePDG supports the reception and handling of overload control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting overload control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Overload Control Information	0

[a] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of load control.

[b] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of overload control.

Update Bearer Request

The PGW sends an Update Bearer Request message to the ePDG requesting a change to the service data flow requirements or QoS parameters.

Table 15-21 lists the Update Bearer Request information elements.

Table 15-21 Update Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Bearer Contexts	Mandatory	Used for bearers that require QoS/TFT modification. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers. If there is no QoS/TFT modification, only one IE with this type and instance value is included.	Bearer Context	93	0
Aggregate Maximum Bit Rate (APN-AMBR)	Mandatory	None	AMBR	72	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
PGW's Node Level Load Control Information[a]	O	The ePDG supports the reception and handling of load control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting load control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Load Control Information	
PGW's Node Level Overload Control Information[b]	O	The ePDG supports the reception and handling of overload control information generated by the PGW over the S2b GTP-C interface. (The ePDG does not support transmitting overload control information.) The ePDG uses this information to select a PGW in subsequent session setups. This functionality is supported at the node level, not at the APN level.		Overload Control Information	0

[a] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of load control.

[b] Please refer to the *Affirmed Networks ePDG Administration Guide* for a description of the full implementation of overload control.

Bearer Contexts Within Update Bearer Request

Within the Update Bearer Request message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 15-22 lists the information elements contained in the Bearer Contexts IE. For more information, see 3GPP TS 29.274

Table 15-22 Bearer Contexts IEs (within Update Bearer Request)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
TFT	Conditional	Included on the S2b interface if the message relates to a bearer modification or TFT change.	Bearer TFT	84	0
Bearer Level QoS	Conditional	Included on the S2b interface if a QoS modification is requested.	Bearer QoS	80	0

Update Bearer Response

The ePDG sends an Update Bearer Response message to the PGW as an answer to an Update Bearer Request message.

Table 15-23 lists the Update Bearer Response information elements.

Table 15-23 Update Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
Bearer Contexts	Mandatory	Used for bearers for which QoS and TFT modification are requested. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0
Recovery	Conditional	Included on the S2b interface if the ePDG is contacting the PGW for the first time.	Recovery	3	0
ePDG-FQ-CSID	Conditional	The ePDG includes this IE on the S2b interface according to the requirements in 3GPP TS 23.007 [17].	FQ-CSID		2

Table 15-23 Update Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
UE Local IP Address	Conditional/ Optional	Included by the ePDG on the S2b interface. When the PGW receives the IKEv2 Tunnel IP Address IE (UE local IP address) from the ePDG via the S2b interface, the PGW includes this information within the Huawei ePDG UE Local IP vendor-specific attribute (VSA) and sends this VSA in all subsequent RADIUS Accounting-Request messages (Start/Interim/Stop) for all bearers belonging to that PDN session. For details, see the *Affirmed Networks RADIUS Interface Specification*.	IP Address	74	0
UE UDP Port	Conditional/ Optional	Included by the ePDG on the S2b interface if NAT is detected and UDP encapsulation is used.	Port Number	126	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Update Bearer Response

Within the Update Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 15-24 lists the information elements contained in the Bearer Contexts IE.

Table 15-24 Bearer Contexts IEs (within Update Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0

1

Gn/Gp Interface

This section contains the following information used to control and monitor the Gn/Gp interface:

Note: *See the Affirmed Networks CLI Reference Guide for more information on the CLI commands used to configure and monitor the Gn/Gp interface.*

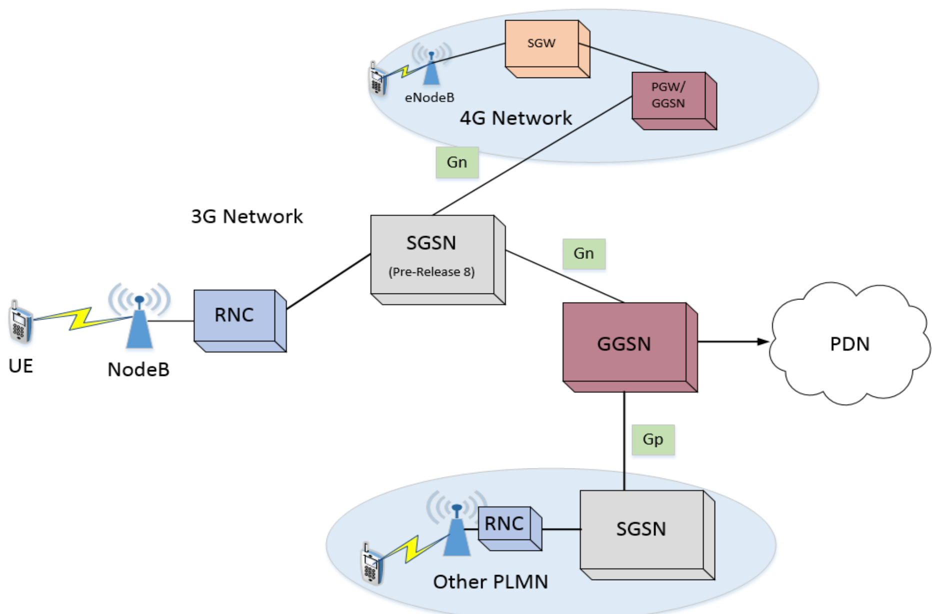
Interface Overview

The Gn/Gp interface provides user plane tunneling and tunnel management between the serving GPRS support node (SGSN) and the Gateway GPRS support node (GGSN). The Gn interface is used in non-roaming scenarios where the SGSN and GGSN are located within the same Public Land Mobile Network (PLMN). The Gp interface is used between the SGSN and GGSN in different PLMNs.

To enable interworking between 3G and 4G networks, the Gn/Gp interface is also used to connect a pre-release 8 SGSN and a Packet Data Network Gateway (PGW), functioning as a GGSN.

Figure 16-1 illustrates the Gn/Gp interface.

Figure 16-1 Gn/Gp Interface



The SGSN and GGSN are the two main components of the General Packet Radio Service (GPRS) network that connect mobile cell phone users to a public data network (PDN), such as the Internet, or private IP networks. GPRS provides packet-switched technology to the circuit-switched network, Global System for Mobile Communications (GSM).

The *Affirmed Networks Mobile Content Cloud* (MCC) acts as a GGSN as part of the UTRAN network to take advantage of existing 3G UTRAN network technology. When enabled, the MCC can also act as a PGW and can be connected to a pre-release 8 SGSN and process 2G and 3G session requests.

This document describes the Gn/Gp interface implementation in the MCC in accordance with 3GPP TS 29.060 V9.3.0 and 3GPP TS 09.60 V7.10.0.

The Gn/Gp interface uses the GPRS tunneling protocol as defined in 3GPP TS 29.060 V9.3.0 and 3GPP TS 09.60 V7.10.0 to transfer signaling messages and user data within the core GPRS network:

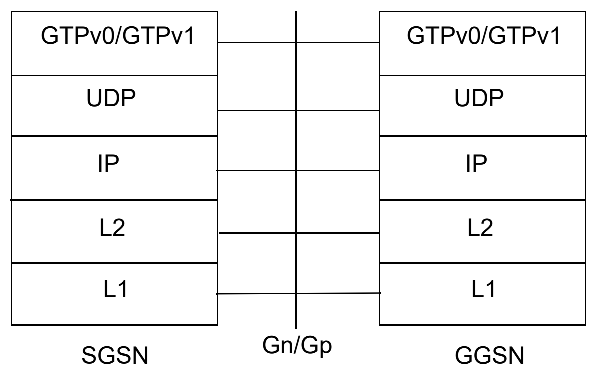
GTPv1-C – Used for transferring signaling messages between the GGSN and SGSN to create, modify and delete Packet Data Protocol (PDP) contexts. The UDP port number for GTPv1-C is 2123.

GTPv1-U – Used for transferring user data over GTP-U tunnels. The UDP port number for GTPv1-U is 2152.

GTPv0 – The GGSN also supports GTPv0 users when a GTPv0-capable SGSN is connected to the GGSN. The UDP port number for GTPv0 is 3386.

Figure 16-2 illustrates the GTPv0/GTPv1 protocol stack for the Gn/Gp interface.

Figure 16-2 GTPv0/GTPv1 Protocol Stack for the Gn/Gp Interface



Direct Tunneling

The SGSN functions as the gateway between the Radio Network Controller (RNC) and the core network. It handles both signaling traffic (to keep track of the location of mobile devices) and the actual data packets being exchanged between a mobile device and the PDN. MCC supports both the two tunnel and direct tunnel modes.

Two Tunnel – SGSN establishes a tunnel between the GGSN and the SGSN and another one between the SGSN and the RNC. The SGSN processes all data packets being exchanged between a mobile device and the PDN.

Direct Tunnel – SGSN establishes a direct user plane tunnel between the RNC and a GGSN thereby bypassing the SGSN and eliminating SGSN tunnel switching latency from the user plane.

References and Specifications

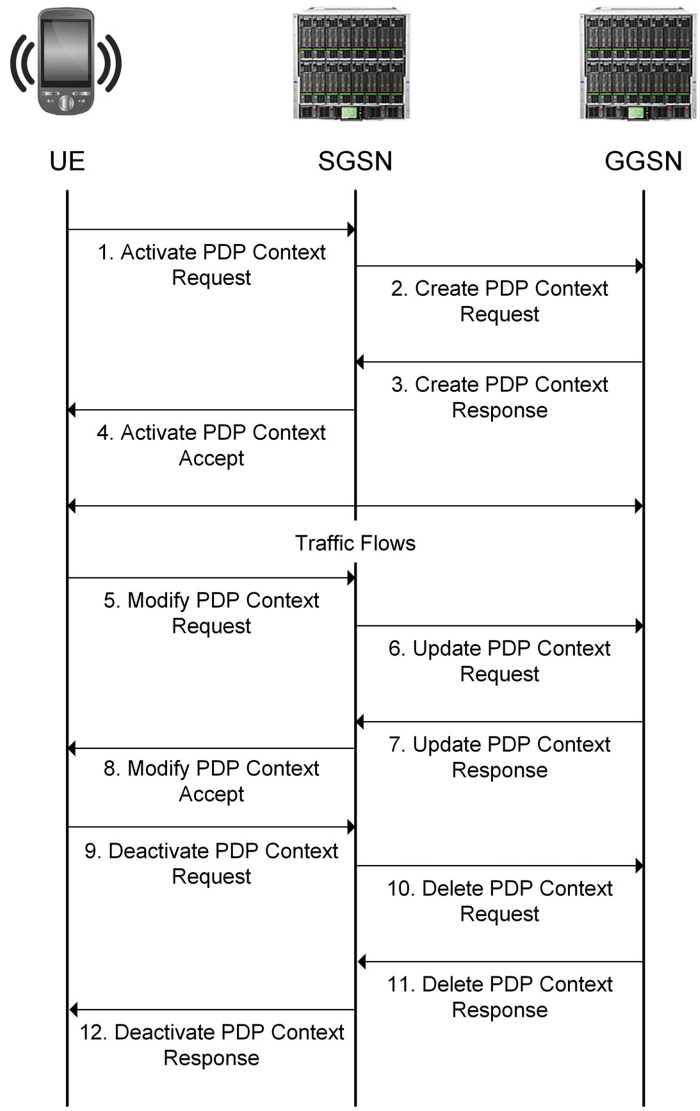
The system supports the following references and specifications for the Gn/Gp interfaces:

- 3GPP TS 29.274 V9.3.0 (2010-06): *3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3*
- 3GPP TS 29.060 V9.3.0 (2010-06): *General Packet Radio Service (GPRS);GPRS Tunneling Protocol (GTP) across the Gn and Gp interface*
- 3GPP TS 23.401 V9.3.0 (2009-12): *General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access*
- 3GPP TS 09.60 V7.10.0 Release 1998: *Digital cellular telecommunications system (Phase 2+); General Packet Radio Service (GPRS); GPRS Tunnelling Protocol (GTP) across the Gn and Gp Interface*

Message Flows

Figure 16-3 illustrates an example of the message flow between the SGSN and GGSN over the Gn/Gp interface for activating, modifying, and deleting a PDP context. In this example, the MS initiates these procedures. For information on message flows when the SGSN or GGSN initiates these procedures, see 3GPP TS 29.060 V9.3.0.

Figure 16-3 Activating, Modifying, and Deleting a PDP Context (MS-Initiated)



1. The GPRS-enabled Mobile Station (MS) initiates the PDP context activation procedure.
2. The SGSN sends a Create PDP Context Request to the GGSN containing information about the PDP context, such as the Tunnel Endpoint Identifier (TEID), negotiated Quality of Service (QoS), Access Point Name (APN), Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) and Network layer Service Access Point Identifier (NSAPI).

3 Upon receipt of the Create PDP Context Request, the GGSN creates the PDP context and returns a Create PDP Context Response message to the SGSN.

4 Upon the successful creation of the PDP context, the SGSN sends an Activate PDP Context Accept message to the MS.

5 While the PDP context is active, the MS sends a Modify PDP Context Request to the SGSN to request a modification to the PDP context parameters, such as a change to the negotiated QoS or the radio priority level or to set up direct tunneling.

6 The SGSN sends an Update PDP Context Request to the GGSN.

7 Upon receipt of the Update PDP Context Request, the GGSN verifies whether the requested parameters are supported and if so, returns an Update PDP Context Response message to the SGSN.

8 The SGSN sends a Modify PDP Context Accept message to the MS.

9 Once the data transfer is complete, the MS sends a Deactivate PDP Context Request to the SGSN.

10 The SGSN sends a Delete PDP Context to the GGSN.

11 Upon receipt of the Delete PDP Context Request, the GGSN deactivates the PDP context and returns a Delete PDP Context Response message to the SGSN.

12 The SGSN sends a Deactivate PDP Context Response to the MS confirming the PDP context deactivation.

Configuring the Gn/Gp Interface

This section describes the steps for configuring the Gn/Gp interface on the MCC GGSN.

Use the following steps to configure the Gn/Gp interface on the MCC GGSN:

1 Create a network-context for the Gn/Gp interface. For information on configuring network-contexts, see *Chapter 8, Network Context Parameters*.

Note: *For dual-stack GTP-U and GTP-C support, configure the Gn/Gp network-context associated with the GGSN with one or more IPv4 and IPv6 loopback addresses.*

---

2 Enable GTP-U services for the loopback address(es) defined for the Gn/Gp network-context.

```
an3000-mcm-slot7cpu1(config)# network-context <name> loopback-ip
<loopback-name> ip-address <ip-address> gtp-u yes
```

where

Parameter	Description
`network-context`	Specifies the name of the network context created for the Gn/Gp interface.
`loopback-ip`	Specifies the name of the loopback address associated with this network context.
`ip-address`	Specifies the IP address of this loopback object. Can be IPv4/IPv6.
`gtp-u yes`	Specifies that this loopback address is used for GTP-U.

3 Configure the Access Point Name(s) (APN) accessible via the GGSN.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
network-context <name> admin-state enabled
```

4 Configure the service profile for GTP-U data path management used to maintain data path status and handle data path failures.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gtp-u-path-management-service-profile
```

The following table describes the configurable parameters.

Parameter	Description
`clear-bearers-on-gtp-u-path-failure`	Specifies whether to clear bearers when a GTP-U path failure occurs. If set to true, the gateway clears the bearers on the peer when a GTP-U path failure occurs. ■ `false` ■ `true`
`gtp-u-echo-interval`	Specifies the amount of time to wait before initiating an Echo Request over the GTP-U path (ms).
`gtp-u-echo-response-timeout`	Specifies the amount of time to wait before retransmitting an Echo Request over the GTP-U path (ms).
`gtp-u-echo-retransmit-count`	Specifies the maximum number of Echo Request retransmission attempts (not including the initial Echo Request) to make before declaring a GTP-U path down.
`initiate-gtp-u-echo-requests`	Specifies whether to send periodic GTP-U Echo Requests to peers upon learning peer information. ■ `true` – (default) The gateway initiates GTP-U path management procedures for each peer. ■ `false` – The gateway does not send any GTP-U Echo Request messages over the GTP-U path.

5 Configure the GTP parameters for the gateway (GGSN) using the Gateway Profile object.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gateway-common <gateway-profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
`allowed-plmnid`	Specifies the policy to authorize the subscriber's PLMN ID during session setup. ■ `any` ■ `homelist` ■ `home-or-roaming-list`
`await-coa-timeout`	Specifies the timeout value (ms) used by the gateway for closed sessions to wait for a Change of Authorization (CoA).
`clear-subscribers-on-path-failure`	Specifies the policy to clear the subscriber when a path failure occurs. ■ `true` – After the system detects the path failure, all calls that go to that remote peer are actively closed. ■ `false` – All calls are retained until they are manually cleared, cleared by another remote device, or the path recovers and restart processing clears the calls.
`default-gtp-peer-classification`	Specifies the default peer classification when not enough information is present to classify a remote GTP peer as local (same administrative domain) or remote (different administrative domain). ■ `local` ■ `remote`
`defer-delete-session-response`	Specifies whether to defer sending a delete session response until after the delete processing is complete. ■ `true` ■ `false`
`dhcp-relay-response-timeout`	Specifies the timeout value used by the gateway (acting as a DHCP Relay) waiting for the user equipment (UE) to complete its DHCP transaction and obtain an IP (ms).
`gtp-echo-interval`	Specifies the amount of time to wait before initiating an Echo Request over the GTP-C path (ms).
`gtp-echo-response-timeout`	Specifies the amount of time to wait before retransmitting an Echo Request over the GTP-C path (ms).
`gtp-echo-retransmit-count`	Specifies the maximum number of Echo Request retransmission attempts (not including the initial Echo Request) to make before declaring a path down.

Parameter	Description
`idle-session-clearing-state`	Specifies whether to enable or disable Idle Session Clearing. Idle Session Clearing enables Operators to clear subscribers who are idle for longer than the configured `idle-timer-duration`. ■ **`enabled`** ■ **`disabled`** ***Note:*** *For PGWs and GGSNs, this value is defined in the gateway apn object if specified, otherwise the system uses the gateway-common settings.*
`idle-timer-duration`	Specifies the idle timer duration for sessions, in minutes. If a session has been idle for longer than the idle timer duration, the session is cleared.
`inactivity-timer-duration`	Specifies the inactivity timer duration for sessions, in minutes. If no activity is detected on the session within the inactivity timer duration, the session is declared idle. ***Note:*** *For PGWs and GGSNs, this value is defined in the gateway apn object if the value is non-zero, and the gateway-common profile if the value is 0 in the gateway apn object.*
`initiate-echo-requests`	Specifies whether to send periodic GTP-C Echo Requests to peers upon learning peer information. ■ **`true`** – (default) When GTP peers are configured and associated with a gateway instance, the gateway begins path management procedures for each peer. ■ **`false`** – The gateway does not send any Echo Request messages to the peer.
`non-matching-plmnid`	Specifies the policy for how to treat subscribers with a home PLMN ID that is neither in the home nor the roaming PLMN ID lists. ■ **`home`** ■ **`visiting`**
`reject-gtpc-messages-from-unconfigured-peers`	Specifies whether to reject GTP-C messages received from non-configured peers. ■ **`true`** ■ **`false`**
`session-max-clearing-state`	Specifies whether to enable or disable PDN Session Clearing for sessions present longer than the maximum time allowed. ■ **`enabled`** ■ **`disabled`** ***Note:*** *For PGWs and GGSNs, this value is defined in the gateway apn object if specified, otherwise the system uses the gateway-common settings.*

Parameter	Description
session-max-timer-duration	Specifies the maximum timer duration for PDN sessions, in minutes. If a PDN session has been present longer than the session maximum timer duration, the PDN session is cleared.

6 Configure the list of PLMN IDs for the home list:

```
an3000-mcm-slot7cpu1(config)# service-construct plmnid-list home-list
plmnid <plmnid-list>
```

7 Configure the Gn/Gp GTP-C endpoint used for the control plane handling of signaling messages on the GGSN.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway ggsn <gateway-name>
admin-state enabled gn-interface-network-context <name>
gateway-profile <name> home-plmnid-list <list> apn-list <apn-name>
[gn-interface|gp-interface] t3-timeout <value> n3-count <value>
dscp-value <value> <gtpc-endpoint-params network-context <name>
loopback-ip <loopback-name> port <primary-loopback-port>
secondary-loopback-ip <secondary-loopback-name> secondary-port
<secondary-loopback-port>
```

where

Parameter	Description
admin-state	Specifies whether to enable or disable the admin state for this service. ■ **enabled** ■ **disabled**
gn-interface-network-context	Specifies the network context containing the Gn/Gp interface.
gateway-profile	Specifies the name of gateway profile used by this service.
home-plmnid-list	Specifies the list of PLMN IDs considered as home.
apn-list	Specifies the list of APN names.
[gn-interface|gp-interface]	Specifies the GTP T3 timeout value and N3 retry value on a per interface basis: ■ **t3-timeout** – Specifies the length of time to wait for an acknowledgment for a message (ms). ■ **n3-count** – Specifies the number of times the gateway retransmits a message if no acknowledgment is received. ■ **dscp-value** – Specifies the Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this Interface. For more information, see Configuring DSCP Values for Control Plane Traffic (page 16-16).

Parameter	Description
`gtpc-endpoint-params` Configure the following Gn/Gp GTP-C endpoint parameters on the GGSN:	
`network-context`	Specifies the name of the Gn/Gp network context on which this GTP-C endpoint is anchored.
`loopback-ip`	Specifies the name of the primary loopback IP address (associated with this network context) used for the GTP-C endpoint. Can be IPv4 or IPv6. ***Note:*** *To use this loopback address for GTP-U, set the gtp-u flag to* `yes`.
`secondary-loopback-ip`	(Optional) Specifies the name of the secondary GTP-C loopback IP address associated with this network context. Can be IPv4 or IPv6. ***Note:*** *For dual-stack GTP-C support, the primary and secondary loopback IP addresses should be different IP address types (IPv4 or IPv6).*

8 (Optional) To define a list of specific loopback addresses (within the Gn/Gp network-context) for the GTP-U endpoint on the GGSN, use the following CLI command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway ggsn <gateway-name>
gtpu-loopback-list <loopback-interface-name>
```

Note: *If the list of specific GTP-U loopback addresses has only IPv4 or IPv6 addresses, only that type is used for the sessions on the gateway.*

9 Configure the GTP peers for each gateway.

Operators can configure GTP peers on the system and associate that peer with any of the gateway objects. A peer is identified by its IP address, network context ID and the version of the GTP-C protocol it supports. A peer can be associated only with a gateway instance whose GTP-C endpoint address is in the same network-context as the peer.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-peer <peer-name>
gtpversion [gtpv0|gtpv1] is-present-in-local-plmn [false|true]
network-context <name> peer-ip-address <ip-address>
```

10 Associate the GTP peer with the gateway instance by adding the peer to the list of GTP peers with which the gateway communicates.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway ggsn <gateway-name>
gtp-peer-list <peer-name>
```

Optional Gn/Gp Interface Configuration Tasks

This section describes optional Gn/Gp interface configuration tasks.

- Configuring the P-CSCF Server IP Addresses (page 16-12)
- Configuring CSG Change Reporting (page 16-15)
- Configuring DSCP Values for Control Plane Traffic (page 16-16)
- Configuring Default Values for the RAT Type, Serving Network PLMN ID MNC/MCC, and the MS Time Zone (page 16-16)

Configuring the P-CSCF Server IP Addresses

The Proxy-Call Session Control Function (P-CSCF) is the UE's access point to the IP Multimedia Subsystem (IMS) network. Operators can configure the MCC GGSN/PGW with a P-CSCF table containing a list of IP addresses (IPv4 and/or IPv6) used for IMS calls. If requested during session creation, the MCC GGSN/PGW sends the P-CSCF IP addresses as part of the Protocol Configuration Options (PCO) Information Element (IE) to a UE.

The MCC supports a maximum of 128 IPv4 and 128 IPv6 address pairs in each P-CSCF table. By default, the MCC GGSN/PGW selects the P-CSCF IP address pair in a round-robin fashion. Operators can allocate a maximum number of P-CSCF IPv4/IPv6 addresses from the P-CSCF table to be included in the PCO IE by the PGW. The maximum is 16 of each type.

Alternatively, Operators can configure the MCC to use a weighted round-robin algorithm to select the P-CSCF IP addresses. For weighted round robin, a maximum of only two IP addresses will be allocated in each category (IPV4/IPV6) even if the maximum number of P-CSCF IP addresses allocated is greater than two. Operators assign a weight to each P-CSCF IP address pair to distribute the IMS calls based on the P-CSCF capacity.

Note: *If you add/remove an IP address from the list or modify the configured weight/algorithm, allocations for subsequent session setups use the new configuration.*

Use the following CLI parameters to configure the P-CSCF IPv4 table entries and weights.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pcscf-table
<pcscf-table-name> address-allocation-mode
[round-robin|weighted-round-robin] ip-v4-list <list-name>
[primary-ipv4-address|secondary-ipv4-address] <ip-address> weight
<value>
```

Use the following CLI parameters to configure the P-CSCF IPv6 table entries and weights.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pcscf-table
<pcscf-table-name> ip-v6-list <list-name>
[primary-ipv6-address|secondary-ipv6-address] <ip-address> weight
<value>
```

where

Parameter	**Description**
pcscf-table	Specifies the name of the P-CSCF table.
address-allocation-mode	Specifies the algorithm to use when allocating P-CSCF IP addresses from the P-CSCF table. ■ **round-robin** – (default) The GGSN/PGW selects the P-CSCF IP addresses in a round-robin fashion from the P-CSCF table. ■ **weighted-round-robin** – The GGSN/PGW selects a P-CSCF IP address pair based on the configured weight assigned to each P-CSCF IP address pair. The IP address pair with the highest weight receives the most distribution of calls.
ip-v4-list|ip-v6-list	Specifies the name of the P-CSCF IPv4 or IPv6 address table. ***Note:*** *The P-CSCF IP address name must be unique across all P-CSCF tables.* ***Note:*** *If you delete an IP address from the list, statistics information is no longer generated for the deleted IP address and is lost after a short period of time.*
primary-ipv4-address| secondary-ipv4-address	Specifies the IPv4 address of a primary and secondary P-CSCF server. The MCC supports a maximum of 128 IPv4 address pairs.
primary-ipv6-address| secondary-ipv6-address	Specifies the IPv6 address of a primary and secondary P-CSCF server. The MCC supports a maximum of 128 IPv6 address pairs.
weight	Specifies the weight allocated to this P-CSCF IP address pair. Value 0 disables the IP address pair and the pair is not used regardless of whether the algorithm is round robin or weighted round robin. Weights 1–100 are applicable for weighted round robin only. ***Note:*** *Applies only when* `address-allocation-mode` *is set to* `weighted-round-robin`. *The IP address pair with the highest weight receives the most distribution of calls.*

Each P-CSCF address table is associated with an APN. When the MCC GGSN/PGW receives a Create PDP Context/Session Request for a P-CSCF address, the configured algorithm (round robin or weighted round robin) is used to select a pair of IP addresses from the P-CSCF table with which the APN is associated. Even if the UE does not request the P-CSCF address, Operators can configure the MCC GGSN/PGW to always send the configured P-CSCF IP addresses in the PCO IE as part of the initial Create PDP Context/Session Response message.

Use the following CLI command to associate a P-CSCF address table with an APN.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
pcscf-table <pcscf-table-name> maxPcscfIpV4Addresses <value>
maxPcscfIpV6Addresses <value> send-pcscf-address-in-pco-always
[true|false]
```

where

Parameter	Description
apn-name	Specifies the name of the APN.
pcscf-table	Specifies the name of the P-CSCF table.
max-pcscf-ipv4-addresses	Specifies the maximum number of P-CSCF IPv4 addresses from the P-CSCF table to be included in the PCO IE. The maximum is 16. ***Note:*** *For weighted round robin, a maximum of only two IP addresses will be allocated in each category (IPV4/IPV6) even if the maximum number of P-CSCF IP addresses allocated is greater than two.*
max-pcscf-ipv6-addresses	Specifies the maximum number of P-CSCF IPv6 addresses from the P-CSCF table to be included in the PCO IE. The maximum is 16. ***Note:*** *For weighted round robin, a maximum of only two IP addresses will be allocated in each category (IPV4/IPV6) even if the maximum number of P-CSCF IP addresses allocated is greater than two.*
send-pcscf-address-in-pco-always	Specifies one of the following options: ■ **false** – (default) The GGSN/PGW only includes the P-CSCF server IP addresses in the PCO IE of the Create PDP Context/Session Response message if the UE requests it. ■ **true** – The GGSN/PGW always includes the P-CSCF server IP addresses in the PCO IE of the Create PDP Context/Session Response message, even if the UE does not request it.

Use the following CLI command to view statistics on the MCC GGSN/PGW to track the number of PDN connections served per allocated P-CSCF IP address per APN. If you delete an IP address from the list, statistics information is no longer generated for the deleted IP address and is lost after a short period of time. For more information on performance statistics, see *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
pcscf-address-in-use-stats
```

Configuring CSG Change Reporting

The MCC supports Closed Subscriber Group (CSG) change reporting on the GGSN as defined in 3GPP TS 29.060 V9.0.0. A CSG is a limited set of UEs with connectivity access to one or more CSG cells within the PLMN for a home Radio Network Controller (RNC). Only those UEs on the RNC's access control list are allowed to use the RNC services. Operators specify the CSG change reporting of different events on the GGSN. Depending on the CSG change reporting configuration, CSG information received from the SGSN is sent to the GGSN where it is processed and reported in Charging Data Records (CDRs) across all interfaces (local, Gz, or Ga).

To configure Closed Subscriber Group (CSG) change reporting on the GGSN, perform the following tasks:

1 Specify the CSG change reporting action on a GGSN.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
uci-change-reporting-action [start-reporting-uci-csg|
start-reporting-uci-shc|start-reporting-uci-uhc]
```

where

Parameter	Description
`uci-change-reporting-action`	Specifies the User Closed Subscriber Group Information (UCI) reporting action. Select one or more of the following options: ***Note:*** *To configure more than one option for this parameter, separate each option by a comma (,).* ■ `start-reporting-uci-csg` – The SGSN starts reporting user CSG information when the UE accesses, enters and leaves a CSG cell. ■ `start-reporting-uci-shc` – The SGSN starts reporting user CSG information when the UE accesses, enters and leaves a subscribed hybrid cell (SHC). ■ `start-reporting-uci-uhc` – The SGSN starts reporting user CSG information when the UE accesses, enters and leaves an unsubscribed hybrid cell (UHC).

2 Ensure the APN is associated with the GGSN.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway ggsn <gateway-name>
apn-list <apn-name>
```

Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on GTP interfaces using Differentiated Services Code Point (DSCP).

GTP interfaces (Gn/Gp, S4, S11, S5/S8, S2a, S2b) – DSCP value configured at the gateway object level per interface (SGW, PGW, GGSN).

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway [sgw|pgw|ggsn]
<name> [gn-interface|gp-interface|s4-interface|s11-interface|
s5-interface|s8-interface|S2a-interface|s2b-interface]
dscp-value <value>
```

Note: *A DSCP value can only be configured for the S2a and the S2b interface on the PGW. You cannot configure a DSCP value for the S2a interface on the MCC TWAG/TWAP or the S2b interface on the MCC ePDG.*

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>
control-dscp-value <value>
```

Note the following guidelines when using DSCP for control plane traffic:

- The DSCP value configured at the gateway object level overrides the network-context DSCP value.
- If a DSCP value is not configured at the gateway object level for GTP-C interfaces (**dscp-value** = **dscp_none**), the network-context DSCP value is used.

Configuring Default Values for the RAT Type, Serving Network PLMN ID MNC/MCC, and the MS Time Zone

The RAT Type IE, Routing Area Identity (RAI) IE and MS Time Zone IE are optional fields within the Create/Update PDP Context message (sent from the SGSN to the GGSN) and are not always present.

This feature enables Operators to configure a GTPv1 signaling information table containing default values for the RAT Type, serving network PLMN ID (MNC/MCC) and the MS Time Zone on an IP/Subnet basis (SGSN IP). If the PDNGW receives a Create PDP Context message without the RAT Type IE, MS Time Zone IE, and/or RAI IE, the MCC looks for the SGSN IP address in the signaling information table and if there is a match, the PDNGW uses the configured default values to populate the RAT type, serving network PLMN ID MNC/MCC, and the MS Time Zone.

Note: *If the RAI IE is missing from the Create/Update PDP Context message, the configured MNC/MCC values in the GTPv1 signaling information table will only be used to supplement the MCC/MNC, which are part of the RAI.*

The MCC only performs a lookup for the Update PDP Context message if a hand-off has taken place. For example, if the SGSN IP address received in the Create PDP Context message is different from the SGSN IP address received in the Update PDP Context message.

Set parameters for this feature using the following commands:

1 Configure the signaling information table and table entries. A maximum of 256 tables can be configured.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sig-info-table
<table-name> sig-info-entry <entry-name>
Possible completions:
admin-state          Admin state
plmn-id-mcc          PLMN ID MCC-3 digits representing Mobile
                     Country Code
plmn-id-mnc          PLMN ID MNC-2 or 3 digits representing Mobile
                     Network Code
rat-type             RAT Type
source-ip-mask       GTP message source IP Mask
source-ip-subnet     GTP message source IP Subnet or Address
time-zone            Time Zone Eg: +5:30, -10:00
time-zone-dst        Adjustment for Daylight Saving Time
```

Note: *If overlapping IP addresses are configured, the entry with the most specific IP address is matched. For example, 1.1.1.1 matches with 1.1.1.1/32 and not 1.1.1.0/24.*

Note: *Any IP addresses contained in the subnet mask are included. For example, 1.1.1.0/24 ranges from 1.1.1.0 to 1.1.1.255.*

Use the `no` form of this command to remove a signaling information table or table entry.

```
an3000-mcm-slot7cpu1(config)# no zone <name> gateway sig-info-table
<table_name>
an3000-mcm-slot7cpu1(config)# no zone <name> gateway sig-info-table
<table_name> sig-info-entry <entry-name>
```

2 Associate the signaling information table with the gateway.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway [pgw|ggsn]
<gateway-name> sig-info-table <table-name>
```

3 Use the following command to display the number of times the signaling information table lookup was successful and the number of times the signaling information table lookup failed.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
cluster-level call-performance pgw <pgw-name>
```

```
num-sig-info-lookup-successful-and-filled-configured-values
```

```
num-sig-info-lookup-failure
```

Monitoring the Gn/Gp Interface

This section describes how to monitor the operational status of the GTP peer via the Gn/Gp interface and how to display statistical information for Gn/Gp communication messages. This section also describes the alarm notification that occurs when the remote GTP peer goes down. For more information on statistics, see *Acuitas Performance Statistics.*

GTP Peer Status

Use the following CLI command to display the status information of a specific GTP peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway gtp-peer-status
<gateway-name> <peer-ip-address>
```

The system displays output similar to the following:

```
gateway gtp-peer-status
PEER
GATEWAY  PEER IP      NETWORK  GTP      RESTART  PEER
NAME     ADDRESS      CONTEXT  VERSION  COUNTER  STATE
--------------------------------------------------------
NYggsn    10.32.34.79 NYC      gtpv1     200      up
```

Gn Interface Statistics

You can display statistical information for Gn/Gp interface communication messages for the grid, cluster, and slot level for a specific GTP peer as shown in the following examples.

Gn/Gp Interface Statistics at the Grid Level

Use the following CLI command to display the Gn/Gp interface GTP grid-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics grid-level
gtp peer <name>
```

Gn/Gp Interface Statistics at the Cluster Level

Use the following CLI command to display the Gn/Gp interface GTP cluster-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
cluster-level gtp peer <name> <cluster-id>
```

Gn/Gp Interface Statistics at the Node Level

Use the following CLI command to display the Gn/Gp interface GTP node-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics node-level
gtp peer <name> <cluster-id> <slot>
```

Alarms

A major-level trap similar to the following occurs when a GTP peer goes down. The alarm clears after connectivity is restored.

```
anGtpPeerDown

Description:        GTP peer down. This indicates that the remote GTP
                    peer is not responding to any GTP messages sent to
                    it. If the peer is statically provisioned, the
                    severity is Major. If the peer is dynamic, the
                    severity is Info.

Trap ID:            60

Status:             Current

Severity:           Info, Major

Auto Clearing:      No

Potential Impact:   Sessions that go to the remote peer may be
                    cleared, resulting in loss of data connectivity
                    for the affect UEs.

Corrective Action:  Verify that the remote peer is up and running.
                    Verify that the remote peer is reachable.Ve rify
                    that the local GTP endpoint is correctly
                    provisioned and running normally.
```

Note: *When the interface connection to a dynamic GTP peer goes down, the system generates an Info alarm.*

Message Definitions

This section lists the supported GTPv1-U/GTPv1-C and GTPv0 message definitions used on the Gn/Gp interface and lists the information elements (IEs) contained in each message type.

- GTPv1-U Message Definitions (page 16-20)
- GTPv1-C Message Definitions (page 16-20)
- GTPv0 Message Definitions (page 16-36)

GTPv1-U Message Definitions

Table 16-1 lists the supported GTPv1-U messages for the Gn/Gp interface. For more information on GTPv1-U messages, see 3GPP TS 29.060 V9.3.0.

Table 16-1 Supported GTPv1-U Gn/Gp Interface Message Definitions

Message Type	Decimal Code	Reference
Echo Request	1	3GPP 29.060 V9.3.0
Echo Response	2	3GPP 29.060 V9.3.0

GTPv1-C Message Definitions

Table 16-2 lists the supported GTPv1-C message definitions for the Gn/Gp interface and lists the information elements (IEs) contained in each message type. For more information on the GTPv1-C messages and the information elements, see 3GPP TS 29.060 V9.3.0.

Table 16-2 Supported GTPv1-C Gn/Gp Interface Message Definitions

Message Type	Decimal Code	Reference
Echo Request	1	3GPP 29.060 V9.3.0
Echo Response	2	3GPP 29.060 V9.3.0
Create PDP Context Request (page 16-21)	16	3GPP 29.060 V9.3.0
Create PDP Context Response (page 16-24)	17	3GPP 29.060 V9.3.0
Update PDP Context Request (SGSN-Initiated) (page 16-27)	18	3GPP 29.060 V9.3.0
Update PDP Context Request (GGSN-Initiated) (page 16-29)	18	3GPP 29.060 V9.3.0
Update PDP Context Response (sent by a GGSN) (page 16-30)	19	3GPP 29.060 V9.3.0
Update PDP Context Response (sent by an SGSN) (page 16-32)	19	3GPP 29.060 V9.3.0

Table 16-2 Supported GTPv1-C Gn/Gp Interface Message Definitions

Message Type	Decimal Code	Reference
Delete PDP Context Request (page 16-33)	20	3GPP 29.060 V9.3.0
Delete PDP Context Response (page 16-34)	21	3GPP 29.060 V9.3.0
Initiate PDP Context Activation Request (page 16-34)	22	3GPP 29.060 V9.3.0
Initiate PDP Context Activation Response (page 16-35)	23	3GPP 29.060 V9.3.0

Create PDP Context Request

The SGSN sends a Create PDP Context Request message to the GGSN as part of the GPRS PDP Context Activation procedure.

Table 16-3 lists the supported Create PDP Context Request information elements.

Table 16-3 Create PDP Context Request IEs

IE Name	Requirement	Condition	IE Type Value (Decimal)
IMSI	Conditional	The international mobile subscriber identity (IMSI) is the subscriber identity of a UE. The IMSI and network layer service access point identifier (NSAPI) together uniquely identify a PDP context. When activating a secondary PDP context, this IE is not included in the message.	2
Routing Area Identity (RAI)	Optional	Specifies the Routing Area Identity (RAI) of the SGSN where the MS is registered. This IE is an optional field that can be configured in a GTPv1 signaling information table containing default values for the IE. For more information, see Configuring Default Values for the RAT Type, Serving Network PLMN ID MNC/MCC, and the MS Time Zone (page 16-16).	3
Recovery	Optional	Included on the Gn/Gp interface if the SGSN is contacting the GGSN for the first time or if the SGSN has restarted.	14
Selection Mode	Conditional	Indicates the origin of the Access Point Name (APN) in the message. When activating a secondary PDP context, this IE is not included in the message.	15
Tunnel Endpoint Identifier Data 1	Mandatory	Specifies a downlink Tunnel Endpoint Identifier (TEID) for G-PDUs. The SGSN chooses the TEID. The GGSN includes this IE in the GTP header of all subsequent downlink G-PDUs related to the requested PDP context.	16

Table 16-3 Create PDP Context Request IEs (Continued)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Tunnel Endpoint Identifier Control Plane	Conditional	Specifies a downlink TEID for control plane messages. The SGSN chooses the TEID. The GGSN includes this IE in the GTP header of all subsequent downlink control plane messages related to the requested PDP context. If the SGSN has already confirmed the successful assignment of its Tunnel Endpoint Identifier Control Plane to the GGSN, this field is not present.	17
NSAPI	Mandatory	The Network layer Service Access Point Identifier (NSAPI) identifies a PDP context in a mobility management context specified by the Tunnel Endpoint Identifier Control Plane.	20
Linked NSAPI	Conditional	The SGSN includes this IE in the message when activating a Secondary PDP context. This IE indicates the NSAPI assigned to any one of the already activated PDP contexts for this PDP address.	20
Charging Characteristics	Conditional	The SGSN includes this IE on the Gn/Gp interface if it is available.	26
Trace Reference	Optional	The SGSN includes this IE on the Gn/Gp interface if GGSN trace is activated. Identifies a record or collection of records for a particular trace.	27
Trace Type	Optional	The SGSN includes this IE on the Gn/Gp interface if GGSN trace is activated. Indicates the type of trace.	28
End User Address	Conditional	If the Mobile Station (MS) requests a dynamic PDP address and a dynamic PDP address is allowed, then the PDP Address field in the End User Address IE is left empty. If the MS requests a static PDP Address, then the PDP Address field in the End User Address IE contains the static PDP Address. The PDP Address in the End User Address IE takes precedence over the address in the Protocol Configuration Option (PCO) IE if the addresses in these two IEs contain contradicting information. When activating a secondary PDP context, this IE is not included in the message.	128
Access Point Name	Conditional	The SGSN includes one of the following: ■ An APN provided by the MS ■ A subscribed APN ■ An APN selected by the SGSN The GGSN may use the APN to differentiate access to different external networks.	131
Protocol Configuration Options	Optional	The SGSN includes this IE on the Gn/Gp interface if it is available.	132

Table 16-3 Create PDP Context Request IEs (Continued)

IE Name	Requirement	Condition	IE Type Value (Decimal)
SGSN Address for signaling	Mandatory	The SGSN includes an SGSN address for control plane. The GGSN stores these SGSN addresses and uses them when sending signaling messages to the SGSN for the MS.	133
SGSN Address for user traffic	Mandatory	The SGSN includes an SGSN address for user traffic. The GGSN stores these SGSN addresses and uses them when sending user traffic to the SGSN for the MS.	133
MSISDN	Conditional	The SGSN includes the Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) of the MS in the Create PDP Context Request.	134
Quality of Service Profile	Mandatory	Specifies the Quality of Service (QoS) values negotiated between the MS and SGSN during PDP context activation.	135
TFT	Conditional	The Traffic Flow Template (TFT) IE is used to distinguish between different traffic flows. If the size of TFT packet filters exceeds the 255 byte limit or if the number of packet filters exceeds 15 in the original Create PDP Context Request or Update PDP Context Request message, the gateway sends the remaining TFT packet filters in subsequent Update PDP Context Request messages.	137
Trigger ID	Optional	The SGSN includes this IE on the Gn/Gp interface if GGSN trace is activated. Identifies the device that triggered the trace.	142
OMC Identity	Optional	The SGSN includes the Operations and Maintenance Center (OMC) identity on the Gn/Gp interface if GGSN trace is activated. Identifies the OMC that receives the trace record.	143
Common Flags	Optional	The Common Flags IE is used to hold values for multiple bit flags. ■ The NRSN bit field indicates whether the SGSN supports the network requested bearer control. ■ The Upgrade QoS Supported bit field indicates whether the SGSN supports the QoS upgrade in response message functionality. ■ The Dual Address Bearer Flag bit field is set to 1 when the MS requests a PDP type IPv4v6 and all SGSNs, to which the MS may be handed over, are Release 9 or above and support dual addressing. This is determined based on the node pre-configuration by the operator.	148
APN Restriction	Optional	The SGSN uses this IE to indicate the highest level of restriction type for all currently active PDP contexts associated with the subscriber. When activating a secondary PDP context, this IE is not included in the message.	149

Table 16-3 Create PDP Context Request IEs (Continued)

IE Name	Requirement	Condition	IE Type Value (Decimal)
RAT Type	Optional	Indicates which RADIO Access Technology (RAT) is currently serving the user equipment (UE). The SGSN includes this IE on the Gn/Gp interface if it is available. This IE is an optional field that can be configured in a GTPv1 signaling information table containing default values for the IE. For more information, see Configuring Default Values for the RAT Type, Serving Network PLMN ID MNC/MCC, and the MS Time Zone (page 16-16).	151
User Location Information	Optional	Indicates the CGI/SAI/RAI of where the MS is currently located. The SGSN includes this IE on the Gn/Gp interface if it is available.	152
MS Time Zone	Optional	Indicates the offset between universal time and local time in steps of 15 minutes of where the MS currently resides. The SGSN includes this IE on the Gn/Gp interface if it is available. This IE is an optional field that can be configured in a GTPv1 signaling information table containing default values for the IE. For more information, see Configuring Default Values for the RAT Type, Serving Network PLMN ID MNC/MCC, and the MS Time Zone (page 16-16).	153
IMEI(SV)	Conditional	The SGSN includes this IE on the Gn/Gp interface if it is available.	154
Additional Trace Info	Optional	The SGSN uses this IE to inform the GGSN of additional trace parameters.	162
Correlation-ID	Optional	The SGSN includes the Correlation-ID on the Gn/Gp interface if the PDP Context Activation procedure is performed as part of the Network-Requested Secondary PDP Context Activation procedure.	183
Evolved Allocation/Retention Priority 1	Optional	The SGSN includes this IE on the Gn/Gp interface if the SGSN supports the subscribed Evolved Allocation/Retention Priority received from the Gr interface.	191
User CSG Information (UCI)	Optional	The SGSN includes this IE on the Gn/Gp interface if it is available.	194

Create PDP Context Response

The GGSN sends a Create PDP Context Response message to the SGSN as an answer to a Create PDP Context Request message.

Table 16-4 lists the supported Create PDP Context Response information elements.

Table 16-4 Create PDP Context Response IEs

IE Name	Requirement	Condition	IE Type Value (Decimal)
Cause	Mandatory	Indicates whether the PDP context creation is successful. If not, this IE gives the reason for the failure.	1
Recovery	Optional	Included on the Gn/Gp interface if the GGSN is contacting the SGSN for the first time or if the GGSN has restarted.	14
Tunnel Endpoint Identifier Data 1	Conditional	Specifies an uplink Tunnel Endpoint Identifier (TEID) for G-PDUs. The SGSN chooses the TEID. The SGSN includes this IE in the GTP header of all subsequent uplink G-PDUs related to the requested PDP context.	16
Tunnel Endpoint Identifier Control Plane	Conditional	Specifies an uplink TEID for control plane messages. The GGSN chooses the TEID. The SGSN includes this IE in the GTP header of all subsequent uplink control plane messages related to the requested PDP context. If the GGSN has already confirmed the successful assignment of its Tunnel Endpoint Identifier Control Plane to the SGSN, this field is not present.	17
NSAPI	Optional	The GGSN might include the NSAPI, received from the SGSN in the Create PDP Context Request message, to facilitate error handling in the SGSN.	20
Charging Id	Conditional	Identifies all charging records produced in the SGSN and GGSN for this PDP context. The GGSN generates the Charging ID that is unique within the GGSN.	127
End User Address	Conditional	If the MS requests a dynamic PDP address with PDP type IPv4, IPv6, or IPv4v6 and a dynamic PDP address is allowed, the GGSN includes the End User Address IE and the PDP Address contains the dynamic PDP address(es) allocated by the GGSN. If the MS requests a static PDP address with PDP type IPv4, IPv6, IPv4v6, or a PDP address is specified with PDP Type PPP, then the GGSN includes the End User Address IE and the PDP Address field is not included. The PDP Address in the End User Address IE is the same if both IEs are present in the Create PDP Context Response. When activating a secondary PDP context, this IE is not included in the message.	128
Protocol Configuration Options	Optional	The SGSN includes this IE on the Gn/Gp interface if it is available.	132
GGSN Address for Control Plane	Conditional	The GGSN includes a GGSN address for control plane that can differ from the one provided by the underlying network service (IP).	133

Table 16-4 Create PDP Context Response IEs (Continued)

IE Name	Requirement	Condition	IE Type Value (Decimal)
GGSN Address for user traffic	Conditional	The GGSN includes a GGSN address for user traffic that can differ from the one provided by the underlying network service (IP).	133
Alternative GGSN Address for Control Plane	Conditional	If the SGSN includes an IPv6 address in the request, an IPv4/IPv6 capable GGSN includes an IPv6 address in the GGSN Address for Control Plane IE and an IPv4 address in the Alternative GGSN Address for Control Plane IE. If the SGSN includes an IPv4 address in the request, an IPv4/IPv6 capable GGSN includes an IPv4 address in the GGSN Address for Control Plane IE and an IPv6 address in the Alternative GGSN Address for Control Plane IE.	133
Alternative GGSN Address for User Traffic	Conditional	If the SGSN includes an IPv6 address, an IPv4/IPv6 capable GGSN includes an IPv6 address in the GGSN Address for user traffic IE and an IPv4 address in the Alternative GGSN Address for User Traffic IE. If the SGSN includes an IPv4 address, an IPv4/IPv6 capable GGSN includes an IPv4 address in the GGSN Address for user traffic IE and an IPv6 address in the Alternative GGSN Address for User Traffic IE.	133
Quality of Service Profile	Conditional	The GGSN can negotiate the QoS values supplied in the Create PDP Context Request if supported by the SGSN.	135
Charging Gateway Address	Optional	Specifies the IP address of the recommended charging gateway functionality to which the SGSN should transfer the Charging Data Records (CDRs) for this PDP context.	251
Alternative Charging Gateway Address	Optional	When both the Charging Gateway Address IE and Alternative Charging Gateway Address IE are present, the Charging Gateway Address IE contains an IPv4 address and the Alternative Charging Gateway Address IE contains an IPv6 address.	251
Common Flags	Optional	The Common Flags IE is used to hold values for multiple bit flags. ■ The Prohibit Payload Compression (PPC) flag indicates whether the SGSN compresses the payload of user data.	148
APN Restriction	Optional	The GGSN uses this IE to indicate the restriction type of the associated PDP context being set up.	149
Bearer Control Mode	Optional	Included on the Gn/Gp interface if the GGSN provides the Bearer Control Mode in the PCO.	184
Evolved Allocation/Retention Priority I	Optional	The GGSN includes this IE on the Gn/Gp interface as the authorized Evolved Allocation/Retention Priority if the GGSN supports this IE and it is received in the corresponding request message.	191

Update PDP Context Request (SGSN-Initiated)

The SGSN sends an Update PDP Context Request message to the GGSN as part of the inter-SGSN Routing Area (RA) Update procedure to modify the QoS profile or to request a RAT change.

Table 16-5 lists the supported SGSN-Initiated Update PDP Context Request information elements.

Table 16-5 Update PDP Context Request IEs (SGSN-Initiated)

IE Name	Requirement	Condition	IE Type Value (Decimal)
IMSI	Optional	Included if the SGSN sends the Update PDP Context Request during an inter-SGSN change.	2
Routing Area Identity (RAI)	Optional	Specifies the Routing Area Identity (RAI) of the SGSN where the MS is registered. This IE is an optional field that can be configured in a GTPv1 signaling information table containing default values for the IE. For more information, see Configuring Default Values for the RAT Type, Serving Network PLMN ID MNC/MCC, and the MS Time Zone (page 16-16).	3
Recovery	Optional	Included if the SGSN is contacting the GGSN for the first time or if the SGSN has restarted.	14
Tunnel Endpoint Identifier Data 1	Mandatory	Specifies a downlink Tunnel Endpoint Identifier (TEID) for G-PDUs. The SGSN chooses the TEID. The GGSN includes this IE in the GTP header of all subsequent downlink G-PDUs related to the requested PDP context.	16
Tunnel Endpoint Identifier Control Plane	Conditional	Specifies a downlink TEID for control plane messages. The SGSN chooses the TEID. The GGSN includes this IE in the GTP header of all subsequent downlink control plane messages related to the requested PDP context. If the SGSN has already confirmed the successful assignment of its Tunnel Endpoint Identifier Control Plane to the GGSN, this field is not present.	17
NSAPI	Mandatory	The NSAPI together with the TEID in the GTP header identifies a PDP context in the GGSN.	20
Trace Reference	Optional	The SGSN includes this IE on the Gn/Gp interface if GGSN trace is activated. Identifies a record or collection of records for a particular trace.	27
Trace Type	Optional	The SGSN includes this IE on the Gn/Gp interface if GGSN trace is activated. Indicates the type of trace.	28
Protocol Configuration Options	Optional	The SGSN includes this IE on the Gn/Gp interface if it is available.	132

Table 16-5 Update PDP Context Request IEs (SGSN-Initiated) (Continued)

IE Name	Requirement	Condition	IE Type Value (Decimal)
SGSN Address for signaling	Mandatory	The SGSN includes an SGSN address for control plane that can differ from the one provided by the underlying network service (IP).	133
SGSN Address for user traffic	Mandatory	The SGSN includes an SGSN address for user traffic that can differ from the one provided by the underlying network service (IP).	133
Alternative SGSN Address for Control Plane	Conditional	If an IPv4/IPv6 capable SGSN receives an IPv4 GGSN address from another SGSN, it includes an IPv4 address in the SGSN Address for Control Plane IE and an IPv6 address in the Alternative SGSN Address for Control Plane IE. Otherwise, an IPv4/IPv6 capable SGSN uses only SGSN IPv6 addresses if it has GGSN IPv6 addresses available.	133
Alternative SGSN Address for User Traffic	Conditional	If an IPv4/IPv6 capable SGSN receives an IPv4 GGSN address from another SGSN, it includes an IPv4 address in the SGSN Address for User Traffic IE and an IPv6 address in the Alternative SGSN Address for User Traffic IE. Otherwise, an IPv4/IPv6 capable SGSN uses only SGSN IPv6 addresses if it has GGSN IPv6 addresses available.	133
Quality of Service Profile	Mandatory	Specifies the QoS values negotiated between the MS and SGSN during PDP context activation, or the new QoS negotiated in the PDP context modification procedure.	135
Trigger ID	Optional	The SGSN includes this IE on the Gn/Gp interface if GGSN trace is activated. Identifies the device that triggered the trace.	142
OMC Identity	Optional	The SGSN includes the OMC identity on the Gn/Gp interface if GGSN trace is activated. Identifies the OMC that receives the trace record.	143
Common Flags	Optional	The Common Flags IE is used to hold values for multiple bit flags. ■ The RAN Procedures Ready bit flag indicates whether the SGSN is ready to receive payload on the PDP context indicated in the message. ■ The NRSN bit field indicates whether the SGSN supports the network requested bearer control. ■ The No QoS negotiation bit flag indicates whether the GGSN can renegotiate the QoS in the corresponding Update PDP Context Response. ■ The Upgrade QoS Supported bit field indicates whether the SGSN supports the QoS upgrade in Response message functionality.	148

Table 16-5 Update PDP Context Request IEs (SGSN-Initiated) (Continued)

IE Name	Requirement	Condition	IE Type Value (Decimal)
RAT Type	Optional	Indicates which RADIO Access Technology (RAT) is currently serving the user equipment (UE). The SGSN includes this IE on the Gn/Gp interface if it is available. This IE is an optional field that can be configured in a GTPv1 signaling information table containing default values for the IE. For more information, see Configuring Default Values for the RAT Type, Serving Network PLMN ID MNC/MCC, and the MS Time Zone (page 16-16).	151
User Location Information	Optional	Indicates the CGI/SAI of where the MS is currently located. The SGSN includes this IE on the Gn/Gp interface if it is available.	152
MS Time Zone	Optional	Indicates the offset between universal time and local time in steps of 15 minutes of where the MS currently resides. The SGSN includes this IE on the Gn/Gp interface if it is available. This IE is an optional field that can be configured in a GTPv1 signaling information table containing default values for the IE. For more information, see Configuring Default Values for the RAT Type, Serving Network PLMN ID MNC/MCC, and the MS Time Zone (page 16-16).	153
Additional Trace Info	Optional	The SGSN uses this IE to inform the GGSN of additional trace parameters.	162
Direct Tunnel Flags	Optional	Indicates whether the SGSN is invoking a direct tunnel.	182
Evolved Allocation/Retention Priority 1	Optional	The SGSN includes the negotiated Evolved Allocation/Retention Priority 1 if the SGSN supports it and if the GGSN indicates support for Evolved allocation retention policy (ARP) or when the SGSN has no information on whether the GGSN supports Evolved ARP.	191
User CSG Information (UCI)	Optional	The SGSN includes this IE on the Gn/Gp interface if it is available.	194

Update PDP Context Request (GGSN-Initiated)

The GGSN sends an Update PDP Context Request message to the SGSN to re-negotiate the QoS parameters of a PDP context or to provide the SGSN with a PDP address.

Table 16-6 lists the supported GGSN-Initiated Update PDP Context Request information elements.

Table 16-6 Update PDP Context Request IEs (GGSN-Initiated)

IE Name	Requirement	Condition	IE Type Value (Decimal)
IMSI	Optional	Included in the message when the GGSN wants to check that the PDP context is still active at the SGSN.	2
Recovery	Optional	Included if the GGSN has restarted recently.	14
NSAPI	Mandatory	The NSAPI together with the TEID in the GTP header identifies a PDP context in the SGSN.	20
End User Address	Conditional	Contains a valid IPv4 or IPv6 address.	128
Protocol Configuration Options	Optional	The GGSN includes this IE to provide the MS with application-specific parameters or to indicate the Bearer Control Mode to the MS.	132
Quality of Service Profile	Optional	Specifies the GGSN-requested QoS.	135
TFT	Optional	The GGSN includes the TFT IE to add, modify, or delete the TFT related to the PDP context for network-requested bearer control. If the size of TFT packet filters exceeds the 255 byte limit or if the number of packet filters exceeds 15 in the original Create PDP Context Request or Update PDP Context Request message, the gateway sends the remaining TFT packet filters in subsequent Update PDP Context Request messages.	137
Common Flags	Optional	The Common Flags IE is used to hold values for multiple bit flags. ■ The Prohibit Payload Compression (PPC) flag indicates whether the SGSN compresses the payload of user data.	148
APN Restriction	Optional	The GGSN uses this IE to indicate the restriction type of the associated PDP context being updated.	149
Direct Tunnel Flags	Optional	The GGSN includes the Direct Tunnel Flags IE when a direct tunnel is used and the GGSN receives an Error Indication message from the RNC.	182
Bearer Control Mode	Optional	Included on the Gn/Gp interface if the GGSN provides the Bearer Control Mode in the PCO.	184
Evolved Allocation/Retention Priority 1	Optional	The GGSN includes the newly authorized Evolved Allocation/Retention Priority 1 if the GGSN supports it and if the SGSN indicates support for Evolved ARP.	191

Update PDP Context Response (sent by a GGSN)

The GGSN sends an Update PDP Context Response message to the SGSN as an answer to an Update PDP Context Request message.

Table 16-7 lists the supported Update PDP Context Response (sent by a GGSN) information elements.

Table 16-7 Update PDP Context Response IEs (Sent by a GGSN)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Cause	Mandatory	Indicates whether the update to the PDP context is successful. If not, this IE gives the reason for the failure.	1
Recovery	Optional	Included if the GGSN is contacting the SGSN for the first time or if the GGSN has restarted.	14
Tunnel Endpoint Identifier Data 1	Conditional	Specifies an uplink Tunnel Endpoint Identifier (TEID) for G-PDUs. The GGSN chooses the TEID. The SGSN includes this IE in the GTP header of all subsequent uplink G-PDUs related to the requested PDP context.	16
Tunnel Endpoint Identifier Control Plane	Conditional	Specifies an uplink TEID for control plane messages. The GGSN chooses the TEID. The SGSN includes this IE in the GTP header of all subsequent uplink control plane messages related to the requested PDP context. If the GGSN has already confirmed the successful assignment of its Tunnel Endpoint Identifier Control Plane to the SGSN, this field is not present.	17
Charging Id	Conditional	Identifies all charging records produced in the SGSN and GGSN for this PDP context. The GGSN generates a Charging ID that is unique for this PDP context. If an inter-SGSN routing area update occurs, the Charging ID is transfered to the SGSN as part of each active PDP context.	127
Protocol Configuration Options	Optional	The GGSN includes this IE to provide the MS with application-specific parameters or to indicate the Bearer Control Mode to the MS.	132
GGSN Address for Control Plane	Conditional	The GGSN includes a GGSN address for control plane that does not differ from the one provided at set up time. This address remains the same for the duration of the PDP context.	133
GGSN Address for user traffic	Conditional	The GGSN includes a GGSN address for user traffic that can differ from the one provided by the underlying network service (IP).	133
Alternative GGSN Address for Control Plane	Conditional	If the Update PDP Context Request received from the SGSN includes IPv6 SGSN addresses, an IPv4/IPv6 capable GGSN includes an IPv6 address in the GGSN Address for Control Plane IE and an IPv4 address in the Alternative GGSN Address for Control Plane IE. If the SGSN includes an IPv4 SGSN address in the request, an IPv4/IPv6 capable GGSN includes an IPv4 address in the GGSN Address for Control Plane IE and an IPv6 address in the Alternative GGSN Address for Control Plane IE.	133

Table 16-7 Update PDP Context Response IEs (Sent by a GGSN) (Continued)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Alternative GGSN Address for User Traffic	Conditional	If the Update PDP Context Request received from the SGSN includes IPv6 SGSN addresses, an IPv4/IPv6 capable GGSN includes an IPv6 address in the GGSN Address for User Traffic IE and an IPv4 address in the Alternative GGSN Address for User Traffic IE. If the SGSN includes an IPv4 SGSN address in the request, an IPv4/IPv6 capable GGSN includes an IPv4 address in the GGSN Address for User Traffic IE and an IPv6 address in the Alternative GGSN Address for User Traffic IE.	133
Quality of Service Profile	Conditional	The GGSN can negotiate the QoS values supplied in the Update PDP Context Request downwards. If the SGSN indicates in the request message support for a QoS upgrade, the GGSN can negotiate the QoS values upwards.	135
Charging Gateway Address	Optional	Specifies the IP address of the recommended charging gateway functionality to which the SGSN should transfer the Charging Data Records (CDRs) for this PDP context.	251
Alternative Charging Gateway Address	Optional	When both the Charging Gateway Address IE and Alternative Charging Gateway Address IE are present, the Charging Gateway Address IE contains an IPv4 address and the Alternative Charging Gateway Address IE contains an IPv6 address.	251
Common Flags	Optional	The Common Flags IE is used to hold values for multiple bit flags. ■ The Prohibit Payload Compression (PPC) flag indicates whether the SGSN compresses the payload of user data.	148
APN Restriction	Optional	The GGSN uses this IE to indicate the restriction type of the associated PDP context being updated.	149
Bearer Control Mode	Optional	Included on the Gn/Gp interface if the GGSN provides the Bearer Control Mode in the PCO.	184
Evolved Allocation/Retention Priority 1	Optional	The GGSN includes the newly authorized Evolved Allocation/Retention Priority 1 if the GGSN supports it and if the SGSN included the Evolved ARP in the corresponding request message.	191

Update PDP Context Response (sent by an SGSN)

The SGSN sends an Update PDP Context Response message to the GGSN as an answer to an Update PDP Context Request message.

Table 16-8 lists the supported Update PDP Context Response (sent by an SGSN) information elements.

Table 16-8 Update PDP Context Response IEs (Sent by an SGSN)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Cause	Mandatory	Indicates whether the update to the PDP context is successful. If not, this IE gives the reason for the failure.	1
Recovery	Optional	Included if the SGSN has restarted recently and the new Restart Counter value has not yet been sent to the GGSN.	14
Tunnel Endpoint Identifier Data 1	Optional	Specifies a downlink Tunnel Endpoint Identifier (TEID) for G-PDUs. The SGSN chooses the TEID. The GGSN includes this IE in the GTP header of all subsequent downlink G-PDUs related to the requested PDP context. If the SGSN sends an Update PDP Context Response to re-establish the user plane tunnel between the SGSN and GGSN, the SGSN only includes the Cause IE, the SGSN Address for User Traffic IE, and Tunnel Endpoint Identifier Data I IE.	16
Protocol Configuration Options	Optional	The SGSN includes this IE if the MS wants to provide the GGSN with application-specific parameters.	132
SGSN Address for User Traffic	Optional	The SGSN includes an SGSN address for user traffic that can differ from the one provided by the underlying network service (IP).	133
Quality of Service Profile	Conditional	The SGSN can negotiate the QoS values supplied in the Update PDP Context Request downwards.	135
User Location Information	Optional	Indicates the CGI/SAI of where the MS is currently located. The SGSN includes this IE on the Gn/Gp interface if it is available.	152
MS Time Zone	Optional	Indicates the offset between universal time and local time in steps of 15 minutes of where the MS currently resides. The SGSN includes this IE on the Gn/Gp interface if it is available.	153
Direct Tunnel Flags	Optional	Indicates whether the SGSN is invoking a direct tunnel.	182
Evolved Allocation/Retention Priority 1	Optional	The SGSN includes the negotiated Evolved Allocation/Retention Priority 1 if the SGSN supports it and if the corresponding request message contained an Evolved Allocation/Retention Priority 1 IE.	191

Delete PDP Context Request

The Delete PDP Context Request is sent from either:

- The SGSN to the GGSN as part of the GPRS Detach procedure or the GPRS PDP Context Deactivation procedure.
- The GGSN to the SGSN as part of the GGSN-initiated PDP Context Deactivation procedure.

Table 16-9 lists the supported Delete PDP Context Request information elements.

Table 16-9 Delete PDP Context Request IEs

IE Name	Requirement	Condition	IE Type Value (Decimal)
Teardown Ind	Conditional	Indicates whether all PDP contexts that share the PDP address with the PDP context identified in the request should also be deactivated.	19
NSAPI	Mandatory	The NSAPI together with the TEID in the GTP header identifies a PDP context.	20
Protocol Configuration Options	Optional	The SGSN includes this IE on the Gn/Gp interface if it is available. The GGSN includes this IE to provide the MS with application-specific parameters.	132

Delete PDP Context Response

A Create PDP Context Response message is sent as an answer to a Create PDP Context Request message.

Table 16-10 lists the supported Delete PDP Context Response information elements.

Table 16-10 Delete PDP Context Response IEs

IE Name	Requirement	Condition	IE Type Value (Decimal)
Cause	Mandatory	Indicates whether the PDP context is successfully deleted. If not, this IE gives the reason for the failure.	1
Protocol Configuration Options	Optional	The SGSN includes this IE on the Gn/Gp interface if it is available. The GGSN includes this IE to provide the MS with application-specific parameters.	132
User Location Information	Optional	Indicates the CGI/SAI of where the MS is currently located. The SGSN includes this IE on the Gn/Gp interface if it is available.	152
MS Time Zone	Optional	Indicates the offset between universal time and local time in steps of 15 minutes of where the MS currently resides. The SGSN includes this IE on the Gn/Gp interface if it is available.	153

Initiate PDP Context Activation Request

The GGSN sends an Initiate PDP Context Activation Request message to the SGSN to initiate the Secondary PDP Context Activation procedure.

Table 16-11 lists the supported Initiate PDP Context Activation Request information elements.

Table 16-11   Initiate PDP Context Activation Request IEs

IE Name	Requirement	Condition	IE Type Value (Decimal)
Linked NSAPI	Mandatory	The GGSN includes this IE in the message when activating a Secondary PDP context. This IE indicates the NSAPI assigned to any one of the already activated PDP contexts for this PDP address.	20
Protocol Configuration Options	Optional	The GGSN includes this IE to provide the MS with application-specific parameters.	132
Quality of Service Profile	Mandatory	Specifies the QoS values requested by the GGSN.	135
TFT	Conditional	The Traffic Flow Template (TFT) IE is used to distinguish between different traffic flows.	137
User Location Information	Optional	Indicates the CGI/SAI/RAI of where the MS is currently located. The SGSN includes this IE on the Gn/Gp interface if it is available.	152
Correlation-ID	Mandatory	Used to correlate the subsequent Secondary PDP Context Activation Procedure with the Initiate PDP Context Activation message.	183
Evolved Allocation/Retention Priority 1	Optional	The GGSN includes this IE if the GGSN supports this IE and if the current SGSN has indicated support of Evolved ARP.	191

Initiate PDP Context Activation Response

The SGSN sends an Initiate PDP Context Activation Response message to the GGSN as an answer to an Update PDP Context Request message.

Table 16-12 lists the supported Initiate PDP Context Activation Response information elements.

Table 16-12   Initiate PDP Context Activation Response IEs

IE Name	Requirement	Condition	IE Type Value (Decimal)
Cause	Mandatory	Indicates whether the secondary PDP context is successfully established. If not, this IE gives the reason for the failure.	1
Protocol Configuration Options	Optional	The SGSN includes this IE in the response message if it is available.	132

GTPv0 Message Definitions

Table 16-13 lists the supported GTPv0 message definitions for the Gn/Gp interface and lists the information elements (IEs) contained in each message type. For more information on the GTPv0 messages and the information elements, see 3GPP TS 09.60 V7.10.0.

Table 16-13 Supported GTPv0 Gn/Gp Interface Message Definitions

Message Type	Decimal Code	Reference
Echo Request	1	3GPP TS 09.60 V7.10.0
Echo Response	2	3GPP TS 09.60 V7.10.0
Create PDP Context Request (GTPv0) (page 16-36)	16	3GPP TS 09.60 V7.10.0
Create PDP Context Response (GTPv0) (page 16-37)	17	3GPP TS 09.60 V7.10.0
Update PDP Context Request (GTPv0) (page 16-39)	18	3GPP TS 09.60 V7.10.0
Update PDP Context Response (GTPv0) (page 16-39)	19	3GPP TS 09.60 V7.10.0
Delete PDP Context Request (GTPv0) (page 16-40)	20	3GPP TS 09.60 V7.10.0
Delete PDP Context Response (GTPv0) (page 16-41)	21	3GPP TS 09.60 V7.10.0

Create PDP Context Request (GTPv0)

The SGSN sends a Create PDP Context Request message to the GGSN as part of the GPRS PDP Context Activation procedure.

Table 16-14 lists the supported Create PDP Context Request information elements.

Table 16-14 Create PDP Context Request IEs (GTPv0)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Routing Area Identity (RAI)	Optional	Specifies the Routing Area Identity (RAI) of the SGSN where the MS is registered.	3
Quality of Service Profile	Mandatory	Specifies the Quality of Service (QoS) values negotiated between the MS and SGSN during PDP context activation.	135
Recovery	Optional	Included on the Gn/Gp interface if the SGSN is contacting the GGSN for the first time or if the SGSN has restarted.	14
Selection Mode	Mandatory	Indicates the origin of the Access Point Name (APN) in the message.	15

Table 16-14 Create PDP Context Request IEs (GTPv0) (Continued)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Flow Label Data I	Mandatory	Specifies a downlink flow label for G-PDUs chosen by the SGSN. The GGSN includes this flow label in the GTP header of all subsequent downlink G-PDUs related to the requested PDP context.	16
Flow Label Signaling	Mandatory	Specifies a downlink flow label for signaling messages chosen by the SGSN. The GGSN includes this flow label in the GTP header of all subsequent downlink signaling messages related to the requested PDP context	17
End User Address	Mandatory	If the Mobile Station (MS) requests a dynamic PDP address and a dynamic PDP address is allowed, then the PDP Address field in the End User Address IE is left empty. If the MS requests a static PDP Address, then the PDP Address field in the End User Address IE contains the static PDP Address. The PDP Address in the End User Address IE takes precedence over the address in the Protocol Configuration Option (PCO) IE if the addresses in these two IEs contain contradicting information.	128
Access Point Name	Mandatory	The SGSN includes one of the following: ■ An APN provided by the MS ■ A subscribed APN ■ An APN selected by the SGSN The GGSN may use the APN to differentiate access to different external networks.	131
Protocol Configuration Options	Optional	The Protocol Configuration Options (PCO) information element is only applicable to the end-user protocol ‘IP’.	132
SGSN Address for signaling	Mandatory	The SGSN includes an SGSN address for control plane. The GGSN stores these SGSN addresses and uses them when sending signaling messages to the SGSN for the MS.	133
SGSN Address for user traffic	Mandatory	The SGSN includes an SGSN address for user traffic. The GGSN stores these SGSN addresses and uses them when sending user traffic to the SGSN for the MS.	133
MSISDN	Mandatory	The SGSN includes the Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) of the MS in the Create PDP Context Request.	134

Create PDP Context Response (GTPv0)

The GGSN sends a Create PDP Context Response message to the SGSN as an answer to a Create PDP Context Request message.

Table 16-15 lists the supported Create PDP Context Response information elements.

Table 16-15 Create PDP Context Response IEs (GTPv0)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Cause	Mandatory	Indicates whether the PDP context creation is successful. If not, this IE gives the reason for the failure.	1
Quality of Service Profile	Conditional	The GGSN can negotiate downwards the QoS values supplied in the Create PDP Context Request.	135
Reordering required	Conditional	Indicates whether reordering of incoming T-PDUs is required.	8
Recovery	Optional	Included on the Gn/Gp interface if the GGSN is contacting the SGSN for the first time or if the GGSN has restarted.	14
Flow Label Data I	Conditional	Specifies an uplink flow label for G-PDUs chosen by the GGSN. The SGSN includes this flow label in the GTP header of all subsequent uplink G-PDUs related to the requested PDP context.	16
Flow Label Signaling	Conditional	Specifies an uplink flow label for signaling messages chosen by the GGSN. The SGSN includes this flow label in the GTP header of all subsequent uplink signaling messages related to the requested PDP context	17
Charging Id	Conditional	Identifies all charging records produced in the SGSN and GGSN for this PDP context. The GGSN generates the Charging ID that is unique within the GGSN.	127
End User Address	Conditional	If the Mobile Station (MS) requests a dynamic PDP address and a dynamic PDP address is allowed, then the PDP Address field in the End User Address IE contains the dynamic PDP address allocated by the GGSN. The PDP Address in the End User Address IE takes precedence over the address in the Protocol Configuration Option (PCO) IE if the addresses in these two IEs contain contradicting information.	128
Protocol Configuration Options	Optional	The SGSN includes this IE on the Gn/Gp interface if it is available.	132
GGSN Address for signaling	Conditional	The GGSN includes a GGSN address for signaling messages that can differ from the one provided by the underlying network service (IP). The SGSN stores these GGSN addresses and uses them when sending signaling messages to the GGSN for the MS.	133
GGSN Address for user traffic	Conditional	The GGSN includes a GGSN address for user traffic that can differ from the one provided by the underlying network service (IP). The SGSN stores these GGSN addresses and uses them when sending G-PDUs to the GGSN for the MS.	133
Charging Gateway Address	Optional	Specifies the IP address of the recommended charging gateway functionality to which the SGSN should transfer the Charging Data Records (CDRs) for this PDP context.	251

Update PDP Context Request (GTPv0)

The SGSN sends an Update PDP Context Request message to the GGSN as part of the inter-SGSN Routing Area (RA) Update procedure to modify the QoS profile or to request a RAT change.

Table 16-16 lists the supported SGSN-Initiated Update PDP Context Request information elements.

Table 16-16 Update PDP Context Request IEs (GTPv0)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Routing Area Identity (RAI)	Optional	Specifies the Routing Area Identity (RAI) of the SGSN where the MS is registered.	3
Quality of Service Profile	Mandatory	Specifies the QoS values negotiated between the MS and SGSN during PDP context activation, or the new QoS negotiated in the PDP context modification procedure.	135
Recovery	Optional	Included if the SGSN is contacting the GGSN for the first time or if the SGSN has restarted.	14
Flow Label Data I	Mandatory	Specifies a downlink flow label for G-PDUs chosen by the SGSN. The GGSN includes this flow label in the GTP header of all subsequent downlink G-PDUs related to the requested PDP context.	16
Flow Label Signaling	Mandatory	Specifies a downlink flow label for signaling messages chosen by the SGSN. The GGSN includes this flow label in the GTP header of all subsequent downlink signaling messages related to the requested PDP context	17
SGSN Address for signaling	Mandatory	The SGSN includes an SGSN address for control plane that can differ from the one provided by the underlying network service (IP).	133
SGSN Address for user traffic	Mandatory	The SGSN includes an SGSN address for user traffic that can differ from the one provided by the underlying network service (IP).	133

Update PDP Context Response (GTPv0)

The GGSN sends an Update PDP Context Response message to the SGSN as an answer to an Update PDP Context Request message.

Table 16-17 lists the supported Update PDP Context Response information elements.

Table 16-17 Update PDP Context Response IEs (GTPv0)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Cause	Mandatory	Indicates whether the update to the PDP context is successful. If not, this IE gives the reason for the failure.	1
Quality of Service Profile	Conditional	The GGSN can negotiate the QoS values supplied in the Update PDP Context Request downwards.	135
Recovery	Optional	Included if the GGSN is contacting the SGSN for the first time or if the GGSN has restarted.	14
Flow Label Data I	Conditional	Specifies an uplink flow label for G-PDUs chosen by the GGSN. The SGSN includes this flow label in the GTP header of all subsequent uplink G-PDUs related to the requested PDP context.	16
Flow Label Signaling	Conditional	Specifies an uplink flow label for signaling messages chosen by the GGSN. The SGSN includes this flow label in the GTP header of all subsequent uplink signaling messages related to the requested PDP context	17
Charging Id	Conditional	Identifies all charging records produced in the SGSN and GGSN for this PDP context. The GGSN generates a Charging ID that is unique for this PDP context. If an inter-SGSN routing area update occurs, the Charging ID is transfered to the new SGSN as part of each active PDP context.	127
GGSN Address for signaling	Conditional	The GGSN includes a GGSN address for signaling messages that can differ from the one provided by the underlying network service (IP). The SGSN stores these GGSN addresses and uses them when sending signaling messages to the GGSN for the MS.	133
GGSN Address for user traffic	Conditional	The GGSN includes a GGSN address for user traffic that can differ from the one provided by the underlying network service (IP). The SGSN stores these GGSN addresses and uses them when sending G-PDUs to the GGSN for the MS.	133
Charging Gateway Address	Optional	Specifies the IP address of the recommended charging gateway functionality to which the SGSN should transfer the Charging Data Records (CDRs) for this PDP context.	251

Delete PDP Context Request (GTPv0)

The Delete PDP Context Request is sent from either:

- The SGSN to the GGSN as part of the GPRS Detach procedure or the GPRS PDP Context Deactivation procedure.
- The GGSN to the SGSN as part of the GGSN-initiated PDP Context Deactivation procedure.

Table 16-18 lists the supported Delete PDP Context Request information elements.

Table 16-18 Delete PDP Context Request IEs (GTPv0)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Private Extension	Optional	Contains vendor or operator-specific information.	255

Delete PDP Context Response (GTPv0)

A Create PDP Context Response message is sent as an answer to a Create PDP Context Request message.

Table 16-19 lists the supported Delete PDP Context Response information elements.

Table 16-19 Delete PDP Context Response IEs (GTPv0)

IE Name	Requirement	Condition	IE Type Value (Decimal)
Cause	Mandatory	Indicates whether the PDP context is successfully deleted. If not, this IE gives the reason for the failure.	1

Chapter 17

S5/S8 and S4/S11 Interfaces

This section contains the following information used to control and monitor the S5/S8 and S4/S11 interfaces:

S5/S8 Interface Overview

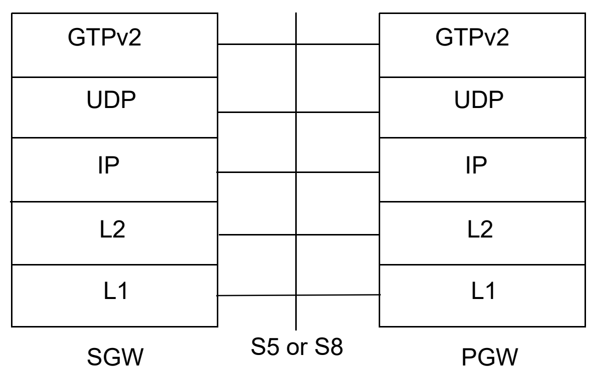
The S5/S8 interface provides user plane tunneling and tunnel management between the SGW and the PGW. The S5 interface is used in non-roaming scenarios where the SGW and PGW are located within the same Public Land Mobile Network (PLMN). The S8 interface is used when roaming between an SGW in a visited PLMN and a PGW in a home PLMN and is the inter-PLMN variant of S5.

This document describes the S5/S8 interface implementation in the *Affirmed Networks Mobile Content Cloud* (MCC) in accordance with 3GPP TS 29.274 V9.3.0. The S5/S8 interface uses the GPRS tunneling protocol (GTPv2-C and GTPv1-U) as defined in 3GPP TS 29.274 V9.3.0 and 3GPP TS 29.281 V9.3.0 to transfer signaling messages and user data over the Evolved Packet Core (EPC) network.

MCC enables the SGW and PGW to operate as stand-alone services on separate devices or as combined services running on the same device. The S5 interface is used when the SGW and PGW operate as stand-alone services on separate devices and is used for control plane signaling when the SGW and PGW are combined.

Figure 17-1 illustrates the GTPv2 protocol stack for the S5/S8 interface.

Figure 17-1 GTPv2 Protocol Stack for the S5/S8 Interface



S4/S11 Interface Overview

The S4 interface provides user plane tunneling and tunnel management between the SGSN and the SGW. The S11 interface provides GTP control signal tunneling between the mobile management entity (MME) and the SGW. The S11 interface coordinates the establishment of SAE bearers within the EPC. The SAE bearer setup can be started by the MME or by the PGW.

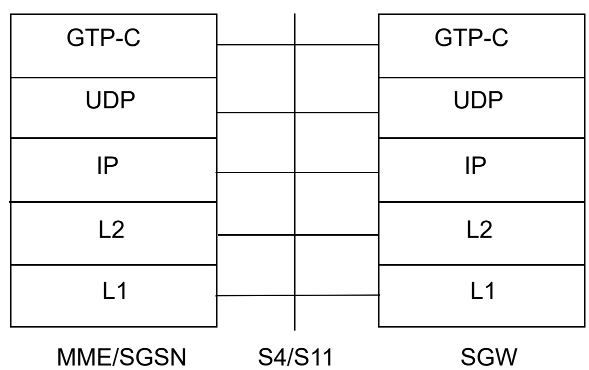
The S4/S11 interface application initiates many events from an SGSN/MME node, such as:

- Creating/deleting a session/default bearer/dedicated bearer
- Adding a rule to create/modify/update a dedicated bearer or perform a user equipment (UE) handover
- X2-based handover with SGW relocation of perform an S1-based UE handover with SGW relocation

This document describes the S4/S11 interface implementation only as it pertains to this release of the *Affirmed Networks Mobile Content Cloud* (MCC) in accordance with 3GPP TS 29.274 V15.3.0.

Figure 17-2 illustrates the GTP-C protocol stack for the S4/S11 interface.

Figure 17-2 GTP-C Protocol Stack for the S4/S11 Interfaces



Paging Overview

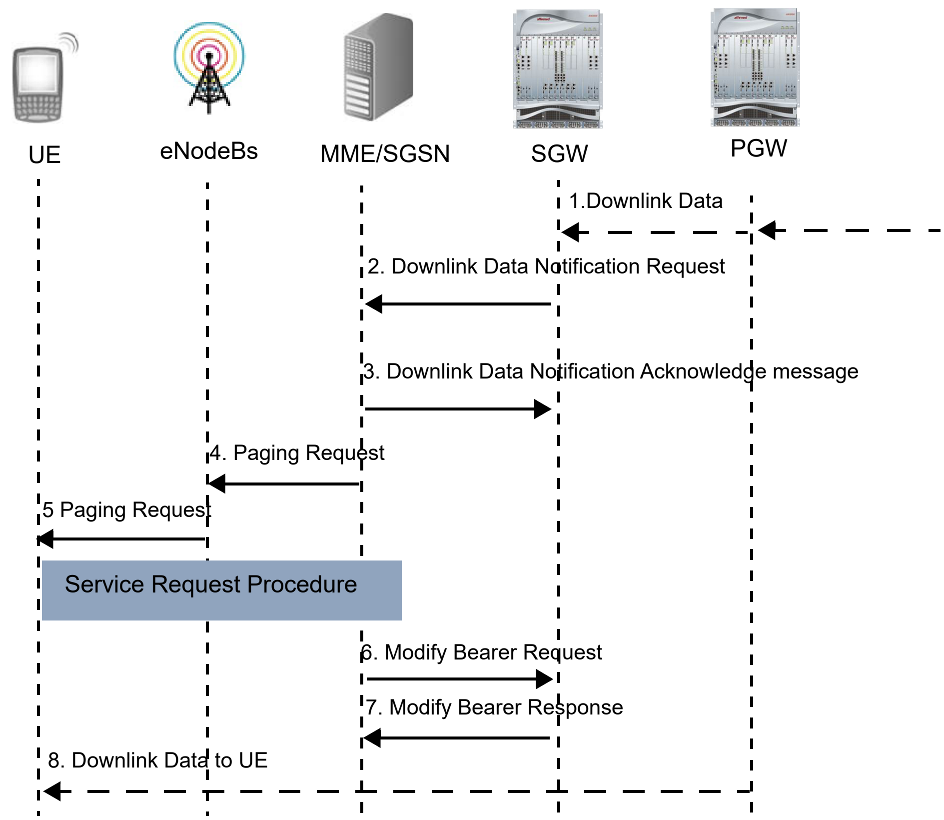
The SGW paging procedure is compliant with 3GPP TS 29.274 V11.0.0 and 3GPP TS 23.401 V11.0.0, including support for the EPS Bearer ID and Allocation/Retention Priority information elements (IEs) in the Downlink Data Notification (DDN) message. When the SGW receives a downlink data packet for a bearer associated with a UE in idle mode, the SGW triggers paging with the MME/SGSN by sending a DDN request. The SGW only triggers the paging procedure if the downlink data packet received is the first packet queued for that bearer.

You can specify whether the SGW triggers paging only for default bearers or for both default bearers and dedicated bearers. You can also configure the maximum number of downlink packets that are buffered on the SGW for a bearer when the UE is being paged. For details, see zone gateway sgw paging-profile (page 12-74).

Message Flow for SGW Paging

Figure 17-3 on page 17-4 illustrates the message flow for the SGW paging procedure. Refer to 3GPP TS 29.274 V11.0.0 and 3GPP TS 23.401 V11.0.0 for more details.

Figure 17-3 Message Flow for SGW Paging Procedure



1 The PGW receives a downlink data packet from an external network for a UE in idle mode and transmits the downlink packet to the SGW.

2 The SGW buffers the packet and sends the MME/SGSN a DDN request to trigger the paging procedure.

3 The MME/SGSN responds with a DDN Acknowledge message.

4 The MME/SGSN sends a paging request to the eNodeBs.

5 Upon receiving a page request from the MME/SGSN, the eNodeBs page the UE.

6 The MME/SGSN sends a Modify Bearer Request message to the SGW to request a modification of bearer services.

7 The SGW responds with a Modify Bearer Response.

8 The PGW transmits the downlink data to the UE.

ARP Priority Level and Paging

Bearers with an ARP priority level less than or equal to the **arp-priority-level-threshold** value are considered high priority (see zone gateway sgw paging-profile (page 12-74)). Bearers with an ARP priority level greater than the **arp-priority-level-threshold** value are considered low priority.

When a downlink data packet for a UE in idle mode is received from an external network for a bearer, the SGW sends a DDN request if a request has not already been sent for that bearer or for any other bearer with higher priority. However, if the MME has requested a delay for the DDN request, the DDN request is not sent until the delay timer expires.

For subsequent downlink data packets for other bearers for the UE, a new DDN request is sent only if the bearer has a higher priority than the bearer for the first request. Once a second DDN request has been sent for high priority bearers, no new DDN requests are sent for that UE.

ARP Priority Level and Throttling

For low priority bearers, the SGW drops downlink packets based on the throttling factor in the throttling IE sent from the MME. The SGW also throttles the DDN requests by the throttling factor and for the length of time specified in the throttling IE.

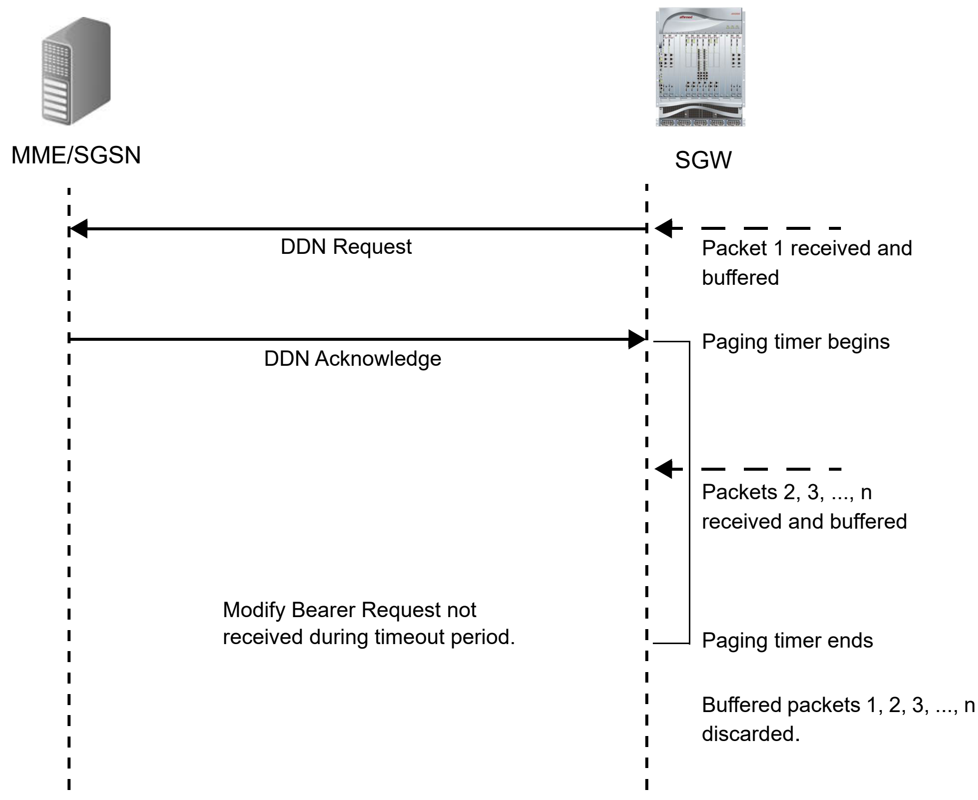
Paging Timers

After the SGW receives a DDN Acknowledge message, the configurable **gtp-page-timeout** and **guard-timeout** (see zone gateway sgw paging-profile (page 12-74)) timers start. If the DDN Acknowledge message does not include cause code 110, the **gtp-page-timeout** timer is used. If the DDN Acknowledge includes cause code 110, the **guard-timeout** timer is used. These timers are used as follows:

- Discard the buffered downlink data if the timer expires before the SGW receives a Modify Bearer Request message.
- Prevent new DDN requests from being sent while the SGW waits for the Modify Bearer Request message.

Figure 17-4 on page 17-6 shows an example of a paging timeout for one bearer.

Figure 17-4 Paging Timeout Failure (one bearer)

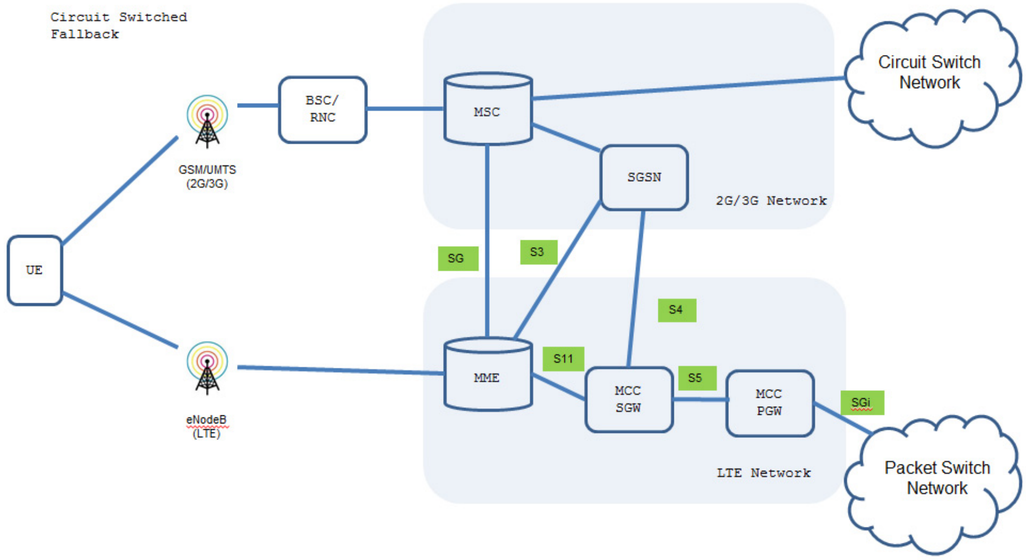


Circuit Switched FallBack

MCC supports Circuit Switched Fallback (CSFB) as defined in 3GPP TS 23.272. CSFB enables the delivery of circuit-switched voice traffic to Long Term Evolution (LTE) mobile devices. LTE is an all-IP packet-switched network and therefore cannot transport circuit-switched services. When a subscriber uses an LTE device to make or receive a voice call, CSFB temporarily switches the subscriber from the LTE packet-based network to the 2G/3G circuit-switched network to complete the call.

In the CSFB network deployment, the MME in the LTE network connects to the mobile switching center (MSC) Server in the 2G/3G network. The MSC server connects to the Operator's circuit-switch network while the MME connects to the SGSN in the Operator's 2G/3G network and the SGW/PGW in the Operator's LTE network as illustrated in Figure 17-5.

Figure 17-5 Circuit Switched Fallback Network Deployment



To support CSFB procedures, MCC supports the following GTPv2-C messages on the S4, S11 and S5/S8 interfaces:

- References and Specifications (page 17-117)
- Suspend Acknowledge (page 17-108)
- Resume Notification (page 17-108)
- References and Specifications (page 17-117)

Note: *For information on CSFB message flows, see the 3GPP TS 23.272 Specification.*

Voice Over LTE

The MCC SGW/PGW supports IP Multimedia Subsystem (IMS) based Voice Over LTE (VoLTE) services:

- Support for setting up a default bearer with Quality of Service Class Identifier (QCI) 5 for IMS sessions. The default bearer is used for SIP signaling messages to the IMS network.
- Support for setting up, updating and tearing down dedicated bearers with QCI 1 for IMS based voice calls based on information received from the PCRF via the Gx interface.

- Support for session setups and various handoff scenarios for multiple Packet Data Network (PDN) connections with multiple dedicated bearers.
- Support for PCRF and MME-signaled termination/downgrade/handover events on the PGW.
- Support for Single Radio Voice Call Continuity (SRVCC) handovers for IMS sessions between circuit-switched (CS) and packet-switched (PS) networks as defined in 3GPP TS 23.216 V11.7.0.
- Support for the GTP-C information elements (IEs) applicable to VoLTE services as defined in 3GPP TS 29.274 V12.6.0 (2014-09).
- Support for dynamically provisioning and activating a Gy online charging session for a dedicated bearer based on PCC rules received from the PCRF via the Gx interface.
- Support for IMS emergency services as specified in 3GPP TS 23.167 V12.0.0 (2014-09) and emergency procedures as specified in 3GPP TS 23.401 V13.1.0 (2014-12).
- VoLTE support for emergency call handling on SGW and PGW as defined in 3GPP TS 23.869.
  - Default bearer establishment for SIP signaling with ARP value reserved for emergency services.
  - Dedicated bearer establishment for the media for the emergency call with ARP value reserved for emergency services and GBR QCI.
  - Emergency sessions identified using IMEI.
- Emergency APN configuration on the SGW and PGW. See Configuring an Emergency APN (page 17-33).
- Support for the configuration and delivery of P-CSCF addresses to the UE using the Protocol Configuration Options (PCO) Information Element (IE) during session setup. See Configuring the P-CSCF Server IP Addresses (page 17-29).

5G Non-Standalone New Radio Solution

This section describes the S5/S8 and S4/S11 interface enhancements added to support the 5G Non-Standalone (5G-NSA) New Radio (NR) solution, as defined in 3GPP TS 29.274 V15.3.0.

Secondary RAT Usage Data Reporting

Secondary radio access technology (RAT) usage data can optionally be reported on a periodic basis.

When the eNB is configured with a minimum time interval for secondary RAT usage data reporting, the eNB sends a radio access network (RAN) request message to the MME when the timer expires for each UE for which the MME has requested reporting. The timer begins with the last usage report for the UE.

Periodic reporting of secondary RAT usage data is provided to the SGW and (optionally) to the PGW. When configured, the MME reports uplink and downlink data volumes to the SGW for the secondary RAT on a per Evolved Packet System (EPS) bearer basis using the secondary RAT usage data reporting procedure. The Secondary RAT IE is present in the GTP-C messages. Several IEs with the same type and instance value may be included to represent multiple usage data reports.

Table 17-1 describes the Secondary RAT Usage Data Report Information Element (IE).

Table 17-1 Secondary RAT Usage Data Report IE

	Bits							
Octets	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**
1	Type = 201 (decimal)							
2 to 3	Length = n							
4	Spare				Instance			
5	Spare						IRSGW	IRPGW
6	Secondary RAT Type							
7	Spare (all bits set to 0)				EPS Bearer ID (EBI)			
8-11	Start Timestamp							
12-15	End Timestamp							
16-23	Usage Data DL							
24-31	Usage Data UL							
32 to (n+4)	These octet(s) is/are present only if explicitly specified							

The following bits within the Secondary RAT Data Report IE indicate:

- Octet 5 bit 2 - IRSGW defines if the Usage Data Report will be used by the SGW or not.
- Octet 5 bit 1 - IRPGW defines if the Usage Data Report will be sent to the PGW or not.

- Octet 6 represents Secondary RAT Type (see Table 17-2).
- Octet 7 represents EPS Bearer ID.
- Octets 8 to 11 and 12 to 15 indicate the UTC time when the recording of the Secondary RAT Usage Data was started and ended.
- Octets 16 to 23 and 24 to 31 represent the Usage Data UL/DL fields.

Table 17-2 Secondary RAT Type Values

RAT Types	Values (Decimal)
NR	0
Unlicensed Spectrum	1
<spare>	2 - 255

Note: Secondary RAT usage information is synchronized with the standby processes. This ensures that if an active GW task fails that it is reported.

UP Function Selection Indication Flags

Based on operator policy, the MME/S4-SGSN includes this IE on the S4/S11 interface if any of the applicable flags is set to 1. The applicable flag is Dual Connectivity with NR (DCNR), which is set to 1 if it is desired to select a specific SGW-U and PGW-U for UEs supporting DCNR and not restricted to use NR by user subscription (for example, due to requirements of higher bit rates). This information is used for the SGW-U, PGW-U or combined SGW-U/PGW-U selection (see Annex B.2 of 3GPP TS 29.244 [80]).

Table 17-3 describes the UP Function Selection Indication Flags IE.

Table 17-3 User Plane Function Selection Indication Flags IE

	Bits							
Octets	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**
1	Type = 202 (decimal)							
2 to 3	Length = n							
4	Spare				Instance			

Table 17-3 User Plane Function Selection Indication Flags IE (Continued)

	Bits	
5	Spare	DCNR
6 to (n+4)	These octet(s) is/are present only if explicitly specified	

For each message, the applicable flags of the UP Function Selection Indication Flags IE are specified in the individual message sub-clause. The remaining flags of the UP Function Selection Indication Flags IE not indicated are discarded by the receiver.

The receiver will consider the value of the applicable flags as “0” if the UP Function Selection Indication Flags IE is applicable for the message but not included in the message by the sender.

The following bits within Octet 5 indicate:

- Bit 8 to 2 – Spare, for future use and set to zero.
- Bit 1 - DCNR (Dual connectivity with NR) – If this bit is set to 1, it indicates to the SGW-C and PGW-C that it is desired to select a specific SGW-U and PGW-U for UEs supporting DCNR and not restricted to use NR by the user subscription.

  The UE signals its support for DCNR to the MME and S4-SGSN with the DCNR of the UE network capability IE in 3GPP TS 24.301 [23]. Subscription restriction to use NR as secondary RAT is specified in “NR as Secondary RAT Not Allowed” bit within Access Restriction Data in 3GPP TS 29.272 [70].

The following bits within the affected octets in this IE indicate:

- Octet 5 bit 2 - IRSGW defines if the Usage Data Report will be used by the SGW or not.

Dedicated Bearer QCI/ARP Modifications

If the PCRF modifies the QoS Class Identifiers/Allocation and Retention Priority (QCI/ARP) values for all charging rules on an existing dedicated bearer, the MCC gateway updates the QCI/ARP value for this dedicated bearer and sends an Update Bearer Request to the MME.

S5/S8 Message Flows

This section provides examples of message flows between the SGW and PGW over the S5/S8 interface. For more information on S5/S8 message flows, see 3GPP TS 23.401 V9.3.0.

- Establishing the Default Bearer Message Flow (page 17-13)
- Establishing and Deleting a Dedicated Bearer Message Flow (page 17-15)

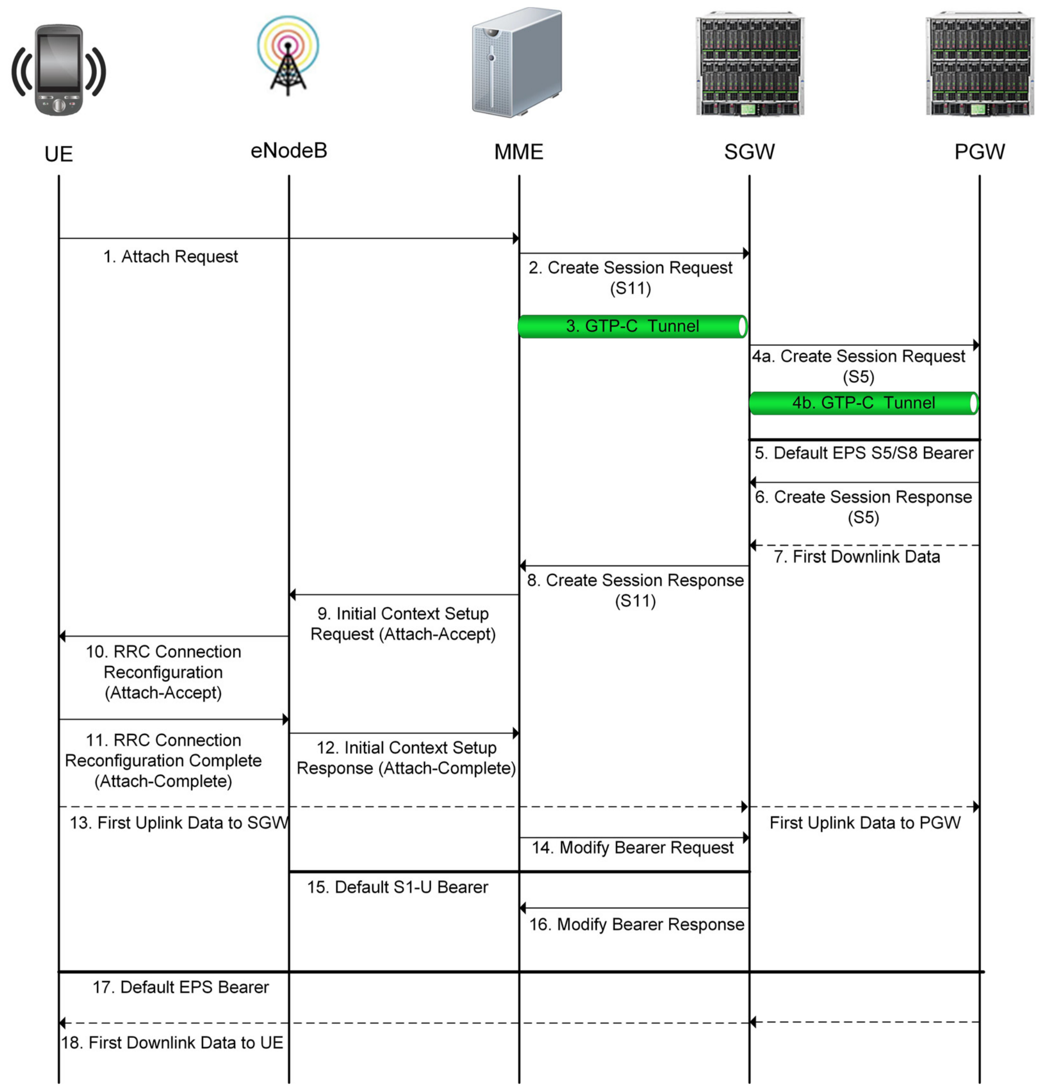
- Emergency Call Default Bearer Message Flow (page 17-16)
- Emergency Call Dedicated Bearer Message Flow (page 17-17)
- UE Initiated Emergency Call Termination Message Flow (page 17-18)

Note: *Refer to 3GPP TS 23.216 V11.7.0. for information on SRVCC handovers for IMS sessions between circuit-switched (CS) and packet-switched (PS) networks.*

Establishing the Default Bearer

Figure 17-6 illustrates an example of the message flow between the SGW and PGW over the S5/S8 interface for establishing the default bearer. The default bearer connects the UE to the PDN and provides the UE with an "always-on" IP connectivity to the PDN.

Figure 17-6 Establishing the Default Bearer Message Flow



1. The UE sends the Attach-Request message to the MME.
2. The MME sends a Create Session Request to the SGW containing information about the default bearer, such as the bearer ID, default QoS and Fully Qualified Tunnel Endpoint Identifier (F-TEID).
3. The MME sets up the GTP-C tunnel toward the SGW used for signaling messages.
4. The SGW passes along the Create Session Request to the PGW and sets up its own GTP-C tunnel toward the PGW for signaling messages.
5. The SGW and PGW establish the default EPS S5/S8 bearer.
6. The PGW responds to the SGW with a Create Session Response containing information about the default bearer, such as the UE’s allocated IP address and the tunnel endpoint IP addresses (TEID).
7. The PGW transmits the first user downlink data (if any) to the SGW.

8 The SGW forwards the Create Session Response to the MME so that the MME can forward the allocated IP address to the UE.

9 The MME sends an Attach-Accept message to the eNodeB containing information about the default bearer context, such as the UE's assigned IP address. The Attach-Accept message is sent in an Initial Context Setup Request message that instructs the eNodeB to create the necessary radio and S1 bearers.

10 The eNodeB configures its radio resources for the default bearer and forwards the radio bearer parameters to the UE in an RRC Connection Reconfiguration message. This message also contains the Attach-Accept message, informing the UE of its IP address.

11 The UE acknowledges the successful setup of the radio bearer with an RRC Connection Reconfiguration Complete message, which also includes an Attach-Complete message for the MME.

12 The eNodeB sends the MME an Initial Context Setup Response message containing the Attach-Complete message and information about its end of the default S1-U bearer.

13 The UE transmits the first uplink data to the SGW and the SGW transmits the first uplink data to the PGW.

14 The MME sends the Default S1-U Bearer configuration to the SGW in a Modify Bearer Request message.

15 The SGW establishes the default S1-U bearer.

16 The SGW responds to the eNodeB with a Modify Bearer Response to acknowledge the successful completion.

17 The Default EPS Bearer is established.

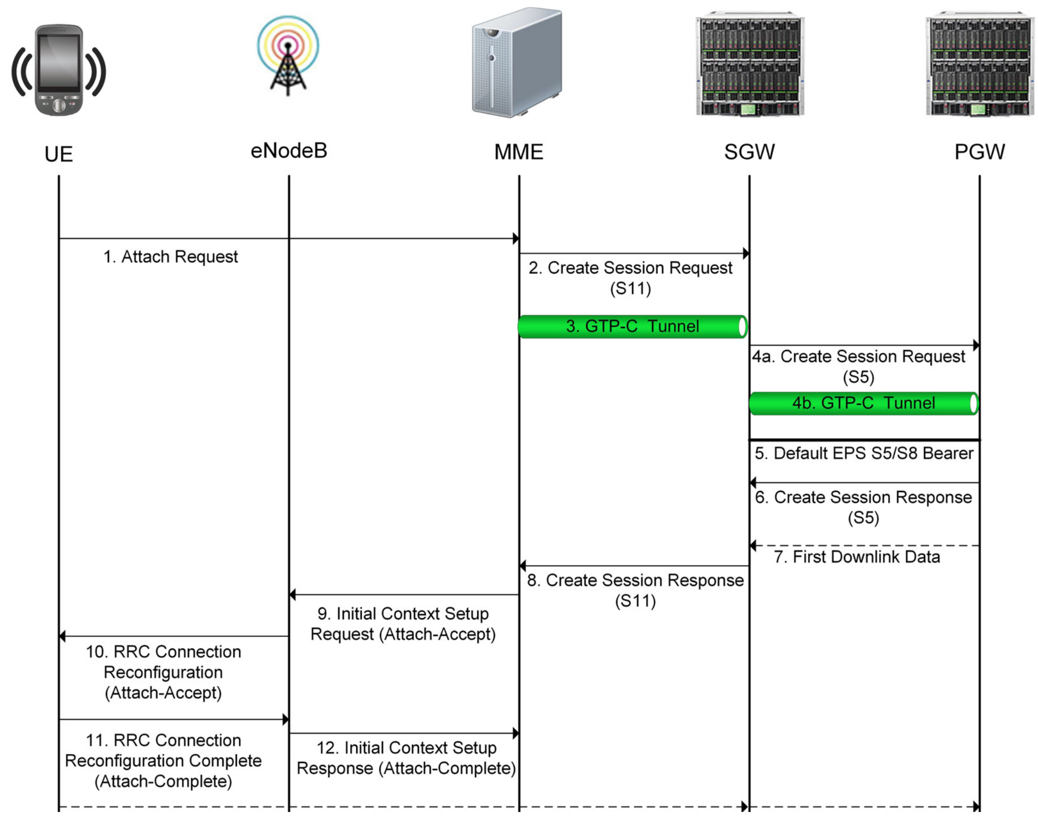
18 The PGW transmits the first downlink data to the UE.

Note: *If the UE requires a service with different QOS requirements than those of the default bearer, a dedicated bearer must be established for that service.*

Establishing and Deleting a Dedicated Bearer

Figure 17-7 illustrates an example of the message flow between the SGW and PGW over the S5/S8 interface for establishing and deleting a dedicated bearer. Dedicated bearers are established when the UE requires a specific QoS for the requested service.

Figure 17-7 Establishing and Deleting a Dedicated Bearer Message Flow



1 The MME sends the Bearer Resource Command message to the SGW to request a dedicated bearer.

2 The SGW forwards the Bearer Resource Command message to the PGW.

3 The PGW responds to the SGW with a Create Bearer Request message containing information about the dedicated bearer, such as the Traffic Flow Template (TFT) and the bearer identifiers.

4 The SGW forwards the Create Bearer Request message to the MME.

5 The MME sends an Activate Dedicated EPS Bearer Context Request message to the UE and sets up the required radio resources and QOS parameters with the eNodeB and UE.

6 The UE responds to the MME with an Activate Dedicated EPS Bearer Context Accept message indicating that the UE is ready to use the new dedicated bearer.

7 The MME sends a Create Bearer Response message to inform the SGW that the S1 tunnel between the MME and SGW is ready.

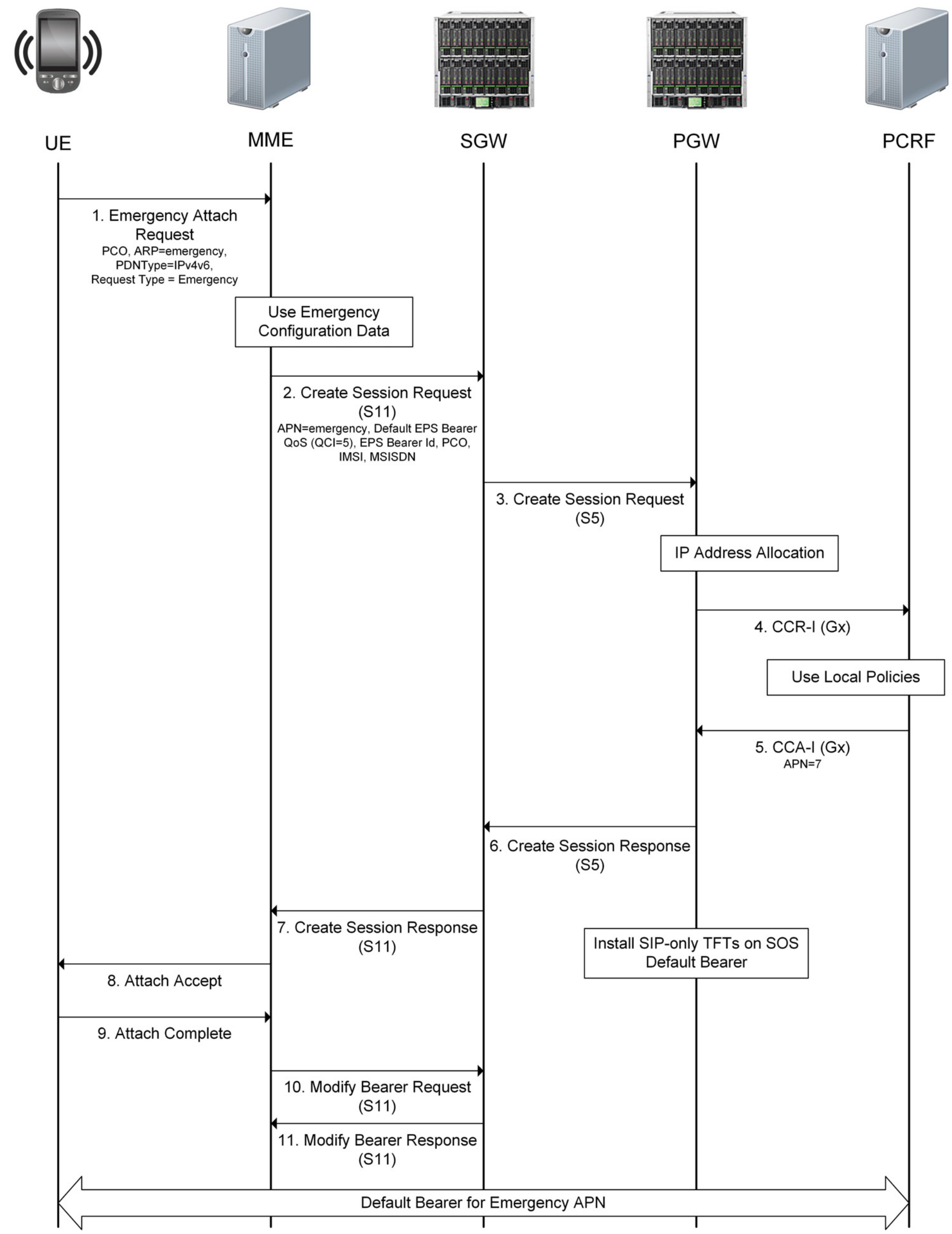
8 The SGW sends the Create Bearer Response message to inform the PGW that the S5/S8 tunnel between the SGW and PGW is ready. The UE can begin using the dedicated EPS bearer to send packets to the network.

Steps 9 through 14 illustrate the message flow where the network initiates the release of the dedicated bearer.

Establishing an Emergency Call Default Bearer

Figure 17-8 illustrates an example of the message flow between the SGW and PGW over the S5/S8 interface for establishing the default bearer for an emergency call.

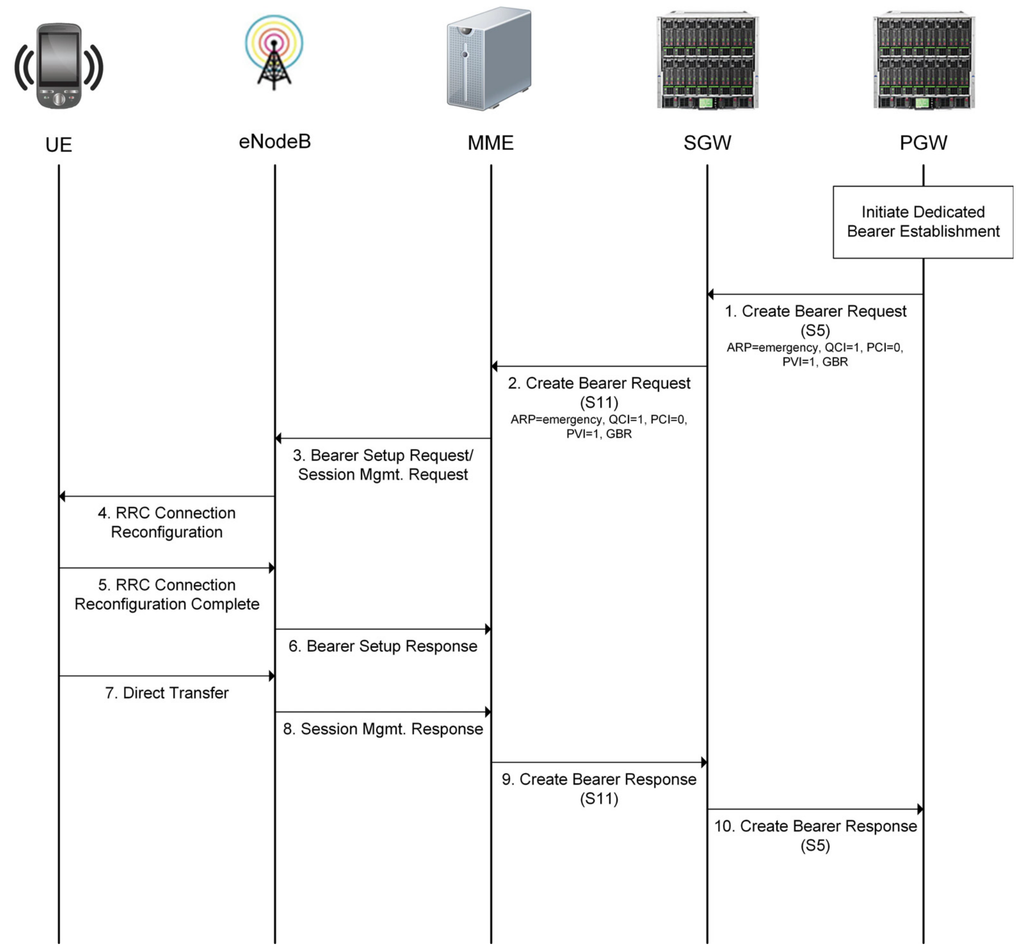
Figure 17-8 Emergency Call Default Bearer Message Flow



Establishing an Emergency Call Dedicated Bearer

Figure 17-9 illustrates an example of the message flow between the SGW and PGW over the S5/S8 interface for establishing the dedicated bearer for an emergency call.

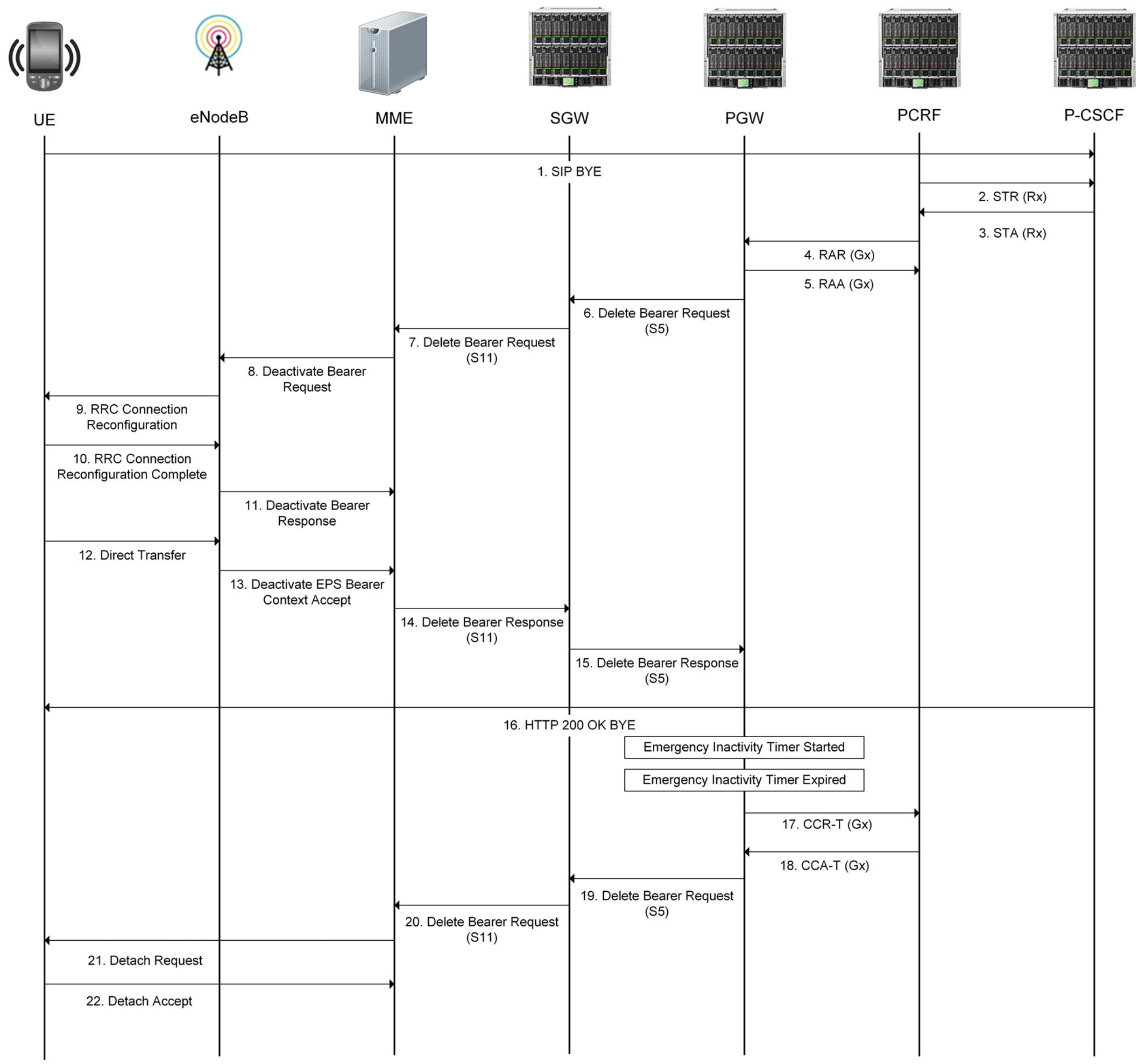
Figure 17-9 Emergency Call Dedicated Bearer Message Flow



UE Initiated Emergency Call Termination

Figure 17-10 illustrates an example of the message flow between the SGW and PGW over the S5/S8 interface for establishing the dedicated bearer for an emergency call.

Figure 17-10 UE Initiated Emergency Call Termination Message Flow



Configuring the S5/S8 Interface

This section describes the steps for configuring the S5/S8 interface on the MCC PGW and SGW.

1 Create a network-context for the S5/S8 interface. For information on configuring network-contexts, see Chapter 8, Network Context Parameters.

Note: *For dual-stack GTP-U and GTP-C support, configure the S5/S8 network-context associated with the gateway object with one or more IPv4 and IPv6 loopback addresses.*

---

2 Enable GTP-U services for the loopback address(es) defined for the S5/S8 network-context.

```
an3000-mcm-slot7cpu1(config)# network-context <name> loopback-ip
<loopback-name> ip-address <ip-address> gtp-u yes
```

where

Parameter	Description
`network-context`	Specifies the name of the network context created for the S5/S8 interface.
`loopback-ip`	Specifies the name of the loopback address associated with this network context.
`ip-address`	Specifies the IP address of this loopback object. Can be IPv4/IPv6.
`gtp-u yes`	Specifies that this loopback address is used for GTP-U.

3 Configure the Access Point Name(s) (APN) accessible via the PGW/SGW.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
network-context <name> admin-state enabled
```

4 Configure the service profile for GTP-U data path management used to maintain data path status and handle data path failures.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gtp-u-path-management-service-profile
```

The following table describes the configurable parameters.

Parameter	Description
`clear-bearers-on-gtp-u-path-failure`	Specifies whether to clear bearers when GTP-U path failure occurs. If set to true, the gateway clears the bearers on the peer when a GTP-U path failure occurs. ■ `false` ■ `true`

Parameter	Description
`gtp-u-echo-interval`	Specifies the amount of time to wait before initiating an Echo Request over the GTP-U path (ms).
`gtp-u-echo-response-timeout`	Specifies the amount of time to wait before retransmitting an Echo Request over the GTP-U path (ms).
`gtp-u-echo-retransmit-count`	Specifies the maximum number of Echo Request retransmission attempts (not including the initial Echo Request) to make before declaring a GTP-U path down.
`initiate-gtp-u-echo-requests`	Specifies whether to send periodic GTP-U Echo Requests to peers upon learning peer information. ■ `true` – (default) The gateway initiates GTP-U path management procedures for each peer. ■ `false` – The gateway does not send any GTP-U Echo Request messages over the GTP-U path.

5 Configure the GTP parameters for the gateway using the Gateway Profile object.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gateway-common <gateway-profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
`allowed-plmnid`	Specifies the policy to authorize the subscriber's PLMN ID during session setup. ■ `any` ■ `homelist` ■ `home-or-roaming-list`
`await-coa-timeout`	Specifies the timeout value (ms) used by the gateway for closed sessions to wait for a Change of Authorization (CoA).

Parameter	Description
`allow-interrat-handover-with-same-remote-fqteid`	Specifies whether the PGW should allow an Inter Radio Access Technologies (IRAT) Handover (SGW–>SGSN or SGSN–>SGW) when the PGW receives a Modify Bearer Request from the SGSN with the same TEID and IP address as the SGW and there is a GTP version change (GTPv1 to GTPv2). ■ `enabled` – When the PGW receives a Modify Bearer Request from the SGSN with the same TEID and IP Address as the SGW and there is a GTP version change (GTPv1 to GTPv2), the PGW allows the IRAT Handover (SGW–>SGSN or SGSN–>SGW). ■ `disabled` – (default) When the PGW receives a Modify Bearer Request from the SGSN with the same TEID and IP Address as the SGW, the PGW considers the peer as unknown and discards the handover message. ***Note:*** *This parameter does not apply to SGW –> SGW handovers with the same TEID and IP address.*
`change-notification`	Specifies whether to enable or disable GTPv2-C Change Notification Request/Response messages. ■ `enabled` (default) ■ `disabled` ***Note:*** *If disabled, the system discards Change Notification messages.*
`clear-subscribers-on-path-failure`	Specifies the policy to clear the subscriber when a path failure occurs. ■ `true` – After the system detects the path failure, all calls that go to that remote peer are actively closed. ■ `false` – All calls are retained until they are manually cleared, cleared by another remote device, or the path recovers and restart processing clears the calls.
`default-gtp-peer-classification`	Specifies the default peer classification when not enough information is present to classify a remote GTP peer as local (same administrative domain) or remote (different administrative domain). ■ `local` ■ `remote`
`defer-delete-session-response`	Specifies whether to defer sending a delete session response until after the delete processing is complete. ■ `true` ■ `false`
`dhcp-relay-response-timeout`	Specifies the timeout value used by the gateway (acting as a DHCP Relay) waiting for the UE to complete its DHCP transaction and obtain an IP (ms).

Parameter	Description
gtp-echo-interval	Specifies the amount of time to wait before initiating an Echo Request over the GTP-C path (ms).
gtp-echo-response-timeout	Specifies the amount of time to wait before retransmitting an Echo Request over the GTP-C path (ms).
gtp-echo-retransmit-count	Specifies the maximum number of Echo Request retransmission attempts (not including the initial Echo Request) to make before declaring a path down.
gtpc-peer-aging-timer	When this timer is configured, MCC deletes GTP-C peer objects when they are not in use. A value of 0 seconds (default) indicates that the feature is disabled and therefore the aging timer does not start for unused GTP-C peers. As a result, peers are never deleted.
idle-session-clearing-state	Specifies whether to enable or disable Idle Session Clearing. Idle Session Clearing enables Operators to clear subscribers who are idle for longer than the configured `idle-timer-duration`. ■ **enabled** ■ **disabled** ***Note:*** *This value is defined in the* `gateway apn` *object if specified, otherwise the system uses the* `gateway-common profile` *settings.*
idle-timer-duration	Specifies the idle timer duration for sessions, in seconds. If a session has been idle for longer than the idle timer duration, the session is cleared. ***Note:*** *This value can also be configured in the* `gateway apn` *object. The value defined in the* `gateway apn` *object takes precedence if the value is non-zero. Otherwise, the value defined in the* `gateway-common profile` *is used.*
inactivity-timer-duration	Specifies the inactivity timer duration for sessions, in seconds. If no activity is detected on the session within the inactivity timer duration, the session is declared idle. ***Note:*** *This value can also be configured in the* `gateway apn` *object. The value defined in the* `gateway apn` *object takes precedence if the value is non-zero. Otherwise, the value defined in the* `gateway-common profile` *is used.*

Parameter	Description
`initiate-echo-requests`	Specifies whether to send periodic GTP-C Echo Requests to peers upon learning peer information. ■ **`true`** – (default) When GTP peers are configured and associated with a gateway instance, the gateway begins path management procedures for each peer. ■ **`false`** – The gateway does not send any Echo Request messages to the peer. The MCC does not initiate GTP-C Echo Requests towards GTP peers added based on the source IP address included in the IP protocol header in incoming Create Session Request/Create PDP Context messages. Instead, if this parameter is set to `true`, the MCC only initiates GTP-C Echo Requests towards: ■ Configured peers (Static GTP peers) ■ GTP peers added based on the IP address encoded in the Sender F-TEID for Control Plane IE of the Create Session Request/Create PDP Context messages.
`non-matching-plmnid`	Specifies the policy for how to treat subscribers with a home PLMN ID that is neither in the home nor the roaming PLMN ID lists. ■ **`home`** ■ **`visiting`**
`parallel-pdn-connections`	Specifies whether to allow multiple PDN connections/default bearers for the same IMSI/APN combination on the SGW/PGW/SAEGW. ■ **`allow-different-ebi`** – A Create Session Request with the same IMSI/APN combination of an existing session will be allowed if the default bearer ID is different from the existing session's default bearer ID. ■ **`allow-different-ebi-pdntype`** – A Create Session Request with the same IMSI/APN combination of an existing session will be allowed if the default bearer ID and PDN type are different from the existing session's default bearer ID and PDN Type. ■ **`disallow`** – (default) Multiple PDN connections/default bearers for the same IMSI/APN combination are not allowed.
`reject-gtpc-messages-from-unconfigured-peers`	Specifies whether to reject GTP-C messages received from non-configured peers. ■ **`true`** ■ **`false`**

Parameter	Description
session-max-clearing-state	Specifies whether to enable or disable PDN Session Clearing for sessions present longer than the maximum time allowed. ■ **enabled** ■ **disabled** ***Note:*** *For PGWs and GGSNs, this value is defined in the gateway apn object if specified, otherwise the system uses the gateway-common settings.*
session-max-timer-duration	Specifies the maximum timer duration for PDN sessions, in minutes. If a PDN session has been present longer than the session maximum timer duration, the PDN session is cleared.

6 Configure the list of PLMN IDs for the home list:

```
an3000-mcm-slot7cpu1(config)# service-construct plmnid-list
home-list-plmnid <plmnid-list>
```

7 Configure the S5/S8 GTP-C endpoint used for the control plane handling of signaling messages on the PGW.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
s5-s8-interface-network-context <name> home-plmnid-list <list>
gateway-profile <name> admin-state enabled
reject-inter-access-handover-request-no-session [true|false] apn-list
<apn-name>; gtpc-endpoint-params network-context <name> loopback-ip
<primary-loopback-name> secondary-loopback-ip
<secondary-loopback-name>
```

where

Parameter	Description
s5-s8-interface-network-context	Specifies the network context containing the S5/S8 interface.
home-plmnid-list	Specifies the list of PLMN IDs considered as home.
gateway-profile	Specifies the name of gateway profile used by the gateway.
admin-state	Specifies whether to enable or disable the admin-state for this service. ■ **enabled** ■ **disabled**

Parameter	Description
reject-inter-access-handover-request-no-session	Specifies whether the PGW accepts or rejects session requests for Wi-Fi to LTE or LTE to Wi-Fi inter-access handovers. ■ **true** – (default) If no prior subscriber PDN connection exists and the PGW receives a Create Session Request with the handover indication set to true, the PGW rejects the request by responding with a Create Session Response with the Cause code as “Context not found.” ■ **false** – The PGW handles the Create Session Request as a normal Create Session Request without the handover request.
apn-list	Specifies the list of APN names.
gtpc-endpoint-params Configure the following S5/S8 GTP-C endpoint parameters on the PGW: ***Note:*** *You must use this parameter to configure the S5/S8 GTP-C endpoint on the PGW. The* `s5s8-gtpc-endpoint-params` *parameter only applies to the SGW.*	
network-context	Specifies the name of the S5/S8 network context on which this GTP-C endpoint is anchored.
loopback-ip	Specifies the name of the primary loopback IP address (associated with this network context) used for the S5/S8 GTP-C endpoint. Can be IPv4 or IPv6. ***Note:*** *To use this loopback address for GTP-U, set the gtp-u flag to* `yes`*.*
secondary-loopback-ip	(Optional) Specifies the name of the secondary S5/S8 GTP-C loopback IP address associated with this network context. Can be IPv4 or IPv6. ***Note:*** *For dual-stack GTP-C support, the primary and secondary loopback IP addresses should be different IP address types (IPv4 or IPv6).*

8 Configure the S5/S8 interface parameters.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
[s5-interface|s8-interface] t3-timeout <value> n3-count <value>
dscp-value <value>
```

where

Parameter	Description
[s5-interface| s8 interface]	Specifies the GTP T3 timeout value and N3 retry value on a per interface basis and specifies the Differentiated Services Code Point (DSCP) value. ■ **t3-timeout** – Specifies the length of time to wait for an acknowledgment for a message (ms). ■ **n3-count** – Specifies the number of times the gateway retransmits a message if no acknowledgment is received. ■ **dscp-value** – Specifies the DSCP value with which to mark all IP packets on this interface. For more information, see Configuring DSCP Values for Control Plane Traffic (page 17-37).

9 Configure the S5/S8 GTP-C endpoint used for the control plane handling of signaling messages on the SGW. To complete the S5/S8 interface configuration on the SGW, you must also configure the GTP endpoint used for the S1 and S11 interfaces on the SGW.

Note: *When executing the following CLI command, configure the* `gtpc-endpoint-params` *parameters on a separate line from the* `s5s8-gtpc-endpoint-params` *parameters and use a semi-colon (;) to separate the two parameters.*

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <gateway-name>
s5-s8-interface-network-context <name> s1u-interface-network-context
<name> home-plmnid-list <list> gateway-profile <name> admin-state
enabled apn-list <apn-name> blocked-apn-list <blocked-apn>;
gtpc-endpoint-params network-context <name> loopback-ip
<primary-loopback-name> secondary-loopback-ip
<secondary-loopback-name>;
s5s8-gtpc-endpoint-params network-context <name> loopback-ip
<primary-loopback-name> secondary-loopback-ip
<secondary-loopback-name>
```

where

Parameter	Description
s5-s8-interface-network-context	Specifies the network context containing the S5/S8 interface.
s1u-interface-network-context	Specifies the network context containing the S1 and S11 interfaces.
home-plmnid-list	Specifies the list of PLMN IDs considered as home.
gateway-profile	Specifies the name of gateway profile used by this service.

Parameter	Description
`admin-state`	Specifies whether to enable or disable the admin-state for this service. ■ `enabled` ■ `disabled`
`apn-list`	Specifies the list of APN names.
`blocked-apn-list`	Specifies the list of APNs whose sessions are rejected by the SGW.
`gtpc-endpoint-params` Configure the following GTP endpoint parameters for the S1 and S11 interfaces on the SGW:	
`network-context`	Specifies the name of the S1 network context on which this GTP endpoint is anchored.
`loopback-ip`	Specifies the name of the primary loopback IP address (associated with this network context) used for the S1/S11 GTP endpoint on the SGW. Can be IPv4 or IPv6. ***Note:*** *If you are using the same network context for both the S1/S11 and S5/S8 interfaces on the SGW, you must use the same loopback IP address for both the S1/S11 and S5/S8 GTP-C endpoints on the SGW.* ***Note:*** *To use this loopback address for GTP-U, set the gtp-u flag to* `yes`.
`secondary-loopback-ip`	(Optional) Specifies the name of the secondary loopback IP address (associated with this network context) used for the S1/S11 GTP endpoint on the SGW. Can be IPv4 or IPv6. ***Note:*** *For dual-stack GTP-C support, the primary and secondary loopback IP addresses should be different IP address types (IPv4 or IPv6).*
`s5s8-gtpc-endpoint-params` Configure the following S5/S8 GTP-C endpoint parameters on the SGW: ***Note:*** *Applies to the SGW only.*	
`network-context`	Specifies the name of the S5/S8 network context on which this GTP-C endpoint is anchored. ***Note:*** *To configure the S1/S11 GTP-C endpoint on the SGW, use the* `gtpc-endpoint-params` *parameter.*
`loopback-ip`	Specifies the name of the primary loopback IP address (associated with this network context) used for the S5/S8 GTP-C endpoint on the SGW. Can be IPv4 or IPv6. ***Note:*** *If you are using the same network context for both the S1/S11 and S5/S8 interfaces on the SGW, you must use the same loopback IP address for both the S1/S11 and S5/S8 GTP-C endpoints on the SGW.* ***Note:*** *To use this loopback address for GTP-U, set the gtp-u flag to* `yes`.

Parameter	Description
secondary-loopback-ip	(Optional) Specifies the name of the secondary S5/S8 GTP-C loopback IP address associated with this network context. Can be IPv4 or IPv6. *Note: For dual-stack GTP-C support, the primary and secondary loopback IP addresses should be different IP address types (IPv4 or IPv6).*

10 (Optional) To define a list of specific loopback addresses (within the S5/S8 network-context) for the GTP-U endpoint on the PGW, use the following CLI command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
gtpu-loopback-list <loopback-interface-name>
```

11 (Optional) To define a list of specific loopback addresses (within the S5/S8 network-context) for the S5/S8 GTP-U endpoint on the SGW, use the following CLI command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <gateway-name>
s5-s8-gtpu-loopback-list <loopback-interface-name>
```

12 (Optional) To define a list of specific loopback addresses (within the S1u network-context) for the S1 GTP-U endpoint on the SGW, use the following CLI command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <gateway-name>
s1u-gtpu-loopback-list <loopback-interface-name>
```

Note: *If the list of specific GTP-U loopback addresses has only IPv4 or IPv6 addresses, only that type is used for the sessions on the gateway.*

13 Configure the GTP peers for each gateway.

Operators can configure GTP peers on the system and associate that peer with any of the gateway objects. A peer is identified by its IP address, network context ID and the version of the GTP-C protocol it supports. A peer can be associated only with a gateway instance whose GTP-C endpoint address is in the same network context as the peer.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-peer <peer-name>
gtpversion gtpv2 is-present-in-local-plmn [false|true] network-context
<name> peer-ip-address <ip-address>
```

14 Associate the GTP peer with the gateway instance by adding the peer to the list of GTP peers with which the gateway communicates.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway [pgw|sgw]
<gateway-name> gtp-peer-list <peer-name>
```

Optional S5/S8 Interface Configuration Tasks

This section describes optional S5/S8 interface configuration tasks.

Configuring the P-CSCF Server IP Addresses

The Proxy-Call Session Control Function (P-CSCF) is the UE's access point to the IMS network. Operators can configure the MCC GGSN/PGW with a P-CSCF table containing a list of IP addresses (IPv4 and/or IPv6) used for IMS calls. If requested during session creation, the MCC GGSN/PGW sends the P-CSCF IP addresses as part of the PCO IE to a UE.

The MCC supports a maximum of 128 IPv4 and 128 IPv6 address pairs in each P-CSCF table. By default, the MCC GGSN/PGW selects the P-CSCF IP address pair in a round-robin fashion. Operators can allocate a maximum number of P-CSCF IPv4/IPv6 addresses from the P-CSCF table to be included in the PCO IE by the PGW. The maximum is 16 of each type.

Alternatively, Operators can configure the MCC to use a weighted round-robin algorithm to select the P-CSCF IP addresses. For weighted round robin, a maximum of only two IP addresses will be allocated in each category (IPv4/IPv6) even if the maximum number of P-CSCF IP addresses allocated is greater than two. Operators assign a weight to each P-CSCF IP address pair to distribute the IMS calls based on the P-CSCF capacity.

Note: *If you add/remove an IP address from the list or modify the configured weight/algorithm, allocations for subsequent session setups use the new configuration.*

Use the following CLI parameters to configure the P-CSCF IPv4 table entries and weights.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pcscf-table
<pcscf-table-name> address-allocation-mode
[round-robin|weighted-round-robin] ip-v4-list <list-name>
[primary-ipv4-address|secondary-ipv4-address] <ip-address> weight
<value>
```

Use the following CLI parameters to configure the P-CSCF IPv6 table entries and weights.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pcscf-table
<pcscf-table-name> ip-v6-list <list-name>
[primary-ipv6-address|secondary-ipv6-address] <ip-address> weight
<value>
```

where

Parameter	Description
pcscf-table	Specifies the name of the P-CSCF table.
address-allocation-mode	Specifies the algorithm to use when allocating P-CSCF IP addresses from the P-CSCF table. ■ **round-robin** – (default) The GGSN/PGW selects the P-CSCF IP addresses in a round-robin fashion from the P-CSCF table. ■ **weighted-round-robin** – The GGSN/PGW selects a P-CSCF IP address pair based on the configured weight assigned to each P-CSCF IP address pair. The IP address pair with the highest weight receives the most distribution of calls.
ip-v4-list|ip-v6-list	Specifies the name of the P-CSCF IPv4 or IPv6 address table. ***Note:*** *The P-CSCF IP address name must be unique across all P-CSCF tables.* ***Note:*** *If you delete an IP address from the list, statistics information is no longer generated for the deleted IP address and is lost after a short period of time.*
primary-ipv4-address| secondary-ipv4-address	Specifies the IPv4 address of a primary and secondary P-CSCF server. The MCC supports a maximum of 128 IPv4 address pairs.

Parameter	Description
primary-ipv6-address|secondary-ipv6-address	Specifies the IPv6 address of a primary and secondary P-CSCF server. The MCC supports a maximum of 128 IPv6 address pairs.
weight	Specifies the weight allocated to this P-CSCF IP address pair. Value 0 disables the IP address pair and the pair is not used regardless of whether the algorithm is round robin or weighted round robin. Weights 1–100 are applicable for weighted round robin only. ***Note:*** *Applies only when* `address-allocation-mode` *is set to* `weighted-round-robin`*. The IP address pair with the highest weight receives the most distribution of calls.*

Each P-CSCF address table is associated with an APN. When the MCC GGSN/PGW receives a Create PDP Context/Session Request for a P-CSCF address, the configured algorithm (round robin or weighted round robin) is used to select a pair of IP addresses from the P-CSCF table with which the APN is associated. Even if the UE does not request the P-CSCF address, Operators can configure the MCC GGSN/PGW to always send the configured P-CSCF IP addresses in the PCO IE as part of the initial Create PDP Context/Session Response message.

Use the following CLI command to associate a P-CSCF address table with an APN.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
pcscf-table <pcscf-table-name> maxPcscfIpV4Addresses <value>
maxPcscfIpV6Addresses <value> send-pcscf-address-in-pco-always
[true|false]
```

where

Parameter	Description
apn-name	Specifies the name of the APN.
pcscf-table	Specifies the name of the P-CSCF table.
max-pcscf-ipv4-addresses	Specifies the maximum number of P-CSCF IPv4 addresses from the P-CSCF table to be included in the PCO IE. The maximum is 16. ***Note:*** *For weighted round robin, a maximum of only two IP addresses will be allocated in each category (IPv4/IPv6) even if the maximum number of P-CSCF IP addresses allocated is greater than two.*
max-pcscf-ipv6-addresses	Specifies the maximum number of P-CSCF IPv6 addresses from the P-CSCF table to be included in the PCO IE. The maximum is 16. ***Note:*** *For weighted round robin, a maximum of only two IP addresses will be allocated in each category (IPv4/IPv6) even if the maximum number of P-CSCF IP addresses allocated is greater than two.*

Parameter	Description
send-pcscf-address-in-pco-always	Specifies one of the following options: ■ **false** – (default) The GGSN/PGW only includes the P-CSCF server IP addresses in the PCO IE of the Create PDP Context/Session Response message if the UE requests it. ■ **true** – The GGSN/PGW always includes the P-CSCF server IP addresses in the PCO IE of the Create PDP Context/Session Response message, even if the UE does not request it.

Use the following CLI command to view statistics on the MCC GGSN/PGW to track the number of PDN connections served per allocated P-CSCF IP address per APN. If you delete an IP address from the list, statistics information is no longer generated for the deleted IP address and is lost after a short period of time. For more information on performance statistics, see *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
pcscf-address-in-use-stats
```

Configuring Traffic Flow Templates on Default Bearers

When no TFT is specified for a default bearer, traffic that does not match the TFT for any dedicated bearer is added to the default bearer. If a TFT is specified for the default bearer, only the traffic matching the TFT is added to the default bearer.

Use the following CLI parameters to configure a TFT for a default bearer on the MCC PGW.

1. Configure a “catch-all” packet filter. The Packet Filter defines the criteria to classify traffic, which can be based on source and destination IP addresses, source port and destination port, and the networking protocol in use.

```
an3000-mcm-slot7cpu1(config)# service-construct packet-filter
<packet-filter-name>
```

2. Configure a Service Rule and associate the “catch-all” Packet Filter with the Service Rule.

```
an3000-mcm-slot7cpu1(config)# service-construct service-rule
<service-rule-name> packet-filter <packet-filter-name>
```

3. Associate the Service Rule with a QoS Flow and set the gate parameter to “block” to block all traffic that does not match the packet filter (TFT).

```
an3000-mcm-slot7cpu1(config)# services quality-of-service qos-flow
<qos-flow-name> service-rule <service-rule-name> gate block
```

4. Use the following command to enable/disable sending TFT packet filters installed for the default bearer to the UE.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gateway-common <name> install-default-bearer-packet-filters-on-ue
[enabled|disabled]
```

Configuring an Emergency APN

The MCC identifies an attach request as an emergency call when it receives a Create Session Request without an IMSI IE present or with an unauthenticated IMSI present. In this case, the MCC uses the UE's IMEI as a key to handle the emergency session on the MCC.

If an authenticated IMSI is present, the MCC matches the full APN name received in the attach request with the names of all APNs marked as providing emergency services with which it is associated. If a match is found, the MCC identifies the session as an emergency call.

To support emergency services, configure an emergency APN and associate it with the MCC gateway (SGW/PGW). Emergency APNs associated with the MCC SGW are used primarily to identify emergency calls if an authenticated IMSI is present in the Create Session Request and to maintain call performance statistics for APNs associated with IMS sessions.

The MCC rejects any UE-requested bearer resource modification for an emergency PDN connection and does not perform any checks of the Maximum APN Restriction for an emergency PDN connection.

Note: *APN mapping is not supported for emergency APNs.*

To configure an emergency APN and associate it with the gateway (SGW/PGW), use the following CLI commands.

1.
```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
emergency-apn true
```
2.
```
an3000-mcm-slot7cpu1(config)# zone <name> gateway [sgw|pgw]
<gateway-name> apn-list <apn-name>
```

An emergency inactivity timer is configurable for APNs marked for emergency services. This timer specifies the number of seconds that the PGW waits before tearing down an emergency session after the established dedicated bearers for that session are released.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
emergency-inactivity-timer-state [enabled|disabled]
emergency-inactivity-timer-duration <value>
```

Use the following CLI command to view APN-level statistics for an APN configured on the SGW or PGW. For more information on performance statistics, see *Acuitas Performance Statistics*.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics grid-level
call-performance [pgw-apn|sgw-apn]
```

Configuring the Cause Code for a Network Initiated Bearer Deactivation

When MCC operates as a stand-alone PGW or combined SGW/PGW, if the default bearer of the IMS PDN is deleted, the PGW sends a Delete Bearer Request with cause “Reactivation Requested”, if configured. The PGW deletes the bearer regardless of the Delete Bearer Response or Delete Bearer Response cause code.

The following cause code configuration is added under APN. By default, no cause code is sent.

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <name>
network-initiated-deactivation-cause [none|reactivation-requested]
```

Use the following CLI command to remove the default bearer (session deactivation).

```
subscriber clear
```

Configuring the Guard Timer and Guard Retry Count on the PGW

When an MCC PGW receives a GTP-C Create/Update/Delete Bearer Response message indicating a rejected request due to cause code 110 (“Temporarily rejected due to handover procedure in progress”) from an MME/S4-SGSN during PGW-initiated procedures, the PGW starts a locally configured guard timer. When the guard timer expires, the PGW retransmits the GTP-C request message up to the maximum number of retransmission attempts configured.

Use the following CLI command to configure the guard timer and guard retry count on the PGW:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <pgw-name>
guard-timeout <value> guard-retry-count <value>
```

where

Parameter	Description
guard-timeout	Specifies the number of milliseconds to wait before the PGW retransmits a network-initiated request after receiving a response indicating a temporary rejection. ***Note:*** *Affirmed Networks recommends setting the guard timer at a higher interval than the T3 timeout value.*
guard-retry-count	Specifies the maximum number of times that the PGW retransmits a network-initiated request when the guard timer is used.

Configuring the Guard Timer on the SGW

When an MCC SGW receives a Downlink Data Notification Acknowledge message indicating a rejected request due to cause code 110 (“Temporarily rejected due to handover procedure in progress”) from an MME/S4-SGSN during paging procedures, the SGW starts a locally configured guard timer, buffers all downlink user packets for the given UE, and waits to receive a Modify Bearer Request message. If the SGW does not receive the Modify Bearer Request message when the guard timer expires, the SGW releases the buffered downlink user packets.

Use the following CLI command to configure the guard timer on the SGW:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <sgw-name>
paging-profile guard-timeout <value>
```

where

Parameter	Description
`guard-timeout`	Specifies the number of milliseconds to wait for the paging procedure to complete after receiving a response indicating temporary rejection. ***Note:*** *Affirmed Networks recommends setting the guard timer at a higher interval than the GTP page timeout value.*

Configuring CSG Change Reporting

The MCC supports Closed Subscriber Group (CSG) change reporting on the SGW/PGW as defined in 3GPP TS 29.274 V11.0.0. A CSG is a limited set of UEs with connectivity access to one or more CSG cells within the PLMN for a home eNB (HeNB). Only those UEs on the HeNB’s access control list are allowed to use the HeNB services. Operators specify the CSG change reporting of different events on the PGW. Depending on the CSG change reporting configuration, CSG information received from the MME is sent to the SGW/PGW where it is processed and reported in Charging Data Records (CDRs) across all interfaces (local, Gz, or Ga).

To configure CSG change reporting on the PGW, perform the following tasks:

1 Specify the CSG change reporting action on a PGW.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
uci-change-reporting-action [start-reporting-uci-csg|
start-reporting-uci-shc|start-reporting-uci-uhc]
```

where

Parameter	Description
`uci-change-reporting-action`	Specifies the User Closed Subscriber Group Information (UCI) reporting action. Select one or more of the following options: ***Note:*** *To configure more than one option for this parameter, separate each option by a comma (,).* ■ `start-reporting-uci-csg` – The MME starts reporting user CSG information when the UE accesses, enters and leaves a CSG cell. ■ `start-reporting-uci-shc` – The MME starts reporting user CSG information when the UE accesses, enters and leaves a subscribed hybrid cell (SHC). ■ `start-reporting-uci-uhc` – The MME starts reporting user CSG information when the UE accesses, enters and leaves an unsubscribed hybrid cell (UHC).

2 Ensure the APN is associated with the PGW.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway pgw <gateway-name>
apn-list <apn-name>
```

Configuring an Idle Timer for Dedicated Bearers Per QCI

To configure an idle timer (`idle-timer-duration`) for dedicated bearers per QCI, Operators configure a QCI profile with an idle timer and associate this QCI profile with an APN. When a dedicated bearer, for a given session, is created with a QCI value that matches the QCI value in a QCI profile (used for an APN), the SGW/PGW/GGSN starts the idle timer configured for the QCI profile. If the gateway terminates a dedicated bearer because the idle timer expired, the gateway notifies the MME so that resources can be released. The `idle-timer-duration` has a range of 30 seconds to 2160000 seconds.

Note: *This idle timer only applies to dedicated bearers on a PGW (standalone or combined SGW/PGW), standalone SGW and standalone GGSN. The idle timer does not apply to dedicated bearers on an SGW operating as a combined SGW/PGW.*

Use the following CLI commands to configure the QCI profile and associate it with an APN:

1 Configure the QCI profile.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile qci-profile
<qci-profile-name> qci-config <qci-value> idle-timer-state
[enable|disable] idle-timer-duration <value-in-seconds>
```

2 Associate the QCI profile with an APN.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
qci-profile <qci-profile-name>
```

Modifying ECGI Information on the S8 interface

You can configure the SGW to modify the E-UTRAN cell global identifier (ECGI) sent in the User Location Information (ULI) IE on the S8 interface. If this feature is enabled on the SGW and the SGW determines that the subscriber's PDN connection is remote, the SGW replaces the ECGI information, received from the MME, with random values and forwards these random values in the ULI IE to the PGW. The SGW uses these random ECGI values in all subsequent messages with the PGW over the S8 interface. However, the SGW uses the real ECGI values in CDRs and log messages.

To determine whether a PDN connection is local or remote, the SGW compares the PGW IP address received in the Create Session Request message with a list of local PGW IP addresses configured on the SGW. If there is not a match and the PGW IP address is not in the list of GTP peers belonging to the local PLMN, the subscriber is considered remote.

Set parameters for this feature using the following commands:

1. Ensure the default peer classification is set to `remote`.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gateway-common <name> default-gtp-peer-classification remote
```

2. Configure the SGW to override the ECGI information on the S8 interface.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <name>
override-ecgi-on-s8 enabled
```

Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on GTP interfaces using Differentiated Services Code Point (DSCP).

GTP interfaces (Gn/Gp, S4, S11, S5/S8, S2a, S2b) – DSCP value configured at the gateway object level per interface (SGW, PGW, GGSN).

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway [sgw|pgw|ggsn]
<name> [gn-interface|gp-interface|s4-interface|s11-interface|
s5-interface|s8-interface|S2a-interface|s2b-interface]
dscp-value <value>
```

Note: *A DSCP value can only be configured for the S2a and the S2b interface on the PGW. You cannot configure a DSCP value for the S2a interface on the MCC TWAG/TWAP or the S2b interface on the MCC ePDG.*

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>
control-dscp-value <value>
```

Note the following guidelines when using DSCP for control plane traffic:

- The DSCP value configured at the gateway object level overrides the network-context DSCP value.
- If a DSCP value is not configured at the gateway object level for GTP-C interfaces (**dscp-value** = **dscp_none**), the network-context DSCP value is used.

Monitoring the S5/S8 Interface

This section describes how to monitor the operational status of the GTP peer via the S5/S8 interface and how to display statistical information for S5/S8 communication messages. This section also describes the alarm notification that occurs when the remote GTP peer goes down. For more information on statistics, see *Acuitas Performance Statistics.*

GTP Peer Status

Use the following CLI command to display the status information of a specific GTP peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway gtp-peer-status
<gateway-name> <peer-ip-address>
```

The system displays output similar to the following:

```
gateway gtp-peer-status
PEER
GATEWAY  PEER IP      Port     NETWORK  GTP      RESTART  PEER
NAME     ADDRESS               CONTEXT  VERSION  COUNTER  STATE
--------------------------------------------------------
NYsgw    10.32.34.79  2152     NYC      gtpv2    0         up
```

Use the following CLI command to display the status information of a GTP peer within a network context.

```
an3000-mcm-slot7cpu1# show zone <name> gateway gtp-peer-status
<gateway-name> <peer-ip> <network-context>
```

S5/S8 Interface Statistics

You can display statistical information for S5/S8 interface communication messages for the grid, cluster, and node level for a specific GTP peer as shown in the following examples.

For information about S5/S8 performance statistics, refer to the *Acuitas Performance Statistics* guide.

S5/S8 Interface Statistics at the Grid Level

Use the following CLI command to display the S5/S8 interface GTP grid-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics grid-level
gtp peer <name>
```

S5/S8 Interface Statistics at the Cluster Level

Use the following CLI command to display the S5/S8 interface GTP cluster-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
cluster-level gtp peer <name> <cluster-id>
```

S5/S8 Interface Statistics at the Node Level

Use the following CLI command to display the S5/S8 interface GTP node-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics node-level
gtp peer <name> <cluster-id> <slot>
```

Alarms

An major-level trap similar to the following occurs when a GTP peer goes down. The alarm clears after connectivity is restored. For more information on alarms, see the *Affirmed Networks SNMP MIB and Alarm Reference Guide*.

```
anGtpPeerDown
Description:          GTP peer down. This indicates that the remote GTP
                      peer is not responding to any GTP messages sent to
                      it. If the peer is statically provisioned, the
                      severity is Major. If the peer is dynamic, the
                      severity is Info.
Trap ID:              60
Status:               Current
Severity:             Info, Major
Auto Clearing:        No
Potential Impact:     Sessions that go to the remote peer may be
                      cleared, resulting in loss of data connectivity
                      for the affect UEs
Corrective Action:    Verify that the remote peer is up and running.
                      Verify that the remote peer is reachable. Verify
                      that the local GTP endpoint is correctly
                      provisioned and running normally.
```

Note: *When the interface connection to a dynamic GTP peer goes down, the system generates an Info alarm.*

Two informational alarms are generated for emergency services.

`anEmergencyCallAttemptFailed` – When the system fails to setup emergency bearers for a session due to a lack of resources/memory or other call processing errors.

`anEmergencyCallDropped` – When active emergency bearers are deactivated due to operator intervention, such as deactivations initiated through the CLI or deactivations due to internal errors.

GTP Message Definitions

This section lists the supported GTPv1-U and GTPv2-C message definitions used on the S5/S8 and S4/S11interfaces and lists the information elements (IEs) contained in each message type.

- GTPv1-U Message Definitions (page 17-40)
- GTPv2-C Message Definitions (page 17-41)

GTPv1-U Message Definitions

Table 17-4 lists the supported GTPv1-U messages for the S5/S8 and S4/S11interfaces. For more information on GTPv1-U messages, see 3GPP TS 29.281 V9.3.0.

Table 17-4 Supported GTPv1-U S5/S8 Interface Message Definitions

Message Type	Decimal Code	Reference
Echo Request	1	3GPP 29.281 V9.3.0
Echo Response	2	3GPP 29.281 V9.3.0
End Marker	254	3GPP 29.281 V9.3.0
Encapsulated T-PDU	255	3GPP 29.281 V9.3.0

GTPv2-C Message Definitions

Table 17-5 describes the supported GTPv2-C message definitions for the S5/S8 and S4/S11 interfaces. For more information on GTPv2-C messages, see 3GPP TS 29.274 V9.3.0 and 3GPP TS 29.274 V12.4.0. The MCC supports the GTP-C information elements (IEs) applicable to VoLTE services as defined in 3GPP TS 29.274 V12.6.0 (2014-09).

Table 17-5 Supported GTPv2-C S5/S8 and S4/S11 Interface Message Definitions

Message Type	Interfaces	Decimal Code	Reference
Echo Request	S5/S8, S4/S11	1	3GPP 29.274 V9.3.0
Echo Response	S5/S8, S4/S11	2	3GPP 29.274 V9.3.0
Bearer Resource Command (page 17-42)	S5/S8, S4/S11	68	3GPP 29.274 V9.3.0
Bearer Resource Failure Indication (page 17-45)	S5/S8, S4/S11	69	3GPP 29.274 V9.3.0
Change Notification Request (page 17-46)	S5/S8, S4/S11	38	3GPP 29.274 V12.4.0
Change Notification Response (page 17-49)	S5/S8, S4/S11	39	3GPP 29.274 V12.4.0
Create Bearer Request (page 17-50)	S5/S8, S4/S11	95	3GPP 29.274 V9.3.0
Create Bearer Response (page 17-52)	S5/S8, S11	96	3GPP 29.274 V9.3.0
Create Indirect Data Forwarding Tunnel Request (page 17-56)	S4/S11	166	3GPP 29.274 V9.3.0
Create Indirect Data Forwarding Tunnel Response (page 17-62)	S4/S11	167	3GPP 29.274 V9.3.0
Create Session Request (page 17-64)	S5/S8, S4/S11	32	3GPP 29.274 V9.3.0
Create Session Response (page 17-70)	S5/S8, S4/S11	33	3GPP 29.274 V9.3.0
Delete Bearer Command (page 17-74)	S5/S8, S4/S11	66	3GPP 29.274 V9.3.0
Delete Bearer Failure Indication (page 17-76)	S5/S8, S4/S11	67	3GPP 29.274 V9.3.0
Delete Bearer Request (page 17-77)	S5/S8, S4/S11	99	3GPP 29.274 V9.3.0
Delete Bearer Response (page 17-79)	S5/S8, S4/S11	100	3GPP 29.274 V9.3.0
Delete Indirect Data Forwarding Tunnel Request (page 17-82)	S4/S11	168	3GPP 29.274 V9.3.0
Delete Indirect Data Forwarding Tunnel Response (page 17-82)	S4/S11	169	3GPP 29.274 V9.3.0
Delete Session Request (page 17-83)	S5/S8, S4/S11	36	3GPP 29.274 V9.3.0
Delete Session Response (page 17-86)	S5/S8, S4/S11	37	3GPP 29.274 V9.3.0

Table 17-5 Supported GTPv2-C S5/S8 and S4/S11 Interface Message Definitions (Continued)

Message Type	Interfaces	Decimal Code	Reference
Downlink Data Notification (page 17-87)	S4/S11	176	3GPP 29.274 V9.3.0
Downlink Data Notification Acknowledge (page 17-91)	S4/S11	177	3GPP 29.274 V9.3.0
Downlink Data Notification Failure Indication (page 17-92)	S4/S11	70	3GPP 29.274 V9.3.0
Modify Bearer Command (page 17-93)	S5/S8, S4/S11	64	3GPP 29.274 V9.3.0
Modify Bearer Failure Indication (page 17-94)	S5/S8, S4/S11	65	3GPP 29.274 V9.3.0
Modify Bearer Request (page 17-95)	S5/S8, S4/S11	34	3GPP 29.274 V9.3.0
Modify Bearer Response (page 17-101)	S5/S8, S4/S11	35	3GPP 29.274 V9.3.0
Release Access Bearers Request (page 17-105)	S4/S11	170	3GPP 23.401
Release Access Bearers Response (page 17-106)	S4/S11	171	3GPP 23.401
Resume Acknowledge (page 17-107)	S5/S8, S4/S11	165	3GPP 29.274 V9.3.0
Resume Notification (page 17-108)	S5/S8, S11	164	3GPP 29.274 V9.3.0
Suspend Acknowledge (page 17-108)	S5/S8, S4/S11	163	3GPP 29.274 V9.3.0
Suspend Notification (page 17-109)	S5/S8, S4/S11	162	3GPP 29.274 V9.3.0
Trace Session Activation (page 17-110)	S5/S8, S4/S11	71	3GPP 29.274 V9.3.0
Trace Session Deactivation (page 17-110)	S5/S8, S4/S11	72	3GPP 29.274 V9.3.0
Update Bearer Request (page 17-111)	S5/S8, S4/S11	97	3GPP 29.274 V9.3.0
Update Bearer Response (page 17-114)	S5/S8, S4/S11	98	3GPP 29.274 V9.3.0

Bearer Resource Command

The SGW sends a Bearer Resource Command message to the PGW on the S5/S8 interface to initiate any of the following procedures:

- UE-requested Dedicated Bearer Allocation
- UE-requested Bearer Resource Modification Procedure (used to also deactivate a dedicated bearer)
- MS-initiated PDP Context Modification
- Secondary PDP Context Activation

The MME sends a Bearer Resource Command message to the SGW on the S11 interface to initiate any of the following procedures:

- UE-requested Dedicated Bearer Allocation
- UE-requested Bearer Resource Modification Procedure (used to also deactivate a dedicated bearer)

The SGSN sends a Bearer Resource Command message to the SGW on the S4 interface to initiate any of the following procedures:

- MS-initiated PDP Context Modification
- Secondary PDP Context Activation

Table 17-6 lists the Bearer Resource Command information elements.

Table 17-6 Bearer Resource Command Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Linked EPS Bearer ID (LBI)	Mandatory	None	EBI	73	0
Procedure Transaction Id (PTI)	Mandatory	None	PTI	100	0
Flow Quality of Service (Flow QoS)	Conditional	The SGW includes this IE on the S5/S8 interface, except for an entire packet filters removal in a bearer resource release. This IE is included on the S4/S11 interface if the “Requested New QoS”/”Required QoS” is included in the corresponding NAS message or the “Required traffic flow QoS” is included in the corresponding NAS message .	Flow QoS	81	0
Traffic Aggregate Description (TAD)		The TAD describes the packet filter(s) for a traffic flow aggregate.	TAD	85	0
	Mandatory	The MME forwards this on the S11 interface.			
	Conditional/ Optional	The S4-SGSN forwards this on the S4 interface if the S4-SGSN receives it.			
	Conditional/ Optional	The SGW forwards this to PGW on the S5/S8 interface if the SGW receives it.			

Table 17-6 Bearer Resource Command Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
RAT Type	Conditional	Included on the S5/S8 and S4 interfaces for the MS-initiated PDP Context Modification and Secondary PDP Context Activation procedures.	RAT Type	82	0
Serving Network	Optional	Included on the S5/S8 and S4 interfaces for the MS-initiated PDP Context Modification procedure.	Serving Network	83	0
User Location Information (ULI)	Optional	Included on the S5/S8 and S4 interfaces for the MS-initiated PDP Context Modification procedure.	ULI	86	0
EPS Bearer ID	Conditional	Indicates the EPS bearer to be modified. Included on the S5/S8 and S4/S11 interfaces for the MS-initiated PDP Context Modification procedure.	EBI	73	1
Indication Flags	Optional	Included on the S5/S8 and S4/S11 interfaces if any of the applicable flags is set to 1: ■ Change Reporting Support Indication ■ Direct Tunnel Flag	Indication	77	0
S4-U SGSN F-TEID	Conditional	Included on the S4 interface when direct tunnel is not established in the MS initiated PDP Context modification procedure.	F-TEID	87	0
S12 RNC F-TEID	Conditional	Included on the S4 interface when direct tunnel flag is set to 1 in the MS initiated PDP Context modification procedure.	F-TEID	87	1
Protocol Configuration Options (PCO)	Optional	Included on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0
Private Extension	Optional	Includes vendor or operator specific information. Included on the S5/S8 and S4/S11 interfaces.	Private Extension	255	Vendor Specific

Bearer Resource Failure Indication

The PGW sends a Bearer Resource Failure Indication message to the SGW on the S5/S8 interface to indicate a failure of any of the following procedures:

- UE-requested Bearer Resource Allocation
- UE-requested Bearer Resource Modification
- MS-initiated PDP Context Modification
- Secondary PDP Context Activation

The SGW sends a Bearer Resource Failure Indication message to the MME on the S11 interface to indicate a failure of any of the following procedures:

- UE-requested Bearer Resource Allocation
- UE-requested Bearer Resource Modification

The SGW sends a Bearer Resource Failure Indication message to the SGSN on the S4 interface to indicate a failure of any of the following procedures:

- MS-initiated PDP Context Modification
- Secondary PDP Context Activation

Table 17-7 lists the Bearer Resource Failure Indication information elements.

Table 17-7 Bearer Resource Failure Indication Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
Linked EPS Bearer ID	Mandatory	None	EBI	73	0
Procedure Transaction Id (PTI)	Mandatory	None	PTI	100	0
Recovery	Optional	None	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Change Notification Request

If the PGW has requested ECGI/TAI/CGI/SAI/RAI Change Reporting and the SGW supports the feature as part of location dependent charging related procedures:

- The MME/S4-SGSN, if it supports the feature, sends a Change Notification Request message to the SGW on the S4/S11 interface.
- The SGW sends a Change Notification Request message to the PGW on the S5/S8 interface.

If the PLMN has configured secondary RAT usage reporting and if PDN GW Secondary RAT reporting is active, the MME sends the Change Notification Request message to the SGW on the S11 interface and includes "Secondary RAT Usage Data Report" when it has received Secondary RAT usage data from the eNodeB in the following procedures, regardless of whether ULI is reported or not:

- Connection Suspend
- S1 release
- E-UTRAN to UTRAN Iu mode Inter RAT handover
- E-UTRAN to GERAN A/Gb mode Inter RAT handover
- MME triggered Serving GW relocation
- MME to 3G SGSN combined hard handover and SRNS relocation

The MME may also send the Change Notification Request message and include "Secondary RAT Usage Data Report" to the SGW on the S11 interface when it has received periodic reporting of Secondary RAT usage data from eNodeB, as described in TS 23.401 [3] subclause 5.7A. The SGW forwards the "Secondary RAT Usage Data Report" to the PGW on the S5/S8 interface if the IRPGW flag is set to "1".

Table 17-8 lists the Change Notification Request information elements as defined in 3GPP TS 29.274 V12.4.0.

Table 17-8 Change Notification Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
IMSI	Conditional	The SGW includes the IMSI on the S5/S8 interface if it is provided by the MME/SGSN. The MME/S4-SGSN includes the IMS on the S4/S11 interface except when the UE is emergency attached and the UE is UICCless.	IMSI	1	0

Table 17-8 Change Notification Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
ME Identity (MEI)	Conditional	The SGW includes the MEI on the S5/S8 interface if it is available. The MME/S4-SGSN includes the MEI on the S4/S11 interface if the UE is emergency attached and the UE is UICCless and the IMSI is not authenticated.	MEI	75	0
Indication Flags	Conditional-Optional	Included on the S5/S8 and S4/S11 interfaces if the Unauthenticated IMSI flag is set to 1. This flag is set to 1 if the IMSI present in the message is not authenticated and is for an emergency attached UE.	Indication	77	0
RAT Type	Mandatory	None.	RAT Type	82	0
User Location Information (ULI)	Conditional-Optional	The SGW includes the ULI on the S5/S8 interface if it is provided by the MME/SGSN and if the SGW supports this feature.	ULI	93	0
	Conditional	The SGSN includes the ULI on the S4 interface if the MS is located in a RAT Type of GERAN, UTRAN or GAN and includes the CGI, SAI and/or RAI.			
	Conditional-Optional	The MME includes the ULI on the S11 interface if the UE is located in a RAT Type of E-UTRAN and includes the ECGI and/or TAI, or TAI and Macro eNB ID, or Macro eNB ID depending on the Change Reporting Action provided to the MME.			
User CSG Information (UCI)	Conditional/ Optional	The MME/S4-SGSN includes the UCI on the S4/S11 interface if the MS is located in the CSG cell or the hybrid cell and the P-GW/PCRF decides to receive the CSG Information. The SGW includes the UCI on the S5/S8 interface if it is provided by the MME/SGSN and if the SGW supports this feature.	UCI	145	0
PGW S5/S8 GTP-C IP Address	Conditional	Included on the S4 interface	EBI	73	0
	Conditional/ Optional	Included on the S11 interface			

Table 17-8 Change Notification Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Linked Bearer Identity (LBI)	Conditional-Optional	The MME/S4-SGSN includes the LBI on the S4/S11 interface. The SGW includes the LBI on the S5/S8 interface if it is provided by the MME/SGSN.	EBI (EPS Bearer ID)	73	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
Secondary RAT Usage Data Report	Conditional-Optional	If PDN GW Secondary RAT reporting is active, the MME includes this IE on the S11 interface if it has received Secondary RAT usage data from eNodeB in a Connection Suspend, or S1 release procedure, or an MME triggered Serving GW relocation, or an E-UTRAN to GERAN A/Gb mode Inter RAT handover, E-UTRAN to UTRAN Iu mode Inter RAT handover, MME to 3G SGSN combined hard handover and SRNS relocation, or Routing Area Update procedures. In this case, the MME also sets the IRSGW flag to "0" and the IRPGW flag to "1". The MME also includes this IE on the S11 interface when it has received periodic reporting of Secondary RAT usage data from eNodeB. Several IEs with the same type and instance value may be included to represent multiple usage data reports.	Secondary RAT Usage Data Report	201	0
	Optional	The SGW forwards this IE on the S5/S8 interface if it receives the Secondary RAT Usage Data Report with the IRPGW flag set to "1" from MME. Several IEs with the same type and instance value may be included to represent multiple usage data reports.			

Change Notification Response

As part of the location dependent charging related procedures to acknowledge the receipt of a Change Notification Request, the Change Notification Response message is sent:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-9 lists the Change Notification Response information elements as defined in 3GPP TS 29.274 V12.4.0.

Table 17-9 Change Notification Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
IMSI	Conditional	The IMSI is included in the message if it is received in the Change Notification Request message.	IMSI	1	0
ME Identity (MEI)	Conditional	The IMSI is included in the message if it is received in the Change Notification Request message.	MEI	75	0
Cause	Mandatory	None	Cause	2	0
Change Reporting Action	Conditional	This IE is included with the appropriate action field if the location change reporting mechanism is started or stopped for this subscriber in the SGSN/MME.	Change Reporting Action	131	0
CSG Information Reporting Action	Conditional/ Optional	Included on the S5/S8 and S4/S11 interfaces with the appropriate Action field if the CSG information reporting mechanism is started or stopped for the subscriber in the SGSN/MME.	CSG Information Reporting Action	146	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Create Bearer Request

As part of the Dedicated Bearer Activation procedure, the Create Bearer Request message is sent:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME on the S4/S11 interface.

Table 17-10 lists the Create Bearer Request information elements.

Table 17-10 Create Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Procedure Transaction Id (PTI)	Conditional	Included on the S5/S8 and S4/S11 interfaces for the UE-requested Bearer Resource Modification, UE-requested Bearer Resource Allocation or Secondary PDP Context Activation procedures.	PTI	100	0
Linked Bearer Identity (LBI)	Mandatory	Indicates the default bearer associated with the PDN connection.	EBI (EPS Bearer ID)	73	0
Protocol Configuration Options (PCO)	Optional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0
Bearer Contexts	Mandatory	Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0
SGW-FQ-CSID	Conditional	Included on the S11 interface according to the requirements in 3GPP TS 23.007.	FQ-CSID	132	1
Change Reporting Action	Conditional	Included on the S5/S8 interface if it is provided by the MME/SGSN.	Change Reporting Action	131	0
CSG Information Reporting Action	Conditional/ Optional	Included on the S5/S8 and S4/S11 interfaces with the appropriate Action field if the CSG information reporting mechanism is started or stopped for the subscriber in the SGSN/MME.	CSG Information Reporting Action	146	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Create Bearer Request

Within the Create Bearer Request message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-11 lists the information elements contained in the Bearer Contexts IE.

Table 17-11 Bearer Contexts IEs (within Create Bearer Request)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Octet 1		Bearer Context IE Type = 93 (decimal)			
Octets 2 and 3		Length = n			
Octet 4		Spare and Instance Fields			
EPS Bearer ID	Mandatory	This IE is set to 0.	EBI	73	0
TFT	Mandatory	The EPS bearer level TFT can contain both uplink and downlink packet filters. If the size of TFT packet filters exceeds the 255 byte limit or if the number of packet filters exceeds 15 in the original Create Bearer/PDP Context Request or Update Bearer/PDP Context Request message, the gateway sends the remaining TFT packet filters in subsequent Update Bearer/PDP Context Request messages. This capability is also supported on the PGW for Create Bearer Request messages sent in handover scenarios, such as 3GPP to non-3GPP access and vice versa.	Bearer TFT	84	0
S1-U SGW F-TEID	Conditional	Included on the S11 interface if the S1-U interface is used.	F-TEID	87	0
S5/S8-U PGW F-TEID	Conditional	Included on S5/S8 and S4/S11 interfaces for GTP-based S5/S8 interface.	F-TEID	87	1
S12 SGW F-TEID	Conditional	Included on the S4 interface if the S12 interface is used.	F-TEID	87	2
S4-U SGW F-TEID	Conditional	Included on the S4 interface if the S4-U interface is used.	F-TEID	87	3
Bearer Level QoS	Mandatory	None	Bearer QoS	80	0

Table 17-11 Bearer Contexts IEs (within Create Bearer Request) (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
Charging Id	Conditional	Included on the S5/S8 interface if available.	Charging Id	94	0
Bearer Flags	Optional	Applicable flags are: ■ Prohibit Payload Compression (PPC) – This flag may be set on the S5/S8 and S4 interfaces. ■ vSRVCC indicator – The PGW may include this IE on the S5/S8 interface according to 3GPP TS 23.216 [43]. When received on the S5/S8, the SGW forwards this IE on the S11 interface.	Bearer Flags	97	0
Protocol Configuration Options (PCO)	Optional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available. This bearer-level PCO IE takes precedence over the message-level PCO IE if they both exist.	PCO	78	0

Create Bearer Response

As an answer to a Create Bearer Request message, the Create Bearer Response message is sent:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME to the SGW on the S11 interface.

Table 17-12 lists the Create Bearer Response information elements.

Table 17-12 Create Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0

Table 17-12 Create Bearer Response Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Bearer Contexts	Mandatory	Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0
Recovery	Conditional	Included on the S5/S8 and S11 interfaces if the SGW is contacting the PGW for the first time.	Recovery	3	0
Protocol Configuration Options (PCO)	Conditional	Included on the S11 interface if the UE includes it. Included on the S5/S8 interface if it is available.	PCO	78	0
UE Time Zone	Conditional/ Optional	The SGW includes this IE on the S5/S8 interface if it is available.	UE Time Zone	114	0
	Optional	The MME optionally includes this IE on the S11 interface.			
User Location Information (ULI)	Conditional/ Optional	The SGW includes this IE on the S5/S8 interface if it is provided by the MME/SGSN.	ULI	86	0
	Conditional/ Optional	The MME includes this IE on the S11 interface. The ECGI is included.			
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Create Bearer Response

Within the Create Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-13 lists the information elements contained in the Bearer Contexts IE.

Table 17-13 Bearer Contexts IEs (within Create Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0

Table 17-13 Bearer Contexts IEs (within Create Bearer Response) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0
S1-U eNodeB F-TEID	Conditional	Included on the S11 interface if the S1-U interface is used.	F-TEID	87	0
S1-U SGW F-TEID	Conditional	Included on the S11 interface to correlate the bearers with those in the Create Bearer Request	F-TEID	87	1
S5/S8-U SGW F-TEID	Conditional	Included on the S5/S8 interface.	F-TEID	87	2
S5/S8-U PGW F-TEID	Conditional	Included on the S5/S8 interface to correlate the bearers with those in the Create Bearer Request.	F-TEID	87	3
S12 SGW F-TEID	Conditional	Included on the S4 interface to correlate the bearers with those in the Create Bearer Request.	F-TEID	87	5
S4-U SGSN F-TEID	Conditional	Included on the S4 interface if the S4-U interface is used.	F-TEID	87	6
S4-U SGW F-TEID	Conditional	Included on the S4 interface to correlate the bearers with those in the Create Bearer Request.	F-TEID	87	7
Protocol Configuration Options (PCO)	Conditional/ Optional	Included by the MME on the S11 interface if the UE includes the PCO IE in the corresponding message. The SGW includes this IE on the S5/S8 interface if it is available. This bearer-level PCO IE takes precedence over the message-level PCO IE if they both exist.	PCO	78	0

Table 17-13 Bearer Contexts IEs (within Create Bearer Response) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
RAN/NAS Cause	Conditional	If the bearer creation failed, the MME includes this IE on the S11 interface to indicate the RAN cause and/or the NAS cause of the bearer creation failure, if available, and if this information is permitted to be sent to the PGW operator according to MME operator's policy. If both a RAN cause and a NAS cause are generated, then several IEs with the same type and instance value is included to represent a list of causes. The SGW includes this IE on the S5/S8 interface if it receives it from the MME.	RAN/NAS Cause	172	0
	Conditional	If the bearer creation failed, the TWAN includes this IE on the S2a interface to indicate the cause of the bearer creation failure, if available, and if this information is permitted to be sent to the PGW operator according to the TWAN operator's policy. When present, the IE is encoded as a Diameter or an ESM cause. The TWAN does not exchange signalling with the 3GPP AAA Server nor with the UE in this procedure. The TWAN may include an internal failure cause for the bearer creation failure. The protocol type used to encode the internal failure cause is implementation-specific.			
	Conditional	If the bearer creation failed, the ePDG includes this IE on the S2b interface to indicate the cause of the bearer creation failure, if available, and if this information is permitted to be sent to the PGW operator according to the ePDG operator's policy. When present, the IE is encoded as a Diameter or an IKEv2 cause.			

Create Indirect Data Forwarding Tunnel Request

The MME/S4-SGSN sends a Create Indirect Data Forwarding Tunnel Request message to the SGW on the S4/S11 interface as part of the handover procedures or TAU/RAU procedure with SGW change and data forwarding.

Table 17-14 lists the supported Create Indirect Data Forwarding Tunnel Request information elements.

Table 17-14 Create Indirect Data Forwarding Tunnel Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
IMSI	Conditional	The IMSI is included if the MME/SGSN selects an SGW for indirect data forwarding that is different from the SGW already in use for the UE as the anchor point.	IMSI	1	0
ME Identity (MEI)	Conditional	The MEI is included if the MME/SGSN selects an SGW for indirect data forwarding that is different from the SGW already in use for the UE as the anchor point and if one of the following conditions exist: ■ The UE is emergency attached and the UE is UICCless ■ The UE is emergency attached and the IMSI is not authenticated	MEI	75	0
Indication Flags	Conditional-Optiona	Included if the Unauthenticated IMSI flag is set to 1. This flag is set to 1 if the IMSI present in the message is not authenticated and is for an emergency attached UE.	Indication	77	0
Sender F-TEID for Control Plane	Conditional	Included if the MME/SGSN selects an SGW for indirect data forwarding that is different from the SGW already in use for the UE as the anchor point. Specifies the F-TEID of the SGW for the GTP-C tunnel.	F-TEID	87	0
Bearer Contexts	Mandatory	Included when needed to represent a list of bearers.	Bearer Context	93	0
Recovery	Conditional-Optiona	Included if the MME/SGSN is contacting the SGW for the first time.	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	VS

Bearer Contexts Created Within Create Indirect Data Forwarding Tunnel Request

Within the Create Indirect Data Forwarding Tunnel Request message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-15 lists the information elements contained in the Bearer Contexts IE.

Table 17-15 Bearer Contexts IEs (within Create Indirect Data Forwarding Tunnel Request)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
eNodeB F-TEID for DL data forwarding	Conditional	Target eNodeB F-TEID. Included in the message sent from the target MME to the SGW selected by the target MME for indirect data forwarding, or in the message sent from the source SGSN/MME to the SGW selected by the source MME for indirect data forwarding if the eNodeB F-TEID for DL data forwarding is included in the Forward Relocation Response message.	F-TEID	8	0
	Conditional/ Optional	Target eNodeB F-TEID. Included in the message sent from the target MME to the SGW selected by the target MME for indirect data forwarding of the DL data buffered in the old SGW during a TAU with SGW change procedure and data forwarding, without Control Plane CIoT EPS optimization.			

Table 17-15 Bearer Contexts IEs (within Create Indirect Data Forwarding Tunnel Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
SGW/UPF F-TEID for DL data forwarding	Conditional	Target SGW F-TEID Included in the message sent from the source MME/SGSN to the SGW selected by the source MME for indirect data forwarding if SGW F-TEID for DL data forwarding is included in the Forward Relocation Response message. This F-TEID is assigned by the SGW that the target MME/SGSN selects for indirect data forwarding.	F-TEID	8	1
	Conditional/ Optional	Target UPF F-TEID Included in the message sent from the source MME to the SGW selected by the source MME for indirect data forwarding if SGW/UPF F-TEID for DL data forwarding is included in the Forward Relocation Response message. This IE contains the target V-UPF F-TEID in home routed roaming scenario, or contains the PGW-U+UPF F-TEID in non-roaming or local breakout scenario			

Table 17-15 Bearer Contexts IEs (within Create Indirect Data Forwarding Tunnel Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
SGSN F-TEID for DL data forwarding	Conditional	Target SGSN F-TEID Included in the message sent from the target SGSN to the SGW selected by the target SGSN for indirect data forwarding in E-UTRAN to GERAN/UTRAN inter RAT handover with SGW relocation procedure, or in the message sent from the source MME to the SGW selected by the source MME for indirect data forwarding if the SGSN F-TEID for DL data forwarding is included in the Forwarding Relocation Response message.	F-TEID	8	2
	Conditional/ Optional	Included in the message sent from the source MME to the SGW selected by the source MME for indirect data forwarding if the SGSN Address for User Traffic and the Tunnel Endpoint Identifier Data II are included in the GTPv1 Forward Relocation Response message. Included when Direct Tunnel is not used, in the message sent from the target SGSN to the SGW selected by the target SGSN for indirect data forwarding of the DL data buffered in the old SGW during a RAU with SGW change procedure and data forwarding.			

Table 17-15 Bearer Contexts IEs (within Create Indirect Data Forwarding Tunnel Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
RNC F-TEID for DL data forwarding	Conditional	Target RNC F-TEID Included in the message sent from the target SGSN to the SGW selected by the target SGSN for indirect data forwarding in E-UTRAN to UTRAN inter RAT handover with SGW relocation procedure, or in the message sent from the source MME to the SGW selected by the source MME for indirect data forwarding if the RNC F-TEID for DL data forwarding is included in the Forwarding Relocation Response message.	F-TEID	8	3
	Conditional/ Optional	Included in the message sent from the source MME to the SGW selected by the source MME for indirect data forwarding if the RNC IP address and TEID are included in the RAB Setup Information and/or the Additional RAB Setup Information in the GTPv1 Forwarding Relocation Response message. Included when Direct Tunnel is used, in the message sent from the target SGSN to the SGW selected by the target SGSN for indirect data forwarding of the DL data buffered in the old SGW during a RAU with SGW change procedure and data forwarding.			

Table 17-15 Bearer Contexts IEs (within Create Indirect Data Forwarding Tunnel Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
eNodeB F-TEID for UL data forwarding	Optional	Target eNodeB F-TEID. Included in the message sent during the intra-EUTRAN HO from the target MME to the SGW selected by the target MME for indirect data forwarding, or in the message sent from the source MME to the SGW selected by the source MME for indirect data forwarding if the eNodeB F-TEID for data UL forwarding is included in the Forward Relocation Response message.	F-TEID	8	4
SGW F-TEID for UL data forwarding	Optional	Target SGW F-TEID Included in the message sent during the intra-EUTRAN HO from the source MME to the SGW selected by the source MME for indirect data forwarding if SGW F-TEID for UL data forwarding is included in the Forward Relocation Response message. This F-TEID is assigned by the SGW that the target MME selects for indirect data forwarding.	F-TEID	8	5
MME F-TEID for DL data forwarding	Conditional/ Optional	Target MME S11-U F-TEID Included in the message sent from the target MME to the SGW selected by the target MME for indirect data forwarding, during a TAU procedure with SGW change and data forwarding, with Control Plane CIoT EPS optimization.	F-TEID	8	6

Create Indirect Data Forwarding Tunnel Response

The SGW sends a Create Indirect Data Forwarding Tunnel Response message to the MME/SGSN on the S4/S11 interface as a response to a Create Indirect Data Forwarding Tunnel Request message.

Table 17-16 lists the supported Create Indirect Data Forwarding Tunnel Response information elements.

Table 17-16 Create Indirect Data Forwarding Tunnel Response Information Request

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
Sender F-TEID for Control Plane	Conditional	Included if the SGW receives a Sender F-TEID for Control Plane IE from an MME/SGSN in a Create Indirect Data Forwarding Tunnel Request message.	F-TEID	87	0
Bearer Contexts	Mandatory	Included when needed to represent a list of bearers.	Bearer Context	93	0
Recovery	Conditional-Optiona	Included if the SGWis contacting the MME/SGSN for the first time.	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	VS
Sender F-TEID for Control Plane	Conditional	Included if the MME/SGSN selects an SGW for indirect data forwarding that is different from the SGW already in use for the UE as the anchor point. Specifies the F-TEID of the SGW for the GTP-C tunnel.	F-TEID	87	0
Bearer Contexts	Mandatory	Included when needed to represent a list of bearers.	Bearer Context	93	0
Recovery	Conditional-Optiona	Included if the MME/SGSN is contacting the SGW for the first time.	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	VS

Bearer Contexts Within Create Indirect Data Forwarding Tunnel Response

Within the Create Indirect Data Forwarding Tunnel Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-17 lists the information elements contained in the Bearer Contexts IE.

Table 17-17 Bearer Contexts IEs (within Create Indirect Data Forwarding Tunnel Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the tunnel setup was successful. If not, this IE provides information on the reason.	Cause	2	0
S1-U SGW F-TEID for DL data forwarding	Conditional	Included in the response sent from the SGW selected by the source MME for indirect data forwarding to the source MME.	F-TEID	8	0
S4-U SGW F-TEID for DL data forwarding	Conditional	S4-U usage only. Included in the response sent from the SGW selected by the source SGSN for indirect data forwarding to the source SGSN.	F-TEID	8	2
SGW F-TEID for DL data forwarding	Conditional	Included in the response message sent from the SGW selected by the target MME/SGSN for indirect data forwarding to the target MME/SGSN.	F-TEID	8	3
S1-U SGW F-TEID for UL data forwarding	Optional	Included in the response sent during the intra-EUTRAN HO from the SGW selected by the source MME for indirect data forwarding to the source MME.	F-TEID	8	4
SGW F-TEID for UL data forwarding	Optional	Included in the response message sent during the intra-EUTRAN HO from the SGW selected by the target MME for indirect data forwarding to the target MME.	F-TEID	8	5

Create Session Request

The Create Session Request message is sent as part of the following procedures:

- Evolved UMTS Terrestrial Radio Access Network (E-UTRAN) Initial Attach
- UE-requested PDN Connectivity
- Packet Data Protocol (PDP) Context Activation

The Create Session Request message is sent:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-18 lists the supported Create Session Request information elements.

Table 17-18 Create Session Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
IMSI	Conditional	The International Mobile Subscriber Identity (IMSI) is included on the S5/S8 and S4/S11 interfaces if it is provided.	IMSI	1	0
MSISDN	Conditional	The Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) is included on the S11 interface if provided by the HSS, and is included on the S5/S8 interface if it is provided by the MME.	MSISDN	76	0
ME Identity (MEI)	Conditional/ Optional	The SGW includes the Mobile Equipment Identity (MEI) on the S5/S8 and S4/S11 interfaces if it is available.	MEI	75	0
User Location Information (ULI)	Conditional/ Optional	The SGW includes the ULI on the S5/S8 and S4/S11 interfaces if it is provided.	ULI	86	0
Serving Network	Conditional	Included on S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	Serving Network	83	0
RAT Type	Mandatory	None	RAT Type	82	0

Table 17-18 Create Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Indication Flags	Conditional	Included on the S5/S8 and S4/S11 interfaces if any of the applicable flags is set to 1: ■ S5/S8 Protocol Type ■ Dual Address Bearer Flag ■ Handover Indication ■ Operation Indication ■ Direct Tunnel Flag ■ Piggybacking Supported ■ Change Reporting Support Indication ■ Unauthenticated IMSI	Indication	77	0
Sender F-TEID for Control Plane	Mandatory	Specifies the F-TEID of the SGW for the GTP-C tunnel.	F-TEID	87	0
Access Point Name (APN)	Mandatory	None	APN	71	0
Selection Mode	Conditional	Included on S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures. Indicates whether the APN is subscribed or a non-subscribed.	Selection Mode	128	0
PDN Type	Conditional	Included on S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures. This IE is set to IPv4, IPv6, or IPv4v6.	PDN Type	99	0
PDN Address Allocation (PAA)	Conditional	Included on S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	PAA	79	0
Maximum APN Restriction	Conditional	Included on S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures. This IE denotes the most stringent restriction as required by any active bearer context.	APN Restriction	127	0

Table 17-18 Create Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Aggregate Maximum Bit Rate (APN-AMBR)	Conditional	Included on S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	AMBR	72	0
Linked EPS Bearer ID	Conditional	Included on S4/S11 interface in RAU/TAU/Handover except in the Gn/Gp SGSN to MME/S4-SGSN RAU/TAU/Handover procedures with SGW change to identify the default bearer of the PDN Connection.	EBI	73	0
Protocol Configuration Options (PCO)	Conditional	The MME/S4-SGSN includes this IE on the S4/S11 interface if it receives it from the UE during attach. The SGW includes this IE on the S5/S8 interface if it is available.	PCO	78	0
Bearer Contexts to be created	Mandatory	Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers on the S5/S8 and S4/S11 interfaces. One bearer is included for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures. One or more bearers are included for the TAU/RAU/Handover procedures with an SGW change.	Bearer Context	93	0
Trace Information	Conditional	Included on the S5/S8 and S4/S11 interface if a PGW or SGW trace is activated.	Trace Information	96	0
Recovery	Conditional	Included on the S5/S8 and S4/S11 interfaces if the peer node is being contacted for the first time.	Recovery	3	0
UE Time Zone	Conditional	The SGW includes this IE on the S5/S8 interface if it is available.	UE Time Zone	114	0
	Conditional/ Optional	The MME includes this IE on the S11 interface during an Initial Attach and during the UE Requested PDN Connectivity procedure. The S4-SGSN includes this IE on the S4 interface during the PDP Context Activation procedure.			

Table 17-18 Create Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
User CSG Information (UCI)	Conditional/ Optional	The SGW includes this IE on the S5/S8 interface if it receives the User CSG Information from the MME/SGSN. The MME/S4-SGSN includes this IE on the S4/S11 interface during an Initial Attach, the UE Requested PDN Connectivity procedure, or the PDP Context Activation procedure.	UCI	145	0
Charging Characteristics	Conditional	Included on the S5/S8 and S4/S11 interfaces.	Charging Characteristics	95	0
Private Extension	Optional	Includes vendor or operator specific information. ■ Origination-TimeStamp ■ Max-Wait-Time	Private Extension	255	Vendor Specific
Secondary RAT Usage Data Report	Conditional	If the PLMN has configured secondary RAT usage reporting and PDN GW Secondary RAT reporting is active, the MME includes this IE on the S11 interface if it has received Secondary RAT usage data from eNodeB in an X2-based handover with Serving GW relocation. The MME also sets the IRSGW flag to "0," to indicate that the Secondary RAT usage data is reported for the Source SGW, and sent via the Target SGW to the PGW. Several IEs with the same type and instance value may be included to represent multiple usage data reports.	Secondary RAT Usage Data Report	201	0

Table 17-18 Create Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
UP Function Selection Indication Flags	Conditional/ Optional	Based on operator policy, the MME/S4-SGSN includes this IE on the S4/S11 interface, if any of the applicable flags is set to 1. Applicable flags are: ■ DCNR: This flag is set to 1 if it is desired to select a specific SGW-U and PGW-U for UEs supporting DCNR and not restricted to use NR by user subscription (e.g., due to requirements of higher bitrates). *Note:* This information is used for the SGW-U, PGW-U or combined SGW-U/PGW-U selection (see Annex B.2 of 3GPP TS 29.244 [80]). The SGW includes this IE on S5/S8 if it receives it from the MME/S4-SGSN.	UP Function Selection Indication Flags	202	0

Bearer Contexts to be Created Within Create Session Request

Within the Create Session Request message, the Bearer Contexts to be Created IE contains other IEs and is considered a Grouped IE.

Table 17-19 lists the information elements contained in the Bearer Contexts to be Created IE.

Table 17-19 Bearer Contexts IEs (within Create Session Request)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
TFT	Optional	May be included on SA4/S11 interfaces.	Bearer TFT	84	0
S1-U eNodeB F-TEID	Conditional	Included on the S11 interface for X2-based handover with SGW relocation.	F-TEID	87	0

Table 17-19 Bearer Contexts IEs (within Create Session Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
S4-U SGSN F-TEID	Conditional	Included on the S4 interface if the S4-U interface is used.	F-TEID	87	1
S5/S8-U SGW F-TEID	Conditional	Included on S5/S8 interface for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	F-TEID	87	2
S5/S8-U PGW F-TEID	Conditional	Included on the S4 and S11 interfaces for the TAU/RAU/Handover cases when the GTP-based S5/S8 is used.	F-TEID	87	3
	Conditional/ Optional	For PMIP-based S5/S8, this IE shall be included on the S11/S4 interface for the TAU/RAU/Handover cases if the PGW provided an alternate address for user plane, that is, an IP address for user plane that is different from the IP address for control plane. When present, this IE shall contain the alternate IP address for user plane and the uplink GRE key.			
S12 RNC F-TEID	Conditional/ Optional	Included on the S4 interface if the S12 interface is used in the Enhanced serving RNS relocation with SGW relocation procedure.	F-TEID	87	4
Bearer Level QoS	Mandatory	None	Bearer QoS	80	0
S11-U MME F-TEID	Conditional/ Optional	Included on the S11 interface, if S11-U is being used, during the E-UTRAN Initial Attach and UE requested PDN connectivity procedures. Also included on the S11 interface, if S11-U is being used, during a Tracking Area Update procedure with Serving GW change if the MME needs to establish the S11-U tunnel.	F-TEID	87	7

Create Session Response

The Create Session Response message is sent as an answer to a Create Session Request message. The Create Session Response message contains information about the default bearer, such as the UE's allocated IP address and the tunnel endpoint IP addresses (TEID). The message is sent:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-20 lists the supported Create Session Response information elements.

Table 17-20 Create Session Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0
Change Reporting Action	Conditional	Included on the S5/S8 interface if it is provided by the MME/SGSN. Included on the S4/S11 interface if the location Change Reporting mechanism is to be started or stopped for the subscriber.	Change Reporting Action	131	0
CSG Information Reporting Action	Conditional/ Optional	Included on the S5/S8 and S4/S11 interfaces with the appropriate Action field if the CSG information reporting mechanism is started or stopped for the subscriber in the SGSN/MME.	CSG Information Reporting Action	146	0
Sender F-TEID for Control Plane	Conditional	Included on the S11/S4 interface.	F-TEID	87	0
PGW S5/S8 F-TEID for PMIP based interface or GTP based Control Plane interface	Conditional	Included on the S5/S8 interface for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures. Included on the S4/S11 interface if received by the SGW.	F-TEID	87	1
PDN Address Allocation (PAA)	Conditional	Included on the S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	PAA	79	0

Table 17-20 Create Session Response Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
APN Restriction	Conditional	Included on the S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	APN Restriction	127	0
Aggregate Maximum Bit Rate (APN-AMBR)	Conditional	Included on the S5/S8 and S4/S11 interfaces if the Policy and Charging Rules Function (PCRF) has modified the APN-AMBR.	AMBR	72	0
Protocol Configuration Options (PCO)	Conditional	Included on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0
Bearer Contexts created	Mandatory	Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers. One bearer is included on the S5/S8 and S4/S11 interface for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	Bearer Context	93	0
Recovery	Conditional	Included on the S5/S8 and S4/S11 interfaces if contacting the peer for the first time.	Recovery	3	0
Charging Gateway Name	Conditional	The PGW includes this IE on the S5/S8 interface when the Charging Gateway Function (CGF) address is configured.	FQDN	136	0
Charging Gateway Address	Conditional	The PGW includes this IE on the S5/S8 interface when the CGF address is configured.	IP Address	74	0
SGW-FQ-CSID	Conditional	Included by the SGW on the S11 interface according to the requirements in 3GPP TS 23.007.	FQ-CSID	132	1
Private Extension	Optional	Includes vendor or operator specific information. ■ Charging-Group-ID	Private Extension	255	Vendor Specific

Table 17-20 Create Session Response Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
PGW Back-Off Time	Optional	This IE may be included on the S5/S8 and S4/S11 interfaces when the PDN GW rejects the Create Session Request with the cause “APN congestion.” It indicates the time during which the MME or S4-SGSN should refrain from sending subsequent PDN connection establishment requests to the PGW for the congested APN for services other than Service Users/emergency services.	EPC Timer	156	0

Bearer Contexts Created Within Create Session Response

Within the Create Session Response message, the Bearer Contexts Created IE contains other IEs and is considered a Grouped IE.

Table 17-21 lists the information elements contained in the Bearer Contexts Created IE.

Table 17-21 Bearer Contexts Created IEs (within Create Session Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling was successful. If not, this IE provides information on the reason.	Cause	2	0
S1-U SGW F-TEID	Conditional	Included on the S11 interface if the S1-U interface is used, that is, if the S11-U Tunnel flag was not set in the Create Session Request.	F-TEID	87	0
S4-U SGW F-TEID	Conditional	Included on the S4 interface if the S4-U interface is used.	F-TEID	87	1

Table 17-21 Bearer Contexts Created IEs (within Create Session Response) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
S5/S8-U PGW F-TEID	Conditional	Included on S5/S8 and S4/S11 interfaces for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	F-TEID	87	2
Bearer Level QoS	Conditional	Included on the S5/S8 and S4/S11 interfaces if the received quality of service (QoS) parameters are modified.	Bearer QoS	80	0
Charging Id	Conditional	Included on the S5/S8 interface for the E-UTRAN initial attach, PDP Context Activation, and UE-Requested PDN Connectivity procedures.	Charging Id	94	0
	Optional	If the S5/S8 interface is GTP, IE is included on the S4 interface to support CAMEL charging at the SGSN, the E-UTRAN initial attach, the PDP Context Activation, and the UE-Requested PDN Connectivity procedures.			
S11-U SGW F-TEID	Conditional	Included on the S11 interface if the S11-U interface is used, that is, if the S11-U Tunnel flag was set in the Create Session Request. If the SGW supports both IP address types, the SGW shall send both IP addresses within the F-TEID IE. If only one IP address is included, then the MME shall assume that the SGW does not support the other IP address type.	F-TEID	87	6

Delete Bearer Command

The Delete Bearer Command message is sent:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-22 lists the Delete Bearer Command information elements.

Table 17-22 Delete Bearer Command Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Bearer Contexts	Mandatory	Used for dedicated bearers. At least one dedicated bearer is present. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
ULI Timestamp	Conditional	This IE is included on the S4/S11 interface if the ULI IE is present. It indicates the time when the User Location Information was acquired. The SGW includes this IE on S5/S8 if the SGW receives it from the MME/SGSN.	ULI Timestamp	170	0
Secondary RAT Usage Data Report	Conditional/ Optional	If the PLMN has configured secondary RAT usage reporting, the MME includes this IE on the S11 interface if it has received Secondary RAT usage data in an MME Initiated Dedicated Bearer Deactivation procedure. Several IEs with the same type and instance value may be included to represent multiple usage data reports.	Secondary RAT Usage Data Report	201	0
	Conditional	The SGW forwards this IE on the S5/S8 interface if it receives the Secondary RAT Usage Data Report with the IRPGW flag set to "1" from MME. Several IEs with the same type and instance value may be included, to represent multiple usage data reports.			

Bearer Contexts Within Delete Bearer Command

Within the Delete Bearer Command message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-23 lists the information elements contained in the Bearer Contexts IE.

Table 17-23 Bearer Contexts IEs (within Delete Bearer Command)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Bearer Flags	Conditional/ Optional	Applicable flags are: ■ VB (Voice Bearer) indicator – Indicates a voice bearer for PS-to-CS (v)SRVCC handover. ■ Vind (vSRVCC indicator) indicator – Indicates a video bearer for PS-to-CS vSRVCC handover.	Bearer Flags	97	0
RAN/NAS Release Cause		The MME includes this IE on the S11 interface to indicate the RAN release cause and/or NAS release cause to release the bearer, if available, and this information is permitted to be sent to the PGW operator according to MME operator's policy. If both a RAN release cause and a NAS release cause are generated, then several IEs with the same type and instance value are included to represent a list of causes. The SGW includes this IE on the S5/S8 interface if it receives it from the MME.	RAN/NAS Cause	172	0

Delete Bearer Failure Indication

The Delete Bearer Failure Indication message is sent:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-24 lists the Delete Bearer Failure Indication information elements.

Table 17-24 Delete Bearer Failure Indication Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
Bearer Contexts	Mandatory	Contains a list of failed bearers.	Bearer Context	93	0
Recovery	Conditional	Included on S5/S8 and S4/S11 if contacting the peer for the first time.	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Delete Bearer Failure Indication

Within the Delete Bearer Failure Indication message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-25 lists the information elements contained in the Bearer Contexts IE.

Table 17-25 Bearer Contexts IEs (within Delete Bearer Failure Indication)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0

Delete Bearer Request

The Delete Bearer Request message is sent:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-26 lists the Delete Bearer Request information elements.

Table 17-26 Delete Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Linked EPS Bearer ID (LBI)	Conditional	Included on the S5/S8 and S4/S11 interfaces to indicate the default bearer associated with the PDN being disconnected. This IE is present only when the EPS Bearer is not present.	EBI	73	0
EPS Bearer ID	Conditional	Included on the S5/S8 and S4/S11 interfaces to indicate the dedicated bearer to be deleted. Several IEs with this type and instance values are included, if necessary, to represent a list of bearers.	EBI	73	1
Failed Bearer Contexts	Optional	Included on the S5/S8 and S4/S11 interfaces if the request corresponds to an MME-initiated bearer deactivation procedure. This IE contains the list of failed bearers if the partial Bearer Contexts included in the Delete Bearer Command message could not be deleted.	Bearer Context	93	0
Procedure Transaction Id (PTI)	Conditional	Included on the S5/S8 and S4/S11 interfaces if the request corresponds to a UE-requested Bearer Resource Modification procedure for E-UTRAN.	PTI	100	0
Protocol Configuration Options (PCO)	Conditional	The PGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0

Table 17-26 Delete Bearer Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Conditional	The PGW includes this IE on the S5/S8 and S4/S11 interfaces if the message is caused by a handover without optimization from 3GPP to non-3GPP.	Cause	2	0
	Optional	This IE may be sent by a PGW on the S5/S8 interface during a PGW initiated bearer deactivation procedure for the default bearer with a value of “Reactivation Requested”. ***Note:*** *To set the configuration to send "Reactivation Requested", see Configuring the Cause Code for a Network Initiated Bearer Deactivation (page 17-34).*			
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Delete Bearer Request

Within the Delete Bearer Request message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-27 lists the information elements contained in the Bearer Contexts IE.

Table 17-27 Bearer Contexts IEs (within Delete Bearer Request)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0

Table 17-27 Bearer Contexts IEs (within Delete Bearer Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
Protocol Configuration Options (PCO)	Conditional/ Optional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available. This bearer-level PCO IE takes precedence over the message-level PCO IE if they are both present.	PCO	78	0

Delete Bearer Response

As an answer to a Delete Bearer Request message, the Delete Bearer Response message is sent:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-28 lists the Delete Bearer Response information elements.

Table 17-28 Delete Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instanc e Value
Cause	Mandatory	None	Cause	2	0
Linked EPS Bearer ID (LBI)	Conditional	Included on the S5/S8 and S4/S11 interfaces to indicate the default bearer associated with the PDN being disconnected.	EBI	73	0
Bearer Contexts	Conditional	Used for dedicated bearers. At least one dedicated bearer is present. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0
Recovery	Conditional	Included on S5/S8 and S4/S11 interfaces if the SGW is contacting the PGW for the first time.	Recovery	3	0

Table 17-28 Delete Bearer Response Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Protocol Configuration Options (PCO)	Conditional	Included on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0
UE Time Zone	Conditional/ Optional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available.	UE Time Zone	114	0
User Location Information (ULI)	Conditional/ Optional	Included on the S5/S8 and S4/S11 interfaces if it is available.	ULI	86	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
ULI Timestamp		This IE is included on the S4/S11 interface if the ULI IE is present. It indicates the time when the User Location Information was acquired. The SGW includes this IE on S5/S8 if the SGW receives it from the MME/SGSN. If ISR is active, after receiving both the Delete Bearer Response messages from the MME and the SGSN, the SGW includes the most recent time and the related User Location Information in the Delete Bearer Response message on S5/S8 interface.	ULI Timestamp	170	
Secondary RAT Usage Data Report	Conditional	If the PLMN has configured secondary RAT usage reporting, the MME includes this IE on the S11 interface if it has received Secondary RAT usage data from eNodeB in a PDN GW initiated bearer deactivation procedure. Several IEs with the same type and instance value may be included to represent multiple usage data reports.	Secondary RAT Usage Data Report	201	0
	Optional	The SGW forwards this IE on the S5/S8 interface if it receives the Secondary RAT Usage Data Report with the IRPGW flag set to "1" from MME. Several IEs with the same type and instance value may be included to represent multiple usage data reports.			

Bearer Contexts Within Delete Bearer Response

Within the Delete Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-29 lists the information elements contained in the Bearer Contexts IE.

Table 17-29 Bearer Contexts IEs (within Delete Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0
Protocol Configuration Options (PCO)	Conditional	The SGW includes this IE on the S5/S8 interface if it is available. This bearer-level PCO IE takes precedence over the message-level PCO IE if they are both present.	PCO	78	0
RAN/NAS Cause	Conditional	The MME includes this IE on the S11 interface to indicate the RAN release cause and/or NAS release cause to release the bearer, if this information is available and is permitted to be sent to the PGW operator according to the MME operator's policy. If both a RAN release cause and a NAS release cause are generated, then several IEs with the same type and instance value is included to represent a list of causes. The SGW includes this IE on the S5/S8 interface if it receives it from the MME.	RAN/NAS Cause	172	0

Delete Indirect Data Forwarding Tunnel Request

The SGSN/MME sends a Delete Indirect Data Forwarding Tunnel Request message to the SGW on the S4/S11 interface to delete the Indirect Forwarding Tunnels in the Source SGW/Target SGW as part of the following procedures:

- S1-based handover
- UTRAN Iu mode to E-UTRAN Inter RAT handover
- GERAN A/Gb mode to E-UTRAN Inter RAT handover
- E-UTRAN to UTRAN Iu mode Inter RAT handover
- E-UTRAN to GERAN A/Gb mode Inter RAT handover
- MME to 3G SGSN combined hard handover and SRNS relocation procedure
- 3G SGSN to MME combined hard handover and SRNS relocation procedure
- Inter RAT handover Cancel
- S1-based handover Cancel
- Optimised Active Handover: E-UTRAN Access to CDMA2000 HRPD Access
- EPS to 5GS handover using N26 interface
- 5GS to EPS handover using N26 interface
- N26 based Handover cancel

Table 17-30 lists the Indirect Data Forwarding Tunnel Request information elements.

Table 17-30 Delete Indirect Data Forwarding Tunnel Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Delete Indirect Data Forwarding Tunnel Response

The SGW sends a Delete Indirect Data Forwarding Tunnel Response message to the SGSN/MME on the S4/S11 interface as a response to a Delete Indirect Data Forwarding Tunnel Request.

Table 17-31 lists the Indirect Data Forwarding Tunnel Response information elements.

Table 17-31 Delete Indirect Data Forwarding Tunnel Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates if the deletion of the indirect tunnel is successful, and if not, gives the reason.	Cause	2	0
Recovery	Conditional	Included if the SGW is contacting the SGSN/MME for the first time.	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Delete Session Request

To request session termination, the Delete Session Request message is sent:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-32 lists the Delete Session Request information elements.

Table 17-32 Delete Session Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Conditional	If Idle State Signaling Reduction (ISR) is deactivated, the Cause value is “ISR deactivation”.	Cause	2	0
Linked EPS Bearer ID	Conditional	Included on the S5/S8 and S4/S11 interfaces to indicate the default bearer associated with the PDN being disconnected except during the TAU/RAU/Handover procedure with an SGW relocation.	EBI	73	0
User Location Information (ULI)	Conditional/ Optional	Included on the S5/S8 and S4/S11 interfaces if it is provided by the MME/SGSN.	ULI	86	0

Table 17-32 Delete Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Indication Flags	Conditional	Included on the S5/S8 and S4/S11 interfaces if any of the applicable flags is set to 1: ■ Operation Indication ■ Scope Indication ***Note:*** *The MCC PGW processes the Delete Session Request message even when the Indication Flag IE is set to 0.*	Indication	77	0
Protocol Configuration Options (PCO)	Conditional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0
Originating Node	Conditional	Included on the S4/S11 interface if the ISR is active in MME/SGSN to denote the type of the node originating the message. The SGW releases the corresponding Originating Node related EPS Bearer contexts information in the PDN Connection identified by the LBI.	Node Type	135	0
Sender F-TEID for Control Plane	Optional	If the PGW receives the Sender F-TEID for Control Plane, the PGW only accepts the Delete Session Request message when the Sender F-TEID for Control Plane in the received message is the same as the Sender F-TEID for Control Plane that was last received in either the Create Session Request message or the Modify Bearer Request message on the given interface.	F-TEID	87	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
RAN/NAS Release Cause	Conditional	The MME includes this IE on the S11 interface to indicate the NAS release cause to release the PDN connection, if available, and this information is permitted to be sent to the PGW operator according to the MME operator's policy. The SGW includes this IE on the S5/S8 interface if it receives it from the MME and if the Operation Indication bit received from the MME is set to 1.	RAN/NAS Cause	172	0

Table 17-32 Delete Session Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
ULI Timestamp	Conditional	This IE is included on the S4/S11 interface if the ULI IE is present. It indicates the time when the User Location Information was acquired. The SGW includes this IE on S5/S8 if the SGW receives it from the MME/SGSN. If ISR is active, after receiving both the Delete Session Request messages from the MME and the SGSN, the SGW includes the most recent ULI timestamp and the related User Location Information in the Delete Session Request message on S5/S8 interface.	ULI Timestamp	170	0
Secondary RAT Usage Data Report	Conditional	If the PLMN has configured secondary RAT usage reporting, the MME includes this IE on the S11 interface if it has received Secondary RAT usage data from eNodeB in a UE-initiated Detach procedure for E-UTRAN, MME-initiated Detach, HSS-initiated Detach, or UE or MME requested PDN disconnection. The MME also includes this IE on the S11 interface to the Source SGW if it has received a Secondary RAT Usage Data Report from the eNB in an S1/X2-based handover with Serving GW relocation, or MME triggered Serving GW relocation procedure, or an E-UTRAN to GERAN A/Gb mode Inter RAT handover, E-UTRAN to UTRAN Iu mode Inter RAT handover, MME to 3G SGSN combined hard handover and SRNS relocation, or Routing Area Update procedures. In this case, the IRPGW flag is set to “0”. Several IEs with the same type and instance value may be included to represent multiple usage data reports.	Secondary RAT Usage Data Report	201	0
	Optional	The SGW forwards this IE on the S5/S8 interface if it receives the Secondary RAT Usage Data Report with the IRPGW flag set to “1” from MME. Several IEs with the same type and instance value may be included to represent multiple usage data reports.			

Delete Session Response

As an answer to a Delete Session Request message, the Delete Session Response message is sent:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-33 lists the Delete Session Response information elements.

Table 17-33 Delete Session Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
Recovery	Conditional	Included on S5/S8 and S4/S11 interfaces if the SGW is contacting the PGW for the first time.	Recovery	3	0
Protocol Configuration Options (PCO)	Conditional	The PGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
PGW's node level Load Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports the load control feature.	Load Control Information	181	0
PGW's APN level Load Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports APN level load control feature.	Load Control Information	181	1
PGW's Overload Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports the overload control feature.	Overload Control Information	180	0

Downlink Data Notification

The SGW sends a Downlink Data Notification message to the MME/S4-4SGSN in the following situations:

- On the S11 interface to the MME as a part of the network triggered service request procedure
- On the S4 interface to the S4-SGSN as part of Paging with no established user plane on S4
- On the S4 interface to the S4-SGSN to re-establish all the previous released bearer(s) for a UE, upon receipt of downlink data for a UE in connected mode but without a corresponding downlink bearer available

Note: *For example, this may occur if the S4-SGSN releases some but not all the bearers of the UE as specified in subclause 12.7.2.2 of 3GPP TS 23.060.*

- On the S11/S4 interface to MME/S4-SGSN if the SGW has received an Error Indication from eNodeB/RNC/MME across S1-U/S12/S11-U interface
- On the S11/S4 interface to the MME/S4-SGSN as part of the network triggered service restoration procedure if both the SGW and the MME/S4-SGSN support this optional feature
- On the S11 interface to the MME as a part of the Mobile Terminated Data Transport in Control Plane CIoT EPS optimization with P-GW connectivity.
- A Downlink Data Notification message may be sent on the S4 interface to the S4-SGSN if the SGW has received an Error Indication from S4-SGSN across the S4-U interface

Table 17-34 lists the Downlink Data Notification information elements.

Table 17-34 Downlink Data Notification Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Conditional/ Optional	If the SGW receives an Error Indication from eNodeB/RNC/S4-SGSN/MME, the Cause IE value "Error Indication received from RNC/eNodeB/S4-SGSN/MME" is included.	Cause	2	0

Table 17-34 Downlink Data Notification Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
EPS Bearer ID	Conditional/ Optional	If the Downlink Data Notification is triggered by the arrival of downlink data packets at the SGW, the EPS Bearer ID stored in the EPS bearer context of the bearer on which the downlink data packet was received is used. If the Downlink Data Notification is triggered by the receipt of an Error Indication from the eNodeB, RNC or S4-SGSN, the EPS Bearer ID stored in the EPS bearer context of the bearer for which the Error Indication was received is used. If the ISR is active and the Downlink Data Notification is triggered by the arrival of control plane signaling, the EPS Bearer ID present in the control plane signaling or derived from the control plane signaling (for PMIP based S5/S8) is used. If the Downlink Data Notification is triggered by a Create Bearer Request message, the EPS Bearer ID of the corresponding PDN connection's default bearer is used. If both the SGW and the MME/S4-SGSN support the network triggered service restoration procedure and the Downlink Data Notification is triggered by the arrival of control plane signaling, the EPS Bearer ID present in the control plane signaling or derived from the control plane signaling (for PMIP based S5/S8) is used. More than one IE with this type and instance value may be included to represent multiple bearers having received downlink data packets or being signaled in the received control plane message.	EBI	73	0

Table 17-34 Downlink Data Notification Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Allocation/ Retention Priority	Conditional/ Optional	If the Downlink Data Notification is triggered by the arrival of downlink data packets at the SGW, the EPS bearer context of the bearer on which the downlink data packet was received is used. If the Downlink Data Notification is triggered by the receipt of an Error Indication from the eNodeB, RNC or S4-SGSN, the ARP stored in the EPS bearer context of the bearer for which the Error Indication was received. is used. If the ISR is active and the Downlink Data Notification is triggered by the arrival of control plane signaling, the the ARP present in the control plane signaling is used. If the ARP is not present, the ARP in the stored EPS bearer context.in the control plane signaling is used. If both the SGW and the MME/S4-SGSN support the network triggered service restoration procedure and the Downlink Data Notification is triggered by the arrival of control plane signaling, the ARP in the control plane signaling is used. If the ARP is not present, the ARP from the stored EPS bearer context is used. If multiple EPS Bearers IDs are reported in the message, the ARP associated with the bearer with the highest priority (the lowest ARP Priority Level value) is used.	ARP	155	0
IMSI	Conditional/ Optional	SGW always includes the IMSI IE if the MME provided the IMSI in the Create Session Request.	IMSI	1	0

Load Control Information within Downlink Data Notification

Table 17-35 lists the information elements contained in the Load Control Information.

Table 17-35 Load Control Information (within Downlink Data Notification)

Octet 1	Load Control Information IE Type = 181 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
Load Control Sequence Number	Mandatory	Contains the sequence number generated for the IE instance.	Sequence Number	183	0
Load Metric	Mandatory	Indicates the current load level of the originating node.	Metric	182	0

Overload Control Information within Downlink Data Notification

Table 17-36 lists the information elements contained in the Overload Control Information message.

Table 17-36 Overload Control Information (within Downlink Data Notification)

Octet 1	Load Control Information IE Type = 181 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
Overload Control Sequence Number	Mandatory	Contains the sequence number generated for the IE instance.	Sequence Number	183	0
Overload Reduction Metric	Mandatory	Indicates the percentage of traffic reduction the sender of the overload control information requests the receiver to apply.	Metric	182	0
Period of Validity	Mandatory	Indicates the length of time during which the overload condition specified by the Overload Control Information IE is to be considered as valid (unless overridden by subsequent new overload control information).	EPC Timer	156	

Downlink Data Notification Acknowledge

The MME/S4-SGSN sends a Downlink Data Notification Acknowledge message to the SGW on the S4/S11 interface in response to a Downlink Data Notification message.

Table 17-37 lists the Suspend Acknowledge information elements.

Table 17-37 Downlink Data Notification Acknowledge Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates whether the Downlink Data Notification was successful. If not, this IE gives the reason for the failure.	Cause	2	0
Data Notification Delay	Conditional	Indicates the delay that the SGW should apply between receiving downlink data and sending Downlink Data Notification for all UEs served by that MME/SGSN if the rate of Downlink Data Notification event occurrence in the MME/SGSN becomes significant (as configured by the operator) and the MME/SGSN's load exceeds an operator configured value.	Delay Value	92	0
Recovery	Conditional	Included if the MME/S4-SGSN is contacting the SGW for the first time.	Recovery	3	0
DL low priority traffic Throttling	Optional	Requests that the SGW reduce the number of Downlink Data Notification requests it sends for downlink low priority traffic for UEs in idle mode served by that MME/SGSN. The reduction is in proportion to the Throttling Factor and is during the Throttling Delay.	Throttling	154	0
IMSI	Conditional/ Optional	The IMSI is included as part of the network triggered service restoration procedure if both the SGW and MME/S4-SGSN support the procedure.	IMSI	1	0

Table 17-37 Downlink Data Notification Acknowledge Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
DL Buffering Duration	Conditional/ Optional	Indicates the duration during which the SGW shall buffer DL data for this UE without sending any further Downlink Data Notification message, if extended buffering in the SGW is required: ■ For a UE in a power saving state that cannot currently be reached by paging ■ For a UE using NB-IoT, WB-EUTRAN or GERAN Extended Coverage with increased NAS transmission delay. If this IE is included, the Cause IE shall be set to "Request Accepted".	EPC Timer	156	0
DL Buffering Suggested Packet Count	Optional	If the DL Buffering Duration IE is included, this IE suggests the maximum number of downlink data packets to be buffered in the SGW for the UE.	Integer Number	187	0
Private Extension	Optional	The MME/S4-SGSN includes this IE if it is available.	Private Extension	255	VS

Downlink Data Notification Failure Indication

The MME/S4-SGSN sends a Downlink Data Notification Failure message to the SGW on the S4/S11 interface to indicate:

- The UE did not respond to paging.
- The UE responded to the page with a Service Request but the MME/SGSN rejected the request (for example, because the requested service was not supported or there was a bearer context mismatch).

Table 17-38 lists the Downlink Data Notification Failure Indication information elements.

Table 17-38 Downlink Data Notification Failure Indication Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0

Table 17-38 Downlink Data Notification Failure Indication Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Originating Node	Conditional/ Optional	Included if the ISR associated GTP entities (MME, S4-SGSN) send the failure indication message to the SGW during the Network Triggered Service Request procedure to denote the type of the node originating the message.	Node Type	135	0
IMSI	Conditional/ Optional	The IMSI is included as part of the network triggered service restoration procedure if both the SGW and MME/S4-SGSN support the procedure.	IMSI	1	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Modify Bearer Command

The Modify Bearer Command requests a modification to the subscribed QoS according to QoS parameters stored in the Home Subscriber Server (HSS), and is sent:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-39 lists the Modify Bearer Command information elements.

Table 17-39 Modify Bearer Command Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Aggregate Maximum Bit Rate (APN-AMBR)	Mandatory	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if received by the MME/SGSN from the HSS.	AMBR	72	0
Bearer Contexts	Mandatory	Used for dedicated bearers. Only one IE with this type and instance value is included.	Bearer Context	93	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Context Within Modify Bearer Command

Within the Modify Bearer Command message, the Bearer Context IE contains other IEs and is considered a Grouped IE.

Table 17-40 lists the information elements contained in the Bearer Context IE.

Table 17-40 Bearer Context IEs (within Modify Bearer Command)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	Contains the default bearer ID.	EBI	73	0
Bearer Level QoS	Conditional	Included if parameters other than the APN-AMBR are changed.	Bearer QoS	80	0

Modify Bearer Failure Indication

The Modify Bearer Failure Indication message indicates a failure of the HSS-Initiated Subscribed QoS Modification procedure, and is sent:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-41 lists the Modify Bearer Failure Indication information elements.

Table 17-41 Modify Bearer Failure Indication Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates that an EPS bearer has not been updated in the PGW.	Cause	2	0
Recovery	Conditional	Included on S5/S8 and S4/S11 interfaces if contacting the peer for the first time.	Recovery	3	0
PGW's Overload Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports the overload control feature.	Overload Control Information	180	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Modify Bearer Request

The Modify Bearer Request message requests a modification of bearer services, and is sent:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-42 lists the Modify Bearer Request information elements.

Table 17-42 Modify Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
ME Identity (MEI)	Conditional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces for the Gn/Gp SGSN to MME TAU procedure.	MEI	75	0
User Location Information (ULI)	Conditional/ Optional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is provided by the MME/SGSN.	ULI	86	0
Serving Network	Conditional/ Optional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is provided by the MME/SGSN during a TAU/RAU/Handover procedure with an associated MME/SGSN change and SGW change.	Serving Network	83	0
RAT Type	Conditional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces for a RAT type change.	RAT Type	82	0
Indication Flags	Conditional	Included on the S5/S8 and S4/S11 interfaces if any of the applicable flags is set to 1: ■ ISRAI ■ Handover Indication ■ Direct Tunnel Flag ■ Change Reporting Support Indication ■ Change F-TEID Support Indication ■ CS to PS SRVCC Indication	Indication	77	0
Sender F-TEID for Control Plane	Conditional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is provided by the MME/SGSN during a TAU/RAU/Handover procedure with an SGW change.	F-TEID	87	0

Table 17-42 Modify Bearer Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Aggregate Maximum Bit Rate (APN-AMBR)	Conditional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is provided by the MME/SGSN.	AMBR	72	0
Delay Downlink Packet Notification Request	Conditional	Included on the S11 interface for a UE triggered Service Request and UE initiated Connection Resume procedures. This IE contains the delay the SGW will apply between receiving downlink data and sending Downlink Data Notification for all UEs served by that MME.	Delay Value	92	0
	Conditional/ Optional	Included on the S4 interface for a UE triggered Service Request. This IE contains the delay the SGW will apply between receiving downlink data and sending Downlink Data Notification for all UEs served by that SGSN.			
Bearer Contexts to be modified	Conditional	This IE is not sent on the S5/S8 and S4/S11 interfaces for a UE-triggered Service request. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers to be modified. During a TAU/RAU/Handover procedure with an SGW change, the SGW sends the PGW all bearers it receives from the MME/SGSN (Bearer Contexts to be created, to be modified and to be removed) into the list of “Bearer Contexts to be Modified” IEs.	Bearer Context	93	0
Bearer Contexts to be removed	Conditional	Included on the S4/S11 interface.	Bearer Context	93	0
Recovery	Conditional	Included on S5/S8 and S4/S11 interfaces if contacting the PGW for the first time.	Recovery	3	0
UE Time Zone	Conditional	Included on the S5/S8 and S4/S11 interfaces if it is available.	UE Time Zone	114	0

Table 17-42 Modify Bearer Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
User CSG Information (UCI)	Conditional/ Optional	The MME/S4-SGSN includes this IE on the S4/S11 interface for Handover procedures, UE initiated Connection Resume and UE-initiated Service Request procedure if the PGW/PCRF has requested CSG Info reporting and the MME/SGSN supports the CSG information reporting and the User CSG information has changed. In TAU/RAU procedure without SGW change, this IE is included on the S4/S11 interface if the PGW/PCRF has requested CSG info reporting, the MME/SGSN supports CSG info reporting, and the User CSG information has changed. The SGW includes this IE on the S5/S8 interface if it receives the User CSG Information from the MME/SGSN.	UCI	145	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific
Secondary RAT Usage Data Report	Conditional	If the PLMN has configured secondary RAT usage reporting, the MME includes this IE on the S11 interface if it has received Secondary RAT usage data from eNodeB in an X2-based handover without Serving GW relocation, S1-based handover without MME or SGW relocation, or E-UTRAN initiated E-RAB modification procedure. The MME also includes this IE on the S11 interface if it has received a Secondary RAT Usage Data Report from the source MME in an S1-based handover with MME relocation procedure. For S1-based handover with SGW relocation, the MME also sets the IRSGW flag to “0”, to indicate that the Secondary RAT usage data is reported for the Source SGW, and sent via the Target SGW to the PGW. Several IEs with the same type and instance value may be included to represent multiple usage data reports.	Secondary RAT Usage Data Report	201	0

Bearer Contexts to be Modified Within Modify Bearer Request

Within the Modify Bearer Request message, the Bearer Contexts to be Modified contains other IEs and is considered a Grouped IE.

Table 17-43 lists the information elements contained in the Bearer Contexts to be Modified IE. For more information, see 3GPP TS 29.274.

Table 17-43 Bearer Contexts IEs (within Modify Bearer Request)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
S1 eNodeB F-TEID	Conditional	Included on the S11 interface if the S1-U is being used: ■ For an E-UTRAN initial attach ■ For a Handover from Trusted or Untrusted Non-3GPP IP Access to E-UTRAN ■ For an UE triggered Service Request; ■ For an UE initiated Connection Resume procedure ■ In all S1-U GTP-U tunnel setup procedures during a TAU procedure /handover ■ In all procedures where the UE is already in ECM-CONNECTED state, for example an E-UTRAN Initiated E-RAB modification procedure ■ In the Establishment of S1-U bearer during Data Transport in Control Plane CIoT EPS optimization procedure	F-TEID	87	0
S5/S8-U SGW F-TEID	Conditional	The SGW includes this IE on the S5/S8 interface if it is provided by the MME/SGSN during a TAU/RAU/Handover procedure with an SGW change.	F-TEID	87	1

Table 17-43 Bearer Contexts IEs (within Modify Bearer Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
S12 RNC F-TEID	Conditional	If available, included on the S4 interface if S12 interface is being used. If an S4-SGSN is aware that the RNC supports both IP address types, the S4-SGSN sends both IP addresses within an F-TEID IE. If only one IP address is included, then the SGW assumes that the RNC does not support the other IP address type.	F-TEID	87	2
S4-U SGSN F-TEID	Conditional	If available, included on the S4 interface if S4-U is being used. If an S4-SGSN supports both IP address types, the S4-SGSN sends both IP addresses within an F-TEID IE. If only one IP address is included, then the SGW assumes that the S4-SGSN does not support the other IP address type.	F-TEID	87	3

Table 17-43 Bearer Contexts IEs (within Modify Bearer Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
S11-U MME F-TEID	Conditional/ Optional	Included on the S11 interface if S11-U is being used for the following procedures: ■ Mobile Originated Data transport in Control Plane CIoT EPS optimization with P-GW connectivity ■ Mobile Terminated Data Transport in Control Plane CIoT EPS optimization with P-GW connectivity ■ All procedures where the S11-U tunnel is already established, for example when reporting a change of User Location Information ■ TAU/RAU with SGW change and data forwarding of DL data buffered in the old SGW for a Control Plane Only PDN connection. Can be included on the S11 interface if S11-U is being used during a E-UTRAN Tracking Area Update without SGW Change and the MME needs to establish the S11-U tunnel.	F-TEID	87	4

Modify Bearer Response

The Modify Bearer Response message is sent as an answer to a Modify Bearer Request message:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-44 lists the Modify Bearer Response information elements.

Table 17-44 Modify Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
MSISDN	Conditional	The PGW includes this IE on the S5/S8 interface if it is stored in its UR context and it is triggered due to a TAU/RAU/Handover procedure with an SGW relocation.	MSISDN	76	0
Linked EPS Bearer ID	Conditional	Included on the S5/S8 and S4/S11 interfaces when the UE moves from a Gn/Gp SGSN to the MME/S4-SGSN to identify the default bearer the PGW selects for the PDN connection.	EBI	73	0
Aggregate Maximum Bit Rate (APN-AMBR)	Conditional	Included on the S5/S8 interface if the PCRF has modified the APN-AMBR.	AMBR	72	0
APN Restriction	Conditional	Included on the S5/S8 interface during Gn/Gp SGSN to S4-SGSN/MME RAU/TAU/Handover procedures. Indicates the restriction on the combination of types of APN for the APN associated with this EPS Bearer Context.	APN Restriction	127	0
Protocol Configuration Options (PCO)	Conditional	Included on the S5/S8 interface if it is available.	PCO	78	0
Bearer Contexts Modified	Conditional	Indicates the EPS bearers corresponding to the Bearer Contexts to be Modified sent in the Modify Bearer Request message. Several IEs with the same type and instance value are included, if necessary, to represent a list of modified bearers.	Bearer Context	93	0

Table 17-44 Modify Bearer Response Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Bearer Contexts Marked for Removal	Conditional	Included if the Modify Bearer Request message included Bearer Contexts to be removed. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers to be removed.	Bearer Context	93	1
Change Reporting Action	Conditional	Included if it is provided by the MME/SGSN.	Change Reporting Action	131	0
CSG Information Reporting Action	Conditional/ Optional	Included with the appropriate Action field if the CSG information reporting mechanism is started or stopped for the subscriber in the SGSN/MME.	CSG Information Reporting Action	146	0
Charging Gateway Name	Conditional	The PGW includes this IE on the S5/S8 interface when the CGF address is configured during an SGW relocation or the UE moves from a Gn/Gp SGSN to an S4-SGSN/MME.	FQDN	136	0
Charging Gateway Address	Conditional	The PGW includes this IE on the S5/S8 interface when the CGF address is configured during an SGW relocation or the UE moves from a Gn/Gp SGSN to an S4-SGSN/MME.	IP Address	74	0
Recovery	Conditional	Included if contacting the peer for the first time.	Recovery	3	0
Private Extension	Optional	Includes vendor or operator specific information. ■ Charging-Group-ID	Private Extension	255	Vendor Specific
PGW’s node level Load Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports the load control feature.	Load Control Information	181	0
PGW’s APN level Load Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports APN level load control feature.	Load Control Information	181	1
PGW’s Overload Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW supports the APN level load control feature.	Overload Control Information	180	0

Bearer Contexts Modified Within Modify Bearer Response

Within the Modify Bearer Response message, the Bearer Contexts Modified IE contains other IEs and is considered a Grouped IE.

Table 17-45 lists the information elements contained in the Bearer Contexts Modified IE.

Table 17-45 Bearer Contexts Modified IEs (within Modify Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0
Charging ID	Conditional	Included on the S5/S8 interface if this message is triggered due to one of the following procedures: ■ TAU/RAU/Handover with an SGW relocation ■ TAU/RAU/Handover from the Gn/Gp SGSN to the S4-SGSN/MME	Charging ID	94	0
	Optional	Included on the S4 interface if S5/S8 interface is GTP in order to support CAMEL charging at the SGSN, for the following procedures: ■ Inter-SGSN RAU/Handover/SRNS Relocation without SGW change. ■ Inter-SGSN Handover/SRNS Relocation with SGW change			
Bearer Flags	Conditional/ Optional	Included on the S5/S8 and S4 interfaces if the Prohibit Payload Compression (PPC) is set to 1.	Bearer Flags	97	0

Table 17-45 Bearer Contexts Modified IEs (within Modify Bearer Response) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
S1-U SGW F-TEID	Conditional	Included on the S11 interface if the S1 interface is used, that is, if the S11-U Tunnel flag was not set in the Modify Bearer Request. If the "Change F-TEID support Indication" flag was set to 1 in the Modify Bearer Request and the SGW needs to change the F-TEID, the SGW will include the new GTP-U F-TEID value. Otherwise, the SGW will return the currently allocated GTP-U F-TEID value.	F-TEID	87	0
S4-U SGW F-TEID	Conditional	Included on the S4 interface if the S4-U interface is being used. If the "Change F-TEID support Indication" flag was set to 1 in the Modify Bearer Request and the SGW needs to change the F-TEID, the SGW will include the new GTP-U F-TEID value. Otherwise, the SGW will return the currently allocated GTP-U F-TEID value.	F-TEID	87	2
S11-U SGW F-TEID	Conditional	Included on the S11 interface if S11-U is being used, that is, if the S11-U Tunnel flag was set in the Modify Bearer Request. If the "Change F-TEID support Indication" flag was set to 1 in the Modify Bearer Request and the SGW needs to change the F-TEID, the SGW will include the new GTP-U F-TEID value. Otherwise, the SGW shall return the currently allocated GTP-U F-TEID value.	F-TEID	87	3

Bearer Contexts Marked for Removal Within Modify Bearer Response

Within the Modify Bearer Response message, the Bearer Contexts Marked for Removal IE contains other IEs and is considered a Grouped IE.

Table 17-46 lists the information elements contained in the Bearer Contexts Marked for Removal IE.

Table 17-46 Bearer Contexts IEs (within Modify Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0

Release Access Bearers Request

The Release Access Bearers Request message is sent on the S11 interface by the MME to the SGW as part of:

- The S1 release procedure and eNodeB-initiated Connection Suspend procedure.
- The Establishment of an S1-U bearer during Data Transport in Control Plane CIoT EPS optimization procedure (see subclause 5.3.4B.4 of 3GPP TS 23.401 [3]).

Note: *The S1 release procedure is also used to release the S11-U bearers for the Control Plane CIoT EPS optimization, except in the case of data buffering in the MME.*

The Release Access Bearers Request message is also sent on the S4 interface by the SGSN to the SGW as part of:

- RAB release using S4
- Iu Release using S4
- READY to STANDBY transition within the network

Table 17-47 describes the new IE for the Release Access Bearers Request message.

Table 17-47 Release Access Bearers Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Secondary RAT Usage Data Report	Conditional	If the PLMN has configured secondary RAT usage reporting, the MME includes this IE on the S11 interface if it has received Secondary RAT usage data from eNodeB in a Connection Suspend, or S1 release procedure. The MME sets the IRPGW flag to "0" to indicate that the IE is not forwarded to the PGW. Several IEs with the same type and instance value may be included, to represent multiple usage data reports.	Secondary RAT Usage Data Report	201	0

Release Access Bearers Response

The SGW sends a Release Access Bearers Response message to the MME/S4-SGSN as an answer to a Release Access Bearers Request.

The Release Access Bearers Response message is sent on the S11 interface to the MME as part of:

- The S1 release procedure and eNodeB-initiated Connection Suspend procedure
- The establishment of S1-U bearer during Data Transport in Control Plane CIoT EPS optimization procedure

The Release Access Bearers Response message is sent on the S4 interface to the SGSN as part of:

- RAB release using S4
- Iu Release using S4
- READY to STANDBY transition within the network

Table 17-48 lists the information elements contained in the Release Access Bearers Response message.

Table 17-48 Release Access Bearers Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0
Recovery	Optional	Included if the SGW is contacting the MME/S4-SGSN for the first time.	Recovery	3	0
Indication Flags	Conditional/ Optional	Included if any of the applicable flags is set to 1: ■ Associate OCI with SGW node's identity. This is set to 1 if the SGW has included the "SGW's Overload Control Information"and if this information is to be associated with the node identity (the FQDN or the IP address received from the DNS during the SGW selection) of the serving SGW.	Indication	77	0
SGW's node level Load Control Information	Optional	Node level load information for the SGW is included if the load control feature is supported by the SGW and is activated in the network. The SGW shall provide only one instance of this IE.	Load Control Information	181	0
SGW's Overload Control Information	Optional	Overload information for the SGW is included if the overload control feature is supported by the SGW and is activated for the PLMN to which the PGW belongs. The SGW shall provide only one instance of this IE.	Overload Control Information	180	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	VS

Resume Acknowledge

The Resume Acknowledge message is sent as an answer to the Resume Notification message:

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-49 lists the Resume Acknowledge information elements.

Table 17-49 Resume Notification Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates whether the return of non-GBR bearers is successful. If not, this IE gives the reason for the failure.	Cause	2	0
Private Extension	Optional	Included if it is available.	Private Extension	255	VS

Resume Notification

The SGW sends a Resume Notification message to the PGW on the S5/S8 interface as part of the resume procedures returning from CSFB to E-UTRAN. After receiving a Resume Notification message, the SGW/PGW forwards packets it receives for the UE.

The MME sends a Resume Notification to the SGW on the S11 interface.

Table 17-50 lists the Resume Notification information elements.

Table 17-50 Resume Notification Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
IMSI	Conditional	The IMSI is included in the message if it is received in the Change Notification Request message.	IMSI	1	0
Linked Bearer Identity (LBI)	Conditional	Indicates the default bearer associated with the PDN connection.	EBI (EPS Bearer ID)	73	0
Private Extension	Optional	Included if it is available.	Private Extension	255	VS

Suspend Acknowledge

The Suspend Acknowledge message is sent as an answer to a Suspend Notification message.

- From the PGW to the SGW on the S5/S8 interface.
- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-51 lists the Suspend Acknowledge information elements.

Table 17-51 Suspend Acknowledge Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	Indicates whether the suspension of non-GBR bearers is successful. If not, this IE gives the reason for the failure.	Cause	2	0
Private Extension	Optional	Includes this IE if it is available.	Private Extension	255	VS

Suspend Notification

The Suspend Notification message is sent as part of the CSFB procedures from E-UTRAN access to UTRAN/GERAN CS domain access as defined in 3GPP TS 23.272:

- The SGW sends a Suspend Notification message to the PGW on the S5/S8 interface. After receiving a Suspend Notification message, the PGW marks all the non-Guaranteed Bit Rate (GBR) bearers with a status of suspended. The PGW discards packets received for the suspended UE.
- The MME/S4-SGSN sends a Suspend Notification message to the SGW on the S4/S11 interface.

Table 17-52 lists the Suspend Notification information elements.

Table 17-52 Suspend Notification Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
IMSI	Conditional	Included on the S11 interface if it is received in the Change Notification Request message.	IMSI	1	0
Linked Bearer Identity (LBI)	Conditional	Included on the S4/S11 interface to indicate the default bearer associated with the PDN connection.	EBI (EPS Bearer ID)	73	0
Originating Node	Conditional Optional	Included on the S11 interface if before MME initiates a Detach procedure the ISR was active in the MME and the MME was in EMM-Connected state. Included on the S4 interface if before S4-SGSN initiates a Detach procedure the ISR was active in the SGSN and the SGSN was in PMM-Connected state.	Node Type	135	0

Table 17-52 Suspend Notification Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Sender F-TEID for Control Plane					
Private Extension	Optional	Includes this IE if it is available.	Private Extension	255	VS

Trace Session Activation

The Trace Session Activation message is sent when session trace is activated for a particular IMSI or IMEI of an attached UE. The UE can be active or idle.

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-53 lists the Trace Session Activation information elements.

Table 17-53 Trace Session Activation Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
IMSI	Mandatory	None	IMSI	1	0
Trace Information	Mandatory	None	Trace Information	96	0
ME Identity (MEI)	Conditional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available.	MEI	75	0

Trace Session Deactivation

A Trace Session Deactivation message is sent when session trace is deactivated for a particular IMSI or IMEI of an attached UE. The UE can be active or idle. The message is sent:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-54 lists the Trace Session Deactivation information elements.

Table 17-54 Trace Session Deactivation Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Trace Reference	Mandatory	None	Trace Reference	115	0

Update Bearer Request

The Update Bearer Request message is sent to request a change to the service data flow requirements or QoS parameters:

- From the PGW to the SGW on the S5/S8 interface.

> ***Note:*** *If the PCRF updates the MBR values associated with the active charging rule for a non-GBR bearer and no other information is updated by the PCRF, the PGW does not send the Update-Bearer-Request message to the SGW.*

- From the SGW to the MME/S4-SGSN on the S4/S11 interface.

Table 17-55 lists the Update Bearer Request information elements.

Table 17-55 Update Bearer Request Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Bearer Contexts	Mandatory	Used for bearers that require QoS/TFT modification. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers. If there is no QoS/TFT modification, only one IE with this type and instance value is included.	Bearer Context	93	0

Table 17-55 Update Bearer Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Procedure Transaction Id (PTI)	Conditional	Included on the S5/S8 and S4/S11 interfaces if the request corresponds to one of the following procedures: ■ UE-requested Bearer Resource Modification procedure for E-UTRAN. ■ MS-initiated EPS bearer Modification procedure This PTI IE is the same IE used in the corresponding Bearer Resource Command.	PTI	100	0
Protocol Configuration Options (PCO)	Conditional	Included this IE on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0
Aggregate Maximum Bit Rate (APN-AMBR)	Mandatory	None	AMBR	72	0
Change Reporting Action	Conditional	Included on the S4/S11 interface if the location Change Reporting mechanism is to be started or stopped for this subscriber in the MME/S4-SGSN. Included on the S5/S8 interfaces if it is provided by the MME/S4-SGSN.	Change Reporting Action	131	0
CSG Information Reporting Action	Conditional/ Optional	Included on the S5/S8 and S4/S11 interfaces with the appropriate Action field if the CSG information reporting mechanism is started or stopped for the subscriber in the MME/S4-SGSN.	CSG Information Reporting Action	146	0
SGW-FQ-CSID	Conditional	Included on S11 interface according to the requirements in 3GPP TS 23.007.	FQ-CSID	132	1
PGW's node level Load Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports the load control feature.	Load Control Information	181	0
PGW's Overload Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports the overload control feature.	Overload Control Information	180	0
PGW's APN level Load Control Information	Conditional/ Optional	Included on the S4/S11 interface if the SGW receives this IE and if it supports APN level load control feature.	Load Control Information	181	1

Table 17-55 Update Bearer Request Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Update Bearer Request

Within the Update Bearer Request message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-56 lists the information elements contained in the Bearer Contexts IE. For more information, see 3GPP TS 29.274

Table 17-56 Bearer Contexts IEs (within Update Bearer Request)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
TFT	Conditional	Included on the S5/S8 and S4/S11 interfaces if the message relates to a bearer modification or TFT change. If the size of TFT packet filters exceeds the 255 byte limit or if the number of packet filters exceeds 15 in the original Create Bearer/PDP Context Request or Update Bearer/PDP Context Request message, the gateway sends the remaining TFT packet filters in subsequent Update Bearer/PDP Context Request messages. This capability is also supported on the PGW for Create Bearer Request messages sent in handover scenarios, such as 3GPP to non-3GPP access and vice versa.	Bearer TFT	84	0
Bearer Level QoS	Conditional	Included on the S5/S8 and S4/S11 interfaces if a QoS modification is requested.	Bearer QoS	80	0

Table 17-56 Bearer Contexts IEs (within Update Bearer Request) (Continued)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
Bearer Flags	Optional	Applicable flags are: ■ Prohibit Payload Compression (PPC) – This flag may be set on the S5/S8 and S4 interfaces. ■ vSRVCC indicator – The PGW may include this IE on the S5/S8 interface according to 3GPP TS 23.216 [43]. When received on the S5/S8, the SGW forwards this IE on the S11 interface.	Bearer Flags	97	0
Protocol Configuration Options (PCO)	Conditional	The SGW includes this IE on the S5/S8 and S4/S11 interfaces if it is available. This bearer-level PCO IE takes precedence over the message-level PCO IE if they are both present.	PCO	78	0

Update Bearer Response

The Update Bearer Response is sent as an answer to an Update Bearer Request message:

- From the SGW to the PGW on the S5/S8 interface.
- From the MME/S4-SGSN to the SGW on the S4/S11 interface.

Table 17-57 lists the Update Bearer Response information elements.

Table 17-57 Update Bearer Response Information Elements

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Cause	Mandatory	None	Cause	2	0

Table 17-57 Update Bearer Response Information Elements (Continued)

IE Name	Requirement	Condition	IE Type	IE Type Value (Decimal)	Instance Value
Bearer Contexts	Mandatory	Used for bearers for which QoS and TFT modification are requested. Several IEs with the same type and instance value are included, if necessary, to represent a list of bearers.	Bearer Context	93	0
Protocol Configuration Options (PCO)	Conditional	Included on the S5/S8 and S4/S11 interfaces if it is available.	PCO	78	0
Recovery	Conditional	Included on the S5/S8 and S4/S11 interfaces if contacting the peer for the first time.	Recovery	3	0
Indication Flags	Optional	Included on the S5/S8 and S4/S11 interfaces if the Direct Tunnel Flag is set to 1.	Indication	77	0
UE Time Zone	Conditional/ Optional	Included on the S5/S8 and S4/S11 interfaces if it is available.	UE Time Zone	114	0
User Location Information (ULI)	Conditional/ Optional	Included on the S5/S8 and S4/S11 interfaces if it is available.	ULI	86	0
Private Extension	Optional	Includes vendor or operator specific information.	Private Extension	255	Vendor Specific

Bearer Contexts Within Update Bearer Response

Within the Update Bearer Response message, the Bearer Contexts IE contains other IEs and is considered a Grouped IE.

Table 17-58 lists the information elements contained in the Bearer Contexts IE.

Table 17-58 Bearer Contexts IEs (within Update Bearer Response)

Octet 1	Bearer Context IE Type = 93 (decimal)				
Octets 2 and 3	Length = n				
Octet 4	Spare and Instance Fields				
IE Name	**Requirement**	**Condition**	**IE Type**	**IE Type Value (Decimal)**	**Instance Value**
EPS Bearer ID	Mandatory	None	EBI	73	0
Cause	Mandatory	Indicates whether the bearer handling is successful. If not, this IE gives the reason for the failure.	Cause	2	0
Protocol Configuration Options (PCO)	Conditional	Included on the S5/S8 and S4/S11 interfaces if it is available. This bearer-level PCO IE takes precedence over the message-level PCO IE if they are both present.	PCO	78	0
RAN/NAS Cause	Conditional	The SGW includes this IE on the S5/S8 interface if it receives it from the MME. If the bearer modification failed, the MME includes this IE on the S11 interface to indicate the RAN and/or NAS cause of the bearer modification failure, if available, and if this information is permitted to be sent to the PGW according to the MME operator's policy. If both a RAN and NAS cause and are generated, then several IEs with the same type and instance value are included to represent a list of causes. The TWAN does not exchange signaling with the 3GPP AAA server nor with the UE in this procedure. The TWAN may include an internal failure cause for the bearer modification failure. The protocol type used to encode the internal failure cause is implementation-specific.	RAN/NAS Cause	172	0

References and Specifications

The system supports the following references and specifications for the S5/S8 interfaces:

- 3GPP TS 23.167 V12.0.0 (2014-09): *IP Multimedia Subsystem (IMS) Emergency Sessions*
- 3GPP TS 23.216 V11.7.0 (2012-12): *Single Radio Voice Call Continuity (SRVCC); Stage 2*
- 3GPP TS 23.401 V9.3.0 (2009-12): *General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access*
- 3GPP TS 23.401 V13.1.0 (2014-12): *General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access*
- 3GPP TS 23.869: *Support for IMS Emergency Calls over GPRS and EPS*
- 3GPP TS 29.060 V9.3.0 (2010-06): *General Packet Radio Service (GPRS);GPRS Tunneling Protocol (GTP) across the Gn and Gp interface*
- 3GPP TS 29.274 V9.3.0 (2010-06): *3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3*
- 3GPP TS 29.274 V12.4.0 (2014-03): *3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3*
- 3GPP TS 29.274 V12.6.0 (2014-03): *3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3*
- 3GPP TS 29.274 V15.3.0 (2018-03): *33GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunnelling Protocol for Control plane (GTPv2-C); Stage 3*
- 3GPP TS 29.281 V9.3.0 (2010-06): *General Packet Radio System (GPRS) Tunnelling Protocol User Plane (GTPv1-U) (Release 9)*

S6b Interface

This section contains the following information used to configure, control, and monitor the Diameter-based S6b interface:

Interface Overview

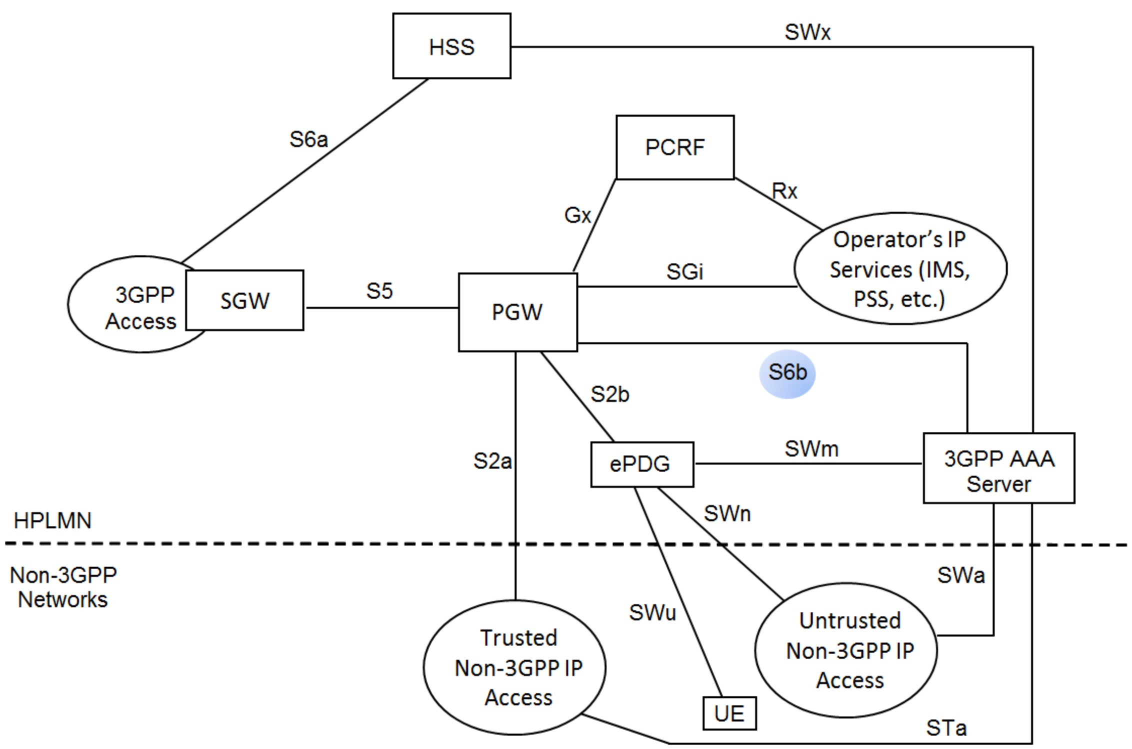
The Affirmed Networks Mobile Content Cloud (MCC) S6b interface is a Diameter-based interface between the 3GPP AAA server and the Packet Data Network Gateway (PGW). The S6b interface complies with 3GPP TS 29.273 V12.5.0 Release 12.

The S6b interface is used to manage address information, retrieve and update mobility-related information, authenticate APNs, and allocate UE IP addresses. For example, when the UE attaches to the Evolved Packet Core (EPC) using S2a/S2b/S5/S8 interface over GTPv2, the S6b interface updates the 3GPP AAA server/proxy with the PGW address and GTPv2 protocol information.

Through the S6b interface, the PGW informs the 3GPP AAA server when the UE disconnects a PDN connection associated with an APN, or when the PDN connection is handed over to the 3GPP access network, and therefore removes the mobility session established for this PDN connection. The S6b interface is also used to download subscriber and equipment-trace information to the PGW.

Figure 18-1 illustrates the position and role of the network components and interfaces for trusted and untrusted Wi-Fi access to the EPC.

Figure 18-1 Trusted and Untrusted Wi-Fi Access



Diameter Routing Agent

A Diameter Routing Agent (DRA) provides routing capabilities between a PGW and 3GPP AAA server when multiple 3GPP AAA servers are deployed in a Diameter realm. The MCC PGW includes the 3GPP AAA server host name and realm within the Destination-Host and Destination-Realm AVPs in Re-Auth-Request/Re-Auth-Answer (RAR/RAA) messages and in Abort-Session-Request/Abort-Session-Answer (ASR/ASA) messages sent over the S6b interface. Only the initial AA-Request (AAR) message includes the DRA host name and realm within the Destination-Host and Destination-Realm AVPs.

If the 3GPP AAA server becomes overloaded, the DRA can throttle S6b requests by sending permanent error code = 5198 (DIAMETER_OVERLOAD_RETRY_NOT_ALLOWED_TO_ANY). In this case, the MCC PGW releases the PDN session.

Configuring the S6b Interface

Use the following sequence to configure the S6b Diameter interface:

1 Create the network context and configure the S6b Diameter interface and GTP-U and GTP-C interface (as applicable) loopback IP addresses.

```
an3000-mcm-slot7cpu1(config)# network-context <name> loopback-ip
<loopback-name> ip-address <ip-address> gtp-u [yes|no]
```

where

Parameter	Description
network-context	Specifies the name of the network context.
loopback-ip	Specifies the name of the loopback address associated with this network context.
ip-address	Specifies a loopback IP address to use for a service endpoint (S6b) interface on the network context (can be IPv4/IPv6).
gtp-u	Specifies whether to use this loopback address for GTP-U. ■ **yes** – (Default) Use yes only when configuring the S6b GTP-U interface loopback IP address. ■ **no** – Use no when configuring the S6b GTP-C interface. This disables GTP-U, which improves security hardening.

2 Configure a local Diameter node on the MCC PGW.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <diameter-node-name>
```

The following table describes the configurable parameters.

Parameter	Description
node-admin	Specifies the name of the local Diameter node on the PGW.
loopback-ip	Specifies the name of the signaling port IP address. ***Note:*** *The network context and loopback IP address must already exist.*
network-context	Specifies the associated local Gi/SGi network context. Restricts the inbound Diameter messages by accepting only inbound messages on the IP interfaces in the interface group. ***Note:*** *The network context and loopback IP address must already exist.*
node-role	Specifies the local node role: ■ **client** ■ **server**
origin-host	Specifies the local host name in the Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the Diameter request message. ***Note:*** *You can configure multiple Diameter server groups with the same origin host.* ***Note:*** *The origin-host configured at the peer level overrides the origin-host configured at the Diameter server-group level.*
origin-realm	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message. ***Note:*** *You can configure multiple Diameter server groups with the same origin realm.* ***Note:*** *The origin-realm configured at the peer level overrides the origin-realm configured at the Diameter server-group level.*
port	Specifies the TCP listening port number used when node-role is set to server. The default is 3868. ***Note:*** *This is the TCP listening port, not the S6b source port used when node-role is set to client. The S6b source port is dynamically assigned and is not configurable.*
transport	Specifies the transport type. ■ **tcp** (default) ■ **sctp** – To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the S6b interface, you must configure an SCTP profile and associate the profile with the local Diameter node. See Configuring an SCTP Profile (page 18-17).
sctp-profile	Specifies the name of the SCTP profile. The SCTP profile must already exist. See Configuring an SCTP Profile (page 18-17). ***Note:*** *The sctp-profile parameter is only visible when the local Diameter node transport type is set to SCTP.*

Parameter	Description
secondary-loopback-ip	Specifies a secondary loopback IP address for the local Diameter node to support SCTP multi-homing. See Configuring an SCTP Profile (page 18-17) for more information. ***Note:*** *The* `secondary-loopback-ip` *parameter is only visible when the local Diameter node transport type is set to SCTP.*

3 Configure the list of peers (remote AAA servers) with which the local Diameter node (S6b interface) on the PGW communicates.

`an3000-mcm-slot7cpu1(config)#` **`service-construct interface aaa-interface-group diameter node-admin <`*`diameter-node-name`*`> peer <`*`peer-host-name`*`>`**

The following table describes the configurable parameters.

Parameter	Description
node-admin	Specifies the name of the local Diameter node on the PGW.
peer	Specifies the peer host name Fully Qualified Domain Name (FQDN). The peer host name is used in the Destination-Host AVP to send Diameter request messages. ***Note:*** *When a Server ID name from a Create Session Request message is received over S2b/S2a interfaces that contains ASCII control characters used as delimiters, the characters are replaced during encoding by the PGW with dot (.) delimiters when it is used as an S6b Diameter-Host AVP. During encoding, any non-printable or space characters that appear at the start or end of the Destination-Host AVP is removed.*
admin-state	Specifies the admin state of the Diameter peer. When enabled, the system attempts to establish peer connection. ■ **enabled** (default) ■ **disabled**
backup-peer-name	Identifies the secondary peer host name used during a failover from the primary to the secondary peer. ***Note:*** *Only primary peers can be configured with a backup peer. If failover to a secondary peer is supported, this parameter must be configured with the host name of a secondary peer.*
dest-ip-address	Specifies the IP address of the peer (can be an IPv4 or IPv6 address).
dest-port	Specifies the signaling port number.

Parameter	Description
dest-realm	Specifies the peer realm name. Realm is used in the Destination-Realm AVP in Diameter requests. ***Note:*** *When a Server ID realm from a Create Session Request message is received over S2b/S2a interfaces that contains ASCII control characters used as delimiters, the characters are replaced during encoding by the PGW with dot (.) delimiters when it is used as a S6b Diameter-Realm AVP. During encoding, any non-printable or space characters that appear at the start or end of the Destination-Host AVP is removed.*
origin-host	Specifies the local host name in Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the request message. See Configuring Multiple Pairs of Primary and Secondary Peers on the S6b Interface (page 18-20) for more information. ***Note:*** *The origin-host configured at the peer level overrides the origin-host configured at the Diameter server-group level.*
origin-realm	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message. See Configuring Multiple Pairs of Primary and Secondary Peers on the S6b Interface (page 18-20) for more information. ***Note:*** *The origin-realm configured at the peer level overrides the origin-realm configured at the Diameter server-group level.*
peer-identifier	Specifies the external identifier for this peer.
role	Specifies the type of Diameter peer. When multiple primary, active peers are configured, the system uses a round-robin approach to select peers. ■ **Primary** – (default) Active peers used for Diameter sessions. ■ **Secondary** – Used for Diameter sessions, only when the primary peers are down. ***Note:*** *If failover to a secondary peer is supported, you must enter the host name of the secondary server in the* `backup-peer-name` *parameter.*
secondary-loopback-ip	Specifies a secondary loopback IP address for the Diameter peer to support SCTP multi-homing. See Configuring an SCTP Profile (page 18-17) for more information. ***Note:*** *The* `secondary-loopback-ip` *parameter is only visible when the local Diameter node transport type is set to SCTP.*

Parameter	Description
weight	Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in "maintenance mode" and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions. *Applies to Gy/Gx/Gz/STa/SWm/S6b interfaces.* See Weighted Round-Robin Distribution to Multiple Diameter Peers (page 18-16) for more information.

4 Configure the Diameter data record templates used to specify which AVPs are sent or suppressed in the S6b messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pdngw S6b diameter [aaa|aar|asa|asr|raa|rar|sta|str] <template-name>
<attribute-name> presentFlag [yes|no|conditional]
```

where

Parameter	Description
[aaa|aar|asa|asr|raa|rar|sta|str]	Specifies the type of data record template.
<template-name>	Specifies the name of the data record template.
<attribute-name>	Specifies the name of the AVP field you are modifying.
presentFlag	Options include: ■ **yes** – The field is sent in Diameter messages and processed when received. ■ **no** – The field is not sent in Diameter messages and is ignored when received. ■ **conditional** – Indicates the field is sent if it is available.

5 Configure the S6b interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
s6b-interface <name>
```

where ***<name>*** specifies the S6b interface name.

The following table describes the configurable parameters.

Parameter	Description
access-type	Specifies the applicable access type for this interface. ■ **non-3gpp-only** (default) ■ **both-non3gpp-and-3gpp**

Parameter	Description
admin-state	Specifies the admin state of the interface. ■ **enabled** (default) ■ **disabled**
auth-application-id	Specifies the application ID (standard or non-standard). This is any value up to 32 bits. The default is **16777272** (default - 3GPP S6b).
authorization-grace-period	Specifies the time interval before the PGW initiates the final reauthorization attempt on the S6b interface, in minutes. This timer starts when the PGW sends the AAR message following the expiration of the authorization-lifetime timer and the AAA message returns an error or the PGW is unable to send the AAR. ***Note:*** *If the system receives the Auth-Grace-Period AVP in the AAA or RAA messages from the 3GPP AAA server, the AVP values received in the AAA or RAA message override the configured values for this parameter.*
authorization-lifetime	Specifies the time interval before the PGW initiates the reauthorization procedure on the S6b interface, in minutes. The PGW starts the authorization-lifetime timer upon receiving the AAA message. ***Note:*** *If the system receives the Authorization-Lifetime AVP in the AAA or RAA messages from the 3GPP AAA server, the AVP values received in the AAA or RAA message override the configured values for this parameter.*
pgw-triggered-reauthorization	Specifies whether to enable or disable the PGW-initiated reauthorization procedure. ■ **enabled** ■ **disabled** (default)
retransmit-count	Specifies the maximum number of retransmit attempts to the AAA server.
retransmit-timeout	Specifies the time interval to wait before attempting to retransmit, in milliseconds.

Parameter	Description
user-name-type	Specifies the User-Name AVP format used in AA-Request messages on the S6b interface if not present in the PCO. This setting overrides the `user-name-type` present at the APN level. However, if the S6b interface `user-name-type` is set to `none`, the APN level `user-name-type` is used. ■ **apn** ■ **default-username** ■ **imsi** ■ **imsi-nai-no-prefix** ■ **imsi-nai-prefix-0** ■ **imsi-plus-apn** ■ **imsi-plus-apnnionly** ■ **msisdn** ■ **msisdn-plus-apn** ■ **msisdn-plus-apnnionly** ■ **none**

6 Associate the Diameter server group and data record templates with the S6b interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
s6b-interface <name> pgw diameter server-group <service-profile>
[aaa-template|aar-template|asa-template|asr-template|raa-template|rar-
template|sta-template|str-template] <template-name>
```

where

Parameter	Description
s6b-interface	Specifies the S6b interface name.
server-group	Specifies the Service Profile for the selected streaming protocol.
aaa-template	Specifies the name of the AA-Answer data record template.
aar-template	Specifies the name of the AA-Request data record template.
asa-template	Specifies the name of the Abort-Session-Answer data record template.
asr-template	Specifies the name of the Abort-Session-Request data record template.
raa-inital	Specifies the name of the Re-Auth-Answer data record template.
rar-template	Specifies the name of the Re-Auth-Request data record template.
sta-update	Specifies the name of the Session-Termination-Answer data record template.
str-final	Specifies the name of the Session-Termination-Request data record template.

7 (Optional) Configure the MIP6-Agent-Info AVP sent in AAR messages.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
s6b-interface <name> pgw diameter mip6-agent-info-type
[mip-home-agent-address|mip-home-agent-host]
mip-home-agent-destination-host <home-agent-host-name>
mip-home-agent-destination-realm <home-agent-realm-name>
```

where

Parameter	Description
s6b-interface	Specifies the S6b interface name.
mip6-agent-info-type	Specifies whether to send the PGW IP address or the PGW FQDN in the S6b MIP6-Agent-Info AVP. ■ **mip-home-agent-address** – (default) Specifies to send the PGW IP address. ■ **mip-home-agent-host** – Specifies to send the PGW's fully-qualified domain name (FQDN).
mip-home-agent-destination-host	Specifies the home agent destination host name in the FQDN. ***Note:*** *This parameter is only applicable when* `mip6-agent-info-type` *is set to* `mip-home-agent-host`. ***Note:*** *If you modify the* `mip-home-agent-destination-host` *name during a session, the subsequent AAR messages include the modified PGW FQDN.*
mip-home-agent-destination-realm	Specifies the home agent destination realm name in the FQDN. ***Note:*** *This parameter is only applicable when* `mip6-agent-info-type` *is set to* `mip-home-agent-host`. ***Note:*** *If you modify* `mip-home-agent-destination-realm` *name during a session, the subsequent AAR messages include the modified PGW FQDN.*

8 (Optional) Use the following command to remove the leading digit in front of the IMSI-NAI in the Username AVP.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn user-name-type
imsi-nai-no-prefix
```

9 Associate the S6b interface with a workflow control profile. The control profile must already exist.

```
an3000-mcm-slot8cpu1(config)# workflow control-profile <profile-name>
default-s6b-interface <s6b-interface-name>
```

where

Parameter	Description
control-profile	Specifies the workflow control profile name.
default-s6b-interface	Specifies the S6b interface name.

Optional S6b Interface Configuration Tasks

This section describes optional S6b interface configuration tasks.

- Configuring Geo-Redundancy Health Checks on the S6b Interface (page 18-12)
- Configuring DSCP Values for Control Plane Traffic (page 18-16)
- Weighted Round-Robin Distribution to Multiple Diameter Peers (page 18-16)
- Configuring an SCTP Profile (page 18-17)
- Using 3GPP AAA Server Identifier IE to Select an S6b Peer (page 18-19)
- Configuring Multiple Pairs of Primary and Secondary Peers on the S6b Interface (page 18-20)

Configuring Geo-Redundancy Health Checks on the S6b Interface

Geographic Redundancy network health checks are supported on RADIUS, Gx, Gy and S6b interfaces and use both shared loopback IPs and private loopback IPs. Operators must enable health-check monitoring on a least one interface.

Note: *It is recommended that you use the Geographic Redundancy Manager application in the Acuitas Service Management System to configure the Geographic Redundancy group, service endpoint, and server parameters (step 1 through step 3). For more information on configuring Geographic Redundancy, see the Acuitas User's Guide.*

To configure Geographic Redundancy network health checks on the S6b interface:

1 Configure the geographic redundancy group.

```
an3000-mcm-slot8cpu1(config)# geographic-redundancy group
<geo-redundancy-group-name> admin-state [enabled|disabled] group-id
<id> manual-state [active|standby] mode
[hot-auto|hot-manual|warm-manual] role [primary|secondary]
```

where

Parameter	Description
`geographic-redundancy group`	Specifies a Geographic Redundancy group name that is unique across the network.

Parameter	Description
`admin-state`	Specifies the administrative state: ■ `enabled` – Initializes and starts the redundancy mechanism (such as synchronization in hot mode). ■ `disabled` – Invalidates any synchronized data and sets the mechanism into a non-ready state.
`group-id`	Specifies the unique group ID that matches with the peer.
`manual-state`	Specifies the manually configured HA state (the Operator current desired state). ■ `active` ■ `standby`
`mode`	Specifies the instance configured HA mode. ■ `warm-manual` – No session synchronization occurs between nodes, failovers are Operator controlled. ■ `hot-auto` – Synchronization of per-session data occurs between nodes, failovers are automatically triggered based on network health monitoring results. ■ `hot-manual` – Synchronization of per-session data occurs.
`role`	Specifies the instance configured HA backup role. ■ `primary` – Specifies which node is “active” under normal network conditions. ■ `secondary` – Specifies which node is “standby” under normal network conditions.

2 Configure the service endpoint for the Geographic Redundancy server.

```
an3000-mcm-slot8cpu1(config)# network-context <network-context-name>
service-endpoint <service-endpoint-name> local-loopback-address
<ip-address> local-port-range-base <port> local-port-range-numports
<value>
```

where

Parameter	Description
`network-context`	Specifies a network context containing this Geographic Redundancy service endpoint.
`service-endpoint`	Specifies the name of the Geographic Redundancy service endpoint.
`local-loopback-address`	Specifies the local node address used for communication between primary and secondary nodes.
`local-port-range-base`	Specifies the base UDP port number used for communication between primary and secondary nodes. Must be the initial value of a contiguous block of UDP ports.

Parameter	Description
`local-port-range-numports`	Specifies the size of the contiguous block of ports.

3 Configure the Geographic Redundancy server parameters.

```
an3000-mcm-slot8cpu1(config)# service-construct geo-redundancy-server
<geo-redundancy-group-name> network-context <network-context-name>
service-endpoint <service-endpoint-name> remote-ip-address
<ip-address> remote-base-port <port> connectivity-monitoring-params
[monitor-AAA|monitor-OCS|monitor-PCRF|monitor-S6B] admin-state
[enabled|disabled]
```

where

Parameter	Description
`geo-redundancy-server`	Specifies the Geographic Redundancy group name to which this Geographic Redundancy server belongs.
`network-context`	Specifies the network context containing this Geographic Redundancy service endpoint.
`service-endpoint`	Specifies the service endpoint for this Geographic Redundancy server.
`remote-ip-address`	Specifies the IP address for the remote peer in the group.
`remote-base-port`	Specifies the port number for the remote peer in the group
`connectivity-monitoring-params`	Specifies network connectivity monitoring for automatic failover. Configure the parameters as follows: ■ `monitor-AAA` – Set to enabled to monitor the RADIUS Accounting Authentication Authorization (AAA) interface to govern fault detection and automatic failover. ■ `monitor-PCRF` – Set to enabled to monitor the Gx interface to govern fault detection and automatic failover. ■ `monitor-OCS` – Set to enabled to monitor the Gy interface to govern fault detection and automatic failover. ■ `monitor-S6B` – Set to enabled to monitor the S6B interface to govern fault detection and automatic failover.

4 Associate the Geographic Redundancy group with the local diameter node associated with the interface and configure health-check parameters on the interface.

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <diameter-node-name>

geo-redundant-group <geo-redundancy-group-name>
net-health-report-enabled [true|false] link-type [private|shared]
```

where

Parameter	Description
node-admin	Specifies the local diameter node associated with the interface. The local diameter node must already exist. See Configuring the S6b Interface (page 18-3).
geo-redundant-group	Specifies the Geographic Redundancy group name.
net-health-report enabled	Specifies whether to monitor network health towards the interface: ■ **true** ■ **false**
link-type	Specifies whether the link used to monitor neighboring network elements is a a private (local) or shared IP address among the primary and secondary NE's: ■ **private** ■ **shared**

5 Specify a default S6b interface in the workflow control-profile.

```
an3000-mcm-slot8cpu1(config)# workflow control-profile <profile-name>
default-s6b-interface <interface-name>
```

Note: *The interface must already exist. See Configuring the S6b Interface (page 18-3).*

6 Use the following command to view the parameters of the diameter node:

```
an3000-mcm-slot8cpu1# show service-construct interface
aaa-interface-group diameter node-admin <diameter-node-name>
```

Example:

```
an3000-mcm-slot8cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin s6b-node2
node-role                  client
origin-host                diameter_client.carrier.com
origin-realm               diameter_server.carrier.com
network-context            NC-AAA
loopback-ip                NC-AAA-SHARED-3
transport                  tcp
port                       9868
connection-attempt-count   0
connection-attempt-timeout 3
cea-timeout                5
dwr-timer                  15
dwr-retry-timer            1
dwr-retry-count            3
```

```
geo-redundant-group         GEO-GRP-2
net-health-report-enabled   true
link-type                   shared
product-name                AN-3000
vendor-id                   10415
send-destination-host-avp   true
peer                        carrier_diameter_server.com
   realm                    carrier.com
   primary-ip-address       10.150.1.13
   port                     9868
   transport                tcp
   role                     primary
   admin-state              enabled
```

Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on Diameter interfaces using Differentiated Services Code Point (DSCP).

Diameter interfaces (S6b, Gx, Gy, Gz/Rf) – DSCP value configured at the Diameter server group level.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <name> dscp-value <value>
```

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>
control-dscp-value <value>
```

Note: *The DSCP value configured at the Diameter server group level overrides the network-context DSCP value.*

Weighted Round-Robin Distribution to Multiple Diameter Peers

Operators can configure the MCC to use a weighted round-robin algorithm to distribute initial session requests among the available Diameter peers within a Diameter server group based on the configured weight assigned to each peer. The Diameter peer with the highest weight (with 10 being the highest and 1 being the lowest) receives the most distribution of new session requests. The MCC only uses the configured weight for the initial peer selection (for example, CCR-I).

For example, if Peer A is set to 4 and Peer B is set 2, the MCC will distribute the initial session requests as ABABAA. In this example, the MCC alternates between both peers starting with Peer A because it has the highest weight.

When a Diameter peer is configured with a weight of 0, the MCC places the peer in "maintenance mode" and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions.

If the Diameter peer with the highest weight is unavailable, the request is sent to the peer with the next highest weight in the same Diameter server group. If multiple Diameter peers are assigned the highest weight (10), the MCC uses a round-robin approach to select from among those peers for new requests.

Operators can configure the MCC to use a weighted round-robin algorithm to select a Diameter peer from among a list of primary peers or from among a list of secondary peers, but not *across* primary and secondary peers.

Use the following command to assign a weight to a Diameter peer:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> weight <value>
```

- **weight** – Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in "maintenance mode" and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions. *Applies to Gy/Gx/Gz/STa/Swm/S6b interfaces.*

Configuring an SCTP Profile

To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the S6b interface, you must configure an SCTP profile and associate the profile with the local Diameter node.

You also have the option of configuring a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. SCTP multi-homing enables the MCC to switchover to an alternate path in the case of a failure on the primary path. The MCC will revert back to the primary path once the primary connection becomes available. The primary and secondary loopback IP addresses can be IPv4 or IPv6. MCC supports socket creation to be associated to a single network context. SCTP multi-homing will be supported in a single network context.

Note: *If using a mix of IPv4/IPv6, ensure the primary loopback IP address is IPv4 and the secondary is IPv6. SCTP Multi-homing will not work if the primary loopback IP address is IPv6 and the secondary is IPv4. For this reason, Affirmed Networks recommends not using a mix of IPv4/IPv6 addresses for SCTP Multi-homing.*

---

1 Create an SCTP profile.

```
an3000-mcm-slot7cpu1(config)# service-construct sctp-profile
<profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
`cookie-life`	Specifies the lifetime of the SCTP cookie, in 100 milliseconds.
`heartbeat-interval`	Specifies the periodic time interval (in milliseconds) at which the SCTP heart beat messages are sent. Value 0 disables SCTP heart beats.
`init-retry`	Specifies the maximum number of attempts to retransmit the INIT message after which the attempted association is terminated.
`path-max-retransmit`	Specifies the maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive.
`max-in-streams`	Specifies the maximum number of incoming SCTP streams.
`max-out-streams`	Specifies the number of outgoing SCTP streams.
`path-mtu`	Specifies the maximum transmission unit (MTU), in bytes, for SCTP streams. Value 0 enables auto-discovery of the path MTU.
`rto-initial`	Specifies the initial retransmission timeout (RTO) value, in milliseconds. Subsequent SCTP RTO will be recomputed starting from this value.
`rto-max`	Specifies the maximum RTO value (in milliseconds) that any subsequent recomputed RTO may never exceed.
`rto-min`	Specifies the minimum RTO value (in milliseconds) that any subsequent recomputed RTO may never go below.
`sack-period`	Specifies the delay before sending the Selective Acknowledgment (SACK), in milliseconds. Value 0 disables the delay value.

2 Specify SCTP as the transport type for the local Diameter node and associate the SCTP profile.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> transport sctp
sctp-profile <profile-name>
```

3 (Optional) Configure a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. The secondary loopback IP address is only configurable for the SCTP transport type.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
secondary-loopback-ip <loopback-name>
```

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> secondary-loopback-ip <loopback-name>
```

Note: *The* `secondary-loopback-ip` *parameter is only visible when the local Diameter node transport type is set to SCTP.*

Using 3GPP AAA Server Identifier IE to Select an S6b Peer

The PGW processes the 3GPP AAA Server Identifier information element (IE) when received on the S2b/S2a interfaces, as defined in 3GPP TS 29.274 V13.6.0. If the PGW receives the 3GPP AAA Server Identifier IE from the ePDG/TWAG in the Create Session Request (CSR) message over the S2b/S2a interfaces, it uses the received server identifier (ID) and realm to select the S6b Diameter peer for authentication.

If the received server ID and realm do not match a configured S6b peer, the PGW uses the existing peer selection algorithm to select an S6b peer, but encodes the received server ID and realm in the Destination-Host and Destination-Realm AVPs sent in the AAR message to the peer.

Note: *If the initial AAR is retransmitted to the same peer or another peer due to any reason, such as the response timed out or the received response has a temporary error code, the Destination-Host and Destination-Realm AVPs in the retransmitted AAR should include the values as received in the CSR from the ePDG/TWAG.*

If the received 3GPP AAA Server Identifier IE contains either the server ID (and not the realm) or the realm (and not the server ID), the PGW will use only the server ID or only the realm to select an S6b peer.

The PGW ignores the 3GPP AAA Server Identifier information received from the ePDG/TWAG in the following cases:

- If the default S6b interface is not configured in the Workflow control profile.

- If the PGW receives the 3GPP AAA Server Identifier IE from ePDG/TWAG during a 3GPP-to-non-3GPP handover and if there is already an existing S6b Diameter connection for that session. Ideally, the IE will have the same server ID and realm as the existing S6b connection's server, but if they are different, the old AAA server should send a redirect indication in the AA-Answer message to the PGW.

If you do not want the PGW to send the Destination-Host AVP in the initial AAR, suppress the Destination-Host AVP in the S6b AAR data record template using the following command.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pdngw S6b diameter aar <template-name> destinationhost presentFlag no
```

Configuring Multiple Pairs of Primary and Secondary Peers on the S6b Interface

The MCC allows for the configuration of multiple primary active peers and secondary peers on the S6b interface within a Diameter server group. Each peer will have a unique peer host name and may have a unique origin-host and origin-realm. All primary peers will have the same destination host name (`override-dest-hostname`) and all secondary peers will have the same destination host name (`override-dest-hostname`).

Figure 18-2 shows an example of the configuration used to configure multiple pairs of primary and secondary peers on the S6b interface. The new CLI parameters added at the peer level appear as red text.

Figure 18-2 Example Configuration for Multiple Pairs of Primary/Secondary S6b Peers

Local Diameter Node Configuration

```
an3000-mcm-slot7cpu1(config)# service-construct interface aaa-interface-group diameter node-admin S6B
 node-role                 client
 origin-host               defaultpeer.affirmed
 origin-realm              affirmed
 network-context           GI
 loopback-ip               S6BLOOP
 transport                 tcp
 port                      3700
 dscp-value                dscp_0_be
 connection-attempt-count 0
 connection-attempt-timeout 3
 cea-timeout               5
 dwr-timer                 15
 dwr-retry-timer           1
 dwr-retry-count           3
 product-name              AN-3000
 vendor-id                 10415
 send-destination-host-avp true
```

Diameter Peer Configuration

Pair 1

```
peer              S6bPeer1.s6bAaaServer.affirmed
  origin-host     primarypeer1.affirmed
  origin-realm    affirmed
  dest-realm      affirmed
  dest-ip-address 10.48.148.3
  dest-port       3868
  transport       tcp
  role            primary
  backup-peer-name   S6bPeer2.s6bAaaServer.affirmed
  admin-state     enabled
  weight          1
  override-dest-hostname  S6bAaaServer1.affirmed

 peer             S6bPeer2.s6bAaaServer.affirmed
  origin-host     primarypeer1.affirmed
  origin-realm    affirmed
  dest-realm      affirmed
  dest-ip-address 10.48.158.3
  dest-port       3869
  transport       tcp
  role            secondary
  admin-state     enabled
  weight          1
  override-dest-hostname  S6bAaaServer2.affirmed
```

Pair 2

```
peer              S6bPeer3.s6bAaaServer.affirmed
  origin-host     primarypeer2.affirmed
  origin-realm    affirmed
  dest-realm      affirmed
  dest-ip-address 10.48.148.3
  dest-port       3868
  transport       tcp
  role            primary
  backup-peer-name   S6bPeer4.s6bAaaServer.affirmed
  admin-state     enabled
  weight          1
  override-dest-hostname  S6bAaaServer1.affirmed

 peer             S6bPeer4.s6bAaaServer.affirmed
  origin-host     primarypeer2.affirmed
  origin-realm    affirmed
  dest-realm      affirmed
  dest-ip-address 10.48.158.3
  dest-port       3869
  transport       tcp
  role            secondary
  admin-state     enabled
  weight          1
  override-dest-hostname  S6bAaaServer2.affirmed
```

Pair 3

```
peer              S6bPeer5.s6bAaaServer.affirmed
  origin-host     primarypeer3.affirmed
  origin-realm    affirmed
  dest-realm      affirmed
  dest-ip-address 10.48.148.3
  dest-port       3868
  transport       tcp
  role            primary
  backup-peer-name   S6bPeer6.s6bAaaServer.affirmed
  admin-state     enabled
  weight          1
  override-dest-hostname  S6bAaaServer1.affirmed

 peer             S6bPeer6.s6bAaaServer.affirmed
  origin-host     primarypeer3.affirmed
  origin-realm    affirmed
  dest-realm      affirmed
  dest-ip-address 10.48.158.3
  dest-port       3869
  transport       tcp
  role            secondary
  admin-state     enabled
  weight          1
  override-dest-hostname  S6bAaaServer2.affirmed
```

Pair 4

```
peer              S6bPeer7.s6bAaaServer.affirmed
  origin-host     primarypeer4.affirmed
  origin-realm    affirmed
  dest-realm      affirmed
  dest-ip-address 10.48.148.3
  dest-port       3868
  transport       tcp
  role            primary
  backup-peer-name   S6bPeer8.s6bAaaServer.affirmed
  admin-state     enabled
  weight          1
  override-dest-hostname  S6bAaaServer1.affirmed

 peer             S6bPeer8.s6bAaaServer.affirmed
  origin-host     primarypeer4.affirmed
  origin-realm    affirmed
  dest-realm      affirmed
  dest-ip-address 10.48.158.3
  dest-port       3869
  transport       tcp
  role            secondary
  admin-state     enabled
  weight          1
  override-dest-hostname  S6bAaaServer2.affirmed
```

Figure 18-2 on page 18-21 uses the following configuration:

- All of the primary peers are configured with the same:
  - Destination host name (`override-dest-hostname`)
  - Destination IP address (`dest-ip-address`)
  - Destination port (`dest-port`)
- All of the secondary peers are configured with the same:
  - Destination host name (`override-dest-hostname`)
  - Destination IP address (`dest-ip-address`)
  - Destination port (`dest-port`)
- Each peer is configured with a unique peer host name (`peer`). The peer host name is used by the MCC for load balancing between peers and for statistics.
- The value configured for the destination host name (`override-dest-hostname`) for each peer overrides the configured peer host name (`peer`) in the Destination-Host AVP within the initial AAR message and in all subsequent outgoing messages (STR/AAR).

- Each primary and secondary peer in a pair is configured with the same:
  - Origin-host name (`origin-host`)
  - Origin-realm (`origin-realm`)

  The origin-host and origin-realm configured at the peer level overrides the origin-host and origin-realm configured at the Diameter server-group level.

- Each primary peer is configured with the secondary peer host name (`peer`) in the `backup-peer-name` parameter.

- **Active secondary/backup peer configured for the pair** – If the primary peer within a pair does not respond to the initial AAR and if there is an active secondary/backup peer configured for the pair, the PGW transmits the initial AAR to the secondary/backup peer within the pair.
  - If the secondary/backup peer responds with a successful AAA, the session is established with the secondary/backup peer.
  - If the link to the secondary/backup peer then goes down after the session has been established and if the secondary does not have a backup configured or if the backup is not in active mode, the PGW terminates the session and will not attempt to connect with the next primary peer.

- **No active secondary/backup peer configured for the pair** – If the primary peer within a pair does not respond to the AAR and if the secondary/backup peer is not in active mode or if there is no backup peer configured for the pair, the PGW transmits the AAR to the primary peer of another pair.
  - If the PGW then receives a delayed AAA response from the previous primary peer, the session is established with the previous primary peer and the PGW drops any subsequent AAA response it receives for the same session.
  - If the link to the selected peer goes down after the session has been established and there is no active secondary/backup peer configured for the pair, the session will eventually time out and will not be moved to the next primary peer.

- If you do not want the PGW to send the Destination-Host AVP in the initial AAR, suppress the Destination-Host AVP in the S6b AAR data record template using the following command.

```
an3000-mcm-slot7cpu1(config)# service-construct
data-record-template pdngw S6b diameter aar <template-name>
destinationhost presentFlag no
```

Monitoring the S6b Interface

This section describes how to monitor the operational status of the 3GPP AAA server via the S6b interface and how to display statistical information for S6b Diameter messages. For more information on statistics, see *Acuitas Performance Statistics*.

Node Status

Use the following CLI command to display the operational status of the local Diameter node.

```
an3000-mcm-slot8cpu1# show service-construct interface
aaa-interface-group diameter node-status <diameter-node-name>
```

The system displays output similar to the following:

NAME	ORIGIN HOST	ORIGIN REALM	HOST IP ADDRESS	TRANSPORT	PORT	GEO REDUN DANT GROUP	STATUS
S6BDIAM	node76.affirm ed.local	node76.affir med	10.48.76.4	tcp	3868		inserv ice

Peer Status

Use the following CLI command to display the operational status of the TCP connection between the MCC PGW and the specified 3GPP AAA server.

```
an3000-mcm-slot8cpu1# show service-construct interface
aaa-interface-group diameter peer-status <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

```
an3000-mcm-slot8cpu1# show service-construct interface
aaa-interface-group diameter peer-status S6BDIAM S6b Peer 1
realm                       s6b3gppAaaServer.affirmed
ip-address                  10.6.1.217
port                        10040
transport                   tcp
role                        primary
backup-host-name            ""
status                      initial
last-state-change-timestamp 2015-10-07T11:32:20-0400
override-hostname           ""
an3000-mcm-slot8cpu1# show service-construct interface
aaa-interface-group diameter peer-status S6BDIAM S6b Peer 2
realm                       s6b3gppAaaServer.affirmed
ip-address                  10.6.1.217
port                        10050
transport                   tcp
role                        primary
backup-host-name            ""
status                      initial
```

```
last-state-change-timestamp 2015-10-01T11:35:42-0400
override-hostname           ""
```

Peer-cmd-Stats

Use the following CLI command to display the number of request/response messages sent per command-code between the MCC PGW and the specified 3GPP AAA server.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-cmd-stats <diameter-node-name>
```

The system displays output similar to the following:

NODE NAME	HOST NAME	COMMAND CODE	REALM	NUM OUT REQ	NUM OUT RSP	NUM IN REQ	NUM IN RSP
S6BDIAM	S6bPeer1	257	s6b3gppAaaServer.affirmed	4647	0	0	144
S6BDIAM	S6bPeer2	258	s6b3gppAaaServer.affirmed	0	3	3	0

Peer Stats

Use the following CLI command to display the number of inbound/outbound requests/responses, protocol errors and failures between the MCC PGW and the specified 3GPP AAA server.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-stats <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-stats S6BDIAM S6b Peer 1
realm                      s6b3gppAaaServer.affirmed
num-protocol-error         0
num-permanent-failure      0
num-routing-failure        0
num-out-request            58440
num-out-response           2662
num-in-request             2662
num-in-response            58440
```

Diameter Message-Level Statistics

Use the following CLI command to display statistics about diameter messages exchanged between the local diameter node and diameter peer via the S6b interface.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level peer
interface-type S6B <diameter-node-name> <peer-host-name>
```

Note: *For more information on S6b interface performance statistics, see Acuitas Performance Statistics.*

Message Definitions

Table 18-1 describes the message definitions for the S6b interface.

Table 18-1 S6b Interface Message Definitions

Command	Acronym	Command Code	Reference
AA-Request (page 18-26)	AAR	265	RFC 4005
AA-Answer (page 18-27)	AAA	265	RFC 4005
Abort-Session-Request (page 18-28)	ASR	274	RFC 3588
Abort-Session-Answer (page 18-28)	ASA	274	RFC 3588
Session-Termination-Request (page 18-29)	STR	275	RFC 3588
Session-Termination-Answer (page 18-29)	STA	275	RFC 3588
Re-Auth-Request (page 18-30)	RAR	258	RFC 4005
Re-Auth-Answer (page 18-30)	RAA	258	RFC 4005

Table 18-2 describes the special symbols in the AVP name for each message definition.

Table 18-2 AVP Name Symbols

Symbol	Description
<>	AVP is mandatory and the AVP with this symbol is at the beginning of a message.
{ }	AVP is mandatory.
[ ]	AVP is optional.
*[ ]	AVP is optional and is repeated.
M	If the Mandatory header bit is set, it indicates that support for this AVP is required. For further details, see RFC 3588.
V	If Vendor-Specific header bit is set, it indicates that the optional Vendor-ID field is present in the AVP header. For further details, see RFC 3588.

AA-Request

The AA-Request (AAR) command, indicated by the Command-Code field set to 265 and the "R" bit set in the Command Flags field, is sent from a PGW to a 3GPP AAA Server. The Command Code and ABNF are re-used from the IETF RFC 4005 [4] AA-Request command.

Message Format:

```
<AA-Request> ::= <Diameter Header: 265, REQ, PXY, 16777272>
                 <Session-Id>
                 {Auth-Application-Id}
                 {Origin-Host}
                 {Origin-Realm}
                 {Destination-Realm}
                 {Auth-Request-Type}
                 [Destination-Host]
                 [User-Name]
                 [Authorization-Lifetime]
                 [Auth-Grace-Period]
                 [MIP6-Agent-Info]
                 [MIP6-Feature-Vector]
                 [Visited-Network-Identifier]
                 [Service-Selection]
                 [Auth-Session-State]
                 [RAT-Type]
                 *[Proxy-Info]
                 *[Route-Record]
                 [Origination-TimeStamp]
                 [Max-Wait-Time]
                 *[AVP]
```

Table 18-3 lists the vendor-specific attributes (VSAs) for AAR messages.

Table 18-3 AAR Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	Format	Description
[Origination-Time-Stamp]	12951	7102	Unsigned64	Indicates the time that the message was sent, representing the number of seconds since January 1, 1900 00:00 UTC. This AVP is only provided with the initial authorization request.

Table 18-3 AAR Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	Format	Description
[Max-Wait-Time]	12951	7103	Unsigned32	Indicates the maximum amount of time (in milliseconds) that the originator is willing to wait for a response since the Origination-TimeStamp. This AVP is only provided with the initial authorization request.

AA-Answer

The AA-Answer (AAA) command, indicated by the Command-Code field set to 265 and the "R" bit cleared in the Command Flags field, is sent from a 3GPP Server to the PGW. The Command Code and ABNF are re-used from the IETF RFC 4005 [4] AA-Answer command.

Message Format:

```
<AA-Answer> ::= <Diameter Header: 265, REQ, PXY, 16777272>
                 <Session-Id>
                 {Auth-Application-Id}
                 {Auth-Request-Type}
                 {Result-Code}
                 {Origin-Host}
                 {Origin-Realm}
                 {Server-Name}
                 [User-Name]
                 *[Class]
                 [Error-Message]
                 [Error-Reporting-Host]
                 *[Failed-AVP]
                 [Idle-Timeout]
                 [Authorization-Lifetime]
                 [Auth-Grace-Period]
                 [Charging-Gateway-Function-Host]
                 [Charging-Group-ID]
                 [MIP6-Feature-Vector]
                 [Session-Timeout]
                 [Auth-Session-State]
                 [Framed-Pool]
                 [Framed-Ipv6-Pool]
                 [Subscription-Id]
                 [3GPP-SGSN-MCC-MNC]
                 [Served-Party-IP-Address]
```

```
*[Proxy-Info]
*[Route-Record]
[Virtual-APN-Name]
*[AVP]
```

Table 18-4 lists the vendor-specific attributes (VSAs) for AAA messages.

Table 18-4 AAA Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	Format	Description
[Virtual-APN-Name]	12951	6116	UTF8String	Used to deliver the virtual APN name of an enterprise subscriber.

Abort-Session-Request

The Abort-Session-Request (ASR) command, indicated by the Command-Code field set to 274 and the "R" bit set in the Command Flags field, is sent from a 3GPP AAA Server to a PGW. The ABNF is based on IETF RFC 4005 [4].

Message format:

```
<Abort-Session-Request> ::= <Diameter Header: 274, REQ, PXY, 16777272>
                 <Session-Id>
                 {Origin-Host}
                 {Origin-Realm}
                 {Destination-Realm}
                 {Destination-Host}
                 {Auth-Application-Id}
                 [Origin-State-Id]
                 *[Class]
                 *[Proxy-Info]
                 *[Route-Record]
                 [Auth-Session-State]
                 *[AVP]
```

Abort-Session-Answer

The Abort-Session-Answer (ASA) command, indicated by the Command-Code field set to 274 and the "R" bit cleared in the Command Flags field, is sent from a PGW to a 3GPP AAA Server. The ABNF is based on IETF RFC 4005 [4].

Message format:

```
<Abort-Session-Answer>  ::= <Diameter Header: 274, PXY, 16777264>
```

```
<Session-Id>
{Result-Code}
{Origin-Host}
{Origin-Realm}
[Error-Message]
[Error-Reporting-Host]
*[Failed-AVP]
*[Proxy-Info]
*[AVP]
```

Session-Termination-Request

The Session-Termination-Request (STR) command, indicated by the Command-Code field set to 275 and the "R" bit set in the Command Flags field, is sent from a PGW to a 3GPP AAA Server. The Command Code and ABNF are re-used from the IETF RFC 3588 [7] Session-Termination-Request command.

Message format:

```
<Session-Termination-Request> ::= <Diameter Header: 275, REQ, PXY,
16777272>
                <Session-Id>
                {Origin-Host}
                {Origin-Realm}
                {Destination-Realm}
                {Auth-Application-Id}
                {Termination-Cause}
                [User-Name]
                [Destination-Host]
                *[Class]
                *[Proxy-Info]
                *[Route-Record]
                *[AVP]
```

Session-Termination-Answer

The Session-Termination-Answer (STA) command, indicated by the Command-Code field set to 275 and the "R" bit cleared in the Command Flags field, is sent from a 3GPP AAA Server to a PGW. The Command Code and ABNF are re-used from the IETF RFC 3588 [7] Session-Termination-Answer command.

Message format:

```
<Session-Termination-Answer>::=<Diameter Header: 275, PXY, 16777272>
                <Session-Id>
                {Result-Code}
                {Origin-Host}
```

```
{Origin-Realm}
*[Class]
[Error-Message]
[Error-Reporting-Host]
*[Failed-AVP]
*[Proxy-Info]
*[Route-Record]
*[AVP]
```

Re-Auth-Request

The Diameter Re-Auth-Request (RAR) command, indicated by the Command-Code field set to 258 and the "R" bit set in the Command Flags field, is sent from a 3GPP AAA Server/Proxy to a PGW. The ABNF is based on IETF RFC 4005 [4].

Message format:

```
<Re-Auth-Request> ::= <Diameter Header: 258, REQ, PXY, 16777272>
       <Session-Id>
       {Origin-Host}
       {Origin-Realm}
       {Destination-Realm}
       {Destination-Host}
       {Auth-Application-Id}
       {Re-Auth-Request-Type}
       [Origin-State-Id]
       {RAR-Flags}
       *[Class]
       *[Proxy-Info]
       *[Route-Record]
       *[AVP]
```

Re-Auth-Answer

The Diameter Re-Auth-Answer (ASA) command, indicated by the Command-Code field set to 258 and the "R" bit cleared in the Command Flags field, is sent from a PGW to a 3GPP AAA Server/Proxy. The ABNF is based on IETF RFC 4005 [4].

Message format:

```
<Re-Auth-Answer> ::= <Diameter Header: 258, PXY, 16777272>
       <Session-Id>
       {Result-Code}
       {Origin-Host}
       {Origin-Realm}
       [Error-Message]
```

```
[Error-Reporting-Host]
*[Failed-AVP]
*[Class}
*[Proxy-Info]
*[AVP]
```

Chapter 19

Flow Authorization Interface

This section contains the following information used to configure, control, and monitor the Diameter-based flow authorization interface:

Interface Overview

The flow authorization interface can be used for subscriber authentication when a NAT IPv4 address is allocated for an IPv6 UE for the configured services/flows. The MCC flow authorization interface is a Diameter-based interface between the Packet Data Network Gateway (PGW) and Diameter server that processes AA-Request (AAR) messages and sends AA-Answer (AAA) messages back to the PGW.

If a NAT IPv4 address is allocated for a subscriber when flow authorization is enabled on the NAT service rule (associated with the subscriber-firewall-flow) and the subscriber's workflow control-profile is configured with a flow authorization interface, the PGW sends an AAR message to the Diameter server and waits for a response. All packets received on the connection are dropped until the PGW receives the AAA message from the Diameter server. If the authorization is successful, the PGW processes the packets on the flow.

If the authorization fails or the PGW does not receive a response within the configured time limit (retransmit-timeout), the PGW retransmits the request until the configured retry limit is reached (retransmit-count). If the configured retry limit is reached and the PGW has not received a response from the Diameter server, the PGW deletes the connection.

Note: *Affirmed Networks recommends using the default values for the retransmit-count (3 retransmit attempts) and retransmit-timeout (3000 milliseconds) parameters. The MCC has an internal timeout timer of 60 seconds after which any received AAA message is ignored. Therefore, the total timeout duration configured for the retransmit-count and retransmit-timeout should be less than 60 seconds. When using the default values for the retransmit-count and retransmit-timeout, the PGW flow authorization process will timeout after 12 seconds.*

The PGW supports 12000 concurrent flow authorization requests for subscribers. Currently, the flow authorization functionality is only supported when NAT is enabled in the services subscriber-firewall subscriber-firewall-flow (page 10-21) and the NAT pool associated with the subscriber firewall flow is configured as statistical 1:1 NAT (see page 8-18).

```
an3000-mcm-slot7cpu1(config)# services subscriber-firewall
subscriber-firewall-flow <service-flow-name> action nat
```

```
an3000-mcm-slot7cpu1(config)# network-context <name> nat-pool <name>
admin-state enabled nat-allocation-policy statistical nat-mode
One-to-One
```

Configuring the Flow Authorization Interface

To configure the flow authorization interface:

To identify packets that require flow authorization, enable flow authorization for the NAT service rule within the subscriber-firewall-flow. By default, flow-auth is disabled for the service rule.

```
an3000-mcm-slot7cpu1(config)# services subscriber-firewall
subscriber-firewall-flow <service-flow-name> service-rule
<service-rule-name> flow-auth enabled
```

1 Configure a local Diameter node on the MCC PGW for the flow authorization interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <diameter-node-name>
loopback-ip <loopback-name> transport [tcp|sctp] port
<signaling-port-number> network-context <network-context-name>
origin-host <local-host-name> origin-realm <local-realm>
```

where

Parameter	Description
node-admin	Specifies the name of the local Diameter node on the PGW.
loopback-ip	Specifies the name of the signaling port IP address. ***Note:*** *The network context and loopback IP address must already exist.*
transport	Specifies the transport type. ■ **tcp** (default) ■ **sctp**
port	Specifies the signaling port number. The default is 3868.
network-context	Specifies the associated local network context. Restricts the inbound Diameter messages by accepting only inbound messages on the IP interfaces in the interface group. ***Note:*** *The network context and loopback IP address must already exist.*
origin-host	Specifies the local host name in the Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the Diameter request message.
origin-realm	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message.

2 Configure the list of peers with which the local Diameter node communicates. Each Diameter node contains a list of peers (remote Diameter servers) with which the PGW communicates.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <diameter-node-name> peer
<peer-host-name> realm <peer-realm> primary-ip-address
<peer-ip-address> transport [tcp|sctp] port <peer-port-number> role
[primary|secondary] backup-host-name <secondary-peer-name> admin-state
[enabled|disabled]
```

where

Parameter	Description
peer	Specifies the peer host name Fully Qualified Domain Name (FQDN). The peer host name is used in the Destination-Host AVP to send Diameter request messages.
realm	Specifies the peer realm name. Realm is used in the Destination-Realm AVP in Diameter requests.
primary-ip-address	Specifies the IP address of the peer (can be an IPv4 or IPv6 address).
transport	Specifies the transport type. ■ **tcp** (default) ■ **sctp**
port	Specifies the signaling port number. The default is 3868.
role	Specifies the type of Diameter peer. When multiple primary, active peers are configured, the system uses a round-robin approach to select peers. ■ **Primary** – (default) Active peers used for Diameter sessions. ■ **Secondary** – Used for Diameter sessions, only when the primary peers are down. ***Note:*** *If failover to a secondary peer is supported, you must enter the host name of the secondary server in the* ***backup-host-name*** *parameter.*
backup-host-name	Identifies the secondary peer used during a failover from the primary to the secondary peer. ***Note:*** *This parameter is only allowed for the secondary peer and must be configured if failover is supported.*
admin-state	Specifies the admin state of the Diameter peer. When enabled, the system attempts to establish peer connection. ■ **enabled** (default) ■ **disabled**

3 Configure the Diameter data record templates used to include/exclude AVPs in the AAR/AAA messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pgw flow-auth diameter [aaa|aar] <template-name> <attribute-name>
presentFlag [yes|no|conditional]
```

where

Parameter	Description
`[aaa|aar]`	Specifies the type of data record template.
`<template-name>`	Specifies the name of the data record template.
`<attribute-name>`	Specifies the name of the AVP field you are modifying.
`presentFlag`	AAR template – Options include: ■ **`yes`** – (default) The field is sent in Diameter messages and processed when received. ■ **`no`** – The field is not sent in Diameter messages and is ignored when received. ■ **`conditional`** – The field is sent if it is available. AAA template – Options include: ■ **`yes`** – (default) The field is processed when received in the AAA message. ■ **`no`** – The field is ignored when received in the AAA message.

4 Configure the flow authorization interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
flow-authorization-interface <name>
```

where ***`<name>`*** specifies the flow-authorization interface name.

The following table describes the configurable parameters.

Parameter	Description
`admin-state`	Specifies the admin state of the interface. ■ **`enabled`** (default) ■ **`disabled`**
`auth-application-id`	Specifies the application ID (standard or non-standard). This is any value up to 32 bits. The default is `16779999`.
`retransmit-count`	Specifies the maximum number of retransmit attempts to the Diameter server. The default is `3` retransmit attempts. ***Note:*** *Affirmed Networks recommends using the default values for the* `retransmit-count` *and* `retransmit-timeout` *parameters. The MCC has an internal timeout timer of 60 seconds after which any received AAA message is ignored. Therefore, the total timeout duration configured for the* `retransmit-count` *and* `retransmit-timeout` *should be less than 60 seconds.*

Parameter	Description
retransmit-timeout	Specifies the time interval to wait before attempting to retransmit, in milliseconds. ***Note:*** *Affirmed Networks recommends using the default values for the* `retransmit-count` *and* `retransmit-timeout` *parameters. The MCC has an internal timeout timer of 60 seconds after which any received AAA message is ignored. Therefore, the total timeout duration configured for the* `retransmit-count` *and* `retransmit-timeout` *should be less than 60 seconds.*

5 Associate the Diameter server group and data record templates with the flow authorization interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
flow-authorization-interface <name> pgw diameter server-group
<service-profile> [aaa-template|aar-template] <template-name>
```

where

Parameter	Description
flow-authorization-interface	Specifies the flow authorization interface name.
server-group	Specifies the Service Profile for the selected streaming protocol.
aaa-template	Specifies the name of the AA-Answer data record template.
aar-template	Specifies the name of the AA-Request data record template.

6 Associate the flow authorization interface with a workflow Control profile. The Control profile must already exist.

```
an3000-mcm-slot8cpu1(config)# workflow control-profile <profile-name>
flow-authorization-interface <interface-name>
```

where

Parameter	Description
control-profile	Specifies the workflow control profile name.
flow-authorization-interface	Specifies the flow authorization interface name.

Viewing Flow Authorization Statistics

The PGW maintains statistics to display information about diameter messages (AAR/AAA) exchanged between the local Diameter node (PGW) and remote Diameter peer via the flow authorization interface.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level peer
interface-type FLOWAUTH <diameter-node-name> <peer-host-name>
```

The system displays output similar to the following:

```
num-aar-initial-attempted          0
num-aaa-initial-success            0
num-aaa-initial-failure            0
num-aar-initial-timeout            0
num-aar-initial-aborts             0
num-aaa-initial-rcvd-first-attempt 0
num-aaa-initial-rcvd-retx          0
num-times-peer-connection-down     0
num-times-peer-flow-off-received   0
```

The following show command displays the list of connection ID/NAT IPv4 address tuples.

```
an3000-mcm-slot7cpu1# show subscriber pdn-session
```

The following show command displays packet drops due to flow authorization at the data-path level.

```
an3000-mcm-slot7cpu1# show data-path statistics workflow data-stats |
nomore | include flow
```

The system displays output similar to the following:

```
uplink-flowauth-pkts-dropped             0
uplink-flowauth-pkts-init-dropped        0
uplink-flowauth-pkts-wait-dropped        0
downlink-flowauth-pkts-dropped           0
downlink-flowauth-pkts-init-dropped      0
downlink-flowauth-pkts-wait-dropped      0
```

The following show command displays flow authorization interface statistics at the module level.

```
an3000-mcm-slot7cpu1# show data-path statistics workflow module-stats
| nomore | include flow
```

The system displays output similar to the following:

```
flow-auth-reqs-sent                      0
flow-auth-rsp-rcvd                       0
flow-auth-rsp-failures                   0
```

```
flow-auth-rsp-success                    0
flow-auth-rsp-ignored                    0
flow-auth-rsp-timeouts                   0
```

The following CLI command displays procedure-level statistics. This CLI command is not applicable to the MCC GiGW.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
[cluster-level|grid-level|node-level] procedure | include flow
```

Message Definitions

Table 19-1 describes the supported message definitions for the flow authorization interface.

Table 19-1 Flow Authorization Interface Message Definitions

Command	Acronym	Command Code	Reference
AA-Request (page 19-8)	AAR	265	RFC 4005
AA-Answer (page 19-9)	AAA	265	RFC 4005

Table 19-2 describes the special symbols in the AVP name for each message definition.

Table 19-2 AVP Name Symbols

Symbol	Description
<>	AVP is mandatory and the AVP with this symbol is at the beginning of a message.
{ }	AVP is mandatory.
[ ]	AVP is optional.
* [ ]	AVP is optional and is repeated.

AA-Request

The AA-Request (AAR) command, indicated by the Command-Code field set to 265 and the "R" bit set in the Command Flags field, is sent from a PGW to a 3GPP AAA Server. The Command Code and ABNF are re-used from the IETF RFC 4005 [4] AA-Request command.

Message Format:

```
<AA-Request> ::= <Diameter Header: 265, REQ, PXY, 16777272>
                 <Session-Id>
```

```
{Auth-Application-Id}
{Origin-Host}
{Origin-Realm}
{Destination-Realm}
{Auth-Request-Type}
[Destination-Host]
[User-Name]
[Auth-Session-State]
[Framed-IP-Address]
*[Framed-IPv6-Prefix]
*[AVP]
```

AA-Answer

The AA-Answer (AAA) command, indicated by the Command-Code field set to 265 and the "R" bit cleared in the Command Flags field, is sent from a 3GPP Server to the PGW. The Command Code and ABNF are re-used from the IETF RFC 4005 [4] AA-Answer command.

Message Format:

```
<AA-Answer> ::= <Diameter Header: 265, REQ, PXY, 16777272>
                 <Session-Id>
                 {Auth-Application-Id}
                 {Auth-Request-Type}
                 {Result-Code}
                 {Origin-Host}
                 {Origin-Realm}
                 {Destination-Realm}
                 [Destination-Host]
                 *[AVP]
```

Chapter 20

Sd Interface

This section contains the following information used to control and monitor the Sd Interface:

Interface Overview

The Sd interface is located between the PCRF and the Traffic Detection Function (TDF) and is used to detect traffic and report information on the detected application traffic. The MCC supports the Sd interface between the PCRF and the GiGW (acting as a TDF). The GiGW has a Diameter session-learner interface (Sd interface) used to establish and terminate TDF sessions to the PCRF.

The MCC supports the following TDF functionality:

- Static, predefined, and dynamic Application Detection and Control (ADC) rule installation to support traffic detection.
- Reporting of detected applications to the PCRF based on the following supported event-triggers:
  - NO_EVENT_TRIGGERS (14)
  - OUT_OF_CREDIT (15)
  - REALLOCATION_OF_CREDIT (16)
  - REVALIDATION_TIMEOUT (17)

  - USAGE_REPORT (33)
  - APPLICATION_START (39)
  - APPLICATION_STOP (40)
  - Support for muting Application Start/Stop notifications about a specific detected application by providing the Mute-Notification AVP within the corresponding Charging-Rule-Definition or ADC-Rule-Definition AVP.

- Usage-Monitoring on installed ADC rules based on the monitoring-key received within the ADC-Rule-Install AVP.

- Time of day procedures:
  - Support for REVALIDATION_TIME_OUT procedures
  - Support for the rule activation/deactivation time as received within the ADC-Rule-Install AVP

- Support online/offline charging on installed ADC rules:
  - If online is enabled, the MCC initiates a session with the OCS to request a grant.
  - If offline is enabled, the MCC initiates a session with the OFCS (Gz interface) for offline statistics reporting.

- Application-based charging:

  A Rating Group can be associated with only one service rule with a protocol-application-rule-group that has only one application enabled. This method allows each identified application to be charged independently.

Note: *Static/predefined rating groups can be configured with a* `charging-method` *of* `offline`*, but when the MCC GiGW receives an ADC rule on the Sd interface with the Offline AVP disabled, the MCC does not support disabling the offline charging interface for the associated PCC rule.*

- Accumulated-usage reporting on ADC rules when the following occurs:
  - PCRF requests accumulated usage
  - Usage threshold is reached
  - Usage-monitoring is disabled
  - All rules associated with the monitoring-key are removed
  - TDF session is terminated

- Solicited application reporting on the Sd interface between the PCRF and GiGW (TDF), as defined in 3GPP 29.212, Release 15.3.0. *Unsolicited application reporting is not supported*.

__Note:__ Simultaneous support for the Sd Interface and Gx Interface is not supported.

Application Detection and Control on the Sd Interface

The Sd interface supports Application Detection and Control (ADC), as defined in 3GPP TS 29.212 Release 15.3.0. The ADC function enables the MCC GiGW (acting as a TDF) to detect applications and send notifications to the PCRF with application identifiers, flow information, and application Start/Stop events.

Based on the reported information, the PCRF can install, remove, or modify ADC rules, and the MCC can perform service control, charging, and bearer management (bearer creation, update, and deletion) based on new ADC rules.

The PCRF can instruct the MCC to detect applications using static, predefined, or dynamic rules:

- Static rules
  - Static rules are pre-configured on the MCC and are always active for the duration of the session (service rules with `always-on` activation).
  - Static rules support charging/metering/QoS when associated with QoS flows/rating groups using the workflow configuration.
- Predefined rules
  - The PCRF can install predefined ADC rules over the Sd interface using the ADC-Rule-Install during the TDF session setup/updates.
  - The PCRF can use the ADC-Rule-Install AVP in the TSR message to activate a predefined service rule on the MCC.
  - The predefined ADC rule must map to a service rule with a protocol-application-rule-group.
  - Predefined rules support charging/metering/QoS when associated with QoS flows/rating groups using the workflow configuration.
  - If the predefined ADC rule is associated with an online rating group, the MCC initiates the Gy session setup with the OCS for quota management.
- Dynamic rules
  - The PCRF can install dynamic ADC rules over the Sd interface using the ADC-Rule-Definition AVP during the TDF session setup/updates.
  - The PCRF includes the TDF-Application-Identifier AVP within the ADC-Rule-Definition AVP.

– If the dynamic ADC rule is associated with an online rating-group (based on the rating-group AVP information), the MCC initiates the Gy session setup with the OCS for quota management.

If any rule activation fails, the MCC generates a TSA/CCR-U message with the ADC-Rule-Report AVP. The PCRF can also mute application Start/Stop notifications about a specific detected application by providing the Mute-Notification AVP within the corresponding ADC-Rule-Definition AVP.

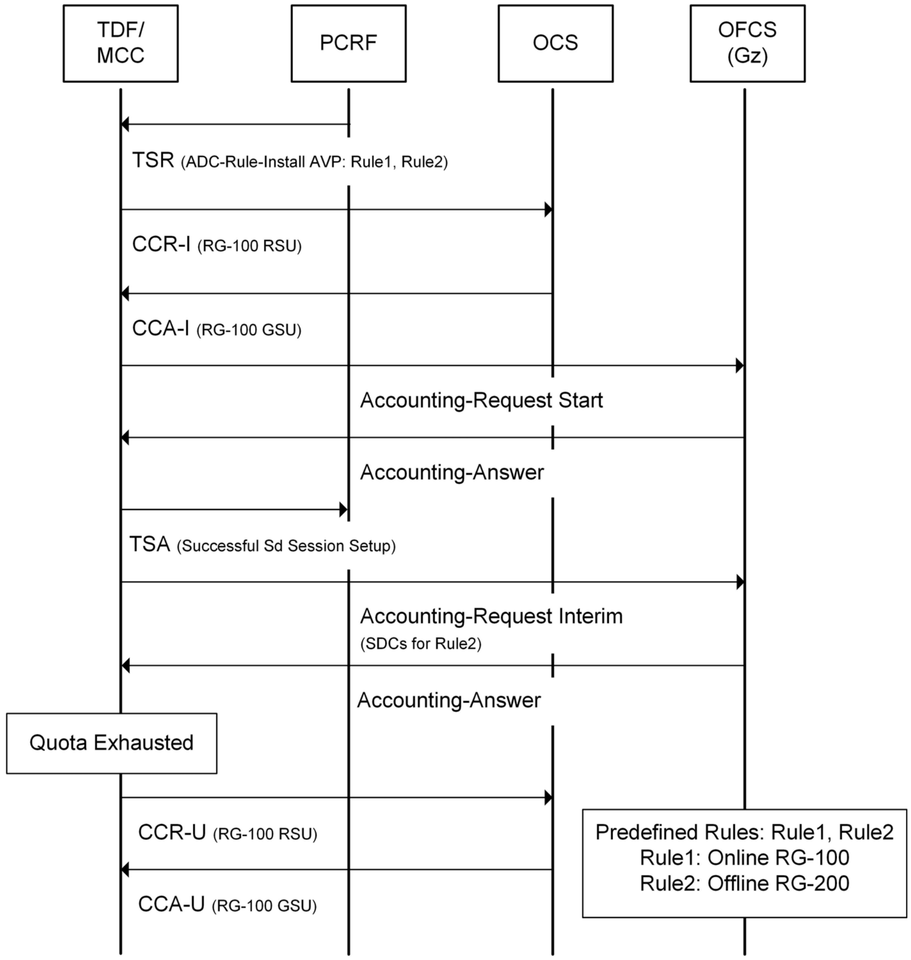
TDF Session Setup

The PCRF initiates an Sd session by sending the TDF-Session-Request (TSR) message with the PDN session information required to setup a subscriber session on the MCC GiGW (TDF).

The Called-Station-ID AVP, the UE IPv4 address within the Framed-IP-Address AVP and/or the UE IPv6 prefix within the Framed-Ipv6-Prefix AVP uniquely identify the session between the PCRF and the MCC GiGW (TDF).

Figure 20-1 illustrates the message flow for setting up a TDF session.

Figure 20-1 TDF Session Setup



Configuring the Sd Interface

This section provides the steps for configuring the Sd interface between the MCC GiGW (acting as a TDF) and PCRF.

1 Configure the local diameter node.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
```

where ***<node-name>*** specifies the name of the Diameter server group on the GiGW. Each Diameter server group contains a list of remote PCRFs with which the GiGW communicates.

Note: *The maximum number of supported Diameter server groups is 128 and the maximum number of supported diameter peers is 32 peers.*

The following table describes the configurable parameters.

Parameter	Description
`cea-timeout`	Specifies the wait time (in seconds) for receiving a Capability-Exchange-Answer (CEA).
`cer-origin-state-id-mode`	Specifies whether to increase the value of the Origin-State-Id AVP, sent in CER messages, for every new Diameter connection. ■ **`static`** – (default) The MCC updates the value of the Origin-State-Id AVP sent in CER messages with the static value, configured using the `cer-origin-state-id-static-value` parameter. ■ **`timestamp`** – For every new Diameter connection, the MCC updates the value of the Origin-State-Id AVP sent in CER messages with the current timestamp. The Diameter peer that receives the message can infer that sessions associated with a lower Origin-State-ID value are no longer active.
`cer-origin-state-id-static-value`	Specifies the static Origin-State-Id value sent in CER messages. The default value is 25. ***Note:*** *This parameter only applies when* `cer-origin-state-id-mode` *is set to* `static`*.*
`connection-attempt-count`	Specifies the maximum number of connection retry attempts from the PGW to the PCRF.
`connection-attempt-timeout`	Specifies the Tc timer, in seconds, as described in RFC 6733. Specifies how frequently to send transport connection attempts to a peer with whom no active transport connection exists.
`dscp-value`	Specifies the Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this Diameter server group.
`dwr-retry-count`	Specifies the maximum number of Device-Watchdog-Request (DWR) attempts to the diameter peer before declaring the peer down.
`dwr-retry-timer`	Specifies the number of seconds the diameter node waits for a Device-Watchdog-Answer (DWA) after retransmitting a DWR once the first dwr-timer expires. ***Note:*** *The* `dwr-retry-timer` *is typically set at a shorter interval than the* `dwr-timer`*.*
`dwr-timer`	Number of seconds the diameter node waits for a DWA after sending a DWR to the diameter peer. ***Note:*** *When the* `dwr-timer` *expires, the diameter node retransmits the DWR to the diameter peer and the* `dwr-retry-timer` *is started.*
`loopback-ip`	Specifies the signaling port IP address.

Parameter	Description
`network-context`	Specifies the network context, selected from the VPN Group list. Restricts the inbound diameter messages by accepting only inbound messages on the IP interfaces in the interface group. The network context must already exist.
`node-role`	Specifies the local node role: ■ **`client`** ■ **`server`**
`origin-host`	Specifies the local host name in Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the request message. ***Note:*** *You can configure multiple Diameter server groups with the same origin host.*
`origin-realm`	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message. ***Note:*** *You can configure multiple Diameter server groups with the same origin realm.*
`port`	Specifies the TCP listening port number used when `node-role` is set to `server`. ***Note:*** *This is the TCP listening port, not the source port used when `node-role` is set to `client`. The source port is dynamically assigned and is not configurable.*
`product-name`	Specifies the diameter product name.
`transport`	Specifies the transport type. ■ **`tcp`** (default) ■ **`sctp`** – To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the Gx interface, you must configure an SCTP profile and associate the profile with the local Diameter node. ***Note:*** *The Sd interface supports TCP only.*
`vendor-id`	Specifies the diameter vendor ID.

2 Configure the list of peers with which the diameter local node (on the GiGW) communicates.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name>
```

The following table describes the configurable parameters.

Parameter	Description
`node-admin`	Specifies the name of the Diameter server group on the GiGW.

Parameter	Description
peer	Specifies the peer host name. Each peer is configured with a unique peer host name. The peer host name is used by the MCC for load balancing between peers and for statistics.
admin-state	Specifies the administrative state of the diameter peer. When enabled, the system attempts to establish peer connection. ■ **enabled** (default) ■ **disabled**
backup-peer-name	Identifies the secondary peer used during a failover from the primary to the secondary peer. ***Note:*** *Only primary peers can be configured with a backup peer. If failover to a secondary peer is supported, this parameter must be configured with the host name of a secondary peer.*
override-dest-hostname	Specifies the host name used to override the configured peer host name in the Destination-Host AVP within CCR-I messages. ***Note:*** *Does not apply to the Sd interface.*
peer-identifier	Specifies the external identifier for this peer.
dest-port	Specifies the signaling port number.
dest-ip-address	Specifies the IP address of the peer (can be an IPv4 or IPv6 address).
dest-realm	Specifies the peer realm name. Realm is used in the Destination-Realm AVP in diameter requests.
origin-host	Specifies the local host name in Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the request message. ***Note:*** *The origin-host configured at the peer level overrides the origin-host configured at the Diameter server-group level.*
origin-realm	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message. ***Note:*** *The origin-realm configured at the peer level overrides the origin-realm configured at the Diameter server-group level.*
role	Specifies the type of peer. When multiple primary, active peers are configured, the system uses a round-robin approach to select peers. ■ **Primary** – (default) Active peers used for diameter sessions. ■ **Secondary** – Used for diameter sessions, only when the primary peers are down. ***Note:*** *If failover to a secondary peer is supported, you must enter the host name of the secondary server in the* *backup-peer-name* *parameter.*

Parameter	Description
`secondary-loopback-ip`	Specifies a secondary loopback IP address for the Diameter peer to support SCTP multi-homing. ***Note:*** *The* `secondary-loopback-ip` *parameter is only visible when the local Diameter node transport type is set to SCTP.*
`weight`	Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions. ***Note:*** *Applies to Gy/Gx/Gz/STa/Swm/S6b interfaces. Does not apply to the Sd interface.*

3 Configure the data record templates to filter optional diameter AVPs in CCR, RAA, and TSA messages transmitted between the GiGW (TDF) and PCRF via the Sd interface.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pgw pcrf diameter [stop|interim|raa|tsa] <template-name>
<attribute-name> presentFlag [yes|no|conditional]
```

where

Parameter	Description
`[pgw|ggsn]`	Specifies the gateway type for the PCRF charging interface.
`[stop|interim|raa|tsa]`	Specifies the type of data record template.
`<template-name>`	Specifies the name of the data record template.
`<attribute-name>`	Specifies the name of the AVP field you are modifying.
`presentFlag`	CCR Stop/Interim and TSA templates: ■ **`yes`** – Indicates the field is sent. ■ **`no`** – Indicates the field is not sent. ■ **`conditional`** – Indicates the field is sent if it is available. RAA template: ■ **`yes`** – Indicates the field is processed when received in the CCA/RAA response. ■ **`no`** – Indicates the field is ignored when received in the CCA/RAA response.

4 Configure the Sd interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface sd-interface
<interface-name>
```

The following table describes the configurable parameters.

Parameter	Description
sd-interface	Specifies the name of the Sd interface on the GiGW.
admin-state	Specifies the administrative state of the Sd interface. ■ **enabled** (default) ■ **disabled**
retransmit-count	Specifies the maximum number of times the GiGW transmits the CCR to the PCRF.
retransmit-timeout	Specifies the time interval, in milliseconds, before the GiGW retransmits the CCR to the PCRF.

5 Associate the Diameter server group and CCR/RAA/TSA data record templates with the Sd interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface sd-interface
<interface-name> server-group <diameter-server-group> sd-template
interim-template <template-name> stop-template <template-name>
raa-template <template-name> tsa-template <template-name>
```

where

Parameter	Description
sd-interface	Specifies the name of the Sd interface on the GiGW.
server-group	Specifies the name of the Diameter server group on the GiGW.
sd-template	Specifies the data record template name: ■ **interim-template** – Credit-Control-Request (CCR) Interim template. ■ **stop-template** – Specifies the CCR Stop template. ■ **raa-template** – Specifies the Re-Auth-Answer (RAA) template. ■ **tsa-template** – Specifies the TDF-Session-Answer (TSA) template.

6 Configure the Sd session learner interface on the GiGW.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
gw-session-learner sd-diameter-client
<sd-diameter-client-interface-name> admin-state [enabled|disabled]
sd-interface <sd-diameter-server-interface-name>
```

where

Parameter	Description
sd-diameter-client	Specifies the name of the Sd diameter client session learner interface on the GiGW.
admin-state	Specifies whether to enable or disable the administrative state for this interface. ■ **enabled** ■ **disabled**
sd-interface	Specifies the name of the Sd interface on the GiGW.

7 Associate the Sd session learner interface with the GiGW service. Each GiGW service must be associated with only one Sd session learner interface.

```
an3000-mcm-slot7cpu1(config)# zone default gateway mcc-gigw
<gigw-service-name> session-learner-interface sd-diameter-client
<sd-diameter-client-interface-name>
```

where

Parameter	Description
mcc-gigw	Specifies the name of the MCC GiGW Service. The MCC GiGW service must already exist.
sd-diameter-client	Specifies the name of the Sd diameter client session learner interface on the GiGW.

Message Definitions

Table 20-1 lists the message definitions for the Sd interface.

Table 20-1 Sd Interface Message Definitions

Command	Acronym	Command Code	Reference
Capabilities-Exchange-Request	CER	257	See the *Affirmed Networks Gx Interface Specification*.
Capabilities-Exchange-Answer	CEA	257	See the *Affirmed Networks Gx Interface Specification*.
Credit-Control-Request	CCR	272	See the *Affirmed Networks Gx Interface Specification*.
Credit-Control-Answer	CCA	272	See the *Affirmed Networks Gx Interface Specification*.
Disconnect-Peer-Request	DPR	282	See the *Affirmed Networks Gx Interface Specification*.
Disconnect-Peer-Answer	DPA	282	See the *Affirmed Networks Gx Interface Specification*.
Device-Watchdog-Request	DWR	280	See the *Affirmed Networks Gx Interface Specification*.
Device-Watchdog-Answer	DWA	280	See the *Affirmed Networks Gx Interface Specification*.
Re-Auth-Request	RAR	258	See the *Affirmed Networks Gx Interface Specification*.
Re-Auth-Answer	RAA	258	See the *Affirmed Networks Gx Interface Specification*.
TDF-Session-Request (page 20-13)	TSR	8388637	3GPP TS 29.212 V15.3.0
TDF-Session-Answer (page 20-14)	TSA	8388637	3GPP TS 29.212 V15.3.0

Table 20-2 describes the special symbols in the AVP name for each message definition.

Table 20-2 AVP Name Symbols

Symbol	Description
<>	AVP is mandatory and the AVP with this symbol is at the beginning of a message.
{ }	AVP is mandatory.
[ ]	AVP is optional.
* [ ]	AVP is optional and is repeated.

TDF-Session-Request

The TDF-Session-Request (TSR) command is sent by the PCRF to the TDF in order to establish the TDF session and to provision the ADC rules. It may also include the requested event triggers. See the *Affirmed Networks Gx Interface Specification* for more information on the AVPs supported in this command.

TSR Message Format:

```
<TS-Request> ::= < Diameter Header: 8388637, REQ, PXY >
                 <Session-Id>
                 {Origin-Host}
                 {Origin-Realm}
                 {Destination-Realm}
                 [Destination-Host]
                 [Origin-State-Id]
                 *[Subscription-Id]
                 [Framed-IP-Address]
                 [Framed-Ipv6-Prefix]
                 [IP-CAN-Type]
                 [RAT-Type]
                 [User-Equipment-Info]
                 [QoS-Information]
                 0*2 [AN-GW-Address]
                 [3GPP-SGSN-Address]
                 [3GPP-SGSN-Ipv6-Address]
                 [3GPP-GGSN-Address]
                 [3GPP-GGSN-Ipv6-Address]
                 [Called-Station-ID]
                 [3GPP-Selection-Mode]
                 [PDN-Connection-Charging-ID]
                 [3GPP-SGSN-MCC-MNC]
                 [RAI]
                 [3GPP-User-Location-Info]
                 [3GPP-MS-TimeZone]
                 [3GPP-Charging-Characteristics]
                 [Charging-Information]
                 *[ADC-Rule-Install]
                 [Revalidation-Time]
                 *[Usage-Monitoring-Information]
                 *[Event-Trigger]
                 *[AVP]
```

TDF-Session-Answer

The TDF-Session-Answer (TSA) command is sent by the TDF to the PCRF in response to the TSR command. See the *Affirmed Networks Gx Interface Specification* for more information on the AVPs supported in this command.

TSA Message Format:

```
<TS-Answer> ::=  < Diameter Header: 8388637, PXY >
                 <Session-Id>
                 {Vendor-Specific-Application-Id}
                 {Origin-Host}
                 {Origin-Realm}
                 [Result-Code]
                 [Origin-State-Id]
                 *[ADC-Rule-Report]
                 *[AVP]
```

Diameter Message-Level Statistics

Use the following CLI command to display statistics about Diameter messages exchanged between the PCRF and MCC GiGW via the Sd interface.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level peer
interface-type SD <diameter-node-name>
<sd-diameter-client-interface-name>
```

The system displays output similar to the following:

```
num-tsr-received                    0
num-tsa-sent                        0
num-ccr-update-attempted            0
num-cca-update-success              0
num-cca-update-failure              0
num-cca-update-timeout              0
num-cca-update-aborts               0
num-ccr-terminate-attempted         0
num-cca-terminate-success           0
num-ccr-terminate-failure           0
num-ccr-terminate-timeout           0
num-ccr-terminate-aborted           0
num-rar-received                    0
num-raa-success                     0
num-raa-failures                    0
num-raa-send-failures               0
```

```
num-raa-failures-session-not-found   0
num-times-peer-connection-down       0
num-times-peer-flow-off-received     0
```

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